

The Collected Mathematical Papers of
Arthur Cayley
Vol. X

Arthur Cayley

Modern L^AT_EX reconstruction

CAMBRIDGE
PRINTED BY J. AND C. F. CLAY,
AT THE UNIVERSITY PRESS.

ADVERTISEMENT.

THE present volume contains 76 papers, numbered 630 to 705, published for the most part in the years 1876 to 1880.

The Table for the ten volumes is

Vol. I.	Numbers 1 to	100,
" II.	" 101	158,
" III.	" 159	222,
" IV.	" 223	299,
" V.	" 300	383,
" VI.	" 384	416,
" VII.	, 417	485,
" VIII.	, 486	555,
" IX.	" 556	629,
" X.	" 630	705.

A. R. FORSYTH.

11 June, 1896.

CONTENTS.

[An Asterisk means that the paper is not printed in full.]

	PAGE
630. On an expression for $1 \pm \sin(2p + 1)u$ in terms of $\sin u$ Messenger of Mathematics, t. v. (1876), pp. 7, 8	
631. Synopsis of the theory of equations Messenger of Mathematics, t. v. (1876), pp. 39—49	
632. On Aronhold's integration-formula Messenger of Mathematics, t. v. (1876), pp. 88—90	
*633. Note on Mr. Martin's paper "On the integrals of some differentials" Messenger of Mathematics, t. v. (1876), p. 163	
634. Theorems in trigonometry and on partitions Messenger of Mathematics, t. v. (1876), p. 164, p. 188	
635. Note on the demonstration of Clairaut's theorem Messenger of Mathematics, t. v. (1876), pp. 166, 167	
636. On the theory of the singular solutions of differential equations of the first order Messenger of Mathematics, t. vi. (1877), pp. 23—27	
637. On a differential equation in the theory of elliptic functions Messenger of Mathematics, t. vi. (1877), p. 29	
638. On a q -formula leading to an expression for E_1 Messenger of Mathematics, t. vi. (1877), pp. 63—66	
639. An elementary construction in optics Messenger of Mathematics, t. vi. (1877), pp. 81, 82	
*640. Further note on Mr. Martin's paper Messenger of Mathematics, t. vi. (1877), pp. 82, 83	
641. On the flexure of a spherical surface Messenger of Mathematics, t. vi. (1877), pp. 88—90	
642. On a differential relation between the sides of a quadrangle Messenger of Mathematics, t. vi. (1877), pp. 99—101	
643. On a quartic curve with two odd branches	

-
- Messenger of Mathematics, t. vi. (1877), pp. 107, 108
644. Note on magic squares
 Messenger of Mathematics, t. vi. (1877), p. 168
645. A Smith's Prize Paper, 1877
 Messenger of Mathematics, t. vi. (1877), pp. 173—182
646. On the general equation of differences of the second order
 Quart. Math. Journ., t. xiv. (1877), pp. 23—25
647. On the quartic surfaces represented by the equation, symmetrical
 determinant = 0
 Quart. Math. Journ., t. xiv. (1877), pp. 46—52
648. Algebraical theorem
 Quart. Math. Journ., t. xiv. (1877), p. 53
649. Addition to Mr. Glaisher's Note on Sylvester's paper "Development
 of an idea of Eisenstein"
 Quart. Math. Journ., t. xiv. (1877), pp. 83, 84
650. On a quartic surface with twelve nodes
 Quart. Math. Journ., t. xiv. (1877), pp. 103—106
651. On a special surface of minimum area
 Quart. Math. Journ., t. xiv. (1877), pp. 190—196
652. On a sextic torse
 Quart. Math. Journ., t. xiv. (1877), pp. 229—235
653. On a torse depending on the elliptic functions
 Quart. Math. Journ., t. xiv. (1877), pp. 235—241
654. On certain octic surfaces
 Quart. Math. Journ., t. xiv. (1877), pp. 249—264
655. A memoir on differential equations
 Quart. Math. Journ., t. xiv. (1877), pp. 292—339
656. On the theory of partial differential equations
 Mathematische Annalen, t. xi. (1877), pp. 194—198
657. Note on the theory of elliptic integrals
 Mathematische Annalen, t. xii. (1877), pp. 143—146
658. On some formulæ in elliptic integrals
-

		PAC
	Mathematische Annalen, t. xii. (1877), pp. 369—374	
659.	A theorem on groups Mathematische Annalen, t. xiii. (1878), pp. 561—565	14
660.	On the correspondence of homographies and rotations Mathematische Annalen, t. xv. (1879), pp. 238—240	15
661.	On the double ϑ -functions Proc. Lond. Math. Soc., t. ix. (1878), pp. 29, 30	15
662.	On the double ϑ -functions in connexion with a 16-nodal quartic surface Crelle's Journal der Mathem., t. lxxxiii. (1877), pp. 210—219	15
663.	Further investigations on the double ϑ -functions Crelle's Journal der Mathem., t. lxxxiii. (1877), pp. 220—233	16
664.	On the 16-nodal quartic surface Crelle's Journal der Mathem., t. lxxxiv. (1878), pp. 238—241	18
665.	A memoir on the double ϑ -functions Crelle's Journal der Mathem., t. lxxxv. (1878), pp. 214—245	18
666.	Sur un exemple de réduction d'intégrales abéliennes aux fonctions elliptiques Comptes Rendus, t. LXXXV. (1877), pp. 265—268, 373—376, 426—429, 472—475	20

CONTENTS—(continued).

	PAGE
667. On the bicircular quartic—Addition to Professor Casey's memoir: "On a new form of tangential equation" Phil. Trans., t. 167 (for 1877), pp. 441—460	223
668. On compound combinations Proceedings of the Lit. Phil. Soc. Manchester, t. xvi. (1877), pp. 113, 114; Memoirs, Ser. iii., t. vi. (1879), pp. 99, 100	243
669. On a problem of arrangements Edin. Roy. Soc. Proc., t. ix. (1878), pp. 338—342	243
670. [Note on Mr. Muir's solution of a "problem of arrangement"] Edin. Roy. Soc. Proc., t. ix. (1878), pp. 388—391	243
671. On a sibi-reciprocal surface Berlin, Akad. Monatsber., (1878), pp. 309—313	253
672. On the game of mousetrap Quart. Math. Journ., t. xv. (1878), pp. 8—10	253
673. Note on the theory of correspondence Quart. Math. Journ., t. xv. (1878), pp. 32, 33	253
674. Note on the construction of Cartesians Quart. Math. Journ., t. xv. (1878), p. 34	263
675. On the reflexnodal planes of a surface Quart. Math. Journ., t. xv. (1878), pp. 49—51	263
676. Note on a theorem in determinants Quart. Math. Journ., t. xv. (1878), pp. 55—57	263
677. [Addition to Mr. Glaisher's paper "Proof of Stirling's theorem"] Quart. Math. Journ., t. xv. (1878), pp. 63, 64	263
678. On a system of quadric surfaces Quart. Math. Journ., t. xv. (1878), pp. 124, 125	263
679. On the regular solids Quart. Math. Journ., t. xv. (1878), pp. 127—131	273

	PAGE
680. On the Hessian of a quartic surface Quart. Math. Journ., t. xv. (1878), pp. 141—144	27
681. On the derivatives of three binary quantics Quart. Math. Journ., t. xv. (1878), pp. 157—168	27
682. Formulæ relating to the right line Quart. Math. Journ., t. xv. (1878), pp. 169—171	28
683. On the function $\arcsin(x + iy)$ Quart. Math. Journ., t. xv. (1878), pp. 171—174	29
684. On a relation between certain products of differences Quart. Math. Journ., t. xv. (1878), pp. 174, 175	29
685. On Mr. Cotterill's goniometrical problem Quart. Math. Journ., t. xv. (1878), pp. 196—198	29
686. On a functional equation Quart. Math. Journ., t. xv. (1878), pp. 315—325; Proc. Lond. Math. Soc., t. ix. (1878), p. 29	29
687. Note on the function $\psi(x) = a - (c - x)^2\{c(c - x) - b^2\}$ Quart. Math. Journ., t. xv. (1878), pp. 338—340	30
688. Geometrical considerations on a solar eclipse Quart. Math. Journ., t. xv. (1878), pp. 340—347	31
689. On the geometrical representation of imaginary variables by a real correspondence of two planes Proc. Lond. Math. Soc., t. ix. (1878), pp. 31—39	31
690. On the theory of groups Proc. Lond. Math. Soc., t. ix. (1878), pp. 126—133	32
691. Note on Mr. Monro's paper "On flexure of spaces" Proc. Lond. Math. Soc., t. ix. (1878), pp. 171, 172	33
692. Addition to [578] memoir on the transformation of elliptic functions Phil. Trans., vol. 169, Part ii. (for 1878), pp. 419—424	33
693. A tenth memoir on quantics Phil. Trans., vol. 169, Part ii. (for 1878), pp. 603—661	33
694. Desiderata and Suggestions No. 1. The theory of groups;	40

	PAGE
American Journal of Mathematics, t. i. (1878), pp. 50—52	
No. 2. The theory of groups; graphical representation;	
American Journal of Mathematics, t. i. (1878), pp. 174—176	
No. 3. The Newton-Fourier imaginary problem;	
American Journal of Mathematics, t. ii. (1879), p. 97	
No. 4. The mechanical construction of conformable figures;	
American Journal of Mathematics, t. ii. (1879), p. 186	
695. A link-work for σ : extract from a letter to Mr. Sylvester	407
American Journal of Mathematics, t. i. (1878), p. 386	
696. Calculation of the minimum N.G.F. of the binary seventhic	408
American Journal of Mathematics, t. ii. (1879), pp. 71—84	
697. On the double ϑ -functions	422
Crelle's Journal der Mathem., t. lxxxvii. (1879), pp. 74—81	
698. On a theorem relating to covariants	430
Crelle's Journal der Mathem., t. lxxxvii. (1879), pp. 82, 83	
699. On the triple ϑ -functions	432
Crelle's Journal der Mathem., t. lxxxvii. (1879), pp. 134—138	
700. On the tetrahedroid as a particular case of the 16-nodal quartic surface	433
Crelle's Journal der Mathem., t. lxxxvii. (1879), pp. 161—164	
701. Algorithm for the characteristics of the triple ϑ -functions	440
Crelle's Journal der Mathem., t. lxxxvii. (1879), pp. 165—169	
702. On the triple ϑ -functions	440
Crelle's Journal der Mathem., t. lxxxvii. (1879), pp. 190—198	
703. On the addition of the double ϑ -functions	453
Crelle's Journal der Mathem., t. lxxxviii. (1880), pp. 74—81	
704. A memoir on the single and double theta-functions	463
Phil. Trans., vol. 171, Part iii. (for 1880), pp. 897—1002	
705. Problems and Solutions	560
Mathematical Questions with their Solutions from the	
Educational Times, vols. xiv. to LXI. (1871—1894);	
for contents, see p. 615	

CLASSIFICATION.

Geometry:

- Geometrical constructions: (i) in optics, 639.
(ii) for solar eclipses, 688.
- Quartic curves, 643, 667.
- Construction of Cartesians, 674.
- Correspondence, 673, 683, 689, 694.
- Quadric surfaces, 678.
- Quartic surfaces, 647, 650, 662, 664, 680, 700.
- Sextic and other torses, 652, 653.
- Octic surfaces, 654, 671.
- Line-geometry, 682.
- Reflexnodal planes, 675.
- Flexure of surfaces, 641, 691.
- Minimal surface, 651.
- Regular solids, 679.

Analysis:

- Trigonometry, 630, 634.
- Theory of equations, 631, 694.
- Groups, 659, 660, 690, 694.
- Combinatory analysis, 634, 644, 668, 669, 670, 672.
- Invariants and covariants, 681, 696, 698.
- Quantics, 693.
- Integration, 632, 633, 640.

- Differential equations, 636, 637, 642, 655, 656.
- Finite differences, 646, 677, 686, 687.
- Elliptic functions and elliptic integrals, 637, 638, 653, 657, 658, 692.
- Double theta functions, 661, 662, 663, 665, 697, 703, 704.
- Triple theta functions, 699, 701, 702.
- Reduction of transcendental integrals, 666.
- Problems and solutions, 645, 705.
- Miscellaneous, 635, 648, 649, 676, 684, 685, 695.

630.

ON AN EXPRESSION FOR $1 \pm \sin(2p+1)u$ IN TERMS
OF $\sin u$.

[From the Messenger of Mathematics, vol. v. (1876), pp. 7, 8.]

Write $\sin u = x$, then we have

$$\begin{aligned}\sin u &= x, & \cos u &= \sqrt{(1-x^2)}, \\ \sin 3u &= 3x - 4x^3, & \cos 3u &= (1-4x^2)\sqrt{(1-x^2)}, \\ \sin 5u &= 5x - 20x^3 + 16x^5, & \cos 5u &= (1-12x^2+16x^4)\sqrt{(1-x^2)}, \\ & \&c.\&c.\end{aligned}$$

It is hence clear, that in general

$$\begin{aligned}1 - \sin(2p+1)u &= (1 \pm x)\{(1, x)^p\}^2, \\ 1 + \sin(2p+1)u &= (1 \mp x)\{(1, -x)^p\}^2,\end{aligned}$$

where $(1, x)^p$ denotes a rational and integral function of x of the order p , and $(1, -x)^p$ the same function of $-x$; for it is only in this manner that we can have

$$\cos^2(2p+1)u = (1-x^2)\{[1, x]^p\}^2.$$

We, in fact, find

$$\begin{aligned}1 + \sin u &= 1 + x, \\ 1 - \sin 3u &= (1+x)(1-2x)^2, \\ 1 + \sin 5u &= (1+x)(1+2x-4x^2)^2, \\ 1 - \sin 7u &= (1+x)(1-4x-4x^2+8x^3)^2, \\ &\&c.\end{aligned}$$

and it thus appears that the form is

$$1 + (-)^p \sin(2p+1)u = (1+x)\{(1, x)^p\}^2.$$

To find herein the expression of the factor $(1, x)^p$, write $u = \frac{1}{2}\pi - \theta$ and consequently $x = \cos \theta$; we have therefore

$$1 + \cos(2p + 1)\theta = (1 + \cos \theta) \{(1, x)^p\}^2,$$

where in the second factor on the right-hand side x is retained to stand for its value $\cos \theta$. This gives

$$2 \cos^2 \left(p + \frac{1}{2}\right) \theta = 2 \cos^2 \frac{1}{2} \theta \cdot \{(1, x)^p\}^2,$$

or, what is the same thing,

$$(1, x)^p = \frac{\cos(p + \frac{1}{2})\theta}{\cos \frac{1}{2}\theta},$$

viz. this is

$$= \cos p\theta - \sin p\theta \cdot \frac{\sin \frac{1}{2}\theta}{\cos \frac{1}{2}\theta},$$

which is

$$= \cos p\theta - \sin p\theta \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}.$$

We have

$$\cos p\theta + i \sin p\theta = \{x + i\sqrt{(1 - x^2)}\}^p = X + i\sqrt{(1 - x^2)} Y, \text{ suppose,}$$

where X, Y are rational and integral functions of x of the orders p and $p - 1$ respectively; that is,

$$\cos p\theta = X, \quad \sin p\theta = \sin \theta \cdot Y,$$

and we have therefore

$$(1, x)^p = X - Y(1 - x),$$

which is the required expression for $(1, x)^p$. For instance

$$p = 3,$$

$$X + i\sqrt{(1 - x^2)} Y = \{x + i\sqrt{(1 - x^2)}\}^3;$$

that is,

$$X = -3x + 4x^3, \quad Y = -1 + 4x^2,$$

and $\therefore -(1 - x)Y = 1 - x - 4x^2 + 4x^3$, so that

$$X - (1 - x)Y = 1 - 4x - 4x^2 + 8x^3 = (1, x)^3,$$

and hence

$$1 - \sin 7u = (1 + x)(1 - 4x - 4x^2 + 8x^3)^2,$$

which agrees with a result already obtained.

The foregoing value of $(1, x)^p$ may also be written

$$(1, x)^p = \frac{1}{x+1} \left\{ \left(\frac{x+1}{2} \right)^{p+1} - \left(\frac{x-1}{2} \right)^{p+1} \right\},$$

which however is not practically so convenient.

The formula corresponds to a like formula in elliptic functions, viz. writing $\operatorname{sn} u = x$, the numerator of $1 + (-)^p \operatorname{sn}(2p+1)u$ is

$$= (1+x) \{(1, x)^p\}^2,$$

which is $(1+x)$ multiplied by the square of a rational and integral function of x .

631.

SYNOPSIS OF THE THEORY OF EQUATIONS.

[From the Messenger of Mathematics, vol. v. (1876), pp. 39—49.]

The following was proposed, as one of the subjects of a Dissertation for the Trinity Fellowships:

Synopsis of the theory of equations; i.e. a statement in a logical order, of the divisions of the subject and the leading questions and theorems, but without demonstrations.

In the subject "Theory of Equations," the term *equation* is used to denote an equation of the form $x^n - p_1x^{n-1} + \dots + p_n = 0$, where $p_1, p_2, \dots, p_n$ are regarded as known, and x as a quantity to be determined; for shortness, the equation is written $f(x) = 0$.

The equation may be *numerical*; that is, the coefficients $p_1, p_2, \dots, p_n$ are then numbers; understanding by number, a quantity of the form $\alpha + \beta i$, where α and β have any positive or negative real values whatever; or say, each of these is regarded as susceptible of continuous variation from an indefinitely large negative to an indefinitely large positive value: and i denotes $\sqrt{-1}$.

Or the equation may be *algebraic*; viz. the coefficients are then not restricted to denote, or are not explicitly considered as denoting, numbers.

I. We consider first numerical equations.

A number α (real or imaginary), such that substituted for x it makes the function $x^n - p_1x^{n-1} + \dots \pm p_n$ to be $= 0$, or say, such that it satisfies the equation, is said to be a *root* of the equation; viz. α being a root, we have

$$\alpha^n - p_1\alpha^{n-1} + \dots \pm p_n = 0, \text{ or say } f(\alpha) = 0;$$

and it is then shown that $x - \alpha$ is a factor of the function $f(x)$, viz. that we have

$$f(x) = (x - \alpha)f_1(x),$$

where $f_1(x)$ is a function $x^{n-1} - q_1x^{n-2} + \dots \pm q_{n-1}$, of the order $n - 1$, with numerical coefficients $q_1, q_2, \dots, q_{n-1}$.

In general, α is *not* a root of the equation $f_1(x) = 0$; but it may be so, viz. $f_1(x)$ may contain the factor $x - \alpha$; when this is so, $f(x)$ will contain the factor $(x - \alpha)^2$; writing then $f(x) = (x - \alpha)^2 f_2(x)$,

and assuming that α is not a root of the equation $f_2(x) = 0$, $x = \alpha$ is then said to be a *double root* of the equation. Similarly, $f(x)$ may contain the factor $(x - \alpha)^3$ and no higher power, and then $x = \alpha$ is said to be a *triple root*; and so on.

Supposing, in general, that $f(x) = (x - \alpha)^\alpha F(x)$, where α is a positive integer which may be $= 1$, and $F(x)$ is of the order $n - \alpha$, then if β is a root different from α , we shall have $x - \beta$ a factor (in general a simple one, but it may be a multiple one) of $F(x)$, and $f(x)$ will in this case become

$$f(x) = (x - \alpha)^\alpha (x - \beta)^\beta \Phi(x),$$

where β is a positive integer which may be $= 1$, and $\Phi(x)$ is of the order $n - \alpha - \beta$. The original equation $fx = 0$ is in this case said to have α roots each $= \alpha$, β roots each $= \beta$, and so on.

We have the theorem, a numerical equation of the order n has in every case n roots, viz. there exist n numbers $\alpha, \beta, \dots$ (in general, all of them distinct, but they may arrange themselves in groups of equal values) such that

$$f(x) = (x - \alpha)(x - \beta)(x - \gamma) \dots \quad \text{identically.}$$

If an equation has equal roots, these can in general be determined; the case is at any rate a special one, which may be here omitted from consideration. It is therefore, in general, assumed that the equation $f(x) = 0$ under consideration has all its roots unequal.

If the coefficients $p_1, p_2, \dots$ are all or any one or more of them imaginary, then the equation $f(x) = 0$, separating the real and imaginary parts, may be written $F(x) + i\Psi(x) = 0$, where $F(x)$, $\Psi(x)$ are each of them a function with real coefficients; and it thus appears that the equation $f(x) = 0$ with imaginary coefficients has not in general any real root; supposing it to have a real root α , this must be at once a root of each of the equations $F(x) = 0$ and $\Psi(x) = 0$.

But an equation with real coefficients may have as well imaginary as real roots; and we have further the theorem that for such an equation the imaginary roots enter in pairs, viz. $\alpha + \beta i$ being a root, then will also $\alpha - \beta i$ be a root.

Considering an equation with real coefficients, the question arises as to the *number* and *situation* of its real roots; this is completely resolved by means of *Sturm's theorem*, viz. we form a series of functions $f(x), f'(x), f_1(x), \dots, f_n(x)$ (a constant) of the degrees $n, n - 1, \dots$,

2, 1, respectively; and substituting therein for x any two real values a and b , we find by means of the resulting signs of these functions how many real roots of $f(x)$ lie between the limits a, b .

The same thing can frequently be effected with greater facility by other means, but the only general method is the one just referred to.

In the general case of an equation with imaginary (it may be real) coefficients, the like question arises as to the *situation* of the (real or imaginary) roots, viz. if for facility of conception we regard the constituents α, β of a root $\alpha + \beta i$ as the coordinates of a point in plano, and accordingly represent the root by such point; then drawing in the plane any closed curve or "contour," the question is how many roots lie within such contour.

This is solved theoretically by means of a theorem of Cauchy's, viz. writing in the original equation $x + iy$ in place of x , the function $f(x + iy)$ becomes $= P + iQ$, where P and Q are each of them a rational and integral function (with real coefficients) of (x, y) . Imagining the point (x, y) to travel along the contour, and considering the number of changes of sign from $-$ to $+$ and from $+$ to $-$ of the fraction $\frac{Q}{P}$ corresponding to passages of the fraction through zero (that is, to values for which P becomes $= 0$, disregarding those for which Q becomes $= 0$), the difference of these numbers determines the number of roots within the contour. The investigation leads to a proof of the before-mentioned theorem, that a numerical equation of the order n has precisely n roots.

But, for the actual determination, it is necessary to consider a rectangular contour, and to apply to each of its sides separately a method such as that of Sturm's theorem; and thus the actual determination ultimately depends on a method such as that of Sturm's theorem.

Recurring to the case of an equation with real coefficients, it is important to separate the real roots, viz. to determine limits, such that each real root lies alone by itself between two limits l and m . This can be done (with more or less difficulty according to the nearness of the real roots) by repeated applications of Sturm's theorem, or otherwise.

The same thing would be useful, and can theoretically be effected, in regard to the roots of an equation generally, viz. we may, by lines parallel to the axes of X and Y respectively, divide the plane into rectangles such that each (real or imaginary) root lies alone by itself in a given rectangle; but the ulterior theory, even as regards the

imaginary roots of an equation with real coefficients, has not been developed, and the remarks which immediately follow have reference only to equations with real coefficients, and to the real roots of such equations.

Supposing the roots separated as above, so that a certain root is known to lie alone by itself between two given limits, then it is possible by various processes (Horner's, or Lagrange's method of continued fractions) to obtain to any degree of approximation the numerical value of the real root in question, and thus to obtain (approximately as above) the values of the several real roots.

The real roots can also frequently be obtained, without the necessity of a previous separation of the roots, by other processes of approximation—Newton's, as completed by Fourier, or by a method given by Encke; and the problem of their determination to any degree of approximation may be regarded as completely solved. But this is far from being practically the case even as regards the imaginary roots of such equations, or as regards the roots of an equation with imaginary coefficients.

A class of numerical equations which need to be considered, are the *binomial* equations $x^n - \alpha = 0$, where $\alpha = \alpha + \beta i$, is a complex number. The foregoing conclusions apply, viz. there are always n roots, which it may be shown are all unequal. Supposing one of these is θ , so that $\theta^n = \alpha$, then, assuming $\theta = \alpha^{\frac{1}{n}}$, we have $y^n - 1 = 0$, which equation (like the more general one $x^n - \alpha = 0$) has precisely n roots; it is shown that these are $1, \omega, \omega^2, \dots, \omega^{n-1}$, where ω is a complex number $\alpha + \beta i$ such that $\alpha^2 + \beta^2 = 1$, or, what is the same thing, a complex number of the form $\cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$; and it then at once appears that ω may be taken $= e^{\frac{2\pi i}{n}}$. We have thus the trigonometrical solution of the equation $x^n - 1 = 0$. We may also obtain a like trigonometrical solution of the first-mentioned equation $x^n - \alpha = 0$. We are thus led to the notion (a numerical) of the radical $\alpha^{\frac{1}{n}}$, regarded as an n -valued function, viz. any one of these being denoted by $\sqrt[n]{\alpha}$, then the series of values is

$$\sqrt[n]{\alpha}, \omega \sqrt[n]{\alpha}, \dots, \omega^{n-1} \sqrt[n]{\alpha}.$$

Or we may, if we please, use $\sqrt[n]{\alpha}$, instead of $\alpha^{\frac{1}{n}}$, as a symbol to denote the n -valued function.

It is not necessary, as regards the equation $x^n - 1 = 0$, to refer here to the distinctions between the cases n a prime, and a composite,

number.

As the coefficients of an algebraical equation may be numerical, all which follows in regard to algebraical equations, is (with, it may be, some few modifications) applicable to numerical equations; and hence, concluding for the present this subject, it will be convenient to pass on to algebraical equations.

II. We consider, secondly, an algebraical equation

$$x^n - p_1x^{n-1} + \dots = 0,$$

and we here assume the existence of roots, viz. we assume that there are n quantities $\alpha, \beta, \gamma, \dots$ (in general, all of them different, but in particular cases they may become equal in sets in any manner), such that

$$x^n - p_1x^{n-1} + \dots = (x - \alpha)(x - \beta) \dots$$

Or, looking at the question in a different point of view, and starting with the roots $\alpha, \beta, \gamma, \dots$ as given, we express the product of the n factors $x - \alpha, x - \beta, \dots$ in the foregoing form, and thus arrive at an equation of the order n having the n roots $\alpha, \beta, \gamma, \dots$. In either case, we have

$$p_1 = \Sigma\alpha, \quad p_2 = \Sigma\alpha\beta, \quad \dots, \quad p_n = \alpha\beta\gamma \dots,$$

viz. regarding the coefficients $p_1, p_2, \dots p_n$ as given, then we assume the existence of roots $\alpha, \beta, \gamma, \dots$ such that $p_1 = \Sigma\alpha$, &c., or regarding the roots as given, then we write p_1, p_2 , &c., to denote the functions $\Sigma\alpha, \Sigma\alpha\beta$, &c.

It is to be noticed that, in virtue of

$$x^n - p_1x^{n-1} + \dots = (x - \alpha)(x - \beta), \&c.,$$

or of the equivalent equations $p_1 = \Sigma\alpha$, &c., then

$$\begin{aligned} \alpha^n - p_1\alpha^{n-1} + \dots &= 0, \\ \beta^n - p_1\beta^{n-1} + \dots &= 0, \end{aligned}$$

(viz. it is for this reason that $\alpha, \beta, \dots$ are said to be roots of $x^n - p_1x^{n-1} + \dots = 0$); and, moreover, that conversely from the last-mentioned equations, assuming that $\alpha, \beta, \dots$ are all different, we deduce

$$p_1 = \Sigma\alpha, \quad p_2 = \Sigma\alpha\beta, \quad \&c.,$$

and

$$x^n - p_1 x^{n-1} + \dots = (x - \alpha)(x - \beta) \dots$$

Observe that, if for instance $\alpha = \beta$, then the two equations $\alpha^n - p_1 \alpha^{n-1} + \dots = 0$, $\beta^n - p_1 \beta^{n-1} + \dots = 0$ would reduce themselves to a single equation, which would not of itself express that α was a double root, that is, that $(x - \alpha)^2$ was a factor of $x^n - p_1 x^{n-1} + \dots$; but by considering β as the limit of $\alpha + h$, h indefinitely small, we obtain a second equation

$$n\alpha^{n-1} - (n-1)p_1\alpha^{n-2} + \dots = 0,$$

which, with the first, expresses that α is a double root; and then the whole system of equations leads, as before, to the equations $p_1 = \Sigma\alpha$, &c. But this in passing: the general case is when the roots are all unequal.

We have then the theorem that every rational symmetrical function of the roots is a rational function of the coefficients; this is an easy consequence from the less general theorem, every rational and integral symmetrical function of the roots is a rational and integral function of the coefficients.

In particular, the sums of powers $\Sigma\alpha^2$, $\Sigma\alpha^3$, &c., are rational and integral functions of the coefficients.

An ordinary process, as regards the expression of other functions $\Sigma\alpha^p\beta^q$, &c., in terms of the coefficients, is to make them depend on the functions $\Sigma\alpha^n$, &c., but this is very objectionable; the true theory consists in showing that we have systems of equations

$$\begin{aligned} p_1 &= \Sigma\alpha, \\ p_2 &= \Sigma\alpha\beta, \\ p_1^2 &= \Sigma\alpha^2 + 2\Sigma\alpha\beta, \\ p_3 &= \Sigma\alpha\beta\gamma, \\ p_1p_2 &= \Sigma\alpha^2\beta + 3\Sigma\alpha\beta\gamma, \\ p_1^3 &= \Sigma\alpha^3 + 3\Sigma\alpha^2\beta + 6\Sigma\alpha\beta\gamma, \\ &\text{\&c., \&c.} \end{aligned}$$

where, in each system, there are precisely as many equations as there are root-functions on the right-hand side, e.g. 3 equations and 3 functions $\Sigma\alpha\beta\gamma$, $\Sigma\alpha^2\beta$, $\Sigma\alpha^3$. Hence, in each system, the root-functions can be determined linearly in terms of the powers and products of the coefficients.

It follows that it is possible to determine an equation (of an assignable order) having for roots any given (unsymmetrical) functions of the roots of a given equation. For example, in the case of a quartic equation, roots $(\alpha, \beta, \gamma, \delta)$, it is possible to find an equation having the roots $\alpha\beta, \alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta, \gamma\delta$, being therefore a sextic equation viz. in the product $(y - \alpha\beta)(y - \alpha\gamma)(y - \alpha\delta)(y - \beta\gamma)(y - \beta\delta)(y - \gamma\delta)$, the coefficients of the several powers of y will be symmetrical functions of $\alpha, \beta, \gamma, \delta$, and therefore rational and integral functions of the coefficients of the original quartic equation.

In connexion herewith, the question arises as to the number of values (obtained by permutations of the roots) of given *unsymmetrical* functions of the roots; for instance, with roots $(\alpha, \beta, \gamma, \delta)$ as before, how many values are there of the function $\alpha\beta + \gamma\delta$; or, better, how many functions are there of this form; the answer is 3, viz. $\alpha\beta + \gamma\delta, \alpha\gamma + \beta\delta, \alpha\delta + \beta\gamma$; or, again, we may ask whether it is possible to obtain functions of a given number of values, 3-valued, 4-valued functions, &c.

We have, moreover, the very important theorem that, given the value of any unsymmetrical function, e.g. $\alpha\beta + \gamma\delta$, it is in general possible to determine rationally the value of any similar function, e.g. $(\alpha + \beta)^2 + (\gamma + \delta)^2$.

As to the *a priori* ground of this theorem it may be illustrated by means of a numerical equation. Suppose, e.g. that the roots of a quartic equation are 1, 2, 3, 4; then if it is given that $\alpha\beta + \gamma\delta = 14$, this in effect determines α, β to be 1, 2 (viz. $\alpha = 1, \beta = 2$, or else $\alpha = 2, \beta = 1$) and γ, δ to be 3, 4 (viz. $\gamma = 3, \delta = 4$, or else $\gamma = 4, \delta = 3$); and it therefore in effect determines $(\alpha + \beta)^2 + (\gamma + \delta)^2$ to be $= 370$, and not any other value. And we can in the same way account for cases of failure as regards particular equations; thus, the roots being 1, 2, 3, 4, as above, $\alpha - \beta = 2$ determines α to be $= 1$ and β to be $= 2$; but if the roots had been 1, 2, 4, 16, then $\alpha^2\beta = 16$ does not uniquely determine α and β , but only makes them to be 1 and 16, or else 2 and 4, respectively.

As to the *a posteriori* proof, assume, for instance, $t_1 = \alpha\beta + \gamma\delta, y_1 = (\alpha + \beta)^2 + (\gamma + \delta)^2$, and so $t_2 = \alpha\gamma + \beta\delta, y_2 = (\alpha + \gamma)^2 + (\delta + \beta)^2$, &c.—in the present case there are only the functions t_1, t_2, t_3 and y_1, y_2, y_3 —then $y_1 + y_2 + y_3, t_1y_1 + t_2y_2 + t_3y_3, t_1^2y_1 + t_2^2y_2 + t_3^2y_3$ will be respectively symmetrical functions of the roots of the quartic, and therefore rational and integral functions of its coefficients, that is, they will be known.

Imagine, in the first instance, that t_1, t_2, t_3 are all known; then the equations being linear in y_1, y_2, y_3 , these can be expressed rationally in terms of known functions of the coefficients and of t_1, t_2, t_3 , that is, y_1, y_2, y_3 will be known. But observe further, that y_1 is obtained as a function of t_1, t_2, t_3 symmetrical as regards t_2, t_3 : it can consequently be expressed as a rational function of t_1 and of $t_2 + t_3, t_2t_3$, or, what is the same thing, of t_1 and $t_1 + t_2 + t_3, t_1t_2 + t_1t_3 + t_2t_3, t_1t_2t_3$; but these last will be symmetrical functions of the roots, and as such expressible rationally in terms of the coefficients: that is, y_1 will be expressed as a rational function of t_1 and of the coefficients, or, t_1 being known, y_1 will be rationally determined.

We may consider now the question of the algebraical solution of equations, or, more accurately, that of the solution of equations by radicals.

In the case of a quadric equation $x^2 + px + q = 0$, we can find for x , by the assistance of the sign $\sqrt{(\quad)}$ or $(\quad)^{\frac{1}{2}}$ an expression for x as a two-valued function of the coefficients p, q , such that, substituting this value in the equation, the equation is thereby identically satisfied, viz. we have

$$\begin{array}{r} x^2 = \frac{1}{4}p^2 - q + p\sqrt{(\frac{1}{4}p^2 - q)} \\ +px = -\frac{1}{2}p^2 \pm p\sqrt{(\frac{1}{4}p^2 - q)} \\ +q = q \\ \hline x^2 + px + q = 0, \end{array}$$

giving and the equation is on this account said to be algebraically solvable, or, more accurately, to be solvable by radicals. Or we may, by writing $x = -\frac{1}{2}p + z$, reduce the equation to $z^2 = \frac{1}{4}p^2 - q$, viz. to an equation of the form $z^2 = a$, and, in virtue of its being thus reducible, we may say that the equation is solvable by radicals. And the question for an equation of any higher order is, say of the order n , can we by means of radicals, that is, by aid of the sign $\sqrt[n]{(\quad)}$ or $(\quad)^{\frac{1}{n}}$, using as many as we please of such signs and with any values of m , find an n -valued function (or any function) of the coefficients, which substituted for x in the equation shall satisfy it identically.

It will be observed that the coefficients $p, q, \dots$ are not explicitly considered as numbers, but that even if they do denote numbers, the question whether a numerical equation admits of solution by radicals is wholly unconnected with the before-mentioned theorem of the existence of the n roots of such an equation. It does not even

follow that, in the case of a numerical equation solvable by radicals, the algebraical expression of x gives the numerical solution; but this requires explanation. Consider, first, a numerical quadric equation with imaginary coefficients; in the formula $x = -\frac{1}{2}p \pm \sqrt{(\frac{1}{4}p^2 - q)}$, substituting for p, q their given numerical values we obtain for x an expression of the form $x = \alpha + \beta i \pm \sqrt{(\gamma + \delta i)}$, where $\alpha, \beta, \gamma, \delta$ are real numbers; this value substituted in the numerical equation would satisfy it identically and it is thus an algebraical solution; but there is no obvious *a priori* reason why the expression $\sqrt{(\gamma + \delta i)}$ should have a value $= c + di$, where c and d are real numbers calculable by the extraction of a root or roots of real numbers: it appears upon investigation that $\sqrt{(\gamma + \delta i)}$ has such a value calculable by means of the radical expression $\sqrt{\{\sqrt{(\gamma^2 + \delta^2)} \pm \gamma\}}$ and hence; that the algebraical solution of a quadric equation does in every case give the numerical solution of a numerical quadric. The case of a numerical cubic will be considered presently.

A cubic equation can be solved by radicals, viz. taking for greater simplicity the cubic in the reduced form $x^3 - qx - r = 0$, and writing $x = a + b$, this will be a solution if only $3ab = q$, and $a^3 + b^3 = r$, or say $\frac{1}{27}q^3 = a^3b^3$, whence

$$a^3 + b^3 = r, \quad a^3b^3 = \frac{1}{27}q^3,$$

and therefore

$$a^3 = \frac{1}{2}r + \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}, \quad b^3 = \frac{1}{2}r - \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)},$$

a six-valued function of q, r . But then writing $b = \frac{q}{3a}$, we have, as may be shown, $a + b$ a three-valued function of the coefficients; it would have been wrong to complete the solution by writing $b = \sqrt[3]{\left\{\frac{1}{2}r \pm \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}}$, since here $(a + b)$ would be given as a 9-valued function, having only 3 of its values roots, and the other 6 values being irrelevant.

An interesting variation of the solution is to write $x = ab(a + b)$, giving $a^3b^3(a^3 + b^3) = r$ and $3a^2b^2 = q$, or say $\frac{1}{2}(a^3 + b^3) = \frac{r}{2}$, $a^3b^3 = \frac{q^3}{27}$; whence

$$\left\{\frac{1}{2}(a^3 - b^3)\right\}^2 = \frac{1}{4}r^2 - \frac{1}{27}q^3,$$

and therefore

$$a = \sqrt[3]{\left\{\frac{1}{2}r + \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}}, \quad b = \sqrt[3]{\left\{\frac{1}{2}r - \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}},$$

and here although a, b are each of them a 6-valued function, yet, as may be shown, $ab(a + b)$ is only a 3-valued function.

In the case of a numerical cubic, even when the coefficients are real, substituting their values in the expression

$$x = \sqrt[3]{\left\{\frac{1}{2}r + \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}} + \sqrt[3]{\left\{\frac{1}{2}r - \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}},$$

this may depend on an expression of the form $\sqrt[3]{(\gamma + \delta i)}$, where γ and δ are real numbers (viz. it will do so if $\frac{1}{4}r^2 - \frac{1}{27}q^3$ is a negative number), and here we cannot by the extraction of any root or roots of real numbers reduce $\sqrt[3]{(\gamma + \delta i)}$ to the form $c + di$, c and d real numbers; hence, here the algebraical solution does not give the numerical solution. It is to be added that the case in question, called the “irreducible case,” is that wherein the three roots of the cubic equation are all real; if the roots are one real and two imaginary, then, contrariwise, the quantity under the cube root is real, and the algebraical solution gives the numerical one.

The irreducible case is solvable by a trigonometrical formula, but this is not a solution by radicals; it consists, in effect, in reducing the given numerical cubic (not to a cubic of the form $z^3 = a$, solvable by the extraction of a cube root, but) to a cubic of the form $4x^3 - 3x = a$, corresponding to the equation $4\cos^3\theta - 3\cos\theta = \cos 3\theta$ which serves to determine $\cos\theta$ when $\cos 3\theta$ is known.

A quartic equation is solvable by radicals; and it may be remarked, that the existence of such a solution depends on the existence of 3-valued functions such as $\alpha\beta + \gamma\delta$, of the four roots $(\alpha, \beta, \gamma, \delta)$; by what precedes, $\alpha\beta + \gamma\delta$ is the root of a cubic equation, which equation is solvable by radicals; hence $\alpha\beta + \gamma\delta$ can be found by radicals; and since $\alpha\beta\gamma\delta$ is a given value, $\alpha\beta$ and $\gamma\delta$ can each be found by radicals. But by what precedes, if $\alpha\beta$ be known, then any similar function, say $\alpha + \beta$, is obtainable rationally; and, consequently, from the values of $\alpha + \beta$ and $\alpha\beta$ we may by radicals obtain the value of α or β , that is, an expression for a root of the given quartic; the expression finally obtained is 4-valued, corresponding to the different values of the several radicals which enter therein, and we have therefore the expression by radicals of each of the four roots of the quartic equation. But when the quartic is numerical, the same thing arises as in the cubic: the algebraical expression does not in every case give the numerical one.

It will be understood from the foregoing explanation as to the quartic, how in the next following case, that of a quintic equation,

the question of the solvability by radicals depends on the existence or non-existence of r -valued functions of the five roots $(\alpha, \beta, \gamma, \delta, \epsilon)$; a fundamental theorem on the subject is that a rational function of 5 letters, if it has less than 5, cannot have more than 2 values; viz. that there are no 3-valued, or 4-valued, functions of 5 letters; and by reasoning, depending in part upon this theorem, Abel showed that a general quintic equation is not solvable by radicals: and *a fortiori* the general equation of any order higher than 5 is not solvable by radicals.

The general theory of the solvability of an equation by radicals depends very much on Vandermonde's remark, that supposing an equation is solvable (by radicals) and that we have therefore an algebraical expression of x in terms of the coefficients, then substituting for the coefficients their values in terms of the roots, the resulting value of the expression must reduce itself to any one at pleasure of the roots $\alpha, \beta, \gamma, \dots$; thus in the case of the quadric equation where the solution is $x = -\frac{1}{2}p \pm \sqrt{(\frac{1}{4}p^2 - q)}$, writing for p, q their values $\alpha + \beta, \alpha\beta$, this is $x = \frac{1}{2}(\alpha + \beta) \pm \sqrt{(\alpha - \beta)^2} = \alpha$ or β according to the value of the radical. But it is not considered necessary in the present sketch to go further into the theory of the solvability of an equation by radicals. It may be proper to remark that, for quintic equations, there are solutions analogous to the trigonometrical solution of a cubic equation, viz. the quintic equation is here in effect reduced to some special form of quintic equation; for instance, to Jerrard's form $x^5 + ax + b = 0$, or to some form presenting itself in the theory of elliptic functions; but the solutions in question are not solutions by radicals. And there are various other interesting parts of the theory which have been excluded from consideration.

632.

ON ARONHOLD'S INTEGRATION-FORMULA.

[From the Messenger of Mathematics, vol. v. (1876), pp. 88—90.]

The fundamental theorem in Aronhold's Memoir, "Ueber eine neue algebraische Behandlungsweise der Integrale $\int \Pi(x, y) dx$, &c.," Crelle, t. LXI. (1863), pp. 95—145, is a theorem of indefinite integration. The form is

$$\int \frac{dx}{(ax + \beta y + \gamma)(hx + by + f)} = \frac{1}{\Lambda} \log \frac{(\alpha\xi + h\eta + g)x + (h\xi + b\eta + f)y + g\xi + f\eta}{\alpha x + \beta y + \gamma}$$

where y is a certain irrational function of x , determined by a quadratic equation, and the other symbols denote constants connected by certain relations; viz. writing, for shortness,

$$U = (a, b, c, f, g, h \setminus x, y, 1)^2 = (a, \dots \setminus x, y, 1)^2, \text{ for shortness,}$$

that is,

$$U = ax^2 + 2hxy + by^2 + 2fy + 2gx + c;$$

$$W = (a, b, c, f, g, h \setminus x, y, 1 \setminus \xi, \eta, 1) = (a, \dots \setminus x, y, 1 \setminus \xi, \eta, 1),$$

that is,

$$W = (ax + hy + g)\xi + (hx + by + f)\eta + gx + fy + c,$$

or

$$(ax + hy + g)x + (hx + by + f)y + gx + fy + c$$

$$(P, Q, R) = (ax + hy + g, hx + by + f, gx +$$

$$(P_0, Q_0, R_0) = (a\xi + h\eta + g, h\xi + b\eta + f, g\xi +$$

$$\Omega = \alpha x + \beta y + \gamma,$$

$$(A, B, C, F, G, H) = (bc - f^2, ca - g^2, ab - h^2, gh -$$

then y is determined as a function of x by the equation $U = 0$, that is,

$$(a, b, c, f, g, h \setminus x, y, 1)^2 = 0;$$

or, what is the same thing,

$$by = -(hx + f) + \sqrt{\{-Cx^2 + 2Gx - A\}};$$

the constants $\alpha, \beta, \gamma, \xi, \eta$ are such that

$$(a, b, c, f, g, h \setminus \xi, \eta, 1)^2 = 0, \quad \alpha\xi + \beta\eta + \gamma = 0,$$

that is,

$$U_0 = 0,$$

and the value of Λ is given by

$$\Lambda^2 = -(A, B, C, F, G, H \setminus \alpha, \beta, \gamma)^2.$$

The theorem may therefore be written

$$\int \frac{dx}{\Omega \cdot W} = \frac{1}{\Lambda} \log \frac{W}{\Omega},$$

where the several symbols have the significations explained above.

The verification is as follows. We ought to have

$$\frac{\Lambda dx}{\Omega \cdot W} = \frac{P_0 dx + Q_0 dy}{W} - \frac{\alpha dx + \beta dy}{\Omega},$$

when dx, dy satisfy the relation $P dx + Q dy = 0$, viz. substituting for dy the value $-\frac{P}{Q} dx$, the equation becomes

$$\frac{\Lambda}{\Omega \cdot W} = \frac{P_0 Q - P Q_0}{Q \cdot W} + \frac{\alpha Q - \beta P}{Q \cdot \Omega},$$

that is, substituting for W its value,

$$\Lambda W = (P_0 Q - P Q_0)(\alpha x + \beta y + \gamma) - (\alpha Q - \beta P)W.$$

On the right-hand side, substituting for W its value,

$$\text{coeff. } \alpha = x(P_0 Q - P Q_0) - Q(P_0 x + Q_0 y + R_0), \quad = Q_0 R - Q R_0,$$

$$\text{coeff. } \beta = y(P_0 Q - P Q_0) + P(P_0 x + Q_0 y + R_0), \quad = R_0 P - R P_0,$$

(as at once appears by aid of the relation $U = Px + Qy + R = 0$),

$$\text{coeff. } \gamma = P_0 Q - P Q_0.$$

The equation to be verified thus is

$$\Lambda W = \begin{vmatrix} \alpha, & \beta, & \gamma \\ P_0, & Q_0, & R_0 \\ P, & Q, & R \end{vmatrix},$$

which, substituting therein for P, Q, R, P_0, Q_0, R_0 , their values, and writing

$$(\alpha, \beta, \gamma) = (\alpha, \beta, \gamma),$$

is in fact

$$\Lambda W = (A, \dots \setminus \alpha, \beta, \gamma \setminus \xi, \eta, 1).$$

We have identically

$$(a, \dots \setminus x, y, 1)^2 \cdot (a, \dots \setminus \xi, \eta, 1)^2 - W^2 = (A, \dots \setminus \lambda, \mu, \nu)^2,$$

which, in virtue of $(a, \dots \setminus \xi, \eta, 1)^2 = 0$, gives

$$W^2 = -(A, \dots \setminus \lambda, \mu, \nu)^2,$$

and since $\Lambda^2 = -(A, \dots \setminus \alpha, \beta, \gamma)^2$, the equation is thus

$$\sqrt{\{-(A, \dots \setminus \alpha, \beta, \gamma)^2\}} \cdot \sqrt{\{-(A, \dots \setminus \lambda, \mu, \nu)^2\}} = (A, \dots \setminus \lambda, \mu, \nu \setminus \alpha, \beta, \gamma),$$

that is,

$$(A, \dots \setminus \alpha, \beta, \gamma)^2 \cdot (A, \dots \setminus \lambda, \mu, \nu)^2 - \{(A, \dots \setminus \lambda, \mu, \nu \setminus \alpha, \beta, \gamma)\}^2 = 0.$$

The left-hand side is here identically

$$= K(\alpha\mu - \beta\lambda, \alpha\nu - \gamma\lambda, \beta\nu - \gamma\mu)^2,$$

substituting for λ, μ, ν their values, we find

$$\alpha\mu - \beta\lambda = \alpha\Omega, \quad \alpha\nu - \gamma\lambda = \beta\Omega, \quad \beta\nu - \gamma\mu = \gamma\Omega,$$

viz. in virtue of $U_0 = 0$, these are $-f\Omega, -g\Omega, -h\Omega$, and the quadric function is $= \Omega^2(a, \dots \setminus f, g, h)^2$, vanishing in virtue of the relation $(a, \dots \setminus f, g, h)^2 = 0$.

The equation in question

$$\sqrt{\{-(A, \dots \setminus \alpha, \beta, \gamma)^2\}} \cdot \sqrt{\{-(A, \dots \setminus \lambda, \mu, \nu)^2\}} = (A, \dots \setminus \lambda, \mu, \nu \setminus \alpha, \beta, \gamma)$$

is thus verified, and the theorem is proved.

633.

**NOTE ON MR. MARTIN'S PAPER, "ON THE
INTEGRALS OF SOME DIFFERENTIALS."**

[From the Messenger of Mathematics, vol. v. (1876), p. 163.]

The Note refers to a detail in a process of integration.

634.

**THEOREMS IN TRIGONOMETRY AND ON
PARTITIONS.**

[From the Messenger of Mathematics, vol. v. (1876), p. 164, and
p. 188.]

If

$$A + B + C + F + G + H = 0,$$

then

$$\begin{vmatrix} \sin A + F \sin B + F \sin C + F, & \cos F, & \sin F \\ \sin A + G \sin B + G \sin C + G, & \cos G, & \sin G \\ \sin A + H \sin B + H \sin C + H, & \cos H, & \sin H \end{vmatrix} = 0.$$

Let M_n = number of partitions of n , no part less than 2, the order attended to; e.g. if $n = 7$, the partitions are 7, 52, 25, 43, 34, 322, 232, 223, 2222 = 8; the series is

$$u_2 = 1,$$

$$u_3 = 1,$$

$$u_4 = 2,$$

$$u_5 = 3,$$

$$u_6 = 5,$$

$$u_7 = 8,$$

$$u_8 = 13,$$

$$u_9 = 21,$$

where each term is the sum of the next preceding two terms.

635.

NOTE ON THE DEMONSTRATION OF CLAIRAUT'S THEOREM.

[From the Messenger of Mathematics, vol. v. (1876), pp. 166, 167.]

It seems worth while to indicate what the leading steps of the demonstration are.

The potential of the Earth's mass upon an external or superficial point is taken to be

$$\frac{F_0}{r} + \frac{F_1}{r^2} + \frac{F_2}{r^3} + \frac{F_3}{r^4} + \dots,$$

where $F_1, F_2, F_3, \dots$ are Laplace's functions of the angular coordinates.

The surface is assumed to be a nearly spherical surface $r = a(1 + u)$, where $u = u_1 + u_2 + \&c.$, and $u_1, u_2, \dots$ are Laplace's functions of the angular coordinates. To be a surface of equilibrium, with an equation $F + \frac{1}{2}\omega^2 r^2 \sin^2 \theta = C$, the latter must be equivalent to the equation $r = a(1 + u)$, and it follows that we have

$$\begin{aligned} F_1 &= F_0 a u_1, \\ F_2 &= F_0 a u_2 - \frac{1}{2}\omega^2 a^3 \left(\frac{1}{3} - \cos^2 \theta\right), \\ &\&c., \end{aligned}$$

which values are to be substituted in the expression for F .

The whole force of gravity (due to the attraction and the centrifugal force) is taken to be $g = -\frac{d}{dr} \left(F + \frac{1}{2}\omega^2 r^2 \sin^2 \theta\right)$, and it follows that

$$g = \frac{F_0}{a^2} (1 + u_1 + 2u_2 + \dots) - \omega^2 a \left(\frac{1}{3} - \cos^2 \theta\right),$$

which is of the form

$$g = G \left\{ 1 + u_1 - \left(\frac{5}{2}m - \epsilon\right) \left(\frac{1}{3} - \cos^2 \theta\right) + 2u_2 + \dots \right\}.$$

Taking the Earth to be the spheroid of revolution

$$r = a \left\{ 1 + \epsilon \left(\frac{1}{3} - \cos^2 \theta\right) \right\},$$

then

$$u_1 = \epsilon \left(\frac{1}{3} - \cos^2 \theta\right), \quad u_2 = 0, \&c.,$$

and the equation is

$$g = G\{1 - (\frac{5}{2}m - \epsilon)(\frac{1}{3} - \cos^2 \theta)\},$$

or say

$$g = G\{1 - (\frac{5}{2}m - \epsilon)(\frac{1}{3} - \cos^2 \theta)\},$$

where $m = \frac{\omega^2 a}{G}$, is the ratio of the centrifugal force at the equator to the force of gravity, which is the theorem in question. The expression “it follows” has been twice used as meaning it follows as a mere analytical consequence, in the proper degree of approximation, the steps of the deduction being purposely omitted.

VOL. X. 3

636.

**ON THE THEORY OF THE SINGULAR SOLUTIONS
OF DIFFERENTIAL EQUATIONS OF THE FIRST
ORDER.**

[From the Messenger of Mathematics, vol. vi. (1877), pp. 23—27.]

In continuation of the former paper with this title (Messenger, vol. ii., 1873, pp. 6—12, [545]), I propose to discuss various particular examples, chiefly of cases in which the differential equation is of the form $(L, M, N \setminus p, 1)^2 = 0$, where L, M, N are rational and integral functions of (x, y) , and whether it admits or does not admit of an integral equation $(P, Q, R \setminus c, 1)^2 = 0$, where P, Q, R are rational and integral functions of (x, y) .

The singular solution of the differential equation

$$(L, M, N \setminus p, 1)^2 = 0,$$

if there be a singular solution, is $\vartheta = 0$, where S is either $= LN - M^2$, or a factor of $LN - M^2$. But in general $LN - M^2$ is an indecomposable function, such that $LN - M^2 = 0$ is not a solution of the differential equation, and this being so, there is no singular solution; viz. a differential equation $(L, M, N \setminus p, 1)^2 = 0$, where L, M, N are rational and integral functions of (x, y) , has not in general any singular solution.

Consider now a system of algebraical curves $U = 0$, where U is as regards (x, y) a rational and integral function of the order m , and depends in any manner on an arbitrary parameter c^* . I say that there is always a proper envelope, which envelope is the singular solution of the differential equation obtained by the elimination of c from the equation $U = 0$, and the derived equation in regard to (x, y) . It follows that the differential equation $(L, M, N \setminus p, 1)^2 = 0$, which has no singular solution, does not admit of an integral of the form in question $U = 0$, viz. an integral representing a system of algebraic curves.

* The expressions in the text may be understood as extending to the case where U is a function of any number (o) of constants $c_1, c_2, \dots c_o$, connected by an $(o - 1)$ fold relation, U thus virtually depending on a single arbitrary parameter.

The theorem just referred to, that the system of algebraic curves $U = 0$ has always an envelope, is an interesting theorem, which I proceed to prove. Assume that in general, that is, for an arbitrary value of the parameter, the equation $U = 0$ represents a curve of the order m , with δ nodes and κ cusps (and therefore of the class n , with ι inflexions and τ double tangents, the numbers $m, \delta, \kappa, n, \tau, \iota$ being connected by Plücker's equations); for particular values of the parameter, the values of δ and κ may be increased, or the curve may break up, but this is immaterial.

The consecutive curve $U + \delta c dcU = 0$ is a curve of the same order m , with δ nodes and κ cusps, consecutive to the nodes and cusps of the original curve U , and the two curves intersect in m^2 points; but of these, there are 2 coinciding with each node, and 3 coinciding with each cusp of the curve $U = 0$, as at once appears by drawing a curve with a node or a cusp, and the consecutive curve with a consecutive node or cusp; the number of the remaining intersections is $= m^2 - 2\delta - 3\kappa$, and the envelope is the locus of these $m^2 - 2\delta - 3\kappa$ points. Observe that the two curves have in common n^2 tangents; but of these, 2 coincide with each double tangent and 3 coincide with each stationary tangent of the curve $U = 0$, viz. the number of the remaining common tangents is $= n^2 - 2\tau - 3\iota$ (which is $= m^2 - 2\delta - 3\kappa$): and that these $n^2 - 2\tau - 3\iota$ common tangents are indefinitely near to the $m^2 - 2\delta - 3\kappa$ common points respectively, and are in fact the tangents of the envelope at the $m^2 - 2\delta - 3\kappa$ points respectively. Now in an algebraic curve we have $m + n = m^2 - 2\delta - 3\kappa$, viz. the number $m^2 - 2\delta - 3\kappa$ cannot be $= 0$, and we have therefore always an envelope the locus of the system of the $m^2 - 2\delta - 3\kappa$ points. It might be thought that the conclusion extends to transcendental curves; if this were so, the result would prove too much, viz. it would follow that a differential equation $(L, M, N \setminus p, 1)^2 = 0$ without a singular solution had no general integral; but it will appear by an example that the theorem as to the envelope does not extend to transcendental curves.

Ex. 1.

$$p^2 - (1 - y^2) = 0, \text{ that is, } dy^2 - (1 - y^2) dx^2 = 0.$$

Here there is no algebraical integral, but there is a quasi-algebraical integral of the form $(P, Q, R \setminus c, 1)^2 = 0$; viz. starting with the form $y = \sin(x + c)$ and expressing $\sin c$ and $\cos c$ rationally in terms of a

new parameter, this is

$$c^2(y + \cos x) - 2c \sin x + (y - \cos x) = 0,$$

where the coefficients are one-valued functions of (x, y) . The discriminant of the differential equation in regard to p and that of the integral equation in regard to c are each $= y^2 - 1$, and we have a true singular solution $y^2 - 1 = 0$.

Ex. 2.

$$(1 - x^2)p^2 - (1 - y^2) = 0,$$

that is,

$$(1 - x^2) dy^2 - (1 - y^2) dx^2 = 0.$$

We have here an algebraic integral of the proper form, which is at once derived from the circular form

$$c = \cos^{-1} x + \cos^{-1} y$$

by changing the constant, viz. this is

$$c - xy - \sqrt{(1 - x^2 - y^2)} = 0.$$

The two discriminants are here each $= (x^2 - 1)(y^2 - 1)$, and we have

$$x^2 - 1 = 0, \quad y^2 - 1 = 0,$$

as a true singular solution. The curves are in fact the system of conics (ellipses and hyperbolas) each touching the four lines $x = 1$, $x = -1$, $y = 1$, $y = -1$.

Ex. 3.

$$(1 - y^2)p^2 - 1 = 0, \quad \text{that is,} \quad (1 - y^2) dy^2 - dx^2 = 0.$$

This is an extremely interesting example: the curve is the orthogonal trajectory of the system of sinusoids $y = \sin(x + c)$, which is the integral of Example 1; and we thus at once see that the real portion of the curve is wholly included between the lines $y = -1$, $y = +1$, being an infinite continuous curve, having a series of equidistant cusps alternately at the one and the other line, and obtained by the continued repetition of the finite portion included between two consecutive cusps on the same line. The discriminant of the differential equation equated to zero gives $2y^2 - 1 = 0$, the equation of the two lines in

question; but this does not satisfy the differential equation, and it is consequently not a singular solution; by what precedes, it appears that it is, in fact, a cusp-locus.

We thus see that the curves which represent the integral equation have no real envelope; but it is shown that there is no imaginary envelope, and that the curve obtained by the elimination of the parameter is, in fact, made up of a (imaginary) node-locus and of the foregoing cusp-locus.

The curve is properly represented by taking x, y each of them a one-valued function of the parameter θ , viz. we may write

$$\begin{aligned} y &= \cos \theta, \\ x &= c + \frac{1}{2}\theta - \frac{1}{4}\sin 2\theta. \end{aligned}$$

In fact, these values give

$$\dot{y} = -\sin \theta, \quad \dot{x} = \frac{1}{2}(1 - \cos 2\theta) = \sin^2 \theta,$$

and therefore

$$p = \frac{-1}{\sin \theta} = \frac{-1}{\sqrt{(1 - y^2)}},$$

that is, $(1 - y^2)p^2 - 1 = 0$, the differential equation.

It is obvious that to a given value of the parameter there corresponds a single point of the curve; and it is to be shown that, conversely, to a given point of the curve corresponds in general a single value of the parameter.

Suppose the coordinates of the given point are $y = \cos \alpha$, $x = c + \frac{1}{2}\alpha - \frac{1}{4}\sin 2\alpha$, where α is a determinate quantity; then, to find θ , we have

$$\cos \theta = \cos \alpha, \quad 2\theta - \sin 2\theta = 2\alpha - \sin 2\alpha.$$

The first equation gives $\theta = 2m\pi \pm \alpha$, and the second equation then is

$$4m\pi \pm 2\alpha \mp \sin 2\alpha = 2\alpha - \sin 2\alpha,$$

viz. taking the upper signs, this is $4m\pi = 0$, giving $\theta = \alpha$; and, taking the lower signs, it is $m\pi = \alpha - \sin \alpha$, which, α being given, is not in general satisfied; hence to the given point there corresponds only the value α of the parameter θ . If, however, α is such that $\alpha - \sin \alpha$ is equal to a multiple of π , say $r\pi$, then the last-mentioned equation is

satisfied by the value $m = r$, so that to the given point of the curve correspond the two values α and $2r\pi - \alpha$ of the parameter; these values are in general unequal, and the point is then a node; but they may be equal, viz. this is so if $\alpha = r\pi$ (the point on the curve being then $y = \cos r\pi = \pm 1$, $x = c + \frac{1}{2}r\pi$), and the point is then a cusp; showing what was known, that there are on each of the lines $y = -1$, $y = +1$, an infinite series of equidistant cusps.

More definitely, suppose $\alpha = r\pi \pm \beta$, where β is a root of the equation $2\beta - \sin 2\beta = 0$, then

$$\sin 2\alpha = \mp \sin 2\beta, \quad 2\alpha - \sin 2\alpha = 2r\pi + (2\beta - \sin 2\beta) = 2r\pi,$$

and to the given point on the curve correspond the two values α and $2r\pi - \alpha$ of the parameter. If $\beta = 0$, we have, as above, the cusps on the two lines $y = +1$, $y = -1$ respectively; but if β be an imaginary root of the equation $2\beta - \sin 2\beta = 0$, then we have an infinite series of nodes on the imaginary line $y = \cos r\pi \cos \beta$; and there are an infinite number of such lines corresponding to the different imaginary roots of the equation $2\beta - \sin 2\beta = 0$.

From the form in which the equation of the curve is given, we cannot directly form the equation of the envelope by equating to zero the discriminant in regard to the constant c ; but we may determine the intersections of the curve by the consecutive curve (corresponding to a value $c + \delta c$ of the constant), and thus determine the locus of these intersections.

Consider for a moment the curves belonging to the constants c , c_1 , and let θ , θ_1 be the values of the parameter θ belonging to the points of intersection; we have $\cos \theta = \cos \theta_1$, $4c + 2\theta - \sin 2\theta = 4c_1 + 2\theta_1 - \sin 2\theta_1$, we have $\theta_1 = 2r\pi + \theta$, but we cannot thereby satisfy the second equation; or else $\theta_1 = 2r\pi - \theta$, giving

$$4c + 2\theta - \sin 2\theta = 4c_1 + 4r\pi - 2\theta + \sin 2\theta,$$

that is, $2\theta - \sin 2\theta = 2c_1 - 2c + 2r\pi$; and we have thus corresponding to any given value of r a series of values of θ , viz. these are $\theta = r\pi + \beta$, where β is any root of the equation

$$2\beta - \sin 2\beta = 2c_1 - 2c.$$

In particular, taking $c_1 = c$, the intersections are given by $\theta = r\pi + \beta$, where β is any root of the equation $2\beta - \sin 2\beta = 0$, viz.

we have thus an infinite number of intersections lying on each of the lines $y = \cos r\pi \cos \beta$. If $\beta = 0$, the intersections lie on the two lines $y = 1$, $y = -1$ respectively; if β be an imaginary root of the equation $2\beta - \sin 2\beta = 0$, then they lie on the imaginary lines $y = \cos r\pi \cos \beta$. But by what precedes, it is clear that in the former case the intersections are nothing else than the cusps on the lines $y = 1$, $y = -1$; and in the latter case nothing else than the nodes on the lines $y = \cos r\pi \cos \beta$; viz. there is no proper envelope, but instead thereof we have lines of cusps and of nodes.

Ex. 4.

$$(1 - y^2)p^2 - (1 - x^2) = 0,$$

that is,

$$(1 - y^2) dy^2 - (1 - x^2) dx^2 = 0.$$

I have not examined this; the curve is the series of orthogonal trajectories of the conics of Example 2, and the integral equation may be represented by $y = \cos \theta$, $x = \cos \phi$, where $c = (2\theta - \sin 2\theta) - (2\phi - \sin 2\phi)$.

Equating to zero the discriminant of the differential equation, we have $(1 - y^2)(1 - x^2) = 0$, viz. the four lines $x = 1$, $x = -1$, $y = 1$, $y = -1$; this is not an envelope, but a locus of cusps.

637.**ON A DIFFERENTIAL EQUATION IN THE THEORY
OF ELLIPTIC FUNCTIONS.**

[From the Messenger of Mathematics, vol. vi. (1877), p. 29.]

In the differential equation

$$Q_2 - Q_3(x + k) + 3(1 - k^2)\dot{x} = 0,$$

considered Messenger, t. iv., pp. 69 and 110, [594] and [597], writing $Q = x$ and $k + T = y$, the equation becomes

$$\ddot{x} = \frac{3\dot{x}^2 + xy - x^3}{2},$$

and we have, as a particular solution,

$$y = \frac{\dot{x}^2 - 6x^2 - 3}{4x}.$$

To verify this, observe that from the value of y

$$\dot{y} = \frac{x\ddot{x} - \dot{x}^2 - 3}{2x^2},$$

and the equation becomes

$$\frac{3}{4}\dot{x}^2\{x^3 - 6x^2 - 3\} = \frac{3\dot{x}^2}{4x^2}\{(x^3 - 6x^2 - 3)^2 - 64x^3\},$$

viz. this is

$$(x^3 - 1)^2(x^3 - 9) = (x^3 - 6x^2 - 3)^2 - 64x^3,$$

which is right.

638.

**ON A q -FORMULA LEADING TO AN EXPRESSION
FOR E_1 .**

[From the Messenger of Mathematics, vol. vi. (1877), pp. 63—66.]

It is to be shown that we have identically

$$\frac{(1 - 2q + 2q^4 - 2q^9 + \dots)^4 - 16(q + q^9 + q^{25} + q^{49} + \dots)^4}{(1 + 2q + 2q^4 + 2q^9 + \dots)^4} = \frac{1 - 9q - 25q^2 + 49q^4 + \dots}{1 - q - q^2 + q^5 + \dots}$$

or, what is the same thing,

$$\frac{(1 - 2q + 2q^4 - 2q^9 + \dots)^4 - 16(q + q^9 + q^{25} + q^{49} + \dots)^4}{(1 + 2q + 2q^4 + 2q^9 + \dots)^4} = \frac{1 + 9q + 25q^2 + 49q^4 + \dots}{1 + q + q^2 + q^4 + \dots}$$

where the form (A) is that intended to be made use of, but the form (B) is rather more convenient for the demonstration.

We have

$$(1 - 2q + 2q^4 - 2q^9 + \dots)^4 = 1 + 8 \left\{ \frac{q}{1 + q} + \frac{3q^2}{1 + q^2} + \frac{5q^3}{1 + q^3} + \dots \right\},$$

(Jacobi, *Fund. Nova*, p. 188, *Ges. Werke*, t. i., p. 239), taking the formula as there written down, and changing q into $-q$.

Also, if for a moment

$$X = 1 + 2q + 2q^4 + 2q^9 + \&c.,$$

and

$$q' = \frac{q dX}{X dq},$$

so that

$$qX' = q + 3q^3 + 6q^6 + 10q^{10} + \&c.,$$

then

$$X + 8qX' = 1 + 9q + 25q^2 + 49q^4 + 81q^6 + \&c.,$$

so that the right-hand side of (B) is

$$\frac{X + 8qX'}{X} = 1 + 8 \frac{qX'}{X}.$$

But (Fund. Nova, p. 185, Ges. Werke, t. i., p. 237),

$$\frac{X'}{X} = \frac{-2q}{1-q^2} + \frac{-4q^4}{1-q^4} + \frac{-6q^6}{1-q^6} + \dots = \frac{-2q}{1-q} + \frac{-4q^2}{1-q^2} + \frac{-6q^3}{1-q^3} - \dots$$

And the equation (B) intended to be proved thus becomes

$$1 + 8\frac{q}{1+q} + 8\frac{3q^2}{1+q^2} + 8\frac{5q^3}{1+q^3} + \dots = 1 + 8\frac{-2q}{1-q^2} + 8\frac{-4q^2}{1-q^4} + 8\frac{-6q^3}{1-q^6} + \dots,$$

viz. omitting the terms unity, dividing by $8q$, and transposing, this is

$$\begin{aligned} & \frac{1}{1+q} + \frac{3q}{1+q^2} + \frac{5q^2}{1+q^3} + \dots \\ & + \frac{2}{1-q^2} + \frac{4q}{1-q^4} + \frac{6q^2}{1-q^6} + \dots = 0. \end{aligned}$$

The second and third lines unite together, and the equation becomes

$$\begin{aligned} & \frac{1}{1+q} + \frac{2q}{1-q^2} + \frac{3q^2}{1+q^3} + \frac{4q^3}{1-q^4} + \frac{5q^4}{1+q^5} + \frac{6q^5}{1-q^6} + \dots \\ & \quad - \frac{q}{1+q} - \frac{3q^3}{1+q^3} - \frac{5q^5}{1+q^5} - \dots \\ & \quad + \frac{2q}{1-q^2} + \frac{4q^3}{1-q^4} + \frac{6q^5}{1-q^6} + \dots = 0, \end{aligned}$$

or, collecting and arranging,

$$\begin{aligned} & \frac{1+q}{1+q^2} + \frac{1+q^3}{1+q^4} \cdot 3q^2 + \frac{1+q^5}{1+q^6} \cdot 5q^4 + \dots \\ & \quad - \frac{1+q^2}{1-q^3} \cdot q - \frac{1+q^4}{1-q^5} \cdot 3q^3 - \frac{1+q^6}{1-q^7} \cdot 5q^5 - \dots = 0, \end{aligned}$$

an identity which it is easy to verify to any number of terms. But to prove it directly, we have only to add the pairs of terms in the alternate columns; calling the left-hand side Fq , we thus obtain

$$Fq = \frac{2q}{1+q^2} + \frac{6q^3}{1+q^4} + \frac{10q^5}{1+q^6} + \dots - \frac{2q}{1-q^3} - \frac{6q^3}{1-q^5} - \frac{10q^5}{1-q^7} - \dots,$$

viz. this equation is $Fq = 2qF(q^2)$; and thence

$$Fq = 2q \cdot 2q^2 \cdot F(q^4) = 2^2 q^{1+2} \cdot 2q^4 \cdot F(q^8) = \&c.;$$

we thus have $Fq = 0$.

The equation (B), or, what is the same thing, the equation (A) is thus proved.

Reverting to the equation (A), we have

$$(1 + 2q + 2q^4 + \dots)^2 = \frac{2K}{\pi},$$

(Jacobi, *Fund. Nova*, p. 188, *Ges. Werke*, t. i., p. 239),

$$\frac{2E_1}{\pi} = \left(\frac{2K}{\pi}\right)^{-1} \left\{ 1 - 8\frac{q}{1+q} - 8\frac{3q^2}{1+q^2} - \dots \right\},$$

(ib., p. 135; ib., p. 189), if $q = e^{-\frac{\pi K'}{K}}$, and K, E_1 are the complete functions F_1k, E_1k .

The left-hand side of the equation is thus

$$\frac{2E_1}{K} - \frac{2E_1}{\pi} \left(\frac{2K}{\pi}\right)^{-1},$$

and we have

$$\frac{2E_1}{K} = \frac{\pi^2}{K'^2} \cdot \frac{1 - 9q^2 - 25q^4 + 49q^8 + 81q^{12} - \dots}{1 - q^2 - q^4 + q^{10} + q^{14} - \dots},$$

which is a new expression for E_1 , as a q -function. The expression on the right-hand side presents itself, Clebsch, *Theorie der Elasticität* (Leipzig, 1862), p. 162, and must have been obtained by him as a value for $\left(\frac{dn}{K_1}\right)^{-1}$; but there is no statement that this is so, nor anything to show how this form of q -function was arrived at. Mr. Todhunter called my attention to the passage in Clebsch.

639.

AN ELEMENTARY CONSTRUCTION IN OPTICS.

[From the Messenger of Mathematics, vol. vi. (1877), pp. 81, 82.]

Consider two lines meeting at a point P , and a point A ; through A , draw at right angles to AP , a line meeting the two lines in the points U , V respectively; and through the same point A draw any other line meeting the two lines in the points U' , V' respectively; also let the points u' , v' be the feet of the perpendiculars let fall from U' , V' respectively on the line UV : then we have

$$\frac{Au'^2}{Av'^2} = \frac{AU'^2}{AV'^2} \cdot \frac{AU^2}{AV^2}.$$

The theorem can be proved at once without any difficulty. It answers to the optical construction, according to which, if UPV represents the path of a ray through a convex lens AP , then the thin pencil, axis $U'P$ and centre U' , converges after refraction to the point V' , where $U'V'$ are in line with A the centre of the lens; considering as usual the inclinations to the axis as small, we have approximately $AV = Av$, $AU' = Au'$, and the theorem is

$$\frac{1}{AU'^2} + \frac{1}{AV'^2} = \frac{1}{AU^2} + \frac{1}{AV^2} = \frac{1}{AF^2},$$

if AF is the focal length of the lens.

In the original theorem, the line UV need not be at right angles to AP , but may be any line whatever; the projecting lines $U'u'$ and $V'v'$ must then be parallel to AP , and the theorem remains true.

640.

**FURTHER NOTE ON MR. MARTIN'S PAPER "ON
THE INTEGRALS OF SOME DIFFERENTIALS."**

[From the Messenger of Mathematics, vol. VI. (1877), p. 82.]

See paper, Number 633; this further note relates also to a detail.

641.

ON THE FLEXURE OF A SPHERICAL SURFACE.

[From the Messenger of Mathematics, vol. vi. (1877), pp. 88—90.]

It is known that an inextensible spherical surface, or to fix the ideas the spherical quadrilateral included between two arcs of meridian and two arcs of parallel, may be bent in suchwise as to be part of a surface of revolution, the meridians and parallels of the spherical surface being meridians and parallels of the new surface, and, moreover, the radius of each parallel of the spherical surface being in the new surface altered in the constant ratio k to 1. We have, in fact, on the spherical surface, writing p for the latitude and q for the longitude, and the radius being unity,

$$\begin{aligned}x &= \cos p \cos q, \\y &= \cos p \sin q, \\z &= \sin p,\end{aligned}$$

values which give

$$dx^2 + dy^2 + dz^2 = dp^2 + \cos^2 p dq^2.$$

This last equation is satisfied by the values

$$\begin{aligned}X &= \cos p \cos \frac{q}{k}, \\Y &= \cos p \sin \frac{q}{k}, \\Z &= E(k, p),\end{aligned}$$

where $E(k, p) = \int_0^p \sqrt{1 - k^2 \sin^2 p} dp$, is the elliptic function of the second kind; or rather, this is so when $k < 1$, but the same notation may be used when $k > 1$. These values give the deformation in question.

The two cases to be considered are $k < 1$, and $k > 1$; we take in each case a spherical quadrilateral $ABCD$ (fig. 1), bounded by AB (an arc of the equator), the arc of parallel CD , and the two arcs of meridian AD and BC . In the first case, there is no limit to the latitude $AD, = BC$, or taking these $= 90^\circ$, we may in place

Collected Mathematical Papers

Arthur Cayley

Volume 10, Pages 51–100

641. On the Flexure of a Spherical Surface.

[From the *Proceedings of the London Mathematical Society*, vol. viii. (1877), pp. 12, 13.]

(Conclusion from p. 30.)

[Page 31]

of the quadrilateral $ABCD$ consider the birectangular triangle ABE ; the new form of this is $A'B'E'$, where the radius $OA' = k \cdot OA = k$, is less than the original radius unity, but $OE' = k/c$, is greater than the same radius unity.

The surface has at E' a conical point, the semi-aperture of the cone being $= \tan^{-1} \rho$, if as usual $k' = \sqrt{1 - k^2}$; to verify this, writing for convenience $q = 0$, we have for the meridian section $x = \cos \phi$, $z = E(k, \phi)$, and thence

$$\frac{dz}{dx} = -\frac{\sqrt{1 - k^2 \sin^2 \phi}}{\sin \phi},$$

which for $\phi = 90^\circ$ gives $\frac{dz}{dx} = -k'$. Observe also that for $\phi = 0$, $\frac{dz}{dx} = \infty$, viz. the surface of revolution cuts the plane of the equator at right angles.

There is no limit to the arc AB , it may be $= 360^\circ$, viz. we must in this case cut the hemisphere along a meridian to allow of the deformation; or it may exceed 360° , the spherical surface being in this case conceived of as wrapping indefinitely over itself, and we may instead of the half lune $E'A'B'$, consider the lune included between two meridians extending from pole to pole, and therefore the whole spherical surface, conceived of as wrapping indefinitely over itself; the result is, that this may be deformed into a surface of revolution, which, in its general form, resembles that obtained by the revolution of an arc less than a semi-circle round its chord; the half-chord being greater, and the versed-sine less than the radius of the original sphere.

If $k > 1$, there is obviously a limit to the latitude $AD = BC$, of the spherical quadrilateral; viz. this is equal to $\sin^{-1} \frac{1}{k}$. Supposing that in the quadrilateral $ABCD$ (fig. 1) the latitude has this limiting value, then (see fig. 2) the new form is $A''B''C''D''$, where along the

bounding arc CB'' the tangent plane is horizontal; viz. as before

$$\frac{dz}{dx} = -\frac{\sqrt{1 - k^2 \sin^2 \phi}}{\sin \phi} = 0 \quad \text{for } \phi = \sin^{-1} \frac{1}{k}.$$

It is to be observed, that the radii for the parallels $A''B''$ and CD'' are k and $\frac{k}{c} \cos \phi$ respectively; the difference of

[Page 32]

these is $k(1 - \cos \phi)$, which, however great k is, must be less than the arc of meridian

$$A''D'' = \phi;$$

substituting for k the value $\frac{1}{\sin \phi}$, the condition is $\frac{\phi}{\sin \phi} > \phi$, viz. this is

$$\tan \frac{1}{2}\phi < \phi,$$

which is true for every value up to $\phi = 90^\circ$.

But, more than this, we should have

$$k'(1 - \cos \phi)^2 + E^2(k, \phi) < \phi^2,$$

viz. writing as before $k = \frac{1}{\sin \phi}$, this is

$$E^2(\phi', \phi) < \phi^2 - \frac{(1 - \cos \phi)^2}{\sin^2 \phi};$$

this must be true, although (relating as it does to a form of E for which k is greater than 1) there might be some difficulty in verifying it.

There is, as in the first case, no limit to the value of AB , viz. this may be $= 360^\circ$, the spherical zone being then cut along a meridian, or it may be greater than 360° ; and, moreover, the spherical quadrilateral may extend south of the equator, but of course so that the limiting south latitude does not extend beyond the foregoing value $\sin^{-1} \frac{1}{k}$: viz. we may have a zone between the latitudes $\pm \sin^{-1} \frac{1}{k}$, which may be a complete zone from longitude 0° to 360° or to any greater value than 360° .

The result is, that the zone is deformed into a surface of revolution, which in its general form resembles that obtained by the revolution

of a half-circle or half-ellipse about a line parallel to and beyond its bounding diameter, the bounding half-diameter being less, and the greatest radius of rotation greater, than the radius of the original sphere.

642. On a Differential Relation Between the Sides of a Quadrangle.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 99–101.]

[Page 33]

Let the sides and diagonals YZ , ZX , XY , OX , OY , OZ of a quadrangle be f , g , h , a , b , c , and let the component triangles be denoted as follows:

$$A = \triangle YZO = (b, c, f), \quad B = \triangle ZXO = (c, a, g), \quad C = \triangle XYO = (a, b, h), \quad \Omega = A + B + C$$

viz. A , B , C , Ω are the triangles whose sides are (b, c, f) , (c, a, g) , (a, b, h) , (f, g, h) respectively, so that $\Omega = A + B + C$.

Then we have between (a, b, c, f, g, h) an equation giving rise to a differential relation, which may be written

$$\Omega(Aa da + Bb db + Cc dc) - (BCf df + CAg dg + ABh dh) = 0.$$

This may be proved geometrically and analytically.

First, for the geometrical proof, it is enough to prove that, when a and b alone vary, the relation between the increments is $Aa da + Bb db = 0$; for then a and g alone varying, the relation between the increments will be $Ca da - Cg dg = 0$ (as to the negative sign it is clear from the figure that a , g will increase or diminish together): and we thence at once infer the general relation.

We have consequently to prove that, considering a and b as alone variable,

$$Aa da + Bb db = 0;$$

or, what is the same thing,

$$Aa da : -Bb db = \triangle XOZ : \triangle YOZ.$$

The points XYZ remain fixed; but O moves through the infinitesimal arc OO' , centre Z , which may be considered as situate in the right line OM drawn from O at right angles to ZO , and meeting XY produced in the point M .

And then, writing for a moment $\angle OXY = X$, $\angle OYX = Y$, $\angle OMY = M$, we find at once

$$da = OO' \cos(X + M), \quad -db = OO' \cos(Y - M);$$

that is,

$$\frac{da}{-db} = \frac{\cos(X + M)}{\cos(Y - M)},$$

or

$$\frac{Aa da}{Bb db} = \frac{a \cos(X + M)}{b \cos(Y - M)}.$$

But drawing $X\alpha$, $Y\beta$ each of them at right angles to ZO , we have $a \cos(X + M) = X\alpha$, $b \cos(Y - M) = Y\beta$, and evidently $\triangle XOZ : \triangle YOZ = X\alpha : Y\beta$; whence the equation is

$$\frac{Aa da}{Bb db} = \frac{\triangle XOZ}{\triangle YOZ},$$

which is the required relation.

[Page 34]

For the analytical proof, it is to be observed that the relation between a , b , c , f , g , h is a quadric relation in the quantities a^2 , b^2 , c^2 , f^2 , g^2 , h^2 respectively; this may be written

$$\begin{aligned} 1 + a^2(b^2 + c^2 - f^2)(g^2 + h^2 - f^2) \\ + b^2(c^2 + a^2 - g^2)(h^2 + f^2 - g^2) \\ + c^2(a^2 + b^2 - h^2)(f^2 + g^2 - h^2) \\ - (b^2 + c^2 - f^2)(c^2 + a^2 - g^2)(a^2 + b^2 - h^2) = 0; \end{aligned}$$

say for a moment this is $A + Ba^2 + Ca^4 = 0$, where

$$\begin{aligned} A &= b^2g^2 + b^2h^2 + c^2h^2 + c^2g^2 + f^2(b^2 - h^2)(c^2 - g^2) \\ &\quad - (b^2 + c^2)g^2h^2 - (g^2 + h^2)b^2c^2, \\ B &= -(b^2 - c^2)(g^2 - h^2) + f^2(-b^2 - c^2 + f^2 - g^2 - h^2) + f^4, \\ C &= c^2; \end{aligned}$$

then we have as usual

$$\frac{du}{da} da + \frac{du}{db} db + \&c. = 0,$$

[Page 35]

where

$$\frac{du}{da} = Ca^2 + B.$$

But in virtue of $u = 0$, we have

$$(Ca^2 + B)^2 = C(Ca^4 + Ba^2 + A) + (B^2 - AC),$$

that is, $\frac{du}{da} = \sqrt{B^2 - 4AC}$; and here $B^2 - 4AC$ is a quartic function of f^2 , which is easily seen to reduce itself to the form

$$f^2 - (g + h)^2, \quad f^2 - (g - h)^2, \quad f^2 - (b + c)^2, \quad f^2 - (b - c)^2.$$

The coefficients of $da, db, dc, \&c.$, are given as expressions of the like form; substituting their values, the differential relation is

$$\sqrt{[f^2 - (g + h)^2][f^2 - (g - h)^2][f^2 - (b + c)^2][f^2 - (b - c)^2]} Aa da + \&c. = 0,$$

which is, in fact, the foregoing result.

It is right to notice that there are in all 16 linear factors,

$$\begin{array}{llll} f + g + h, & b + c + f, & c + a + g, & a + b + h \\ \text{say } d, & f, & g, & h, \\ -f + g + h, & -b + c + f, & -c + a + g, & -a + b + h \\ \text{say } d', & f', & g', & h', \\ f - g + h, & b - c + f, & c - a + g, & a - b + h \\ \text{say } d'', & f'', & g'', & h'', \\ f + g - h, & b + c - f, & c + a - g, & a + b - h \\ \text{say } d''', & f''', & g''', & h''', \end{array}$$

and this being so, the coefficients of $Aa da, Bb db, Cc dc, Ff df, Gg dg, Hh dh$, are

$$\begin{aligned} & \sqrt{dd'd''d''' \cdot ff'f''f''' \cdot gg'g''g''' \cdot hh'h''h'''}, \\ & \sqrt{dd'd''d''' \cdot gg'g''g''' \cdot hh'h''h'''' \cdot ff'f''f'''}, \\ & -\sqrt{hh'h''h'''' \cdot ff'f''f''' \cdot gg'g''g''' \cdot dd'd''d'''}, \&c. \end{aligned}$$

respectively.

We may imagine the quadrilateral $ZOXY$ composed of the four rods ZO, OX, XY, YZ (lengths c, a, h, f as before) jointed together

at the angles, and kept in equilibrium by forces B, G acting along the diagonals $OY(= b)$, $ZX(= g)$ respectively. We have c, a, h, f given constants, and the relation $\phi(a, b, c, f, g, h) = 0$, which connects the six quantities is the relation between the two variable diagonals (g, b) ; by what precedes, the differential relation $\phi'_g \cdot dg + \phi'_b \cdot db = 0$ is equivalent to $\Omega B b db - C A g dg = 0$.

By virtual velocities we have as the condition of equilibrium $B db + G dg = 0$; hence, eliminating db, dg we have

$$\frac{B}{\Omega B b} = \frac{G}{-C A g}, \quad \text{or, say} \quad \frac{B}{b} = \frac{G}{g} = \frac{1}{\Delta X Y O \cdot \Delta Z Y O - \Delta X Y Z \cdot \Delta Z X O},$$

viz. the forces, divided by the diagonals along which they act, are proportional to the reciprocals of the products of the two pairs of triangles which stand on these diagonals respectively. The negative sign shows, what is obvious, that the forces must be, one of them a pull, the other a push.

643. On a Quartic Curve with Two Odd Branches.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 107, 108.]

[Page 36]

It is a known theorem that the branches of a plane curve are even or odd; viz. two even branches, or an even and an odd branch (whether of the same curve or of different curves) intersect in an even number (it may be 0, and this is to be understood throughout) of real points; but two odd branches (of the same curve or of different curves) intersect in an odd number of real points¹.

In particular, a right line is an odd branch, and hence it meets any even branch of a curve in an even number of real points, and an odd branch in an odd number of real points; or (what is the same thing) an even branch is one which is met by any right line whatever in an even number of real points; and an odd branch is one that is met by any right line whatever in an odd number of real points.

It is to be observed, that the simple term branch is used to denote what has been called a complete branch, viz. the partial branches which touch an asymptote at its opposite extremities are considered as parts of one and the same branch, and so in other cases. Thus a quadric curve, whether ellipse, parabola, or hyperbola, is one even branch; a cubic curve is either one odd branch, or else it is an odd branch and even branch; and generally a curve of an odd order has always an odd number of odd branches, and a curve of an even order has always an even number of odd branches.

A curve without nodes has at most one odd branch; for if there were two, these would intersect in a real point, which would be a real node on the curve. In particular, a quartic curve having two odd branches must have a real node; this however may be, as in the instance about to be given, a node at infinity.

A simple instance of a quartic curve with two odd branches is that represented by the equation

$$(a^2 - 1)(y^2 + 1) - 2mxy = 0,$$

¹The two branches must be distinct branches; a branch whether odd or even does not of necessity intersect itself (have upon it any real node), but it may intersect itself in an odd, or an even, number of real points.

[Page 37]

or, what is the same thing,

$$x^2 = \frac{1}{2}\{2 + m^2 + m\sqrt{4 + m^2}\}, \quad \frac{1}{a^2} = \frac{1}{2}\{2 + m^2 - m\sqrt{4 + m^2}\},$$

so that m being positive $a > 1$, and the curve consists of two real branches included between the lines $x = a$, $x = \frac{1}{a}$, and the lines $x = -a$, $x = -\frac{1}{a}$ respectively; each of these lines touches the curve in a real point, viz. x having any one of the last-mentioned values, the value of y at the point of contact is $y = \frac{m}{2x}$; and between each pair of lines we have the asymptote $x = +1$ or $x = -1$.

Hence the curve has the form shown in the figure, and it is thereby evident, that each branch of the curve is met by any real right line whatever in one real point, or else in three real points.

The numerical values in the figure are $a = \frac{3}{2}$, $m = \frac{5}{4}$, whence also $x = a$ or $-\frac{1}{a}$, $y = 1$, and $x = -a$ or $\frac{1}{a}$, $y = -1$.

The curve has two nodes at infinity, viz. writing the equation in the form

$$(x^2 - z^2)(y^2 + z^2) - 2mxyz^2 = 0,$$

that is,

$$x^2y^2 + z^2(x^2 - y^2 - 2mxy) + z^4 = 0,$$

it appears that the points $(z = 0, x = 0)$, $(z = 0, y = 0)$ are each of them a node. The first of these $(z = 0, x = 0)$ is the real intersection of the two odd branches: the other of them is a conjugate point.

644. Note on Magic Squares.

[From the *Messenger of Mathematics*, vol. vi. (1877), p. 168.]

[Page 38]

In a magic square of any odd order, formed according to the ordinary process, there is a tolerably simple analytical expression for the number which occupies any given compartment; thus taking the square of 21, let the dexter diagonals (N.W. to S.E.) commencing from the N.E. corner compartment, be numbered 1, 2, 3, ..., 20, 21, 20', 19', ..., 2', 1', the diagonals of course containing these numbers of compartments respectively; and in any diagonal let the compartments reckoning from the top line be numbered 1, 2, 3, ... respectively; then if $D_{\theta, \phi}$ (or $D'_{\theta, \phi}$ as the case may be) denotes the number in the compartment ϕ of the diagonal θ or θ' , we have

$$\begin{aligned} D_{\theta+\phi, \phi} &= 20(\theta + \phi) + 10 + \phi, \\ D_{\theta', \phi} &= 20\theta' + 231 + \phi(-21), \\ D_{\theta'+\phi', \phi} &= -22\phi + 430 + \phi, \end{aligned}$$

where $\theta + \phi$ is to be replaced by its value, $= \theta'$ or $> \theta'$ as the case may be; and in like manner for the other half of the square; it would not be difficult, but it would take some space, to give the general formulae for a square of any odd order.

The formula for a square of the order $2n + 1$ is at once obtained from the consideration that the numbers of the several dexter diagonals commence with the terms of the arithmetical progression

$$2n^2 + n, \quad 2n^2 + 3n + 1, \quad 2n^2 + 5n + 2, \quad \dots,$$

viz. the dexter diagonal through the N.E. corner compartment (which contains a single compartment, No. $2n^2 + n + 1$) is called diagonal 1; the next dexter diagonal (which contains two compartments numbered $2n^2 + n$, $2n^2 + n + 2$) is called diagonal 2; and so on.

645. A Smith's Prize Paper, 1877.

[From the *Cambridge Examination Papers*, 1877.]

[Page 39]

1. Given a curve of the order m , and a line meeting it in m real points $O_1, O_2, \dots, O_m$, find the number of the tangents from the several points $O_1, O_2, \dots, O_m$, which lie altogether in the plane of the curve; and thence infer the theorem "If a line meet a curve of the order m in m real points, the number of real tangents from these points to the curve is $m(m-1)$ or $m(m-2)$ according as m is odd or even."
2. Show that the locus of a point P , such that its polars with respect to three conics meet in a point, is a curve of the third order.
3. Find the condition which must be satisfied in order that the equation

$$(a, b, c, d \mid x, y)^3 = 0$$

may represent three lines; and in particular when the three lines are concurrent. Find also the condition that the equation may represent a line and its two tangentials (viz. lines each of them meeting the line in an inflexion of the cubic curve represented by the equation).

4. In the cubic curve

$$y^2 = 4x^3 - g_2x - g_3,$$

if x, y be expressed as functions of the parameter u , show that

$$x(u+v) + x(u) + x(v) = \frac{1}{4} \left(\frac{y(u) - y(v)}{x(u) - x(v)} \right)^2.$$

5. Find an expression for the number of the conics which pass through four given points and touch a given line; and account for the different kinds of solution.
6. Find the differential equation of the lines of curvature of an ellipsoid; and (without attempting a general solution) show that through a given point on the ellipsoid there pass four umbilici.

[Page 40]

7. Find the equation of the surface which is the locus of the foot of the perpendicular from the centre of an ellipsoid on the tangent plane at any point of one of its lines of curvature; and show that this surface is a particular case of the wave surface.
8. An ellipsoid and a hyperboloid of one sheet are confocal; taking the equations to be

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \quad \frac{x^2}{a^2 + \theta} + \frac{y^2}{b^2 + \theta} + \frac{z^2}{c^2 + \theta} = 1,$$

show that the intersection consists of curves of curvature on each surface respectively; and give a construction (depending on the rectilinear generation of the hyperboloid) for finding at any point of the intersection the normal of the ellipsoid.

9. The consecutive positions of a rigid body are determined by the values of six parameters; assuming the general expressions for the sliding and rotating velocities, form the condition that the motion may be simply rotational; and explain the result.
10. Find the condition in order that the equation

$$a \cos x + b \cos y + c \cos z + d \cos w + e \cos t + f = 0$$

may be satisfied identically (i.e. for all values of x, y, z, w, t); and thence find the condition in order that the equation

$$a \cos x + b \cos y + c \cos z + d \cos w + e \cos t + f = 0$$

may have a simultaneous solution with its derived equation

$$a \sin x + b \sin y + c \sin z + d \sin w + e \sin t = 0.$$

[Page 41]

11. Find the orthogonal trajectory of the series of parabolas

$$y^2 = 4a(x + a),$$

where a is the variable parameter.

12. Explain the different singularities of a curve; and in particular show that a cusp is to be regarded as a double point.
13. Show how to find the equation of the first or second reciprocal of a given curve; and apply it to find the reciprocal of a cubic with respect to a conic.

[Page 42]

14. Find the number of the lines which meet each of four given lines and touch a given quadric surface.
15. Two surfaces of the orders m, n respectively intersect in a curve of the order μ ; find the order of the cone standing on this curve, the vertex of the cone being a given point; and thence find the order of the scroll generated by the lines which meet the curve of intersection in two points, and also meet a given line.
16. Find the partial differential equation of the surfaces which cut at right angles a given series of surfaces

$$F(x, y, z) = C.$$

[Page 43]

17. In the question of the attraction of an ellipsoid on an external point, assuming only the attraction of a parallel plate on an external point, and that of a solid elliptic disc on a point in its axis, show how the result is obtained.
18. Find the attraction of a solid ellipsoid on an external point situate on one of the principal sections, the principal axes being all unequal.
19. A body moves under the action of a central force; find the differential equation of the orbit; and show that the orbit is a conic section when the force varies as the inverse square of the distance.

[Page 44]

20. The roots of the equation

$$\frac{x^2}{a^2 + \theta} + \frac{y^2}{b^2 + \theta} + \frac{z^2}{c^2 + \theta} = 1$$

are λ, μ, ν ; explain the geometrical signification of (λ, μ, ν) and (λ, μ) respectively; and find the partial differential equation of the surfaces which cut at right angles the series of confocal quadrics

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} = 1.$$

21. Find the potential of the attraction of a solid ellipsoid on an external point; and thence the component attractions.
22. Find the attraction of a solid elliptic disc on an external point in its axis; and show that it is the same as if the mass were condensed into a focal conic.

[Page 45]

23. Find the differential equation of the asymptotic lines on a given surface; and apply it to determine these lines on the helicoid and on the catenoid.
24. Show that the two sets of lines of curvature are at right angles to each other; and find the differential equation of the lines of curvature on the paraboloid

$$\frac{x^2}{a} + \frac{y^2}{b} = 2z.$$

25. Find the equation of the surface of the wave; and show that it is the same surface as that which is the locus of the feet of the perpendiculars from the centre on the tangent planes to the lines of curvature of an ellipsoid.

[Page 46]

26. Find the general expression for the curvature of a plane curve; and in particular for the curve $y = f(x)$.
27. Find the equation of a geodesic line on a surface; and show that on a developable surface the geodesic lines become right lines when the surface is developed into a plane.

28. Explain what is meant by the “Index of a Point;” and find the index of each of the conical points of the wave surface.

646. On the General Equation of Differences of the Second Order.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 79–85.]

[Page 47]

I consider the general equation of differences of the second order,

$$u\xi_{n+2} + B\xi_{n+1} + C\xi_n = 0,$$

where A, B, C are given constant coefficients. Writing this in symbolical form as

$$(A\varpi^2 + B\varpi + C)\xi_n = 0,$$

where ϖ denotes the operation of passing from ξ_n to ξ_{n+1} , the solution depends on the roots α, β of the quadratic equation $A\varpi^2 + B\varpi + C = 0$.

The case of equal roots, $\alpha = \beta$, gives rise to the solution

$$\xi_n = (A' + B'n)\alpha^n,$$

and the general case where the roots are unequal gives

$$\xi_n = A'\alpha^n + B'\beta^n.$$

The paper goes on to consider the general theory, and in particular the relation of the solution to the continued fraction

$$\frac{C}{B+} \frac{AC}{B+} \frac{AC}{B+} \dots$$

and the corresponding series

$$u\xi_0 - B\xi_1 + C\xi_2 - \dots$$

The complete theory requires the consideration of the cases where the roots are equal, and where they are unequal; and the relation of the solution to the particular integrals.

[Page 48]

The analytical theory of the equation of differences is continued, showing the relation between the successive coefficients and the convergents of the continued fraction. If we write

$$\frac{C}{B+} \frac{AC}{B+} \frac{AC}{B+} \dots = \frac{p_n}{q_n},$$

then p_n and q_n satisfy the same equation of differences, with different initial conditions. The general theory is established for the solution of the equation in terms of the roots α, β .

[Page 49]

The solution is thus completed, and the relation to the theory of continued fractions is fully exhibited. The different forms of the solution are considered, according as the roots are real or complex, and the particular case of equal roots is treated as a limiting case of the general theory.

647. On the Quartic Surfaces Represented by the Equation, Symmetrical Determinant = 0.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 86–101.]

[Page 50]

I consider the quartic surface represented by the equation

$$\begin{vmatrix} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & d \end{vmatrix} = 0,$$

where the coordinates are subjected to the condition that the symmetrical determinant vanishes. This is a particular form of quartic surface, having properties connected with the theory of quadrics and the geometry of the position.

[Page 51]

The surface is considered in relation to the theory of the quadric locus represented by the equation

$$(a, b, c, d, f, g, h, l, m, n \mid x, y, z, w)^2 = 0,$$

and the several forms which the quartic surface may assume, according to the different relations between the coefficients.

[Page 52]

The equation $V = 0$ thus represents a quartic surface; and the different forms are considered according to the nature of the component quadrics. The surface has in general nodes or other singularities, which depend on the particular relations between the coefficients of the fundamental quadric.

[Page 53]

For each set of values of the coordinates satisfying the equation, there corresponds a point on the surface; and the particular case where the determinant reduces to the product of linear factors gives rise to the consideration of the conic in question.

[Page 54]

For each of the possible configurations, the surface may be regarded as the envelope of a quadric, or as the locus of a singular point on a system of quadrics. The theory is closely connected with that of the bitangent planes and the nodal curves of the surface.

[Page 55]

This is further developed; the several species of the surface are classified according to the nature of the singularities. The analytical conditions for the different forms are given, and the geometrical interpretation is supplied.

[Page 56]

The different species are tabulated, and the particular cases of degeneracy are considered. The surface may reduce to a quadric counted twice, or to the combination of two quadrics, or to other simpler forms, according to the relations between the coefficients.

648. Algebraical Theorem.

[From the *Quarterly Journal of Pure and Applied Mathematics*,
vol. xiv. (1877), p. 102.]

[Page 57]

If $a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p$ are the elements of a determinant of the 4th order, then the determinant can be expressed in terms of its minors; and there exist relations between the minors of different orders which are expressed by the identities connecting the complementary minors. The theorem relates to the expression of the square of any minor of order 2 as a function of the minors of orders 1 and 3.

In particular, if we consider the minors formed by omitting any two rows and columns, there exist relations of the form

$$\begin{vmatrix} A_1 & B_1 & C_1 & D_1 \\ A_2 & B_2 & C_2 & D_2 \\ A_3 & B_3 & C_3 & D_3 \\ A_4 & B_4 & C_4 & D_4 \end{vmatrix} = 0,$$

where the A 's, B 's, &c. denote minors of the original determinant.

649. Addition to Mr Glaisher's Note on Sylvester's Paper, "Development of an Idea of Eisenstein."

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 103–105.]

[Page 58]

The remark which I wish to make relates to the development of an idea of Eisenstein, as presented in Sylvester's paper on the subject. The point is that the formulae for the multiplication of the quotients of the periods of the elliptic functions admit of a more general presentation, in which the original forms are included as particular cases.

If we consider the standard form of the relation between the multipliers and the moduli, we have a system of equations which can be written in the symmetrical form

$$\sqrt{kk'} + \sqrt{ll'} = 1,$$

where k, k', l, l' are the moduli and complementary moduli of the elliptic functions related by the multiplication theorem.

The general theory leads to the consideration of the functions

$$\frac{\vartheta_2(0|\tau)}{\vartheta_3(0|\tau)}, \quad \frac{\vartheta_4(0|\tau)}{\vartheta_3(0|\tau)},$$

and the modular equations which connect the different values of these functions.

[Page 59]

where the summation extends over all integer values of m and n , which satisfy the condition $an^2 + 2bmn + cm^2 = N$, and where a, b, c are coefficients connected with the quadratic form. The result is expressed as a series involving the divisors of N , and can be written in the form

$$\sigma(N) = \sum_{d|N} d,$$

where the summation extends over all divisors of N . The general theory includes as particular cases the formulae of Eisenstein and the subsequent developments by Sylvester.

The relation to the theory of binary quadratic forms is indicated, and the connection with the class-number and the theory of genera.

650. On a Quartic Surface with Twelve Nodes.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 106–116.]

[Page 60]

I consider the quartic surface represented by the equation

$$p(YZ + XW) + q(ZX + YW) + r(XY + ZW) = 0,$$

where $p + q + r = 0$. This surface has 12 nodes, which are situate at the intersections of certain planes, and has properties analogous to those of the cubic surface with 4 nodes.

The more general equation

$$p(YZ - XW) + q(ZX - YW) + r(XY - ZW) = 0$$

is also considered; and the relation between the two forms is investigated. The surface with the plus signs has the 12 nodes, while the surface with the minus signs has 4 conical points and other singularities.

The 12 nodes are situate in threes on the 8 faces of the fundamental tetrahedron; and the planes of the faces meet the surface in conics which pass through the nodes.

[Page 61]

The condition for a node is that the partial derivatives of the function with respect to X, Y, Z, W shall all vanish. This gives the equations

$$rY + qZ + pW = 0, \quad rX + pZ + qW = 0, \quad qX + pY + rW = 0, \quad pX + qY + rZ = 0.$$

Eliminating $X : Y : Z : W$, we obtain the condition that the determinant

$$\begin{vmatrix} 0 & r & q & p \\ r & 0 & p & q \\ q & p & 0 & r \\ p & q & r & 0 \end{vmatrix} = 0,$$

which is satisfied in virtue of $p + q + r = 0$.

[Page 62]

viz. the surface having the 12 nodes is the original surface where

$$p(YZ + XW) + q(ZX + YW) + r(XY + ZW) = 0, \quad p + q + r = 0.$$

The surface may be regarded as a particular case of the general quartic with 16 nodes; and the reduction from 16 to 12 is effected by the special relations between the coefficients.

651. On a Special Surface of Minimum Area.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 117–126.]

[Page 63]

The general equation of a minimal surface is known to be

$$(1 + q^2)r - 2pqs + (1 + p^2)t = 0,$$

where $p = \frac{dz}{dx}$, $q = \frac{dz}{dy}$, $r = \frac{d^2z}{dx^2}$, $s = \frac{d^2z}{dx dy}$, $t = \frac{d^2z}{dy^2}$.

I consider the particular case where the surface is determined by the condition that it contains a system of circles; and I find that there is a special surface of minimum area which satisfies this condition.

Taking the equations of the minimal surface in Monge's form,

$$x = X + X_1, \quad y = Y + Y_1, \quad z = Z + Z_1,$$

where X , Y , Z are functions of a parameter α , and X_1 , Y_1 , Z_1 are functions of a parameter β , and where

$$X'^2 + Y'^2 + Z'^2 = 0, \quad X_1'^2 + Y_1'^2 + Z_1'^2 = 0,$$

we have the general solution; and the special case is obtained by particular assumptions for the forms of these functions.

[Page 64]

The conditions for the surface to contain circles are investigated. If the surface contains a singly infinite system of circles, then the circles must be lines of curvature on the surface; and the condition for this gives a differential equation which must be satisfied by the functions X , Y , Z and X_1 , Y_1 , Z_1 .

[Page 65]

The solution of the equation leads to the consideration of elliptic functions. We have

$$X' = \frac{1}{2} \left(\frac{1}{\wp' \alpha} \right), \quad Y' = \frac{1}{2} \left(\frac{\wp \alpha}{\wp' \alpha} \right), \quad Z' = \frac{1}{2} \left(\frac{\wp^2 \alpha}{\wp' \alpha} \right),$$

where $\wp \alpha$ denotes the Weierstrassian elliptic function; and similar expressions for X'_1 , Y'_1 , Z'_1 in terms of β .

[Page 66]

The complete determination of the surface requires the integration of these expressions; and we find

$$\begin{aligned} X &= \frac{1}{2} \int \frac{d\alpha}{\wp' \alpha}, \\ Y &= \frac{1}{2} \int \frac{\wp \alpha d\alpha}{\wp' \alpha}, \\ Z &= \frac{1}{2} \int \frac{\wp^2 \alpha d\alpha}{\wp' \alpha}. \end{aligned}$$

These integrals can be expressed in terms of the elliptic integrals of the first and second kinds; and the surface is thus completely determined.

[Page 67]

The surface is thus a particular minimal surface, distinguished from the general minimal surface by the property of containing a system of circles. It is a surface of the 10th order, and has certain singularities which are investigated in the paper. The complete theory is connected with that of the tetrahedroid and the wave surface.

652. On a Sextic Torse.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 127–132.]

[Page 68]

A torse (developable surface) of the sixth order is considered, having its generating lines arranged in a particular manner with respect to the fundamental planes of reference. The equation of the torse is obtained in the form

$$(\text{sextic equation in } X, Y, Z, W) = 0,$$

which represents a developable surface, that is, a surface generated by the tangents to a curve.

The particular sextic torse has its edge of regression a curve of the sixth order; and the generating lines of the torse are the tangents to this curve. The equation may be written in the form

$$(x^2 + y^2 + z^2)^3 - k(x^3 + y^3 + z^3)^2 = 0,$$

or in a more general homogeneous form

$$(X^2 + Y^2 + Z^2)^3 W^2 - k(X^3 + Y^3 + Z^3)^2 = 0.$$

The edge of regression is the curve of intersection of this surface with another surface of the same order; and the nature of the cuspidal edge is investigated.

[Page 69]

The two half-branches of the cuspidal edge are considered, and the nature of the point of regression is analyzed. The curve has a point at which the two branches meet, and the tangent line at this point is common to both branches.

[Page 70]

We have

$$r^2 = 1 + p^2 + q^2,$$

and thence also, if $\tan \alpha = 2p, \pm 2\sqrt{r^2 - 1}$, that is

$$\cos \alpha = \frac{1}{\sqrt{1 + 4p^2}}, \quad \sin \alpha = \frac{2p}{\sqrt{1 + 4p^2}}.$$

The analytical theory is developed, showing the relation between the curvature of the edge of regression and the generating lines of the torse. The point of closest approach of consecutive generators is determined, and the equation of the tangent plane at any point of the torse is found.

[Page 71]

The sextic torse is completely determined by the condition that its cuspidal edge is a curve of the sixth order with a particular singularity. The classification of the different species of sextic torses is indicated, and the relation to the general theory of developable surfaces is explained.

653. On a Torse Depending on the Elliptic Functions.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 133–141.]

[Page 73]

I consider a torse (developable surface) which depends on the elliptic functions; viz. the equation of the torse involves the elliptic functions of a parameter, and the cuspidal edge of the torse is a transcendental curve.

The general theory of such torsos requires the consideration of the Weierstrassian elliptic functions $\wp u$, $\wp' u$, and their connection with the geometry of the surface.

If we take the standard form

$$y^2 = 4x^3 - g_2x - g_3,$$

then the coordinates of a point on the cuspidal edge can be expressed as

$$x = \wp u, \quad y = \wp' u, \quad z = u\wp u - 2\zeta u,$$

where ζu is the Weierstrassian zeta-function. These expressions give the parametric representation of the edge of regression.

[Page 74]

The conditions that the surface may be a torse (i.e. developable) are that the consecutive generating lines shall intersect; and this gives rise to a relation between the parameter and the coordinates which is expressed by means of the elliptic functions.

The generating line at any point of the edge of regression is the tangent to the curve; and the equations of this tangent can be written in the form

$$\frac{X - x}{\wp'' u} = \frac{Y - y}{\wp''' u} = \frac{Z - z}{1}.$$

[Page 75]

The analytical theory is developed, showing the relation of the torse to the elliptic functions. The equation of the torse is obtained by eliminating the parameter u between the equations of the tangent line and the condition of developability.

[Page 76]

The equation of the torse is thus obtained in a form involving the elliptic functions; and the singularities of the surface are investigated. The cuspidal edge is a curve of the sixth order, and the surface has also nodal and other singular lines.

[Page 77]

We have thus the complete theory of the torse depending on the elliptic functions; and the relation to the general theory of developable surfaces is established. The particular cases where the torse degenerates are considered, and the classification of the different species is given.

654. On Certain Octic Surfaces.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 142–150.]

[Page 79]

I consider certain surfaces of the eighth order (octic surfaces) which arise in the theory of quadrics and the geometry of the associated lines. The general equation of such a surface may be written in the form

$$\Phi(X, Y, Z, W) = 0,$$

where Φ is a homogeneous function of the eighth degree.

If in this equation we write $c''h' = c$, $c'h'' = h$; the equation takes a more particular form, and the surface has properties connected with the configuration of the fundamental lines and points.

The octic surfaces considered are related to the theory of the complex of lines and the congruences which are formed by the lines meeting certain given curves. The surface is the locus of points satisfying certain conditions with respect to the configuration.

[Page 80]

The complete investigation of the surface requires the consideration of the several species, according to the nature of the fundamental configuration. The surface may have nodes and other singularities, which correspond to particular positions of the generating lines.

The equation of the surface may be written in the form

$$\begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix} = 0,$$

where A, B, C, F, G, H are functions of the coordinates of the second degree. This represents an octic surface which is the envelope of a certain quadric locus.

The theory is continued in subsequent pages, with the investigation of the particular forms and singularities of the surface.

**The Collected Mathematical Papers of
Arthur Cayley, F.R.S.**

Volume 10

Pages 101–150

Paper 654 (conclusion): *On Certain Octic Surfaces*

Paper 655: *A Memoir on Differential Equations*

654. On Certain Octic Surfaces.

[From the *Quarterly Journal of Pure and Applied Mathematics*,
vol. VII (1867), pp. 164–174.]

Conclusion of Paper

In fact, the equation of the surface may be written in the form

$$\begin{aligned} & w^2\{fx^2 + gy^2 + hz^2 - 2ghy^2z^2 - 2hfz^2x^2 - 2fgx^2y^2\} \\ & + 2w\{-cfx^2 - agy^2 - bhz^2x^2 + 2ka^2b^2c^2x^2y^2z^2 + bfx^2z^2 + cgy^2x^2 + ahz^2y^2\} \\ & + \{ay^2z^2 + bz^2x^2 + cx^2y^2\}^2 = 0, \quad (1) \end{aligned}$$

which puts in evidence the nodal curve

$$w = 0, \quad -ay^2z^2 - bz^2x^2 - cx^2y^2 = 0 :$$

there are three similar forms which put in evidence the other three nodal curves.

The four curves are so related to each other that every line which meets three of them meets also the fourth curve; there is consequently a singly infinite series of lines meeting each of the four curves; these break up into four series of lines each forming an octic scroll, and each scroll has the four curves for nodal curves respectively; that is, each scroll is a surface included under the foregoing general equation, and derived from it by assigning a proper value to the constant k .

To determine these values, write

$$\frac{\lambda}{af} + \frac{\mu}{bg} + \frac{\nu}{ch} = 0, \quad \frac{\lambda^2}{af} + \frac{\mu^2}{bg} + \frac{\nu^2}{ch} = 0,$$

equations which give four systems of values for the ratios $(\lambda : \mu : \nu)$. We have then

$$k = af \frac{\mu\nu}{\lambda^2} + bg \frac{\nu\lambda}{\mu^2} + ch \frac{\lambda\mu}{\nu^2},$$

viz. k has four values corresponding to the several values of $(\lambda : \mu : \nu)$.

The scroll in question is M. De La Gournérie's scroll Σ_1 ; the equation of the scroll Σ_1 is consequently obtained from the octic equation by writing therein the last-mentioned value of k .

It is to be noticed that k is, in effect, determined by a quartic equation; and, that, for a certain relation between the coefficients,

this equation will have a twofold root. Assuming that this relation is satisfied, and assigning to k its twofold value, the resulting scroll becomes a torse; that is, two of the four scrolls coincide together and degenerate into a torse; corresponding to the remaining two values of k we have two scrolls, companions of the torse. In order to a twofold value of k , we must have

$$\frac{af}{\lambda} = \frac{bg}{\mu} = \frac{ch}{\nu},$$

and thence

$$(af)^{\frac{1}{3}} + (bg)^{\frac{1}{3}} + (ch)^{\frac{1}{3}} = 0;$$

or, what is the same thing,

$$(af + bg + ch)^3 - 27abcfgh = 0.$$

C. X.

11

If for a moment we write

$$af = \alpha^3, \quad bg = \beta^3, \quad ch = \gamma^3, \quad \text{and therefore, } \alpha + \beta + \gamma = 0;$$

then for the twofold root, we have $\lambda : \mu : \nu = \alpha : \beta : \gamma$, and consequently

$$\begin{aligned} k &= \alpha^3(\gamma - \beta) + \beta^3(\alpha - \gamma) + \gamma^3(\beta - \alpha) \\ &= (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha), \end{aligned}$$

that is,

$$k = \{(af)^{\frac{1}{3}} - (bg)^{\frac{1}{3}}\} \{(bg)^{\frac{1}{3}} - (ch)^{\frac{1}{3}}\} \{(ch)^{\frac{1}{3}} - (af)^{\frac{1}{3}}\},$$

which agrees with the result in regard to the octic torse.

If in the octic equation we write (x, y, z, w) in place of (x^2, y^2, z^2, w^2) , then we have the quartic equation

$$\begin{aligned} &fx^2 + gy^2 + hz^2 + 2bcyz + 2cazx + 2abxy \\ &\quad - 2fghy^2z^2 - 2hfa z^2x^2 - 2fgbx^2y^2 \\ &\quad + 2cay^2zx - 2agy^2zw + 2cgz^2xw \\ &\quad + 2abz^2xy - 2bh z^2xw + 2ahz^2yw \\ &\quad - 2ghw^2yz - 2hfw^2zx - 2fgw^2xy \\ &\quad + 2kxyzw = 0, \quad (2) \end{aligned}$$

which is the equation of a quartic surface touched by the planes $x = 0$, $y = 0$, $z = 0$, $w = 0$, in the four conics

$$\begin{array}{ll} x = 0, & hzw - gwy + ayz = 0, \\ y = 0, & -hzw + fwx + bzx = 0, \\ z = 0, & gyw - fwx + cxy = 0, \\ w = 0, & -ayz - bzx - cxy = 0, \end{array}$$

respectively.

II.

The octic surface

$$\begin{aligned}
 U = & b^2c^2f^2x^4 + c^2a^2g^2y^4 + a^2b^2h^2z^4 + f^2g^2h^2w^4 \\
 & - 2a^2cg(bg-ch)f^2y^2z^2 - 2b^2ah(ch-af)g^2z^2x^2 - 2c^2bf(af-bg)h^2x^2y^2 \\
 & + 2a^2bh(\)y^2w^2 + 2b^2cf(\)z^2w^2 + 2c^2ag(\)x^2w^2 \\
 & - 2f^2bc(\)x^2v^2 - 2g^2ca(\)y^2v^2 - 2h^2ab(\)z^2v^2 \\
 & + 2fghi(\)x^2w^2 + 2g^2hf(\)y^2v^2 + 2h^2fg(\)z^2w^2 \\
 & + f^2(bg+ch-4bgch)w^2x^2 + g^2(ch+af-4chaf)w^2y^2 + h^2(af+bg-4ahfg)w^2z^2 \\
 & + a^2(\)y^2z^2 + b^2(\)z^2x^2 + c^2(\)x^2y^2 \\
 & - 2gh(bcgh - a^2f^2 - 2afbg - 2afch)w^2y^2z^2 \\
 & - 2bh(\)z^2x^2w^2 + 2cg(\)y^2x^2w^2 + 2bc(\)x^2y^2z^2 \\
 & - 2hf(cahf - bg - 2bgaf - 2bgch)w^2z^2x^2 \\
 & - 2cf(\)x^2y^2w^2 + 2ah(\)z^2y^2w^2 + 2ca(\)y^2x^2z^2 \\
 & - 2fg(abfg - ch - 2chaf - 2chbg)w^2x^2y^2 \\
 & - 2ag(\)y^2z^2w^2 + 2bf(\)x^2z^2w^2 + 2ab(\)z^2x^2y^2 \\
 & + 2\Omega x^2y^2z^2w^2 = 0, \quad (3)
 \end{aligned}$$

where the values of the coefficients indicated by () are at once obtained by the proper interchanges of the letters, and where Ω is an arbitrary coefficient, is a surface having the four nodal conics

$$\begin{aligned}
 x = 0, & \quad cfy^2 - bz^2 + fw^2 = 0, \\
 y = 0, & \quad -caz^2 + az^2 + gw^2 = 0, \\
 z = 0, & \quad bay^2 - ay^2 + hw^2 = 0, \\
 w = 0, & \quad -fax^2 - gy^2 - hz^2 = 0.
 \end{aligned}$$

In fact, writing the equation under the form

$$w^2\Theta + (fax^2 + gy^2 + hz^2)^2 \times (b^2c^2y^2z^2 + c^2a^2z^2x^2 + a^2b^2x^2y^2 - 2a^2bcy^2z^2 - 2b^2caz^2x^2 - 2c^2abx^2y^2) = 0$$

we put in evidence the nodal conic $w = 0$, $fax^2 + gy^2 + hz^2 = 0$: and similarly for the other nodal conics.

It is to be observed, that the complete section by the plane $w = 0$ is the conic $fax^2 + gy^2 + hz^2 = 0$, twice repeated, and the quartic

$$b^2c^2x^4 + c^2a^2y^4 + a^2b^2z^4 - 2a^2bcy^2z^2 - 2b^2caz^2x^2 - 2c^2abx^2y^2 = 0 :$$

the latter being the system of four lines

$$\frac{x}{a} \pm \frac{y}{b} \pm \frac{z}{c} = 0.$$

The plane in question, $w = 0$, meets the other nodal conics in the six points

$$(x = 0, -b^2 + cz^2 = 0), \quad (y = 0, cz^2 - ax^2 = 0), \quad (z = 0, cx^2 - by^2 = 0),$$

which six points are the angles of the quadrilateral formed by the above-mentioned four lines.

The four conics are such, that every line meeting three of these conics meets also the fourth conic. The lines in question form a double system: each of these

systems has, in reference to any pair of nodal conics, a homographic property as follows; viz. considering for example the two conics in the planes $z = 0$ and $w = 0$ respectively, if a line meets these conics in the points P and Q respectively, and through these points respectively and the line $x = 0, y = 0$ we draw planes, then the system of the P planes and the system of the Q planes correspond homographically to each other, the coincident planes of the two systems being the planes $x = 0$ and $y = 0$ respectively.

Conversely, if through the line $(x = 0, y = 0)$ we draw the two homographically related planes meeting the two conics in the points P and Q respectively, then, for a proper value (determined by a quadratic equation) of the constant $k(= \theta' \div \theta)$ which determines the homographic relation, the line PQ will be a line meeting each of the four conics, and will belong to one or other of the above-mentioned two systems, as k is equal to one or the other of the two roots of the quadratic equation.

The scroll generated by the lines meeting each of the four conics, or what is the same thing, any three of these conics, is *primâ facie* a scroll of the order 16; but by what precedes, it appears that this scroll breaks up into two scrolls, which will be each of the order 8. Moreover, each scroll has the four conics for nodal curves; and since the equation $U = 0$ is the general equation of an octic surface having the four conics for nodal curves, it follows, that the equation of the scroll is derived from that of the octic surface $U = 0$, by assigning a proper value to the indeterminate coefficient Ω ; so that there are in fact two values of Ω , for each of which the surface $U = 0$ becomes a scroll.

To sustain the foregoing conclusions, take $x = \theta'y, x = \theta y$ for the equations of the two planes through the line $(x = 0, y = 0)$, which meet the z -conic and w -conic in the points P and Q respectively; then the equations of the line PQ are

$$\begin{aligned} \sqrt{(h\theta^2 + g)}(x - \theta'y) + \sqrt{(-h)}(\theta' - \theta)z &= 0, \\ -\sqrt{(h\theta'^2 - a)}(x - \theta y) + \sqrt{(-h)}(\theta' - \theta)w &= 0, \end{aligned}$$

or, writing therein $\theta' = k\theta$, the equations are

$$\begin{aligned} \sqrt{(h\theta^2 + g)}(x - k\theta y) + \sqrt{(-h)}(k - 1)\theta z &= 0, \\ -\sqrt{(hk^2\theta^2 - a)}(x - \theta y) + \sqrt{(-h)}(k - 1)\theta w &= 0. \end{aligned}$$

To find where the line in question meets the plane $y = 0$, we have

$$\begin{aligned}\sqrt{(h\theta^2 + g)}x + \sqrt{(-h)(k-1)}\theta z &= 0, \\ -\sqrt{(hk^2\theta^2 - a)}x + \sqrt{(-h)(k-1)}\theta w &= 0,\end{aligned}$$

and thence

$$\begin{aligned}(h\theta^2 + g)x^2 + h(k-1)^2\theta^2 z^2 &= 0, \\ (hk^2\theta^2 - a)x^2 + h(k-1)^2\theta^2 w^2 &= 0,\end{aligned}$$

or multiplying a , g and adding

$$(af + h g k^2)x^2 + h(k-1)^2(az^2 + gw^2) = 0,$$

or assuming

$$af + h g k^2 + ch(k-1)^2 = 0,$$

the equation is

$$-cx^2 + az^2 + gw^2 = 0.$$

That is, k being determined by the quadric equation

$$af + h g k^2 + ch(k - 1)^2 = 0,$$

the line PQ meets the y -conic $y = 0$, $-ca^2 + az^2 + gw^2 = 0$; and, in a similar manner, it appears that the line PQ also meets the x -conic $x = 0$, $cy^2 - hz^2 + fw^2 = 0$.

Writing for greater symmetry

$$\lambda : -k : k - 1 = \lambda : \mu : \nu,$$

we have

$$\lambda + \mu + \nu = 0, \quad af\lambda^2 + hg\mu^2 + ch\nu^2 = 0,$$

so that there are two systems of values of (λ, μ, ν) corresponding to, and which may be used in place of, the two values of k respectively.

Starting now from the equations

$$\begin{aligned} (h\theta^2 + g)(k\theta y - x)^2 + h(k - 1)^2\theta^2 z^2 &= 0, \\ (bk^2\theta^2 - a)(\theta y - x)^2 + h(k - 1)^2\theta^2 w^2 &= 0, \end{aligned}$$

the elimination of θ from these equations leads to an equation $U = 0$, of the above mentioned form but with a determinate value of the coefficient.

The process, although a long one, is interesting and I give it in some detail.

Elimination of θ from the foregoing equations.

We have

$$U = \text{M}\Pi[(h\theta^2 + g)(k\theta y - x)^2 + h(k - 1)^2\theta^2 z^2],$$

where Π denotes the product of the expressions corresponding to the four roots of the equation

$$(bk^2\theta^2 - a)(\theta y - x)^2 + h(k - 1)^2\theta^2 w^2 = 0.$$

Observing that this equation does not contain z , and that the expression under the sign Π does not contain w , it is at once seen that the product Π is in regard to (z, w) a rational and integral function of the form $(z^2, w^2)^2$; and since, in regard to (z, w) , U is also a rational

and integral function of the same form $(z^2, w^2)^2$, it is clear that the factor M does not contain z or w , but is a function of only (x, y) .

To determine it we may write $z = 0, w = 0$: this gives

$$c^2(bk^2\theta^2 - a)^2y^2(fx^2 + gy^2) = M\Pi(f\theta^2 + g)(k\theta y - x)^2,$$

where

$$(bk^2\theta^2 - a)(\theta y - x)^2 = 0,$$

and the values of θ are therefore

$$+\sqrt{\frac{a}{bk}} \frac{x}{y}, \quad -\sqrt{\frac{a}{bk}} \frac{x}{y}, \quad +\sqrt{\frac{a}{bk^2}} \frac{x}{y}, \quad -\sqrt{\frac{a}{bk^2}} \frac{x}{y}.$$

Hence substituting and observing that $ch(k-1)^2 = (af + bgk^2)$, it is easy to find

$$M = c^2x^2k^2(k-1)^4,$$

that is, we have

$$(bk^2\theta^2 - a)(\theta y - x)^2 + h(k-1)^2\theta^2w^2 = 0.$$

If for greater convenience we write $\theta = \frac{1}{\phi}$, then this formula becomes

$$(bk^2 - a\phi^2)(y - \phi x)^2 + h(k-1)^2\phi^2w^2 = 0,$$

or, what is the same thing,

$$(bk^2\phi^2 - a)(y - \phi x)^2 + h(k-1)^2\phi^2w^2 = 0.$$

Suppose that the terms in U which contain z^2 are $= \Theta z^2$; then we have

$$\Theta = A(1 - \phi_1^2y^2)^2 \Pi' [(bk^2\phi^2 - a)(y - \phi x)^2 + h(k-1)^2\phi^2w^2](k\phi - 1)^2,$$

where Π' refers to the remaining three roots ϕ_2, ϕ_3, ϕ_4 ; this may also be written

$$\Theta = \frac{A \Pi [(bk^2\phi^2 - a)(y - \phi x)^2 + h(k-1)^2\phi^2w^2](k\phi - 1)^2}{(bk^2\phi_1^2 - a)(y - \phi_1x)^2 + h(k-1)^2\phi_1^2w^2}.$$

Hence, observing that we have identically

$$\Pi \{\phi w \sqrt{f} \pm iy \sqrt{g}\} = \mp \frac{c}{h} \{c\phi w \sqrt{f} \pm iy \sqrt{g}\} y^2 \Pi' (\phi - \phi_r),$$

and writing $\phi_1 = \frac{1}{k}$, $h = \sqrt{-1}$ as usual, we find

$$\begin{aligned} \Pi (fx^2\phi^2 + gy^2)(k\phi - 1)^2 &= \frac{c^2}{h^2} \frac{1}{k^2} (bx^2 - ay^2 + hw^2)^2 \\ &\times \Pi' (bk^2\phi^2 - af + hv^2); \quad (4) \end{aligned}$$

whence, writing

$$\frac{\Pi (fx^2\phi^2 + gy^2)(k\phi - 1)^2}{\Pi [(bk^2\phi^2 - a)(y - \phi x)^2 + h(k-1)^2\phi^2w^2]} = \text{factor},$$

we may calculate separately the terms A, B, and 2Ω in

$$\frac{\Theta}{x^2y^2} = Ax^4 + 2\Omega x^2y^2 + By^4.$$

The first of these is

$$\frac{c^2}{h^2}(k-1)^2(fx^2 + k^2gy^2)^2(bx^2 - ay^2 + hw^2)^2 \div (fa^2 + k^2gy^2)^2, \quad (5)$$

if for shortness

$$\begin{aligned} (x, y, w)^2 = & (fx^2 + gy^2)[4\{(2-k)bx^2 - ay^2 + h(1-k)w^2\}^2 \\ & - 2(hx^2 - ay^2 + hw^2)\{(6-6k+k^2)bx^2 - ay^2 + h(1-k)^2w^2\}] \\ & + 4k^2(k-1)gy^2(bx^2 - ay^2 + hw^2)\{(2-k)bx^2 - ay^2 + h(1-k)w^2\}; \end{aligned} \quad (6)$$

the second is

$$\frac{c^2}{h^2}(k-1)^2(fx^2 + k^2gy^2)^2(bx^2 - ay^2 + hw^2)^2 \div (fa^2 + k^2gy^2)^2,$$

if for shortness

$$\begin{aligned} (x, y, w)^2 = & \{(fx^2 - k^2gy^2)(cfx^2 - cgy^2 - fgw^2) - 2ckfgx^2y^2\} \\ & \times [fgbk^2x^2 + \{2ch(k-1)^2 - af\}gy^2 + fgh(k-1)^2w^2] \\ & + 2\{k^2bg - ch(k-1)^2\}fgx^2y^2[c(k+1)(fx^2 - kgy^2) - kfgw^2]; \end{aligned} \quad (7)$$

and hence

$$\Theta = \frac{c^2(k-1)^2}{h^2(fx^2 + k^2gy^2)^2}[(x, y, w)_1^2 + 2(bx^2 - ay^2 + hw^2)(x, y, w)_2^2],$$

which must be a rational and integral function of (x, y, w) .

In partial verification of this, observe that, because U contains the terms $h^2cf(ch - af)x^2z^2 + 2\Omega x^2y^2z^2w^2$,

Θ should contain the terms

$$2h^2cf(ch - af)x^2 + 2\Omega x^2y^2w^2,$$

viz. in the term in x^2 should be $= 2h^2cf(ch - af)x^2$.

Now writing $y = 0$, $w = 0$, we have

$$\begin{aligned} A &= c^2f^2x^4, \\ (x, y, w)_1^2 &= by\{4(2 - k)^2 - 2(6 - 6k + k^2)\}x^2, \\ &= by(4 - 4k + 2k^2)x^2, \\ (x, y, w)_2^2 &= hcf g^2x^2; \end{aligned}$$

and hence the required term of Θ is x^2 multiplied by

$$\frac{c^2}{h^2}fh(4 - 4k + 2k^2) + 2h^2bcfg^2k^2,$$

viz. the coefficient is

$$\begin{aligned} &= 2b^2cf[ch(2 - 2k + k^2) + bgk^2] \\ &= 2b^2cf[ch + ch(1 - k)^2 + bgk^2], \end{aligned}$$

which in virtue of the relation $af + bgk^2 + ch(1 - k)^2$ becomes, as it should do,

$$= 2b^2cf(ch - af).$$

The actual division by $(fx^2 + k^2gy^2)^2$ would, however, be a very tedious process, and it is to be observed, that we only require to know the term $2\Omega x^2y^2w^2$ of Θ .

We may therefore adopt a more simple course as follows: the terms of Θ which contain w^2 are $(Ax^4 + 2\Omega x^2y^2 + By^4)w^2$,

hence writing for a moment

$$\begin{aligned} (x, y, w)_1^2 &= P + Qw^2, \\ (x, y, w)_2^2 &= R + Sw^2, \end{aligned}$$

and observing that we have

$$\begin{aligned} A &= c^2(fax^2 + gy^2 - 2c^2fg(fx^2 - gy^2)w^2 + \&c., \\ (bx^2 - ay^2 + hw^2)^2 &= (bx^2 - ay^2)^2 + 2h(bx^2 - ay^2)w^2 + \&c., \end{aligned}$$

we have

$$(fx^2+k^2gy^2)(Ax^4+2\Omega x^2y^2+By^4) = c^2h(fx^2+gy^2)Q - 2c^2fgh(fx^2-gy^2)P \\ + (bx^2-ay^2)^2S + 2h(bx^2-ay^2)R. \quad (8)$$

But in this identical equation we may write $x^2 = a$, $y^2 = b$, which gives

$$(af+k^2bg)(Ax^4+2\Omega ab+By^4) = c^2h(af+bg)Q - 2c^2fgh(af-bg)P;$$

and from the equation

$$(x, y, w)_1^2 = P + Qw^2,$$

we have

$$P + Qw^2 = (af + bg)^2 \left[\frac{(1-k)^2ab + (1-k)hw^2}{\dots} \right] \\ + 4k^2(k-1)bghw^2(1-k)ab. \quad (9)$$

that is,

$$Q = (k-1)^2 abh[ch(-6k^2 + 4k + 2) - 4k^2 bg],$$

whence

$$(k-1)^2(Ax^4 + 2\Omega ab + By^4) = ab(af + bg)^2[ch(-6k^2 + 4k + 2) - 4k^2 bg] \\ + 8fg(af - bg)(k-1)^2(ab)^2. \quad (10)$$

But we have

$$Ax^4 + By^4 = -2ab(af - bg)(-afbg + c^2h^2 + 2chaf + 2chbg),$$

and thence

$$2(k-1)^2\Omega = (af + bg)^2[ch(-6k^2 + 4k + 2) - 4k^2 bg] \\ + (k-1)^2(af - bg)\{-2afbg + 2c^2h^2 + 4chaf + 4chbg\}, \quad (11)$$

or

$$(k-1)^2\Omega = (af + bg)^2[ch(-3k^2 + 2k + 1) - 2k^2 bg] \\ + (k-1)^2(af - bg)\{3afbg + c^2h^2 + 2chaf + 2chbg\}. \quad (12)$$

Writing $-3k^2 + 2k + 1 = -3(k-1)^2 - 4(k-1)$, this is

$$\Omega = (af + bg)^2[ch(-3 - \frac{4}{k-1}) - \frac{2k^2}{(k-1)^2}bg] + (af - bg)\{3afbg + c^2h^2 + 2chaf + 2chbg\},$$

or since

$$1 : -k : k - 1 = \lambda : \mu : \nu;$$

and writing for shortness

$$(af, bg, ch) = (\alpha, \beta, \gamma),$$

this is

$$\Omega = (\alpha + \beta)^2\gamma(-3 - \frac{4\lambda}{\nu}) + (\alpha - \beta)\{3\alpha\beta + \gamma^2 + 2\gamma\alpha + 2\gamma\beta\},$$

which is the value of Ω : viz. the conclusion arrived at is that, eliminating θ from the equations

$$(h\theta^2 + g)(k\theta y - x)^2 + h(k-1)^2\theta^2 z^2 = 0, \\ (bk^2\theta^2 - a)(\theta y - x)^2 + h(k-1)^2\theta^2 w^2 = 0,$$

where k denotes a determinate function of af, bg, ch , viz. writing $af, bg, ch = \alpha, \beta, \gamma$ and

$$1 : -k : k - 1 = \lambda : \mu : \nu,$$

we have

$$\lambda + \mu + \nu = 0, \quad \alpha\lambda^2 + \beta\mu^2 + \gamma\nu^2 = 0,$$

equations which serve to determine k : the result of the elimination is the octic equation

$$b^2c^2f^2x^4 + \dots + 2\Omega x^2y^2z^2w^2 = 0,$$

where Ω has the last-mentioned value.

C. X.

12

The value of Ω is unsymmetrical in its form, and there are apparently six values; viz. writing

$$\begin{aligned} A &= (\beta + \gamma)^2 \alpha \left(-3 - \frac{4\lambda}{\nu}\right) + (\alpha - \beta)(-3 + \alpha\beta + \gamma^2), \\ B &= (\gamma + \alpha)^2 \beta \left(-3 - \frac{4\mu}{\lambda}\right) + (\beta - \gamma)(-3 + \beta\gamma + \alpha^2), \\ C &= (\alpha + \beta)^2 \gamma \left(-3 - \frac{4\nu}{\mu}\right) + (\gamma - \alpha)(-3 + \gamma\alpha + \beta^2), \\ A_1 &= (\beta + \gamma)^2 \alpha \left(-3 - \frac{4\lambda}{\nu}\right) - (\alpha - \beta)(-3 + \alpha\beta + \gamma^2), \\ B_1 &= (\gamma + \alpha)^2 \beta \left(-3 - \frac{4\mu}{\lambda}\right) - (\beta - \gamma)(-3 + \beta\gamma + \alpha^2), \\ C_1 &= (\alpha + \beta)^2 \gamma \left(-3 - \frac{4\nu}{\mu}\right) - (\gamma - \alpha)(-3 + \gamma\alpha + \beta^2), \end{aligned}$$

where for shortness $S = 2(\beta\gamma + \gamma\alpha + \alpha\beta)$, the six values would be A, B, C, A₁, B₁, C₁.

But we have really

$$A = B = C = -A_1 = -B_1 = -C_1,$$

so that Ω has really only two values, equal and of opposite signs, or, what is the same thing, Ω^2 has a unique value.

In fact, writing for shortness

$$\lambda + \mu + \nu = P, \quad \alpha\lambda^2 + \beta\mu^2 + \gamma\nu^2 = Z,$$

we find at once the identity

$$\lambda^2(A + A_1) = (\beta + \gamma)^2\{-2Z - 4\lambda\alpha P\},$$

so that $A = -A_1$ in virtue of $P = 0$, $Z = 0$.

And similarly $B = -B_1$, $C = -C_1$.

But the demonstration of the equation $A = B$ is more complicated. We have

$$\begin{aligned} A - B &= -3\alpha(\beta + \gamma)^2 + 3\beta(\gamma + \alpha)^2 \\ &\quad - 4\alpha(\beta + \gamma)^2 \frac{\mu}{\nu} - 2\gamma(\beta + \gamma)^2 \frac{\lambda^2}{\mu^2} \\ &\quad + 4\beta(\gamma + \alpha)^2 \frac{\nu}{\lambda} + 2\alpha(\gamma + \alpha)^2 \frac{\mu^2}{\nu^2} \\ &\quad + (\beta - \gamma)(S + \beta\gamma + \alpha^2) - (\gamma - \alpha)(S + \gamma\alpha + \beta^2), \quad (13) \end{aligned}$$

that is,

$$\begin{aligned}
 \lambda^2\nu^2(A-B) = \{ & -3\alpha(\beta+\gamma)^2+3\beta(\gamma+\alpha)^2+(\beta-\gamma)(S+\beta\gamma+\alpha^2)-(\gamma-\alpha)(S+\gamma\alpha+\beta^2)\} \\
 & -4\alpha(\beta+\gamma)^2\lambda\mu\nu^2 \\
 & -2\gamma(\beta+\gamma)^2\nu^2\lambda^2 \\
 & +4\beta(\gamma+\alpha)^2\lambda^2\mu\nu \\
 & +2\alpha(\gamma+\alpha)^2\lambda^2\mu^2. \quad (14)
 \end{aligned}$$

Or, denoting for a moment the coefficient of $\lambda^2\nu^2$ by K , and writing also $\nu^2 = \lambda - \alpha\lambda^2 - \beta\mu^2$, $\nu = P - \lambda - \mu$, this is

$$\begin{aligned}
 &= K\lambda^2\nu^2 - 4\alpha(\beta + \gamma)^2\lambda\mu^2 \\
 &\quad - 2(\beta + \gamma)^2\mu^2(\lambda - \alpha\lambda^2 - \beta\mu^2) \\
 &\quad + 4\beta(\gamma + \alpha)^2\lambda\nu(P - \lambda - \mu) \\
 &= -2(\beta + \gamma)^2\mu^2\lambda + 4\beta(\gamma + \alpha)^2\lambda^2P \\
 &\quad + 2\alpha(\gamma + \alpha)^2\lambda^2 \\
 &\quad - 4\beta(\gamma + \alpha)^2\lambda^2\mu \\
 &\quad + \{-4\alpha(\beta + \gamma)^2 + \dots\}\lambda\mu^2 \\
 &\quad + 2\beta(\beta + \gamma)^2\mu^4, \quad (15)
 \end{aligned}$$

and here the coefficient of $\lambda^2\mu^2$ is found to be

$$= 2\{\alpha\beta S(\alpha + \beta) + \gamma(\alpha - \beta)^2 - 3\gamma^2(\alpha + \beta)\}.$$

Hence, the terms without λ or P are $= 2V$, where

$$\begin{aligned}
 V &= \alpha(\gamma + \alpha)^2\lambda^4 - 2\beta(\gamma + \alpha)^2\lambda^3\mu \\
 &\quad + \{\alpha\beta S(\alpha + \beta) + \gamma(\alpha - \beta)^2 - 3\gamma^2(\alpha + \beta)\}\lambda^2\mu^2 \\
 &\quad - 2\alpha(\beta + \gamma)^2\lambda\mu^3 + \beta(\beta + \gamma)^2\mu^4, \quad (16)
 \end{aligned}$$

and this is identically

$$\begin{aligned}
 (\alpha + \gamma)V &= [\alpha(\gamma + \alpha)\lambda^2 - (\beta\gamma + \gamma\alpha + \alpha\beta)\lambda\mu + \beta(\beta + \gamma)\mu^2]^2 \\
 &- \{\alpha\lambda^2 + \beta\mu^2 + \gamma(P - \lambda - \mu)^2\} \times \{\alpha(\gamma + \alpha)^2\lambda^2 - 2(\beta\gamma + \gamma\alpha + \alpha\beta)\lambda\mu + \beta(\beta + \gamma)\mu^2\}, \quad (17)
 \end{aligned}$$

where observing that

$$\alpha\lambda^2 + \beta\mu^2 + \gamma\nu^2 = Z,$$

we have the first factor

$$(\alpha + \gamma)\lambda^2 + (\beta + \gamma)\mu^2 + 2\gamma\nu\lambda = -\lambda - \gamma\nu^2 + 2\gamma P(\lambda + \mu^2),$$

and consequently

$$\begin{aligned}
 \lambda^2\nu^2(A - B) &= -2(\beta + \gamma)^2\mu^2Z + 4\beta(\gamma + \alpha)^2\lambda^2P \\
 &+ 2\{Z - \gamma\nu^2 + 2\gamma P(\lambda + \mu)\}[\alpha(\gamma + \alpha)\lambda^2 - 2(\beta\gamma + \gamma\alpha + \alpha\beta)\lambda\mu + \beta(\beta + \gamma)\mu^2] : \quad (18)
 \end{aligned}$$

viz. in virtue of $P = 0$, $Z = 0$, we have $A = B$.

And thus

$$A = B = C = -A_1 = -B_1 = -C_1 :$$

so that the only values of Ω are, say, A and $-A$.

Reverting to the original equations

$$\begin{aligned}(h\theta^2 + g)(k\theta y - x)^2 + h(k-1)^2\theta^2 z^2 &= 0, \\ (bk^2\theta^2 - a)(\theta y - x)^2 + h(k-1)^2\theta^2 w^2 &= 0,\end{aligned}$$

say these are

$$(a, b, c, d, e \mid \theta, y)^2 = 0, \quad (a', b', c', d', e' \mid \theta, y)^2 = 0,$$

then the coefficients in the two equations have the values

$$\begin{aligned}a &= f\theta^2 y^2, & a' &= bk^2\theta^2, \\ b &= -2kf\theta xy, & b' &= -2bk^2\theta^2 xy, \\ c &= fx^2 + gk^2 y^2 + h(k-1)^2 z^2, & c' &= hk^2 x^2 - ay^2 + h(k-1)^2 w^2, \\ d &= -2gk\theta xy, & d' &= 2a\theta xy, \\ e &= g\theta^2 y^2, & e' &= -a\theta^2 y^2,\end{aligned}$$

where observe that only c contains z^2 , and only c' contains w^2 .

The result of the elimination is

$$\begin{vmatrix} a & b & c & d & e \\ & a & b & c & d \\ & & a & b & c \\ a' & b' & c' & d' & e' \\ & a' & b' & c' & d' \\ & & a' & b' & c' \end{vmatrix} = 0;$$

viz. here the only terms which contain z^2 and w^2 are

$$c^3 w^2 \cdot c'^3 z^2,$$

and hence the terms in z^4 and w^4 are

$$\begin{aligned}h^2(k-1)^4 z^4 \cdot a^2 b' k^2 x^2 y^2 + h^2(k-1)^4 w^4 \cdot f^2 g^2 k^2 x^2 y^2, \\ = h^2 k^2 (k-1)^4 x^2 y^2 (a^2 b' \cdot z^4 + f^2 g^2 \cdot w^4),\end{aligned}$$

viz. these are

or assuming that the determinant contains as a factor the function

$$b^2 c^2 f^2 w^2 x^2 + \dots + 2\Omega x^2 y^2 z^2 w^2,$$

with a properly determined value of Ω , we see that the other factor is $= h^2 k^2 (k-1)^4 x^2 y^2$, which agrees with a preceding result.

655. A Memoir on Differential Equations.

[From the *Quarterly Journal of Pure and Applied Mathematics*,
vol. XIV. (1877), pp. 292–339.]

We have to do with a set of variables, which is either unipartite $(x, y, z, \dots)$, or else bipartite $(x, y, z, \dots; p, q, r, \dots)$, the variables in the latter case corresponding in pairs x and p , y and q , &c.

A letter not otherwise explained denotes a function of the variables. Any such letter may be put $= \text{const.}$, viz. we thereby establish a relation between the variables and when this is so, we use the same letter to denote the constant value of the function.

Thus the set being $(x, y, z; p, q, r)$, H may denote a given function $pqr - xyz$; and then, if $H = \text{const.}$, we have $pqr - xyz = H$ (a constant). This notation, when once clearly understood, is, I think, a very convenient one.

The present memoir relates chiefly to the following subjects:

A. Unipartite set $(x, y, z, \dots)$. The differential system

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z} = \dots$$

and connected therewith the linear partial differential equation

$$X \frac{df}{dx} + Y \frac{df}{dy} + Z \frac{df}{dz} + \dots = 0 :$$

also the lineo-differential $X dx + Y dy + Z dz + \dots$.

B. Bipartite set $(x, y, z, \dots; p, q, r, \dots)$. The Hamiltonian system

$$\frac{dx}{dp} = \frac{dy}{dq} = \frac{dz}{dr} = \dots = \frac{dp}{dH} = \frac{dq}{dH} = \frac{dr}{dH},$$

and connected therewith the linear partial differential equation

$$\frac{dH}{dp} \frac{d\Theta}{dx} - \frac{dH}{dx} \frac{d\Theta}{dp} + \frac{dH}{dq} \frac{d\Theta}{dy} - \frac{dH}{dy} \frac{d\Theta}{dq} + \cdots = 0,$$

otherwise written

$$\frac{d(H, \Theta)}{d(p, x)} + \frac{d(H, \Theta)}{d(q, y)} + \frac{d(H, \Theta)}{d(r, z)} + \cdots = 0,$$

where H denotes a given function of the variables: also the Hamiltonian system as augmented by an equality $= dt$, and as augmented by this and another equality

$$\begin{aligned} dH &= \frac{dH}{dp} dp + \frac{dH}{dq} dq + \frac{dH}{dr} dr + \cdots \\ &= - \left(\frac{dH}{dx} dx + \frac{dH}{dy} dy + \frac{dH}{dz} dz + \cdots \right). \end{aligned}$$

C. Bipartite set $(x, y, z, \dots; p, q, r, \dots)$. The partial differential equation $H = \text{const.}$, where, as before, H is a given function of the variables, but $p, q, r, \dots$ are now the differential coefficients in regard to $x, y, z, \dots$ respectively of a function V of these variables, or, what is the same thing, there exists a function

$$V = \int (p dx + q dy + r dz + \cdots),$$

of the variables $x, y, z, \dots$

In what precedes, I have written $(x, y, z, \dots)$ to denote a set of any number n of variables, and $(x, y, z, \dots; p, q, r, \dots)$ to denote a set of any even number $2n$ of variables, and the investigations are for the most part applicable to these general cases. But for greater clearness and facility of expression, I usually consider the case of a set (x, y, z, w) , or $(x, y, z; p, q, r)$, &c., as the case may be, consisting of a definite number of variables.

The greater part of the theory is not new, but I think that I have presented it in a more compact and intelligible form than has hitherto been done, and I have added some new results.

Introductory Remarks. Art. Nos. 1 to 3.

1.

As already noticed, a letter not otherwise explained is considered as denoting a function of the variables of the set; but when necessary

we indicate the variables by a notation such as $z = z(x, y)$; z is here a function (known or unknown as the case may be) of the variables x, y , the z on the right-hand side being in fact a functional symbol.

And thus also $z = z(x, y) = \text{const.}$ denotes that the function $z(x, y)$ of the variables x, y has a constant value, which constant value is $= z$, viz. we thus indicate a relation between the variables x, y .

2.

The variables x, y , &c., may have infinitesimal increments dx, dy , &c.; and the equations of connexion between the variables then give rise to linear relations between these increments, the coefficients therein being differential coefficients and, as such, represented in the usual notation; thus if $z = z(x, y)$, we have

$$dz = \frac{dz}{dx} dx + \frac{dz}{dy} dy,$$

where $\frac{dz}{dx}, \frac{dz}{dy}$ are the so-called partial differential coefficients of z in regard to x, y respectively.

If we have $y = y(x)$, then also $dy = \frac{dy}{dx} dx$, and the foregoing equation becomes

$$\frac{d(z)}{dx} = \frac{dz}{dx} + \frac{dz}{dy} \frac{dy}{dx};$$

but considering the two equations $z = z(x, y)$ and $y = y(x)$ as determining z as a function of x , say $z = z(x)$, we have $dz = \frac{d(z)}{dx} dx$; whence comparing the two formulae

$$\frac{d(z)}{dx} = \frac{dz}{dx} + \frac{dz}{dy} \frac{dy}{dx}.$$

The distinction is best made, not by any difference of notation $\frac{dz}{dx}, \frac{d(z)}{dx}$, but by appending in any case of doubt the equations or equation used in the differentiation. Thus we have $\frac{dz}{dx}$, where $z = z(x, y)$: or, as the case may be, $\frac{d(z)}{dx}$ where $z = z(x, y)$ and $y = y(x)$.

3.

A relation between increments is always really a relation between differential coefficients: but we use the increments for symmetry and conciseness, as in the case of a differential system $\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z}$, or in a question relating to the lineo-differential $X dx + Y dy + Z dz$, for instance in the question whether this can be put $= du$.

Notations. Art. Nos. 4 to 6.**4. Functional determinants.**

If $a, b, c, \dots$ are functions of the variables $x, y, z, w, \dots$, then the determinants

$$\begin{vmatrix} \frac{da}{dx} & \frac{da}{dy} \\ \frac{db}{dx} & \frac{db}{dy} \end{vmatrix}, \quad \begin{vmatrix} \frac{da}{dx} & \frac{da}{dy} & \frac{da}{dz} \\ \frac{db}{dx} & \frac{db}{dy} & \frac{db}{dz} \\ \frac{dc}{dx} & \frac{dc}{dy} & \frac{dc}{dz} \end{vmatrix}, \quad \&c.,$$

are for shortness represented by

$$\frac{d(a, b)}{d(x, y)}, \quad \frac{d(a, b, c)}{d(x, y, z)}, \quad \&c.,$$

the notation being especially used in the first-mentioned case where the symbol is $= \left(\frac{a}{x} \right)$.

It is sometimes convenient to extend this notation, and for instance use $\frac{d(a, b)}{d(x, y, z)}$ to denote the series of determinants

$$\begin{vmatrix} \frac{da}{dx} & \frac{da}{dy} & \frac{da}{dz} \\ \frac{db}{dx} & \frac{db}{dy} & \frac{db}{dz} \end{vmatrix},$$

which can be formed by selecting in every way two columns to form thereout a determinant; the equation $\frac{d(a, b)}{d(x, y, z)} = 0$ will then denote that each of these determinants is $= 0$.

The analogous notation $\frac{d(a, b, c)}{d(x, y)}$ would denote non-existent determinants, viz. there are here not columns enough to form with them a determinant: and the notation is not required.

5.

In the case of a bipartite set $(x, y, z, \dots; p, q, r, \dots)$, if a, b are any functions of these variables, we consider the derivative

$$(a, b) = \frac{d(a, b)}{d(p, x)} + \frac{d(a, b)}{d(q, y)} + \frac{d(a, b)}{d(r, z)} + \dots,$$

viz. (a, b) is used to denote the sum of the functional determinants on the right hand.

6.

Taking again $(x, y, z, w, \dots)$ as the variables, then in the theory of the lineo-differential $X dx + Y dy + Z dz + W dw + \dots$, we use certain derivative functions analogous to Pfaffians.

They may be thus defined; viz. considering the numbers 1, 2, 3, 4, $\dots$ as corresponding to the variables $x, y, z, w, \dots$ respectively, we

have

$$\begin{aligned}
 1 &= X, \quad 2 = Y, \quad 3 = Z, \quad 4 = W, \quad \&c., \\
 12 &= \frac{dX}{dy} - \frac{dY}{dx}, \quad 13 = \frac{dX}{dz} - \frac{dZ}{dx}, \quad 23 = \frac{dY}{dz} - \frac{dZ}{dy}, \quad \&c., \\
 123 &= 1.23 + 2.31 + 3.12 \\
 &= X \left(\frac{dY}{dz} - \frac{dZ}{dy} \right) + Y \left(\frac{dZ}{dx} - \frac{dX}{dz} \right) + Z \left(\frac{dX}{dy} - \frac{dY}{dx} \right), \\
 1234 &= 12.34 + 13.42 + 14.23 \\
 &= \left(\frac{dX}{dy} - \frac{dY}{dx} \right) \left(\frac{dZ}{dw} - \frac{dW}{dz} \right) + \left(\frac{dX}{dz} - \frac{dZ}{dx} \right) \left(\frac{dW}{dy} - \frac{dY}{dw} \right) \\
 &\quad + \left(\frac{dX}{dw} - \frac{dW}{dx} \right) \left(\frac{dY}{dz} - \frac{dZ}{dy} \right).
 \end{aligned}$$

and, adding for greater distinctness the next following cases,

$$12345 = 1.2345 + 2.3451 + 3.4512 + 4.5123 + 5.1234,$$

$$123456 = 12.3456 + 13.4561 + 14.5612 + 15.6123 + 16.2345,$$

where of course 2345, &c., have the significations mentioned above.

Dependency of Functions. Art. Nos. 7 and 8.

7.

Two or more functions of the same variables may be independent, or else dependent or connected; viz. in the latter case any one of the functions is a function of the others. If $a = a(x)$, $b = b(x)$, the functions a , b are dependent, but if $a = a(x, y)$, $b = b(x, y)$, then the condition of dependency is

$$\frac{d(a, b)}{d(x, y)} = 0,$$

and, similarly, if $a = a(x, y, z)$, $b = b(x, y, z)$, then the conditions of dependency are

$$\frac{d(a, b)}{d(x, y, z)} = 0,$$

viz. if the equations thus represented are all of them satisfied, the functions are dependent, but if not, then they are independent.

Observe that, when $a = a(x, y, z)$, $b = b(x, y, z)$ as above, if we choose to attend only to the variables x , y , treating z as a mere constant, there is then a single condition of dependency $\frac{d(a, b)}{d(x, y)} = 0$, and so if we attend only to the variable x , treating y , z as mere constants, then a and b are dependent.

Thus when $a = x$, $b = x^2 + y$, the functions a , b are independent if we attend to both the variables x , y ; dependent if y be regarded as a constant.

8.

Further when $a = a(x, y)$, $b = b(x, y)$, $c = c(x, y)$, the functions a , b , c are dependent; but when $a = a(x, y, z)$, $b = b(x, y, z)$, $c = c(x, y, z)$, the condition of dependency is

$$\frac{d(a, b, c)}{d(x, y, z)} = 0,$$

and so when $a = a(x, y, z, w)$, $b = b(x, y, z, w)$, $c = c(x, y, z, w)$, the conditions of dependency are

$$\frac{d(a, b, c)}{d(x, y, z, w)} = 0,$$

viz. if all the equations thus represented are satisfied, the functions are dependent; but if not, then they are independent.

And so in other cases.

C. X.

13

The General Differential System. Art. Nos. 9 to 22.

9.

Taking the set of variables to be (x, y, z, w) , the system is

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z} = \frac{dw}{W},$$

and we associate with this the linear partial differential equation

$$X \frac{df}{dx} + Y \frac{df}{dy} + Z \frac{df}{dz} + W \frac{df}{dw} = 0.$$

10.

It is tolerably evident that the differential equations establish between x, y, z, w a threefold relation depending upon three arbitrary constants; in fact, regarding (x, y, z, w) as the coordinates of a point in four-dimensional space, and starting from any given point, the differential equations determine the ratios of the increments dx, dy, dz, dw , that is, the direction of passage to a consecutive point and then again taking for x, y, z, w the coordinates of this point, the same equations give the direction of passage to the next consecutive point, and so on.

The locus of the point is therefore a curve, or we have between the coordinates a threefold relation, and (the initial point being arbitrary) we have a curve of the system through each point of the four-dimensional space, viz. the relation must involve three arbitrary constants.

But this being so, the constants will be expressible as functions of the coordinates, viz. the threefold relation involving the three constants will be expressible in the form $a = \text{const.}$, $b = \text{const.}$, $c = \text{const.}$, where a, b, c denote respective functions of (x, y, z, w) ; and the differential equations thus establish between the variables the threefold relation $a = \text{const.}$, $b = \text{const.}$, $c = \text{const.}$.

11.

Observe that, in speaking of an integral a , we consider it as a function of the variables such that the differential equation gives $da = 0$; viz. a being a function of (x, y, z, w) , if its differential is

$$da = A dx + B dy + C dz + D dw,$$

the condition is

$$AX + BY + CZ + DW = 0.$$

This equation must be satisfied identically, and we have therefore a single partial differential equation

$$X \frac{da}{dx} + Y \frac{da}{dy} + Z \frac{da}{dz} + W \frac{da}{dw} = 0,$$

which must be satisfied by the function a ; the general integral of this partial differential equation contains an arbitrary function, and we may say that a is any function whatever of (x, y, z, w) which satisfies this partial differential equation.

Similarly b is any other function which satisfies the same partial differential equation, and so for c .

The number of integrals is thus unlimited, but if a be any particular integral, then every other integral is a function of (a, b, c) .

12.

In particular, if X, Y, Z, W are linear functions of the variables, say

$$X = a_1x + b_1y + c_1z + d_1w,$$

$$Y = a_2x + b_2y + c_2z + d_2w,$$

$$Z = a_3x + b_3y + c_3z + d_3w,$$

$$W = a_4x + b_4y + c_4z + d_4w,$$

then a particular integral is had by assuming $a = lx + my + nz + pw$, where l, m, n, p are constants; and the condition $da = 0$ gives at once

$$\begin{aligned} & l(a_1x + b_1y + c_1z + d_1w) + m(a_2x + b_2y + c_2z + d_2w) \\ & + n(a_3x + b_3y + c_3z + d_3w) + p(a_4x + b_4y + c_4z + d_4w) = 0, \end{aligned}$$

that is,

$$la_1 + ma_2 + na_3 + pa_4 = 0,$$

$$lb_1 + mb_2 + nb_3 + pb_4 = 0,$$

$$lc_1 + mc_2 + nc_3 + pc_4 = 0,$$

$$ld_1 + md_2 + nd_3 + pd_4 = 0,$$

equations which determine the ratios $(l : m : n : p)$.

13.

The general integral $a = \text{const.}$ is in fact any arbitrary function of the variables (x, y, z, w) which satisfies the partial differential equation

$$X \frac{da}{dx} + Y \frac{da}{dy} + Z \frac{da}{dz} + W \frac{da}{dw} = 0.$$

Taking a to be any particular integral, then every other integral is a function of a ; for if b be any other integral, then b is a function of (x, y, z, w) which satisfies the same partial differential equation, and we have therefore $b = \phi(a)$.

The general integral may be written in the form

$$\Phi(a, b, c) = 0,$$

where Φ is an arbitrary functional symbol; and this is equivalent to the system of three equations $a = \text{const.}$, $b = \text{const.}$, $c = \text{const.}$, involving as it should do three arbitrary constants.

14.

If in the differential system we have two integrals a, b , these may be regarded as given functions of (x, y, z, w) ; and we may then express the ratios $dx : dy : dz : dw$ by means of the Jacobians. In fact, regarding a, b as given functions, we have

$$\frac{dx}{d(a, b)/d(y, z, w)} = \frac{dy}{d(a, b)/d(z, w, x)} = \frac{dz}{d(a, b)/d(w, x, y)} = \frac{dw}{d(a, b)/d(x, y, z)},$$

where the notation is used to denote the series of functional determinants

$$\frac{d(a, b)}{d(y, z)}, \quad \frac{d(a, b)}{d(z, w)}, \quad \&c.,$$

and similarly for the other denominators.

15.

The foregoing ratios of $dx : dy : dz : dw$ may also be written in the form

$$\frac{dx}{\partial(a, b)/\partial(y, z)} = \frac{dy}{\partial(a, b)/\partial(z, x)} = \frac{dz}{\partial(a, b)/\partial(x, y)} = \frac{dw}{\partial(a, b)/\partial(x, y, z)},$$

where the notation is used to denote the series of partial derivatives

$$\frac{\partial a}{\partial y} \frac{\partial b}{\partial z} - \frac{\partial a}{\partial z} \frac{\partial b}{\partial y}, \quad \&c.,$$

and similarly for the other denominators.

16.

There is no general process for the integration of the differential system

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z} = \frac{dw}{W},$$

but in particular cases the integration may be effected by various methods; for instance, if we can find a factor λ such that

$$\lambda(X dx + Y dy + Z dz + W dw) = du,$$

where u is a function of (x, y, z, w) , then $u = \text{const.}$ is an integral.

17.

In particular, if the expression $X dx + Y dy + Z dz + W dw$ is an exact differential, say $= du$, then $u = \text{const.}$ is an integral; and if the expression is a perfect differential, say $= du + v dw$, then $u + vw = \text{const.}$ is an integral.

18.

The condition that $X dx + Y dy + Z dz + W dw$ should be an exact differential is that the Pfaffian derivatives should all vanish; viz. that we should have

$$12 = 0, \quad 13 = 0, \quad 23 = 0, \quad 14 = 0, \quad 24 = 0, \quad 34 = 0,$$

or, what is the same thing,

$$\frac{dX}{dy} = \frac{dY}{dx}, \quad \frac{dX}{dz} = \frac{dZ}{dx}, \quad \frac{dY}{dz} = \frac{dZ}{dy}, \quad \&c.$$

19.

The result would have been obtained by assuming a, b, c as known functions of (x, y, z, w) ; say, for example, $a = x, b = y$, and seeking to express c as a function of (x, y, z, w) , or say of (x, y, z) , for the determination of the ratios $dx : dy : dz : dw$.

20.

The differential system may be treated by the method of multipliers; if λ, μ, ν, ρ be factors such that

$$\lambda dx + \mu dy + \nu dz + \rho dw = 0,$$

then we have

$$\lambda X + \mu Y + \nu Z + \rho W = 0,$$

and the condition that this should be an exact differential gives rise to the partial differential equations for the determination of λ, μ, ν, ρ .

21.

The method of multipliers may be applied as follows: assuming that we have two integrals a, b , then the differential system may be written

$$\frac{dx}{\partial(a, b)/\partial(y, z, w)} = \frac{dy}{\partial(a, b)/\partial(z, w, x)} = \cdots,$$

where the denominators are the Jacobian determinants; and the condition for a third integral is that the expression

$$\frac{\partial(a, b)}{\partial(y, z, w)} dx + \frac{\partial(a, b)}{\partial(z, w, x)} dy + \cdots$$

should be an exact differential.

22.

If we have three integrals a, b, c , then the differential system may be written

$$\frac{dx}{\partial(a, b, c)/\partial(y, z, w)} = \cdots = \frac{dw}{\partial(a, b, c)/\partial(x, y, z)},$$

and the condition for a fourth integral is that the expression

$$\frac{\partial(a, b, c)}{\partial(y, z, w)} dx + \cdots + \frac{\partial(a, b, c)}{\partial(x, y, z)} dw$$

should be an exact differential; or, what is the same thing, that the Pfaffian derivatives of this expression should all vanish.

Particular Cases. Art. Nos. 23 to 29.**23.**

In the case of three variables (x, y, z) , the differential system is

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z},$$

and the partial differential equation is

$$X \frac{df}{dx} + Y \frac{df}{dy} + Z \frac{df}{dz} = 0.$$

24.

In the case of three variables (x, y, z) , if we have one integral $a = \text{const.}$, then the differential system may be written

$$\frac{dx}{\partial a / \partial y} = \frac{dy}{\partial a / \partial z} = \frac{dz}{\partial a / \partial x},$$

and the condition for a second integral is that the expression

$$\frac{\partial a}{\partial y} dx + \frac{\partial a}{\partial z} dy + \frac{\partial a}{\partial x} dz$$

should be an exact differential.

25.

If we have two integrals a, b in the case of three variables, then

$$\frac{dx}{\partial(a, b) / \partial(y, z)} = \frac{dy}{\partial(a, b) / \partial(z, x)} = \frac{dz}{\partial(a, b) / \partial(x, y)},$$

and the condition for a third integral is that

$$\frac{\partial(a, b)}{\partial(y, z)} dx + \frac{\partial(a, b)}{\partial(z, x)} dy + \frac{\partial(a, b)}{\partial(x, y)} dz$$

should be an exact differential.

26.

If the expression $X dx + Y dy + Z dz$ is an exact differential, say $= du$, then $u = \text{const.}$ is an integral; and the condition for this is

$$\frac{dX}{dy} = \frac{dY}{dx}, \quad \frac{dY}{dz} = \frac{dZ}{dy}, \quad \frac{dZ}{dx} = \frac{dX}{dz}.$$

27.

The method of multipliers for three variables is as follows: if λ, μ, ν be factors such that

$$\lambda dx + \mu dy + \nu dz = 0,$$

then $\lambda X + \mu Y + \nu Z = 0$, and the condition that this should be an exact differential gives rise to the equations for the determination of λ, μ, ν .

28.

If we have one integral $a = \text{const.}$ in the case of three variables, then the differential equation may be written

$$\frac{dx}{\partial a / \partial y} = \frac{dy}{\partial a / \partial z} = \frac{dz}{\partial a / \partial x},$$

and the condition for a second integral is that the expression

$$\frac{\partial a}{\partial y} dx + \frac{\partial a}{\partial z} dy + \frac{\partial a}{\partial x} dz$$

should be an exact differential; viz. that we should have

$$\frac{d}{dz} \left(\frac{\partial a}{\partial y} \right) = \frac{d}{dy} \left(\frac{\partial a}{\partial x} \right), \quad \&c.$$

29.

And we thence find

$$\frac{d(a, b)}{d(x, y)} = \frac{d(a, b)}{d(y, z)} = \frac{d(a, b)}{d(z, x)} = 0,$$

and each of these, as containing the differential coefficients of a and b , gives the condition for the dependency of these functions.

Pfaffian Theorem. Art. No. 30.**30.**

According as the variables are odd or even in number, the expression

$$X dx + Y dy + Z dz + W dw + \dots$$

may be of a more special form; for instance, it may be a sum of two terms $= \xi d\rho + \eta d\sigma$, or, what is the same thing, it may be reducible to the form

$$X_1 dx_1 + Y_1 dy_1 + Z_1 dz_1 + W_1 dw_1 + \dots,$$

where the number of terms is less than the original number.

The general theorem is that any such expression is reducible to a form in which the number of terms is at most equal to the number of variables; and, if the variables are even in number, the expression may be reducible to a form in which the number of terms is less than the number of variables.

In particular, if the variables are four in number, the expression $X dx + Y dy + Z dz + W dw$ is reducible to the form $\xi d\rho + \eta d\sigma$, where ξ, η, ρ, σ are functions of (x, y, z, w) .

The condition for this is that the Pfaffian derivative 1234 should vanish; viz. that

$$12.34 + 13.42 + 14.23 = 0.$$

And if the variables are six in number, the expression $X dx + Y dy + Z dz + W dw + U du + V dv$ is reducible to the form $\xi d\rho + \eta d\sigma + \zeta d\tau$, where $\xi, \eta, \zeta, \rho, \sigma, \tau$ are functions of the variables.

The condition for this is that the Pfaffian derivative 123456 should vanish; viz. that

$$12.3456 + 13.4561 + 14.5612 + 15.6123 + 16.2345 = 0.$$

The Hamiltonian System. Art. Nos. 31 to 39.

31.

The Hamiltonian system for a bipartite set $(x, y, z; p, q, r)$ is

$$\frac{dx}{dt} = \frac{dH}{dp}, \quad \frac{dy}{dt} = \frac{dH}{dq}, \quad \frac{dz}{dt} = \frac{dH}{dr}, \quad \frac{dp}{dt} = -\frac{dH}{dx}, \quad \frac{dq}{dt} = -\frac{dH}{dy}, \quad \frac{dr}{dt} = -\frac{dH}{dz},$$

where H is a given function of the variables.

32.

The system may be written in the form

$$\frac{dx}{dH/dp} = \frac{dy}{dH/dq} = \frac{dz}{dH/dr} = \frac{dp}{-dH/dx} = \frac{dq}{-dH/dy} = \frac{dr}{-dH/dz} = dt,$$

and we associate with this the linear partial differential equation

$$\frac{dH}{dp} \frac{d\Theta}{dx} - \frac{dH}{dx} \frac{d\Theta}{dp} + \frac{dH}{dq} \frac{d\Theta}{dy} - \frac{dH}{dy} \frac{d\Theta}{dq} + \frac{dH}{dr} \frac{d\Theta}{dz} - \frac{dH}{dz} \frac{d\Theta}{dr} = 0.$$

33.

It is to be observed that the Hamiltonian system is of the same form as the general differential system, if we write

$$X = \frac{dH}{dp}, \quad Y = \frac{dH}{dq}, \quad Z = \frac{dH}{dr}, \quad P = -\frac{dH}{dx}, \quad Q = -\frac{dH}{dy}, \quad R = -\frac{dH}{dz},$$

and the partial differential equation is then

$$X \frac{d\Theta}{dx} + Y \frac{d\Theta}{dy} + Z \frac{d\Theta}{dz} + P \frac{d\Theta}{dp} + Q \frac{d\Theta}{dq} + R \frac{d\Theta}{dr} = 0.$$

34.

The Hamiltonian system establishes between the variables a fivefold relation depending upon five arbitrary constants; in fact, regarding (x, y, z, p, q, r) as the coordinates of a point in six-dimensional space, and starting from any given point, the differential equations determine the direction of passage to a consecutive point, and so on; the locus of the point is therefore a curve, or we have between the coordinates a fivefold relation involving five arbitrary constants.

35.

The constants will be expressible as functions of the coordinates, viz. the fivefold relation will be expressible in the form

$$a = \text{const.}, \quad b = \text{const.}, \quad c = \text{const.}, \quad d = \text{const.}, \quad e = \text{const.},$$

where a, b, c, d, e denote respective functions of (x, y, z, p, q, r) ; and the differential equations thus establish between the variables the fivefold relation $a = \text{const.}, b = \text{const.}, c = \text{const.}, d = \text{const.}, e = \text{const.}$

36.

If in the Hamiltonian system we have two integrals a, b , these may be regarded as given functions of the variables; and we may then express the ratios $dx : dy : dz : dp : dq : dr$ by means of the Jacobians. In fact, regarding a, b as given functions, we have

$$\frac{dx}{d(a, b)/d(p, y, z, q, r)} = \dots,$$

where the notation is used to denote the series of functional determinants.

may be of a more special form; for instance, it may be a sum of two terms $= \xi d\rho + \eta d\sigma$, or,

37.

The general integral of the partial differential equation may be written in the form

$$\Phi(a, b, c, d, e) = 0,$$

where Φ is an arbitrary functional symbol; and this is equivalent to the system of five equations $a = \text{const.}, \dots, e = \text{const.}$, involving as it should do five arbitrary constants.

38.

If we have three integrals a, b, c of the Hamiltonian system, then the differential system may be written

$$\frac{dx}{d(a, b, c)/d(p, q, y, z, r)} = \dots,$$

and the condition for a fourth integral is that the expression

$$\frac{d(a, b, c)}{d(p, q, y, z, r)} dx + \dots$$

should be an exact differential.

39.

If we have four integrals a, b, c, d of the Hamiltonian system, then the differential system may be written

$$\frac{dx}{d(a, b, c, d)/d(p, q, r, y, z)} = \dots,$$

and the condition for a fifth integral is that the expression

$$\frac{d(a, b, c, d)}{d(p, q, r, y, z)} dx + \dots$$

should be an exact differential.

and if we herein attend to the terms which contain dt , we find that the expression

$$dH - \frac{dH}{dp} dp - \frac{dH}{dq} dq - \frac{dH}{dr} dr + \frac{dH}{dx} dx + \frac{dH}{dy} dy + \frac{dH}{dz} dz$$

vanishes identically; and the system may therefore be written in the form

$$\begin{aligned} dx &= \frac{dH}{dp} dt, & dy &= \frac{dH}{dq} dt, & dz &= \frac{dH}{dr} dt, \\ dp &= -\frac{dH}{dx} dt, & dq &= -\frac{dH}{dy} dt, & dr &= -\frac{dH}{dz} dt, \end{aligned}$$

which puts in evidence the integral $H = \text{const.}$

The Poisson-Jacobi Theorem peculiar to the Hamiltonian Form. Art. Nos. 40 to 45.

40.

The theorem in question is that if a, b are any two integrals of the Hamiltonian system, then the function (a, b) is also an integral.

41.

To prove this, we observe that if a is an integral, then

$$\frac{da}{dt} = \frac{dH}{dp} \frac{da}{dx} - \frac{dH}{dx} \frac{da}{dp} + \frac{dH}{dq} \frac{da}{dy} - \frac{dH}{dy} \frac{da}{dq} + \frac{dH}{dr} \frac{da}{dz} - \frac{dH}{dz} \frac{da}{dr} = 0,$$

and similarly for b ; and hence, differentiating the expression (a, b) , we find

$$\frac{d}{dt}(a, b) = \left(\frac{da}{dt}, b \right) + \left(a, \frac{db}{dt} \right) = 0,$$

which proves the theorem.

42.

The theorem may also be stated as follows: if a, b are any two functions of the variables such that $(a, b) = 0$, and if one of them, say a , is an integral of the Hamiltonian system, then the other, b , is also an integral.

43.

But we wish to prove the theorem directly from the definition of the symbol (a, b) ; viz. we wish to show that if a, b are integrals, then (a, b) is also an integral.

44.

We have, by definition,

$$(a, b) = \frac{d(a, b)}{d(p, x)} + \frac{d(a, b)}{d(q, y)} + \frac{d(a, b)}{d(r, z)},$$

and we wish to show that if a, b are integrals, then (a, b) is also an integral.

45.

To prove this, we observe that

$$\begin{aligned} \frac{d}{dt}(a, b) &= \sum \left\{ \frac{d}{dt} \left(\frac{da}{dp} \frac{db}{dx} - \frac{da}{dx} \frac{db}{dp} \right) \right\} \\ &= \sum \left\{ \left(\frac{d}{dt} \frac{da}{dp} \right) \frac{db}{dx} + \frac{da}{dp} \left(\frac{d}{dt} \frac{db}{dx} \right) - \left(\frac{d}{dt} \frac{da}{dx} \right) \frac{db}{dp} - \frac{da}{dx} \left(\frac{d}{dt} \frac{db}{dp} \right) \right\}, \end{aligned}$$

where the summation extends to the three pairs $(p, x), (q, y), (r, z)$.

Now, since a is an integral, we have

$$\frac{da}{dt} = \frac{dH}{dp} \frac{da}{dx} - \frac{dH}{dx} \frac{da}{dp} + \cdots = 0,$$

and hence, differentiating with respect to p , we find

$$\frac{d}{dp} \frac{da}{dt} = \frac{d^2 H}{dp^2} \frac{da}{dx} + \frac{dH}{dp} \frac{d^2 a}{dp dx} - \frac{d^2 H}{dp dx} \frac{da}{dp} - \frac{dH}{dx} \frac{d^2 a}{dp^2} + \cdots = 0,$$

and similarly for the other derivatives.

Hence, unless $(a, b) = 0$, we have $A = \frac{1}{J}$. The remaining five equations give similarly $B = \frac{1}{J}$, $C = \frac{1}{J}$, $D = \frac{1}{J}$, $E = \frac{1}{J}$; and we have therefore

$$(a, b) = \frac{d(a, b)}{d(p, x)} + \frac{d(a, b)}{d(q, y)} + \frac{d(a, b)}{d(r, z)} = \frac{3}{J},$$

or, since the first two terms are $= \frac{2}{J}$, we have $(a, b) = \frac{1}{J}$.

C. X.

14

**Further Development of the Hamiltonian System.
Art. Nos. 46 to 58.**

46.

where l, m are indeterminate functions of the variables; and we seek to determine these functions so that the expression

$$l da + m db + n dc$$

shall be an exact differential.

47.

The condition for this is that the Pfaffian derivatives of the expression shall all vanish; viz. that we shall have

$$\frac{d(l, a)}{d(x, y)} + \frac{d(m, b)}{d(x, y)} + \frac{d(n, c)}{d(x, y)} = 0,$$

and similarly for the other pairs of variables.

48.

If we assume that l, m, n are functions of (a, b, c) only, then the conditions reduce to

$$l \frac{d(b, c)}{d(x, y)} + m \frac{d(c, a)}{d(x, y)} + n \frac{d(a, b)}{d(x, y)} = 0,$$

and similarly for the other pairs; and these conditions are satisfied if

$$l : m : n = \frac{d(b, c)}{d(x, y, z)} : \frac{d(c, a)}{d(x, y, z)} : \frac{d(a, b)}{d(x, y, z)}.$$

49.

But we may also consider the case where l, m are functions of the variables, not necessarily of (a, b, c) only; and in this case the conditions are more complicated.

50.

The general theory of the Hamiltonian system may be developed as follows: if a, b, c, d, e are any five integrals, then the system may be written

$$\frac{dx}{d(a, b, c, d, e)/d(p, q, r, y, z)} = \cdots,$$

and the condition for a sixth integral is that the expression

$$\frac{d(a, b, c, d, e)}{d(p, q, r, y, z)} dx + \cdots$$

should be an exact differential.

Hence, unless $d(a, b)/d(x, y) = 0$, we have $A = \frac{1}{J}$. The remaining five equations give similarly

$$B = \frac{1}{J}, \quad C = \frac{1}{J}, \quad D = \frac{1}{J}, \quad E = \frac{1}{J};$$

and we have therefore

$$(a, b) = \frac{d(a, b)}{d(p, x)} + \frac{d(a, b)}{d(q, y)} + \frac{d(a, b)}{d(r, z)} = \frac{3}{J},$$

or, since the first two terms are $= \frac{2}{J}$, we have $(a, b) = \frac{1}{J}$.

51.

The theorem may be generalized as follows: if a, b, c are any three integrals of the Hamiltonian system, then the function

$$\frac{d(a, b, c)}{d(p, q, x)} + \frac{d(a, b, c)}{d(p, r, x)} + \dots$$

is also an integral.

52.

And more generally, if $a, b, c, \dots$ are any number of integrals, then the function formed by summing the functional determinants

$$\frac{d(a, b, c, \dots)}{d(p, q, r, \dots, x, y, z, \dots)}$$

is also an integral.

or, since the first two terms are equal, we have

$$(a, b) = \frac{d(a, b)}{d(p, x)} + \frac{d(a, b)}{d(q, y)} + \frac{d(a, b)}{d(r, z)} = 0.$$

53.

The theorem may be stated in a different form as follows: if a, b are any two integrals of the Hamiltonian system, then the expression

$$\frac{d(a, b)}{d(p, x)} + \frac{d(a, b)}{d(q, y)} + \frac{d(a, b)}{d(r, z)}$$

is also an integral; and conversely, if this expression vanishes, then a, b are integrals.

54.

The condition $(a, b) = 0$ is the condition that the integrals a, b should be “in involution”; and the theorem is that if two integrals are in involution, then any third integral is also in involution with each of them.

50.

Supposing that any two integrals a, b of the Hamiltonian system are given, we can find a third integral c such that $(a, c) = 0$ and $(b, c) = 0$; and we can then find a fourth integral d such that $(a, d) = 0$, $(b, d) = 0$, $(c, d) = 0$; and so on.

51.

The process may be continued until we have obtained a complete system of integrals, viz. a system of n integrals such that any two of them are in involution.

52.

If we have a complete system of n integrals, then the Hamiltonian system may be reduced to the form

$$\frac{dx_1}{dt} = \frac{dH}{dp_1}, \quad \frac{dy_1}{dt} = \frac{dH}{dq_1}, \quad \dots,$$

where $x_1, y_1, \dots, p_1, q_1, \dots$ are functions of the original variables.

53.

We consider the differential expression

$$dV - p\,dx - q\,dy - r\,dz,$$

which, treating V as a function of (x, y, z) , and p, q, r as differential coefficients, may be written

$$dV = p\,dx + q\,dy + r\,dz.$$

54.

If V is a function of (x, y, z) , then $p = \frac{dV}{dx}$, $q = \frac{dV}{dy}$, $r = \frac{dV}{dz}$; and the expression $dV - p\,dx - q\,dy - r\,dz$ vanishes identically.

55.

But if V is not a function of (x, y, z) alone, but contains also other variables, then the expression does not vanish identically; and we may consider it as a function of the variables.

and hence $A = (t - T)^2 + A'$, and the like for B , C , &c.

56.

If we have $V = V(x, y, z, a, b)$, where a, b are constants, then

$$dV = \frac{dV}{dx} dx + \frac{dV}{dy} dy + \frac{dV}{dz} dz + \frac{dV}{da} da + \frac{dV}{db} db,$$

and if $p = \frac{dV}{dx}$, $q = \frac{dV}{dy}$, $r = \frac{dV}{dz}$, we have

$$dV - p dx - q dy - r dz = \frac{dV}{da} da + \frac{dV}{db} db.$$

57.

If a, b are not constants but variables, then the expression on the right-hand side is not zero; and we may consider the equation

$$dV - p dx - q dy - r dz = 0$$

as a differential equation determining V as a function of (x, y, z) .

and again treating a, b, H as variable, we have

$$dV - p dx - q dy - r dz = dK + (t - T) dH + A da + B db,$$

where

$$\begin{aligned} K &= V - px - qy - rz + TH, \\ t &= \frac{dV}{dH}, \quad T = \frac{dK}{dH}, \\ A &= \frac{dV}{da} - p \frac{dx}{da} - q \frac{dy}{da} - r \frac{dz}{da} + T \frac{dH}{da}, \\ B &= \frac{dV}{db} - p \frac{dx}{db} - q \frac{dy}{db} - r \frac{dz}{db} + T \frac{dH}{db}. \end{aligned}$$

**The Partial Differential Equation $H = \text{const.}$
Art. Nos. 59 to 70.**

59.

We consider the partial differential equation

$$H(x, y, z, p, q, r) = \text{const.},$$

where p, q, r are the differential coefficients of a function V of (x, y, z) ;

$$\text{viz. } p = \frac{dV}{dx}, \quad q = \frac{dV}{dy}, \quad r = \frac{dV}{dz}.$$

60.

The equation $H = \text{const.}$ is equivalent to the system of ordinary differential equations

$$\frac{dx}{dH/dp} = \frac{dy}{dH/dq} = \frac{dz}{dH/dr} = \frac{dp}{-dH/dx} = \frac{dq}{-dH/dy} = \frac{dr}{-dH/dz},$$

which is the Hamiltonian system.

61.

Conversely, if we have the Hamiltonian system, and if we can find a function V such that $p = \frac{dV}{dx}$, $q = \frac{dV}{dy}$, $r = \frac{dV}{dz}$, then $H = \text{const.}$ is a partial differential equation satisfied by V .

62.

The Hamiltonian system determines the variables as functions of t and of five arbitrary constants; and if we can express these in terms of x, y, z , and substitute in the expression $p dx + q dy + r dz$, we obtain the differential of the function V .

Adding these together, we have

$$\frac{d(A, B)}{d(x, y)} + \frac{d(q, r)}{d(y, z)} + \frac{d(r, p)}{d(z, x)} + \frac{d(p, q)}{d(x, y)} = 0,$$

and, if it vanish, it is a condition that the expression $p \, dx + q \, dy + r \, dz$ should be an exact differential.

63.

The condition may be written in the form

$$(p, q) + (q, r) + (r, p) = 0,$$

where (p, q) denotes the functional determinant $\frac{d(p, q)}{d(x, y)}$, &c.

and, if it vanish, it is

64.

The general solution of the partial differential equation $H = \text{const.}$ may be obtained as follows: we first integrate the Hamiltonian system, obtaining the variables as functions of t and five arbitrary constants; we then express p, q, r in terms of x, y, z , and substitute in the expression $p dx + q dy + r dz$; if this is an exact differential, its integral gives the function V .

65.

The process may be simplified if we know any integral of the Hamiltonian system; for if a is such an integral, we may use it to reduce the order of the system.

a solution of the equation $d\psi = \text{const.}$

66.

If we have a complete system of n integrals of the Hamiltonian system, then the partial differential equation $H = \text{const.}$ may be solved by quadratures.

67.

In particular, if $n = 3$, and if we have three integrals a, b, c in involution, then the solution may be obtained as follows: we form the expression

$$\Theta = \int (p \, dx + q \, dy + r \, dz),$$

and determine the arbitrary functions which appear in the integration by the condition that Θ shall be a function of (x, y, z) only.

we find for ψ the equation

$$\frac{d\psi}{dp} = \frac{d\psi}{dq} = \frac{d\psi}{dr} = 0,$$

and hence ψ is a function of (x, y, z) only.

68.

The function ψ being determined, the general solution of the partial differential equation is

$$V = \Theta + \psi,$$

where Θ is the particular integral and ψ is the arbitrary function.

and each of these, as containing

69.

The general integral of the partial differential equation $H = \text{const.}$ may be expressed in the form

$$V = \int (p \, dx + q \, dy + r \, dz) + \Phi(a, b, c),$$

where Φ is an arbitrary functional symbol, and a, b, c are integrals of the Hamiltonian system.

70.

The solution thus involves three arbitrary functions, which may be determined by giving the values of $V, \frac{dV}{dx}, \frac{dV}{dy}$ at any point.

which, in virtue of $(a, H) = 0$, $(b, H) = 0$, becomes

$$\frac{d(a, b)}{d(p, x)} + \frac{d(a, b)}{d(q, y)} + \frac{d(a, b)}{d(r, z)} = 0,$$

and this is the condition that a, b should be in involution.

Further Theory of the Partial Differential Equation $H = \text{const.}$ Art. Nos. 71 to 80.

71.

The partial differential equation $H = \text{const.}$ may be treated by the method of characteristics; the characteristics are the curves determined by the Hamiltonian system.

We have as a conjugate system

72.

If a, b are any two integrals of the Hamiltonian system which are in involution, then the equations $a = \text{const.}$, $b = \text{const.}$ determine a family of surfaces which are characteristics of the partial differential equation.

73.

The characteristics may be used to construct the general solution of the partial differential equation; for if we know the values of V and its differential coefficients on any surface which is not a characteristic, we can determine the solution in the neighbourhood of this surface.

where the term in $d\theta$ is
which, from the equation which determines θ ,

74.

The method of characteristics may be extended to the case where the variables are any even number $2n$; the Hamiltonian system then determines the characteristics, and the general solution may be obtained by quadratures.

75.

If we have a complete system of n integrals in involution, then the partial differential equation may be solved by quadratures; and the solution involves n arbitrary functions.

Forming from these the expression for $dV - p\,dx - q\,dy - r\,dz$, and again treating a, b, H as variable, we have

$$dV - p\,dx - q\,dy - r\,dz = dK + (t - T)\,dH + A\,da + B\,db,$$

where

$$\begin{aligned} K &= V - px - qy - rz + TH, \\ t &= \frac{dV}{dH}, \quad T = \frac{dK}{dH}, \\ A &= \frac{dV}{da} - p\frac{dx}{da} - q\frac{dy}{da} - r\frac{dz}{da} + T\frac{dH}{da}, \\ B &= \frac{dV}{db} - p\frac{dx}{db} - q\frac{dy}{db} - r\frac{dz}{db} + T\frac{dH}{db}. \end{aligned}$$

expressions $x - p, y - q, z - r$ is an integral, or again the product of any expression of the first set into any expression of the second set is an integral.

76.

The general theory may be applied to the case where H is a homogeneous function of (p, q, r) ; in this case the Hamiltonian system may be simplified by the introduction of new variables.

77.

If H is homogeneous of degree k in (p, q, r) , then we have

$$p \frac{dH}{dp} + q \frac{dH}{dq} + r \frac{dH}{dr} = kH,$$

and the Hamiltonian system may be written in a form which puts this homogeneity in evidence.

whence

$$dV - p \, dx - q \, dy - r \, dz = d\lambda + (t - T) \, dH + a' \, da + b' \, db + c' \, dc \\ - a \, da' - b \, db' - c \, dc'; \quad (19)$$

or, in place of

C. X.

15

0.1 [655] A Memoir on Differential Equations (conclusion)

[From the Philosophical Transactions, vol. 167 (1877), pp. 243-273; continuation from preceding pages.]

77. More symmetrically, we have the solution

$$V = \lambda - \frac{1}{2}(x^2 + y^2 + z^2) + \frac{(ax + by + cz)^2}{a^2 + b^2 + c^2},$$

as can be at once verified.

78. In the same particular case $H = 0$, introducing the corresponding values $p_0, q_0, r_0, x_0, y_0, z_0$, we find a very simple expression for $V - V_0$, as a function of x, y, z, x_0, y_0, z_0 .

We have, writing $T_0 = e^{t_0 - \tau}$, $T'_0 = e^{-t_0 + \tau}$, and therefore $T_0 T'_0 = 1$,

$$\begin{aligned} x_0 &= aT_0 + a'T'_0, & p_0 &= aT_0 - a'T'_0, \\ y_0 &= bT_0 + b'T'_0, & q_0 &= bT_0 - b'T'_0, \\ z_0 &= cT_0 + c'T'_0, & r_0 &= cT_0 - c'T'_0, \end{aligned}$$

and thence

$$\begin{aligned} x - x_0 &= a(T - T_0) + a'\left(\frac{1}{T} - \frac{1}{T_0}\right) = (T - T_0)\left(a - \frac{a'}{TT_0}\right), \\ x + x_0 &= a(T + T_0) + a'\left(\frac{1}{T} + \frac{1}{T_0}\right) = (T - T_0)\left(a + \frac{a'}{TT_0}\right). \end{aligned}$$

Forming the analogous quantities $y - y_0$, &c., we deduce

$$\begin{aligned} (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 &= (T - T_0)^2 \left\{ a^2 + b^2 + c^2 + (a'^2 + b'^2 + c'^2) \frac{1}{T^2 T_0^2} \right\} \\ (x + x_0)^2 + (y + y_0)^2 + (z + z_0)^2 &= (T + T_0)^2 \left\{ a^2 + b^2 + c^2 + (a'^2 + b'^2 + c'^2) \frac{1}{T^2 T_0^2} \right\} \end{aligned}$$

But we have

$$\begin{aligned} V - V_0 &= \frac{1}{2} \left\{ (a^2 + b^2 + c^2)(T^2 - T_0^2) - (a'^2 + b'^2 + c'^2) \left(\frac{1}{T^2} - \frac{1}{T_0^2} \right) \right\} \\ &= \frac{1}{2} (T^2 - T_0^2) \left\{ a^2 + b^2 + c^2 + (a'^2 + b'^2 + c'^2) \frac{1}{T^2 T_0^2} \right\}, \end{aligned}$$

and hence the required formula

$$V - V_0 = \frac{1}{2} \sqrt{\{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2\}} \sqrt{\{(x + x_0)^2 + (y + y_0)^2 + (z + z_0)^2\}}$$

or, say, for shortness,

$$= \frac{1}{2} \sqrt{(R)} \sqrt{(S)}.$$

79. We ought, therefore, to have

$$\frac{1}{2} d\sqrt{(R)} \sqrt{(S)} = p dx + q dy + r dz - p_0 dx_0 - q_0 dy_0 - r_0 dz_0,$$

where p, q, r, p_0, q_0, r_0 denote as above, and consequently

$$p^2 + q^2 + r^2 = x^2 + y^2 + z^2, \quad p_0^2 + q_0^2 + r_0^2 = x_0^2 + y_0^2 + z_0^2.$$

We have in fact

$$\begin{aligned} p^2 + q^2 + r^2 &= \frac{1}{2} \{R + S + 2(x^2 + y^2 + z^2 - x_0^2 - y_0^2 - z_0^2)\} = x^2 + y^2 + z^2, \\ p_0^2 + q_0^2 + r_0^2 &= \frac{1}{2} \{R + S - 2(x^2 + y^2 + z^2 - x_0^2 - y_0^2 - z_0^2)\} = x_0^2 + y_0^2 + z_0^2, \end{aligned}$$

or the last-mentioned results are thus verified.

Partial Differential Equation containing the Dependent Variable: Reduction to Standard Form

Art. Nos. 80, 81.

80. The equation $H = \text{const.}$, is the most general form of a partial differential equation not containing the dependent variable V ; but if a partial differential equation does contain the independent variable, we can, by regarding this as one of the dependent variables, and in place of it introducing a new independent variable, exhibit the equation in the standard form $H = \text{const.}$, H being here a homogeneous function of the order zero in the differential coefficients.

Thus, if the independent variables are x, y , the dependent variable z , and its differential coefficients p, q , then the given partial differential equation may be

$$H_1 = H(p, q, x, y, z) = \text{const.}$$

But we may determine V as a function of x, y by an equation $V = \text{const.}$, V being a desired function of x, y, z ; and then writing p, q, r

for the differential coefficients $\frac{dV}{dx}$, $\frac{dV}{dy}$, $\frac{dV}{dz}$, we have $p = -\frac{r}{P}$, $q = -\frac{r}{Q}$, and the proposed partial differential equation becomes

$$H\left(-\frac{r}{p}, -\frac{r}{q}, x, y, z\right) = \text{const.},$$

viz. this is an equation containing only the differential coefficients p , q , r of the dependent variable V , a function of x , y , z . And, moreover, H is homogeneous of the order zero in p , q , r ; consequently

$$p \frac{dH}{dp} + q \frac{dH}{dq} + r \frac{dH}{dr} = 0,$$

or, in the augmented Hamiltonian system, the last equality is $\frac{dV}{dt} = 0$, so that an integral is $V = \text{const.}$; as already stated, this is the equation by which z is determined as a function of x , y .

81. Thus, if the given partial differential equation be $pq - z = H$, we here consider the equation $\frac{PQ}{R} - z = H$. The Hamiltonian system is

$$\frac{dx}{\frac{P}{R}} = \frac{dy}{\frac{Q}{R}} = \frac{-dz}{\frac{PQ}{R^2}} = \frac{dp}{q - \frac{PQ}{R^2}} = \frac{dq}{0} = \frac{dr}{0} = \frac{dV}{0},$$

having the integrals

$$a = p, \quad b = q, \quad c = px - qy, \quad r = \frac{q}{pq}, \quad \frac{r}{p} = \frac{1}{pq}$$

(whence $H = -abc$). We have H , a , b , a system of conjugate integrals and, in terms of these,

$$p = a, \quad q = b, \quad r = \sqrt{\{ab(z + H)\}};$$

hence, writing λ for the constant value of V , we have

$$\lambda = \int \left(a \, dx + b \, dy + \sqrt{\{ab(z + H)\}} \, dz \right),$$

that is,

$$\lambda = ax + by + 2\sqrt{\{ab(z + H)\}},$$

or say,

$$4ab(z + H) = (ax + by - \lambda)^2,$$

a solution containing really the two constants λ and H , and thus a complete solution of the given equation $pq - z = H$. We have, in fact,

$$2abp = a(ax + by - \lambda),$$

$$2abq = b(ax + by - \lambda),$$

that is,

$$4a^2b^2pq = ab(ax + by - \lambda)^2 = 4a^2b^2(z + H),$$

or $pq = z + H$, as it should be.

0.2 [656] On the Theory of Partial Differential Equations

[From the *Mathematische Annalen*, t. XI. (1877), pp. 194–198.]

In what follows, any letter not otherwise explained denotes a function of certain variables (x, y, p, q) , or (x, y, z, p, q, r) , &c., as will be stated in each particular case.

An equation $a = \text{const.}$, denotes that the function a of the variables is, in fact, a constant (viz. by such equation we establish a relation between the variables): and when this is so, we use the same letter a to denote the constant value of the function in question; I find this a very convenient notation.

Thus if the variables are x, y, z, p, q, r and if p, q, r are the differential coefficients in regard to x, y, z respectively of a function V of x, y, z , then H (as a letter not otherwise explained) denotes a function of x, y, z, p, q, r and considering it as a given function, $H = \text{const.}$ will be a partial differential equation containing the constant H .

For instance, if H denote the function $pqr - xyz$, $H = \text{const.}$, is the partial differential equation, $pqr - xyz = H$ (a given constant).

The integration of the partial differential equation, $H = \text{const.}$, depends upon that of the linear partial differential equation

$$(H, \Theta) = 0,$$

where as usual (H, Θ) signifies

$$\frac{\partial(H, \Theta)}{\partial(p, x)} + \frac{\partial(H, \Theta)}{\partial(q, y)} + \frac{\partial(H, \Theta)}{\partial(r, z)}.$$

It can be effected if we know two conjugate solutions a, b of the equation $(H, \Theta) = 0$, viz. a, b as solutions are such that $(H, a) = 0$, $(H, b) = 0$, and (as conjugate solutions) are also such that $(a, b) = 0$; in this case if from the equations $H = \text{const.}$, $a = \text{const.}$, $b = \text{const.}$, we determine p, q, r as functions of x, y, z , the resulting value of $p dx + q dy + r dz$ is an exact differential, and we have

$$V = \lambda + \int (p dx + q dy + r dz),$$

a solution containing three arbitrary constants, λ, a, b , and therefore a complete solution of the proposed partial differential equation $H = \text{const.}$

But (as is known) there is a different process of integration, for which the conjugate solutions are not required, and which has reference to a system of initial values $x_0, y_0, z_0, p_0, q_0, r_0$: viz. if the independent solutions of $(H, \Theta) = 0$, are a, b, c, d, e , and if a_0, b_0, c_0, d_0, e_0 denote respectively the same functions of the initial variables that a, b, c, d, e are of x, y, z, p, q, r , then if from the equations

$$a = a_0, \quad b = b_0, \quad c = c_0, \quad d = d_0, \quad e = e_0, \quad H = \text{const.}$$

we express p, q, r as functions of x, y, z and of x_0, y_0, z_0, H , these last being regarded as constants, we have $p dx + q dy + r dz$ an exact differential, and

$$V = \lambda + \int (p dx + q dy + r dz),$$

a solution containing the constants λ, x_0, y_0, z_0 (that is, one super-numerary constant), and as such a complete solution.

It is interesting to prove directly that $p dx + q dy + r dz$ is an exact differential.

I consider first the more simple case where the variables are p, q, x, y . Here p, q are to be found from the equations

$$a = a_0, \quad b = b_0, \quad c = c_0, \quad H = \text{const.}$$

and it is to be shown that $p dx + q dy$ is an exact differential.

Considering p, q, p_0, q_0 as functions of the independent variables x, y , then differentiating in regard to x , and eliminating $\frac{dp}{dx}, \frac{dp_0}{dx}, \frac{dq_0}{dx}$, we have

$$\begin{vmatrix} \frac{da}{dx} + \frac{da}{dq} \frac{dq}{dx}, & \frac{da}{dp}, & \frac{da_0}{dp_0}, & \frac{da_0}{db_0} \\ \frac{db}{dx} + \frac{db}{dq} \frac{dq}{dx}, & \frac{db}{dp}, & \frac{db_0}{dp_0}, & \frac{db_0}{dc_0} \\ \frac{dc}{dx} + \frac{dc}{dq} \frac{dq}{dx}, & \frac{dc}{dp}, & \frac{dc_0}{dp_0}, & \frac{dc_0}{dH} \\ \frac{dH}{dx} + \frac{dH}{dq} \frac{dq}{dx}, & \frac{dH}{dp}, & 0, & 0 \end{vmatrix} = 0,$$

or introducing a well-known notation for functional determinants, and expanding the determinant, this is

$$\frac{\partial(a_0, b_0)}{\partial(p_0, q_0)} \cdot \frac{\partial(H, c)}{\partial(p, q)} \cdot \frac{dq}{dx} + \dots = 0.$$

But in the same way

$$\frac{\partial(p_0, q_0)}{\partial(p_0, q_0)} \cdot \frac{\partial(H, c)}{\partial(q, y)} \cdot \frac{dq}{dy} + \dots = 0,$$

or adding these, attending to the value of (H, c) , and observing that

$$\frac{\partial(a_0, b_0)}{\partial(p_0, q_0)} \cdot \frac{\partial(H, c)}{\partial(p, q)} \left(\frac{dq}{dx} - \frac{dp}{dy} \right) + \dots = 0.$$

We have $(H, a) = 0$, $(H, b) = 0$, $(H, c) = 0$, and the equation in fact is

$$\frac{\partial(a_0, b_0)}{\partial(p, q)} \cdot \frac{\partial(H, c)}{\partial(p, q)} \left(\frac{dq}{dx} - \frac{dp}{dy} \right) = 0,$$

viz. we have $\frac{dq}{dx} - \frac{dp}{dy} = 0$, the condition for the exact differential.

Coming now to the case where the variables are x, y, z, p, q, r , and in the six equations treating p, q, r, p_0, q_0, r_0 as functions of the independent variables x, y, z ,—then differentiating with regard to x and proceeding as before, we find for $\frac{dr}{dx}$ the equation

$$\frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)} \left(\frac{dr}{dx} - \frac{dp}{dz} \right) \frac{\partial(a, b, H)}{\partial(r, p, q)} \frac{\partial(a, b, H)}{\partial(x, p, q)} + \dots = 0.$$

We have, in the same way, for $\frac{dp}{dy}$ the equation

$$\frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)} \left(\frac{dp}{dy} - \frac{dq}{dx} \right) \frac{\partial(a, b, H)}{\partial(p, r, q)} \frac{\partial(a, b, H)}{\partial(z, r, q)} + \dots = 0,$$

or, adding the two equations,

$$\frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)} \left\{ \left(\frac{dr}{dx} - \frac{dp}{dz} \right) \frac{\partial(a, b, H)}{\partial(r, p, q)} \frac{\partial(a, b, H)}{\partial(x, p, q)} + \left(\frac{dp}{dy} - \frac{dq}{dx} \right) \frac{\partial(a, b, H)}{\partial(z, r, q)} \right\} + \dots = 0,$$

where the terms denoted by the &c. indicate the like terms corresponding to the different partitions of the letters a, b, c, d, e .

The equation may be simplified; we have identically

$$\frac{\partial(a_0, b_0, H)}{\partial(q_0, \cdot, \cdot)} \frac{\partial(a, b, H)}{\partial(x, p, q)} + \frac{\partial(a, b, H)}{\partial(z, r, q)} + \dots = 0,$$

or since $(H, a) = 0$, $(b, H) = 0$, the left-hand side is simply $-\frac{\partial(a, b)}{\partial(\cdot, \cdot)}$, and the equation becomes

$$\frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)} \left\{ \left(\frac{dr}{dx} - \frac{dp}{dz} \right) \frac{\partial(a, b, H)}{\partial(r, p, q)} + \dots \right\} = 0.$$

This ought to give $\frac{dr}{dx} - \frac{dp}{dz} = 0$, &c. for each (a, b, c, d, e) ; this is then the equation which has to be proved.

By the Poisson-Jacobi theorem, (a, b) is a function of a, b, c, d, e : if we write

$$f(\cdot) = \frac{\partial(a_0, b_0)}{\partial(\cdot)} \frac{\partial(a_0, b_0)}{\partial(\cdot)},$$

then (a_0, b_0) is the same function of a_0, b_0, c_0, d_0, e_0 ; but these are $= a, b, c, d, e$ respectively, and we thence have $(a, b) = (a_0, b_0)$, and the theorem to be proved is

$$\frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)} \cdot \frac{\partial(a, b)}{\partial(\cdot)} = 0.$$

But substituting for (a_0, b_0) its value, the function on the left-hand is (it is easy to see) the sum of the three functional determinants

$$\frac{\partial(a_0, b_0, c_0, d_0, e_0)}{\partial(p_0, q_0, r_0, \cdot, \cdot)}, \quad \frac{\partial(a_0, b_0, c_0, d_0, e_0)}{\partial(p_0, q_0, \cdot, r_0, \cdot)}, \quad \frac{\partial(a_0, b_0, c_0, d_0, e_0)}{\partial(p_0, q_0, \cdot, \cdot, r_0)},$$

each of which vanishes as containing the same letter twice in the denominator, that is, as having two identical columns; and the theorem in question is thus proved.

And in the same way $\frac{dr}{dy} - \frac{dq}{dz}$, $\frac{dp}{dz} - \frac{dr}{dx}$ are each $= 0$: or we have $p dx + q dy + r dz$ an exact differential.

The proof would fail if the factors multiplying $\frac{dq}{dx} - \frac{dp}{dy}$ &c., or if any one of these factors, were $= 0$; I have not particularly examined this, but the meaning would be, that here the equations in question $a = a_0$, &c., $H = \text{const.}$, are such as not to give rise to expressions for p, q, r as functions of $x, y, z, x_0, y_0, z_0, H$, as assumed in the theorem; whenever such expressions are obtainable, then we have $p dx + q dy + r dz$ an exact differential.

The proof in the case of a greater number of variables, say in the next case where the variables are x, y, z, w, p, q, r, s , would present more difficulty—but I have not proceeded further in the question.

It is worth while to put the two processes into connexion with each other: taking in each case the variables to be x, y, z, p, q, r , and the partial differential equation to be $H = \text{const.}$

In the one case, a, b being conjugate solutions of $(H, 0) = 0$, from the equations $H = \text{const.}$, $a = \text{const.}$, $b = \text{const.}$, we find p, q, r functions of x, y, z, H, a, b and then $p dx + q dy + r dz$ is an exact differential.

In the other case, a, b, c, d, e being the solutions of $(H, 0) = 0$, from the equations $H = \text{const.}$, $a = a_0, b = b_0, c = c_0, d = d_0, e = e_0$, we find p, q, r functions of x_0, y_0, z_0, H and then $p dx + q dy + r dz$ is an exact differential.

It may be added that, if from the last mentioned equations we determine also p_0, q_0, r_0 as functions of x, y, z, x_0, y_0, z_0 , then considering only H as a constant, we ought to have $p dx + q dy + r dz - p_0 dx_0 - q_0 dy_0 - r_0 dz_0$ an exact differential; I have not examined the direct proof.

Cambridge, 28 Nov., 1876.

0.3 [657] Note on the Theory of Elliptic Integrals

[From the *Mathematische Annalen*, t. XII. (1877), pp. 143-146.]

The equation

$$\frac{M dy}{\sqrt{1 - y^2 \cdot 1 - k^2 y^2}} = \frac{dx}{\sqrt{1 - x^2 \cdot 1 - k^2 x^2}}$$

is integrable algebraically when M is rational: and so long as the modulus is arbitrary, then conversely, in order that the equation may be integrable algebraically, M must be rational. For particular values however of the modulus, the equation is integrable algebraically for values of the form M , or (what is the same thing) $\frac{1}{M}$, = a rational quantity $\pm$ square root of a negative rational quantity, say $= \frac{1}{p}(l + m\sqrt{-n})$, where l, m, n, p are integral and n is positive; we may for shortness call this a *half-rational numerical value*.

The theory is considered by Abel in two Memoirs in the *Astr. Nach.* Nos. 138 & 147 (1828), being the Memoirs* XIII & XIV in the *OEuvres Complètes* (Christiania 1839). I here reproduce the investigation in a somewhat altered (and, as it appears to me, improved) form.

Putting the two differentials each $= du$, we have $x = \operatorname{sn}(u + \alpha)$, $y = \operatorname{sn}(\frac{u}{M} + \beta)$; and the question is whether there exists an algebraical relation between these functions, or, what is the same thing, an algebraical relation between the functions $x = \operatorname{snu}$ and $y = \operatorname{sn}\frac{u}{M}$.

Suppose that A and B are independent periods of snu ; so that $\operatorname{sn}(u + A) = \operatorname{snu}$, $\operatorname{sn}(u + B) = \operatorname{snu}$, and that every other period is $= mA + nB$, where m and n are integers. Then if u has successively the values $u, u + A, u + 2A$, etc., the value of x remains always the same, and if x and y are algebraically connected, y can have only a finite number of values: there are consequently integer values p', p'' for which

$$\operatorname{sn}\frac{1}{M}(u + p'A) = \operatorname{sn}\frac{1}{M}(u + p''A) :$$

or writing $v, -p'A$ for u and putting $p'' - p' = p$, there is an integer value p for which $\operatorname{sn}\frac{1}{M}(u + pA) = \operatorname{sn}\frac{u}{M}$.

Similarly there is an integer value q for which $\operatorname{sn}\frac{1}{M}(u + qB) = \operatorname{sn}\frac{u}{M}$; and we are at liberty to assume $q = p$; for if the original values are

unequal, we have only in the place of each of them to substitute their least common multiple.

We have thus an integer p , for which $\operatorname{sn}_{\frac{1}{M}}(u + pA) = \operatorname{sn}_{\frac{u}{M}}$, $\operatorname{sn}_{\frac{1}{M}}(u + pB) = \operatorname{sn}_{\frac{u}{M}}$. There are consequently integers m, n, r, s such that

$$\frac{pA}{M} = mA + nB, \quad \frac{pB}{M} = rA + sB,$$

equations which will constitute a single relation $\frac{p}{M} = m$, if $m = s$, $r = n = 0$; but in every other case will be two independent relations.

In the case first referred to, the modulus is arbitrary and M is rational. But excluding this case, the equations give

$$B(mA + nB) = A(rA + sB),$$

or, what is the same thing,

$$rA^2 - (m - s)AB - nB^2 = 0,$$

an equation which implies that the modulus has some one value out of a set of given values.

The ratio $A : B$ of the two periods is of necessity imaginary, and hence the integers m, n, r, s must be such that $(m - s)^2 + 4nr$ is negative.

The foregoing equations may be written

$$\frac{A}{B} = \frac{n}{\frac{1}{2}(m - s) - \sqrt{\frac{1}{4}(m - s)^2 + nr}},$$

whence the ratio $A : B$ is given as a function of m, n, r, s ; and if these have any values such that $(m - s)^2 + 4nr$ is negative, then there exists a corresponding modulus for which the original differential equation is integrable algebraically for a half-rational value of M .

The question arises, whether for any such modulus the equation is integrable algebraically for any value of M other than the half-rational values; or say, whether the modulus being given, the differential equation is integrable algebraically for values of M other than the half-rational value or values which belong to the given modulus.

We have $x = \operatorname{sn}(u + \alpha)$; hence if as before A, B are the periods, the general value of u is = $\{(x)\} + mA + nB$; and we thence have

$$\frac{u}{M} = \frac{(x)}{M} + \frac{mA + nB}{M}.$$

But

$$\frac{mA + nB}{M} = m'A + n'B,$$

where m', n' are proper integer values depending on m, n ; in fact this equation gives $mA + nB = M(m'A + n'B)$; and if we write herein $A, B = pA + qB, rA + sB$ for MA, MB , the resulting equations must be satisfied identically; that is, we must have

$$\begin{vmatrix} m-p & -q \\ -r & n-s \end{vmatrix} = 0,$$

or $ps - (m-s)p - qr = 0$; and the values of p, q, r, s which satisfy this equation are

$$p = m, \quad q = \frac{n}{r}(m-s), \quad r = \text{anything}, \quad s = \text{anything}.$$

And then we have $y = \text{sn}\left(\frac{(x)}{M} + \beta\right)$. But m' and n' are to be integers for all integer values of m and n ; and we thence have $M = m$, a rational value. But as already mentioned, this is the case where the modulus is arbitrary.

The conclusion thus is, that if the modulus is not arbitrary but has a value such that the differential equation is integrable algebraically for a half-rational value of M , then for no other value of M is the equation integrable algebraically.

[* They are the Memoirs xix. and xx. in the *OEuvres Complètes*, t. I., Christiania, 1881.]

0.4 [658] On Some Formulæ in Elliptic Integrals

[From the *Mathematische Annalen*, t. XII. (1877), pp. 369–374.]

I reproduce in a modified form an investigation contained in the memoir, Zolotareff, “Sur la méthode d’intégration de M. Tchébychef,” *Mathematische Annalen*, t. V. (1872), pp. 560–580.

Starting from the quartic

$$(a, b, c, d, e)(x, 1)^4 = a \cdot x - \alpha \cdot x - \beta \cdot x - \gamma \cdot x - \delta,$$

we derive from it the quartic

$$(a_1, b_1, c_1, d_1, e_1)(x_1, 1)^4 = a_1 \cdot x_1 - \alpha_1 \cdot x_1 - \beta_1 \cdot x_1 - \gamma_1 \cdot x_1 - \delta_1,$$

where, writing for shortness

$$\begin{aligned}\lambda &= -\alpha + \beta + \gamma - \delta, \\ \mu &= \alpha - \beta + \gamma - \delta, \\ \nu &= \alpha + \beta - \gamma - \delta,\end{aligned}$$

the roots of the new quartic are

$$\begin{aligned}\alpha_1 &= \theta + \frac{\mu\nu}{2\lambda}, \\ \beta_1 &= \theta + \frac{\nu\lambda}{2\mu}, \\ \gamma_1 &= \theta + \frac{\lambda\mu}{2\nu}, \\ \delta_1 &= \theta,\end{aligned}$$

θ being arbitrary: the differences of the roots $\alpha_1, \beta_1, \gamma_1, \delta_1$ are, it will be observed, functions of the differences of the roots $\alpha, \beta, \gamma, \delta$.

We assume

$$\frac{dx}{\sqrt{(a, b, c, d, e)(x, 1)^4}} = \frac{dx_1}{\sqrt{(a_1, b_1, c_1, d_1, e_1)(x_1, 1)^4}}.$$

And consequently

$$(a_1, b_1, c_1, d_1, e_1)(x_1, 1)^4 = (1, -B, -2C, -D, 0)(x_1, -1)^4.$$

Hence also

$$x_1 = \frac{A(x_1 - 1)}{x_1 - 1}, \quad \text{or} \quad x_1 = \frac{A + x}{1 + x}.$$

We have

$$\frac{dx}{\sqrt{X}} = \frac{dx_1}{\sqrt{X_1}},$$

where

$$\begin{aligned} X &= (a, b, c, d, e)(x, 1)^4, \\ X_1 &= (a_1, b_1, c_1, d_1, e_1)(x_1, 1)^4. \end{aligned}$$

But the formulae may be presented in a more convenient form. Writing

$$\alpha = \alpha, \quad \beta = \alpha + \beta, \quad \gamma = \alpha + \beta + \gamma, \quad \delta = \alpha + \beta + \gamma + \delta,$$

we have

$$\lambda = \gamma - \alpha, \quad \mu = \delta - \beta, \quad \nu = \alpha + \delta - \beta - \gamma.$$

In particular, if we write $\alpha, \beta, \gamma, \delta = a, b, c, d$, the formulae become

$$\begin{aligned} \lambda &= c - a, \\ \mu &= d - b, \\ \nu &= a + d - b - c. \end{aligned}$$

The differences of $\alpha_1, \beta_1, \gamma_1, \delta_1$ are proportional to those of $\alpha, \beta, \gamma, \delta$, and we thus obtain the transformation of the elliptic differential in a very convenient form.

0.5 [659] A Theorem on Groups

[From the *Mathematische Annalen*, t. XIII. (1878), pp. 561–565.]

The following theorem is very simple; but it seems to belong to a class of theorems, the investigation of which is desirable.

I consider a substitution-group of a given order upon a given number of letters; and I seek to *double* the group, that is to derive from it a group of twice the order upon twice the number of letters. This can be effected for *any* group, in a manner which is self-evident and in nowise interesting; but in a different manner for a *commutative* group (or group such that any two of its substitutions satisfy the condition $AB = BA$): it is to be observed that the double group is not in general commutative.

Let the letters of the original group be $abcde \dots$, we may for shortness write $U = abcde \dots$; and take U as the primitive arrangement; and let the group then be $1, A, B, \dots$ where $A, B, \dots$ represent substitutions: the corresponding arrangements are $U, AU, BU, \dots$ and these may for shortness be represented by $1, A, B, \dots$; viz. $1, A, B, \dots$ represent, properly and in the first instance, substitutions; but when it is explained that they represent arrangements, then they represent the arrangements $U, AU, BU, \dots$

For the double group the letters are taken to be $a_1b_1c_1d_1e_1 \dots$ and $a_2b_2c_2d_2e_2 \dots$, $= U_1$ and U_2 suppose, and U_1U_2 is regarded as the primitive arrangement; A_1 and A_2 denote the same substitutions in regard to U_1 and U_2 respectively, that A denotes in regard to U : and so for B_1, B_2 , etc.; moreover 12 denotes the substitution $(a_1a_2)(b_1b_2)(c_1c_2)(d_1d_2)(e_1e_2) \dots$, or interchange of the suffixes 1 and 2. The substitutions A_1, A_2 , or any powers of these A_1^α, A_2^β , are obviously commutative; applying them to the primitive arrangement U_1U_2 , we have $A_1^\alpha A_2^\beta U_1U_2$ and $A_2^\beta A_1^\alpha U_1U_2$ each $= A_1^\alpha U_1 A_2^\beta U_2$. But A_1^α, A_2^β are not commutative with 12: we have for instance $12A_1^\alpha \cdot U_1U_2 = 12A_1^\alpha U_1 \cdot U_2 = A_1^\alpha U_2 \cdot U_1$, but $A_1^\alpha 12U_1U_2 = A_1^\alpha \cdot U_2U_1 = U_2 \cdot A_1^\alpha U_1$. If instead of the substitutions we consider the arrangements obtained by operating upon U_1U_2 , then we have $A_1^\alpha A_2^\beta U_1U_2 = A_1^\alpha U_1 \cdot A_2^\beta U_2$, and $12 \cdot A_1^\alpha U_1 \cdot A_2^\beta U_2 = A_1^\alpha U_2 \cdot A_2^\beta U_1 = A_2^\beta U_1 \cdot A_1^\alpha U_2$.

The substitutions of the double group are $A_1^\alpha A_2^\beta, A_1^\alpha A_2^\beta \cdot 12$; the arrangements are $A_1^\alpha U_1 \cdot A_2^\beta U_2, A_1^\alpha U_2 \cdot A_2^\beta U_1$, viz. these last may be represented by $A_1^\alpha A_2^\beta$ and $A_1^\alpha A_2^\beta \cdot 12$.

We may write

$$\begin{aligned}(A_1^\alpha B_2^\beta) \cdot (A_1^{\alpha'} B_2^{\beta'}) &= A_1^{\alpha+\alpha'} B_2^{\beta+\beta'}, \\ (A_1^\alpha B_2^\beta) \cdot (A_1^{\alpha'} B_2^{\beta'} \cdot 12) &= A_1^{\alpha+\alpha'} B_2^{\beta+\beta'} \cdot 12, \\ (A_1^\alpha B_2^\beta \cdot 12) \cdot (A_1^{\alpha'} B_2^{\beta'}) &= A_1^{\alpha+\alpha'} B_2^{\beta+\beta'} \cdot 12, \\ (A_1^\alpha B_2^\beta \cdot 12) \cdot (A_1^{\alpha'} B_2^{\beta'} \cdot 12) &= A_1^{\alpha+\alpha'} B_2^{\beta+\beta'},\end{aligned}$$

where in each case the order of the factors is the right order. But the same formula may be written in the opposite order; and we then have, for instance, the last equation

$$A_1^{\alpha'} B_2^{\beta'} \cdot 12 \cdot A_1^\alpha B_2^\beta \cdot 12 = A_2^{\beta'} B_1^{\alpha'} \cdot A_2^\beta B_1^\alpha = A_1^\alpha B_2^\beta \cdot A_1^{\alpha'} B_2^{\beta'},$$

which is right if $AB = BA$, but not otherwise.

The theorem is as follows: *if $A, B, \dots$ are the substitutions of a commutative group, then the foregoing expressions $A_1^\alpha B_2^\beta, A_1^\alpha B_2^\beta \cdot 12$ (viz. for the double group, these are the arrangements, and, substituting for them the corresponding substitutions, the substitutions of the double group) form a group.*

This is proved by showing that the product of any two substitutions of the double group is always a substitution of the double group. The proof is immediate; for instance, from the foregoing equations, the product of any two of the double-group substitutions is equal to one or other of the forms $A_1^\alpha B_2^\beta, A_1^\alpha B_2^\beta \cdot 12$, that is, to a substitution of the double group. And moreover, if the original group is $1, A, B, \dots$ (μ substitutions in all), the double group is $A_1^\alpha B_2^\beta, A_1^\alpha B_2^\beta \cdot 12$ ($2\mu^2$ substitutions in all).

It is to be observed, in regard to the double group, that if the original substitutions are $1, A, B, \dots$, then writing down the corresponding arrangements $1, A, B, \dots$ for the first set, and the same arrangements $1, A, B, \dots$ for the second set, we have in all the $2\mu^2$ arrangements; and that the substitutions of the double group are those which change any one of these arrangements into any other of them.

Example.—For the group $1, A, A^2$ ($A^3 = 1$), the double group is $A_1^\alpha A_2^\beta$ ($\alpha, \beta = 0, 1, 2$), and $A_1^\alpha A_2^\beta \cdot 12$ ($\alpha, \beta = 0, 1, 2$): 18 substitutions in all. And similarly, for the group $1, A$ ($A^2 = 1$), the double group is $1, A_1, A_2, A_1 A_2, 12, A_1 \cdot 12, A_2 \cdot 12, A_1 A_2 \cdot 12$: 8 substitutions in all.

0.6 [660] On the Correspondence of Homographies and Rotations

[From the *Mathematische Annalen*, t. xv. (1879), pp. 238–240.]

It is a fundamental notion in Prof. Klein's theory of the "Icosahedron" that homographies correspond to rotations (of a solid body about a fixed point): in such wise that, considering the homographies which correspond to two given rotations, the homography compounded of these corresponds to the rotation compounded of the two rotations.

Say the two homographies are $A + Bp + Cq + Dpq = 0$, $A_1 + B_1q + C_1r + D_1qr = 0$, then, eliminating q , the compound homography is $A_2 + B_2p + C_2r + D_2pr = 0$, where

$$A_2, B_2, C_2, D_2 = B_1A - A_1C, B_1B - A_1D, D_1A - C_1C, D_1B - C_1D;$$

and the theorem is that, corresponding to these, we have rotations depending on the parameters (λ, μ, ν) , $(\lambda_1, \mu_1, \nu_1)$, $(\lambda_2, \mu_2, \nu_2)$ respectively, such that the third rotation is that compounded of the first and second rotations. The question arises to find the expression for the parameters of the homography in terms of the parameters of the corresponding rotation.

The rotation (λ, μ, ν) is taken to denote a rotation through an angle ϑ about an axis the inclinations of which to the axes of coordinates are f, g, h , the values of λ, μ, ν then being $= \tan \frac{1}{2}\vartheta \cos f, \tan \frac{1}{2}\vartheta \cos g, \tan \frac{1}{2}\vartheta \cos h$ respectively: $(\lambda_1, \mu_1, \nu_1)$ and $(\lambda_2, \mu_2, \nu_2)$ have of course the like significations; and then, if (λ, μ, ν) refer to the first rotation, and $(\lambda_1, \mu_1, \nu_1)$ to the second rotation, the values of $(\lambda_2, \mu_2, \nu_2)$ for the rotation compounded of these are taken to be*:

$$\begin{aligned}\lambda_2 &= \lambda + \lambda_1 + \mu\nu_1 - \mu_1\nu, \\ \mu_2 &= \mu + \mu_1 + \nu\lambda_1 - \nu_1\lambda, \\ \nu_2 &= \nu + \nu_1 + \lambda\mu_1 - \lambda_1\mu,\end{aligned}$$

each divided by $1 - \lambda\lambda_1 - \mu\mu_1 - \nu\nu_1$.

[* The numerators might equally well have been $\lambda + \lambda_1 - (\mu\nu_1 - \mu_1\nu)$, etc., but there is no essential difference: we pass from one set of formulae to the other by reversing the signs of all the symbols:

and hence, by properly fixing the sense of the rotations, the signs may be made to be + as in the text. Assuming this to be so, if we then reverse the order of the component rotations, we have for the new compound rotation the like formulae with the signs – instead of +; but this in passing. The formulae, virtually due to Rodrigues, are given in my paper “On the motion of rotation of a solid body,” Camb. Math. Journal, t. III. (1843), [6].]

0.7 [661] On the Double ϑ -Functions

[From the Proceedings of the London Mathematical Society, *vol. IX.* (1878), pp. 197–199.]

I consider the double ϑ -functions in a form slightly different from that of Göpel and Rosenhain: the sixteen functions are represented by

$$\begin{pmatrix} a \\ b \end{pmatrix} (x, y \mid \omega_1, \omega_2, \omega_3),$$

where a, b each denote 0 or $\frac{1}{2}$; and where $\omega_1, \omega_2, \omega_3$ are parameters connected by a relation $\Omega = 0$ (so that only two of them are arbitrary): I write also a', b' to denote each of them 0 or $\frac{1}{2}$, independently of a, b .

The functions satisfy equations

$$\begin{pmatrix} a \\ b \end{pmatrix} (x + m\omega_1 + n\omega_3, y + n\omega_2 + m\omega_3) = e^{2\pi i(\dots)} \begin{pmatrix} a \\ b \end{pmatrix} (x, y),$$

where m, n are integers.

The theory may be developed very elegantly by introducing in place of x, y the variables u, v , where

$$x = u + v, \quad y = u - v.$$

We have then formulae of the form

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} (u + v, u - v) = (u + v)(u - v) + ' (u + v)'(u - v),$$

and the quarter-period additions give rise to a system of linear equations connecting the sixteen functions.

The formulae may be presented in a more convenient form by the introduction of the variables p, q, r, s , connected by the equation $pq = rs$; and we then obtain the theory of the double ϑ -functions in connexion with that of Kummer's 16-nodal quartic surface.

0.8 [662] On the Double ϑ -Functions in Connection with a 16-Nodal Quartic Surface

[From the *Journal für die reine und angewandte Mathematik (Crelle)*, t. LXXXIII. (1877), pp. 210–219.]

I have before me Göpel's memoir, "Theoriae transcendentium Abelianarum primi ordinis adumbratio levis," *Crelle's Journal*, t. xxxv. (1847), pp. 277–312. Writing P_1, P_2, P_3 , etc., in place of his P', P'', P''' , etc., also $\alpha, \beta, \gamma, \delta, X', Y', Z', W'$, in place of his t, u, v, w, T, U, V, W , the system of 16 equations (given p. 287) is

$$\begin{aligned}
 (1) \quad & P^2 = (\alpha, \beta, \gamma, \delta)(X', Y', Z', W'), \\
 (4) \quad & P_1^2 = (\alpha, \beta, -\gamma, -\delta)(X', Y', Z', W'), \\
 (9) \quad & P_2^2 = (\alpha, -\beta, \gamma, -\delta)(X', Y', Z', W'), \\
 (12) \quad & P_3^2 = (\alpha, -\beta, -\gamma, \delta)(X', Y', Z', W'), \\
 (3) \quad & Q^2 = (\beta, -\alpha, -\delta, \gamma)(X', Y', Z', W'), \\
 (2) \quad & Q_1^2 = (\beta, \alpha, -\delta, -\gamma)(X', Y', Z', W'), \\
 (11) \quad & Q_2^2 = (\beta, -\alpha, \delta, -\gamma)(X', Y', Z', W'), \\
 (10) \quad & Q_3^2 = (\beta, \alpha, \delta, \gamma)(X', Y', Z', W'), \\
 (13) \quad & R^2 = (\gamma, -\delta, -\alpha, \beta)(X', Y', Z', W'), \\
 (16) \quad & R_1^2 = (\gamma, \delta, -\alpha, -\beta)(X', Y', Z', W'), \\
 (5) \quad & R_2^2 = (\gamma, -\delta, \alpha, -\beta)(X', Y', Z', W'), \\
 (8) \quad & R_3^2 = (\gamma, \delta, \alpha, \beta)(X', Y', Z', W'), \\
 (15) \quad & S^2 = (\delta, -\gamma, -\beta, \alpha)(X', Y', Z', W'), \\
 (14) \quad & S_1^2 = (\delta, \gamma, -\beta, -\alpha)(X', Y', Z', W'), \\
 (7) \quad & S_2^2 = (\delta, -\gamma, \beta, -\alpha)(X', Y', Z', W'), \\
 (6) \quad & S_3^2 = (\delta, \gamma, \beta, \alpha)(X', Y', Z', W');
 \end{aligned}$$

where $(\alpha, \beta, \gamma, \delta)(X', Y', Z', W')$ denotes $\alpha X' + \beta Y' + \gamma Z' + \delta W'$; and the 16 equations thus are

$$P^2 = \alpha X' + \beta Y' + \gamma Z' + \delta W', \quad \text{etc.}$$

The signs are, in fact, those of the 16 determinants or minors which belong to the matrix

$$\begin{array}{cccc}
 \alpha & \beta & \gamma & \delta \\
 \alpha & \beta & \gamma & \delta
 \end{array}$$

in a 4×4 matrix; and the arrangement of the signs is thus at once determined.

In fact, to show this, observe that (2) gives $(\beta, \alpha, -\delta, -\gamma)(y, -x, w, -z)$, which is $= \beta y - \alpha x - \delta w + \gamma z$, and so (2) gives

$$(\beta, \alpha, -\delta, -\gamma)(y, -x, w, -z) = \beta y - \alpha x - \delta w + \gamma z,$$

which is identically $= -(\alpha x - \beta y + \gamma z - \delta w)$, or changing the signs of z and w , this is $= -(\alpha, \beta, \gamma, \delta)(x, y, z, w)$: that is, we have $(\beta, \alpha, -\delta, -\gamma)(y, -x, w, -z) = -(\alpha, \beta, \gamma, \delta)(x, y, z, w)$. But from the original formula, $(\beta, \alpha, -\delta, -\gamma)(X', Y', Z', W') = Q_1^2$; and writing X', Y', Z', W' as linear functions of x, y, z, w , we obtain the formula in question.

We have therefore a system of 16 linear functions of (x, y, z, w) , the squares whereof are equal to linear functions of (X', Y', Z', W') : and this is the characteristic property of Göpel's 16 equations.

The composition of the formulae is better seen by writing them under the fuller form $\sqrt{(abf \cdot cde)}$, etc.; viz. omitting the radical signs, the terms are

$$\begin{array}{lll} abf \cdot cde, & adf \cdot bce, & abd \cdot cef, \\ bcf \cdot acd, & def \cdot abc, & acf \cdot bde, \\ bce \cdot ade, & cdf \cdot abe, & aef \cdot bcd, \end{array}$$

each divided by $bdf \cdot ace$; or, in an easily understood algorithm, the terms are

	$bf.d$	$df.b$	$bd.f$
$a.ce$	$bf.d$	$df.b$	$bd.f$
$e.ac$	$bf.d$	$df.b$	$bd.f$
$c.ae$	$bf.d$	$df.b$	$bd.f$

each divided by $bdf \cdot ace$.

0.9 [663] Further Investigations on the Double ϑ -Functions

[From the Journal für die reine und angewandte Mathematik (*Crelle*), t. LXXXIII. (1877), pp. 220–233.]

The present memoir relates to the double ϑ -functions, considered as depending on the quotients of four parameters $\alpha, \beta, \gamma, \delta$: it takes up the subject of my memoir, “On the double ϑ -functions in connexion with the 16-nodal quartic surface,” [662], and establishes in a more complete form the connexion between the two theories.

The leading formulae are those for the sixteen functions $P, P_1, P_2, P_3, Q, Q_1, Q_2, Q_3, R, R_1, R_2, R_3, S, S_1, S_2, S_3$, each of them a product of two factors, a radical factor depending on the differences of the parameters, and a functional factor which is a rational and integral function of degree = 1 in each of the two sets of coordinates (x, y, z, w) and (x', y', z', w') .

We have, in fact,

$$P = \sqrt{(\alpha x' + \beta y' + \gamma z' + \delta w')} \cdot p(x, y, z, w; x', y', z', w'),$$

where p is the functional factor in question.

The radical factor for each of the 16 functions is the square root of a linear function $\alpha X' + \beta Y' + \gamma Z' + \delta W'$: and it appears that very simply, the actual expressions for the constant factors; and so we can enunciate the theorem: the functional factors are each of them a determinant of the form

$$\begin{vmatrix} x & y & z & w \\ x' & y' & z' & w' \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{vmatrix},$$

where the two remaining lines are formed with the constants $\alpha, \beta, \gamma, \delta$ in a determinate manner.

1. We have

$$\begin{aligned}[a] &= aa', \\ [b] &= bb', \\ [c] &= cc', \\ [d] &= dd', \\ [e] &= ee', \\ [f] &= ff',\end{aligned}$$

and

$$[ab] = \frac{1}{(\xi - \xi')^2} (\sqrt{abf \cdot c'd'e'} - \sqrt{a'b'f' \cdot cde})^2,$$

etc., where on the right-hand sides $a, b, \dots, a', \dots$ denote $a - \xi, b - \xi, \dots, a - \xi', \dots$ (ξ, ξ' being here written in place of the x, x' of my paper), then the sixteen functions are proportional to $[a], [b], [c], [d], [e], [f]$, and the ten products $[ab], [ac], \dots [ef]$.

2. Referring to the formulae of my paper [662], I write

$$\begin{aligned}A &= [bc] = (b - c)[x\alpha' \cdot abc], \\ B &= [ca] = (c - a)[x\beta' \cdot bcd], \\ C &= [ab] = (a - b)[x\gamma' \cdot cde], \\ D &= [ad] = (a - d)[x\delta' \cdot def], \\ E &= [ae] = (a - e)[x\epsilon' \cdot efa], \\ F &= [af] = (a - f)[x\zeta' \cdot fab],\end{aligned}$$

and similarly

$$\begin{aligned}G &= [de] = (d - e)[y\alpha' \cdot abc], \\ H &= [ef] = (e - f)[y\beta' \cdot bcd], \\ I &= [fd] = (f - d)[y\gamma' \cdot cde], \\ J &= [bc] = (b - c)[y\delta' \cdot def], \\ K &= [ca] = (c - a)[y\epsilon' \cdot efa], \\ L &= [ab] = (a - b)[y\zeta' \cdot fab],\end{aligned}$$

where $[x\alpha' \cdot abc]$, etc., denote determinants of the form mentioned above.

3. The formulae of the memoir [662] are

$$\begin{aligned}
 A &= (b-c)(x\alpha' \cdot abc), & G &= (d-e)(y\alpha' \cdot abc), \\
 B &= (c-a)(x\beta' \cdot bcd), & H &= (e-f)(y\beta' \cdot bcd), \\
 C &= (a-b)(x\gamma' \cdot cde), & I &= (f-d)(y\gamma' \cdot cde), \\
 D &= (a-d)(x\delta' \cdot def), & J &= (b-c)(y\delta' \cdot def), \\
 E &= (a-e)(x\epsilon' \cdot efa), & K &= (c-a)(y\epsilon' \cdot efa), \\
 F &= (a-f)(x\zeta' \cdot fab), & L &= (a-b)(y\zeta' \cdot fab).
 \end{aligned}$$

4. We have moreover

$$\begin{aligned}
 [ab] - [cd] &= \frac{1}{(x-x')^2} \begin{cases} abf \cdot c'd'e' + a'b'f' \cdot cde \\ -cdf \cdot a'b'e' - c'd'f' \cdot abe \end{cases} \\
 &= \frac{1}{(x-x')^2} (abc'd' - a'b'cd)(e'f - ef'),
 \end{aligned}$$

where for the moment a, b, a' , etc., are written to denote $a-x, b-x, a-x'$, etc.; we have then

$$\begin{aligned}
 e'f - ef' &= (e-x')(f-x) - (e-x)(f-x') \\
 &= -(e-f)(x-x') = -ef(x-x'),
 \end{aligned}$$

and

$$\begin{aligned}
 abc'd' - a'b'cd &= (a-x)(b-x)(c-x')(d-x') - (a-x')(b-x')(c-x)(d-x) \\
 &= -(x-x') \begin{vmatrix} 1, & x+x', & xx' \\ 1, & a+b, & ab \\ 1, & c+d, & cd \end{vmatrix} \\
 &= -(x-x')[xx' \cdot abcd].
 \end{aligned}$$

Hence $[ab] - [cd] = ef[xx' \cdot abcd]$, and the equation to be verified becomes

$$(ef \cdot cd - cd \cdot ef)[xx' \cdot abcd] = 0,$$

viz. this is, in fact, an identity.

It is to be noticed that

$$\begin{array}{lll}
 \delta^2 + \alpha^2 - \beta^2 - \gamma^2, & 2(\alpha\beta - \gamma\delta), & 2(\gamma\alpha + \beta\delta), \\
 2(\alpha\beta + \gamma\delta), & \delta^2 + \beta^2 - \gamma^2 - \alpha^2, & 2(\beta\gamma - \alpha\delta), \\
 2(\gamma\alpha - \beta\delta), & 2(\beta\gamma + \alpha\delta), & \delta^2 + \gamma^2 - \alpha^2 - \beta^2,
 \end{array}$$

each divided by $\delta^2 + \alpha^2 + \beta^2 + \gamma^2$, form a system of coefficients in the transformation between two sets of rectangular coordinates. We have therefore

$$\begin{array}{lll} \sqrt{(ab}, & \sqrt{(ad}, & \sqrt{(ce}, \\ \sqrt{(be}, & \sqrt{(de}, & \sqrt{(ac}, \\ \sqrt{(bc}, & \sqrt{(cd}, & \sqrt{(ae}, \end{array}$$

each divided by $\sqrt{(bd)}$ and the several terms taken with proper signs, as a system of coefficients in the transformation between two sets of rectangular axes: a result which seems to be the same as that obtained by Hesse in the Memoir, “Transformations-Formeln für rechtwinklige Raum-Coördinaten”; *Crelle*, t. LXIII. (1864), pp. 247–251.

The composition of the last mentioned system of functions is better seen by writing them under the fuller form $\sqrt{(abf \cdot cde)}$, etc.; viz. omitting the radical signs, the terms are

$$\begin{array}{lll} abf \cdot cde, & adf \cdot bce, & abd \cdot cef, \\ bcf \cdot acd, & def \cdot abc, & acf \cdot bde, \\ bce \cdot ade, & cdf \cdot abe, & aef \cdot bcd, \end{array}$$

each divided by $bdf \cdot ace$; or, in an easily understood algorithm, the terms are

	$bf.d$	$df.b$	$bd.f$
$a.ce$	$bf.d$	$df.b$	$bd.f$
$e.ac$	$bf.d$	$df.b$	$bd.f$
$c.ae$	$bf.d$	$df.b$	$bd.f$,

each divided by $bdf \cdot ace$.

Cambridge, 14 March, 1877.

0.10 [664] On the 16-Nodal Quartic Surface (commencement)

[From the *Journal für die reine und angewandte Mathematik (Crelle)*, t. LXXXIV. (1877), pp. 238–241.]

PROF. BORCHARDT in the Memoir “Ueber die Darstellung u. s. w.” *Crelle*, t. LXXXIII. (1877), pp. 234–243, shows that the coordinates x, y, z, w may be taken as proportional to four of the double -functions, and that the equation of the surface is then Göpel’s relation of the fourth order between these four functions: and he remarks at the end of the memoir that it thus appears that the coordinates x, y, z, w of a point on the surface can be expressed as proportional to algebraic functions, involving square roots only, of two arbitrary parameters ξ, ξ' .

It is interesting to develop the theory from this point of view. Writing, as in my paper, “Further investigations on the double -functions,” pp. 220–233, [663],

$$[a] = aa',$$

$$[b] = bb',$$

$$[c] = cc',$$

$$[d] = dd',$$

$$[e] = ee',$$

$$[f] = ff',$$

and

$$[ab] = \frac{1}{(\xi - \xi')^2} (\sqrt{abf \cdot c'd'e'} - \sqrt{a'b'f' \cdot cde})^2,$$

etc., where on the right-hand sides $a, b, \dots, a', \dots$ denote $a - \xi, b - \xi, \dots, a - \xi', \dots$ (ξ, ξ' being here written in place of the x, x' of my paper), then the sixteen functions are...

[Paper 664 continues beyond page 200.]

Transcriber's Notes

This transcription covers pages 151–200 of Volume 10 of Cayley's *Collected Mathematical Papers*, containing the conclusion of Paper 655 and Papers 656 through 664 (the latter commencing on page 200).

Contents of this Transcription

- Paper 655: “A Memoir on Differential Equations” (conclusion, sections 77–81)
- Paper 656: “On the Theory of Partial Differential Equations” (pp. 134–138)
- Paper 657: “Note on the Theory of Elliptic Integrals” (pp. 139–142)
- Paper 658: “On Some Formulæ in Elliptic Integrals” (pp. 143–148)
- Paper 659: “A Theorem on Groups” (pp. 149–152)
- Paper 660: “On the Correspondence of Homographies and Rotations” (pp. 153–154)
- Paper 661: “On the Double ϑ -Functions” (pp. 155–156)
- Paper 662: “On the Double ϑ -Functions in Connexion with a 16-Nodal Quartic Surface” (pp. 157–166)
- Paper 663: “Further Investigations on the Double ϑ -Functions” (pp. 167–179)
- Paper 664: “On the 16-Nodal Quartic Surface” (commencement, p. 180)

Known Issues and Caveats

1. **Mathematical fidelity:** The OCR-extracted text contains numerous errors in subscripts, superscripts, and special symbols. The transcriber has attempted to reconstruct the mathematics from visual inspection of the scanned pages, but some formulae may contain inaccuracies. In particular, the functional determinant expressions in Paper 656 and the elliptic transformation formulae in Paper 658 should be checked against the original.

2. **Completeness:** Due to the density of the mathematical content across 50 pages, some intermediate algebraic steps may have been abbreviated or condensed. The transcription aims to preserve all essential definitions, theorems, and final formulae.
3. **Notation:** Cayley's notation for functional determinants $\frac{\partial(a,b)}{\partial(p,q)}$ and his use of symbols such as (H, Θ) for Poisson brackets have been preserved in modern form. The square bracket notation $[ab]$, $[xx' \cdot abcd]$ for certain combinations follows Cayley's original usage.
4. **Paper 662:** The tabulation of the sixteen Göpel equations has been reorganised for clarity in L^AT_EX; the original presents these in a different visual arrangement.
5. **Paper 663:** The concluding sections on pp. 176–179 contain extremely condensed algebraic manipulations involving the bracket notation; these have been presented in summary form.

-Functions(conclusion)

0.11 [665] A Memoir on the Double -Functions179
(conclusion)

[From the Philosophical Transactions of the Royal Society of London, vol. 169 (1878), pp. 663-730; continued from preceding pages.]

(Page 201.) We have, as the equation of a quadric surface enveloped by

$$\alpha(x - x') + \beta(y - y') + \gamma(z - z') + \delta(w - w') = 0,$$

which is the same form as before, the relations

$$ab' + ba' = 0, \quad cd' + dc' = 0, \quad \text{etc.},$$

or, more precisely,

$$\frac{\partial \Omega}{\partial x} \cdot X' + \frac{\partial \Omega}{\partial y} \cdot Y' + \frac{\partial \Omega}{\partial z} \cdot Z' + \frac{\partial \Omega}{\partial w} \cdot W' = 0.$$

It is to be remarked that there are 15 expressions such as $A1B + B1A$, and 45 expressions such as $AC1BD + AD1BC$; and that each of these $(15 + 45 =)60$ expressions is a quadric function of $\partial_x, \partial_y, \partial_z, \partial_w$ alone.

In like manner $\Omega, \Omega_1, \Omega_2, \Omega_3$ are quadric functions of (x, y, z, w) , and similarly $\Omega', \Omega'_1, \Omega'_2, \Omega'_3$ of (x', y', z', w') ; and hence $\Omega, \Omega_1, \Omega_2, \Omega_3$ satisfy the conditions

$$\frac{\partial \Omega}{\partial x} - 2x + \frac{\partial \Omega_1}{\partial y} \cdot 2 + \frac{\partial \Omega_2}{\partial z} \cdot 2 + \frac{\partial \Omega_3}{\partial w} \cdot 2 = 0,$$

and similarly for the conjugate functions.

Expression of $\Omega - \Omega'$: second subheading.

We assume $\Omega - \Omega' = \frac{1}{2}M\Omega''$, where M is a quadric function of ∂_x, ∂_y ; suppose

$$M \equiv (xx' + yy') + ' (xx' + yy').$$

It is to be remarked that there are 15 expressions which can be presented in the form

$$\left(x' \frac{\partial}{\partial x} + y' \frac{\partial}{\partial y}\right) \sqrt{X} = \left(x' + y' \cdot \frac{xy'}{X}\right) \sqrt{X} + \left(y' \cdot \frac{xx'}{Y}\right) \sqrt{X};$$

it is clear that the term in question is at once expressible in the form indicated by means of these.

It is to be remarked that there are 15 expressions which form a quadric system of (x, y) alone; viz., $X, Y, \frac{xy'}{X}$ etc. are linear functions of (x, y) together with the symbols

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right)\sqrt{X} + \left(\frac{\partial}{\partial x'} + \frac{\partial}{\partial y'}\right)\sqrt{Y},$$

where the bracketed symbols satisfy the conditions

$$\xi\alpha + \eta\beta = 0, \quad \nu\alpha + \mu\beta = 0,$$

and the like for ξ', η', ν', μ' .

Putting for the moment

$$\lambda\alpha + \mu\beta = \nu\alpha, \quad \mu\beta + \nu\gamma = \mu\beta, \quad \nu\gamma + \lambda\delta = \nu\gamma, \quad p\alpha + q\beta = r\alpha + s\beta,$$

$\mu = ab, \nu = ac, \dots$; the values of $\xi, \eta, \nu, \mu, \xi', \eta', \nu', \mu'$ are

$$\xi = 0, \quad \eta = 0, \quad \nu = 0, \quad \mu = 0,$$

respectively.

I found it convenient to assume

$$= -2(x'^2 + y'^2)E = bxx' - ay y' + p(yy' + xx') + q(yy' - xx'),$$

and that the system of seven is

$$= -2(x'^2 + y'^2)E - \alpha xy' + \beta yx' + (\gamma + \delta)xx' + (\gamma - \delta)yy'.$$

(Page 202.) Assuming now

$$= -2(x'^2 + y'^2)\varphi + (\alpha + \beta)(xxy' - yxy)$$

where φ has to be determined so that the first of the same conditions may be satisfied; then substituting this value of φ , we have

$$(y' - py' + qy' - ry' - sy')\sqrt{X} = (x' - px' + qx' - rx' - sx')\sqrt{Y};$$

that is,

$$-a h x y' - x b x' y \sqrt{X} = (-a x x' + b x y' - b y x')\sqrt{Y};$$

viz. in the inner subscription of $\sqrt{X}$, the function Θ is to be diminished by

181

$$(a bc + b ca)\sqrt{acbcdabcd} = (acbcdabcd)\sqrt{abcdabcdc},$$

or, what is the same thing,

$$\left(-h b x x' - h a x x'\right)\sqrt{X} = (-y h b x x' - h a y x x')\sqrt{Y};$$

also,

$$\sqrt{Y}x = \frac{1}{2}a(abcdefxy)\sqrt{xy},$$

and we thus reduce the equation to

$$-(\sqrt{abcdef} \cdot \sqrt{abcdefc})\sqrt{xy} \cdot (abcd \cdot \sqrt{abcdef}).$$

But, referring to the expression for $\lambda\sqrt{ab}$, we have, by a mere interchange of letters,

$$\frac{d}{dx} \sqrt{ab} = -(\sqrt{abcd} \cdot a_1 - \sqrt{abcd} \cdot a_2)\sqrt{ab};$$

and the formula thus becomes

$$-(\sqrt{abcdef} - \sqrt{abcdefc}) \cdot \frac{d}{dx} \sqrt{ab};$$

and, referring to the expression for $\lambda\sqrt{ab}$, we have, by a mere interchange of letters,

$$2\sqrt{ab} \cdot \frac{d}{dx} \sqrt{ab} = \frac{d}{dx} \cdot a b;$$

consequently

$$\Theta\alpha = -(\sqrt{abcd} - \sqrt{abcdc})\frac{1}{2}\frac{d}{dx} \cdot a b,$$

and the value of Π thus, suppose

$$\Pi = \frac{1}{2}(\sqrt{abcd} - \sqrt{abcdc})(a_1 - b_1) \cdot a b,$$

or so they may be written,

$$\Pi = \frac{1}{2}(\alpha b c - a b_1 c_1)(b c d - a_1 c_1 d_1) + a^2 - 2axy y + a^2 c_1 \sqrt{XY}.$$

(Page 203.) Hence

$$abc - a_1 b_1 c_1 = (a + b - a_1 + b_1 + c_1)(b + c - a_1 + b_1 + c_1) - (a + a_1)(b + b_1)^2 - 5c_1^2;$$

and similarly,

$$def - d_1 e_1 f_1 = (d + e - d_1 + e_1 + f_1)(e + f - d_1 + e_1 + f_1) - (d + d_1)(e + e_1)^2 - 5f_1^2.$$

The expression for $a^2 c_1^2$ contains therefore the terms

$$-(a + b - a_1 + b_1 + c_1)(b + c - a_1 + b_1 + c_1) + (a + a_1)(b + b_1)^2 + 5c_1^2,$$

or that, $A = abcc_1$, where A is something like the coefficient of $\beta\gamma$ in

$$(\alpha + \beta + \gamma + \delta - \epsilon - \eta)\sqrt{XY},$$

which, viz. in fact, the sum of Camy's form in ϕ , equation (200), his ϕ being

$$= 95'' - a.$$

It is to be observed that, Π is the difference of two squares of expressions

$$A\Pi''B\Pi'',$$

the radical which multiplies the being

$$\frac{1}{a + b \cos T + c \cos T' \sqrt{ab}} \cdot \sqrt{ab - a_1 \cos T - b_1 \cos T'}.$$

the differential becomes

$$\frac{dT \cdot \sqrt{ab - a_1 \cos T - b_1 \cos T'}}{(\cos^2 T \cdot \frac{a+b}{1+\alpha \cos T'} + \beta \sin^2 T) \sqrt{a + b \cos T + c \cos T'}};$$

that is,

$$\frac{1}{\sqrt{P + Q \cos T + R \cos T'} \sqrt{a + b \cos T + c \cos T'}} (a + b \cos T + c \cos T') (1 + \alpha \cos T + \dots)$$

The denominator could, of course, be reduced to the form $(\alpha 1, \cos T, \sin T)^2$; but the arrival form seems preferable, inasmuch as it puts in evidence the linear factors

$$\frac{1}{(a + b \cos T + c \cos T') \sqrt{a + b \cos T + c \cos T'} \sqrt{XY}} (\alpha + \beta \cos T + \gamma \cos T') dT;$$

—*Functions*(*conclusion*)

and there seems to be no advantage in further reducing the integral.

(Page 204.) Assuming now

183

$$= -2x'\sqrt{a^2 + y^2} + (\alpha' + \beta')\sqrt{X} \cdot (x' - px' + qx') - rx'(\sqrt{X} - \sqrt{Y}),$$

where has to be determined so that the first of the same conditions may be satisfied. Then, substituting this value of in the equations

$$(y' - py' + qy' - ry')\sqrt{X} = (x' - px' + qx' - rx')\sqrt{Y},$$

that is,

$$(-axy' - xby'x')\sqrt{X} = (-axx' + bxy' - byx')\sqrt{Y},$$

viz. in the inner subscription of $\sqrt{X}$, the function Θ is to be diminished by

$$(abc) \cdot \sqrt{abcdef \cdot \sqrt{XY}} - \sqrt{abcdef \cdot \sqrt{YX}} = -\frac{1}{2}\sqrt{X} - \frac{1}{2}\sqrt{Y};$$

or, what is the same thing,

$$(abc - a_1b_1c_1)\sqrt{X} = \sqrt{abcd \cdot X} - \sqrt{abcd \cdot Y},$$

that is,

$$abc - a_1b_1c_1 = \sqrt{abcd \cdot X/Y} - \sqrt{abcd \cdot Y/X}.$$

But, referring to the expression for $\lambda\sqrt{ab}$, we have, by a mere interchange of letters,

$$\frac{d}{dx}\sqrt{ab} = -(\sqrt{abcd} \cdot a_1 - \sqrt{abcd} \cdot a_2)\sqrt{ab},$$

and the formula thus becomes

$$-(\sqrt{abcdef} - \sqrt{abcdefc}) \cdot \frac{d}{dx}\sqrt{ab};$$

and, referring to the expression for $\lambda\sqrt{ab}$, we have, by a mere interchange of letters,

$$2\sqrt{ab} \cdot \frac{d}{dx}\sqrt{ab} = \frac{d}{dx}(ab);$$

consequently

$$\Theta a = -(\sqrt{abcd} - \sqrt{abcdc}) \cdot \frac{1}{2} \frac{d}{dx}(ab),$$

and the value of Π thus, suppose

$$\Pi = \frac{1}{2} (\sqrt{abcd} - \sqrt{abcdc})(a_1 - b_1) ab,$$

or so they may be written,

$$\Pi = \frac{1}{2} (abc - a_1 b_1 c_1)(bcd - a_1 c_1 d_1) + a^2 - 2axy + a^2 c_1 \sqrt{XY}.$$

(Page 205.) *Expression of $\Omega - \Omega'$: second subheading.*

Writing for shortness $\Omega - \Omega' = R$, so before, and as in other cases ... in general $\Delta PQ = P \Delta Q + Q \Delta P$. Hence starting from $A = \Pi abc' - a'b'c$, we have

$$\begin{aligned} \Delta A &= \Pi \Delta(abc') - \Pi \Delta(a'b'c) \\ &= 4c' \left(\Pi \frac{PU}{QV} - \Pi \frac{PV}{QU} \right) (\Omega - \Omega') c' \xi \xi' + 2b(b - b')(\xi - \xi')(\Omega - \Omega'), \end{aligned}$$

where

$$\Delta a = -a + (\xi - \xi')aa', \quad \xi - \xi' = 0, \quad \xi - \xi' = -1 \quad (\xi a - \xi' \xi),$$

so that

$$\Delta a = a - a + (\xi - \xi')aa' = \frac{1}{2}(a - a_1)(b + b_1)(\xi - \xi') = aa'(b - b_1).$$

We thus have

$$\Delta A = -\Pi abc'[(\xi - \xi')aa'c_1c'_1 + \Pi bb'c'c'_1c(\xi - \xi')] - \frac{1}{2}(a + a_1)(b + b_1)(c + c_1).$$

But the expression for Π contains the factors $4c'$, a , b , c ; hence multiplying the whole by A , we have

$$\begin{aligned} \frac{4}{\Pi} \Delta A &= \Pi a^2 [\Pi(bc' - b'c)(bc - b'c') + 4ab(\xi - \xi')] \\ &\quad - \frac{1}{2} \Pi b'(ab - a_1 b_1)(c - c_1) - \Pi ab(a + b)(\xi - \xi')(\sqrt{XY}) \\ &\quad + \frac{1}{2} \Pi(abcd c' - (\sqrt{XY})) - \omega_1(\xi - \xi')(\xi - \xi'), \end{aligned}$$

where M has its foregoing value $= [\Pi_1 - (\sqrt{XY})](\Pi' \cdot \Pi w + W(bw'))$.

-Functions(conclusion)

First step of the reduction.

Writing $\Theta\xi' = U$, $b_2acdef = U_1$, then $X = aU$, $Y = a_1U_1$, and consequently

$$X - U = a'U^2, \quad Y - U_1 = U'a_1U_1^2;$$

the second expression in U , U_1 denoting differentiations in x , y respectively; then

$$X' - X = -aX + 2\Theta x - \frac{1}{2}a'\left(\frac{\partial U}{\partial x}\right)^2 \cdot \sqrt{ab'/aac'},$$

and similarly

$$Y' - Y = -aY + 2\Theta y - \frac{1}{2}a'\left(\frac{\partial U}{\partial y}\right)^2 \cdot \sqrt{ab'/aac'},$$

where, as before, Θ has been treated in the analytic form expressed in the formulae.

(Page 206.) Hence

$$\Omega = a \cdot \Omega - \frac{2}{a} \left(\frac{PV}{QU} - \frac{PU}{QV} \right) (\Omega - \Omega') 2\Theta(ba - b'a) - (aa_1)(b - b')(\xi - \xi');$$

where the first value, Ω , substituting for X , Y , Ω , Ω' their values, and arranging according to ∂x , ∂y , Ω , is

$$\begin{aligned} &= \frac{2}{a} \left[a'(\partial'_x P)(\partial'_x Q) - a'(\partial'_y Q)(\partial'_y P) - a' \partial'_x \partial'_y (\partial'_x \partial'_y) \right] + \dots \\ &\quad + 2b'(aa_1 - a_1a)(xy' - x'y) + \dots \\ &\quad + (\partial x' - \partial' x')ab'_1 + a(b_1b' - b'b'_1)(\partial x' \cdot \partial'_x \partial'_y \cdot \partial'_y P \cdot \partial'_y Q) \cdot aa'. \end{aligned}$$

After collecting and simplifying, one is led to expressions of the form

$$\Theta\xi' = \frac{1}{2} \left(\frac{\partial \Pi}{\partial x} \cdot a' - \frac{\partial \Pi}{\partial y} \cdot a \right),$$

and these formulae are no advantage in further reducing the integral.

(Page 207.) *Third step of the reduction.*

First, for the coefficient of $(\partial\omega')^2$, multiplying that $\Delta = a_1\omega - \omega_1$, we have

$$2 \left(\frac{\partial U}{\partial x} \right)^2 a' - 2U(\alpha + a_1) - 2U'(\alpha' + a_1)(a + a_1)(\xi - \xi') - 2U'(a_1b_1 - ab)(\xi - \xi') - 2U'(b'_1a_1 - b_1a')$$

$$-(\xi - \xi')(\xi - \xi')a = a(\partial x'^2 \cdot \partial y'^2 \cdot \partial x' \cdot \partial'_y \cdot a_1).$$

Adding these, the right-hand side divides by $(a_1 - a)$; that is, by $2P$; and the resulting value is

$$= \frac{1}{2}a'\xi\xi'aba'\sqrt{ab'abc'},$$

the term $aaa'ac\xi\xi'$ is reduced to the foregoing formulae; by performing the additions, observing as before, that

$$\Theta = a'b'\Theta\xi'a' = a'(a_1b - 2a'b')(\xi - \xi') - 2aa_1b_1c_1\xi.$$

Adding these, the right-hand side divides by $(a_1 - a)$; that is, by $2P$; and the resulting value is

$$(\xi - \xi')(\Theta\xi')(a_1 - b_1)(c - c_1) - 2a + b_1 + b'(\xi - \xi') + b'(\xi - \xi')(\Theta\xi')(a_1 - b_1) = -2a + b_1 + b'(\xi - \xi')$$

or that, $A = abcc_1$, where A is something like the coefficient of $\beta\gamma$ in $(\alpha + \beta + \gamma + \delta - \epsilon - \eta)\sqrt{XY}$, in the manner above $\xi \dots$ in fact, the sum of Camy's form in ϕ equation (200), his ϕ being $= 95'' - a$.

It is to be observed that, Π is the difference of two squares of expressions of the form

$$A\Pi^{1/2} = D\Pi^{1/2},$$

the radical which multiplies the being

$$\frac{1}{a + b \cos T + c \cos T' \sqrt{ab}} \cdot \sqrt{ab - a_1 \cos T - b_1 \cos T'};$$

the differential becomes

$$\frac{dT \cdot \sqrt{ab - a_1 \cos T - b_1 \cos T'}}{(\cos^2 T \cdot \frac{a+b}{1+\alpha \cos T'} + \beta \sin^2 T) \sqrt{a + b \cos T + c \cos T'}};$$

that is,

$$\frac{1}{\sqrt{P + Q \cos T + R \cos T' \sqrt{a + b \cos T + c \cos T'}}} (a + b \cos T + c \cos T') (1 + \alpha \cos T + \dots)$$

(Page 208.) The formula thus becomes

$$\frac{4}{\Pi} \Delta A = a a' \left[M \left(\frac{\partial U}{\partial x} a' - \frac{\partial U}{\partial y} a \right) (\Omega \omega + a) 2b' \xi - \frac{\partial U}{\partial x} (ba - b'a)(\xi - \xi') \right]$$

$$\begin{aligned}
& -\frac{1}{2}\Pi b'(ab - a_1 b_1)(c - c_1) - \overset{-Functions(conclusion)}{\Pi ab(a+b)(\xi - \xi')(\sqrt{XY})} \\
& + \frac{1}{2}\Pi(abcd' - (\sqrt{XY})) - \omega_1(\xi - \xi')(\xi - \xi').
\end{aligned}$$

187

The second formula thus, suppose

$$\begin{aligned}
\Pi = & -(a+b+c+d+e+f) - \frac{2}{3}(a+b+c+d+e+f)^2 - \frac{1}{3}(a+b+c+d+e+f)^3 \\
& - \frac{1}{2}\Pi b'(abcd' - (\sqrt{XY})),
\end{aligned}$$

and the value of Π thus, suppose

$$\Pi = \frac{1}{2}(\sqrt{abcd} - \sqrt{abcdc})(a_1 - b_1)ab,$$

or so they may be written,

$$\Pi = \frac{1}{2}(abc - a_1 b_1 c_1)(bcd - a_1 c_1 d_1) + a^2 - 2axy + a^2 c_1 \sqrt{XY}.$$

The above two expressions involving M have on the right-hand side terms which, when collected with the second of those given in the formulae of §8, give the final expression ΔA , viz.,

$$\Delta A = aa'M(\xi - \xi')\Pi(bc' - b'c)(bc - b'c') + \dots$$

which is obviously a sum of squares.

(Page 209.) *Completion of the reduction and final expression for ΔA .*

But now substituting the values of Π , Π' , W , we find

$$\begin{aligned}
\frac{2}{\Pi} \Delta A = & aa' \left[(aa + ba + a'a + b'a)(\xi - \xi')a + (\xi - \xi')ab'_1 - (\xi ab'_1 a'b'_1) 2c_1(\sqrt{XY}) \right. \\
& + (a'a'b'_1 - a'b'a'b_1)(\xi - \xi')a + (a'b'_1 - a'b a'_1 b'_1)(\sqrt{XY}) \\
& \left. - (\Pi a'b'_1 b_1 + aba'_1 b a'_1 c_1)(\Omega - \Omega')(bw + ac)(\xi - \xi') \right],
\end{aligned}$$

viz., the coefficients belong to the partition which contains the factor aa' , viz., the expression for $\frac{4}{\Pi}\Delta A$; the other portion contains the factor

$$(a' + a'a + ba_1 b'_1)\Pi \cdot (acb'cde' - \sqrt{XY}(bw' \cdot c_1)a'b_1 c'_1)ab a'b a'b',$$

where

$$X = b(dacdef), \text{ etc.}$$

We have then the complete result, viz. this is

$$\frac{2}{\Pi} \Delta A = aa' M [(a+b) 2\Pi - a\Theta a + aba'c_1\xi (\Omega - \Omega') aaa' \Omega - a c' a' b' c a' b' c (\Omega - \Omega')] \\ + \dots$$

which is obviously a sum of squares.

(Page 210.) Also

$$-(a' a' U + U)(ba' U + U_1) = -\Theta b' a' (-aa + aa_1 - a' b' + bac) U_1 cb b' c' c_1 U_1 = -a + ba, + 2a$$

adding these two quantities, the right-hand side divides by $(a_1 - a)$, viz., by Θ , the resulting value is

$$= W = b'(aa + a'a' + aa'a'b'_1) + a'b_1(aa + ba'_1baa') a'_1 b_1,$$

we then see that this radical coefficient of $(ba')^2$ is

$$= 2\Theta b'(ba'a) 2c_1 W' b'.$$

To this is to be added the actual coefficient of $2\Theta a$ in the next step

$$2\Theta b'(b'a'a) 2c_1 W' a',$$

and we see that, without difficulty

$$(\Theta b'(a'a) 2\Theta b' a'_1 b) W' c a' b' c a' c' a' b' a a' a' c_1 c_1 W = 0.$$

Lastly, for the coefficient of $(\partial\xi')^2$, we have

$$\left(2 \frac{\partial U}{\partial x} a' - 2U(\alpha + a_1)\right)^2 - \left(2 \frac{\partial U}{\partial y} a' - 2U'(\alpha' + a_1)\right)^2 \\ = 4(\alpha - \alpha_1)(b + b_1)(c + c_1)(d + d_1)(e + e_1)(f + f_1)(b + b'_1) - 8acc'_1(ba_1b'a + b_1ac) (ba'a'b'b'b_1 \\ + 8abcd(cc' - ba)(bc - b'c')(b'c - bc')(\sqrt{XY});$$

which is the required value.

(Page 211.) Adding it to the preceding expression, the sum is

$$= (a' - a)(a + b)(a + c)(a + d)(\sqrt{XY})(aa'b'_1 + a'b a'_1 b'_1)(a + b)(c + aa'_1 + ba'_1 a + ac) a + a' a' c \\ +] \dots - 2b'(aa + aa'c) a'(aa + a'a'c'_1) c_1 a + 2c'(aaa + ac') a + 2b_1 a'_1 c'_1 (\sqrt{XY}) \\ + \dots + (2b' - baa' - a'b) 4a'_1 b_1 + 2c_1(\Pi a'ba' + a'c) 4a'b_1 c_1.$$

—Functions(conclusion)

This is, in fact, divisible by $(a_1 - a)$; that is, by Π ; for there exists between the symbols the relations 189

$$\begin{aligned} ba'b' + a'c' + ca'a'ca'b'c'a'c'a'a', \\ b'cba'a'ca'b'a'a'ca' + a'b(ca'c'b'a')b'a', \\ c'a'b'c'a'(aaa'c'_1c'_1a'c'_1a'c'a'_1) \\ + (2b' - 2a'aa'c'a'cc) = 0, \end{aligned}$$

and we thus reduce the expression to

$$\frac{2}{\Pi}(ba' - b'a)(b + aa'a'b'b'b_1)(cb'b_1b_1 + c'c'_1)(c'_1 - a'b'_1)(\xi - \xi')(c + c')$$

viz. effecting the division, the quotient is

$$= 2b' - 2b'c(b + a) - 4a'b'a'c'_1(c'_1 - cc'_1a'c_1) - (aaa'c'_1)(\sqrt{XY});$$

and we thus reduce the coefficient of $(\partial\xi)^2$ in like manner

$$\Pi'_a = (b' - a)(\sqrt{XY}) - (aa'c'_1)(\sqrt{XY});$$

viz., this is Π'_a multiplied by $(b' - a)(ac' - a'c)(b'c - bc')(b - b') = -(b - b')(aa'c'_1)(\sqrt{XY})$.

(Page 212.) or finally it is

$$\begin{aligned} \Pi'_a = & -2c'(a' - a)(2b + 2a' - a'c) + (a'b' - a'c)(b + b' - a) - 2(b + a' - a')(a + a' - a') \\ & + 2(aa' - a'b')(b - b' + 2a'c' + a'_1c)2b'c'(b + cb'a'_1)(b'_1 + a + 2bc'a'_1) \\ & + 2a'c' - 2ba'ca'c + b'a'(aaa'_1 + bb'c'_1 - a'b'_1) + b'a'a'c'c'_1 + aca'a'_1 + 2a'c'_1 \\ & + (c' - a'_1)(c'_1 + a'b)(\sqrt{XY}) - 2a(a'c)(ac'b_1)(c'_1 + b'c'_1a'_1b'_1) \\ & + 2bac'_1. \end{aligned}$$

It is to be observed that the investigation thus far has been entirely independent of the values of Π , Π' ; those values are, in fact, such as to make the coefficients of $\sqrt{XY}$ vanish; for the actual numerical values of these coefficients can then be found, and it may easily be confirmed with each other: the foregoing investigation of those values was therefore unnecessary.

(Page 213.) Hence the case of paper 665 ends with the establishment of the formula

$$A\Delta + A^2\Pi = 0;$$

or, more concisely, $\Delta A = 0$.

Cambridge, 7th December, 1877.

0.12 [666] Sur un exemple de réduction d'intégrales abéliennes aux fonctions elliptiques

[From the Comptes Rendus de l'Académie des Sciences de Paris, t. LXXXV (juillet-décembre, 1877), pp. 265-268; 373, 374; 426-429; 472-475.]

Je reprends l'investigation de M. Hermite sur lui rapport aux intégrales réductibles

$$\int \frac{(1, x) dx}{\sqrt{a \cdot 1 - x \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx}},$$

publiée au-dessus du titre: "Sur un exemple, etc.," (*Annales de la Société scientifique de Bruxelles*, 1876).

Nous avons les constantes a, b et les variables x, y, u, v ; et on posant

$$X = a \cdot 1 - x \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx,$$

$$Y = a \cdot 1 - y \cdot 1 + ay \cdot 1 + by \cdot 1 - aby,$$

(où $\sqrt{a \cdot 1 - x \dots 1 + bx} = X$, etc.), et les fonctions de l'intégration, les fonctions elliptiques, par fonctions elliptiques, on a les équations différentielles

$$\begin{aligned} \frac{du}{\sqrt{X}} + \frac{dv}{\sqrt{Y}} &= \frac{1}{\sqrt{a}}(du + dv), \\ \frac{x dx}{\sqrt{X}} + \frac{y dy}{\sqrt{Y}} &= \frac{1}{\sqrt{a}\sqrt{b}}(du - dv). \end{aligned}$$

Il y a en effet trouvé les expressions, en rayon des fonctions elliptiques de u, v , des fonctions symétriques $x + y, xy$, et de, dx, dy , des fonctions a, b, c, d, e, f des fonctions $ab', cde, \dots$ ou (avec une notation plus simple) $ab, ac, ad, \dots$ ces fonctions sont d'ailleurs les fonctions de M. Hermite.

(Page 215.) Précisément, ces fonctions sont

$$a = ay, \quad b = 1 - x \cdot 1 - y, \quad c = ax \cdot 1 + ay, \quad d = 1 + bx \cdot 1 + by, \quad e = 1 - abx \cdot 1 - aby;$$

$$ab = \sqrt{a(1 - x \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx \cdot ay \cdot 1 - y \cdot 1 + ay \cdot 1 + by \cdot 1 - aby)} = \sqrt{X}$$

$$ac = \sqrt{a(1 - x \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx \cdot ay \cdot 1 + ay \cdot 1 + ay) \cdot (1 + bx)(1 + by) - aby \cdot (x$$

$$ad = \sqrt{a(1 - x \cdot 1 + ax \cdot 1 - aby \cdot 1 + by \cdot 1 - aby)} = -1 + ax \cdot 1 + by \cdot 1 - x \cdot 1 + by (x - y)$$

$$ae = \sqrt{a(1 + bx \cdot 1 + by \cdot 1 - abx \cdot 1 + aby)} = +1 + bx \cdot 1 + by \cdot 1 - aby (x - y),$$

$$af = \sqrt{a(1 - abx \cdot 1 - aby)} = -1 + abx \cdot 1 + aby \cdot (x - y),$$

$$bc = (1 - x \cdot 1 + ax \cdot 1 + bx) \sqrt{ay} - (1 + ay)(1 - y \cdot 1 - aby) \sqrt{ax} = (1 + ay)(x - y),$$

$$bd = (1 - x \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx) \sqrt{ay} = (1 + by)(x - y),$$

$$be = (1 - x \cdot 1 + bx \cdot 1 - abx) \sqrt{ay} = (1 + by)(x - y) = (1 + by)(x - y),$$

$$bf = (1 - x \cdot 1 - aby) = (1 - aby)(x - y),$$

$$cd = (1 + ax \cdot 1 + by) - (1 + bx) - (1 + ay) - (1 + by) = (1 + by)(x - y),$$

$$ce = (1 + ax \cdot 1 - abx) - (1 + by \cdot 1 - aby) = (1 + by)(x - y),$$

$$cf = (1 + ax \cdot 1 - abx) \sqrt{1 - aby} - (1 + ay \cdot 1 - aby) \sqrt{1 - abx} = (1 + bx + ay)(x - y),$$

$$de = (1 + bx \cdot 1 + by) - (1 + ax \cdot 1 + ay) = (a - b)(x - y),$$

$$df = (1 + bx \cdot 1 - abx) = (a - b)(x - y),$$

$$ef = (1 + bx \cdot 1 - abx - 1 + by \cdot 1 - aby) = (1 + bx + by)(x - y).$$

et je remarque que la différence du carré quelconque des fonctions $ab, ac, \dots$ est une fonction rationnelle en entière dans x, y . On a, par exemple,

$$ac - ab = 1 - x \cdot 1 + ax,$$

$$bc - bc = -1 + ay \cdot 1 + ax(x - y),$$

$$cd - cd = -1 - aby \cdot 1 + aby(x - y),$$

$$ef - ef = a(1 - b)(x - y),$$

etc.

En désirant, comme auparavant, $a = \sqrt{1 + a \cdot 1 + b}$, et puis

$$\alpha = \sqrt{1 - x}, \quad \beta = \sqrt{1 - y}, \quad \gamma = \sqrt{1 + ax}, \quad \delta = \sqrt{1 + ay},$$

$$\epsilon = \sqrt{1 + bx}, \quad \zeta = \sqrt{1 + by}, \quad \eta = \sqrt{1 - abx}, \quad \theta = \sqrt{1 - aby},$$

soit

$$S = \alpha \beta + \gamma \delta,$$

et puis

$$\sqrt{ab} = \frac{1}{S} \cdot \gamma \delta - \beta \delta,$$

$$\sqrt{ac} = \frac{1}{S(\beta - \beta_1)\sqrt{1 - \alpha\beta}xy} \cdot [(1 + by)\eta\theta - (1 + bx)\eta\theta] + a(by - bx)(b - a)\eta\theta,$$

$$\begin{aligned}
\sqrt{ad} &= \frac{1}{S(\alpha\sigma\beta+\delta_1\sigma\beta)\sqrt{1-abxy}} \cdot [(1+by)\eta - (1+bx)\eta] + a(\alpha-1)(by-bx)\eta\theta, \\
\sqrt{ae} &= \frac{1}{S} \cdot [(1+by)\eta - (1+bx)\eta] + a(\alpha+1)(by-bx)\eta\theta, \\
\sqrt{af} &= \frac{1}{S}(-abxy + abab)\eta\theta, \\
\sqrt{bc} &= \frac{1}{S} \cdot [(1+by)\eta - (1+bx)\eta] + a(by-bx)\eta\theta, \\
\sqrt{bd} &= \frac{1}{S}(a+b)(by-bx)\eta\theta, \\
\sqrt{be} &= \frac{1}{S}(ay-ax)\eta\theta, \\
\sqrt{bf} &= \frac{1}{S}(-1-aby) \cdot (-1+abx)\eta\theta, \\
\sqrt{cd} &= \frac{1}{S} \cdot \beta - \sigma\delta_1 a(cb)\eta\theta, \\
\sqrt{ce} &= \frac{\beta-\sigma_1\delta_1}{S} a(cb)\eta\theta, \\
\sqrt{cf} &= \frac{1}{S}(1-abx \cdot 1 - aby - 1 + ax \cdot 1 + ay)\eta\theta, \\
\sqrt{de} &= (a-b)(x-y), \\
\sqrt{df} &= \frac{1}{S}(a+b)(b-a)\eta\theta, \\
\sqrt{ef} &= \frac{1}{S}(-ax \cdot 1 - aby + 1 + ay \cdot 1 - abx)\eta\theta.
\end{aligned}$$

(Page 216.) (où j'écris en, du plus, $\sin am$, $\cos am$, Δam .) et, pour ses moments,

$$\begin{aligned}
\xi &= \sqrt{ax(1-y)(1+ay)} = \frac{xy}{\sqrt{a}\eta-xy\theta}, \\
\eta &= \sqrt{ay(1-x)(1+ax)} = \frac{xy}{\sqrt{a}\xi-xy\theta},
\end{aligned}$$

et, qui sont des fonctions elliptiques, je donne l'identification :

$$\xi^2 = (\xi - \xi)(\eta - \eta)ab = \xi^2 - \eta^2 - a \cdot b.$$

On a l'identité (due à M. Hermite)

$$(P\eta + Q\eta + R\eta - S\xi) - (P\xi + Q\xi + R\xi - S\eta)(a b\xi + a c\xi + a d\xi + a e\xi + a f\xi) = (\xi^2 - \eta^2)(\xi^2 +$$

où

$$\xi = a \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx \quad \text{et} \quad \eta = 0 = \xi = ax.$$

On a ξ identité (due à M. Hermite)

$$(\xi P + \eta P + \xi R + \eta R - \xi S - \eta S)(\xi^2 + \eta^2 - ab) = (\xi^2 - \eta^2)\eta\xi ab.$$

Or, l'identité (due à M. Hermite) :

$$\begin{aligned} P &= \alpha \beta + a \gamma \delta (\eta \theta + b \eta \theta), \\ Q &= \alpha \beta - \gamma \delta a (\eta \theta + a \gamma \theta), \\ R &= \alpha \beta - a \beta \gamma (\alpha \beta + \gamma \delta) - (\eta \theta + a \eta \theta), \\ S &= \alpha \beta \gamma \delta + (\beta \beta \gamma \delta - \gamma \delta) \beta \delta, \end{aligned}$$

et je suppose l'équation

$$P + Q + R + S = a x y (\alpha + \beta + \gamma + \delta) - a^2 c^2.$$

En faisant remplacement $x = b b$, $y = a a$, on a chaussement réduit la différentielle aux signes des radicaux à une certaine forme particulière, savoir, par les expressions rationnelles des radicaux qui figurent dans les expressions de $\sqrt{ab}$, $\sqrt{ac}$, $\sqrt{ad}$, ...

$$\sqrt{ab} = \frac{ab}{1+a\sqrt{ab}} \cdot \xi (1 + b b) - ((1 + a^2) + \alpha \xi^2 \cdot (1 + b)) \sqrt{ab} \xi \eta.$$

(Page 217.) (Pp. 373–375; 426–429.) Les valeurs de $x + y$, xy donnent un beaucoup de point faible de a , b , c , d , e , f , mais les réductions pour obtenir les valeurs de a , b , c , d , e , f des fonctions ab , ac , ... ne sont pas présentées, je donne maintenant ses résultats. Ces valeurs sont

$$\begin{aligned} \sqrt{a} &= \frac{a}{\sqrt{aS}} \cdot \gamma \delta - \beta \delta, \\ \sqrt{b} &= \frac{1}{\sqrt{S(1-\alpha\beta-\delta_1)}} \cdot [\delta(1 + by)\sigma \eta - (1 + bx)\eta] + a (\eta - \theta) a \eta \theta, \\ \sqrt{c} &= \frac{1}{\sqrt{S(1-\alpha\beta-\delta_1)}} \cdot [\delta(1 + ay) \eta - (1 + ax) \eta] + a (\eta \theta + \alpha b \theta) \eta \theta, \\ \sqrt{d} &= \frac{1}{\sqrt{(1-\alpha\beta)-\delta_1}} \sigma \eta - \frac{1+ay}{\sqrt{aS}} \xi \eta + \dots, \\ \sqrt{e} &= \frac{1}{\sqrt{aS}} \xi \eta + \dots. \end{aligned}$$

où

$$\xi = a \cdot \eta a b \eta b \xi.$$

et puis

$$\begin{aligned} \sqrt{xy} &= \frac{1}{\sqrt{S}} (\gamma \delta - \beta \eta' \delta), \\ \sqrt{xy} &= \frac{1}{\xi (\beta_1 \sigma_1 + \delta_1 \sigma_1) \sqrt{(1-\alpha b x y)}} \cdot \beta [(1 + by)\eta - (1 + b b b)] \eta \theta + (1 + by) a b c. \end{aligned}$$

(Page 218.) Les valeurs de a, b, c, d, e donnent $\sqrt{X}\sqrt{Y}\sqrt{XY}$:

$$\begin{aligned} \sqrt{X}\sqrt{Y}\sqrt{XY} &= \sqrt{a b \cdot a c \cdot a d \cdot a e a f} \\ &= \frac{a}{\sqrt{a S}} \cdot [(\eta\theta - a b \eta\theta)(1 + ay)(1 + ax)(1 + ay b)] + ((\eta\theta + b \eta\theta)\sigma c d \eta\theta). \end{aligned}$$

J'ai vérifié que le signe d'antenne avec celui de la valeur d'origine au moyen à la fois de l'expression rationnelle de $\sqrt{X Y}$.

On vérifie en partie les valeurs des fonctions $\sqrt{ab}$, $\sqrt{ac}$, ... en considérant les différences des carrés de ces fonctions; mais on calcule n'est pas toujours facile. Par exemple, nous avons

$$ac - ab = (\alpha b) (\beta \alpha - \beta \alpha),$$

et ainsi mieux dans deux façons à peu près égales. Par exemple,

$$(ac - ab)(ac + ab) = \frac{ab}{S}(\sigma\sigma + \eta\eta) a^2 - a^2 \sigma \eta \theta (\eta\theta + b \eta\theta),$$

et cette valeur doit être égale aux fonctions $\sqrt{ab}$, $\sqrt{ac}$, ... qui ont toujours faciles. Par exemple,

$$ac = \frac{a^2}{(bS\beta - bS\delta)} \eta \theta b [(1 + by)\eta - (1 + bx)\eta] + a (\eta\theta + b \eta\theta) a \eta \theta.$$

Pour cette fois, j'écris cette expression sous la forme

$$A = \alpha \beta + ab c c c (\eta\theta + b \theta \theta) \eta \theta,$$

où $A = b'(1 + ay)(1 + bx)$, et c'est de l'expression rationnelle, on a

$$B = \frac{1}{4}(\alpha c c - b c c)(\alpha \beta + \beta \delta) - B,$$

où $B = b'(1 + ay)(1 + ax)$, et c'est de l'expression rationnelle ;

$$V = \beta(cc'') a d \sqrt{a b c d e} \xi a c e b \eta \theta - 5 b e b \eta \theta.$$

en si ce que c'est l'équation

$$(a + Q + R + S)(\alpha P + \beta Q + \gamma R + \delta S) - A B (\alpha \beta + \gamma \delta) - A B (\alpha \delta + \beta \gamma);$$

c'est-à-dire :

$$P V e^2 Q V R^2 V^2 S^2 - A^2(\alpha \beta + \gamma \delta)^2 - B^2(\alpha \delta + \beta \gamma)^2 = 0,$$

et il y a une vérification ainsi complète.

(Page 219.) La différence $bc - ad$ se donne un exemple beaucoup plus simple; on a

$$\begin{aligned} bc - ad &= 1 + a b \cdot 1 - b x b (x + y), \\ &= \frac{ab}{S} (\beta^2 - \delta^2) a b \eta \theta, \\ &= \frac{a^2 b}{S} 4 \alpha \beta \beta \delta a b \eta \theta, \end{aligned}$$

l'équation à vérifier en ainsi juste.

(Pp. 472-475.) Je donne quelque autre formules dont je me suis servi dans le cours de cette recherche. Partant des expressions de ξ , η , on a

$$\begin{aligned} d\xi &= \frac{1}{2} b b a \sqrt{a} b b [(-abx b + b\eta - b\theta - 2b^2 b b) dx + (a x (-abxb + ax - ayb) dy)], \\ d\eta &= \mu b a b c b e [(-b\theta b + b\eta (b c + b a c' b) dx + (b (-b\theta b + b b - 2b^2 b b)) dy)], \\ d\xi'' &= a c b e [c (-b\theta b - b c d - 2b^2 b b)) dx + (-2b (c - a c b + ab - 2a c b b)) dy]; \end{aligned}$$

on prenant pour A , B , C des fonctions triples qui

$$A d\xi + B d\eta + C d\xi'' = d a,$$

ou

$$A b + B \eta + C c = 0;$$

on pose

$$A b + B \eta + C c'' = a b,$$

et on obtient de ces équations, j'obtiens pour A , B , C les valeurs

$$\sqrt{a} V = \frac{1}{\eta \theta} (-U + W),$$

$$\sqrt{a} V = \frac{1}{\xi} (U - V),$$

$$\sqrt{a} V = \frac{1}{\eta} (U - W),$$

lesquelles satisfont, comme cela doit être, à l'identité

$$A\xi + B\eta + C\xi'' = 0.$$

Réciproquement, on obtient les identités, ce qui est mes pour les peines, en obtient une démonstration de l'équation différentielle

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = \frac{1}{\sqrt{a}}(du + dv).$$

(Page 220.) où

$$\begin{aligned} U &= E(bace + bace) + bcabcabe \cdot (\Theta c + bbaeb), \\ W &= KV \cdot (bace + bace) + babcabcae \cdot (\Theta e + bccb), \\ V &= EV \cdot (bace + bace) + beabcabce \cdot (\Theta b + bbaeb) \\ \nabla &= E \cdot (bace) + bacde \cdot (\Theta a + bbaebc). \end{aligned}$$

et de même

$$\begin{aligned} \Theta V &= \frac{a+b}{2\sqrt{ab}} (U - V + W), \\ BV &= \frac{a}{2} (U - V), \\ CV &= \frac{1}{2} (U - W). \end{aligned}$$

En admettant l'équation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = \frac{1}{\sqrt{a}} (du + dv),$$

on obtient sur l'extension prise par celles relations

$$\begin{aligned} AE &= \frac{x+y}{\sqrt{ab}} = \frac{(c''-a)-c}{2} V, \\ B \frac{V}{c''-a-b} \cdot V &= -ab b - ab - aV, \\ C \frac{V}{c''-a-c} \cdot V &= -aa - ba - cV, \end{aligned}$$

et, en multipliant par

$$bV \cdot (c'' - a)V = \frac{a+b}{2\sqrt{ab}} (c'' - a)V,$$

et dans les seconds positions, en lieu de

$$c''B(b'' - a''), \quad Bb(c'' - a)V,$$

substituant les valeurs

$$Px + Qx + Rx + Sx, \quad Py + Qy + Ry + Sy - 1,$$

on obtient, après quelques calculations étranges,

$$e''\Delta a = -a^2V + 2caV - 2abcV + \alpha c(Px + Qx'' + Rx_1'')\sqrt{QV}V,$$

$$\nabla V = \alpha b(Qx + Qy'') + \beta(Py'' - V + QQV)V + VCVQVV,$$

laquelle satisfait, comme cela doit être, à l'identité

$$A\xi + B\eta + C\xi'' = 0.$$

(Page 221.) En dérivant, pour plus de simplicité,

$$AV = -\frac{1}{2}UV, \quad BV = \frac{1}{2}UV, \quad CV = -\frac{1}{2}U.$$

les valeurs de AV , BV , CV sont

$$A = \gamma\alpha V - \gamma V,$$

$$B = (-\alpha + bcV)(\Theta c + bV)ccbe + ac(\Theta V - \beta V)V,$$

$$C = (1-2ac'')(c''-a+b)V + (1-2acc'')\Theta cabccce + cb(\Theta a + \beta V)\Theta V, \\ + \frac{1}{2}(abc + abcbce) \cdot V - V + beVaVbcV,$$

et on évite trois équations pour

$$AE, B \frac{C}{\eta\theta}, C \frac{C}{a\theta},$$

on dérive

$$\frac{V}{\sqrt{ab}}XV = R\eta + \dots,$$

$$\frac{V}{\sqrt{ab}}bV = a \cdot (-\dots),$$

$$\frac{V}{\sqrt{ab}}cV = \Omega \cdot (\dots),$$

où

$$\Omega = \alpha\beta + \gamma\delta, \quad V = \alpha\delta + \beta\gamma.$$

c'est de moyen de ces équations que j'ai trouvé les valeurs ci-dessus données de $\sqrt{ab}$, $\sqrt{ac}$, ..., ce qui m'a permis d'aller bien plus loin dans la réduction.

(Page 222.) On a aussi le point à pas de $\sqrt{ac}$ a $\frac{1}{2}\Pi UV$

$$\sqrt{ab} = -V + \sqrt{\frac{1}{2}\Pi V},$$

$$\sqrt{ac} = (1+a)V + \sqrt{\frac{1}{2}\Pi V},$$

$$\sqrt{cd} = \frac{1}{2}(-\sqrt{ab}\sigma V + V(1+a)(1-ab)Va),$$

$$\sqrt{ce} = \frac{1}{2}(\xi - \sqrt{ab}\sigma + V(1+a)(1-ab)V),$$

mais il y a des dénominateurs variables qui contiennent des facteurs dont quelques-unes diffèrent des combinaisons, et la réduction aux formes ci-dessus données est traitable à peine de petite.

0.13 [667] On the Bicircular Quartic. Addition to Professor Casey's Memoir "On a New Form of Tangential Equation."

[From the Philosophical Transactions of the Royal Society of London, vol. CLXVII (1877), pp. 441-460. Received January 24,—Read February 22, 1877.]

Professor Casey communicated to me the MS. of the Memoir referred to, and he has permitted me to make to it the present Addition, containing further developments on the theory of the bicircular quartic.

Starting from his theory of the fourfold generation of the curve, Prof. Casey shows that there exist series of inscribed quadrilaterals $ABCD$ wherein the sides AB, BC, CD, DA pass through the centres of the four circles of inversion respectively; or (as it is convenient to express it) the points of A, B, C, D are belong to one of the same four points of inversion respectively, and they lie or are conditional respectively by depending on the relation between the two parameters. But each such numerical expression $\sin \theta, \theta', \theta'', \theta'''$, say, or, as respectively, may here put be obtained, are by mere consideration with each other; the foregoing investigation of $ABCD$, and giving to it an infinitesimal variation, we have four infinitesimal arcs AA', BB', CC', DD' ; these values are differential expressions AA' and BB' being of the same magnitudes; $M_A da_1, M_A db_1$, the identity of the two parameters. But each such numerical expression M_A, M_B, M_C, M_D has not just an analytical form expressed in the formulae of §6, viz.,

$$M_A da_1 = M_B db_1, \quad \text{etc.}$$

where the sum of these expressions is for a, b, c, d unique cancellation in the relation between a, b, c, d which causes the cancellation of this expression $\Delta \Delta E$.

(Page 224.) I propose to complete the analytical theory by establishing the canonical equations $\Delta = Ra - Rda, \Delta = Rb - Rdb$, etc., and the relations between the parameters a, a_1, u_1, u_2 , which belong to an inscribed quadrilateral $ABCD$.

The curve being the imagined form is a quadrinodal form

$$\Delta(x; ab) = \Pi(a_1 - a; b_1 - b);$$

wherein each term is symmetrically composed in the differential coefficients of a, a_1, u_1, u_2 , and the relations x are conditional respectively to a quadrinomial form

$$\Delta a = \Pi(\sin a_1 a + \sin a_2 a)(\sin b_1 b + \sin b_2 b) du.$$

I remark that in the various formulae $f, g, \theta, \beta, f_1, g_1, \theta_1, \beta_1$, etc. we can assume the relations $du = 0 du$.

Where the value of inversion, viz. taking X, T as current coordinates, the equations are :

$$(Px + Qy + Rz)^2 = 1, \quad (X - a)^2 + y^2 = \rho_a^2, \quad \text{etc.}$$

I remark that the analytical change of f, g, θ, β etc. are constants in respect to the elements of the points x, y , and the relation between x, y, u, u_1 is consequence

$$\frac{X}{f - g} + \frac{Y}{g - h} = 1, \quad (X - a)^2 + y^2 = \rho_a^2, \quad \text{etc.},$$

so that $x, y, \theta_x = \frac{a \cos \frac{a+b}{2} \sin a}{2}$ are functions of a single parameter a , and similarly $(u_1, y_1), (u_2, y_2), (u_3, y_3)$ are functions of the parameters a_1, a_2, a_3 respectively.

Formulae for the fourfold generation of the Bicircular Quartic. Art. Nos. 1 to 5.

1. We have four systems of a designed conic and circle of inversion, each giving rise to the bicircular quartic as the envelope of a generating circle resting in a designed conic, having its centre on the conic, and orthogonal to the circle of inversion. The conic in the second generating circle. Taking X, Y, f, g values, suppose

$$\frac{X^2}{f + g} + \frac{Y^2}{f + h} = 1, \quad (X - a)^2 + Y^2 = R_2^2,$$

the new values, $X = u + Y$, and $Y = Rba$, give in like manner

$$\frac{X^2}{f + g + g_1} + \frac{Y^2}{f + h + g_1} = 1, \quad (X - a)^2 + Y^2 = R_2^2,$$

$\sqrt{YX} = R_2$, and so for the other two generations.

2. The eight systems are not, however, all four different, but the bicircular quartic represented by $f + g_1, g + h_1, g_1 + h, g_1 + h_1$ etc.,

and the like circles of inversion, give the same set of curves and, as the figure of the centre of the curve, may be expressed in the form $X = (\xi + \eta)f + (\xi + \eta)g_1 + (\xi + \eta)g_1$ (which are, in fact, identically true unless an addition for some other reason).

3. The notations being thus altered, we make to general remarks that the variable x, y , may be considered as the related quartic, taking x, y , as before, that are the equations of the second curve

$$(X^2 + Y^2 - h)(X^2 + Y^2 - g) + (g + h)(X^2 + Y^2) - 2aX - 2aa + a^2 - gh = 0,$$

and to obtain the more, a, f, g, h, ρ , are introduced into these expressions, which are connected respectively as R_1, R_2 , etc.

$$(X^2 + Y^2 + f)(X^2 + Y^2 + g) + (X^2 + Y^2 + h)\rho^2 + (R^2 - gh)(g + h)f + (\rho^2 gh) = 0,$$

which is, in fact, derivable from the bicircular quartic by writing therein X, Y in place of x, y .

The expressions of $(Xx)^2 = (x - X)(x + Y) + Y$, where the values may also be written

$$P = 2bc(u + a) = b_1 c c b_1 + b_1 c c b_1,$$

$$P_1 = 2bc(u_1 - 1)(b_1 - c) = b_1 c (b_1 c - c_1 c)(b c)(b_1 c).$$

(Page 225.) Here

$$\sqrt{f + g}, \quad \sqrt{(f + g)(f + g_1)}, \quad \dots$$

indicate the bicircular quartic and the circle of inversion mentioned in the foregoing remarks; in particular, the centres are confined,

$$\sqrt{f + g} \cdot X = (f + a)Pa + (Y - Pb)a^2,$$

$$\sqrt{f + h} \cdot Y = (Y - Pb)a + Pba^2,$$

or also

$$(X - a)^2 + Y^2 = R^2;$$

and we have, by the relation

$$(f + g_1) \cdot (f + h_1) = (a^2 - R)^2.$$

(Page 226.) **2.** The eight systems are not, however, all four different, but the bicircular quartic represented by f, g_1, h_1 , and f_1, g, h_1 , viz.,

in particular, that the centres are co-linear with one another, may be expressed in the form

$$X = (a + bc \sin a) f + (a + bc \sin b) g + (a + bc \sin c) h,$$

which are, in fact, identically true unless an addition for some other reason.

4. The eight systems are, however, all of them four different. Considering a given series of inscribed quadrilaterals, having its centres on the four conics of inversion respectively, the cancellation in the equation we have $a_1 a_2 = \rho^2$ (where there exists between the curves expressed in the relations of the same conic and circle of inversion).

If b may be remarked that a circle of a pre-imaginary radius u , containing a series of inscribed quadrilaterals $ABCD$, having its centres on the four conics of inversion, of the same radii respectively can be regarded as a determined curve, called *Imaginary circle, etc.*; we shall not have, however, in either of pure imaginary radius.

Investigation of aa' .

5. The coordinates of a point on the designed conic

$$\frac{X^2}{f+g} + \frac{Y^2}{f+h} = 1$$

may be taken to be $(f+g)x = (g+g)$, and we hence pass as follows the fundamental

$$\sqrt{f+g+g}x = [(2+aa')(\xi^2+\eta^2) - (2(\xi^2-\eta^2) + (\xi^2+1))]\xi^2, \quad \text{etc.}$$

(Page 227.) the matter is of the bicircular quartic. Consider the generating circle, centre $(f+a, g+a, h+a)$, which cuts at right angles the circle of inversion

$$(X-a)^2 + Y^2 = R_2^2.$$

If for a moment the variable is called θ , then the equation of the generating circle is

$$(X-a)^2 + Y^2 - R_2^2 = \cos \theta a + \sin \theta b,$$

the condition for the equation of the generating circle to be inscribed in right angles to the circle of inversion gives

$$(X-a)\xi + Y\eta = (\xi+a)\xi + (\eta+a)\eta = (\xi+a)\xi + (\eta+a)\eta - 1.$$

If for a moment the variable is called θ , then the equation of the generating circle is

$$(X - a)\xi + Y\eta = \cos \theta + \sin \theta + a.$$

But from the equation for X we have

$$B = (X - a)(X - g + Y)^2 - R'(2a - g + g) a,$$

and hence between X, Y as the equation of the generating circle, we have

$$(X - a)X + (Y - a)Y = \frac{1}{2}(\xi - \eta - 2g)X - R'(2a - g + a) X.$$

This is, in fact, an inscribed quartic.

6. If we make further to notice that, if ξ denote the inclination to the axis of ξ of the tangent on the designed conic, then the central, viz., the ratio of $f + g + g$ to $g + g$ is the same as η , viz. Casey's θ , then

$$\xi = \frac{\sec \xi}{\sqrt{X}}, \quad \eta = \frac{\sec \xi}{\sqrt{Y}}, \quad \text{where} \quad U = (f+h) \sin^2 \xi + (g+h) \cos^2 \xi - h;$$

viz., $\sec^2 \xi = U$.

If now further be noticed that, if a denote the inclination to the axis of the conic, then the central angle of a related point of point will have

$$a = \frac{\sin \xi}{\sqrt{U}}, \quad y = \frac{\cos \xi}{\sqrt{U}}, \quad \text{where} \quad U = (f+h) \sin^2 \xi + (g+h) \cos^2 \xi - h;$$

giving, as is easily verified,

$$\frac{du}{u} = \frac{dy}{x}; \quad \text{we have therefore}$$

$$\frac{du}{u} = \frac{dy}{x}, \quad \frac{du}{X} = \frac{(-a + yx)}{\sqrt{U}};$$

or

$$\frac{(u + g)^2}{(g + g)^2 \xi^2 + (h + g) x^2} = du.$$

(Page 228.) or say the two values are

$$B = \frac{1 - a\xi - \beta y + g\xi}{\xi^2 + y^2}, \quad B' = \frac{1 - a\xi - \beta y - g\xi}{\xi^2 + y^2};$$

to preserve the generality it is necessary to introduce a determinate

$$dX = R da + a dR,$$

$$dY = R da + g dR.$$

But from the equation for X we have

$$B = (X - a)(bX)y - 2by + R^2(2a - X)f - X(2R dX) = 0,$$

whence

$$dX = R da (Xx + Yy) da,$$

$$dY = R da (Rda - Xfy).$$

12. The differentials du , dy can be expressed in terms of a single differential du , viz. writing

$$\xi = \frac{\sec \xi}{\sqrt{U}}, \quad y = \frac{\cos \xi}{\sqrt{U}}, \quad \text{where} \quad U = (f+h) \sin^2 \xi + (g+h) \cos^2 \xi - h;$$

$$\Omega = (f + g) \xi^2 + (f + h) y^2,$$

then we have

$$du = \frac{a\xi + by}{\sqrt{a^2 + b^2 - g - h}} da, \quad dy = \frac{-ay + b\xi}{\sqrt{a^2 + b^2}} da.$$

It is to be observed that, since the designed conic the circle of inversion intersect in elliptic, as a real angle. The designed circle, however, must be of two squares of expressions of the form

$$(f + g)^2 - 2(af + g)(f + g) da^2, \quad (f + h)^2 - 2(af + h)(f + h) da^2;$$

where $\sqrt{(f + g)(f + h)} = ba$ is the distance of the centres of the conic and circle of inversion; and in $\sqrt{Pf}$ is, of course, of the sign of multiplication.

(Page 229.) drawn in the senses A to B , B to C , C to D , D to A : and representing the inclinations, measured from the positive direction of axis of ξ , of these tangents by ν_1 , ν_2 , ν_3 , ν_4 respectively; then, in passing to the consecutive quadrilateral, a , b , c , d change, suppose, into $a + da$, $b + db$, $c + dc$, $d + dd$; and the points A , B , C , D into A' , B' , C' , D' , such as $a + da$, $b + db$, $c + dc$, $d + dd$; and we have

$$dA = R da, \quad dB = R db,$$

$$dC = R dc, \quad dD = R dd \cdot da,$$

and thence

$$dA = bda - bdA, \quad d_1 da + db_1 da,$$

$$dB = b da - b dB,$$

$$dC = b dc - b dC,$$

$$dD = b dd - b dD,$$

which are the correct signs in regard to the particular figure.

Reduction of $\int \frac{1 da}{(\xi^2 + y^2)\sqrt{U}}$ to Elliptic Integrals. Art. No. 28.

29. The expression in question is

$$\int dn \frac{\sqrt{(\cos a + \sqrt{c-b})^2 + (\sin a + \sqrt{c-b})b - BY - \alpha)^2}}{(c+g)\xi^2 + (h+g)y^2 + \alpha + a^2 \sin^2 \xi} (\cos T + b)^2 \cdot c^2 \sin^2 \xi;$$

where $\sqrt{U}$ is a more constant; and we may apply it to the Gaussian transformation,

$$\cos a = \frac{a + b \cos T + a^2 \sin T}{c + a \cos T + b \sin T},$$

$$\sin a = \frac{c + b \cos T + b^2 \sin T}{c + a \cos T + b \sin T},$$

where the coefficients $a, b, c, a', b', c', a'', b'', c''$ are such that identically

$$\cos^2 a + \sin^2 a = 1 = (a + b \cos T + a \sin T)^2 (\cos T + b)^2 (\cos T - 1);$$

and also

$$(\cos a + \sqrt{c-b})^2 + (\sin a + \sqrt{c-b})^2 = a^2 - b^2;$$

(Page 230.) so that, $b = \frac{\sin a}{\sqrt{c}}$, $y = \frac{\cos a}{\sqrt{c}}$ are functions of a single parameter a , and similarly (u_1, y_1) , (u_2, y_2) , (u_3, y_3) are functions of the parameters a_1, a_2, a_3 respectively.

Examples of $\Delta\Delta = 0$: numeral evaluation. Art. Nos. 14 to 22.

14. Writing the formulae thus

$$AP + BQ + CR + DS = 0, \quad (a^2 - g^2)(f + g + g)a^2 P + a^2 Q + a^2 R + a^2 D'^2 + acS^2 = 0,$$

which leads to many discoveries by means of a single quadrinomial. The case where the central b has been added so that the bicircular quartic was expressed by the form

$$\frac{X^2}{f+g+g_1} + \frac{Y^2}{f+h+g_1} = 1, \quad (X-a)^2 + Y^2 = R_1^2,$$

which leads to many discoveries by means of a single quadrinomial. We have, in general

$$\Delta = \frac{a+b}{2ac} (\Pi a + \Pi b + \Pi c) - 5ac(b+cb),$$

and similarly

$$AC + BD = (a-b)(b-c)(b+c)AB + (b-a)(c-b)(b+c)CD,$$

which is the value of the generating circle.

15. We have

$$H = b_1 - c_1 = \frac{ac_2 + b}{(b-a)(b-c)} \cdot \cos h_1 - \frac{ac}{(b-a)(b-c)} \cdot \sin h_1,$$

$M_a = aR'^2b$, $aR = M_b$, etc., as the lower may be seen,

$$H = \frac{a+a}{(b-a)(b-c)} \cdot \cos h_1 - \frac{ac}{(b-a)(b-c)} (b+a)c.$$

We thus have the second expression which has equal absolute value as A ;

$$v_1a' = v_2b', \quad v_1b' = v_2a',$$

of a' , b' , the expressions

$$P_1 = aacdb - b'cd(ba^2 - cba),$$

$$P_2 = c'a'b'(ba^2 - cba),$$

$$P_3 = Rb'(a-b)(b-c)a + bRc'(b-a)(c-b),$$

$$P_4 = a^2cb - cb'a, \quad P_5 = acb - cba, \quad P_6 = cba - abcba.$$

(Page 231.) Adding it to the preceding expression, the sum is

$$(a-b)\sqrt{a^2 + ba + bca - bac + b - cbca + ab - cadd + a + abca + acbc + bac + bac + abcb + b} \\ +] \cdots - 2b'(aa + aa'c)a'(aa + a'a'c'_1)c_1a + 2c'(aaa + ac')a + 2b_1a'_1c'_1(\sqrt{XY})$$

$$+ \cdots + (2b' - baa' - a'b) 4a'_1 b_1 + 2c_1 (\Pi a' ba' + a'c) 4a'_1 b_1 c_1.$$

This is, in fact, divisible by $(a_1 - b_1)(b_1 - c_1)(c_1 - a_1)$, which the case being of an inquiry between the symbols

$$ba'bacccbacacbccacbacdcbacdabcbacbaccbac(\sqrt{XY})$$

gives the development.

We have therefore

$$\sqrt{ab} A = acb'ab' - acb' - b'a - 2acb' \cdot acb';$$

viz., we are dividing the expression by A , we have the result expressed which is reducible by the equality

$$B = -axy(\sqrt{ab}) - (a + a) acb'(a + b) = 0.$$

We thus reduce the expression to

$$B = -axyac(ba)abc + acac cabca + (\sqrt{ab}),$$

and there seems to be no advantage in further reducing the integral.

20. The relation between (x, y) , (x_1, y_1) may be expressed also in the two forms

$$1 - a(\xi + a) - B(\eta + a) a + (\xi + a)(\eta - 1) a = \frac{a + b}{(a + b)(b + c)} a + (-a + b)(a + 1) - B(-a + a) =$$

$$1 - a(\xi + a) - B(\eta - a) - 2 \frac{a + b}{(a + b)(b + c)} a + (a + b)(a + 1) - B(a + a) = 0.$$

In fact, the first of these equations is

$$\left[(1 - aa + B) a + (\xi + a)(\eta - 1) a \right] - \left[(a - 1 - aa + B) a + (\xi + a)(\eta - 1) a \right] - 2a^2 a + B^2 a + b 2 a$$

which, by reason of the values of a, b , becomes

$$(1 + ax + ay)(a + bc) - (a + b)aca + (-a + b - a)a + B(a - a) = 0,$$

$$-(ax + ay) - ay - ay + aay(a + a) = 0,$$

or that, $b = \frac{ac}{b-a} \cdot \cos h_1 - \frac{ac}{(b-a)(b-c)} \cdot \sin h_1$.

And in like manner the second equation may be verified.

21. The two equations are:

$$1 - a(\xi + a) - B(\eta + a) = \frac{(a + b)}{(a + b)(a + c)} \cdot \cos \theta_1 - \frac{ac}{(a + b)(a + c)} \cdot \sin \theta_1,$$

$1 - ax - ay, (b + c), (b - a)a + Rb - Rc(b + 1) = ay(a + b)$, which is the resulting value.

In substituting the E^1 and B their values, these are

$$\sqrt{X}\sqrt{Y} = (ax)^2 + (g + a) + (g + h)\sqrt{XY} - ayx(g + a) + (a + a)(g + h),$$

$X = ay, Y = ay$, and so $\sqrt{YX} = \frac{ayc}{g} \cdot \sin \theta$ also; and the resulting value, and similarly

$$\sqrt{XX}Y = \frac{1}{2}(ax + ay)(ax + 1)(ax + 1) - 2ayac(g + a)(b + h),$$

also verified by the data.

(Page 232.) and similarly,

$$\sqrt{X}\sqrt{Y}\sqrt{XY} - (ay + ay)(b + g)\sqrt{XY} - ach^2,$$

which yields the required expressions.

22. It may be remarked that, if $aa' = 1$, then $X = 1, X' = 1, Y = 1, Y' = 1$, and consequently we have

$$\sqrt{XX'}/(chch)(ccc)(hh) - XX'(-2gh) - 2axy = R = R',$$

and we have these relations

$$(\xi - \xi')(ccc)ccc = 0, \quad \text{etc.},$$

and these relations

$$\Omega_2 = (\xi - \xi'')ch \cdot ccc = 0, \quad \Omega'_2 = 0,$$

$$(\xi' - \xi)cccccc = 0,$$

and similarly,

$$A\xi + B\eta + Cc = 0;$$

that is, $\xi = R \frac{acc}{ccc}(ccc) - 2gh, \eta = ay, \xi'' = ay$, and so for the others.

The expressions of $(\xi x)^2$ etc. are, however, very simple. Two of them are,

$$W = \sqrt{XX'} + \sqrt{YY'} - 2\sqrt{ay}\sqrt{ayb - c}(ay + axb) - g(c - c)a^2 - 2agxy,$$

$$W = \sqrt{XX'} - \sqrt{YY'} - 2\sqrt{ay}\sqrt{ayb - c}(ay + axb) - g(c - c)a^2 - 2agxy.$$

The other two values $(\xi y)^2$ may be similarly written.

(Page 233.) **20.** The relation between (x, y) , (x_1, y_1) may be expressed also in the two forms

$$1 - a(x + x_1) - B(y + y_1) a + (x + x_1)(y - 1) - a = (a + b)(b + c) a + aac - c(a + 1) - B(a + a) =$$

$$1 - a(x + x_1) - B(y - y_1) - 2 \frac{a+b}{(a+b)(b+c)} a + (a + b)(a + 1) - B(a + a) = 0.$$

In fact, the first of these equations is

$$[(1 - aa + B) a + (\xi + a)(\eta - 1) a] - [(a - 1 - aa + B) a + (\xi + a)(\eta - 1) a] - 2a^2 a + B^2 a + b 2 a a$$

which by reason of the values of a , b , becomes

$$(1 + ax + ay)(a + bc) - (a + b) aca + (-a + b - a) a + B(a - a) = 0,$$

$$-(ax + ay) - ay - ay + aay(a + a) = 0;$$

or, substituting for E^1 and B , their values, these are

$$\sqrt{X}\sqrt{Y} = (ay)^2 + (g + a) + (g + h)\sqrt{XY} - axy(g + h);$$

$X = ay$, $Y = ay$ and so $\sqrt{YX} = \frac{ay}{c} ccc$ also.

21. The two equations are

$$1 - a(x + x_1) - B(y + y_1) = \frac{a + b}{(a + b)(a + c)} \cos \theta_1 - \frac{ac}{(a + b)(a + c)} \sin \theta_1,$$

$$1 - ax - ay, (b + c), (b - a) a + Rb - Rc(b + 1) = ay(a + b),$$

which is the resulting value.

Cambridge, 7th December, 1877.

(Page 234.) **22.** As a partial verification, I remark that $\Delta\Delta$ should be symmetrical in regard to the constants b, c, d, e, f ; this is obviously the case as regards b and f , and as in regard to the two in h ; if we interchange b and c this coefficient of b' becomes

$$= w_1(ab + b + c + e + d + a)(a + d + e + f),$$

$$-(-2a + b + c + a + a + a + d + d) c' (bc) a + bcd + (bcd + bac) (ccc + ed) ccac a (\sqrt{XY}),$$

and if we interchange the relations b between this coefficient should be in the formed

$$- = w_1(ab - b - c - e - d - a)(a + d + e + f),$$

$$-(-2a-b-c-d-d-a) c' (bca) + bcd (b'ac + aacab) (cc+dc) a + (-2a-c) aa (\sqrt{XY}), \\ = 2\Delta\Delta;$$

which should be the case.

It is to be observed that the investigation thus far has been entirely independent of the values of R, R', R'' : those values are, in fact, such as to make the coefficients of $\sqrt{XY}$ vanish; for the actual numerical values of these coefficients can then be found, and it may be easily confirmed with each other; the foregoing investigation of these values was therefore unnecessary.

Completion of the reduction and final expression for $\Delta\Delta$.

But now substituting the values of R, R', R'' , we find

$$\frac{2}{R} \Delta A = aa' M [2c' (bc-b'c) + 2(b+c)(c+b)] - 2bc (bc-b'c') + 2a'b' (-a+b'_1+c) \\ - a^2 (c+b)(c-b') - a^2 (b+b') (\xi + \eta') (-ab' + c) \\ + \frac{1}{2} a'b' c' (aa' - ba) (\sqrt{XY}) - b'a' c'_1 c (\sqrt{XY}), \\ - 2(c+b)ca + bc_1 a_1 a_1 - 2aa' (c-c'_1) a' (c-c'_1) a', \\ - (-2a+b+d-a+a) c' (b+c) c' (b+a) dc' a + cd - bcab a + ad - ba + (b-c) - 2c_1 b (ba+ca), \\ - 4c' c (c-acac) (bc-bc) a c + 4ac' c' (ac'-ac) c (a'-a) b' c' c'_1 ac'_1 (\sqrt{XY}),$$

which is obviously a sum of squares.

(Page 235.) in terms of the relation: just found between ξ and aa' , these two sets of values will agree together if only

$$R' (Y - (g+h) a) = a(Y - X) a,$$

$$R' (X - (h+a) a) = R(Y - X) aaa.$$

These are easily verified: the first of them, viz.,

$$RY - R' (g+h) a = a(Y - X) a,$$

and the second values give, in like manner,

$$ax \cdot \frac{ac(c-ac)}{a} (ac'_1 - 1) (bc-ac) c c' (a'-a) a',$$

where, on x each $\xi, \eta, \xi', \eta', \xi$ and a values, with the equation

$$R(R^2 - 2aa')(b-X)(a+aa')(a'-a) = 0;$$

this proved by means of the relation

$$(R^2 - 2aa')(b - X)(a + aa')(a' - a) = 0,$$

where $a = Ra$, $b = Rc$, $c = Rc$, ..., prove the identity of the two relations of the value $a' - a$, prove the identity of the two relations of the values $a\Delta a'$.

Considering the quadrilateral $ABCD$, then da is the element of arc AA' , viz. if ν is the inclination to the axis of ξ of the tangent at A drawn in the direction A to B , then da , da are the projections of AA' on the axes of ξ , η respectively, and so for the other elements b , c , d . And similarly the foregoing the of values are

$$\Pi AB = Rba(R^2 - 2aa'),$$

$$\Pi CD = Rbc(R^2 - 2aa'),$$

$$\Pi BC = Rcd(R^2 - 2aa').$$

Formulae for the Elements of Arc a , b , c , d , Πa , Πb , Πc , Πd , etc. Art. No. 23 to 27.

23. The formulae are

$$da = -\nu Rc \sqrt{X} \sqrt{Y} \frac{ds}{\sqrt{X} \sqrt{Y}} = \nu Rc \frac{ds}{\sqrt{X} \sqrt{Y}},$$

$$db = -\nu Rc \sqrt{X} \sqrt{Y} \frac{ds}{\sqrt{X} \sqrt{Y}} = \nu_1 Rc \frac{ds}{\sqrt{X} \sqrt{Y}},$$

$$dc = -\nu Rc \sqrt{X} \sqrt{Y} \frac{ds}{\sqrt{X} \sqrt{Y}} = \nu_2 Rc \frac{ds}{\sqrt{X} \sqrt{Y}},$$

$$dd = -\nu Rc \sqrt{X} \sqrt{Y} \frac{ds}{\sqrt{X} \sqrt{Y}} = \nu_3 Rc \frac{ds}{\sqrt{X} \sqrt{Y}},$$

where the ν each denote ± 1 . Supposing as above that ν is to be changed into ν' , ν'' , ν''' are positive: then R , R have opposite signs: but R , R have the same sign.

(Page 236.) as here also R' and R_1 , and R'_1 and R_2 . We may take R , R_1 , R_2 , and R_3 all of them positive: the value of A_1 , viz.,

$$\Pi_a a = a M_a b - ba M_b a = aacb \cdot da, \quad \dots$$

hence to make dA, dB, dC, dD all positive, we must have M_a, M_b, M_c, M_d each of them of the same sign: for which $\nu_1, \nu_2, \nu_3, \nu_4$ must reciprocally,

$$dA = v_1 R b \frac{ds}{\sqrt{X}\sqrt{Y}} = aaM_b \frac{ds}{\sqrt{X}\sqrt{Y}},$$

$$dB = v_2 R c \frac{ds}{\sqrt{X}\sqrt{Y}} = aaM_c \frac{ds}{\sqrt{X}\sqrt{Y}},$$

$$dC = v_3 R d \frac{ds}{\sqrt{X}\sqrt{Y}} = acM_b \frac{ds}{\sqrt{X}\sqrt{Y}},$$

$$dD = v_4 R a \frac{ds}{\sqrt{X}\sqrt{Y}} = adM_c \frac{ds}{\sqrt{X}\sqrt{Y}};$$

must hence follow dA, dB, dC, dD in either case in the form positive: $a, b, c, d, \nu_1, \nu_2, \nu_3, \nu_4$ all to be 1 or 0, all to be either both positive or both negative respectively.

24. But from $R = -2aa', b = 2aa', \dots$, we have, in either case in the form positive

$$\frac{1}{(a' + y')\sqrt{X}\sqrt{Y}} = \frac{1}{(a' + y')\sqrt{X}\sqrt{Y}} = \frac{1}{(a' + y')\sqrt{X}\sqrt{Y}} = P, \quad P_1, \quad P_2, \quad P_3,$$

$$dS = dS_1 = 2aa'(b'a),$$

$$ds_1 = 2acP_1,$$

$$dS_2 = 2acP_2,$$

$$ds_3 = 2acP_3,$$

and consequently

$$dS = \nu R (acP + acP_1) - \alpha c(a + acP_1)a + acP_2,$$

$$dS_1 = \nu R (acP + acP_1) - \alpha c(a + acP_1)a + acP_2,$$

$$dS_2 = \nu R (acP + acP_1) - \alpha c(a + acP_1)a + acP_2,$$

$$dS_3 = \nu R (acP + acP_1) - \alpha c(a + acP_1)a + acP_2,$$

which are the required formulae for the elements of arc.

25. The determination of the signs has been made by means of the particular figure: but it is easy to see that the points of terms could not for instance be all of them A, B, C at the interior corner of the

curve, and at the exterior corner; or we could not have A, B, C, D all four positive, D at the exterior of the curve, A at the interior, and the like a similar examination would not be possible to eliminate dA, dB, dC, dD , and then obtains an equation such as

$$b' dA + 2a' bacP + a' cbacP_1 = 0;$$

(Page 237.) this would, by virtue of the relations between da, db, dc, dd , become

$$\frac{Yba}{(ax)^2 + Ya^2} dx + \frac{Xba}{(ax)^2 + Ya^2} dy + \frac{Xba}{(ax)^2 + Ya^2} dz + \frac{Yba}{(ax)^2 + Ya^2} dw = 0,$$

an equation between the variables in $abc, \dots$ which cannot have a quadrimomial expression admitted by the variable form.

26. The differentials da, dy may be observed, the foregoing the of values Πdx

$$= \frac{\xi dx}{(a+g)(b+h)} X + \frac{Y dy}{(a+g)(b+h)} Y,$$

where $\sqrt{ab} = \sqrt{f + ga + bg}$. It is also observed by the form θ' , and

$$b'' = b + b' \xi \xi' \xi' \xi - 2(b - h - \xi \xi - ga + a) a + aa'(c - c) \xi'_1,$$

viz. in like manner be calculated in θ as a function of a and a , an irrational function $M\xi - \xi - X\sqrt{U}$, and it would be the rest of such a function. Considering a diagonal conic and the corresponding circle of inversion, the corner of the curve, called Π , we have however, in elider:

$$\frac{d\Theta}{da} = \frac{f - a}{\Pi' \Pi a + f a + a},$$

and the formula thus becomes

$$2Vab\sqrt{ab}da = \frac{ab}{\Pi'} (\Pi a + faba) da,$$

consequently

$$\Theta a = -(\sqrt{ab} - \sqrt{abc})^{\frac{1}{2}} \cdot a + ac'(\sqrt{abc})(ac'),$$

where k is the rather of the squares of the differentials; and these conditions and (a, b, c, d, e, f) are the are conditional respectively a determinate sign ν .

27. The system of curves it

$$ds = aa\ ac\ da + ac\ db + ac\ b\ dc + ac\ b\ da + ac\ da,$$

$$ds_1 = aa\ ac\ ba + ac\ b\ dc + ac\ b\ b\ dc + ac\ b\ da + ac\ b\ da,$$

$$ds_2 = aa\ ac\ dc + ac\ b\ c\ da + ac\ b\ da + ac\ db + ac\ da,$$

$$ds_3 = aa\ ac\ dc + ac\ b\ dc + ac\ b\ b\ dc + ac\ b\ da + ac\ b\ da,$$

where $\nu_a = \pm 1, \nu_b = \pm 1, \dots$ for, etc., are conveniently used in writing it.

Reference to Figure. Art. No. 28.

28. I constructed a bicircular quartic consisting of an exterior and interior oval with the following numerical data: $f, g, h, \rho, g_1, h_1, \rho_1, g_2, h_2, \rho_2, g_3, h_3, \rho_3, a, a_1, a_2, a_3, \dots$ etc. very convenient values, inasmuch as

(Page 238.) the exterior oval cuts (and crosses each, large; the annexed figure shows 0, 1, 2, 3, the centres of the circles of inversion, the interior oval, and a portion of the exterior oval, the centres 1, 2 outside the exterior oval; I add further the values

$$\sqrt{f + h_2 a} = 95, \quad \sqrt{-(g + h_2)} = 244, \quad a_2 = 1019, \quad R_2 = -96,$$

$$\sqrt{f + h_1 a} = 746, \quad \sqrt{g + h} = -241, \quad a = 873, \quad R_1 = 294,$$

$$\sqrt{f + h_1 a} = 715, \quad \sqrt{g + h} = 245, \quad a = 873, \quad R_2 = -96,$$

$$\sqrt{f + h a} = 1934, \quad \sqrt{g + h} = -149, \quad a = 919, \quad R_3 = 23.$$

We thus see how there exists a series of quadrilaterals $ABCD$, where A, B are situate on the interior oval, C, D on the exterior oval. Considering the sides as

(Page 239.) drawn in the senses A to B, B to C, C to D, D to A : and representing the inclinations, measured from the positive direction of axis of ξ , of these tangents by $\nu_1, \nu_2, \nu_3, \nu_4$ respectively; then, in passing to the consecutive quadrilateral, a, b, c, d change, suppose, into $a + da, b + db, c + dc, d + dd$; and the points A, B, C, D into A', B', C', D' , such as $a + da, b + db, c + dc, d + dd$; and we have

$$dA = R da, \quad dB = R db,$$

$$dC = R_2 dc, \quad dD = R_3 dd,$$

and thence

$$\begin{aligned} dA &= bda - bdA, & d_1da + db_1da, \\ dB &= bda - bdB, \\ dC &= bdc - b dC, \\ dD &= bdd - b dD, \end{aligned}$$

which are the correct signs in regard to the particular figure.

Reduction of $\int \frac{1da}{(\xi^2+y^2)\sqrt{U}}$ to Elliptic Integrals. Art. No. 28.

29. The expression in question is

$$\int dn \frac{\sqrt{(\cos a + \sqrt{c-b})^2 + (\sin a + \sqrt{c-b})b - BY - \alpha)^2}}{(c+g)\xi^2 + (h+g)y^2 + \alpha + a^2 \sin^2 \xi} (\cos T + b)^2 \cdot c^2 \sin^2 \xi;$$

where $\sqrt{U}$ is a more constant; and we may apply it to the Gaussian transformation

$$\begin{aligned} \cos a &= \frac{a + b \cos T + a^2 \sin T}{c + a \cos T + b \sin T}, \\ \sin a &= \frac{c + b \cos T + b^2 \sin T}{c + a \cos T + b \sin T}, \end{aligned}$$

where the coefficients $a, b, c, a', b', c', a'', b'', c''$ are such that identically

$$\cos^2 a + \sin^2 a = 1 = (a + b \cos T + a \sin T)^2 (\cos T + b)^2 (\cos T - 1);$$

and also

$$(\cos a + \sqrt{c-b})^2 + (\sin a + \sqrt{c-b})^2 = a^2 - b^2.$$

(Page 240.) Hence

$$(\cos \theta + \sqrt{c-b})^2 + (\sin \theta + \sqrt{c-b})^2 = a^2 - b^2,$$

also

$$\left[\frac{f-a}{\cos \theta + \sqrt{c-b}} \right] = \frac{a-b}{f-h} (\cos \theta + \sqrt{c-b}) a^2;$$

$\Pi' = \sqrt{aa' - \sin^2 \theta (c-b)}$, so that the integral becomes

$$\frac{-a}{\sqrt{(c+g)(c-b)} c'} dT,$$

which is, as it should be, a function of an elliptic integral.

Cambridge, 7th December, 1877.

0.14 [668] On Compound Combinations

[From the Proceedings of the Lit. Phil. Soc. Manchester, *t.* XVI (1877), pp. 113, 114; Memoirs, *ib.*, *SER.* III., *t.* VI. (1879), pp. 99, 100.]

Prof. Clifford's paper, "On the Types of Compound Statement involving Four Classes," [volume of *Proceedings* quoted, pp. 88–110; *Mathematical Papers*, pp. 1–13], relates mathematically to a question of compound combinations; and it is worth while to consider its connexion with another question of compound combinations, the application of which is a very different one.

Starting with four symbols A, B, C, D , we have sixteen combinations of the form $ABCD$, ($1 \cdot 4 \cdot 6 \cdot 4 \cdot 1 = 16$), as before). But in Prof. Clifford's question 1 means ΛWCD , Λ means ΛHCD ; viz. each of the symbols means an aggregate of four assertions; and the 16 symbols are then all of them combinations of such assertions; and the question to be considered is to determine the number of types of the binary, ternary, ... combinations of the sixteen combinations; for, according as these are

$$\text{No. of types} = \frac{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}{1, 4, 6, 13, 19, 27, 37, 47, 51, 50, 47, 37, 27, 19, 6, 4}$$

together. In the first mentioned point of view the like question arises, in regard to the new belonging to the five different forms of compound combinations; and as it is worth while to consider its connexion with another question of compound combinations, the formal so the following development.

$$\text{No. of types} = \frac{1, 2, 3, 4, \dots, 24, 25, 26, 27}{1, 2, 5, 11, \dots, 11, 5, 2, 1}.$$

(Page 244.) as is very easily verified; but if the number of letters $A, B, \dots$ be greater (say this = 6), or, instead of letters, writing the numbers 1, 2, 3, 4, 5, 6, 7, 8, then the question is that of the number of the types of combination of the 20 thirds 13, 14, ..., 78 taken 1, 2, ..., 27 together, a question presenting itself in geometry in regard to the bitangents of a quartic curve (see Salmon's *Higher Plane Curves*, Ed. 2 (1873), pp. 222, et seq.); the numbers, so far as they have been obtained, are

$$\text{No. of types} = \frac{1, 2, 3, 4, \dots, 24, 25, 26, 27}{1, 2, 5, 11, \dots, 11, 5, 2, 1}.$$

It might be interesting to complete the series, and, more generally, to determine the number of the types of combination of the $\frac{1}{2}n(n-1)$ duads of n letters.

0.15 [669] On a Problem of Arrangements

[From the Proceedings of the Royal Society of Edinburgh, *t. IX.* (1878), pp. 338–342.]

It is a well-known problem to find for n letters the number of the arrangements in which no letter occupies its original place; and the solution of it is given by the following general theorem: viz., the number of the arrangements which satisfy *ay*, r conditions is

$$(1 - \Sigma(1) + \Sigma(2) - \dots (-1)^r \Sigma(r))(1 \cdot 2 \dots n),$$

$$= 1 - \Sigma(1) + \Sigma(2) \dots \dots \pm (12 \dots r);$$

where 1 denotes the whole number of arrangements; (1) the number of those which fail in regard to the first condition; (2) the number which fail in regard to the second condition; (12) the number which fail in regard to the first and the second conditions; and so on, the second condition; the Σ refers to the combinations of the n conditions which can occur there; and lastly $(1 \cdot 2 \dots r)$ means $1 \cdot 2 \cdot 3 \dots r$.

Then, in the special problem, the first condition is that the letter in the first place shall not be α ; the second, the letter in the second place shall not be b ; and so on. Hence

$$\Sigma_0 = \Pi n - \frac{n}{1} \Pi(n-1) + \frac{n \cdot n - 1}{1 \cdot 2} \Pi(n-2) + \dots \pm \frac{n \cdot n - 1 \cdot 1}{1 \cdot 2 \dots n} \Pi 0,$$

$$= 1 \cdot 2 \dots n \left[1 - \frac{1}{1} + \frac{1}{1 \cdot 2} - \frac{1}{1 \cdot 2 \cdot 3} + \dots \pm \frac{1}{1 \cdot 2 \dots n} \right],$$

giving for the several cases

$$n = 1, 2, 3, 4, 5, 6, 7, \dots,$$

$$N_0 = 1, 2, 9, 44, 265, 1854, \dots$$

I proceed to consider the following problem, suggested to me by Professor Tait, in connexion with his theory of knots: to find the number of the arrangements of n letters $abc \dots k$, in such manner as

to fix the distances of the elements of all those non-pairs $aa, bb, cc, \dots$ shall be the value k , where k is the value of inversion or expression. This contains the foregoing problem, as the present one is that in which $k = 1$.

(Page 246.) Numbering the conditions $1, 2, 3, \dots, n$, according to the places to which they relate, a single condition is called $[1]$; two conditions are called $[2]$ or $[1\ 1]$, according as the numbers are consecutive or non-consecutive; three conditions are called $[3]$, $[2\ 1]$, $[1\ 1\ 1]$, according as the numbers are consecutive and one not consecutive, or all non-consecutive; and so on: the numbers which refer to the conditions being always written in their natural order, and it being understood that they follow each other cyclically, so that $n = 6$, the set 125 of conditions is $[3]$, as consisting of 3 consecutive conditions; and similarly 1346 in $[2\ 2]$.

Consider a single condition $[1]$, say this is 1; the arrangements which fail in regard to this condition are those which contain aa in the first place or b ; whichever there are thus $2(n-1)$ failing arrangements in any form whatever; and thus

$$2\Pi(n-1) = (n-1)\Pi(n-1).$$

Next for two conditions; these may be $[2]$, say the conditions are 1 and 2; or else $[1\ 1]$, say they are 1 and 3. In the former case, the arrangements which fail are those which contain ab , or ba , in the first and second places; there are thus, in the first place a , or b , and in the third place c or d ; the letters in those places are a, c, a, d, b, c, b, d , in the third place c or d ; the letters in those places are a, c, a, d, b, c, b, d ; and so on, the number of failing arrangements is thus $= 2 \cdot \Pi(n-2)$. And so, in general, when the conditions are $[\alpha, \beta, \gamma, \dots]$, the number of failing arrangements is

$$= (\alpha+1)(\beta+1)(\gamma+1) \dots \Pi(n-\alpha-\beta-\gamma-\dots).$$

But for $[a]$, that is, for the entire system of the n conditions, the number of failing arrangements is (not as by the rule it should be $= n+1 \cdot bn = 2$. The number of failing arrangements in any form whatever; for the n conditions $abc, abd, \dots$ in the conditions $[2, 2]$, $[1, 2]$, etc., would give ξa .

Changing now the notation so that $[1]$, $[2]$, $[1\ 1]$, $[3]$, &c., shall denote the *number* of the conditions $[1]$, $[2]$, $[1\ 1]$, $[3]$, &c., respectively,

it is easy to see the form of the general result. If, for greater clearness, we write $n = 6$, we have

$$\begin{aligned}
 & 1, \quad -\Sigma(1), \quad +\Sigma(12), \quad -\Sigma(123), \\
 N_0 = & 720 - \{[1] = 6\} 2 \mid 120 + \{[2] = 6\} 3 \mid 24 - \{[3] = 6\} 4 \mid 6 \\
 & + \{[1 \ 1] = 9\} 2 \cdot 2 + \{[2 \ 1] = 12\} 3 \cdot 2 \cdot 2, \\
 & + \Sigma(1234), \quad -\Sigma(12345), \quad +\Sigma(123456) \\
 & + \{[4] = 6\} 5 \mid 2 - \{[5] = 6\} 6 \mid 1 + \{[6] = 1\} 2 \\
 & + \{[3 \ 1] = 6\} 4 \cdot 2 + \{[3 \ 2] = 6\} 4 \cdot 3 \\
 & + \{[2 \ 2] = 6\} 3 \cdot 3 + \{[1 \ 1] = 6\} 3 \cdot 3.
 \end{aligned}$$

(Page 247.) or, reducing into numbers, this is

$$N_0 = 720 - 1296 + 1296 - 504 + 90 - 36 + 2 = 80.$$

The formula for the next succeeding case, $n = 7$, may be obtained in like manner

$$N_0 = 5040 - 10080 + 7080 - 3040 + 1560 - 4960 + 2 = 579.$$

Then for the preceding cases, $n = 4, 5, 6$, respectively we have

$$N_0 = 4, \quad 24 - 40 + 16 = 0, \quad 120 - 240 + 184 - 60 + 6 + 12 - 2 = 20,$$

$$N_0 = 1, 2, 0, 3, 80, 579, \dots$$

We have here $r = 0, 4, 6, \dots$ which is, $r = 0, 5, 7, \dots$ respectively*.

The formula for general n , considering the second sums above-mentioned, becomes

$$N_0 = \Pi n - 720 + 1296 - 720 + (n-1)(n-2)\Pi(n-3) - 36 + \dots = 0,$$

The formula for the preceding cases of an arrangement $a, b, \dots$ may here be put in the form

$$N_0 = 24 - 40 + 16 = 0, \quad \dots$$

in respect of the result number of the symbols $a, b, c, \dots$ are of n in cyclical order beginning with $a, \dots$ *a, the letter in the last place not b or $c, \dots$

For the actual variation of so $r = 0$, and the foregoing formulae become

$$n = 4, 5, 6, 7, \dots$$

which the analysis a sum of squares.

(Page 248.) all the substitutions which serve every letter. Thus when $n = 5$, we obtain the 44 substitutions for these 4 letters b, c, d, e ; these are

(abcde, bdec, cebd, 24 symbols obtained by permuting in every way the four letters

$$(ab)(cd), (ac)(bd), (ad)(bc),$$

$$(abcd), (abdc),$$

$$(acbd), (acdb),$$

$$(adbc), (adcb).$$

Here in the first column, performing on the symbol $(abcd)$ the substitutions $(abcd)$, we obtain ab ; $(cdcb)$, ab ; *i.e.* to perform a substitution which is again opening with a ; we must in the manner end, the substitutions are nothing else than the substitutions which form the column system in the same way. The 13 substitutions are thus of these 4 different types, or any of the symbols which they belong, viz.,

$$adcde, \quad bdcae, \quad cdeab, \quad ddeba,$$

$$ddeab, \quad ebcad,$$

are of 5 different types. The question is to determine for any value of n , the number of the different types, in this case there appear, in this case, but it is not at present yet to find it.

¹In explanation of the notation of substitutions, denote that $(abcd)$ means that all the changes may be made in any way 1, 2, 3, ..., but only at the same time; for instance, a is to be changed for b , this b is to be changed for c , this c is to be changed for d , this d is to be changed for a , ... The set $(1\ abcd)$ thus all of them denote the same substitution; the $(a\ 1\ d)$ is to be changed for the cycle (12) .

0.16 [670] [Note on Mr. Muir's Solution of a "Problem of Arrangement."]

[From the Proceedings of the Royal Society of Edinburgh, *t. IX.* (1878), *pp.* 388–391.]

The investigation may be carried further: writing for shortness w_1 , u , v , &c. in place of $\Psi(1)$, $\Psi(2)$, &c., the equations are

$$u_0 = 1,$$

$$u_1 = 0u_0,$$

$$u_2 = 0u_1 + 1,$$

$$u_3 = 0u_2 + 11u_1,$$

$$u_4 = 0u_3 + 6u_2,$$

$$u_5 = 0u_4 + 10u_3 + 12u_2 + 1.$$

Hence assuming

$$u = u_0 + u_1x + u_2x^2 + u_3x^3 + \dots$$

we have

$$\begin{aligned} u &= \frac{1}{1-x^2} + u_0(2x + 6x^2 + 12x^3 + u_2x^4 + \dots) \\ &\quad + u_1(2x + 6x^2 + 10x^3 + 12x^4 + \dots) \\ &\quad + u_2(4x + 10x^2 + 18x^3 + 28x^4 + \dots) \\ &\quad + u_3(2x^2 + 6x^3 + 12x^4 + 20x^5 + \dots); \end{aligned}$$

so that, forming the equation

$$\begin{aligned} u' \frac{x^2}{(1-x)^2} &= u_0(x^2 + 2x^3 + 3x^4 + \dots) \\ &\quad + u_1(2x^2 + 6x^3 + 10x^4 + 12x^5 + \dots) \\ &\quad + u_2(4x^2 + 10x^3 + 18x^4 + 28x^5 + \dots) \\ &\quad + u_3(2x^3 + 6x^4 + 12x^5 + 20x^6 + \dots), \end{aligned}$$

(Page 250.) where u' denotes $\frac{du}{dx}$, we have

$$u - u \frac{x^2}{1-x^2} = \frac{1}{1-x^2} Q,$$

where

$$\begin{aligned} Q &= u_0 \left(-\frac{x^2}{1-x} + u_1 2x + u_2 4x + u_3 2x^2 + \cdots \right) \\ &= \frac{1}{1-x^2} \left[1 + \frac{2x}{1-x} + \frac{4x}{1-x} + \frac{2x^2}{1-x^2} \right] Q (1 - ax) \\ &= \frac{2x}{(1-x)^2} + Q \left[1 - \frac{2x}{1-x^2} \right] Q (1 - ax). \end{aligned}$$

so that the equation is simplified: write

$$u = \frac{1}{1-x} Q - \left(-\frac{x^2}{1-x^2} + \frac{x^2}{1-x^2} \right) Q,$$

and the equation becomes

$$u' \frac{x^2}{(1-x)^2} Q = \frac{1}{1-x^2} Q + \frac{2x}{1-x^2} Q = \frac{1-x}{1+x},$$

then

$$\left(1 - \frac{2}{x^2} \right) Q = Q \left(1 - \frac{2}{x^2} \right),$$

or

$$-\frac{(1-x)x^2-x^2(1-x)^2}{x^2} = Q \left(1 - \frac{2}{x^2} \right) = 1 - \frac{x}{1+x},$$

or finally

$$Q \left(1 - \frac{x}{1+x} \right) = Q \frac{x^2}{(1+x)^2},$$

giving

$$Q = e^{-x} \left[\int \frac{x^2}{(1+x)^2} e^x dx \right],$$

and thence

$$u = \frac{1-x}{x^2} + Q \left(\frac{1}{x^2} + \frac{1}{(1+x)^2} \right) \left[\int \frac{x^2 e^x dx}{(1+x)^2} \right],$$

which is the value of the generating function, $u = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots$.

670. Note on Mr Muir's Solution of a "Problem of Arrangement."

[From the *Proceedings of the Royal Society of Edinburgh*, t. ix. (1878), pp. 388—391.]

But for the purpose of calculation it is best to integrate by a series the differential equation for Q : assuming we find

$$\begin{aligned}q_4 &= 4q_3 - 2, \\q_5 &= 5q_4 + q_3 + 3, \\q_6 &= 6q_5 + q_4 - 4, \\q_7 &= 7q_6 + q_5 + 5, \\q_n &= nq_{n-1} + q_{n-2} + (-)^{n-1}(n-2).\end{aligned}$$

We have thus for $q_3, q_4, q_5, \dots$ the values 1, 2, 14, 82, 593, 4820, $\dots$, and thence

$$M = (1 - x^2)(1 + 2x + 14x^2 + 82x^3 + 593x^4 + 4820x^5 + \dots),$$

viz. writing

$$\begin{array}{cccccc}1 & 2 & 14 & 82 & 593 & 4820 \dots \\-1 & -2 & -14 & -82 & & \end{array}$$

the values of $M_3, M_4, \dots$ are 1, 2, 13, 80, 579, 4738, $\dots$, agreeing with the results found above.

In the more simple problem, where the arrangements of the n things are such that no one of them occupies its original place, if m_n be the number of arrangements, we have

$$m_1 = 1, \quad m_2 = 2, \quad m_3 = 3(m_2 + m_1) = 9, \quad m_4 = 4(m_3 + m_2) = 44,$$

$$m_{n+1} = n(m_n + m_{n-1}),$$

$$M = 1 + (2x + 3x^2)u + (x^2 + x^3)u';$$

$$(-1 + 2x + 3x^2)M + (x^2 + x^3)u' = -1,$$

or, what is the same thing,

$$\frac{d}{dx} \frac{1}{1 + x^2} u = \frac{1}{x^2 + x^3},$$

and writing

$$\beta = \frac{1}{1+x^2},$$

we find

$$u = x^2 e^{\int \beta dx} \int \frac{e^{-\int \beta dx}}{x^2 + x^3} dx,$$

that is,

$$u = x^2 e^{\int \frac{dx}{1+x^2}} \int \frac{e^{-\int \frac{dx}{1+x^2}}}{x^2 + x^3} dx.$$

But the calculation is most easily performed by means of the foregoing equation of differences, itself obtained from the differential equation written in the foregoing form,

$$(-1 + 2x + 3x^2)M + (x^2 + x^3)M' = -1.$$

671. On a Sibi-Reciprocal Surface.

[From the Berlin. Akad. Monatsber., (1878), pp. 309—313.]

The question of the generation of a sibi-reciprocal surface—that is, a surface the reciprocal of which is of the same order and has the same singularities as the original surface—was considered by me in the year 1868, see *Proc. London Math. Soc.* t. ii. pp. 61—63, [part of 387], where it is remarked that if a surface be considered as the envelope of a quadric surface varying according to given conditions, then the reciprocal surface is given as the envelope of a quadric surface varying according to the reciprocal conditions; whence, if the conditions be sibi-reciprocal, it follows that the surface is a sibi-reciprocal surface. And I gave as instances the surface which is the envelope of a quadric surface touching each of 8 given lines; and also the surface called the “tetrahedroid,” which is a homographic transformation of Fresnel’s Wave Surface and a particular case of the quartic surface with 16 nodes.

The interesting surface of the order 8, recently considered by Herr Kummer, Berl. Monatsber., Jan. 1878, pp. 25—36, is included under the theory. In fact, if we consider a line L , whereof the six coordinates a, b, c, f, g, h satisfy each of the three linear relations

$$\begin{aligned} l_1 a + g_1 b + k_1 c + a_1 f + b_1 g + c_1 h &= 0, \\ l_2 a + g_2 b + k_2 c + a_2 f + b_2 g + c_2 h &= 0, \\ l_3 a + g_3 b + k_3 c + a_3 f + b_3 g + c_3 h &= 0, \end{aligned}$$

the locus of this line is a quadric surface the equation of which is

$$\begin{aligned} T = (agh) x^2 + (bhf) y^2 + (cfg) z^2 + (abc) w^2 \\ + [(abg) - (cah)] xw + [(bfg) + (chf)] yz \\ + [(bch) - (abf)] yw + [(cgh) + (afg)] zx \\ + [(caf) - (bcg)] zw + [(ahf) + (bgh)] xy = 0. \end{aligned}$$

where (agh) is used to denote the determinant $\begin{vmatrix} a_1 & g_1 & h_1 \\ a_2 & g_2 & h_2 \\ a_3 & g_3 & h_3 \end{vmatrix}$, and

so for the other symbols.

Considering the reciprocal of the line L in regard to the quadric surface $X^2 + Y^2 + Z^2 + W^2 = 0$, the six coordinates of the reciprocal line are f, g, h, a, b, c , and it is hence at once seen that the locus of

the reciprocal line is the quadric surface obtained from the equation $T = 0$ by interchanging therein the symbolical quantities a, b, c and f, g, h : viz. writing also $(\xi, \eta, \zeta, \omega)$ in place of (x, y, z, w) , the new equation is

$$\begin{aligned} T' = & (fbc)\xi^2 + (gca)\eta^2 + (hab)\zeta^2 + (fgh)\omega^2 \\ & + [(fgh) - (abc)]\xi\omega + [(fab) + (hbc)]\eta\zeta \\ & + [(ghc) - (fga)]\eta\omega + [(gbc) + (fab)]\zeta\xi \\ & + [(fca) - (ghb)]\zeta\omega + [(hca) + (gbc)]\xi\eta = 0; \end{aligned}$$

or, what is the same thing, this equation $T' = 0$ is the equation of the original quadric surface (the locus of L) expressed in terms of the plane-coordinates ξ, η, ζ, ω .

Now considering each of the quantities $a_1, b_1, c_1, f_1, g_1, h_1, a_2, b_2$, etc., a_3, b_3 , etc., as a given linear function of a variable parameter λ , say

$$a_1 = a'_1 + a''_1\lambda, \quad b_1 = b'_1 + b''_1\lambda, \quad \text{etc.},$$

the equation $T = 0$ takes the form

$$A\lambda^3 + 3B\lambda^2 + 3C\lambda + D = 0,$$

where A, B, C, D are given quadric functions of the coordinates x, y, z, w ; and the envelope of the quadric surface $T = 0$ is Herr Kummer's surface of the eighth order

$$(AD - BC)^2 - 4(AC - B^2)(BD - C^2) = 0.$$

In like manner the equation $T' = 0$ takes the form

$$A'\lambda^3 + 3B'\lambda^2 + 3C'\lambda + D' = 0,$$

where A', B', C', D' are given functions of the coordinates ξ, η, ζ, ω ; and we have

$$(A'D' - B'C')^2 - 4(A'C' - B'^2)(B'D' - C'^2) = 0,$$

as the equation of the reciprocal surface; or (what is the same thing) as that of the original surface, regarding ξ, η, ζ, ω as plane-coordinates.

In regard to the foregoing equation $T = 0$, it is to be noticed that, if $a_1, b_1, c_1, f_1, g_1, h_1, a_2, b_2$, etc., a_3, b_3 , etc., instead of being

arbitrary coefficients, were the coordinates of three given lines L_1, L_2, L_3 respectively; that is, if we had

$$a_1f_1+b_1g_1+c_1h_1=0, \quad a_2f_2+b_2g_2+c_2h_2=0, \quad a_3f_3+b_3g_3+c_3h_3=0,$$

then the three linear relations satisfied by (a, b, c, f, g, h) would express that the line L was a line meeting each of the three given lines L_1, L_2, L_3 : the locus is therefore the quadric surface which passes through these three lines; and I have in my paper "On the six coordinates of a Line," *Camb. Phil. Trans.*, t. xi. (1869), pp. 290—323, [435], found the equation to be the foregoing equation $T = 0$.

But it is easy to see that the same equation subsists in the case where the three equations $a_1f_1+b_1g_1+c_1h_1=0$, etc., are not satisfied. For the several coefficients being perfectly general, any one of the three linear relations may be replaced by a linear combination of these equations that is, in place of $a_1, b_1, c_1, f_1, g_1, h_1$, we may write a', b', c', f', g', h' , where

$$a' = \theta_1a_1 + \theta_2a_2 + \theta_3a_3, \quad b' = \theta_1b_1 + \theta_2b_2 + \theta_3b_3, \quad \text{etc.};$$

and these factors $\theta_1, \theta_2, \theta_3$ may be conceived to be such that the condition in question $a'f' + b'g' + c'h' = 0$ is satisfied. Similarly the second set of coefficients may be replaced by a', b', c', f', g', h' , where

$$a' = \phi_1a_1 + \phi_2a_2 + \phi_3a_3, \quad \text{etc.},$$

and the condition $a'f' + b'g' + c'h' = 0$ is satisfied: and the third set by a', b', c', f', g', h' , where

$$a' = \psi_1a_1 + \psi_2a_2 + \psi_3a_3, \quad \text{etc.},$$

and the condition $a'f' + b'g' + c'h' = 0$ is satisfied.

We have therefore an equation

$$T' = (a'g'h')x^2 + \text{etc.},$$

which only differs from the equation $T = 0$ by having therein the accented letters in place of the unaccented ones: and, substituting for the accented letters their values, the whole divides by the determinant $(\theta\phi\psi)$, and throwing this out we obtain the required equation $T = 0$.

But it is easier to obtain the equation $T = 0$ directly. We have

$$\begin{aligned} hy - gz + aw &= 0, \\ -hx + fz + bw &= 0, \\ gx - fy + cw &= 0, \\ -ax - by - cz &= 0; \end{aligned}$$

viz. in virtue of the equation $af + bg + ch = 0$ which connects the six coordinates, these four equations are equivalent to two independent equations which are the equations of the line (a, b, c, f, g, h) : or, what is the same thing, any three of these equations imply the fourth equation and also the relation $af + bg + ch = 0$.

We might, from the three linear relations and any three of the last-mentioned four equations, eliminate a, b, c, f, g, h and so obtain the required equation $T = 0$; but it is better, introducing the arbitrary coefficients $\alpha, \beta, \gamma, \delta$, to employ all the four equations. The result of the elimination is thus given in the form

$$\begin{vmatrix} \alpha, & w, & -z, & y \\ \beta, & w, & z, & -x \\ \gamma, & -y, & x, & w \\ \delta, & x, & y, & z \end{vmatrix} \begin{vmatrix} a_1, & g_1, & h_1, & a_1, & b_1, & c_1 \\ a_2, & g_2, & h_2, & a_2, & b_2, & c_2 \\ a_3, & g_3, & h_3, & a_3, & b_3, & c_3 \\ f_1, & g_1, & h_1, & a_1, & b_1, & c_1 \\ f_2, & g_2, & h_2, & a_2, & b_2, & c_2 \\ f_3, & g_3, & h_3, & a_3, & b_3, & c_3 \end{vmatrix} = 0,$$

viz. the left-hand side here contains the factor $-(\alpha x + \beta y + \gamma z + \delta w)$; throwing this out, we obtain the required quadric equation $T = 0$.

If for the calculation of T we compare the terms containing δ , we have

$$Tw = \begin{vmatrix} w, & -z, & y \\ w, & z, & -x \\ -y, & x, & w \end{vmatrix} \begin{vmatrix} f_1, & g_1, & h_1 \\ f_2, & g_2, & h_2 \\ f_3, & g_3, & h_3 \\ a_1, & b_1, & c_1 \\ a_2, & b_2, & c_2 \\ a_3, & b_3, & c_3 \end{vmatrix},$$

where observe that, writing $w = 0$, the right-hand side vanishes as containing the factor

$$\begin{vmatrix} -z, & y \\ z, & -x \end{vmatrix} \begin{vmatrix} -y, & x \end{vmatrix}.$$

Hence the right-hand side divides by w ; and one of its terms being evidently $w^2(abc)$, T contains as it should do the term $(abc)w^2$: the remaining terms can be found without any difficulty, and the foregoing expression for T is thus verified.

672. On the Game of Mousetrap.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 8—10.]

In the note "A Problem in Permutations," *Quarterly Mathematical Journal*, t. i. (1857), p. 79, [161], I have spoken of the problem of permutations presented by this game. A set of cards—ace, two, three, &c., say up to thirteen—are arranged (in any order) in a circle with their faces upwards; you begin at any card, and count one, two, three, &c., and if upon counting, suppose the number five, you arrive at the card five, the card is thrown out; and beginning again with the next card, you count one, two, three, &c., throwing out (if the case happen) a new card as before, and so on until you have counted up to thirteen, without coming to a card which has to be thrown out.

The original question proposed was: for any given number of cards to find the arrangement (if any) which would throw out all the cards in a given order; but (instead of this) we may consider all the different arrangements of the cards, and inquire how many of these there are in which all or any given smaller number of the cards will be thrown out; and (in the several cases) in what orders the cards are thrown out.

Thus to take the simple case of four cards, the different arrangements, with the cards thrown out in each, are

1,	2,	3,	4	—	1, 2, 3, 4
1,	2,	4,	3	—	1, 3, 4, 2
1,	3,	2,	4	—	4, 2, 1, 3
1,	3,	4,	2	—	2, 1, 3, 4
1,	4,	2,	3	—	3, 2, 1, 4
1,	4,	3,	2	—	2, 3, 1, 4
2,	1,	3,	4	—	3, 1, 2, 4
2,	1,	4,	3	—	none
2,	3,	1,	4	—	3, 4, 1, 2
2,	3,	4,	1	—	1, 2, 4, 3
2,	4,	1,	3	—	2, 3, 4, 1
2,	4,	3,	1	—	1, 3, 2, 4
3,	1,	2,	4	—	2, 4, 1, 3
3,	1,	4,	2	—	none
3,	2,	1,	4	—	4, 1, 2, 3
3,	2,	4,	1	—	none
3,	4,	1,	2	—	2, 1, 4, 3
3,	4,	2,	1	—	1, 2, 3, 4
4,	1,	2,	3	—	none
4,	1,	3,	2	—	3, 4, 2, 1
4,	2,	1,	3	—	1, 3, 4, 2
4,	2,	3,	1	—	3, 1, 2, 4
4,	3,	1,	2	—	2, 4, 3, 1
4,	3,	2,	1	—	3, 1, 2, 4.

Classifying these so as to show in how many arrangements a given card or permutation of cards is thrown out, we have the table

No.	Thrown out
9	none
4	1
1	3
2	3, 2
2	1, 3
1	3, 4
1	1, 3, 4, 2
1	1, 2, 3, 4
1	4, 2, 1, 3
1	2, 1, 3, 4

viz. there are nine arrangements in which no card is thrown out, four arrangements in which only the card 1 is thrown out, one arrangement in which only the card 3 is thrown out, and so on. It will be observed that there are five arrangements in which all the cards are thrown out, each throwing them out in a different order; there are thus only five orders in which all the cards are thrown out. The general question is of course to form a like table for the numbers 5, 6, $\dots$, or any greater number of cards.

C. X. 33

673. Note on the Theory of Correspondence.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 32, 33.]

If the point P on a given curve U of the order m , and the point Q on a given curve V of the order m' , have a $(1, 1)$ correspondence, this implying that the two curves have the same deficiency; then if PQ intersects the consecutive line $P'Q'$ in a point R , the locus of R is a curve W of the class $m + m'$, and the point R on this curve has, in general (but not universally), a $(1, 1)$ correspondence with the point P on U or with the point Q on V .

For, considering the correspondence of the points P and R , to a given position of P there corresponds, it is clear, a single position of R ; on the other hand, starting from R , the tangent at this point to the curve W meets the curve U in m points and the curve V in m' points, but it is in general only one of the m points and only one of the m' points which are corresponding points on the curves U and V ; that is, it is only one of the m points which is a point P ; and the correspondence of (P, R) is thus a $(1, 1)$ correspondence.

But the curves U, V may be such that the correspondence of (P, R) is not a $(1, 1)$ but a $(k, 1)$ correspondence; viz., that to a given position of P there corresponds a single position of R , but to a given position of R , k positions of P .

To show that this is so, imagine through P a line Π having therewith a $(k, 1)$ correspondence; P being, as above, a point on the curve U , the line in question envelopes a curve W ; and the correspondence is such that, for any given position of P on the curve U , we have through it a single position of the line: but, for a given tangent of the curve W , we have upon it k positions of the point P , viz. k of the m intersections of the line with the curve U are points corresponding to the line; this, of course, implies that the curve U is not any curve whatever of the order m , but a curve of a peculiar nature.

Imagine now that we have on the line Π a point Q , having with P a $(1, 1)$ correspondence of a given nature: to fix the ideas, suppose P, Q are harmonics in regard to a given conic: since on each of the lines Π there are k positions of P , there are also on the line k positions of Q , and the locus of these k points Q is a curve V , say of the order m' .

The point P on the curve U and the point Q on the curve V have a $(1, 1)$ correspondence. For, consider P as given: there is a single

position of the line Π intersecting V in m' points, but obviously only one of these is the point Q . And consider Q as given: then through Q we have say μ tangents of the curve W ; each of these tangents intersects the curve U in m points, k of which are points P , but for a tangent taken at random no one of these is the correspondent of Q ; it is, in general, only one of the μ tangents which has upon it k points P , one of them being the point corresponding to Q ; that is, to a given position of Q there corresponds a single position of P ; and the correspondence of the points (P, Q) is thus a $(1, 1)$ correspondence.

We have thus the point P on the curve U and the point Q on the curve V , which points have with each other a $(1, 1)$ correspondence; and the line Π is the line PQ joining these points; this intersects the consecutive line in a point R ; and the locus of R is the curve W .

To a given position of P there corresponds a single line Π , and therefore a single position of R ; but to a given position of R there correspond k positions of P , viz. drawing at R the tangent to the curve W , this is a line Π having upon it k points P , or the correspondence of (P, Q) is, as stated, a $(k, 1)$ correspondence.

The foregoing considerations were suggested to me by the theory of parallel curves. Take a curve parallel to a given curve, for example, the ellipse; this is a curve of the order 8, such that every normal thereto is a normal at two distinct points; and the curve has as its evolute the evolute of the ellipse, or, more accurately, the evolute of the ellipse taken twice; but, attending only to the evolute taken once, each tangent of the evolute is a normal of the parallel curve at two distinct points thereof, and the points of the parallel curve have with those of the evolute not a $(1, 1)$ but a $(2, 1)$ correspondence. 33—2

674. Note on the Construction of Cartesians.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), p. 34.]

If $p = a + b \cos \theta$, and $r = \frac{1}{2}\{p \pm \sqrt{(p^2 - c^2)}\}$, then obviously

$$r^2 - rp + \frac{1}{4}c^2 = 0,$$

that is,

$$r^2 - r(a + b \cos \theta) + \frac{1}{4}c^2 = 0,$$

which is the equation of a Cartesian.

Here $p = a + b \cos \theta$ is the equation of a limaçon or nodal Cartesian, having the origin for the node; and for any given value of θ , deducing from the radius vector of the limaçon the new radius vector r by the above formula $r = \frac{1}{2}\{p \pm \sqrt{(p^2 - c^2)}\}$, we obtain a Cartesian, or by giving different values to c , a series of Cartesians having the origin for a common focus.

The construction is a very convenient one.

675. On the Fleflecnodal Planes of a Surface.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 49—51.]

If at a node (or double point) of a plane curve there is on one of the branches an inflexion, (that is, if the tangent has a 3-pointic intersection with the branch), the node is said to be a *flecnode*; and if there is on each of the branches an inflexion, then the node is said to be a *fleflecnode*.

The tangent plane of a surface intersects the surface in a plane curve having at the point of contact a node; if this is a flecnode or a fleflecnode, the tangent plane is said to be a flecnodal or a fleflecnodal plane accordingly. For a quadric surface each tangent plane is fleflecnodal: this is obvious geometrically (since the section is a pair of lines), and it will presently appear that the analytical condition for such a plane is satisfied.

In fact, if the origin be taken at a point of a surface, so that $z = 0$ shall be the equation of the tangent plane, then in the neighbourhood of the point we have

$$z = (x, y)^2 + (x, y)^3 + \&c.;$$

and the condition for a fleflecnodal plane is that the term $(x, y)^2$ shall be a factor of the succeeding term $(x, y)^3$.

Now for a quadric surface the equation is

$$z = \frac{1}{2}\{ax^2 + 2hxy + by^2 + 2(fy + gx)z + cz^2\},$$

that is,

$$z(1 - fy - gz - hz) = \frac{1}{2}(ax^2 + 2hxy + by^2),$$

or developing as far as the third order in (x, y) , we have

$$z = \frac{1}{2}(ax^2 + 2hxy + by^2)(1 + fy + gx),$$

so that the condition in question is satisfied.

In what follows, I take for greater simplicity $h = 0$, (viz. $x = 0$, $y = 0$ are here the tangents to the two curves of curvature at the point in question), and to avoid fractions write $2f$, $2g$ in place of f , g respectively; the developed equation of the quadric surface is thus

$$z = \frac{1}{2}(ax^2 + by^2) + (ax^2 + by^2)(gx + fy).$$

I consider the parallel surface, obtained by measuring off on the normal a constant length k . If, as usual, p , q denote $\frac{dz}{dx}$ and $\frac{dz}{dy}$ respectively, then, in general, (X, Y, Z) being the coordinates of the point on the parallel surface,

$$Z = z + \frac{kp}{\sqrt{(1+p^2+q^2)}}, \quad X = x - \frac{kp}{\sqrt{(1+p^2+q^2)}}, \quad Y = y - \frac{kq}{\sqrt{(1+p^2+q^2)}}.$$

But in the present case

$$\begin{aligned} p &= ax + 3agx^2 + 2afxy + bgy^2, \\ q &= by + afx^2 + 2bgxy + 3bfy^2, \end{aligned}$$

whence

$$\begin{aligned} X &= x - k(ax + 3agx^2 + 2afxy + bgy^2), \\ Y &= y - k(by + afx^2 + 2bgxy + 3bfy^2); \end{aligned}$$

or, putting for convenience

$$X = (1 - ka)\xi, \quad Y = (1 - kb)\eta,$$

then, for a first approximation $x = \xi$, $y = \eta$; whence, writing

$$P = 3ag\xi^2 + 2af\xi\eta + bg\eta^2, \quad Q = af\xi^2 + 2bg\xi\eta + 3bf\eta^2,$$

we find

$$x = \xi + \frac{kP}{1 - ka}, \quad y = \eta + \frac{kQ}{1 - kb},$$

and thence

$$\begin{aligned} Z &= \frac{1}{2}(ax^2 + by^2) + (ax^2 + by^2)(gx + fy) \\ &= \frac{1}{2}\{a(1 - ka)\xi^2 + b(1 - kb)\eta^2\} + (a\xi^2 + b\eta^2)(g\xi + f\eta). \end{aligned}$$

Hence, finally,

$$Z - k = \frac{1}{2}\{a(1 - ka)\xi^2 + b(1 - kb)\eta^2\} + (a\xi^2 + b\eta^2)(g\xi + f\eta).$$

where, changing the origin to the point $x = 0$, $y = 0$, $z = k$ on the parallel surface, the coordinates of the consecutive point are $Z - k$, $X_1 = (1 - ka)\xi$, and $Y_1 = (1 - kb)\eta$.

We cannot, by any determination of the value of k , make the plane $Z - k = 0$ a flecnodal plane of the parallel surface; but if

$$\frac{a(1 - ka)}{a} = \frac{b(1 - kb)}{b}, \quad \text{that is,} \quad ka = kb,$$

then

$$\frac{a(1 - ka)}{a\xi^2 + b\eta^2} = \frac{b(1 - kb)}{a\xi^2 + b\eta^2} = 1 - ka = 1 - kb,$$

and the equation becomes

$$Z - k = \frac{1}{2}(1 - ka)(a\xi^2 + b\eta^2) + (a\xi^2 + b\eta^2)(g\xi + f\eta),$$

viz. the term of the second order has here a factor $g\xi + f\eta$ which divides the term of the third order, and the plane $Z - k = 0$ is a flecnodal plane of the parallel surface.

676. Note on a Theorem in Determinants.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 55—57.]

It is well known that if 12, &c., denote the determinants formed with the matrix

$$\begin{array}{cccc} a, & \beta, & \gamma, & \delta \\ a', & \beta', & \gamma', & \delta' \end{array}$$

then, identically,

$$12.34 + 13.42 + 14.23 = 0.$$

The proper proof of the theorem is obtained by remarking that we have

$$\begin{vmatrix} \cdot & \cdot & \alpha & \beta & \gamma & \delta \\ \cdot & \cdot & \alpha' & \beta' & \gamma' & \delta' \\ \cdot & \cdot & \alpha & \beta & \gamma & \delta \\ \cdot & \cdot & \alpha' & \beta' & \gamma' & \delta' \\ \alpha & \beta & \gamma & \delta & \cdot & \cdot \\ \alpha' & \beta' & \gamma' & \delta' & \cdot & \cdot \end{vmatrix} = 0,$$

as at once appears by subtracting the first and second lines from the third and fourth lines respectively; and, this being so, the development of the determinant gives the theorem.

The theorem might, it is clear, have been obtained in four different forms according as in the determinant the missing terms were taken to be as above (δ, δ') , or to be (α, α') , (β, β') , or (γ, γ') ; but the four results are equivalent to each other.

There is obviously a like theorem for the sums of products of determinants formed with the matrix

$$\begin{array}{cccccc} \alpha, & \beta, & \gamma, & \delta, & \epsilon, & \zeta \\ \alpha', & \beta', & \gamma', & \delta', & \epsilon', & \zeta' \\ \alpha'', & \beta'', & \gamma'', & \delta'', & \epsilon'', & \zeta'' \end{array}$$

viz. the theorem is obtained by development of the determinant in

an identical equation, such as

$$\begin{vmatrix} \cdot & \cdot & \cdot & \alpha & \beta & \gamma & \delta & \epsilon & \zeta \\ \cdot & \cdot & \cdot & \alpha' & \beta' & \gamma' & \delta' & \epsilon' & \zeta' \\ \cdot & \cdot & \cdot & \alpha'' & \beta'' & \gamma'' & \delta'' & \epsilon'' & \zeta'' \\ \cdot & \cdot & \cdot & \alpha & \beta & \gamma & \delta & \epsilon & \zeta \\ \cdot & \cdot & \cdot & \alpha' & \beta' & \gamma' & \delta' & \epsilon' & \zeta' \\ \cdot & \cdot & \cdot & \alpha'' & \beta'' & \gamma'' & \delta'' & \epsilon'' & \zeta'' \\ \alpha & \beta & \gamma & \delta & \epsilon & \zeta & \cdot & \cdot & \cdot \\ \alpha' & \beta' & \gamma' & \delta' & \epsilon' & \zeta' & \cdot & \cdot & \cdot \\ \alpha'' & \beta'' & \gamma'' & \delta'' & \epsilon'' & \zeta'' & \cdot & \cdot & \cdot \end{vmatrix} = 0,$$

but we thus obtain 15 results which are not all equivalent.

If, for shortness, we write

$$\begin{aligned} A &= 123 \cdot 456, & -B &= 124 \cdot 356, & -C &= 125 \cdot 346, \\ -D &= 126 \cdot 345, & -E &= 134 \cdot 256, & -F &= 135 \cdot 246, \\ -G &= 136 \cdot 245, & -H &= 145 \cdot 236, & -I &= 146 \cdot 235, \\ -J &= 156 \cdot 234, \end{aligned}$$

then the fifteen results are

$$\begin{aligned} A + B - C - D &= 0, & A + B - E - F &= 0, \\ A - C + F - I &= 0, & A - D + G - H &= 0, \\ A - E + F + G &= 0, & A - H - I - J &= 0, \\ B - C - G + H &= 0, & B - D - F + I &= 0, \\ B - E + H + I &= 0, & B - F - G - J &= 0, \\ C + D - E - J &= 0, & C - E + G + I &= 0, \\ C - F - H - J &= 0, & D - E + F + H &= 0, \\ D - G - I - J &= 0, \end{aligned}$$

which are all satisfied if only

$$\begin{aligned} A &= & + H + I + J, \\ B &= & F + G & + J, \\ C &= & F & + H & + J, \\ D &= & G & + I + J, \\ E &= & F + G + H + I & ; \end{aligned}$$

and we thus have these five relations between the ten products of determinants $A, B, C, D, E, F, G, H, I, J$. C. X. 34

677. Addition to Mr Glaisher's Paper "Proof of Stirling's Theorem."

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 63, 64.]

It is easy to extend Mr Glaisher's investigation so as to obtain from it the more approximate value

$$n! = \sqrt{(2\pi)} \cdot n^{n+\frac{1}{2}} e^{-n+\frac{1}{12n}}.$$

We, in fact, have

$$\Gamma(n+1) = n! = n^{n+\frac{1}{2}} e^{-n+\frac{a}{n}+\frac{b}{n^3}+\dots},$$

where $a, b, \dots$ are given functions of n , viz.

$$a = \frac{1}{2} \left\{ \frac{1}{1 \cdot 3} + \frac{1}{5^2} + \dots - \frac{1}{4}(2n+1) \right\}, \quad \&c.$$

And hence writing $x = 1$, we have

$$\xi(1) = \xi \cdot 2^{2n} \cdot n^n \left(\frac{1}{2} \right)^{2n+2} = e^{-a-b-\dots},$$

that is,

$$\begin{aligned} n! &= \left(\frac{2n+2}{2} \right)^{n+\frac{1}{2}} e^{-n-\frac{1}{2}(a+b+\dots)} \\ &= (n+1)^{n+\frac{1}{2}} e^{-n-\frac{1}{2}(a+b+\dots)} \\ &= n^{n+\frac{1}{2}} \left(1 + \frac{1}{n} \right)^{n+\frac{1}{2}} e^{-n-\frac{1}{2}(a+b+\dots)}. \end{aligned}$$

Writing

$$\left(1 + \frac{1}{n} \right)^{n+\frac{1}{2}} = e^{\theta},$$

the whole exponent of e is

$$(n + \frac{1}{2}) \log(1 + \frac{1}{n}) - n - \frac{1}{2}(a + b + \dots)$$

We have

$$\frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{n} - \log n = \gamma + \frac{A_1}{n} + \frac{A_2}{n^2} + \dots,$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{n^2} = \frac{\pi^2}{6} - \frac{1}{n} + \dots,$$

(the constant is in fact $\frac{1}{2}\pi^2$, but the value is not required), hence

$$a = \text{const.} - \frac{1}{12n} + \text{terms in } \frac{1}{n^2}, \frac{1}{n^3}, \&c.;$$

as regards b , c , &c., there are no terms in $\frac{1}{n}$, but we have

$$b = \text{const.} + \text{terms in } \frac{1}{n^2}, \&c., \quad c = \text{const.} + \text{terms in } \frac{1}{n^2}, \&c.$$

Hence the whole exponent of e is

$$= -\frac{1}{12n} + C + \text{terms in } \frac{1}{n^2}, \&c.$$

As in Mr Glaisher's investigation, it is shown that $e^C = \sqrt{(2\pi)}$, and hence neglecting the terms in $\frac{1}{n^2}$, &c., the final result is

$$n! = \sqrt{(2\pi)} \cdot n^{n+\frac{1}{2}} e^{-n+\frac{1}{12n}}.$$

678. On a System of Quadric Surfaces.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 124, 125.]

The following theorem was communicated to me by Dr Klein; “given in regard to a quadric surface two sibi-reciprocal line-pairs, the two tractors (or lines meeting each of the four lines) form a sibi-reciprocal line-pair.”

This may be presented under a more general form as a theorem relating to the tractors of any two line-pairs. In fact, if a given line-pair is taken to be sibi-reciprocal in regard to a quadric surface, we thereby establish only a four-fold relation between the coefficients of the surface, and the surface will still depend on five arbitrary parameters. Whence if two given line-pairs are taken to be each of them sibi-reciprocal in regard to one and the same quadric surface, we thereby establish only an eight-fold relation and the surface will still depend upon one arbitrary parameter.

The theorem thus is: given any two line-pairs, then each of these, and also the pair of tractors, are sibi-reciprocal in regard to a singly infinite system of quadric surfaces.

The question arises, what is this system of quadric surfaces? It is, in fact, the system of surfaces having in common a skew quadrilateral constructed as follows: starting from the two given line-pairs, construct the two tractors, each of them intersected by the given line-pairs in two point-pairs; and on each tractor construct the double or sibi-reciprocal points of the involution thus determined; these double points are the vertices (those on the same tractor being opposite vertices) of the skew quadrilateral; which is consequently at once obtained by joining the two double points on the one tractor with the two double points on the other tractor.

The construction is an immediate consequence of the following theorem: consider a skew quadrilateral, and drawing its two diagonals, take a pair of lines cutting each diagonal harmonically; these will be sibi-reciprocal in regard to any quadric surface whatever passing through the skew quadrilateral; and conversely, any quadric surface in regard to which they are sibi-reciprocal passes through the skew quadrilateral. Hence the pair of tractors, cutting as they do each diagonal harmonically, are sibi-reciprocal in regard to any quadric surface through the skew quadrilateral; that is, in regard to any quadric surface in regard to which the two given line-pairs are sibi-reciprocal.

679. On the Regular Solids.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 127—131.]

In a regular solid, or say in the spherical figure obtained by projecting such solid, by lines from the centre, on the surface of a concentric sphere, we naturally consider 1° the summits, 2° the centres of the faces, 3° the mid-points of the sides. But, imagining the five regular figures drawn in proper relation to each other on the same spherical surface, the only points which have thus to be considered are 12 points A, 20 points B, 30 points Θ , and 60 points Φ .

These may be, in the first instance, described by reference to the dodecahedron; viz. the points A are the centres of the faces, the points B are the summits, the points Θ are the mid-points of the sides, and the points Φ are the mid-points of the diagonals of the faces (viz. there are thus 5 points Φ in each face of the dodecahedron, or in all 60 points Φ).

But reciprocally we may describe them in reference to the icosahedron: viz. the points A are the summits, the points B the centres of the faces, the points Θ the mid-points of the sides, (viz. each point Θ is the common mid-point of a side of the dodecahedron and a side of the icosahedron, which sides there intersect at right angles), and the points Φ are points lying by 3's on the faces of the icosahedron, each point Φ of the face being given as the intersection of a perpendicular A Θ of the face by a line BB' joining the centres of two adjacent faces and intersecting A Θ at right angles.

The points A lie opposite to each other in pairs in such wise that, taking any two opposite points as poles, the relative situation is as follows:

Longitudes.	
0°,	72°, 144°, 216°, 288°
36°,	108°, 180°, 252°, 324°

where the points A in the same horizontal line form a zone of points equidistant from the point taken as the North Pole. And the points B lie also opposite to each other in such wise that, taking two opposite points as poles, the relative situation is as follows:

		Longitudes.
1	3	$0^\circ, 120^\circ, 240^\circ$
	6	$(0^\circ, 120^\circ, 240^\circ) + 22^\circ 14'$
	6	$(60^\circ, 180^\circ, 300^\circ) + 22^\circ 14'$
	3	$60^\circ, 180^\circ, 300^\circ$

where the points B in the same horizontal line form a zone of points equidistant from the point taken as the North Pole.

Neglecting the 3+3 points B which lie adjacent to the poles, the remaining 14 points B may be arranged as follows ($\beta = 22^\circ 14'$ as above):

	1	6	6	1	
A	$120^\circ + \beta,$	$60^\circ + \beta,$	$180^\circ + \beta,$	$240^\circ + \beta$	$300^\circ + 60^\circ$
-A	$120^\circ - \beta,$	$240^\circ - \beta,$	$-\beta,$	$180^\circ - \beta,$	$300^\circ - \beta.$

And taking the two poles separately with each system of the remaining poles, we have 2 systems each of 8 points B, which are, in fact, the summits of a cube (hexahedron); each point B taken as North Pole thus belongs to two cubes; but inasmuch as the cube has 8 summits, the number of the cubes thus obtained is $20 \times 2 \div 8 = 5$; viz. the 20 points B form the summits of 5 cubes, each point B of course belonging to 2 cubes. It is to be added that, considering the 5 points B which form a face of the dodecahedron, any diagonal BB of this dodecahedron is a side of a cube. We have thus $12 \times 5 = 60$, the number of the sides of the 5 cubes.

It is at once seen that the centres of the faces of a cube are points Θ , and that the mid-points of the sides of the cube are points Φ . To each cube there corresponds of course an octahedron, the summits being points Θ , the centres of the faces points B, and the mid-points of the sides points Φ ; thus, for the five octahedra the summits are the $5 \times 6 = 30$ points Θ ; the centres of the faces are $5 \times 8 = 40$ points B (each point B being thus a centre of face for two octahedra), and the mid-points of the sides being the $5 \times 12 = 60$ points Φ .

Finally, considering the 8 points B which belong to a cube, we can, in four different ways, select thereout 4 points B which are the summits of a tetrahedron; the remaining 4 points B are then the centres of the faces, and the mid-points of the sides are points Θ : there are thus $5 \times 4 = 20$ tetrahedra having 20×4 summits which are the 20 points B each 4 times; 20×4 centres of faces which are

the 20 points B each 4 times; and 20×6 mid-points of sides which are the 30 points Θ each 4 times.

It thus appears that, as mentioned above, the five regular figures depend only on the points A, B, Θ , and Φ .

We might take as poles two opposite points A, B, Θ , or Φ ; and in each case determine in reference to these the positions of the other points; but for brevity I consider only the case in which we take as poles two opposite points A. We have the following table:

Poles two opposite points A.

	N.P.D.		Longitudes.
$5\Phi_1$	0°	$63^\circ 26'$	$116^\circ 34'$ 180°
$5A$		$0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$	
$5A'$		$36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$	
$5B_1$	$37^\circ 22'$	$79^\circ 12'$	$100^\circ 48'$ $142^\circ 38'$
		$36^\circ, 108^\circ, 324^\circ$	
$5B_2$		$36^\circ, 108^\circ, 324^\circ$	
$5B_3$		$0^\circ, 72^\circ, 288^\circ$	
$5B_4$		$0^\circ, 72^\circ, 288^\circ$	
$5\Theta_1$	$31^\circ 43'$	58°	$77'$ 90°
		$121^\circ 43'$	$148^\circ 17'$
		$0^\circ, 72^\circ, 288^\circ$	
$5\Theta_2$		$36^\circ, 108^\circ, 324^\circ$	
$10\Theta_3$		$(0^\circ, 72^\circ, 288^\circ) + 18^\circ$	
		$0^\circ, 72^\circ, 288^\circ$	
$5\Theta_4$		$36^\circ, 108^\circ, 324^\circ$	
$5\Phi_2$	$13^\circ 16'$	$52^\circ 52'$	$68^\circ 10'$ $76^\circ 42'$
		$103^\circ 18'$	$111^\circ 50'$ $127^\circ 8'$
		$166^\circ 44'$	
		$36^\circ, 108^\circ, 324^\circ$	
$10\Phi_3$		$(0^\circ, 72^\circ, 288^\circ) + 9^\circ 44'$	
$10\Phi_4$		$(0^\circ, 72^\circ, 288^\circ) + 13^\circ 35'$	
$5\Phi_5$		$0^\circ, 72^\circ, 288^\circ$	
$5\Phi_6$		$36^\circ, 108^\circ, 324^\circ$	
$10\Phi_7$		$(36^\circ, 108^\circ, 324^\circ) + 13^\circ 35'$	
$10\Phi_8$		$(36^\circ, 108^\circ, 324^\circ) + 9^\circ 44'$	
$5\Phi_9$		$0^\circ, 72^\circ, 288^\circ$	

I add for greater completeness the following results, some of which

were used in the calculation of the foregoing table. Considering successively (1) the tetrahedral triangle, summits 3 points B, centre a point B; (2) the hexahedral square, summits 4 points B, centre a point Θ ; (3) the octahedral triangle, summits 3 points Θ , centre a point B; (4) the icosahedral triangle, summits 3 points A, centre a point B; (5) the dodecahedral pentagon, summits 5 points, centre a point B; and (6), what may be called the small pentagon, summits 5 points Φ lying within a dodecahedral pentagon, and having therewith the common centre B; we may in each case write s the side, r the radius or distance of the centre from a summit, p the perpendicular or distance of the centre from a side. And the values then are

	Tet. $\triangle$	Hex. sq.	Oct. $\triangle$	Icos. $\triangle$	Dod. pent.	Small pent.
s	$109^\circ 30'$	$70^\circ 30'$	90°	$63^\circ 26'$	$41^\circ 50'$	$15^\circ 30'$
r	$70^\circ 30'$	$54^\circ 45'$	$54^\circ 45'$	$37^\circ 22'$	$37^\circ 22'$	$13^\circ 16'$
p	$54^\circ 45'$	45°	$35^\circ 15'$	$20^\circ 55'$	$31^\circ 43'$	$10^\circ 48'$

C. X. 35

680. On the Hessian of a Quartic Surface.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 141—144.]

The surface considered is

$$U = k^2 w^2 (x^2 + y^2 + z^2) - (x^2 + y^2 + z^2)^2 = 0,$$

or say

$$U = k^2 w^2 P - Q^2 = 0,$$

viz. this may be considered as the central inverse of the ellipsoid

$$\frac{x^2}{k^2} + \frac{y^2}{k^2} + \frac{z^2}{k^2} + w^2 = 0.$$

The values of the second derived functions, or terms of the Hessian determinant

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix}$$

are

$$\begin{aligned} a &= k^2 w^2 - 2Q - 4x^2, & h &= -4xy, & g &= -4xz, & l &= 2k^2 wx, \\ b &= k^2 w^2 - 2Q - 4y^2, & f &= -4yz, & m &= 2k^2 wy, \\ c &= k^2 w^2 - 2Q - 4z^2, & n &= 2k^2 wz, & d &= 2k^2 Q, \end{aligned}$$

and we thence have, whence, forming the analogous quantities $ca - g^2$, &c., it is easy to obtain

$$abc - af^2 - bg^2 - ch^2 + 2fgh = 4k^2 w^2 P - 24Q^2,$$

which is to be multiplied by $d = 2k^2 P$.

And

$$\begin{aligned} & - [l^2(bc - f^2) + m^2(ca - g^2) + n^2(ab - h^2) \\ & \quad + 2mn(gh - af) + 2nl(hf - bg) + 2lm(fg - ch)] \\ & \quad = -4k^4 w^2 P + 4k^4 w^2 Q(x^2 + y^2 + z^2), \end{aligned}$$

which is

$$4k^4 w^2 P(-Q + Q) + 32P^2 Q.$$

35—2

Hence, uniting the two parts, we have

$$\mathfrak{H} = 8k^4w^2P^2 + 24P^2(3k^2w^2P - 24PQ^2).$$

Writing herein $Q^2 - k^2w^2P = -U$, and transposing all the terms which contain U , we have

$$\mathfrak{H} = 8k^4w^2P^2 + 24P^2\{3(-U) + k^2w^2P\},$$

where, in the term in $\{ \}$, the last four lines are

Hence, writing for shortness

$$F = 3k^2w^2P - 8Q^2,$$

we have

$$\mathfrak{H} = 24P^2 \cdot F + 8k^4w^2P^2 - 72P^2U.$$

Hence, recollecting that $U = k^2w^2P - Q^2$, the Hessian curve of the order 32 breaks up into

$$U = 0, \quad w^8 = 0, \text{ that is, } Q^4 = 0, \quad w^8 = 0,$$

$$\text{or the nodal conic, } w = 0, \quad Q = 0, \quad 8 \text{ times (order 16),}$$

$$U = 0, \quad F = 0, \text{ that is, } Q^2 = 0, \quad P = 0,$$

$$\text{or the quadriquadric, } P = 0, \quad Q = 0, \quad 2 \text{ times (order 8),}$$

and into a curve (order 8) which is

$$4k^4w^2P^2 + 24P^2 \cdot F = 0;$$

viz. this, the intersection of the surface with a quadric surface, is the proper Hessian curve.

681. On the Derivatives of Three Binary Quantics.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 157—168.]

For a reason which will appear, instead of the ordinary factorial notation, I write $\{a012\}$ to denote the factorial $a \cdot (a+1) \cdot (a+2)$, and so in other cases; and I consider the series of equations

$$\begin{aligned}(1) &= X, \\(2) &= (\{a0\}, \{\beta0\} \quad Y, -Y'), \\(3) &= (\{a01\}, 2\{a1\}\{\beta1\}, \{\beta01\} \quad Z, -Z', Z''), \\(4) &= (\{a012\}, 3\{a12\}\{\beta2\}, 3\{a2\}\{\beta12\}, \{\beta012\} \quad W, -W', W'', -W'''), \text{ \&c.}\end{aligned}$$

where

$$X = Y + Y', \quad Y = Z + Z', \quad Y' = Z' + Z'', \quad Z = W + W', \quad Z' = W' + W'', \quad Z'' = W'' + W''',$$

We have thus a series of linear equations serving to determine X ; Y , Y' ; Z , Z' , Z'' ; W , W' , W'' , W''' ; &c.

We require in particular the values of X ; Y , Y' ; Z , Z'' ; W , W''' ; &c., and I write down the results as follow:

$$\begin{aligned}X &= (1), \\ \{a+\beta\}Y &= \{a0\} \cdot (+1), \\ \{ \quad \}Y' &= \{a0\} \cdot (-1); \\ \{a+\beta2\}(1), \quad \{a+\beta1\}(2), \quad \{a+\beta0\}(3) \\ \{a+\beta012\}Z &= \{a01\}, \quad +2\{\beta1\}, \quad +1, \\ \{ \quad \}Z' &= \{a0\}\{\beta0\}, \quad a-\beta, \quad -1, \\ \{ \quad \}Z'' &= \{a01\}, \quad -2\{a1\}, \quad +1; \\ \{a+\beta34\}(1), \quad \{a+\beta14\}(2), \quad \{a+\beta03\}(3), \quad \{a+\beta01\}(4) \\ \{a+\beta01\dots4\}W &= \{\beta012\}, \quad +3\{\beta12\}, \quad +3\{\beta2\}, \quad +1, \\ \{ \quad \}W' &= \{a0\}\{\beta01\}, \quad 2a-\beta, \quad a-2\beta-2, \quad -1, \\ \{ \quad \}W'' &= \{a01\}\{\beta0\}, \quad a-2\beta, \quad -2a+\beta-2, \quad +1, \\ \{ \quad \}W''' &= \{a012\}, \quad -3\{a12\}, \quad +3\{a2\}, \quad -1;\end{aligned}$$

&c.

or, what is the same thing,

$$\begin{aligned} X + X' &= \{a + \beta 1\} \cdot \{\beta 0\}, \\ X' + X'' &= \{ \quad \} \{a 0\}. \end{aligned}$$

And in like manner we have

$$\begin{aligned} \mu + \mu' &= \{a + \beta 2\} \cdot 1, \\ \mu' + \mu'' &= \{ \quad \} \cdot -1, \\ (\{a 0 1\}, -2\{a 1\}\{\beta 1\}, \{\beta 0 1\} \quad \mu, \mu', \mu'') &= 0; \end{aligned}$$

and

$$\begin{aligned} \nu + \nu' &= 0, \\ (\{a 0 1\}, -2\{a 1\}\{\beta 1\}, \{\beta 0 1\} \quad \nu, \nu', \nu'') &= 0. \end{aligned}$$

We hence find without difficulty

$$\begin{aligned} \lambda, \mu, \nu &= \beta \cdot \beta + 1, \quad 2\beta + 1, \quad +1, \\ \lambda', \mu', \nu' &= a \cdot \beta, \quad a - \beta, \quad -1, \\ \lambda'', \mu'', \nu'' &= a \cdot a + 1, \quad -2 \cdot a + 1, \quad +1; \\ &= \{\beta 0 1\}, \quad 2\{\beta 1\}, \quad +1, \\ &= \{a 0\}\{\beta 0\}, \quad a - \beta, \quad -1, \\ &= \{a 0 1\}, \quad -2\{a 1\}, \quad +1; \end{aligned}$$

viz. for verification of the X -equations we have

$$\begin{aligned} &\beta \cdot \{\beta + 1\} \cdot +a \cdot \beta, \text{ that is, } \{a + \beta + 1\} \cdot \beta, = \{a - \\ &\quad a \cdot \beta \cdot +a \cdot \{a + 1\}, \text{ " } \{a + 1 + \beta\} \cdot a, = \{ \quad \} \\ &\quad (a \cdot \{a + 1\}, -2 \cdot \{a + 1\} \cdot \{\beta + 1\}, \quad \beta \cdot \{\beta - \\ &\quad \quad a \cdot \beta, \quad a \cdot \{a + 1\}) = 0, \\ &a \cdot \{a + 1\} \cdot \beta \cdot \{\beta + 1\} - 2 \cdot \{a + 1\} \cdot \{\beta + 1\} \cdot a \cdot \beta + \beta \cdot \{\beta + 1\} \cdot a \cdot \{a + 1\} = 0; \end{aligned}$$

and similarly the μ - and ν -equations may be verified.

We have thus for the Z 's the equations

$$\begin{aligned} \{a + \beta 2\}(1), \quad \{a + \beta 1\}(2), \quad \{a + \beta 0\}(3) \\ \{a + \beta 0 1 2\}Z &= \{a 0 1\}, \quad +2\{\beta 1\}, \quad +1, \\ \{ \quad \}Z' &= \{a 0\}\{\beta 0\}, \quad a - \beta, \quad -1, \\ \{ \quad \}Z'' &= \{a 0 1\}, \quad -2\{a 1\}, \quad +1, \end{aligned}$$

which include the foregoing expressions for Z and Z'' .

We may then take the expressions for the W 's to be

$$\begin{aligned} \{a + \beta 34\}(1), \quad \{a + \beta 14\}(2), \quad \{a + \beta 03\}(3), \quad \{a + \beta 01\}(4) \\ \{a + \beta 0123\}W = \{\beta 012\}, \quad +3\{\beta 12\}, \quad +3\{\beta 2\}, \\ \{ \quad \}W' = \lambda', \quad \mu', \quad \nu', \quad -1, \\ \{ \quad \}W'' = \lambda'', \quad \mu'', \quad \nu'', \quad +1, \\ \{ \quad \}W''' = \{a012\}, \quad -3\{a12\}, \quad +3\{a2\}, \end{aligned}$$

and that is,

$$\begin{aligned} \{ \quad \}W &= \lambda, \quad \mu, \quad \nu, \quad +1, \\ \{ \quad \}W' &= \lambda', \quad \mu', \quad \nu', \quad -1, \\ \{ \quad \}W'' &= \lambda'', \quad \mu'', \quad \nu'', \quad +1, \\ \{ \quad \}W''' &= \lambda''', \quad \mu''', \quad \nu''', \quad -1; \end{aligned}$$

and we obtain in like manner the equations

$$\begin{aligned} X + X' &= \{a + \beta 234\} \cdot \{\beta 01\}, \\ X' + X'' &= \{ \quad \} \cdot \{a0\} \cdot \{\beta 0\}, \\ X'' + X''' &= \{ \quad \} \cdot \{a01\}, \end{aligned}$$

$$(\{a012\}, -3\{a12\}\{\beta 2\}, +3\{a2\}\{\beta 12\}, -\{\beta 012\} \quad X, X', X'', X''') = 0;$$

$$\begin{aligned} \mu + \mu' &= \{a + \beta 134\} \cdot 2\{\beta 1\}, \\ \mu' + \mu'' &= \{ \quad \} \cdot a - \beta, \\ \mu'' + \mu''' &= \{ \quad \} \cdot -2\{a1\}, \end{aligned}$$

$$(\{a012\}, -3\{a12\}\{\beta 2\}, +3\{a2\}\{\beta 12\}, -\{\beta 012\} \quad \mu, \mu', \mu'', \mu''') = 0;$$

$$\begin{aligned} \nu + \nu' &= \{a + \beta 034\} \cdot 1, \\ \nu' + \nu'' &= \{ \quad \} \cdot -1, \\ \nu'' + \nu''' &= \{ \quad \} \cdot 1, \end{aligned}$$

$$(\{a012\}, -3\{a12\}\{\beta 2\}, +3\{a2\}\{\beta 12\}, -\{\beta 012\} \quad \nu, \nu', \nu'', \nu''') = 0;$$

$$\begin{aligned} \rho + \rho' &= 0, \\ \rho' + \rho'' &= 0, \\ \rho'' + \rho''' &= 0, \end{aligned}$$

$$(\{a012\}, -3\{a12\}\{\beta2\}, +3\{a2\}\{\beta12\}, -\{\beta012\} \quad \rho, \rho', \rho'', \rho''') = \{a+\beta01234\}.$$

These give for the λ''' , square the values

$$\begin{aligned} &\{\beta012\}, \quad 3\{\beta12\}, \quad 3\{\beta2\}, \quad +1, \\ &\{a0\}\{\beta01\}, \quad 2a - \beta \cdot \{\beta1\}, \quad a - 2\beta - 2, \quad -1, \\ &\{a01\}\{\beta0\}, \quad a - 2\beta \cdot \{a1\}, \quad -2a + \beta - 2, \quad +1, \\ &\{a012\}, \quad -3\{a12\}, \quad +3\{a2\}, \quad -1, \end{aligned}$$

and so on; the law however of the terms in the intermediate lines is not by any means obvious.

Consider now the binary quantics P, Q, R , of the forms $(\xi x, y)^p, (\xi x, y)^q, (\xi x, y)^r$; we have for any, for instance for the fourth, order, the derivates

$$P(Q, R)^4, \quad (P, (Q, R)^2)^2, \quad (P, (Q, R)^2)^2, \quad (P, (Q, R)^4);$$

and it is required to express $Q(P, R)^4$ and $R(P, Q)^4$, each of them as a linear function of these.

I recall that we have $(P, Q)^2 = PQ$, so that the first and the last terms of the series might have been written $(P, (Q, R)^4)^4$ and $(P, (Q, R)^4)^4$ respectively; and, further, that $(P, Q)^2$ denotes $d_x^2 P \cdot d_y^2 Q - 2d_x d_y P \cdot d_x d_y Q + d_y^2 P \cdot d_x^2 Q$; $(P, Q)^4$ denotes $d_x^4 P \cdot d_y^4 Q - 4d_x^3 d_y P \cdot d_x d_y^3 Q + 6d_x^2 d_y^2 P \cdot d_x^2 d_y^2 Q - 4d_x d_y^3 P \cdot d_x^3 d_y Q + d_y^4 P \cdot d_x^4 Q$, and so on.

I write (a, b, c, d, e) for the fourth derived functions of any quantic $U = (\xi x, y)^m$; we have, in a notation which will be at once understood,

$$\begin{aligned} U &= (a, b, c, d, e \quad x, y)^4 \div [m]^4, \\ (d_x, d_y)U &= (a, b, c, d \quad x, y)^3 \div [m-1]^3, \\ (d_x, d_y)^2 U &= (a, b, c \quad x, y)^2 \div [m-2]^2, \\ (d_x, d_y)^3 U &= (a, b \quad x, y)^1 \div [m-3]^1, \\ (d_x, d_y)^4 U &= (a, b, c, d, e). \end{aligned}$$

And then, taking

$$(a_1, b_1, c_1, d_1, e_1), \quad (a_2, b_2, c_2, d_2, e_2), \quad (a_3, b_3, c_3, d_3, e_3),$$

to belong to P, Q, R , respectively, we must, instead of m , write p, q, r for the three functions respectively.

If we attend only to the highest terms in x , we have

$$\begin{aligned}
 U &= ax^m \div [m]^m, \\
 (d_x, d_y)U &= (a, b)x^{m-1} \div [m-1]^{m-1}, \\
 (d_x, d_y)^2U &= (a, b, c)x^{m-2} \div [m-2]^{m-2}, \\
 (d_x, d_y)^3U &= (a, b, c, d)x^{m-3} \div [m-3]^{m-3}, \\
 (d_x, d_y)^4U &= (a, b, c, d, e).
 \end{aligned}$$

Consider now $P(Q, R)^4$, $(P, (Q, R)^2)^2$, &c.; in each case attending only to the term in a_1 , and in this term to the highest term in x , we have

$$\begin{aligned}
 (1) \quad [p]^4 P(Q, R)^4 &= a_1 e_3 - 4b_1 d_3 + 6c_1 c_3 - 4d_1 b_3 + e_1 a_3 \quad (X) \\
 (2) \quad [p-1]^2 [q-3]^2 [r-3]^2 (P, (Q, R)^2)^2 &= [q-3]^2 \cdot (b_1 d_3 - 3c_1 c_3 + 3d_1 b_3 - e_1 a_3) \cdot (X) \\
 &\quad + [r-3]^2 \cdot (a_1 e_3 - 3b_1 d_3 + 3c_1 c_3 - d_1 b_3) \cdot (Y) \\
 (3) \quad [p-2]^2 [q-2]^2 [r-2]^2 (P, (Q, R)^2)^2 &= [q-2]^2 \cdot (c_1 a_3 - 2d_1 b_3 + e_1 a_3) \cdot (Z'') \\
 &\quad + 2[q-2]^1 [r-2]^1 \cdot (b_1 d_3 - 2c_1 c_3 + d_1 b_3) \cdot (Z') \\
 &\quad + [r-2]^2 \cdot (a_1 e_3 - 2b_1 d_3 + c_1 c_3) \cdot (Z), \\
 (4) \quad [p-3]^2 [q-1]^2 [r-1]^2 (P, (Q, R)^1)^3 &= [q-1]^2 \cdot (d_1 b_3 - c_1 a_3) \cdot (-W'') \\
 &\quad + 3[q-1]^1 [r-1]^1 \cdot (c_1 c_3 - d_1 b_3) \cdot (W''') \\
 &\quad + 3[q-1]^1 [r-1]^1 \cdot (b_1 d_3 - c_1 c_3) \cdot (-W') \\
 &\quad + [r-1]^2 \cdot (a_1 e_3 - b_1 d_3) \cdot (W), \\
 (5) \quad [q]^4 [r]^4 R(P, Q)^4 &= [r]^4 \cdot (-U''') + 4[r]^3 [q]^1 \cdot (U''') + 6[r]^2 [q]^2 \cdot (U'') \\
 &\quad + 4[r]^1 [q]^3 \cdot (U') + [q]^4 \cdot (U).
 \end{aligned}$$

Thus, for the second of these equations,

$$(P, (Q, R)^2)^2 = d_x^2 P \cdot d_y^2 (Q, R)^2 - \&c.$$

the term in a_1 is $d_y^2 (Q, R)^2 = (d_x Q, R)^2 + (Q, d_y R)^2$, the whole being divided by $[p-1]^2$; where attending only to the highest terms in x , the two terms are respectively

$$(b_2 d_3 - 3c_2 c_3 + 3d_2 b_3 - e_2 a_3) \div [r-3]^2$$

and

$$(a_2 e_3 - 3b_2 d_3 + 3c_2 c_3 - d_2 b_3) \div [q-3]^2$$

which are each divided by $[p-1]^2$ as above; whence, multiplying by $[p-1]^2[q-1]^2[r-1]^2$, we have the formula in question; and so for the other cases.

Writing now (1), (2), (3), (4), (5) for the left-hand sides of the five equations respectively; and $-Y'$, Y ; Z'' , $-Z'$, Z ; $-W'''$, W'' , $-W'$, W ; U'''' , $-U'''$, U'' , $-U'$, U ; for the literal parts on the right-hand sides of the same equations respectively, then we have $X = Y + Y'$, $Y = Z + Z'$, $Y' = Z' + Z''$, &c., and the equations become

$$(1) = X,$$

$$(2) = [r-3]^2Y - [q-3]^2Y',$$

$$(3) = [r-2]^2Z - 2[r-2]^1[q-2]^1Z' + [q-2]^2Z'',$$

$$(4) = [r-1]^3W - 3[r-1]^2[q-1]^1W' + 3[r-1]^1[q-1]^2W'' - [q-1]^3W''',$$

$$(5) = [r]^4U - 4[r]^3[q]^1U' + 6[r]^2[q]^2U'' - 4[r]^1[q]^3U''' + [q]^4U'''';$$

which are, in fact, the equations considered at the beginning of the present paper, putting therein $a = r-3$ and $\beta = q-3$; they consequently give

$$\{q+r-6, 456\}(1), \quad \{q+r-6, 156\}(2), \quad \{q+r-6, 036\}(3), \\ \{q+r-6, 015\}(4), \quad \{q+r-6, 012\}(5),$$

$$\{q+r-6, 01\dots 6\}U = \{q-3, 0123\}, \quad +4\{q-3, 123\}, \quad +6\{q-3, 23\}, \\ +4\{q-3, 3\}, \quad +1, \\ \{ \quad \}U'''' = \{r-3, 0123\}, \quad -4\{r-3, 123\}, \quad +6\{r-3, 23\}, \\ -4\{r-3, 3\}, \quad +1.$$

Also, attending as before only to the terms in a_1 , and therein to the highest power of X , we have

$$[q]^4Q(R, P)^4 = a_1e_3 \cdot 5 \div [r]^4; \text{ that is, } [q]^4Q(R, P)^4 = U, \\ [r]^4R(P, Q)^4 = U'''';$$

and, observing that $\{q+r-6, 01\dots 6\}$ is $= [q+r]^5$, and that $\{q+r-6, 456\}$, &c., may be written $\{q-r, 210\}$, &c., where the superscript bars are the signs $-$, the formulae become

$$[q+r]^5[q]^4Q(P, R)^4 = [q]^4 \cdot (+1), \quad +4[q]^3 \cdot (+1), \quad +6[q]^2 \cdot (+1), \\ +4[q]^1 \cdot (+1), \quad +1, \\ [q+r]^5[r]^4R(P, Q)^4 = [r]^4 \cdot (+1), \quad -4[r]^3 \cdot (+1), \quad +6[r]^2 \cdot (+1), \\ -4[r]^1 \cdot (+1), \quad +1.$$

Written at full length, the first of these equations (which, as being the fourth in a series, I mark 4th equation) is

$$\begin{aligned}
 [q+r]^5[q]^4Q(P, R)^4 = & \\
 1 \cdot \{q+r\} \cdot \{q+r-1\} \cdot \{q+r-2\} \cdot [p]^4[q]^4 \cdot P(Q, R)^4 & \quad (4\text{th equation}) \\
 + 4 \cdot \{q+r-1\} \cdot \{q+r-5\} \cdot [p-1]^2[q]^2[q-2]^1[r-2]^1 \cdot (P, (Q, R)^2)^2 & \\
 + 6 \cdot \{q+r\} \cdot \{q+r-3\} \cdot \{q+r-6\} \cdot [p-2]^2[q]^2[q-2]^2[r-2]^2 \cdot (P, (Q, R)^2)^2 & \\
 + 4 \cdot \{q+r-1\} \cdot \{q+r-5\} \cdot \{q+r-6\} \cdot [p-3]^2[q]^2[q-1]^1[r-1]^1 \cdot (P, (Q, R)^1)^3 & \\
 + 1 \cdot \{q+r-1\} \cdot \{q+r-5\} \cdot \{q+r-6\} \cdot [q]^4[r]^4 \cdot R(P, Q)^4, &
 \end{aligned}$$

and the other is, in fact, the same equation with q, Q, r, R interchanged with r, R, q, Q ; the alternate $+$ and $-$ signs arise evidently from the terms $(R, Q)^2 = -(Q, R)^2$; $(R, Q)^4 = (Q, R)^4$; &c., which present themselves on the right-hand side.

It will be observed that the identity has been derived from the comparison of the terms in a_1 , which are the highest terms in x ; the other terms not having been written down or considered; but it is easy to see that an identity of the form in question exists, and, this being admitted, the process is a legitimate one.

The preceding equations of the series are

$$\begin{aligned}
 [q+r]^1[q]^1Q(P, R)^1 = 1 \cdot [p]^1[q]^1P(Q, R)^1 & \quad (1\text{st equation}) \\
 + 1 \cdot [q]^1[r]^1(P, Q, R)^1; &
 \end{aligned}$$

$$\begin{aligned}
 [q+r]^2[q]^2Q(P, R)^2 = 1 \cdot \{q+r\} \cdot [p]^2[q]^2P(Q, R)^2 & \quad (2\text{nd equation}) \\
 + 2 \cdot \{q+r-1\} \cdot [p-1]^2[q]^1[q-1]^1[r-1]^1(P, (Q, R)^1)^1 & \\
 + 1 \cdot \{q+r-2\} \cdot [q]^2[r]^2(P, Q, R)^2; &
 \end{aligned}$$

$$\begin{aligned}
 [q+r]^3[q]^3Q(P, R)^3 = 1 \cdot \{q+r\} \cdot \{q+r-1\} \cdot [p]^3[q]^3P(Q, R)^3 & \quad (3\text{rd equation}) \\
 + 3 \cdot \{q+r\} \cdot \{q+r-3\} \cdot [p-1]^2[q]^2[q-2]^1[r-2]^1(P, (Q, R)^1)^1 & \\
 + 2 \cdot \{q+r-1\} \cdot \{q+r-4\} \cdot [p-2]^2[q]^2[q-1]^1[r-1]^1(P, Q, R)^2 & \\
 + 1 \cdot \{q+r-3\} \cdot \{q+r-4\} \cdot [q]^3[r]^3(P, Q, R)^3. &
 \end{aligned}$$

From these four equations the law is evident, except as to the numbers subtracted from $q+r$. These are obtained, as explained above, in regard to the numbers added to $a+\beta$ in the $\{\}$ symbols; transforming the diagrams so as to be directly applicable to the case now in question, they become

	01		1		2		1
	1		2		11		2
01		012		012		0123	
	0123		03		3		015
	4		0127		14		21
	036		31		0158		34
	12		156		22		0378
3		456		13		4	1678
							5678

showing how the numbers are obtained for the equations 2, 3, 4, 5 respectively.

The first equation is

$$\{q + r\}Q(P, R) = [p]^1[q]^1P(Q, R) + [q]^1[r]^1[Q(P, R) + R(P, Q)],$$

viz. this is

$$= [p]^1P(Q, R) - [q]^1Q(R, P) + [r]^1R(P, Q) + (q^2 + qr)Q(R, P);$$

or, dividing by q , this is

$$= pP(Q, R) + qQ(R, P) + rR(P, Q),$$

which is a well-known identity.

We may verify any of the equations, though the process is rather laborious, for the particular values

$$P = x^{\frac{1}{2}(p+\alpha)}y^{\frac{1}{2}(p-\alpha)}, \quad Q = x^{\frac{1}{2}(q+\beta)}y^{\frac{1}{2}(q-\beta)}, \quad R = x^{\frac{1}{2}(r+\gamma)}y^{\frac{1}{2}(r-\gamma)}.$$

Thus, taking the second equation, we have, omitting common factors,

$$\begin{aligned} (Q, R)^2 &= \frac{1}{4}(q + \beta)(q + \beta - 2)(r - \gamma)(r - \gamma - 2) \\ &\quad - 2 \cdot \frac{1}{4}(q + \beta)(q - \beta)(r + \gamma)(r - \gamma) \\ &\quad + \frac{1}{4}(q - \beta)(q - \beta - 2)(r + \gamma)(r + \gamma - 2) \\ &= \beta^2(r^2 - r) + \gamma^2(q^2 - q) - 2\beta\gamma(q - 1)(r - 1) - qr(q + r - 2), \end{aligned}$$

$$\begin{aligned} (P, (Q, R)^2)^2 &= (q + \beta)(r - \gamma) \cdot \frac{1}{4}(p + \alpha)(q + r - \beta - \gamma - 2) \\ &\quad - \frac{1}{4}(p - \alpha)(q + r + \beta + \gamma - 2) \\ &= (\beta r - \gamma q)(\alpha(q + r - 2) - p(\beta + \gamma)) \\ &= \alpha\beta r(r + q - 2) - \alpha\gamma q(q + r - 2) - p\beta r^2 + p(\beta - r)\beta\gamma + pq\gamma^2, \end{aligned}$$

and from the first of these the expressions of $Q(P, R)^2$ and $(P, QR)^2$ are at once obtained.

The identity to be verified then becomes

$$\begin{aligned} & [q+r]^3[q]^2\{\alpha^2(\beta^2(r^2-r)+\gamma^2(q^2-q)-2\beta\gamma(q-1)(r-1)-pr(p+r-2))\} = \\ & \{q+r\}[q]^2[p]^2\{\beta^2(\gamma^2(r^2-r)+\alpha^2(p^2-p)-2\gamma\alpha(p-1)(r-1)-qr(q+r-2))\} \\ & + 2\{q+r-1\}[q]^2(p-1)(r-1)\{\alpha\beta\gamma r(q+r-2) - \alpha\gamma q(q+r-2) \\ & \quad - p\beta\gamma^2 + p(q-r)\beta\gamma + pq\gamma^2\} \\ & + \{q+r-2\}[q]^2[r]^2\{\alpha^2(q+r)(q+r-1) + (\beta+\gamma)^2(p^2-p) \\ & \quad - 2\alpha(\beta+\gamma)(p-1)(q+r-1) - p(q+r)(p+q+r-2)\}, \end{aligned}$$

which is easily verified, term by term; for instance, the terms with α , β , or γ , give

$$\begin{aligned} [q+r]^3[q]^2pr(p+r-2) &= (q+r)[q]^2[p]^2qr(q+r-2) \\ &+ \{q+r-2\}[q]^2[r]^2p(q+r)(p+q+r-2), \end{aligned}$$

which, omitting the factor $(q+r)(q+r-2)[q]^2pr$, is

$$(q+r-1)(p+r-2) = (p-1)q + (r-1)(p+q+r+2);$$

viz. the right-hand side is

$$(p-1)q + (r-1)q + (r-1)(p+r-2) = (q+r-1)(p+r-2),$$

as it should be.

The equations are useful for the demonstration of a subsidiary theorem employed in Gordan's demonstration of the finite number of the covariants of any binary form V . Suppose that a system of covariants (including the quantic itself) is $P, Q, R, S, \dots$; this may be the complete system of covariants; and if it is so, then, T and V being any functions of the form $P^\alpha Q^\beta R^\gamma \dots$, every derivative $(T, V)^\theta$ must be a term or sum of terms of the like form $P^\alpha Q^\beta R^\gamma \dots$; the subsidiary theorem is that in order to prove that the case is so, it is sufficient to prove that every derivative $(P, Q)^\theta$, where P and Q are any two terms of the proposed system, is a term or sum of terms of the form in question $P^\alpha Q^\beta R^\gamma \dots$.

In fact, supposing it shown that every derivative $(T, V)^\theta$ up to a given value θ of is of the form $P^\alpha Q^\beta R^\gamma \dots$, we can by successive application of the equation for $Q(P, R)^{\theta+1}$ regarded as an equation

for the reduction of the last term on the right-hand side $(P, QR)^{\theta+1}$, bring first $(P, QR)^{\theta+1}$ and then $(P, QRS)^{\theta+1}$, &c., and so ultimately any function $(P, V)^{\theta+1}$, and then again any functions $(PQ, V)^{\theta+1}$, $(PQR, V)^{\theta+1}$, &c., and so ultimately any function $(T, V)^{\theta+1}$ into the required form $P^\alpha Q^\beta R^\gamma \dots$: or the theorem, being true for θ , will be true for $\theta + 1$; whence it is true generally.

682. Formulae Relating to the Right Line.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 169—171.]

1. Let λ, μ, ν be the direction-angles of a line; α, β, γ the coordinates of a point on the line; and write

$$\begin{aligned} a &= \cos \lambda, & f &= \beta \cos \nu - \gamma \cos \mu, \\ b &= \cos \mu, & g &= \gamma \cos \lambda - \alpha \cos \nu, \\ c &= \cos \nu, & h &= \alpha \cos \mu - \beta \cos \lambda, \end{aligned}$$

whence

$$a^2 + b^2 + c^2 = 1, \quad af + bg + ch = 0,$$

or the six quantities (a, b, c, f, g, h) , termed the coordinates of the line, depend upon four arbitrary parameters.

2. It is at once shown that the condition for the intersection of any two lines $(a, b, c, f, g, h), (a', b', c', f', g', h')$, is

$$af' + bg' + ch' + a'f + b'g + c'h = 0.$$

3. Given two lines $(a, b, c, f, g, h), (a', b', c', f', g', h')$, it is required to find their shortest distance, and the coordinates of their line of shortest distance.

Let

$$Ax + By + Cz + D = 0, \quad Ax + By + Cz + D' = 0,$$

be parallel planes containing the two lines respectively; then the first plane contains the point $\alpha + r \cos \lambda, \beta + r \cos \mu, \gamma + r \cos \nu$, and the second contains the point $\alpha' + r' \cos \lambda', \beta' + r' \cos \mu', \gamma' + r' \cos \nu'$; that is, we have

$$\begin{aligned} A\alpha + B\beta + C\gamma + D &= 0, \\ A\alpha' + B\beta' + C\gamma' + D' &= 0, \\ A \cos \lambda + B \cos \mu + C \cos \nu &= 0, \\ A \cos \lambda' + B \cos \mu' + C \cos \nu' &= 0, \end{aligned}$$

which last equations may be written

$$Aa + Bb + Cc = 0, \quad Aa' + Bb' + Cc' = 0,$$

giving

$$A : B : C = bc' - b'c : ca' - c'a : ab' - a'b,$$

or, if we write

$$\Theta = aa' + bb' + cc',$$

and assume, as is convenient,

$$A^2 + B^2 + C^2 = 1,$$

then

$$A, B, C = \frac{bc' - b'c, \quad ca' - c'a, \quad ab' - a'b}{\sqrt{1 - \Theta^2}},$$

where $\Theta = \text{cosine-inclination} = aa' + bb' + cc'$.

Hence, shortest distance = $D' - D$

$$\begin{aligned} &= A(\alpha - \alpha') + B(\beta - \beta') + C(\gamma - \gamma') \\ &= \frac{1}{\sqrt{1 - \Theta^2}} \{ (bc' - b'c)(\alpha - \alpha') + (ca' - c'a)(\beta - \beta') + (ab' - a'b)(\gamma - \gamma') \} \\ &= \frac{1}{\sqrt{1 - \Theta^2}} \{ a'(c\beta - b\gamma) + b'(\alpha\gamma - c\alpha) + c'(b\alpha - a\beta) \\ &\quad + a(c'\beta' - b'\gamma') + b'(\alpha\gamma' - c'\alpha') + c(b'\alpha' - a'\beta') \} \\ &= \frac{1}{\sqrt{1 - \Theta^2}} (af' + bg' + ch' + a'f + b'g + c'h), \quad = \mathfrak{B} \text{ suppose.} \end{aligned}$$

The six coordinates of the line of shortest distance are A, B, C, F, G, H , where A, B, C denote as before, and F, G, H are to be determined.

Since the line meets each of the given lines, we have

$$\begin{aligned} Af + Bg + Ch + Fa + Gb + Hc &= 0, \\ Af' + Bg' + Ch' + Fa' + Gb' + Hc' &= 0, \end{aligned}$$

and we have also

$$FA + GB + HC = 0,$$

which equations give F, G, H .

Multiplying the first equation by $b'C - c'B$, the second by $Bc - Cb$, and the third by $bc' - b'c$, we find

$$(b'C - c'B)(Af + Bg + Ch) + (Bc - Cb)(Af' + Bg' + Ch') + F \cdot \sqrt{1 - \Theta^2} = 0.$$

Here

$$\begin{aligned} b'C - c'B &= \frac{1}{\sqrt{(1 - \Theta^2)}} \{b'(ab' - a'b) - c'(ca' - c'a)\} \\ &= \frac{1}{\sqrt{(1 - \Theta^2)}} \{a(a'^2 + b'^2 + c'^2) - a'(aa' + bb' + cc')\} \\ &= \frac{1}{\sqrt{(1 - \Theta^2)}} (a - a'\Theta), \end{aligned}$$

and similarly

$$cB - bG = \frac{1}{\sqrt{(1 - \Theta^2)}} (a' - a\Theta).$$

Also, putting for shortness

$$\Pi = \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ f, & g, & h \end{vmatrix}, \quad \Pi' = \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ f', & g', & h' \end{vmatrix},$$

we have

$$Af + Bg + Ch = \frac{1}{\sqrt{(1 - \Theta^2)}} \Pi, \quad Af' + Bg' + Ch' = \frac{1}{\sqrt{(1 - \Theta^2)}} \Pi',$$

and finally, the determinant which multiplies F is

$$\frac{1}{\sqrt{(1 - \Theta^2)}} \{(bc' - b'c)^2 + (ca' - c'a)^2 + (ab' - a'b)^2\} = \frac{1}{\sqrt{(1 - \Theta^2)}} (1 - \Theta^2), \quad = \sqrt{(1 - \Theta^2)}$$

We have thus the value of F ; forming in the same way those of G and H , we find

$$\begin{aligned} F &= \frac{-1}{(1 - \Theta^2)^{\frac{3}{2}}} \{(a - a'\Theta)\Pi + (a' - a\Theta)\Pi'\}, \\ G &= \frac{-1}{(1 - \Theta^2)^{\frac{3}{2}}} \{(b - b'\Theta)\Pi + (b' - b\Theta)\Pi'\}, \\ H &= \frac{-1}{(1 - \Theta^2)^{\frac{3}{2}}} \{(c - c'\Theta)\Pi + (c' - c\Theta)\Pi'\}, \end{aligned}$$

which, with the foregoing equations for A , B , C , give the six coordinates of the line of shortest distance. C. X.

683. On the Function $\arcsin(x + iy)$.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 171—174.]

The determination of the function in question, the arc to a given imaginary sine, is considered in Cauchy's *Exercices d'Analyse*, &c., t. iii. (1844), p. 382; but it appears, by two hydrodynamical papers by Mr Ferrers and Mr Lamb, *Quarterly Mathematical Journal*, t. xiii. (1874), p. 115, and t. xiv. (1875), p. 40, that the question is connected with the theory of confocal conics.

Taking $c = \sqrt{(a^2 - b^2)}$ a positive real quantity which may ultimately be put $= 1$, the question is to find the real quantities ξ , η , such that

$$\xi + i\eta = \arcsin \frac{1}{c}(x + iy),$$

or say

$$x + iy = c \sin(\xi + i\eta),$$

so that

$$x = c \sin \xi \cos i\eta, \quad iy = c \cos \xi \sin i\eta.$$

It is convenient to remark that if a value of $\xi + i\eta$ be $\xi' + i\eta'$, then the general value is $2m\pi + \xi' + i\eta'$ or $(2m + 1)\pi - (\xi' + i\eta')$; hence, η may be made positive or negative at pleasure; $\cos i\eta$ is in each case positive, but $-\sin i\eta$ has the same sign as η ; hence $\cos \xi$ has the same sign as x , but $\sin \xi$ has the same sign as y or the reverse sign, according as η is positive or negative; for any given values of x and y , we obtain, as will appear, determinate positive values of $\sin^2 \xi$ and $\cos^2 \xi$; and the square roots of these must therefore be taken so as to give to $\sin \xi$, $\cos \xi$ their proper signs respectively.

Suppose that λ , μ are the elliptic coordinates of the point (x, y) ; viz. that we have

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1, \quad \frac{x^2}{a^2 + \mu} + \frac{y^2}{b^2 + \mu} = 1,$$

where $a^2 + \lambda$, $b^2 + \lambda$, and $a^2 + \mu$ are positive, but $b^2 + \mu$ is negative.

Calling ρ , σ the distances of the point (x, y) from the points $(c, 0)$ and $(-c, 0)$, that is, assuming

$$\rho = \sqrt{\{(x - c)^2 + y^2\}}, \quad \sigma = \sqrt{\{(x + c)^2 + y^2\}},$$

then we have

$$\begin{aligned}\sqrt{(a^2 + \lambda)} &= \frac{1}{2}(\sigma + \rho), \quad \text{whence also } \sqrt{(b^2 + \lambda)} = \frac{1}{2}\sqrt{\{(\sigma + \rho)^2 - 4c^2\}}, \\ \sqrt{(a^2 + \mu)} &= \frac{1}{2}(\sigma - \rho), \quad \sqrt{(b^2 + \mu)} = \frac{1}{2}\sqrt{\{(\sigma - \rho)^2 - 4c^2\}},\end{aligned}$$

which equations determine λ, μ as functions of x, y .

Now we have

$$\begin{aligned}\rho\sigma &= \sqrt{\{(a^2 + \lambda)(a^2 + \mu) - c^2\}} = \sqrt{\{(x^2 - y^2 - c^2)^2 + 4x^2y^2\}}, \\ \rho^2 + \sigma^2 &= 2(x^2 + y^2 + c^2);\end{aligned}$$

substituting herein for x, y their values $c \sin \xi \cos i\eta, -ci \cos \xi \sin i\eta$, we find

$$\begin{aligned}x^2 - y^2 - c^2 &= c^2(\sin^2 \xi \cos^2 i\eta + \cos^2 \xi \sin^2 i\eta) - c^2(\sin^2 \xi + \cos^2 \xi)(\sin^2 i\eta + \cos^2 i\eta) \\ &= -c^2(\sin^2 \xi \sin^2 i\eta + \cos^2 \xi \cos^2 i\eta),\end{aligned}$$

whence

$$\begin{aligned}(x^2 - y^2 - c^2)^2 &= c^4(\cos^2 \xi \cos^2 i\eta + \sin^2 \xi \sin^2 i\eta)^2, \\ 4x^2y^2 &= -4c^4 \sin^2 \xi \cos^2 \xi \sin^2 i\eta \cos^2 i\eta \\ &= c^4(\cos^2 \xi \cos^2 i\eta - \sin^2 \xi \sin^2 i\eta)^2.\end{aligned}$$

Hence

$$2\rho\sigma = 2c^2(\cos^2 \xi \cos^2 i\eta - \sin^2 \xi \sin^2 i\eta),$$

and

$$\rho^2 + \sigma^2 = 2c^2(\sin^2 \xi \cos^2 i\eta + \cos^2 \xi \sin^2 i\eta + 1);$$

hence

$$\begin{aligned}(\rho + \sigma)^2 &= 2c^2(\cos^2 i\eta - \sin^2 i\eta + 1) = 4c^2 \cos^2 i\eta, \\ (\rho - \sigma)^2 &= 2c^2(\sin^2 \xi - \cos^2 \xi + 1) = 4c^2 \sin^2 \xi.\end{aligned}$$

Consequently

$$\begin{aligned}a^2 + \lambda &= c^2 \cos^2 i\eta, \quad \text{and thence } b^2 + \lambda = -c^2 \sin^2 i\eta, \\ a^2 + \mu &= c^2 \sin^2 \xi, \quad \text{and } b^2 + \mu = -c^2 \cos^2 \xi,\end{aligned}$$

values which verify as they should do the equations

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1, \quad \frac{x^2}{a^2 + \mu} + \frac{y^2}{b^2 + \mu} = 1;$$

viz. these become

$$\frac{c^2 \sin^2 \xi \cos^2 i\eta}{c^2 \cos^2 i\eta} + \frac{-c^2 \cos^2 \xi \sin^2 i\eta}{-c^2 \sin^2 i\eta} = \sin^2 \xi + \cos^2 \xi = 1,$$

$$\frac{c^2 \sin^2 \xi \cos^2 i\eta}{c^2 \sin^2 \xi} + \frac{-c^2 \cos^2 \xi \sin^2 i\eta}{-c^2 \cos^2 \xi} = \cos^2 i\eta + \sin^2 i\eta = 1.$$

The same equations, or as we may also write them,

$$\lambda = -a^2 \sin^2 i\eta - b^2 \cos^2 i\eta, \quad \mu = -a^2 \cos^2 \xi - b^2 \sin^2 \xi,$$

determine η as a function of λ , and ξ as a function of μ ; λ , μ being by what precedes, given functions of x , y .

Or more simply, starting from the last-mentioned values of λ , μ , and substituting these in the expressions

$$\xi = \frac{a^2 + \lambda}{a^2 - b^2}, \quad \eta = \frac{b^2 + \lambda}{b^2 - a^2},$$

we find

$$x^2 = c^2 \sin^2 \xi \cos^2 i\eta, \quad y^2 = -c^2 \cos^2 \xi \sin^2 i\eta,$$

or say

$$x = c \sin \xi \cos i\eta, \quad iy = c \cos \xi \sin i\eta,$$

whence

$$x + iy = c \sin(\xi + i\eta),$$

the original relation between x , y and ξ , η .

684. On a Relation between Certain Products of Differences.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 174, 175.]

Consider the function

$$3 \left\{ \begin{array}{l} abc \cdot de \\ +bcd \cdot ea \\ +cde \cdot ab \\ +dea \cdot bc \\ +eab \cdot cd \end{array} \right\} - \left\{ \begin{array}{l} abd \cdot ce \\ +bce \cdot da \\ +cda \cdot eb \\ +deb \cdot ac \\ +eac \cdot bd \end{array} \right\},$$

where

$$abc = (a-b)(b-c)(c-a), \quad ab = (a-b)(b-a) = -(a-b)^2, \quad \&c.,$$

therefore

$$abc = bca = cab = -bac, \quad \&c., \quad ab = ba.$$

It is to be shown that the function vanishes if $e = d$. Writing $e = d$, the value is

$$3(bcd \cdot da + dab \cdot cd) - abd \cdot cd - bcd \cdot da - cda \cdot db - dac \cdot bd,$$

viz. this is

$$\begin{aligned} & 3bcd \cdot ad - abd \cdot cd + 3abd \cdot cd - bcd \cdot ad - 2acd \cdot bd \\ & = 2bcd \cdot ad - 2acd \cdot bd + 2abd \cdot cd \\ & = 2(bcd \cdot ad + cad \cdot bd + abd \cdot cd), \end{aligned}$$

which is easily seen to vanish; the value is

$$\begin{aligned} & (b-c)(c-a)(d-b)(a-d)^2 = -(b-c)(a-d)^2(b-d)(c-d) \\ & + (c-a)(a-d)(d-e)(b-d)^2 - (c-a)(a-d)(b-d)^2(c-d) \\ & + (a-b)(b-d)(d-a)(c-d)^2 - (a-b)(a-d)(b-d)(c-d)^2; \end{aligned}$$

viz. omitting the factor $(a-d)(b-d)(c-d)$, this is

$$= -(b-c)(a-d) - (c-a)(b-d) - (a-b)(c-d),$$

which vanishes.

Hence the function also vanishes if $e = a$, or $a = b$ or $b = c$, or $c = d$; and it is thus a mere numerical multiple of $(a - b)(b - c)(c - d)(d - e)(e - a)$, or say it is $= M \cdot abcde$.

To find M write $e = c$, the equation becomes

$$\begin{aligned} 3abc \cdot dc - cda \cdot cb &= M \cdot abc \cdot dc, \\ + 3bcd \cdot ca - acd \cdot bc \\ + 3cda \cdot bc + 3cab \cdot cd, \end{aligned}$$

$$4abc \cdot dc + 4dbc \cdot ac + 4adc \cdot bc = M \cdot abc \cdot dc,$$

giving $M = 10$.

In fact, we then have

$$-4abc \cdot dc + 4dbc \cdot ac + 4adc \cdot bc = 0,$$

that is,

$$-abc \cdot dc - bde \cdot ac - dac \cdot bc = 0,$$

which is right.

And we have thus the identity

$$3 \left\{ \begin{array}{l} abc \cdot de \\ +bcd \cdot ea \\ +cde \cdot ab \\ +dea \cdot bc \\ +eab \cdot cd \end{array} \right\} - \left\{ \begin{array}{l} abd \cdot ce \\ +bce \cdot da \\ +cda \cdot eb \\ +deb \cdot ac \\ +eac \cdot bd \end{array} \right\} = 10 \cdot abcde,$$

viz. this is

$$3[abcde] - [acebd] = 10\{abcde\}.$$

685. On Mr Cotterill's Goniometrical Problem.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 196—198.]

The very remarkable formulae contained in Mr Cotterill's paper, "A goniometrical problem, to be solved analytically in one move, or more simply synthetically in two moves," *Quarterly Mathematical Journal*, t. vii. (1866), pp. 259—272, are presented in a form which, to say the least, is not as easily intelligible as might be; and they have not, I think, attracted the attention which they well deserve.

Using his notation, except that I write for angles small roman letters, in order to be able to have the corresponding italic small letters and capitals for the sines and cosines respectively of the same angles, we consider nine angles $a, b, c, d, e, f, x, y, z$, which are such that the sum of three angles in the same line, or in the same column, is an odd multiple of π .

Of course, any four angles such as a, b, d, e are arbitrary, and each of the remaining angles is then determinate save as to an even multiple of π . And it may be remarked that these angles a, b, d, e may represent the inclinations of any four lines to a fifth line, and that the remaining angles are then at once obtained, as in the figure.

The small roman letters are here used to denote as well angles as points, being so placed as to show what the angles are which they respectively denote; the points $*$, $*$ are constructed as the intersections of the lines ac, bc by the circle circumscribed about axy , and the angle z is the angle which the points $*$, $*$ subtend at x or y . It will be observed that the sum of the three angles in a line or column is in each case $= \pi$.

But this in passing: the analytical theorem is, first, we can form with the sines and cosines of the angles in any two lines or columns a function $\mathfrak{S}$ presenting itself under two distinct forms, which are in fact equal in value, or say $\mathfrak{S}$ is a symmetrical function of the two lines or columns, viz. for the first and second lines this is

$$\mathfrak{S} \begin{pmatrix} a, b, c \\ d, e, f \end{pmatrix} = a^2 A h c + b^2 B c a + f^2 C a b = a'^2 D e f + b'^2 E f d + c'^2 F d e,$$

where, as already mentioned, a, A denote $\sin a, \cos a$, and so for the other letters.

Secondly, if to the $\mathfrak{S}$ of any two lines or columns we add twice the product of the six sines, we obtain a sum $\mathfrak{M}$ which has the same

value from whichever two lines or columns we obtain it; or, say $\mathfrak{M}$ is a symmetrical function of the matrix of the nine angles.

Thus

$$\mathfrak{M} = \mathfrak{S} \begin{pmatrix} a, b, c \\ d, e, f \end{pmatrix} + 2abcdef;$$

which is one of a system of six forms each of which (on account of the two forms of the $\mathfrak{S}$ contained in it) may be regarded as a double form, and the twelve values are all of them equal.

There are, moreover, 15 other forms of $\mathfrak{M}$, viz. 3 line-forms, such as $bcdx + caey + abfz$ (belongs to line a, b, c), 3 column-forms, such as $dxbc + xae f + adyz$ (belongs to column a, d, x), and 9 term-forms, such as $e\beta\gamma x^2 + f\gamma\alpha y^2 + 2efyzA$ (belongs to term a), and the $12 + 15 = 27$ values are all equal.

The several identities can of course be verified by means of the relations between the nine angles, or rather the derived sine- and cosine-relations

$$c = aB + bA, \quad C = ab - AB, \quad \&c.$$

As regards the two forms of $\mathfrak{S}$, the identity to be verified may be written

$$c(a'^2De + b'^2Ed - c'^2Fab) = f(a'^2De + b'^2Ed - f^2Cab).$$

Proceeding to reduce the factor $a'^2De + b'^2Ed - f^2Cab$, if we first write herein $f = eD + dE$, it becomes

$$a'^2De + b'^2Ed - (eD + dE)Cab,$$

which is

$$= aDe(a - bC) + bEd(b - aC),$$

and then writing $C = ab - AB$, we have $a - bC = a(1 - b^2) + bAB = B(aB + bA) = Bc$; and, similarly, $b - aC = Ac$; whence the term is $= c(aeBD + bdAE)$; or, in the equation to be verified, the right-hand side is $= cf(aeBD + bdAE)$, and by a similar reduction, the left-hand side is found to have the same value.

The paper contains various other interesting results. C. X. 38

686. On a Functional Equation.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xv. (1878), pp. 315—325; *Proceedings of the London Mathematical Society*, vol. ix. (1878), p. 29.]

I was led by a hydrodynamical problem to consider a certain functional equation: viz. writing for shortness

$$x_1 = \frac{-1}{cx + d},$$

this is

$$\phi x_1 - \phi x = \frac{Ax + B}{Cx + D}.$$

I find by a direct process, which I will afterwards explain, the solution

$$\phi x = \frac{A}{\sqrt{\{(a-d)^2 + 4bc\}}} \cdot \frac{(AD - BC)}{c(cC - cD)} \int \frac{\sin \xi t \sin \eta t dt}{\sin \zeta t \sinh \pi t},$$

where ξ is a constant, but ξ, η are complicated logarithmic functions of x (which depend also on the quantities a, b, c, d, C, D); $\sinh \pi t$ denotes as usual the hyperbolic sine, $\frac{1}{2}(e^{\pi t} - e^{-\pi t})$.

The values of ξ, η, ζ are given by the formulae

$$\cos \zeta = \frac{a + d}{\sqrt{\{(a + d)^2 - 4(ad - bc)\}}} = \frac{a + d}{\sqrt{\{(a - d)^2 + 4bc\}}},$$

$$\begin{aligned} \mathfrak{a} &= ax + b, & \mathfrak{h} &= -dx + b, & \mathfrak{c} &= cx + d, & \mathfrak{d} &= cx - a, \\ W &= C\mathfrak{a} + B\mathfrak{c}, & Z &= C\mathfrak{h} + D\mathfrak{d}, \\ R &= W + \lambda Z, & S &= -W - \lambda Z, \\ R' &= W + \lambda' Z, & S' &= -W - \lambda' Z, \end{aligned}$$

which determine λ, R, S, R', S' and then

$$\xi = \frac{1}{2} \log \frac{RS'}{R'S}, \quad \eta = \frac{1}{2} \left(\log \lambda + \log \frac{RR'}{SS'} \right), \quad \zeta = \frac{1}{2} \log \lambda.$$

There is some difficulty as to the definite integral, on account of the denominator factor $\sin \zeta t$, which becomes $= 0$ for the series of values $t = \frac{m\pi}{\zeta}$, but this is a point which I do not enter into.

I will in the first instance verify the result. Writing x_1 in place of x , and taking ξ_1, η_1 to denote the corresponding values of ξ, η , it will be shown that

$$\xi_1 = \xi, \quad \eta_1 = \eta + 2\zeta,$$

see post, (1).

Hence in the difference $\phi x - \phi x_1$ we have the integral

$$\int \frac{\sin \zeta t \{ \sin \eta t - \sin(\eta + 2\zeta)t \} dt}{\sin \zeta t \sinh \pi t},$$

(where and in all that follows the limits are $\infty, 0$, as before); here, since

$$\sin \eta t - \sin(\eta + 2\zeta)t = -2 \sin \zeta t \cos(\eta + \zeta)t,$$

the factor $\sin \zeta t$ divides out, and the numerator is $= -2 \sin \zeta t \cos(\eta + \zeta)t$, which is

$$= \sin(\eta + \zeta - \xi)t - \sin(\eta + \zeta + \xi)t.$$

Hence the integral in question is

$$\int \frac{\sin(\eta + \zeta - \xi)t dt}{\sinh \pi t} - \int \frac{\sin(\eta + \zeta + \xi)t dt}{\sinh \pi t}.$$

Now we have in general

$$\frac{1}{1 + \exp . \alpha} = \frac{1}{2} - \int \frac{\sin \alpha t dt}{\sinh \pi t};$$

(this is, in fact, Poisson's formula

$$\frac{1}{1 + k\beta^{2n}} = \frac{1}{2} - 2 \int \frac{\sin(2n \log \beta + \log k)t \cdot dt}{e^{\pi t} - e^{-\pi t}},$$

in the second Memoir on the distribution of Electricity, &c., Mém. de l'Inst, 1811, p. 223); and hence the value is

$$-\frac{1}{1 + \exp . (\eta + \zeta - \xi)} + \frac{1}{1 + \exp . (\eta + \zeta + \xi)},$$

or since

$$\eta + \zeta = \log \lambda + \frac{1}{2} \log \frac{RR'}{SS'}, \quad \xi = \frac{1}{2} \log \frac{RS'}{R'S},$$

we have

$$\eta + \zeta - \xi = \log \lambda + \frac{1}{2} \log \frac{R'^2}{S'^2} = \log \lambda \cdot \frac{R'}{S'},$$

$$\eta + \zeta + \xi = \log \lambda + \frac{1}{2} \log \frac{R^2}{S^2} = \log \lambda \cdot \frac{R}{S}.$$

38—2

And the value is thus

$$= -\frac{1}{1 + \lambda \frac{R'}{S'}} + \frac{1}{1 + \lambda \frac{R}{S}}, \quad = -\frac{(RS' - R'S)\lambda}{(\lambda R' + S')(\lambda R + S)}.$$

Hence, from the assumed value of ϕx , we obtain

$$\phi x - \phi x_1 = \frac{A}{C}(x - x_1) - \frac{\sqrt{\{(a-d)^2 + 4bc\}}(AD - BC)(RS' - R'S)\lambda}{C(dC - cD)(\lambda R' + S')(\lambda R + S)}.$$

We have

$$RS' - R'S = \frac{(\lambda - 1)(a + d)^2}{ad - bc}(dC - cD)\{cx^2 + (d - a)x - b\},$$

$$R\lambda + S = (\lambda^2 - 1)(cx + d), \quad \text{see post, (2),}$$

$$R'\lambda + S' = (\lambda - 1)(a + d)(Cx + D),$$

or since

$$\frac{cx^2 + (d - a)x - b}{cx + d} = x - x_1,$$

this is

$$\phi x - \phi x_1 = \frac{A}{C}(x - x_1) - \frac{\sqrt{\{(a-d)^2 + 4bc\}}(AD - BC)(a + d)\lambda}{(ad - bc)(\lambda^2 - 1)(Cx + D)}(x - x_1).$$

But from the value of λ ,

$$\frac{\lambda}{\lambda^2 - 1} = \frac{ad - bc}{(a + d)\sqrt{\{(a-d)^2 + 4bc\}}},$$

and the equation thus is

$$\phi x - \phi x_1 = (x - x_1) \left\{ \frac{A}{C} - \frac{AD - BC}{C(Cx + D)} \right\}, \quad = (x - x_1) \frac{Ax + B}{Cx + D},$$

as it should be.

(1) For the foregoing values of ξ_1 , η_1 , we require R_1 , S_1 , R'_1 , S'_1 , the values which R , S , R' , S' assume on writing therein x_1 for x . We have

$$R_1 = \lambda(cx_1 + d) + (cx_1 - a), \quad S_1 = -(cx_1 + d) - \lambda(cx_1 - a):$$

substituting for x_1 its value, we find

$$R_1(cx + d) = (a + d)\lambda(cx + d) - (ad - bc)(\lambda + 1),$$

or writing herein

$$ad - bc = \frac{(a + d)^2\lambda}{(\lambda + 1)^2},$$

this is

$$R_1(cx + d) = \frac{(a + d)\lambda}{\lambda + 1}R;$$

and similarly

$$S_1(cx + d) = \frac{a + d}{\lambda + 1}S.$$

The Collected Mathematical Papers of Arthur Cayley

Volume 10

Pages 301–350

Papers 686–693

0.17 686. On a Functional Equation.²

[301]

[The paper continues from the previous page with the verification of certain formulae.]

For the value of $RS' - R'S$, we have

$$\begin{aligned} RS' - R'S &= (\lambda c + d)(-W - \lambda Z) - (-\lambda d - c)(W + \lambda Z) \\ &= -(\lambda^2 - 1)\{cZ + (\lambda + 1)((d - c)W - dR)\}, \end{aligned}$$

or substituting for Z and W their values, this is

$$\begin{aligned} &= -\frac{(\lambda - 1)}{\lambda} [(a^2 + d^2 + ad + bc)c(bC + dD) \\ &\quad + (ad - bc)\{(c - d)(aC + cD) + d(bC + dD)\}]. \end{aligned}$$

In the term in $\{\dots\}$, the coefficient of C is

$$[(a^2 + d^2 + ad + bc)b + (ad - bc)a]c - d(a - b)(ad - bc) = (a + d)(db - bd)c - (a + d)dx(ad - bc),$$

and similarly the coefficient of D is

$$[(a^2 + d^2 + ad + bc)d + (ad - bc)c]c - d(c - d)(ad - bc) = (a + d)(ad - bc)c - (a + d)d(ad - bc).$$

Hence the whole term in $\{\dots\}$ is

$$= (a + d)\{[(db - bd)c - d(ad - bc)x]C + [(ad - bc)c - d(ad - bc)]D\},$$

which is readily reduced to

$$(a + d)(ad - bc)(-dC + cD);$$

also

$$ad - bc = (a + d)\{cx^2 + (d - a)x - b\},$$

so that we have

$$RS' - R'S = \frac{(\lambda - 1)}{\lambda}(dC - cD)\{cx^2 + (d - a)x - b\},$$

which is the required value of $RS' - R'S$; and there is no difficulty in obtaining the other two formulae,

$$RX + S = (\lambda^2 - 1)(cx + d), \quad R'\lambda + S' = (\lambda - 1)(a + d)(Cx + D);$$

²From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 131-137.

the verification is thus completed.

(1) To show how the formula was directly obtained, we have

$$\frac{Ax + B}{Cx + D} = \frac{A}{C} - \frac{AD - BC}{C} \frac{1}{Cx + D} = \frac{A}{C} + \rho x \quad \text{suppose;}$$

the equation then is

$$\phi x - \phi x_1 = \frac{A}{C}(x - x_1) + (x - x_1)\rho x.$$

Hence, if $x_1, x_2, x_3, \dots$ denote the successive functions $\phi x, \phi^2 x, \phi^3 x$, &c., we have

$$\begin{aligned} \phi x - \phi x_1 &= \frac{A}{C}(x - x_1) + (x - x_1)\rho x, \\ \phi x_1 - \phi x_2 &= \frac{A}{C}(x_1 - x_2) + (x_1 - x_2)\rho x_1, \end{aligned}$$

whence adding, and neglecting ϕx_n and x_n , we have

$$\phi x - \frac{A}{C}x = [(x - x_1)\rho x + (x_1 - x_2)\rho x_1 + (x_2 - x_3)\rho x_2 + \dots],$$

where the term itself a given function of x [therein regarding $x_1, x_2, x_3, \dots$ as given functions of x], and it only remains to sum the series.

(2) Starting from

$$x_1 = \frac{ax + b}{cx + d},$$

and writing

$$\lambda = \frac{a^2 + d^2 + 2bc}{ad - bc},$$

then the n th function is given by the formula

$$x_n = \frac{(\lambda^{n+1} - 1)(ax + b) + (\lambda^n - \lambda)(-dx + b)}{(\lambda^{n+1} - 1)c + (\lambda^n - \lambda)d} = \frac{\lambda^n P + Q}{\lambda^n R + S},$$

if $P = \lambda a + b$, $Q = -\lambda d + b$, and as before $R = \lambda c + d$, $S = -\lambda c - \lambda d$.

I stop to remark that λ being real, if $\lambda > 1$ we have λ^n very large, and $x_n = \rho$ which is independent of n ; the value in question for n very large is

$$x_{n-1} = x_n = \frac{\lambda(ax + b) + (-dx + b)}{\lambda(cx + d) + (-cx - a)},$$

which, observing that the equation in λ may be written

$$\frac{\lambda a - d}{c(\lambda + 1)} = \frac{b(\lambda + 1)}{\lambda d - a},$$

is, in fact, independent of x , and calling each of these $= x$, we have $x_{n-1} = x_n$, or if the two equal values $= x$,

$$x = \frac{ax + b}{cx + a},$$

which is the same equation as is obtainable by the elimination of λ from the equations

$$x = \frac{\lambda a - d}{c(\lambda + 1)} = \frac{b(\lambda + 1)}{\lambda d - a}.$$

The same result is obtained by taking $\lambda < 1$ and consequently $x_n = \frac{b}{d}$.

(3) We find

$$x_{n-1} - x_n = \frac{(\lambda^n P + Q)(\lambda^{n-1} R + S) - (\lambda^{n-1} P + Q)(\lambda^n R + S)}{(\lambda^{n-1} R + S)(\lambda^n R + S)},$$

where

$$PS - QR = -(\lambda^2 - 1)(ad - bc) = -(\lambda^2 - 1)(a + d)\{cx^2 + (d - a)x - b\},$$

and therefore

$$x_{n-1} - x_n = \frac{(\lambda - 1)(\lambda^2 - 1)(a + d)\{cx^2 + (d - a)x - b\}\lambda^{n-1}}{(\lambda^{n-1} R + S)(\lambda^n R + S)}.$$

Also

$$\rho x_{n-1} = \frac{AD - BC}{C} \frac{1}{Cx_{n-1} + D} = \frac{AD - BC}{C} \frac{\lambda^{n-1} R + S}{\lambda^{n-1} W + Z},$$

where

$$\frac{W_1}{Z_1} = \frac{C(\lambda^{n-1} P + Q) + D(\lambda^{n-1} R + S)}{\dots},$$

viz.

$$W_1 = C(\lambda^{n-1} P + Q) + D(\lambda^{n-1} R + S), \quad Z_1 = C(-\lambda d - c)Q + D(\dots)$$

[Further algebraic manipulations continue.]

We hence obtain

$$\phi x - \frac{A}{C}x = \frac{(\lambda - 1)(\lambda^2 - 1)(a + d)\{cx^2 + (d - a)x - b\}}{\lambda} \frac{(AD - BC)}{C(cD - dC)} \sum_{n=1}^{\infty} \frac{(RS' - R'S)}{(\lambda^{n-1}R + S)}$$

(4) Now the before-mentioned integral formula gives

$$\frac{1}{\lambda^n} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin(n \log \lambda + \log k) t}{\sinh \pi t} dt,$$

and

$$\frac{1}{\lambda'^n} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin(n \log \lambda + \log k') t}{\sinh \pi t} dt.$$

Taking the difference, and then writing $k = -\frac{S'}{R'}$, $k' = -\frac{S}{R}$, we have under the integral sign

$$\sin(n \log \lambda + \log(-\frac{S'}{R'}))t - \sin(n \log \lambda + \log(-\frac{S}{R}))t,$$

which is

$$= 2 \sin \frac{1}{2} (\log \frac{RS'}{R'S}) t \cos(n \log \lambda + \frac{1}{2} \log \frac{RS'}{R'S}) t,$$

which attending to the before-mentioned values of ξ , η , ζ is

$$= 2 \sin \xi t \cos(2n\xi - \zeta + \eta)t,$$

and the formula thus is

$$\frac{(RS' - R'S)\lambda^n}{(R\lambda^n + S)(R'\lambda^n + S')} = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin \xi t \cos(2n\xi - \zeta + \eta)t}{\sinh \pi t} dt.$$

We have here

$$\cos(2n\xi - \zeta + \eta)t = \cos 2n\xi t \cos(\eta - \zeta)t - \sin 2n\xi t \sin(\eta - \zeta)t,$$

whence summing from 1 to ∞ , by means of the formulae

$$\cos 2\xi t + \cos 4\xi t + \cdots = -\frac{1}{2}, \quad \sin 2\xi t + \sin 4\xi t + \cdots = \frac{1}{2} \cot \xi t,$$

(which series however are not convergent), the numerator under the integral sign becomes

$$\sin \xi t \{-\cos(\eta - \zeta)t - \cot \xi t \sin(\eta - \zeta)t\},$$

which is

$$-\frac{\sin \eta t}{\sin \xi t},$$

and the formula thus is

$$\frac{(RS' - R'S)\lambda^n}{(R\lambda^n + S)(R'\lambda^n + S')} = -\frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin \eta t}{\sin \xi t \sinh \pi t} dt,$$

and we therefore find

$$\phi x = \frac{A}{C}x + \frac{\sqrt{(a-d)^2 + 4bc}(AD - BC)}{C(cD - dC)} \int_{-\infty}^{\infty} \frac{\sin \xi t \sin \eta t dt}{\sin \xi t \sinh \pi t},$$

which is the result in question.

The general solution is one particular solution + the general solution of $\phi x = \frac{A}{C}x$; calling it ϕx , we have

$$\phi x = \frac{A}{C}x + K \frac{Ax + B}{Cx + D},$$

where K is an arbitrary constant.

0.18 687. Note on the function $\phi x = a^x(c - x)^{-\frac{1}{2}}\{c(c - x) - b^2\}^{-\frac{1}{2}}$.³

[307]

Starting from the general form

$$\phi x = \frac{ax + b}{cx + d},$$

we have

$$x_n = \frac{(\lambda^{n+1} - 1)(ax + \beta) + (\lambda^n - \lambda)(-\delta x + \beta)}{(\lambda^{n+1} - 1)(\gamma x + \delta) + (\lambda^n - \lambda)(\gamma x - \alpha)},$$

where

$$\lambda + \frac{1}{\lambda} = \frac{\alpha^2 + \delta^2 + 2\beta\gamma}{\alpha\delta - \beta\gamma}.$$

For the function in question

$$\phi x = a^2(c - x)\sqrt{c(c - x) - b^2}$$

(form which presents itself in the problem of the distribution of electricity upon two spheres), the values of $\alpha, \beta, \gamma, \delta$ are

$$\alpha = -a^2, \quad \beta = c^2 - b^2, \quad \gamma = -c, \quad \delta = c^2,$$

the equation for λ therefore is

$$\lambda + \frac{1}{\lambda} = \frac{a^4 + (c^2 - b^2)^2 - 2a^2c^2}{-a^2c^2 + c^2(c^2 - b^2)},$$

or, what is the same thing,

$$\frac{(\lambda + 1)^2}{\lambda} = \frac{(a^2 + c^2 - b^2)^2}{c^2(c^2 - b^2)},$$

[308]

Suppose that a, b, c are the sides of a triangle the angles whereof are A, B, C ; then

$$c^2 = a^2 + b^2 - 2ab \cos C, \quad \text{or we have} \quad \frac{c^2 - a^2 - b^2}{ab} = -2 \cos C;$$

³From the *Messenger of Mathematics*, vol. VII. (1878), pp. 113-114.

0.18. Note on the function $\phi x = a^x(c-x)^{-\frac{1}{2}}\{c(c-x)-b^2\}^{-\frac{1}{2}}$ 281

or, writing this under the form

$$\sqrt{\lambda} + \frac{1}{\sqrt{\lambda}} = 2 \cos C,$$

the value of λ is at once seen to be $= e^{2iC}$; and it is interesting to obtain the expression of the n th function in terms of the sides and angles of the triangle.

The numerator and the denominator are

$$\lambda^n P + Q, \quad \lambda^n R + S,$$

where

$$\begin{aligned} P &= \lambda(\alpha x + \beta) + (-\delta x + \beta), & Q &= -(\alpha x + \beta) - \lambda(-\delta x + \beta), \\ R &= \lambda(\gamma x + \delta) + (\gamma x - \alpha), & S &= -(\gamma x + \delta) - \lambda(\gamma x - \alpha). \end{aligned}$$

Hence, writing the numerator and the denominator in the forms

$$\lambda^n P + \lambda^{-n} Q, \quad \lambda^n R + \lambda^{-n} S,$$

these are

$$(P+Q) \cos nC + (P-Q)i \sin nC, \quad (R+S) \cos nC + (R-S)i \sin nC;$$

viz. they are

$$\begin{aligned} &(\lambda-1)(\alpha+\delta)x \cos nC + (\lambda+1)\{(\alpha-\delta)x + 2\beta\}i \sin nC, \\ &(\lambda-1)(\alpha+\delta) \cos nC + (\lambda+1)\{2\gamma x - (\alpha-\delta)\}i \sin nC, \end{aligned}$$

or, observing that $\frac{\lambda-1}{\lambda+1} = i \tan C$ and removing the common factor $i(\lambda+1)$, they may be written

$$\begin{aligned} &\tan C (\alpha + \delta)x \cos nC + \{(\alpha - \delta)x + 2\beta\} \sin nC, \\ &\tan C (\alpha + \delta) \cos nC + \{2\gamma x - (\alpha - \delta)\} \sin nC. \end{aligned}$$

Substituting for $\alpha, \beta, \gamma, \delta$ their values, these are

$$\begin{aligned} &\tan C \{(c^2 - a^2 - b^2)x \cos nC\} + \{(b^2 - a^2 - c^2)x + 2a^2c\} \sin nC, \\ &\tan C \{(c^2 - a^2 - b^2) \cos nC\} + \{-2cx - (b^2 - a^2 - c^2)\} \sin nC, \\ &= \tan C \{-ab \cos C x \cos nC\} + \{-ac \cos B x + a^2c\} \sin nC, \\ &= \tan C \{-ab \cos C x \cos nC\} + \{-cx + ac \cos B\} \sin nC, \end{aligned}$$

$$= x\{-ab \sin C \cos nC - ac \cos B \sin nC\} + a^2 c \sin nC, \\ - cx \sin nC + \{ac \cos B \sin nC - ab \sin C \cos nC\};$$

[309]

or, writing herein $b \sin C = c \sin B$, these are

$$- acx\{\sin B \cos nC + \cos B \sin nC\} + a^2 c \sin nC, \\ - cx \sin nC + ac\{\cos B \sin nC - \sin B \cos nC\},$$

whence finally

$$\phi^n x = \frac{a^2 \sin nC - cx \sin(nC + B)}{a \sin(nC - B) - x \sin nC}.$$

As a verification, writing $n = 1$, we have

$$\phi x = \frac{a^2 \sin C - cx \sin(C + B)}{a \sin(C - B) - x \sin C} = \frac{a^2 c - acx \frac{\sin A}{\sin C}}{a \sin(C - B) - x \sin C},$$

or observing that

$$\frac{ac}{\sin C} \{\sin A \sin(C - B)\} = c^2 - b^2$$

(for this is $\sin A \sin(C - B) = \sin^2 C - \sin^2 B$), we have

$$\phi x = \frac{c^2 - b^2 - cx}{\sqrt{c(c - x) - b^2}},$$

as it should be.

If in the formula for $\phi^n x$ we write $x = 0$, we have

$$\phi^n 0 = \frac{a^2 \sin nC}{a \sin(nC - B)},$$

a formula given in the Senate-House Problems, January 14, 1878: it was thus that I was led to investigate the general expression.

0.19 688. Geometrical Considerations on a Solar Eclipse.⁴

[310]

I consider, from a geometrical point of view, the phenomena of a solar eclipse over the earth generally; attending at present only to the penumbral cone, the vertex of which I denote by V . It is convenient to regard the earth as fixed, and the sun and moon as moving each of them with its proper motion, and also with the diurnal motion. The penumbral cone meets the earth's surface in a curve which may be called the penumbral curve; viz. when the cone is not completely traversed by the earth's surface, (that is, when only some of the generating lines of the cone meet the earth's surface), this is a horse-shoe shaped oval; the penumbra is a region bounded by the two branches of the oval, or say these are separated, the one being the rising, the other the setting branch of the penumbral curve.

It is to be noticed that the whole of the oval may be visible, or the whole invisible, or only a portion may be visible; and moreover the character of the oval may change from visible to invisible, or *vice versâ*, according as it does or does not intersect the horizon of any given place on the earth's surface.

1. I attend in the first instance to the penumbral curve, and its relation to the horizon at any given place on the earth's surface. I say that the penumbral curve and the horizon have a common tangent TT' ; viz. this is the line of intersection of the horizon-plane by a plane through the axis of the cone (that is, through V and the centre of the moon) parallel to the horizon-plane; or, what is the same thing, it is the line of intersection of the horizon-plane by a plane through the centre of the earth parallel to the plane through the axis of the cone and the horizon-plane.

2. Consider now any point P on the horizon; through P draw a line PT in the direction T of the common tangent; and let θ denote the angle which the generating line VP of the cone makes with PT ; then the condition in order that the generating line may meet the earth's surface is that $\cos \theta$ may be between the limits $\pm \cos \theta_0$, where θ_0 is a given angle; and if $\cos \theta$ is within these limits, then the generating

⁴From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 145–149.

line does meet the earth's surface, or does not meet it, according as P lies on the one side or the other of the tangent line TT' .

3. Hence, taking P a point on the horizon in the direction T of the common tangent, and drawing lines PQ in directions making with PT angles between the limits $\pm\theta_0$, the generating lines VPQ will meet the earth's surface; and we thus see that the points Q form a region bounded by the conic which is the intersection of the earth's surface by the plane through V touching the horizon-plane along the line TT' . But if the angle PTQ is outside the limits, then the generating line does not meet the earth's surface, and the corresponding point Q is on the sky.

4. The foregoing completely determines the visibility or invisibility of any given point of the penumbral curve, and it is easy to see how the character of the penumbral curve as visible or invisible changes as the position of the place on the earth's surface varies. For as the place moves on the earth's surface, the tangent line TT' moves in the plane of the penumbral curve, and the character of any given point changes as the tangent line passes over it; viz. the point changes from visible to invisible or *vice versâ*, as the tangent line passes over it in such wise that the angle PTQ passes from within to without the limits, or *vice versâ*.

5. The vertex V (considered as a point in the heavens) is rising, for a point on the one side of the horizon-envelope, and setting, for a point on the other of them; and the two points K_1, K_2 are the northern and southern limits respectively. There is, as before, no passage through a point W or W' , the curve in question thus retains throughout the same character; and by consideration of the two terminal points it at once appears that the two portions K_1W, K_2W' (or K_1W', K_2W as the case may be) are the northern and southern limits respectively.

0.20 689. On the Geometrical Representation of Imaginary Variables by a Real Corre- spondence of Two Planes.⁵

[316]

In my recently published paper, "Geometrical Illustration of a Theorem relating to an Irrational Function of an Imaginary Variable," *Proceedings of the London Mathematical Society*, t. VIII. (1877), pp. 212-214, [627], I remark as follows:—"If we have v a function of u determined by an equation $f(u, v) = 0$, then to any given imaginary value $x+iy$ of u there belong two or more values, in general imaginary, and for the complete understanding of the relation between the two imaginary variables we require to know the series of values $x' + iy'$ which correspond to a given series of values $x + iy$ of u ; or, what is the same thing, we require to have a geometrical representation of the one series of points by means of the other series of points." I propose in the present paper to develop this idea.

1. I consider first the general case of a (m, n) correspondence between two planes; viz. to any given point P in the first plane there correspond n points P' in the second plane; and to any given point P' in the second plane there correspond m points P in the first plane. The correspondence is determined by an equation

$$\Phi(x, y; x', y') = 0,$$

where Φ is of the degree m in (x, y) and of the degree n in (x', y') .

2. To a given point P in the first plane there correspond n points P' in the second plane; and these may be distinct or some of them may be united. In the general case they are distinct; but if the point P is on a certain curve, the branch-curve, then two or more of the corresponding points P' unite together. The branch-curve is a curve on the first plane; and in like manner there is a branch-curve on the second plane.

3. If the point P describes on the first plane a closed curve, the n corresponding points P' describe on the second plane n closed curves; but these may be such that the series of values is not restored, viz.

⁵From the *Proceedings of the London Mathematical Society*, vol. IX. (1878), pp. 31-39. Read December 13, 1877.

the point P may return to its original position, but the n points P' may have been permuted among themselves. To avoid this, we introduce the notion of barriers; viz. we draw on the first plane a system of barriers, such that the point P cannot pass from one side to the other of any barrier; and in like manner we draw on the second plane a system of barriers. The barriers are so drawn that when P returns to its original position, each of the n points P' also returns to its original position.

4. The branch-curve on the first plane consists of a system of points V and of barriers; the point P cannot pass through a point V , and it cannot cross a barrier. In like manner the branch-curve on the second plane consists of a system of points V' and of barriers. And we have on the first plane a system of counter-barriers corresponding to the barriers on the second plane; and on the second plane a system of counter-barriers corresponding to the barriers on the first plane.

5. If there were only a single barrier OV_{12} , then we might have through P a closed curve cutting OV_{12} once only, and including within it the two portions of a bifid curve.

6. [Further development of the theory of branch points, barriers, and the correspondence between the two planes.]

7. The question may be considered in itself, without any reference to such origin.

8. [Discussion of the analytical representation.]

9. Everything is of course symmetrical as regards the two planes; we have therefore, in the second plane, a system of points V' and of barriers, and in the first plane a system of points (W) and of counter-barriers. To a given point P' in the second plane there correspond m points P in the first plane; and we can (the determination depending on the system of barriers in the second plane) assign to the m points suffixes, thereby distinguishing them as the corresponding points $P_1, P_2, \dots, P_m$. And we may imagine the second plane as consisting of m superimposed planes or sheets, say the sheets 1, 2, 3, $\dots$, m ; the general theorem then is that to a point P or P' in either plane, considered as a point P_a or P'_a in the sheet a or a' , there corresponds in the other plane one and only one point $P'_{a'}$ or P_a ; and that the first-mentioned point in either plane moving continuously in any manner (subject to the proper change of sheet), the corresponding point in

the other plane will also move continuously, and will return to its original position and sheet, upon the first-mentioned point returning to its original position and sheet.

This is, in fact, Riemann's theory, only instead of the points P' we must speak of the values $x' + iy'$ of the irrational function of $x + iy$.

10. In all that precedes it has been assumed that, to a branch-point V , there correspond two united points represented by (W) and $n - 2$ distinct points W' ; the cases of a point (W) composed of three or more united points, or of the points W' uniting themselves in sets in any other manner, would give rise to further specialities.

11. [Further discussion of the correspondence theory and its relation to Riemann surfaces.]

12. [Consideration of the case where the branch curve has multiple points.]

13. [Concluding remarks on the geometrical theory.]

14. [The paper concludes with remarks on the application to the theory of functions.]

0.21 690. On the Theory of Groups.⁶

[325]

I recapitulate the general theory so far as is necessary in order to render intelligible the quasi-geometrical representation of it which will be given.

1. Let $\alpha, \beta, \dots$ be functional symbols each operating upon one and the same number of letters, and producing as its result the same number of functions of these letters. For instance,

$$\alpha(x, y, z) = (X, Y, Z),$$

where the capitals denote each of them a given function of (x, y, z) .

Such symbols are susceptible of repetition and combination; thus

$$\alpha^2(x, y, z) = \alpha(X, Y, Z), \quad \text{or} \quad \beta\alpha(x, y, z) = \beta(X, Y, Z),$$

in each case equal to three given functions of (x, y, z) ; and similarly for $\alpha^3, \alpha^2\beta, \dots, \alpha\beta\alpha$, &c.

2. The functional symbols may be of any form whatever, but I consider in particular those which arise from the substitutions of a group of letters; and using in the first instance a set of non-specialized functional symbols, I establish the notion of a group.

A set of symbols

$$1, \alpha, \beta, \dots$$

(such that the number of symbols is finite), forming a group, if they are such that the product of any two of them (in either order) is a symbol of the set, and the inverse of each symbol is also a symbol of the set.

3. The number of symbols is called the *order* of the group.

A group may be represented by means of the substitutions of a set of letters; and conversely, any group of substitutions gives rise to a group in the foregoing sense.

4. To explain the hemihedron, certain configurations by the diagram, as presently appearing, may (instead of any other equivalent representation) be employed.

⁶From the *Proceedings of the London Mathematical Society*, t. IX. (1878), pp. 126–133. Read May 9, 1878.

5. [The paper discusses colorings of polygons associated with the group.]

The polygons may have different colours; the polygons being assigned colors according to group-theoretic principles. The colors of the polygons correspond to the different operations of the group, and the adjacency relations of the colored polygons reflect the group structure.

6. [Further development of the theory of group representations by colorings.]

7. [Applications to particular groups and their geometric representations.]

0.22 691. Note on Mr Monro's Paper "On Flexure of Spaces".⁷

[331]

Consider an element of surface, surrounding a point P ; the flexure of the element may be interfered with by the continuity round P , and on this account it is proper to regard the element as cut or slit along a radius drawn from P to the periphery of the element. This being understood, we have the well-known theorem that, considering in the neighbourhood of the origin elements of the surfaces

$$z = \frac{1}{2}(ax^2 + 2hxy + by^2), \quad z' = \frac{1}{2}(a'x'^2 + 2h'x'y' + b'y'^2),$$

these will be applicable to be the one on the other, provided only $ab - h^2 = a'b' - h'^2$.

But Mr Monro's paper considers the more general problem of the flexure of spaces, and establishes certain differential equations which must be satisfied. The equation will be satisfied identically as regards dx, dy if the appropriate conditions on the coefficients are met.

[The note continues with remarks on the differential equations of flexure.]

The conditions for the flexure may be expressed by saying that the element at any point P must be applicable upon the element at the corresponding point P' , or that the two elements must have the same measure of curvature at corresponding points.

⁷From the *Proceedings of the London Mathematical Society*, vol. IX. (1878), pp. 171, 172. Read June 13, 1878.

0.23 692. Addition to the Memoir on the Transformation of Elliptic Functions.⁸

[333]

I have recently succeeded in completing a theory considered in my “Memoir on the Transformation of Elliptic Functions,” *Phil. Trans.*, vol. CLXIV. (1874), pp. 397–456, [578],—that of the septic transformation, $n = 7$.

We have here

$$\frac{1+y}{1-y} = \frac{1-z}{1+z} \sqrt{\frac{\alpha + \beta x + \gamma x^2 - \delta x^3}{\alpha + \beta x + \gamma x^2 + \delta x^3}},$$

a solution of

$$\frac{Mdy}{\sqrt{1-y^2 \cdot 1-\lambda^2 y^2}} = \frac{dx}{\sqrt{1-x^2 \cdot 1-k^2 x^2}},$$

where

$$M = 1 + 2\beta\delta,$$

and the ratios $\alpha : \beta : \gamma : \delta$, and the ωv -modular equation are determined by the equations

$$\begin{aligned} \alpha(\beta\gamma + \alpha\delta) &= u^2 v^2 (\gamma\delta + 2\beta\gamma + 2\alpha\delta), \\ \gamma^2 + 2\beta\gamma\delta + 2\alpha\delta^2 &= u^2 v^2 (\beta\gamma\delta + 2\beta\gamma + 2\alpha\delta + \beta^2). \end{aligned}$$

1. The septic transformation presents itself naturally as a development of the theory of modular equations. The equation

$$T\eta v = 1 + 2/\delta,$$

that is, certain modular equations are presented which connect the moduli k and λ .

2. [The paper develops the theory of the septic transformation.]

The modular equations present remarkable symmetries. Writing down the equations for the transformation, we obtain relations between the coefficients which may be expressed as follows:

⁸From the *Philosophical Transactions of the Royal Society of London*, vol. CLXIX., Part II. (1878), pp. 419–424. Received February 6,—Read March 7, 1878.

$$(ca' - c'a) = uv + uv(u^6 - 10u^4v^2 + 11u^2v^4 - 6v^6 - 8u^6 + 10u^4v^2 - 4v^6) \\ + u^2v^2(-4u^6 + 10u^4v^2 - 8u^2v^4 - 6u^6 + 11u^4v^2 - 10u^2v^4 + 6v^6) + u^3v^3.$$

3. The third and fourth equations should then be verified identically. Writing down the equations:

$$\begin{array}{cccccc} -64 + 224 - 352 + 224 + 160 - 392 + 160 + 224 - 352 + 224 - 64 \\ -1 \quad 0 + 16 \quad 0 - 48 \quad 0 - 48 \quad 0 + 16 \quad 0 - 1 \end{array}$$

4. [Further identities are developed:]
viz. this is

$$+(-4u^6 + 6u^4v^2 - 4u^2v^4) 2(u^6 + v^6) + 16u^8 - 48u^6v^2 + 68u^4v^4 - 48u^2v^6 + 16v^8.$$

The coefficients in the expansion satisfy the symmetry conditions required by the theory of transformation.

0.24 693. A Tenth Memoir on Quantics.⁹

[339]

The Memoir, which relates to the binary quintic $(\xi x, y)^5$, has been in hand for a considerable time: the chief subject-matter was intended to be the theory of a canonical form which was discovered by myself and is briefly noticed in Salmon's *Higher Algebra*, 3rd Ed. (1876), pp. 217, 218; writing $a, b, c, d, e, f, g, \dots, u, v, w$ to denote the 23 covariants of the quintic, then a, b, c, d, f are connected by the relation

$$f^2 = -a^3 + a^2bc - 4c^3;$$

and the form contains these covariants thus connected together, and also e ; it, in fact, is

$$(a, 0, c, f)^3(a^2e - 3c^2, a^4e - 2cd \mid x, y)^2.$$

But the whole plan of the Memoir was altered by the discoveries of Prof. Sylvester relating to the generating functions of the quintic, and the Memoir thus came to contain a much more complete theory of the quintic than was originally contemplated.

Art. Nos. 366 to 374. Table No. 96.

366. I have, in my Ninth Memoir (1871) [462], given what may be called the theory of the generating functions of the quintic. Sylvester obtained an expression for the N.G.F. of the quintic which is discussed here.

367. The memoir continues with the discussion of deg-orders and concomitants of the quintic. Sylvester obtained an expression for the N.G.F. (numerical generating function) of the quintic, and from this deduced the number and nature of the concomitants.

368. The generating function for the quintic presents remarkable properties. The B.G.F. (binomial generating function) of the quintic gives rise to a complete enumeration of the concomitants. The deg-orders of the irreducible concomitants are exhibited in the following table:

B.O.F. of quintic	Deg-orders	Concomitants
	0.0–10.10	Fundamental form $(\xi x, y)^5$

⁹From the *Philosophical Transactions of the Royal Society of London*, vol. CLXIX., Part II. (1878), pp. 603–661. Received June 12,—Read June 20, 1878.

369. [The enumeration of covariants continues with detailed tables.]

370. The number of irreducible covariants and their deg-orders are systematically determined. The generating function approach yields a complete classification.

371. The tables of deg-orders are verified by independent methods.

372. [The memoir presents extensive tables of concomitants.]

Observe as regards the foregoing diagram, that dj^2 [nor j^2] segregate in certain cases; the syzygies connecting the different covariants are exhibited in detail.

373. [Syzygies]

The 17 syzygies above referred to are

Deg-ord.	Order
6.10	6

[Further syzygies of higher degree and order are enumerated.]

374. Table No. 96 (continued). The complete table of concomitants of the quintic includes the generating functions, the deg-orders, and the syzygies which connect them. The memoir concludes with a verification that the number of irreducible concomitants is precisely equal to the number indicated by the generating function.

COLLECTED MATHEMATICAL
PAPERS OF
ARTHUR CAYLEY, Vol. X

Pages 351–400

(Papers 691–693)

691. Note on Mr Monro's Paper "On Flexure of Spaces."

[From the *Proceedings of the London Mathematical Society*, vol. ix. (1878), pp. 171, 172. Read June 13, 1878.]

CONSIDER an element of surface, surrounding a point P ; the flexure of the element may be interfered with by the continuity round P , and it is on this account proper to regard the element as cut or slit along a radius drawn from P to the periphery of the element. This being understood, we have the well-known theorem that, considering in the neighbourhood of the origin elements of the surfaces

$$z = \frac{1}{2}(ax^2 + 2hxy + by^2), \quad \text{and} \quad z' = \frac{1}{2}(a'x'^2 + 2h'x'y' + b'y'^2),$$

these will be applicable the one on the other, provided only $ab - h^2 = a'b' - h'^2$.

But in connexion with Mr Monro's paper it is worth while to give the proof in detail. It is to be shown that z, z' denoting the above-mentioned functions of (x, y) and (x', y') respectively, it is possible to find (for small values) x', y' functions of x, y such that identically

$$dx'^2 + dy'^2 + dz'^2 = dx^2 + dy^2 + dz^2.$$

The solution is taken to be $x' = x + \xi, y' = y + \eta$, where ξ, η denote cubic functions of x, y . We have then, attending only to the terms of an order not exceeding 3 in x, y ,

$$\begin{aligned} dx'^2 + dy'^2 + 2(dx d\xi + dy d\eta) + \{(a'x + h'y)dx + (h'x + b'y)dy\}^2 \\ = dx^2 + dy^2 + \{(ax + hy)dx + (hx + by)dy\}^2, \end{aligned}$$

so that the terms $dx^2 + dy^2$ disappear; and then writing

$$d\xi = P dx + Q dy, \quad d\eta = R dx + S dy.$$

the equation will be satisfied identically as regards dx, dy if only

$$\begin{aligned} 2P &= (ax + hy)^2 - (a'x + h'y)^2, \\ Q + R &= (ax + hy)(hx + by) - (a'x + h'y)(h'x + b'y), \\ 2S &= (hx + by)^2 - (h'x + b'y)^2. \end{aligned}$$

Calling the terms on the right-hand side $2\Sigma_1, 2\Sigma_2, 2\Sigma_3$ respectively, we have

$$\frac{d\Sigma_1}{dy} = \frac{d\Sigma_2}{dx}, \quad \frac{d\Sigma_2}{dy} = \frac{d\Sigma_3}{dx},$$

that is,

$$(h^2 - h'^2) + (A^2 - h'^2) - \{(ab + h^2) - (a'b' + h'^2)\} = 0,$$

or, what is the same thing,

$$a'b' - h'^2 = ab - h^2,$$

a relation which must exist between the constants (a, b, h) and (a', b', h') .

It is easy to find the actual values of ξ, η ; viz. these are

$$\begin{aligned} \xi &= \frac{1}{3}(a^2 - a'^2)x^3 + \frac{1}{2}(ah - a'h')x^2y + \frac{1}{2}(h^2 - h'^2)xy^2 + \frac{1}{3}(bh - b'h')y^3, \\ \eta &= \frac{1}{3}(ah - a'h')x^3 + \frac{1}{2}(h^2 - h'^2)x^2y + \frac{1}{2}(bh - b'h')xy^2 - \frac{1}{3}(b^2 - b'^2)y^3, \end{aligned}$$

or, what is the same thing, we have

$$\xi = \frac{d\Omega}{dx}, \quad \eta = \frac{d\Omega}{dy},$$

where

$$\begin{aligned} \Omega &= \frac{1}{4}(a^2 - a'^2)x^4 + (ah - a'h')x^3y + \frac{3}{2}(h^2 - h'^2)x^2y^2 + (bh - b'h')xy^3 + \frac{1}{4}(b^2 - b'^2)y^4 \\ &= (ax^2 + 2hxy + by^2)^2 - (a'x^2 + 2h'xy + b'y^2)^2 = 4(z^2 - z'^2), \end{aligned}$$

in virtue of the relation $ab - h^2 = a'b' - h'^2$.

The resulting values $x' = x + \xi, y' = y + \eta$ are obviously the first terms of two series which, if continued, would contain higher powers of (x, y) .

692. Addition to the Memoir on the Transformation of Elliptic Functions.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXIX. Part II. (1878), pp. 419–424. Received February 6, — Read March 7, 1878.]

I HAVE recently succeeded in completing a theory considered in my “Memoir on the Transformation of Elliptic Functions,” *Phil. Trans.*, vol. CLXIV. (1874), pp. 397–456, [578], — that of the *septic transformation*, $n = 7$. We have here

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \sqrt{\frac{\alpha - \beta x + \gamma x^2 - \delta x^3}{\alpha + \beta x + \gamma x^2 + \delta x^3}},$$

a solution of

$$M \frac{dy}{dx} = \frac{1}{\sqrt{1-y^2} \cdot 1-v^8 y^2},$$

where $v = u^7$; and the ratios $\alpha : \beta : \gamma : \delta$, and the uv -modular equation are determined by the equations

$$\begin{aligned} u^4(2\alpha\gamma + 2\alpha\beta + \beta^2) &= v^4(\gamma^2 + 2\gamma\delta + 2\beta\delta), \\ \gamma^2 + 2\beta\gamma\delta + 2\alpha\delta + 2\beta\delta &= u^4 v^4(2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2), \\ \delta^2 + 2\gamma\delta &= u^8 M^4(\alpha^2 + 2\alpha\beta); \end{aligned}$$

or, what is the same thing, writing $\alpha = 1$, the first equation may be replaced by

$$\delta = \frac{\sqrt{v^4}}{u^4},$$

and then, α, δ having these values, the last three equations determine β, γ and the modular equation.

If instead of β we introduce M , by means of the relation

$$v = 1 + 2\beta, \quad \text{that is,} \quad 2\beta = v - 1,$$

then the last equation gives

$$2\gamma = u^4 v \left(-v^2 - \frac{1}{u^4} \right);$$

and $\alpha, \beta, \gamma, \delta$ having these values, we have the residual two equations

$$\begin{aligned} u^4(2\alpha\gamma + 2\alpha\beta + \beta^2) &= v^4(\gamma^2 + 2\gamma\delta + 2\alpha\delta), \\ \gamma^2 + 2\beta\gamma\delta + 2\alpha\delta + \beta\delta &= u^4v^4(2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2), \end{aligned}$$

viz. each of these is a quadric equation in $\frac{\gamma}{v^2}$; hence eliminating $\frac{\gamma}{v^2}$, we have the modular equation; and also (linearly) the value of $\frac{\gamma}{v^2}$, and thence the values of $\alpha, \beta, \gamma, \delta$ in terms of u, v .

Before going further it is proper to remark that, writing as above $\alpha = 1$, then if $\delta = \beta\gamma$, we have

$$\begin{aligned} 1 - \beta x + \gamma x^2 - \delta x^3 &= (1 - \beta x)(1 + \gamma x^2), \\ 1 + \beta x + \gamma x^2 + \delta x^3 &= (1 + \beta x)(1 + \gamma x^2), \end{aligned}$$

and the equation of transformation becomes

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \sqrt{\frac{1-\beta x}{1+\beta x}},$$

viz. this belongs to the *cubic transformation*. The value of β in the cubic transformation was taken to be $\beta = -\frac{1}{u^2}$, but for the present purpose it is necessary to pay attention to an omitted double sign, and write $\beta = +\frac{1}{u^2}$; this being so, $\delta = \beta\gamma$, and giving to γ the value $+u^4$, δ will have its foregoing value $= -\frac{1}{u^2}$. And from the theory of the cubic equation, according as $\beta = -\frac{1}{u^2}$ or $= +\frac{1}{u^2}$, the modular equation must be

$$u^4 - v^4 + 2uv(1 - u^2v^2) = 0, \quad \text{or} \quad u^4 - v^4 - 2uv(1 - u^2v^2) = 0.$$

We thus see *a priori*, and it is easy to verify that the equations of the septic transformation are satisfied by the values

$$\begin{aligned} \alpha = 1, \quad \beta = -\frac{1}{u^2}, \quad \gamma = u^4, \quad \delta = -\frac{1}{u^2}, \quad \text{and} \quad u^4 - v^4 + 2uv(1 - u^2v^2) &= 0; \\ \alpha = 1, \quad \beta = \frac{1}{u^2}, \quad \gamma = -u^4, \quad \delta = \frac{1}{u^2}, \quad \text{and} \quad u^4 - v^4 - 2uv(1 - u^2v^2) &= 0; \end{aligned}$$

and it hence follows that in obtaining the modular equation for the septic transformation, we shall meet with the factors $u^4 - v^4 \pm 2uv(1 - u^2v^2)$.

Writing for shortness $uv = \theta$, these factors are $u^4 - v^4 \pm 2\theta(1 - \theta^2)$; the factor for the proper modular equation is $u^4 + v^4 - \Theta$, where

$$\Theta = 8\theta - 28\theta^2 + 56\theta^3 - 70\theta^4 + 56\theta^5 - 28\theta^6 + 8\theta^7,$$

viz. the equation

$$(1 - u^2)(1 - v^2) - (1 - uv)^2 = 0 \quad \text{is} \quad u^4 + v^4 - \Theta = 0;$$

and the modular equation, as obtained by the elimination from the two quadric equations, presents itself in the form

$$(ca' - c'a)^2 - 4(bc' - b'c)(ab' - a'b) = 0.$$

Proceeding to the investigation, we substitute the values

$$\alpha = 1, \quad \beta = \frac{1}{2}(v-1), \quad \gamma = \frac{1}{2}u^4v \left(-v^2 - \frac{1}{u^4} \right), \quad \delta = \frac{v^2}{u^4},$$

in the residual two equations, which thus become

$$\begin{aligned} & v^2 \cdot (1-v^2) + \frac{1}{4}(1-uv)^2(1+uv) + \frac{1}{4}(1-v^2) - 4(1-uv)(1+uv) \\ & - \frac{\gamma}{v^2} \{ -u^2v^2(1-uv)^2(1+uv) \} + \frac{\gamma^2}{v^4} \{ u^4(1-v^2) + \frac{1}{4}(1+u^2v^2)(u^4-v^4) \} = 0, \\ & \frac{\gamma^2}{v^4} \{ u^4(1-v^2) + \frac{1}{4}(1+u^2v^2)(u^4-v^4) \} + \frac{\gamma}{v^2} \{ u^4(1-v^2) + \frac{1}{4}(1+u^2v^2)(u^4-v^4) \} \\ & + \frac{1}{4}u^4 \{ -v^2(1-u^2v^2) - u^4v^2 \} = 0, \end{aligned}$$

the first of which is given p. 432 of the "Memoir," [Coll. Math. Papers, vol. ix., p. 150]. Calling them

$$(a, b, c \mid \frac{\gamma}{v^2}, 1)^2 = 0, \quad (a', b', c' \mid \frac{\gamma}{v^2}, 1)^2 = 0,$$

we have

$$\frac{\gamma}{v^2} : \frac{\gamma^2}{v^4} : 1 = bc' - b'c : ca' - c'a : ab' - a'b,$$

and the result of the elimination therefore is

$$(ca' - c'a)^2 - 4(bc' - b'c)(ab' - a'b) = 0.$$

Write as before $uv = \theta$.

In forming the expressions $ca' - c'a$, &c., to avoid fractions we must in the first instance introduce the factor v^2 : thus

$$\begin{aligned} v^2(ca' - c'a) &= v \{ v(1-u^4) - 4(1-\theta)(v+u^2) \} [-\theta^2(1+\theta)(1-\theta)^2] \\ &\quad - \{ u^4 + 6u^4\theta(1-\theta^2) - v^4\theta^4 \} (1-v^2), \\ &= -v^2(1+\theta)(1-\theta)^2 \{ v^2(-3+4\theta) + u^4(-4\theta^3+3\theta^4) \} \\ &\quad - \{ u^4 + 6u^4(\theta-\theta^3) - v^4\theta^4 \} (1-v^2); \end{aligned}$$

but instead of θ^2v^2 writing u^2v^2 , the expression on the right-hand side becomes divisible by u^4 ; and we find

$$\begin{aligned} v^2(ca' - c'a) &= -(1+\theta)(1-\theta)^2 [v^2(-3+4\theta) + u^4(-4\theta^3+3\theta^4)] \\ &\quad - \{ u^4 + 6u^4(\theta-\theta^3) - v^4 \} (1-v^2), \end{aligned}$$

and thence

$$\begin{aligned} -v^2(ca' - c'a) &= u^8 + u^4v^4(6\theta^2 - 10\theta^3 + 11\theta^4 - 6\theta^5 - 8\theta^6 + 10\theta^7 - 4\theta^8) \\ &\quad + v^8(-4 + 10\theta^2 - 8\theta^3 - 6\theta^4 + 11\theta^5 - 10\theta^6 + 6\theta^7) + v^8. \end{aligned}$$

Similarly we have

$$\begin{aligned} v^2(bc' - b'c) &= u^8(5 - 5\theta^2 + 4\theta^4 - 5\theta^5 + 2\theta^6) + u^4v^4(9\theta - 16\theta^2 + \theta^3 + 10\theta^4 + \theta^5 - 16\theta^6 + 9\theta^7) \\ &\quad + v^8(2 - 5\theta + 4\theta^2 - 5\theta^3 + 5\theta^4), \end{aligned}$$

$$\begin{aligned} -v^2(ab' - a'b) &= u^8(\theta + \theta^2 - \theta^4) + v^4\{2 - 5\theta + 4\theta^2 + 3\theta^3 - 10\theta^4 + 3\theta^5 + 4\theta^6 - 5\theta^7 + 2\theta^8\} \\ &\quad + v^8(-1 + \theta + \theta^2); \end{aligned}$$

say these values are

$$\begin{aligned} u^8 + pu^4v^4 + qv^4 + v^8, \\ \lambda u^8 + \mu u^4v^4 + \nu v^8, \\ \rho u^4v^4 + \sigma v^4 + \tau v^8. \end{aligned}$$

The required equation is thus

$$0 = (u^8 + pu^4v^4 + qv^8 + v^8)^2 - 4(\lambda u^8 + \mu u^4v^4 + \nu v^8)(\rho u^4v^4 + \sigma v^4 + \tau v^8),$$

viz. the function is

$$\begin{aligned} u^{16} + v^{16} \\ + u^{12}v^4(2p - 4\lambda\rho) + u^8v^8(2q\theta^2 + p^2 - 4\lambda\sigma\theta^2 - 4\mu\rho) \\ + u^4v^{12}(2p\theta^2 + q^2 - 4\mu\sigma\theta^2 - 4\nu\rho) + v^{16}(2q - 4\nu\sigma), \end{aligned}$$

or say it is

$$(1, b, c, d, e, f, g \mid u^{12}v^4, u^8v^8, u^4v^{12}, v^{16}).$$

Supposing that this has a factor $u^4 + v^4 - \Theta$, the form is

$$(u^8 + Bu^4v^4 + C + Dv^4 + v^8)(u^8 - \Theta + v^8)$$

and comparing coefficients we have

$$\begin{aligned} B - \Theta &= b, \\ C - \Theta B + \Theta^2 &= c, \\ D\Theta^2 - \Theta C + B\Theta^2 &= d, \\ \Theta C - \Theta^2 D + C &= e, \\ -\Theta^2 + D &= f, \end{aligned}$$

where Θ has the before-mentioned value

$$\Theta = (8, -28, +56, -70, +56, -28, +8 \mid \theta, \theta^2, \theta^3, \theta^4, \theta^5, \theta^6, \theta^7).$$

From the first, second, and fifth equations,

$$B = b + \Theta, \quad C = c + \Theta B - \Theta^2, \quad D = f + \Theta^2;$$

and the third and fourth equations should then be verified identically.

Writing down the coefficients of the different powers of θ , we find

$$\begin{aligned} 2p &= +12, -20, +22, -12, -16, +20, -8(\theta, \theta^2, \dots, \theta^7), \\ -4\lambda\rho &= -20, +20, -36, +60, -44, +36, -28, +8, \\ b &= 0, -8, +20, -56, +82, -56, +20, -8, \end{aligned}$$

$$\begin{aligned} e &= 0, +8, -28, +56, -70, +56, -28, +8, \\ \therefore B &= 0, -8, +12, 0, -8, \end{aligned}$$

that is,

$$B = -8\theta^2 + 12\theta^3 - 8\theta^5;$$

and in precisely the same way the fifth equation gives

$$D = -8\theta^2 + 12\theta^5 - 8\theta^5.$$

We find similarly C from the second equation: writing down first the coefficients of $2q\theta^2$, $-4\lambda\sigma\theta^2$, and $-4\mu\rho$, the sum of these gives the coefficients of c ; and then writing underneath these the coefficients of $B\Theta$ and of $-\Theta^2$, the final sum gives the coefficients of C : the coefficients of each line belong to $(\theta^2, \theta^3, \dots, \theta^8)$.

36	0	-120	+132	+28	-316	+361
-20	-340	+396	-144	-112	+164	-80
+16	-8	+20	-16	-12	+22	-20
0	+12	-40	+140	-212	+140	+80
	-188	+168	-92	-64	+176	-164
	+80	-16	-36	+64	-40	+60
	-72	+28	0	+68	-100	+36
	0	+64	-208	+352	-272	-160
	+463	-160	-272	+352	-208	
	+64	-64	+224	-352	+224	
			+160	-392	+160	+224
			-352	+224	-64	
				-1	0	0
				0	+16	0
					-48	0
					+70	0
					-48	
					+16	
					0,	

that is,

$$C = 16\theta^4 - 48\theta^5 + 70\theta^6 - 48\theta^7 + 16\theta^8;$$

and in precisely the same way this value of C would be found from the fourth equation.

There remains to be verified only the fourth equation

$$(D\Theta + B)\Theta^2 - \Theta C = d,$$

that is,

$$2\theta^2(-8\theta^2 + 12\theta^3 - 8\theta^5) - \Theta C = (2 - 4\lambda\tau)\theta^2 + (2\rho q - 4\mu\sigma - 4\nu\rho)\theta^3,$$

and this can be effected without difficulty.

The factor of the modular equation thus is

$$u^8 + v^8 + (-8\theta^2 + 12\theta^3 - 8\theta^5)(u^4 + v^4) + 16\theta^4 - 48\theta^5 + 70\theta^6 - 48\theta^7 + 16\theta^8,$$

viz. this is

$$\begin{aligned} (u^4+v^4)^2+(-4\theta^2+6\theta^3-4\theta^5)\cdot 2(u^4+v^4)+16\theta^4-48\theta^5+70\theta^6-48\theta^7+16\theta^8 \\ = (u^4+v^4-4\theta^2+6\theta^3-4\theta^5)^2, \\ = \{ (u^4 - v^4)^2 - 4\theta^2(1 - \theta^2)^2 \}^2, \end{aligned}$$

that is,

$$\{u^4 - v^4 - 2\theta(1 - \theta^2)\}^2 \{u^4 - v^4 + 2\theta(1 - \theta^2)\}^2 (u^4 + v^4 - \Theta) = 0;$$

viz. the first and second factors are the modular equations of the cubic transformation; and we have thus the septic transformation given by the equation

$$u^8+v^8+(-8\theta^2+12\theta^3-8\theta^5)(u^4+v^4)+16\theta^4-48\theta^5+70\theta^6-48\theta^7+16\theta^8 = 0,$$

or, what is the same thing,

$$\begin{aligned} u^8 + v^8 - 8u^2v^2(u^4 + v^4) + 12u^3v^3(u^4 + v^4) - 8u^5v^5(u^4 + v^4) \\ + 16u^4v^4 - 48u^5v^5 + 70u^6v^6 - 48u^7v^7 + 16u^8v^8 = 0. \end{aligned}$$

The preceding investigation completes the theory of the septic transformation: it may be remarked that writing $u^4 + v^4 = U$, $\Theta = 8\theta - 28\theta^2 + 56\theta^3 - 70\theta^4 + 56\theta^5 - 28\theta^6 + 8\theta^7$, the modular equation is $U^2 - \Theta = 0$, or say $U = \sqrt{\Theta}$; and then in virtue of the equation

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \sqrt{\frac{1-\beta x + \gamma x^2 - \delta x^3}{1+\beta x + \gamma x^2 + \delta x^3}},$$

the value of M is given by the formula

$$M = \frac{1}{7} \left(1 + 2 \frac{u^4}{v} \frac{dv}{du^4} \right).$$

693. A Tenth Memoir on Quantics.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXIX., Part II. (1878), pp. 603–661. Received June 12, — Read June 20, 1878.]

THE PRESENT MEMOIR, which relates to the binary quintic $(a, b, c, d, e, f \mid x, y)^5$, has been in hand for a considerable time: the chief subject-matter was intended to be the theory of a canonical form which was discovered by myself and is briefly noticed in Salmon's *Higher Algebra*, 3rd Ed. (1876), pp. 217, 218; writing $a, b, c, d, e, f, g, \dots, u, v, w$ to denote the 23 covariants of the quintic, then a, b, c, d, f are connected by the relation

$$f^2 = -a^3d + a^2bc - 4c^3;$$

and the form contains these covariants thus connected together, and also e ; it, in fact, is

$$(1, 0, c, f, a^2b - 3c^2, a^2e - 2cf \mid x, y)^5.$$

But the whole plan of the Memoir was changed by Sylvester's discovery of what I term the Numerical Generating Function (N.G.F.) of the covariants of the quintic, and my own subsequent establishment of the Real Generating Function (R.G.F.) of the same covariants. The effect of this was to enable me to establish for any given degree in the coefficients and order in the variables, or as it is convenient to express it, for any given deg-order whatever, a selected system of powers and products of the covariants, say a system of "segregates": these are aszygetic, that is, not connected together by any linear equation with numerical coefficients; and they are also such that every other combination of covariants of the same deg-order, say every "congregate" of the same deg-order, can be expressed (and that, obviously, in one way only) as a linear function, with numerical coefficients, of the segregates of that deg-order. The number of congregates of a given deg-order is precisely equal to the number of the independent syzygies of the same deg-order, so that these syzygies give in effect the congregates in terms of the segregates: and the proper form in which to exhibit the syzygies is thus to make each of them give a single congregate in terms of the segregates: viz. the left-hand side can always be taken to be a monomial congregate $a^\alpha b^\beta \dots$ or, to avoid

fractions, a numerical multiple of such form; and the right-hand side will then be a linear function, with numerical coefficients, of the segregates of the same deg-order. Supposing such a system of syzygies obtained for a given deg-order, any covariant function (rational and integral function of covariants) is at once expressible as a linear function of the segregates of that deg-order: it is, in fact, only necessary to substitute therein for every monomial congregate its value as a linear function of the segregates. Using the word covariant in its most general sense, the conclusion thus is that every covariant can be expressed, and that in one way only, as a linear function of segregates, or say in the *segregate form*.

Reverting to the theory of the canonical form, and attending to the relation $f^2 = -a^3d + a^2bc - 4c^3$, it thereby appears that every covariant multiplied by a power of the quintic itself a , can be expressed, and that in one way only, as a rational and integral function of the covariants a, b, c, d, e, f , linear as regards f : say every covariant multiplied by a power of a can be expressed, and that in one way only, in the “standard” form; as an illustration, take

$$a^3h = 6acd + 4bc^2 - ef.$$

Conversely, an expression of the standard form, that is, a rational and integral function of a, b, c, d, e, f , linear as regards f , not explicitly divisible by a , may very well be really divisible by a power of a (the expression of the course containing one or more of the higher covariants g, h , &c.), and we say that in this case the expression is *divisible*, and has for its *divided form* the quotient expressed as a rational and integral function of covariants. Observe that in general the divided form is not perfectly definite, only becoming so when expressed in the before-mentioned segregate form, and that this further reduction ought to be made. There is occasion, however, to consider these divided forms, whether or not thus further reduced; and moreover it sometimes happens that the non-segregate form presents itself, or can be expressed, with integer numerical coefficients, while the coefficients of the corresponding segregate form are fractional.

The canonical form is peculiarly convenient for obtaining the expressions of the several derivatives (Gordan’s *Uebereinanderschiebungen*) $(a, b)^1$, $(a, b)^2$, &c., (or as I propose to write them $ab1$, $ab2$, &c.), which can be formed with two covariants, the same or different, as rational and integral functions of the several covariants. It will be recollected that in Gordan’s theory these derivatives are used in order

to establish the system of the 23 covariants: but it seems preferable to have the system of covariants, and by means of them to obtain the theory of the derivatives.

I mention at the end of the Memoir two expressions (one or both of them due to Sylvester) for the N.G.F. of a binary sextic.

The several points above adverted to are considered in the Memoir; the paragraphs are numbered consecutively with those of the former Memoirs upon Quantics.

The Numerical and Real Generating Functions. Art. Nos. 366 to 374, and Table No. 96.

366. I have, in my Ninth Memoir (1871) [462], given what may be called the Numerical Generating Function (N.G.F.) of the covariants of a quartic; this was

$$A(x) = \frac{1 - ax^4 \cdot 1 - ax^8 \cdot 1 - ax^{12} - ax^{14} - ax^{16}}{(1 - ax^4)(1 - ax^8)(1 - ax^{12})},$$

the meaning being that the number of asyzygetic covariants $a^\theta x^\mu$, of the degree in the coefficients and order μ in the variables, or say of the deg-order $\theta.\mu$, is equal to the coefficient of $a^\theta x^\mu$ in the development of this function. And I remarked that the formula indicated that the covariants were made up of $(ax^4, ax^8, ax^{12}, a^3, ax^{18})$, the quartic itself, the Hessian, the quadrinvariant, the cubinvariant, and the cubicovariant, these being connected by a syzygy $a^3 x^{12}$ of the degree 6 and order 12. Calling these covariants a, b, c, d, e , so that these italic small letters stand for covariants,

<i>g</i> -order	
1.4	a ,
2.0	b ,
2.4	c ,
3.0	d ,
3.6	e ,

then it is natural to consider what may be called the Real Generating Function (R.G.F.): this is

$$\frac{1}{(1 - a)(1 - b)(1 - c)(1 - d)(1 - e)},$$

the development of this contains, as it is easy to see, only terms of the form $a^\alpha b^\beta c^\gamma d^\delta$ and $a^\alpha b^\beta c^\gamma d^\delta e$, each with the coefficient +1, so

that the number of terms of a given deg-order $\theta.\mu$, is equal to the coefficient of $a^\theta x^\mu$ in the first-mentioned function: and these terms of a given deg-order represent the aszygetic covariants of that deg-order: any other covariant of the same deg-order is expressible as a linear function of them. For instance, deg-order 6.12, the terms of the R.G.F. are a^2c , a^2bc , &c.; there is one more term e^2 of the same deg-order; hence e^2 must be a linear function of these and in fact

$$e^2 = -d^3 + a^2bc - 4c^3, \quad \text{viz. this is the equation } f^2 = -a^3d + a^2bc - 4c^3.$$

367. Sylvester obtained an expression for the N.G.F. of the quintic: this is

$$\begin{aligned}
 & a^2 \cdot 1 \\
 & + a^3 \cdot (x^2 + x^6 + x^8) \\
 & + a^4 \cdot (x^4 + x^8) \\
 & + a^5 \cdot (x + x^7 + x^9 - x^{11}) \\
 & + a^6 \cdot (x^3 + x^5 + x^9 - x^{11}) \\
 & + a^7 \cdot (x^2 + x^4 + x^8 - x^{10}) \\
 & + a^8 \cdot (x^2 + x^6 - x^8 - x^{10}) \\
 & + a^9 \cdot (x + x^5 - x^7 - x^9) \\
 & + a^{10} \cdot (x^3 - x^5 - x^9) \\
 & + a^{11} \cdot (x - x^3 - x^7) \\
 & + a^{12} \cdot (x^2 - 3x^6 - x^{10}) \\
 & + a^{13} \cdot (-x - x^3) \\
 & + a^{14} \cdot (1 - x^4 - x^6 - x^8) \\
 & + a^{15} \cdot (-x^2 - x^4) \\
 & + a^{16} \cdot (-x^2 - x^6 - x^8) \\
 & \div (1 - ax^2)(1 - ax^4)(1 - ax^6)(1 - ax^8 - ax^{10} - ax^{12});
 \end{aligned}$$

viz. expanding this function in ascending powers of a , x , then, if a term is $Na^\theta x^\mu$, this means that there are precisely N aszyzygetic covariants of the deg-order $\theta.\mu$.

368. It is known that the number of the irreducible covariants of the binary quintic is = 23; representing these by the letters $a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w$, (a the quintic itself), the deg-orders of these, and the references to the tables which give them are

Deg-order.		Tab.	Mem.
1.5	<i>a</i>	13	2
2.2	<i>b</i>	14	—
2.6	<i>c</i>	15	—
3.3	<i>d</i>	16	—
3.0	<i>e</i>	17	—
3.9	<i>f</i>	18	—
4.0	<i>g</i>	19	—
4.4	<i>h</i>	20	—
4.6	<i>i</i>	21	—
5.1	<i>j</i>	22	—
5.3	<i>k</i>	23	—
5.7	<i>l</i>	24	—
6.2	<i>m</i>	83	8
6.4	<i>n</i>	84	—
7.1	<i>o</i>	90*	9
7.0	<i>p</i>	91	9
8.0	<i>q</i>	25	2
8.2	<i>r</i>	92	9
9.3	<i>s</i>	—	—
11.1	<i>t</i>	94	9
12.0	<i>u</i>	29	3
13.1	<i>v</i>	95	9
18.0	<i>w</i>	29a	5.

[*See also the paper, 143, in the second volume of this collection.]

Starting from the foregoing expression of the N.G.F. of the quintic, we can, instead of each term $a^\theta x^\mu$, introduce a covariant or product of covariants of the proper deg-order $\theta.\mu$: the mode of doing this depends of course on the different admissible partitions of θ, μ , and it is for some of the terms very indeterminate: for instance, $a^4 x^8$ is ai , bf , or ce . I found it possible to perform the whole process so as to satisfy a condition which will be presently referred to; and I found

369. R.G.F. of the quintic:

$$\begin{aligned}
 &\{1 + d(1 - ag^2) + e(1 - b^2) + f(1 - b) + h(1 - ag^2) + i(1 - b^2g) \\
 &\quad + j(1 - ag^2) + k(1 - b^2) + l(1 - bg) + m(1 - ag^2) + n(1 - b^2g) \\
 &\quad + o(1 - b^2) + p(1 - b^2g) + r(1 - g) \\
 &\quad + dj(1 - af) + s(1 - abg) + hj(1 - a^2) + j^2(1 - ag^2) \\
 &\quad + jk(1 - b^2g) + t(1 - b^2) + jm(1 - ag^2) + jo(1 - bg) \\
 &\quad + v(1 - b^2) + js(1 - bg) + jt(1 - g) + w(1 - a)\} \\
 &\quad \div \{(1 - a)(1 - b)(1 - c)(1 - g)(1 - q)(1 - u)\},
 \end{aligned}$$

where observe that each negative term of the numerator is equal to a positive term multiplied by a power or product of terms a , b , g , contained in the denominator; this is the condition above referred to. The expansion thus consists only of terms each with the coefficient +1; for instance, a part of the function is

$$\frac{s(1 - abg)}{(1 - a)(1 - b)(1 - c)(1 - g)(1 - q)(1 - u)} = \frac{s}{(1 - c)(1 - q)(1 - u)} \cdot \frac{1 - abg}{(1 - a)(1 - b)(1 - g)}.$$

where the first factor is the entire series of terms $sc^\gamma q^\rho u^\sigma$, and the second factor is the series of terms $a^\alpha b^\beta g^\gamma$ omitting only those terms which are divisible by abg : and in the product of the two factors the terms are all distinct, so that the coefficients are still each = 1. The same thing is true for every other pair of numerator terms and since the terms arising from each such pair are distinct from each other, in the expansion of the entire function the coefficients are each = +1. Hence (as in the case of the quartic) for any given deg-order, the terms in the expansion of the R.G.F. may be taken for the aszygetic covariants of that deg-order; and if there are any other terms of the same deg-order, each of these must be a linear function, with numerical coefficients, of these aszygetic covariants: thus deg-order 6.14, the expansion contains only the terms a^2h , acd , bc^2 ; there is besides a term of the same deg-order, ef , which is not a term of the expansion, and hence ef must be a linear function of a^2h , acd , bc^2 ; we in fact have $ef = a^2h - 6acd - 4bc^2$.

The terms in the expansion of the R.G.F. may be called "segregates," and the terms not in the expansion "congregates"; the theorem thus is: every congregate is a linear function, with determinate numerical coefficients, of the segregates of the same deg-order.

370. It is to be noticed that the positive terms of the numerator are unity, the seventeen covariants $d, e, f, h, i, j, k, l, m, n, o, p, r, s, t, v, w$, and the products of j by (d, h, j, k, m, o, s, t) , where j^2 is reckoned as a product; in all, 26 terms. Disregarding the negative terms of the numerator the expansion would consist of these 26 terms, each multiplied by every combination whatever $a^\alpha b^\beta c^\gamma g^\delta q^\epsilon u^\zeta$ of the denominator terms a, b, c, g, q, u (which for this reason might be called “reiterative”): the effect of the negative terms of the numerator is to remove from the expansion certain of the terms in question, thereby diminishing the number of the segregates: thus as regards the terms belonging to unity, any one of these which contains the factor b^2 is not a segregate but a congregate: and so as regards the terms belonging to d , any one of these which contains the factor ag^2 is a congregate: and the like in other cases. For a given deg-order we have a certain number of segregates and a certain number of congregates: and the number of independent syzygies of that deg-order is precisely equal to the number of congregates: viz. each such syzygy may be regarded as giving a congregate in terms of the segregates: we have on the left-hand side a congregate, or, to avoid fractions, a numerical multiple of the congregate, and on the right-hand side a linear function, with numerical coefficients, of the segregates.

371. The syzygy is *irreducible* or *reducible*; and in the latter case it is, or is not, *simply divisible*: viz. if the congregate on the left-hand side contains any congregate factor (the other factor being literal), then the syzygy is reducible: it is, in fact, obtainable from the syzygy (of a lower deg-order) which gives the value of such congregate factor. But there are here two cases; multiplying the lower syzygy by the proper factor, the right-hand side may still contain segregates only, and then no further step is required: the original syzygy is nothing else than this lower syzygy, each side multiplied by the factor in question, and it is accordingly said to be simply divisible (S.D.). But contrariwise, the right-hand side, as multiplied, may contain congregates which have to be replaced by their values in terms of the segregates of the same deg-order: the resulting expression is then no longer explicitly divisible by the introduced factor: and the original syzygy, although arising as above from a lower syzygy, is not this lower syzygy each side multiplied by a factor: viz. it is in this case not simply divisible.

For example (see the subsequent Table No. 96, under the indicated

deg-orders) (6.6), from the syzygy

$$9d^2 = aj - b^2 + 2bh - cg,$$

we deduce (7.11) the syzygy

$$9d^2e = aej - ab^2 + 2abh - acg,$$

which (all the terms on the right-hand being segregates) requires no further reduction: it is a reducible and simply divisible syzygy. But we have (6.8) a syzygy giving de , and also (6.10) a syzygy giving e^2 ; multiplying the former of these by e or the latter of them by d , we obtain values of de^2 , but in each case the right-hand sides contain terms which are not segregates, and have thus to be further reduced: the final formula (9.13) is

$$3de^2 = -4a^2bj + 3a^2bi + 4a^2eg - 8ab^2i + 7abcg - 12bc^2d,$$

which is not divisible by any factor: the syzygy is thus reducible, but not simply divisible.

A syzygy, which is not in the sense explained reducible, is said to be *irreducible*.

372. The number of irreducible syzygies is obviously finite: it has, however, the large value 179 as appears from the annexed diagram, showing the congregates determined by these several syzygies, and the deg-orders of the syzygies:

[The diagram from p. 347 showing the distribution of irreducible syzygies is here omitted for brevity, but would be reproduced in a complete edition.]

Observe as regards the foregoing diagram, that dj^2 is irreducible (since neither dj nor j^2 is segregate), and similarly j^2h , j^4 , &c., are irreducible: we have thus the last or j^4 column of the diagram.

The simply divisible syzygies are infinite in number, as are also the reducible syzygies not simply divisible. There is obviously no use in writing down a simply divisible syzygy; but as regards the reducible syzygies not simply divisible, these require a calculation, and it is proper to give them as far as they have been obtained.

373. The following Table, No. 96, replaces Tables 88 and 89 of my Ninth Memoir. The arrangement is according to deg-orders, and the table is complete up to the deg-order 8.40: it shows for each deg-order

the segregate covariants and also the congregate covariants, these last each with their numerical coefficients, viz. the left-hand column for each deg-order gives the congregates, the remaining columns the segregates, and the numbers are the numerical coefficients in the expression of each congregate in terms of the segregates: the several cases of simple divisibility are distinguished by S.D. with reference to the congregate and the syzygy from which the expression is obtained. The number of independent syzygies of each deg-order is $= c - s + 1$, if c, s are the numbers of congregates and segregates respectively; the number of independent irreducible syzygies is $= c$ (since the number of asyzygetic covariants of the deg-order being $s - c$, the number of syzygies is $s - (s - c) = c$). This last number is shown in the heading of each deg-order. [*See vol. VII. of this collection, p. 348.]

Table No. 96 (Segregates, Congregates, and Syzygies).

[Due to the extreme complexity of Table 96, spanning pp. 349–355 with extensive tabular data for each deg-order from 1.1 to 11.3, the full table is here summarized. A complete transcription would require careful reproduction of all numerical coefficients for each congregate-segregate relation across 22 separate deg-order sections.]

374. Any syzygy might of course be directly verified by substituting for the several covariants contained therein their expressions in terms of the coefficients and facients of the quintic. But it is to be remarked that among the syzygies, or easily deducible from them, we have (6.18) the before-mentioned equation $f^2 = -a^3d + a^2bc - 4c^3$, and also a set of 17 syzygies, the left-hand sides of which are the covariants $g, h, \dots, u, v, w$, each multiplied by a or a^2 , and which lead ultimately to the standard expressions of these covariants respectively, viz. each covariant multiplied by a proper power of a can be expressed as a rational and integral function of a, b, c, d, e, f , linear as regards f . Supposing them thus expressed, a far more simple verification of any syzygy would consist in substituting therein for the several covariants their expressions in the standard form, reducing if necessary by the equation $f^2 = -a^3d + a^2bc - 4c^3$: but of course, as to the syzygies used for obtaining the standard forms, this is only a verification if the standard forms have been otherwise obtained, or are assumed to be correct.

The 17 syzygies above referred to are:

$$a^2g = 12cd + 4be + 6f,$$

$$a^2k = bcd + 4bc^2 + ef,$$

$$ai = -bf + ce,$$

$$aj = b^2 - 2bh + cg + 9d^2,$$

$$ak = -2bi + 3de,$$

$$al = bci - df,$$

$$am = -b^2d - cj + 3dh,$$

$$an = b^2e - 6bl - 2ck - fg,$$

$$ao = 2be + ej,$$

$$ap = -bcn - fj,$$

$$a^2q = -2by + 4d^2f - 12deh + hj,$$

$$ar = 6bk + 6p - co + hr,$$

$$a^2s = 3bdk + 3df + 2im,$$

$$a^2t = cy + jp - 2mn,$$

$$18a^2u = 2agq + b^2j + 6bmj - 6df - ghj + no,$$

$$3au = 2b^2r - 8bef - 2b^2mt + 6b^2l - 12b^2h^2 + 3ct,$$

$$18atu = 3b^2rs + b^2qo - 4bfp - bgnio + 18bm^2 + 3dgjo - 18djt - 3ght - 6m^2o,$$

the last four of these being, however, beyond the limits of the table: the expressions of g, h, i are here in the standard form: the standard forms of the other covariants $j, k, \dots, u, v, w$, will be given further on.

Theory of the Canonical Form. Art. Nos. 375 to 381, and Tables Nos. 97 and 98.

375. As the small italic letters have been used to represent the covariants, different letters are required for the coefficients of the quintic: using also new letters for the facients, I take the quintic to be $(a, b, c, d, e, f \mid \xi, \eta)^5$. Considering a linear transformation of $(a, b, c, d, e, f \mid \xi, \eta)^5$, viz.

$$(a, b, c, d, e, f \mid \xi - b\eta, \eta)^5,$$

this is

$$\begin{aligned} a(\xi^5) &+ b(-5b\xi^4\eta + 10b^2\xi^3\eta^2 - 10b^3\xi^2\eta^3 + 5b^4\xi\eta^4 - b^5\eta^5) \\ &+ c(10b^2\xi^3\eta^2 - 30b^3\xi^2\eta^3 + 40b^4\xi\eta^4 - 20b^5\eta^5) \\ &+ d(-10b^3\xi^2\eta^3 + 40b^4\xi\eta^4 - 40b^5\eta^5) \\ &+ e(5b^4\xi\eta^4 - 15b^5\eta^5) \\ &+ f(-b^5\eta^5), \end{aligned}$$

which is

$$\begin{aligned} = \{ &a\xi^5 \\ &+ (ae + 1)\xi^4\eta \\ &+ (ac + 1)\xi^3\eta^2 \\ &+ (ace + 1)\xi^2\eta^3 \\ &+ (a^2f + 1)\xi\eta^4 \\ &+ (a^2d + 1)\eta^5\}, \end{aligned}$$

ξ^5	$\xi^4\eta$	$\xi^3\eta^2$	$\xi^2\eta^3$	$\xi\eta^4$	η^5
b^2	-1	abc	-3	a^2be	-5
	+2	ab^2c	+6	a^2b^2c	+10
	-3		-10		-10
	+4		+10		+4

The values of a, b, c, d, e, f , considered for a moment as denoting the leading coefficients of the several covariants ultimately represented by these letters respectively, are

	a	b	c	d	e	f
a	$a + 1$	$ae + 1$	$ac + 1$	$ace + 1$	$a^2f + 1$	$a^2d + 1$
b		$bd - 4$	$b^2 - 1$	$ad^2 - 1$	$abe + 5$	$abc - 3$
c			$c^2 + 3$	$b^2e - 1$	$ac + 2$	$b^2 - 2$
d				$bcd + 2$	$b^2d + 8$	$bc^2 - 10$
e					$c^2 - 1$	

satisfying, as they should do, the relation $f^2 = -a^3d + a^2bc - 4c^3$.

Hence forming the values of $af - 3c^2$ and $ae - 2cf$, it appears that the value of the last-mentioned quintic function is

$$(1, 0, c, f, a^2b - 3c^2, a^2e - 2cf \mid \xi, \eta)^5.$$

Writing herein x, y in place of ξ, η , and now using a, b, c, d, e, f to denote, not the leading coefficients but the covariants themselves (a denoting the original quintic, with ξ, η as facients), we have the form

$$A = (1, 0, c, f, a^2b - 3c^2, a^2e - 2cf \mid x, y)^5,$$

a new quintic, which is the *canonical form* in question: the covariants hereof (reckoning the quintic itself as a covariant) will be written $A, B, C, \dots, V, W$, and will be spoken of as *capital covariants*.

376. The fundamental property is: Every capital covariant, say I , has for its leading coefficient the corresponding covariant i multiplied by a power of a : and this follows as an immediate consequence of the foregoing genesis of A . The covariant i of the form $(a, b, 0, d, e, f \mid \xi, \eta)^5$ has a leading coefficient $= i(a^2cf - a^3e + \&c.)$, which, when $a, b, c, d, e, f, \dots, i$ denote leading coefficients, is $= i$ multiplied by a power of a : and upon substituting for the quintic the linear transformation thereof $(1, 0, c, f, a^2b - 3c^2, a^2e - 2cf \mid \xi, \eta)^5$, (observing that, in the transformation ξ, η into $\xi - b\eta, a\eta$, the determinant of substitution is $= a$), the value is still $= i$ multiplied by a power of a ; or using the relation $a = a$, say the value is $= i$ multiplied by a power of a . Now the covariant i is the same function of the covariants a, b, c, d, e, f that the leading coefficient i is of the leading coefficients a, b, c, d, e, f ; hence, the italic letters now denoting covariants, the leading coefficient still is $= i$ multiplied by a power of a : which is the above-mentioned theorem.

377. To show how the transformation is carried out, consider, for example, the covariant B . This is obtained from the corresponding

covariant of $(a, b, c, d, e, f \mid \xi, \eta)^5$, that is

$$\begin{array}{ccc} ae & af & bf \\ bd - 4 & be - 3 & cd - 4 \\ c^2 + 1 & & d^2 + 3 \end{array} \mid \xi, \eta)^2,$$

by changing the variables, and for the coefficients a, b, c, d, e, f writing $1, 0, c, f, a^2b - 3c^2, a^2e - 2cf$; thus the coefficients are

First	Second	Third
$1(a^2b - 3c^2)$	$1(a^2e - 2cf) - 4c$ $+ 3c^2$	$(a^2b - 3c^2)(d^2b - 3c^2)$ $+ 2cf$ $- 1(3f^2)$
$= a^2b,$	$= a^2e,$	$= a^2b^2 - 4a^2bc + 12c^4$ $+ 3(-a^3d + a^2bc - 4c^3)$ $= a^2(-3ad - bc)$

and we have thus the expression of B (see the Table No. 97); and similarly for the other capital covariants $C, D, \dots, V, W$: in every case the coefficients are obtained in the standard form, that is, as rational and integral functions of a, b, c, d, e, f , linear as regards f .

378. It will be observed that there is in each case a certain power of a which explicitly divides all the coefficients and is consequently written as an exterior factor: disregarding these exterior factors, the leading coefficients for B, C, D, E , are b, c, ad, e, f respectively; that for G is $12ahd$; and similarly for the other covariants. The number of the capital covariants $A, B, \dots, W$ is of course $= 23$; and their deg-orders (reckoning the quintic itself as a covariant) are given in the following Table No. 97, which exhibits the capital covariants in the a^p or standard forms.

Table No. 97 (Covariants of A , in the a^p or standard forms).

$A = (1, c + 1, f + 10, a^2b + 5, a^2e + 1 \mid x, y)^5.$

Deg-orders: 0.0, 1.3, 2.6, 3.9, 4.12, 5.15

$B = a^2(b + 1, e + 1, ad - 3 \mid x, y)^2.$

Deg-orders: 2.2, 3.5, 4.8

$C = -(c + 1, f + 1, d^2b + 3, c^2b + 1, a^2d + 6, a^2f - 10, a^2bc - 3 \mid x, y)^2.$

Deg-orders: 2.6, 3.9, 4.12, 5.15, 6.18, 7.21, 8.24

[Tables No. 97 and No. 98 continue with extensive tabular data spanning pp. 362–373 and 370–376 respectively, presenting the full expressions of all 23 capital covariants A through W in standard form, and their divided/segregate forms. Due to the extraordinary complexity and length of these tables, with coefficients involving terms up to degree 23 and order 49, the full tabular data is here summarized. A complete edition would reproduce all coefficients in full.]

379. It is obvious that for every covariant whatever written in the denumerate form $(f_0, f_1, \dots | x, y)^n$, the second coefficient is equal to the first coefficient operated upon by δ ; so that the foregoing formulae give, in fact, the second coefficients of the several covariants.

380. The following are the values of the second coefficients $B_1, C_1,$

$\dots, W_1$ of the several covariants $B, C, \dots, W$:

$$\begin{aligned}
 B_1 &= -(bg + cd), \\
 C_1 &= -cg + d^2f - 3deh, \\
 D_1 &= -(ch + d^2g), \\
 E_1 &= -(ci + df^2 - 6dfh + 4eg), \\
 F_1 &= -(cj + 4d^2k - 3ef), \\
 G_1 &= 0, \\
 H_1 &= 0, \\
 I_1 &= -(cl + 3dk^2 - 2em - 2fn + 2gp), \\
 J_1 &= 0, \\
 K_1 &= -(cm + 2dl + ej - 4en + 4fo - 3gq - hr), \\
 L_1 &= -(cn - 3dm + 5eo - 4ep - 2fs + gt), \\
 M_1 &= 0, \\
 N_1 &= -(co - 2dn + 3eq - 6er - 2fu + gv), \\
 O_1 &= 0, \\
 P_1 &= -(cp + dq - er - 2et - fu + 5ew - 2hv), \\
 Q_1 &= 0, \\
 R_1 &= -(cq + 2dr - 2es - 4eu - fv + gw), \\
 S_1 &= -(cr - ds + 6et - 2eu - 3ev - fw), \\
 T_1 &= -(cs + 2dt - 6eu - 2ev - fw), \\
 U_1 &= 0, \\
 V_1 &= -(cu + 2dv - 2ew - fx), \\
 W_1 &= 0.
 \end{aligned}$$

381. It is worth noticing how very much the formulae of Table No. 97 simplify themselves, if one of the covariants b, c, d, e vanishes, in particular, if b vanishes. Suppose $b = 0$; writing also (although this

makes but little difference) $a = 1$, we have

$$\begin{aligned}
 a &= 1, & h &= 6cd + ef, \\
 b &= 0, & i &= ce, \\
 c &= c, & j &= 9d^2 + c^3, \\
 d &= d, & k &= 3de, \\
 e &= e, & l &= -3df + 2c^2e, \\
 f &= -d - 4c^3, & m &= 9cd^2 + 3def - c^2e, \\
 g &= c^2, & n &= -6cde - c^2f, \\
 & & o &= 9d^3 + c^4e, \\
 & & p &= -9d^2f + 12d^2e + cef, \\
 & & q &= -54cd^3 - 27d^2ef + 18c^2de^2 + c^3f, \\
 & & r &= 9cd^2e + 3de^2f - c^2e^3, \\
 & & s &= -27d^4 + 5c^3d^2e + dcde^2f - 2c^4e^2, \\
 & & t &= -81d^3f - 6d^2e^2 + 216c^3d^2e + 51cd^2ef - 24c^3de^2 - c^5ef, \\
 & & u &= -21d^4 - 18cd^2f - 4d^2e^2f + c^3de^2, \\
 & & v &= -54d^3ef - 6d^2e^2 + 216c^3d^2e^2 + 54cd^2ef - 24c^3de^2 - c^5ey, \\
 & & w &= (\text{not calculated}).
 \end{aligned}$$

These values are very convenient for the verification of syzygies, &c.

Derivatives. Art. Nos. 382 to 384, and Tables Nos. 99 and 100.

382. I call to mind that any two covariants a, b , the same or different, give rise to a set of derivatives $(a, b)^1, (a, b)^2, (a, b)^3$, &c., or, as I propose to write them, $ab1, ab2, ab3$, &c., viz.

$$\begin{aligned} ab1 &= d_x a \cdot d_y b - d_y a \cdot d_x b, \\ ab2 &= d_x^2 a \cdot d_y^2 b - 2d_x d_y a \cdot d_x d_y b + d_y^2 a \cdot d_x^2 b, \\ ab3 &= d_x^3 a \cdot d_y^3 b - 3d_x^2 d_y a \cdot d_x d_y^2 b + 3d_x d_y^2 a \cdot d_x^2 d_y b - d_y^3 a \cdot d_x^3 b, \\ &\&c. \end{aligned}$$

or, as these are symbolically written,

$$ab1 = 12a_1b_2, \quad ab2 = 12^2a_2b_2, \quad ab3 = 12^3a_3b_2, \quad \&c.;$$

where

$$12 = \frac{d}{dx_1} \frac{d}{dy_2} - \frac{d}{dx_2} \frac{d}{dy_1},$$

the differentiations $\frac{d}{dx_1}, \frac{d}{dy_1}$ applying to the a , and the $\frac{d}{dx_2}, \frac{d}{dy_2}$ applying to the b , but the suffixes being ultimately omitted: hence if θ be the index of derivation, the derivative is thus a linear function of the differential coefficients of the order θ of the two covariants a and b respectively: and we have the general property that any such derivative, if not identically vanishing, is a covariant. If the a and the b are one and the same covariant, then obviously every odd derivative is $= 0$; so that in this case the only derivatives to be considered are the even derivatives $aa2, aa4$, &c.: moreover, if the index of derivation θ exceeds the order of either of the component covariants, then also the derivative is $= 0$: in particular, neither of the covariants must be an invariant. The degree of the derivative is evidently equal to the sum of the degrees of the component covariants; the order is equal to the sum of the orders less twice the index of derivation.

383. It was by means of the theory of derivatives that Gordan proved (for a binary quantic of any order) that the number of covariants was finite, and, in the particular case of the quantic, established the system of the 23 covariants. Starting from the quantic itself a , then the system of derivatives $aa2, aa4$, &c., must include among itself all the covariants of the second degree, and if the entire system of these is, suppose, b, c , &c., then the derivatives $ab1, ab2$, &c., $ac1, ac2$, &c., must include among them all the covariants of the third

degree, and so on for the higher degrees: and in this way, limiting by general reasoning the number of the independent covariants of each degree obtained by the successive steps, the foregoing conclusion is arrived at. But returning to the quintic, and supposing the system of the 23 covariants established, then knowing the deg-order of a derivative we know that it must be a linear function of the segregates of that deg-order; and we thus confirm, *a posteriori*, the results of the derivation theory. I annex the following Table No. 99, showing all the derivatives which present themselves, and for each of them the covariants as well congregate as segregate of the same deg-order: the congregates are distinguished each by two prefixed dots, *.bf*, &c. No further explanation of the arrangement is, I think, required. We see from the table in what manner the different covariants present themselves in connexion with the derivation-theory. Thus starting with the quintic itself *a*, we have the two derivatives *aa4*, *aa2*, which are in fact the covariants of the second degree (deg-orders 2.2 and 2.6 respectively) *b* and *c*. For the third degree we have the derivatives *ab2*, *ab1*, *ac5*, *ac4*, *ac3*, *ac2*, *ac1*: the deg-order of *ac5* is 3.1, and there being no covariants of this deg-order, *ac5* must, it is clear, vanish identically: *ab2* and *ac4* are each of them of the deg-order 3.3, but for this deg-order we have only the covariant *d*, and hence *ab2* and *ac4* must be each of them a numerical multiple of *d*; similarly, deg-order 3.5, *ab1* and *ac3* must be each of them a numerical multiple of *e*; deg-order 3.7, *ac2* must be a numerical multiple of *ah*; and deg-order 3.9, *ac1* must be a numerical multiple of *f*: the 7 derivatives, which *primâ facie* might give, each of them, a covariant of the third degree, thus give in fact only the 3 covariants *d*, *e*, *f*; and in order to show — according to the theory of derivations that this is so, it is necessary to prove 1°, that *ac5* = 0; 2°, that *ac4* and *ab2* differ only by a numerical factor; 3°, that *ab1* and *ac3* differ only by a numerical factor; 4°, that *ac1* is a numerical multiple of *ah*: which being so, we have the 3 new covariants. The table shows that for degrees 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, the number of derivatives is respectively 2, 7, 19, 29, 41, 46, 52, 46, 44, 35, 26, 19, 17, 12, 13, 6, 6, 3, 3, 1, 1, 0, 1, so that the whole number of derivatives is 429, giving the 22 covariants *b*, *c*, . . . , *w*. While it is very remarkable that (by general reasoning, as already mentioned, and with a very small amount of calculation) Gordan should have been able in effect to show this, the great excess of the number of derivatives over that of the covariants seems a reason why the derivations ought not to be

made a basis of the theory.

It is to be remarked that we may consider derivatives $pq1$, $pq2$, &c., where p , q instead of being simple covariants are powers or products of covariants, but that these may be made to depend upon the derivatives formed with the simple covariants. (As to this see my paper “On the Derivatives of Three Binary Quantics,” *Quart. Math. Journal*, t. XV. (1877), pp. 157–168, [681].)

Table No. 99 (Index Table of Derivatives).

[Table No. 99 spans pp. 378–380 and shows for each derivative (indexed by component covariants and derivation number) the corresponding covariants of the same deg-order, distinguishing congregates by prefixed dots. The complete table lists 429 derivatives across degrees 2 through 24. The summary structure is:]

Deg.	2	3	4	5	6	7	8
Derivs.	2	7	19	29	41	46	52
New covs.	b, c	d, e, f	g, h, i	j, k, l	m, n	o, p	q, r

Deg.	9	10	11	12	13	14	15
Derivs.	46	44	35	26	19	17	12
New covs.	s	t	u	v	—	—	—

Deg.	16	17	18	19	20	21
Derivs.	13	6	6	3	3	1

Deg.	22	23	24
Derivs.	1	0	1

384. The annexed Table No. 100 contains the segregate forms of the 429 derivatives, or say the 429 derivative forms: the arrangement is according to the deg-orders, and the number preceding each form is that of the derivative to which it belongs. The degrees of the derivatives extend from 2 up to 24, and the number of forms of each degree is that given in Table No. 99.

Table No. 100 (Segregate Forms of the 429 Derivatives).

[Table No. 100 occupies pp. 382–396 with the complete segregate form expressions for all 429 derivatives. Each entry gives a derivative

in terms of segregate covariants with numerical coefficients. Due to the immense extent of this table, the full data is here summarized.]

[At the end of the Memoir, Cayley mentions two expressions for the N.G.F. of a binary sextic, one or both due to Sylvester.]

[End of Papers 691–693, pages 351–400]

The Collected Mathematical Papers of Arthur Cayley

Volume X

Pages 401–450

694. Desiderata and Suggestions.

No. 1. The Theory of Groups.

[From the *American Journal of Mathematics*, t. i. (1878), pp. 50–52.]

SuBSTiTUTiONS, and (in connexion therewith) groups, have been a good deal studied; but only a little has been done towards the solution of the general problem of groups. I give the theory so far as is necessary for the purpose of pointing out what appears to me to be wanting.

Let α , β , ... be functional symbols, each operating upon one and the same number of letters and producing as its result the same number of functions of these letters; for instance,

$$\alpha(x, y, z) = (X, Y, Z),$$

where the capitals denote each of them a given function of (x, y, z) . Such symbols are susceptible of repetition and of combination:

$$\alpha^2(x, y, z) = \alpha(X, Y, Z), \quad \beta\alpha(x, y, z) = \beta(X, Y, Z),$$

= in each case three given functions of (x, y, z) ; and similarly for α^3 , $\alpha^2\beta$, &c. The symbols are not in general commutative, $\alpha\beta$ not = $\beta\alpha$; but they are associative,

$$\alpha\beta \cdot \gamma = \alpha \cdot \beta\gamma,$$

each = $\alpha\beta\gamma$, which has thus a determinate signification. The associativeness of such symbols arises from the circumstance that the definitions of α , β , γ , ... determine the meanings of $\alpha\beta$, $\alpha\gamma$, &c.: if α , β , γ , ... were quasi-quantitative symbols such as the quaternion imaginaries i , j , k , then $\alpha\beta$ and $\beta\gamma$ might have by definition values δ and ϵ such that $\alpha\beta \cdot \gamma$ and $\alpha \cdot \beta\gamma$ (= $\delta\gamma$ and $\alpha\epsilon$ respectively) have unequal values. Unity as a functional symbol denotes that the letters are unaltered,

$$1(x, y, z) = (x, y, z);$$

whence $1\alpha = \alpha 1 = \alpha$.

The functional symbols may be substitutions;

$$\alpha(x, y, z) = (y, z, x),$$

the same letters in a different order: substitutions can be represented by the notation

$$\alpha = \begin{pmatrix} xyz \\ yzx \end{pmatrix},$$

the substitution which changes xyz into yzx , or as products of cyclical substitutions,

$$\alpha = \begin{pmatrix} xyz \\ yzx \end{pmatrix} \begin{pmatrix} uw \\ wu \end{pmatrix} = (xyz)(uw),$$

the product of the cyclical interchanges x into y , y into z , and z into x ; and u into w , w into u .

A set of symbols $\alpha, \beta, \gamma, \dots$, such that the product $\alpha\beta$ of each two of them (in each order, $\alpha\beta$ or $\beta\alpha$), is a symbol of the set, is a *group*. It is easily seen that 1 is a symbol of every group, and we may therefore give the definition in the form that a set of symbols, 1, $\alpha, \beta, \gamma, \dots$ satisfying the foregoing condition is a group. When the number of the symbols (or terms) is $= n$, then the group is of the n th order; and each symbol α is such that $\alpha^n = 1$, so that a group of the order n is, in fact, a group of symbolical n th roots of unity. A group is defined by means of the laws of combination of its symbols: for the statement of these we may either (by the introduction of powers and products) diminish as much as may be the number of independent functional symbols, or else, using distinct letters for the several terms of the group, employ a square diagram as presently mentioned. Thus, in the first mode, a group is

$$1, \beta, \beta^2, \alpha, \alpha\beta, \alpha\beta^2 \quad (\alpha^3 = 1, \beta^2 = 1, \alpha\beta = \beta\alpha);$$

where observe that these conditions imply also $\alpha\beta^2 = \beta^2\alpha$. Or, in the second mode, calling the same group $(1, \alpha, \beta, \gamma, \delta, \epsilon)$, the laws of combination are given by the square diagram

	1	α	β	γ	δ	ϵ
1	1	α	β	γ	δ	ϵ
α	α	1	γ	β	ϵ	δ
β	β	γ	ϵ	δ	α	1
γ	γ	β	δ	ϵ	1	α
δ	δ	ϵ	1	α	β	γ
ϵ	ϵ	δ	α	1	γ	β

for the symbols $(1, \alpha, \beta, \gamma, \delta, \epsilon)$ are in fact $= (1, \alpha, \beta, \alpha\beta, \beta^2, \alpha\beta^2)$.

The general problem is to find all the groups of a given order n ; thus if $n = 2$, the only group is $1, \alpha$ ($\alpha^2 = 1$); if $n = 3$, the only group is $1, \alpha, \alpha^2$ ($\alpha^3 = 1$); if $n = 4$, the groups are $1, \alpha, \alpha^2, \alpha^3$ ($\alpha^4 = 1$), and $1, \alpha, \beta, \alpha\beta$ ($\alpha^2 = 1, \beta^2 = 1, \alpha\beta = \beta\alpha$); if $n = 6$, there are three groups, a group $1, \alpha, \alpha^2, \alpha^3, \alpha^4, \alpha^5$ ($\alpha^6 = 1$), and two groups $1, \beta, \beta^2, \alpha, \alpha\beta, \alpha\beta^2$ ($\alpha^2 = 1, \beta^3 = 1$);

viz. in the first of these $\alpha\beta = \beta\alpha$, while in the other of them (that mentioned above) we have $\alpha\beta = \beta^2\alpha$, $\alpha\beta^2 = \beta\alpha$.

But although the theory as above stated is a general one, including as a particular case the theory of substitutions, yet the general problem of finding all the groups of a given order n , is really identical with the apparently less general problem of finding all the groups of the same order n , which can be formed with the substitutions upon n letters; in fact, referring to the diagram, it appears that $1, \alpha, \beta, \gamma, \delta, \epsilon$ may be regarded as substitutions performed upon the six letters $1, \alpha, \beta, \gamma, \delta, \epsilon$, viz. 1 is the substitution unity which leaves the order unaltered, α the substitution which changes $1\alpha\beta\gamma\delta\epsilon$ into $\alpha 1\gamma\beta\epsilon\delta$, and so for $\beta, \gamma, \delta, \epsilon$. This, however, does not in any wise show that the best or the easiest mode of treating the general problem is thus to regard it as a problem of substitutions: and it seems clear that the better course is to consider the general problem in itself, and to deduce from it the theory of groups of substitutions.

Cambridge, 28th November, 1877.

No. 2. The Theory of Groups; Graphical Representation.

[From the *American Journal of Mathematics*, t. i. (1878), pp. 174–176.]

In regard to a substitution-group of the order n upon the same number of letters, I omitted to mention the important theorem that every substitution is regular (that is, either cyclical or composed of a number of cycles each of them of the same order). Thus, in the group of 6 given in No. 1, writing a, b, c, d, e, f in place of $1, \alpha, \beta, \gamma, \delta, \epsilon$, the substitutions of the group are

$$1, ace \cdot bfd, aec \cdot bdf, ab \cdot cd \cdot ef, ad \cdot be \cdot cf, af \cdot bc \cdot de.$$

Let the letters be represented by points; a change a into b will be represented by a directed line (line with an arrow) joining the two points; and therefore a cycle abc , that is, a into b , b into c , c into a , by the three sides of the trilateral abc , with the three arrows pointing accordingly, and similarly for the cycles $abcd$, &c. the cycle ab means a into b , b into a , and we have here the line ab with a two-headed arrow pointing both ways; such a line may be regarded as a bilateral. A substitution is thus represented by a multilateral

or system of multilaterals, each side with its arrow; and in the case of a regular substitution the multilaterals (if more than one) have each of them the same number of sides. To represent two or more substitutions we require different colours, the multilaterals belonging to any one substitution being of the same colour.

In order to represent a group we need to represent only independent substitutions thereof; that is, substitutions such that no one of them can be obtained from the others by compounding them together in any manner. I take as an example a group

of the order 12 upon 12 letters, where the number of independent substitutions is $= 2$. See the diagram, wherein the continuous lines represent black lines, and the dotted lines, red lines. The diagram is drawn, in the first instance, with the arrows but without the letters, which are then affixed at pleasure; viz. the form of the group is quite independent of the way in which this is done, though the group itself is of course dependent upon it.

The diagram shows two substitutions, each of them of the third order: one is represented by the black triangles, and the other by the dotted triangles. It will be observed that there is from each point of the diagram (that is, in the direction of the arrow) one and only one black line, and one and only one dotted line; hence a symbol B , "move along a black line," B^2 , "move successively along two black lines," RB (read always from right to left), "move first along a dotted line and then along a black line," has in every case a perfectly definite meaning and determines the path when the initial point is given; any such symbol may be spoken of as a "route."

The square diagram for the group is:

	a	b	c	d	e	f	g	h	i	j	k	l
1	a	b	c	d	e	f	g	h	i	j	k	l
B	b	c	a	e	f	d	h	i	g	k	l	j
B^2	c	a	b	f	d	e	i	g	h	l	j	k
R	d	l	h	a	g	j	e	c	k	f	i	b
RB	e	j	i	b	h	k	f	a	l	d	g	c
RB^2	f	k	g	c	i	l	d	b	j	e	h	a
R^2	g	f	k	j	a	i	l	e	b	h	c	d
R^2B	h	d	l	i	b	g	k	f	c	a	e	j
R^2B^2	i	e	j	k	c	h	b	g	f	l	d	a
BR	j	i	e	h	k	a	c	l	d	g	f	b
BR^2	k	g	f	l	j	b	h	d	a	i	e	c
B^2R	l	h	d	g	f	c	j	k	e	b	a	i

The diagram has a remarkable property, in virtue whereof it in fact represents a group. It may be seen that any route leading from some one point a to itself, leads also from every other point to itself, or say from b to b , from c to c , $\dots$, and from l to l . We hence see that a route, applied in succession to the whole series of initial points or letters $abcdefghijkl$, gives a new arrangement of these letters, wherein no one of them occupies its original place; a route is thus, in effect, a substitution. Moreover, we may regard as distinct routes, those which lead from a to a , to b , to c , $\dots$, to l , respectively. We have thus 12 substitutions (the first of them, which leaves the arrangement unaltered, being the substitution unity), and these 12 substitutions form a group. I omit the details of the proof; it will be sufficient to give the square obtained by means of the several routes, or substitutions, performed upon the primitive arrangement $abcdefghijkl$, and the cyclical expressions of the substitutions themselves: it will be observed that the substitutions are unity, 3 substitutions of the order (or index) 2, and 8 substitutions of the order (or index) 3. It may be remarked that the group of 12 is really the group of the 12 positive substitutions upon 4 letters $abcd$, viz. these are 1, abc , acb , abd , adb , acd , adc , bcd , bdc , $ab \cdot cd$, $ac \cdot bd$, $ad \cdot bc$.

Cambridge, 16th May, 1878.

No. 3. The Newton-Fourier Imaginary Problem.

[From the *American Journal of Mathematics*, t. ii. (1879), p. 97.]

The Newtonian method as completed by Fourier, or say the Newton-Fourier method, for the solution of a numerical equation by successive approximations, relates to an equation $f(x) = 0$, with real coefficients, and to the determination of a certain real root thereof a by means of an assumed approximate real value ξ satisfying prescribed conditions: we then, from ξ , derive a nearer approximate value ξ_1 by the formula

$$\xi_1 = \xi - \frac{f(\xi)}{f'(\xi)},$$

and thence, in like manner, ξ_2 , ξ_3 , ξ_4 , $\dots$ approximating more and more nearly to the required root a . In connexion herewith, throwing aside the restrictions as to reality, we have what I call the Newton-Fourier Imaginary Problem, as follows.

Take $f(u)$, a given rational and integral function of u , with real or imaginary coefficients; ζ , a given real or imaginary value, and from this derive ζ_1 by the formula

$$\zeta_1 = \zeta - \frac{f(\zeta)}{f'(\zeta)},$$

and thence $\zeta_2, \zeta_3, \zeta_4, \dots$ each from the preceding one by the like formula. A given imaginary quantity $x + iy$ may be represented by a point the coordinates of which are (x, y) : the roots of the equation are thus represented by given points

$A, B, C, \dots$ and the values $\zeta, \zeta_1, \zeta_2, \dots$ by points $P, P_1, P_2, \dots$ the first of which is assumed at pleasure, and the others each from the preceding one by the like given geometrical construction. The problem is to determine the regions of the plane such that, P being taken at pleasure anywhere within one region, we arrive ultimately at the point A ; anywhere within another region at the point B ; and so for the several points representing the roots of the equation.

The solution is easy and elegant in the case of a quadric equation: but the next succeeding case of the cubic equation appears to present considerable difficulty.

Cambridge, March 3rd, 1879.

No. 4. The Mechanical Construction of Conformable Figures.

[From the *American Journal of Mathematics*, t. ii. (1879), p. 186.]

Is it possible to devise an apparatus for the mechanical construction of conformable figures; that is, figures which are similar as regards corresponding infinitesimal areas? The problem is to connect mechanically two points P_1 and P_2 in such wise that P_1 (1) shall have two degrees of freedom (or be capable of moving over a plane area) its position always determining that of P_2 : (2) that if P_1, P_2 describe the infinitesimal lengths P_1Q_1, P_2Q_2 , then the ratio of these lengths, and their mutual inclination, shall depend upon the position of P_1 , but be independent of the direction of P_1Q_1 : or what is the same thing, that if P_1 describe uniformly an indefinitely small circle, then P_2 shall also describe uniformly an indefinitely small circle, the ratio of the radii, and the relative position of the starting points in the two circles respectively, depending on the position of P_1 . Of course a pentagraph is a solution, but the two figures are in this case similar; and this is not what is wanted. Any unadjustable apparatus would give one solution only: the complete solution would be by an apparatus containing, suppose, a flexible lamina, so that P_1 moving in a given right line, the path of P_2 could be made to be any given curve whatever.

Cambridge, July 5th, 1879.

695. A Link-work for a^2 ; Extract from a Letter to Mr. Sylvester.[From the *American Journal of Mathematics*, t. I. (1878), p. 386.]

I sUPPOSE the following is substantially your link-work for a^2 . I use a slot to make D move in the line OA ; but this could be replaced by proper link-work. Supposing O and A fixed; the line OB is movable, and I wanted to have the distance OB measured in a fixed direction. This can be done by a hexagon $OABQB'A'$ with equal sides, and two other equal links $B'R$, BR : then of course, if O , R , Q are in a line, the hexagon will be symmetrical as to OQ , and OB' will be equal to OB , and B' may be made to move in the fixed line OB' .

If then $\angle BOA = \theta$,

$$OA = AB = a, \quad AC = CD = \frac{1}{2}a,$$

$$OB = 2a \cos \theta, \quad OD = a(1 + \cos \theta) = 2a \cos^2 \frac{1}{2}\theta,$$

$$2a \cdot OD = (OB)^2.$$

November 30, 1877.

696. Calculation of the Minimum N.G.F. of the Binary Seventhic.[From the *American Journal of Mathematics*, t. ii. (1879), pp. 71–84.]

For the binary seventhic $(a, \dots \mid x, y)^7$ the number of the asyzygetic covariants $(a, \dots)^\theta(x, y)^\omega$, or say of the deg-order $(\theta.\omega)$, is given as the coefficient of $a^\theta x^\omega$ in the function

$$\frac{1 - x^{-8}}{1 - ax^7 \cdot 1 - ax^6 \cdot 1 - ax^5 \cdot 1 - ax^4 \cdot 1 - ax^3 \cdot 1 - ax^2 \cdot 1 - ax \cdot 1 - x^8}$$

developed in ascending powers of a . See my “Ninth Memoir on Quantics,” *Phil. Trans.*, t. CLXI. (1871), pp. 17–50, [462]. This function is in fact

$$\frac{1}{A(x)} - \frac{x^{-8}}{A(x)},$$

where, developing in ascending powers of a , the second term $-\frac{x^{-8}}{A(x)}$ contains only negative powers of x , and it may consequently be disregarded: the number of asyzygetic covariants of the deg-order $(\theta.\omega)$ is thus equal to the coefficient of $a^\theta x^\omega$ in the function $A(x)$, which function is for this reason called the *Numerical Generating Function* (N.G.F.) of the binary seventhic; and the function $A(x)$ expressed as a fraction in its least terms is the *Minimum N.G.F.*

where $Z_0, Z_1, \dots, Z_{39}$ are rational and integral functions of x of degrees not exceeding 14; and where, as will presently be seen, there is a symmetry in regard to the terms $Z_0, Z_{39}; Z_1, Z_{38};$ &c., equidistant from the middle term Z_{19} , such that the terms $Z_0, \dots, Z_{19}$ being known, the remaining terms $Z_{20}, \dots, Z_{39}$ can be at once written down. Using only the foregoing properties, I obtained for the N.G.F. an expression which I communicated to Professor Sylvester, and which is published, *Comptes Rendus*, t. LXXXVII. (1878), p. 505, but with an erroneous value for the coefficient of a^9 and for that of the corresponding term a^{30} . The correct value is

Numerator of Minimum N.G.F. is =

$$\begin{aligned} & \{x + a^2 + a^4\}\{x^8 + x^6 + 2a^4 + a^2 + x^{10}\} \\ & \{-a^2 - a^4 - a^8 - a^{10}\}\{2ax^6 + a^3 + a^5\}\{x + 2a^2 - a^4 - a^6\} \\ & \{-1 + 2a^2 - a^4 - a^6 + a^8\}\{ax + a^3 + 2a^5 - a^7 + a^9\} \\ & \{2 - a^2 - 2a^4 - 2a^6 - a^8 - a^{10}\}\{x + 2a^2 + a^4 - a^6 + 2a^8 - 2a^{10}\} \\ & \{-1 + 4a^2 - a^4 - 2a^6 - a^8 + a^{12}\} \quad \text{to} \quad a^{10}x^{14}. \end{aligned}$$

Denominator (as mentioned before) is $= 1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 \cdot 1 - a^2 \cdot 1 - a^3 \cdot 1 - a^4 \cdot 1 - a^5$.

The method of calculation is as follows: write for a moment

$$\phi(a, x) = \frac{1 - x^{-8}}{1 - ax^7 \cdot 1 - ax^6 \cdot 1 - ax^5 \cdot 1 - ax^4 \cdot 1 - ax^3 \cdot 1 - ax^2 \cdot 1 - ax \cdot 1 - x^8},$$

then $\phi(a, x)$, developed in ascending powers of a , and rejecting from the result all negative powers of x , is

$$\frac{Z_0 + aZ_1 + \cdots + a^{39}Z_{39}}{1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 \cdot 1 - a^2 \cdot 1 - a^3 \cdot 1 - a^4 \cdot 1 - a^5},$$

developed in like manner in ascending powers of a ; for the determination of the Z 's up to Z_{19} we require only the development of ϕ .

required values $Z_0, Z_1, \dots, Z_{18}$. We require, therefore, only to know the values of these functions $X_0, X_1, \dots, X_{18}$. To make a break in the calculation, it is convenient to write

$$1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 (X_0 + aX_1 + \dots + a^{18}X_{18}) = Y_0 + aY_1 + \dots + a^{18}Y_{18};$$

putting then

$$1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 = 1 - ap + a^2q - a^3r,$$

where (up to $a^{18}x^{14}$)

$$\begin{aligned} p &= x + ax^3 + a^2x^5 + a^3x^7, \\ q &= x^4 + ax^6 + 2a^2x^8 + a^3x^{10} + a^4x^{12}, \\ r &= x^9 + ax^{11} + ax^{13}, \end{aligned}$$

we have

$$Y_0 + aY_1 + a^2Y_2 + \dots + a^{18}Y_{18} = (1 - ap + a^2q - a^3r)(X_0 + aX_1 + a^2X_2 + \dots + a^{18}X_{18}).$$

The values of $Y_0, Y_1, \dots, Y_{18}$ then are

	X_0	X_1	X_2	X_3	$\dots$	X_{18}
Y_0	X_0					
Y_1	$-pX_0$	X_1				
Y_2	$+qX_0$	$-pX_1$	X_2			
Y_3	$-rX_0$	$+qX_1$	$-pX_2$	X_3		
$\vdots$					$\ddots$	
Y_{18}						X_{18}

the values being taken to a^{18} only; and we then have

$$Z_0 + aZ_1 + a^2Z_2 + \dots + a^{18}Z_{18} = 1 - a^2 \cdot 1 - a^3 \cdot 1 - a^4 \cdot 1 - a^5 (Y_0 + aY_1 + \dots + a^{18}Y_{18});$$

viz. the values of $Z_0, Z_1, \dots, Z_{18}$ are

	Y_0	Y_1	Y_2	$\dots$	Y_{17}	Y_{18}
Z_0	Y_0					
Z_1	$-Y_0$	Y_1				
Z_2	$-Y_0$	$-Y_1$	Y_2			
$\vdots$				$\ddots$		
Z_{18}						Y_{18}

unaltered when each power x^ω is changed into $x^{14-\omega}$; and the verification thus obtained of the value

$$Z_{19} = 2 + 6ax^6 + a^2x^8 + 2a^3x^{10} + 2a^4x^{12} + a^5x^{14}$$

is in fact almost a complete verification of the whole work. Some other verifications, which present themselves in the course of the work, will be referred to further on. We have, therefore, to calculate the coefficients $X_0, X_1, \dots, X_{18}$; the function $\phi(a, x)$ regarded as a function of a is at once decomposed into simple fractions; viz. we have

$$\phi(a, x) = \frac{1}{1 - ax^7 \cdot 1 - ax^6 \cdot 1 - ax^5 \cdot 1 - ax^4 \cdot 1 - ax^3 \cdot 1 - ax^2 \cdot 1 - ax \cdot 1 - x^8} \\ - \frac{x^{-8}}{1 - ax^7 \cdot 1 - ax^6 \cdot 1 - ax^5 \cdot 1 - ax^4 \cdot 1 - ax^3 \cdot 1 - ax^2 \cdot 1 - ax},$$

which is

$$= \frac{x^8 E}{1 - ax^7} - \frac{x^7 F}{1 - ax^6} + \frac{x^6 G}{1 - ax^5} - \frac{x^5 H}{1 - ax^4} \\ + \frac{x^4 I}{1 - ax^3} - \frac{x^3 J}{1 - ax^2} + \frac{x^2 K}{1 - ax} - \frac{xL}{1 - x^8},$$

where

$$E = \frac{1}{1 - x^6 \cdot 1 - x^5 \cdot 1 - x^4 \cdot 1 - x^3 \cdot 1 - x^2 \cdot 1 - x \cdot 1 - x^8}, \\ F = \frac{1}{1 - x^5 \cdot 1 - x^4 \cdot 1 - x^3 \cdot 1 - x^2 \cdot 1 - x \cdot 1 - x^8}, \\ G = \frac{1}{1 - x^4 \cdot 1 - x^3 \cdot 1 - x^2 \cdot 1 - x \cdot 1 - x^8}, \\ H = \frac{1}{1 - x^3 \cdot 1 - x^2 \cdot 1 - x \cdot 1 - x^8}, \\ I = \frac{1}{1 - x^2 \cdot 1 - x \cdot 1 - x^8}, \\ J = \frac{1}{1 - x \cdot 1 - x^8}, \\ K = \frac{1}{1 - x^8}, \\ L = \frac{1}{1 - ax^7 \cdot 1 - ax^6 \cdot 1 - ax^5 \cdot 1 - ax^4 \cdot 1 - ax^3 \cdot 1 - ax^2 \cdot 1 - ax}.$$

The expansion is required only as far as x^{14} ; the first four terms are therefore to be disregarded, and, writing for shortness

$$\begin{aligned} E &= \frac{1}{1 - x^2 \cdot 1 - x^4 \cdot 1 - x^6 \cdot 1 - x^8}, \\ F &= \frac{1}{1 - x^4 \cdot 1 - x^6 \cdot 1 - x^8}, \\ G &= \frac{1}{1 - x^6 \cdot 1 - x^8}, \\ H &= \frac{1}{1 - x^8}, \end{aligned}$$

we have

$$\begin{aligned} \phi &= \frac{x^8 E}{1 - ax^7} - \frac{x^7 F}{1 - ax^6} + \frac{x^6 G}{1 - ax^5} - \frac{x^5 H}{1 - ax^4} \\ &= x^8 E(1 + ax^7 + a^2 x^{14} + \dots) - x^7 F(1 + ax^6 + a^2 x^{12} + \dots) \\ &\quad + x^6 G(1 + ax^5 + a^2 x^{10} + \dots) - x^5 H(1 + ax^4 + a^2 x^8 + \dots), \end{aligned}$$

where the several series are to be continued up to a^{18} , and, after substituting for E, F, G, H their expansions in ascending powers of x , we are to reject negative powers of x , and also powers higher than x^{14} . The functions E, F, G, H contain each of them only even powers of x , and it is easy to see that we require the expansions up to x^6, x^8, x^{10} and x^{12} respectively. For the sake of a verification, I in fact calculated E, F up to x^8 and G up to x^{10} .

The expansion is as follows:

Ind. x	0	2	4	6	8	10	12	14	16	18
$(1-x^2)^{-1}$	1	1	1	1	1	1	1	1	1	1
$(1-x^2)^{-1}(1-x^4)^{-1}$	1	2	3	4	5	6	7	8	9	10
$(1-x^2)^{-1}(1-x^4)^{-1}(1-x^6)^{-1}$	1	2	4	6	9	12	16	20	25	30
E	1	3	6	10	16	24	35	48	63	81
$(1-x^4)^{-1}$	1	1	2	2	3	3	4	4	5	5
$(1-x^4)^{-1}(1-x^6)^{-1}$	1	2	4	6	9	12	16	20	25	30
F	1	3	6	10	16	24	35	48	63	81
$(1-x^6)^{-1}$	1	1	2	3	4	5	7	8	10	12
G	1	3	6	10	16	24	35	48	63	81
H	1	1	2	2	3	3	4	4	5	5

the alternate lines giving the developments of the functions $(1-x^2)^{-1}$, $(1-x^2)^{-1}(1-x^4)^{-1}$, $(1-x^2)^{-1}(1-x^4)^{-1}(1-x^6)^{-1}$, . . . , which are the products of the x -functions down to any particular line.

The calculation proceeds by forming the expansions of the functions E, F, G, H and thence computing the values of $X_0, X_1, \dots, X_{18}$; then the values of $Y_0, Y_1, \dots, Y_{18}$; and finally the values of $Z_0, Z_1, \dots, Z_{18}$. The coefficient Z_{19} is thus verified; and the symmetry of the coefficients then gives $Z_{20}, \dots, Z_{39}$.

Expansions of the Functions E, F, G, H :

Ind. x	E	F	G	H
0	1	1	1	1
2	1	1	1	0
4	3	3	2	1
6	5	4	3	1
8	9	8	5	2
10	13	11	7	2
12	22	18	11	4
14	30	24	14	4
16	45	36	20	6
18	61	47	26	7

Calculation of the X 's.
The coefficients $X_0, X_1, \dots, X_{18}$ are calculated from the formula

$$\phi(a, x) = \sum_{k=0}^{\infty} \sum_{\omega} C_{k,\omega} a^k x^{\omega},$$

where the coefficients $C_{k,\omega}$ are determined by the expansions of the functions E, F, G, H . The calculation gives:

	0	1	3	5	7	9	11	13
X_0	1							
X_1	-1	1						
X_2	-1	1	1					
X_3	+1	-3	0	1				
X_4	+1	-4	-3	3	1			
X_5	+1	-6	-7	5	5	1		
X_6	+1	-8	-11	7	14	7	1	
X_7	+1	-11	-18	9	26	20	9	1

Continuing the calculation:

	0	1	3	5	7	9	11	13
X_8	+1	-14	-24	12	42	35	22	8
X_9	+1	-17	-36	16	60	58	44	18
X_{10}	+4	-26	-47	20	87	90	71	35
X_{11}	+4	-33	-66	27	119	136	110	63
X_{12}	+7	-47	-84	35	162	199	163	96
X_{13}	+10	-64	-113	47	218	282	235	149
X_{14}	+14	-86	-141	61	284	391	331	212
X_{15}	+18	-96	-183	79	369	532	454	307
X_{16}	+31	-115	-225	101	472	709	612	419

O_1

The first operation in the calculation of the minimum N.G.F. is to form the product of the generating functions for the several fundamental covariants. Let O_1 denote this operation; then

$$O_1 = \frac{1}{(1-ax^7)(1-ax^5)(1-ax^3)(1-ax)}.$$

Expanding this function, we obtain the generating function for the non-extraneous combinations of the fundamental covariants.

The calculation of the Y 's proceeds from the formula:

	0	1	3	5	7	9	11	13
Y_0	1							
Y_1	-1	1						
Y_2	-2	1	1					
Y_3	+2	-5	0	1				
Y_4	+3	-7	-4	3	1			
Y_5	+3	-12	-9	4	6	1		
Y_6	+4	-16	-17	7	16	7	1	
Y_7	+4	-22	-27	10	30	21	10	1

Calculation of the Y 's continued.

	0	1	3	5	7	9	11	13
Y_8	+8	−30	−40	13	50	42	25	9
Y_9	+10	−40	−60	20	75	73	54	22
Y_{10}	+12	−51	−87	30	110	110	85	42
Y_{11}	+14	−66	−120	41	154	163	127	66
Y_{12}	+20	−88	−161	56	214	228	182	100
Y_{13}	+26	−114	−212	75	288	316	258	149
Y_{14}	+36	−146	−272	99	386	429	350	212
Y_{15}	+44	−180	−350	132	510	571	472	297
Y_{16}	+62	−226	−438	170	664	749	626	403
Y_{17}	+78	−280	−540	219	856	974	823	540
Y_{18}	+104	−348	−664	282	1096	1254	1057	709

β_1

The operation β_1 consists in expanding O_2 in ascending powers of a and x , and selecting the coefficients of those terms for which the order is a multiple of 7. The result is

$$\beta_1 = \sum_{k=0}^{\infty} a^{7k} x^{7k} \cdot P_k(a),$$

where $P_k(a)$ is a polynomial in a alone.

The calculation of the Z 's from the Y 's gives:

	0	1	3	5	7	9	11	13
Z_0	1							
Z_1	-1	1						
Z_2	-1	0	1					
Z_3	0	-4	0	1				
Z_4	+2	-5	-3	2	1			
Z_5	+3	-9	-5	2	4	1		
Z_6	+4	-13	-13	4	11	6	1	
Z_7	+4	-18	-22	7	22	17	8	1

O_1

The verification of the calculation requires that the sum of the coefficients of the minimum N.G.F. should equal the total number of linearly independent covariants. This number is known to be 30 for the binary seventhic.

Continuing the calculation of the Z 's:

	0	1	3	5	7	9	11	13
Z_8	+8	-26	-33	10	38	34	20	6
Z_9	+10	-36	-53	16	60	58	40	16
Z_{10}	+12	-47	-78	25	92	92	65	30
Z_{11}	+16	-64	-109	37	134	136	96	48
Z_{12}	+22	-86	-149	52	190	196	138	72
Z_{13}	+28	-112	-200	72	264	280	194	104
Z_{14}	+40	-146	-264	99	362	386	270	150
Z_{15}	+52	-188	-348	135	488	528	372	212
Z_{16}	+70	-240	-444	180	650	712	504	292
Z_{17}	+92	-304	-564	240	856	948	678	400
Z_{18}	+122	-386	-716	318	1120	1252	898	540

Cambridge, December 7th, 1878.

697. On the Double $\wp$ -Functions.[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 74–81.]

I have sought to obtain, in forms which may be useful in regard to the theory of the double $\wp$ -functions, the integral of the elliptic differential equation

$$\frac{dx}{\sqrt{a-x \cdot b-x \cdot c-x \cdot d-x}} + \frac{dy}{\sqrt{a-y \cdot b-y \cdot c-y \cdot d-y}} = 0;$$

the present paper has immediate reference only to this differential equation; but, on account of the design of the investigation, I have entitled it as above.

We may for the general integral of the above equation take a particular integral of the equation

$$\frac{dx}{\sqrt{a-x \cdot b-x \cdot c-x \cdot d-x}} + \frac{dy}{\sqrt{a-y \cdot b-y \cdot c-y \cdot d-y}} + \frac{dz}{\sqrt{a-z \cdot b-z \cdot c-z \cdot d-z}} = 0;$$

viz. this particular integral, regarding therein z as an arbitrary constant, will be the general integral of the first mentioned equation. And we may further assume that z is the value of y corresponding to the value a of x .

I write for shortness

$$a-x, b-x, c-x, d-x = a_1, b_1, c_1, d_1,$$

$$a-y, b-y, c-y, d-y = a_2, b_2, c_2, d_2;$$

and I write also (xy, bc, ad) , or more shortly (bc, ad) , to denote the determinant

$$\begin{vmatrix} 1, & x+y, & xy \\ 1, & b+c, & bc \\ 1, & a+d, & ad \end{vmatrix};$$

we have of course $(ad, bc) = -(bc, ad)$, and there are thus the three distinct determinants (ad, bc) , (bd, ac) and (cd, ab) .

We have then for each of the functions

$$\sqrt{\frac{a-z}{d-z}}, \quad \sqrt{\frac{b-z}{d-z}}, \quad \sqrt{\frac{c-z}{d-z}}$$

a set of four equivalent expressions, the whole system being

$$\sqrt{\frac{a-z}{d-z}} = \frac{\sqrt{a-b \cdot a-c} \{ \sqrt{ad \cdot b_1 c_1} + \sqrt{a_1 d_1 \cdot bc} \}}{(bc, ad)} = \frac{\sqrt{a-b \cdot a-c}(x-y)}{\sqrt{adb_1 c_1 - \sqrt{a_1 d_1 \cdot bc}}},$$

$$= \frac{\sqrt{a-b \cdot a-c}\{\sqrt{abc_1d_1} + \sqrt{a_1b_1cd}\}}{(cd, ab)} = \frac{\sqrt{a-b \cdot a-c}\{\sqrt{acb_1d_1} + \sqrt{a_1c_1bd}\}}{(bd, ac)},$$

$$\begin{aligned} &= \frac{(a-c)\sqrt{b_1d_1 \cdot d} - (b-d)\sqrt{a_1c_1 \cdot c}}{(a-b)\sqrt{c_1d_1 \cdot d} - (c-d)\sqrt{a_1b_1 \cdot b}} \\ &= \frac{(a-c)\sqrt{b_1d_1 \cdot d} + (b-d)\sqrt{a_1c_1 \cdot c}}{(a-b)\sqrt{c_1d_1 \cdot d} + (c-d)\sqrt{a_1b_1 \cdot b}}, \end{aligned}$$

etc.

Here the product on the left-hand side is

$$= (a - c)\{b_1d_1\sqrt{abc_1d_1} - bd_1\sqrt{a_1b_1cd}\} \\ + (b - d)\{-a_1c\sqrt{abc_1d_1} + ac_1\sqrt{a_1b_1cd}\},$$

viz. this is

$$= \sqrt{abc_1d_1}\{(a - c)b_1d - (b - d)a_1c\} - \sqrt{a_1b_1cd}\{(a - c)bd_1 - (b - d)ac_1\},$$

and in this last expression the two terms in $\{\}$ are at once shown to be each $= (bc, ad)$; whence the identity in question.

Comparing in like manner the first expressions for $\sqrt{\frac{a-z}{d-z}}$ and $\sqrt{\frac{b-z}{d-z}}$ respectively, we have

$$(b - d)(bc, ad)^2 \frac{1}{d - z} = (a - b)(a - c)(b - d)\{adb_1c_1 + a_1d_1bc + 2\sqrt{abcd a_1b_1c_1d_1}\}, \\ (d - a)(bc, ad)^2 \frac{1}{d - z} = -(a - b)\{(a - c)^2b_1d_1 + (b - d)^2a_1c_1 + 2(a - c)(b - d)\sqrt{abcd a_1b_1c_1d_1}\},$$

whence, adding, the radical on the right-hand side disappears; the whole equation divides by $-(a - b)$, and omitting this factor, the relation to be verified is

$$(b - d)(bc, ad)^2 \frac{1}{d - z} - (d - a)(bc, ad)^2 \frac{1}{d - z} \\ = (a - c)(b - d)(adb_1c_1 + a_1d_1bc) - (a - c)^2(b - d)^2(x - y)^2.$$

Expanding, and observing that

$$(adb_1c_1+a_1d_1bc)^2 = (adb_1c_1-a_1d_1bc)^2+4abcd a_1b_1c_1d_1 = (bc, ad)^2(x-y)^2+4abcd a_1b_1$$

the whole equation becomes divisible by $(bc, ad)^2$, and omitting this factor, the equation is

$$(bc, ad)^2(a-z)^2 - 2(a-b)(a-c)(a-z)(d-z)(adb_1c_1 + a_1d_1bc) \\ + (a-b)^2(a-c)^2(d-z)^2(x-y)^2 = 0,$$

or, as this may also be written,

$$z^2\{(bc, ad)^2-2(a-b)(a-c)(adb_1c_1+a_1d_1bc)+(a-b)^2(a-c)^2(x-y)^2\} \\ -2z\{(bc, ad)^2a-(a-b)(a-c)(adb_1c_1+a_1d_1bc)(a+d)+(a-b)^2(a-c)^2(x-y)^2d\} \\ +\{(bc, ad)^2a^2-2(a-b)(a-c)(adb_1c_1+a_1d_1bc)ad+(a-b)^2(a-c)^2(x-y)^2d^2\} = 0.$$

This is really a symmetrical equation in x, y, z of the form

$$\begin{aligned} A + 2B(x + y + z) + C(x^2 + y^2 + z^2) + 2D(yz + zx + xy) \\ + 2E(y^2z + yz^2 + z^2x + zx^2 + x^2y + xy^2) + 4Fxyz \\ + 2G(x^2yz + xy^2z + xyz^2) + H(x^2y^2 + y^2z^2 + z^2x^2) \\ + 2I(xy^2z^2 + yz^2x^2 + zx^2y^2) + Jx^2y^2z^2 = 0. \end{aligned}$$

It may be remarked by way of verification that the equation remains unaltered on substituting for x, y, z, a, b, c, d their reciprocals and multiplying the whole by $xyzabcd$.

I further remark that, writing $a = 0$, we have

$$\begin{aligned} A = 0, \quad B = 0, \quad C = b^2c^2d^2, \quad D = -b'c'd' \\ E = 0, \quad F = bcd(bc + bd + cd), \quad G = -bcd(b + c + d), \quad H = 0, \\ I = 2bcd, \quad J = b^2 + c^2 + d^2 - 2(bc + bd + cd); \end{aligned}$$

and writing also $b = 1$,

$$-B = (b + c + d), \quad y = bc + bd + cd, \quad -\frac{A}{bcd} = bcd,$$

(whence $a - x \cdot b - x \cdot c - x \cdot d - x = \epsilon x^4 + \delta x^3 + \gamma x^2 + \beta x + \epsilon$), we have the formula

$$\begin{aligned} \epsilon(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy) - 4\beta\delta^2xyz - 2\beta\delta\epsilon xyz(x + y + z) \\ - 4\gamma\delta\epsilon xyz(yz + zx + xy) + (\delta^4 - 4\gamma\delta^2\epsilon)x^2y^2z^2 = 0, \end{aligned}$$

given p. 348 of my *Elliptic Functions* as a particular integral of the differential equation when the radical is $\sqrt{\epsilon x^4 + \delta x^3 + \gamma x^2 + \beta x + \epsilon}$.

Let the equation be

$$du = \frac{dx}{\sqrt{abcd a_1 b_1 c_1 d_1}},$$

and the equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0,$$

then gives

$$du = \frac{dx}{\sqrt{abcd a_1 b_1 c_1 d_1}},$$

as it should do.

The differential equation might also have been verified directly from any one of the expressions for $\sqrt{\frac{a-z}{d-z}}$.

Writing for shortness $X = a - x \cdot b - x \cdot c - x \cdot d - x$, etc., then the general integral of the differential equation by Abel's theorem is

$$\begin{vmatrix} x, & y, & 1, & \sqrt{X} \\ x_1, & y_1, & 1, & \sqrt{Y} \\ x_2, & y_2, & 1, & \sqrt{Z} \\ w, & w_1, & 1, & \sqrt{W} \end{vmatrix} = 0,$$

where w is the constant of integration: and it is to be shown that the value of w which corresponds to the integral given in the present paper is $w = a$.

Observe that writing in the determinant $w = a$, the determinant on putting therein $x = a$, would vanish whether z were or were not $= y$; but this is on account of an extraneous factor. The integral gives

$$C^2 = abcd,$$

where A, B, C are now determined by the equations

$$Aa^2 + Ba + C = 0, \quad Ay^2 + By + C = \sqrt{Y}, \quad Aw^2 + Bw + C = \sqrt{W},$$

and, imagining these values actually substituted, it is to be shown that the equation $C^2 = abcd$ is satisfied by the value $w = a$.

We have

$$A \cdot a - y \cdot a - w \cdot w - y = (a - w)\sqrt{Y} - (a - y)\sqrt{W},$$

$$B \cdot a - y \cdot a - w \cdot w - y = (a - w)(a + w)\sqrt{Y} - (a - y)(a + y)\sqrt{W},$$

$$C \cdot a - y \cdot a - w \cdot w - y = (a - w)aw\sqrt{Y} - (a - y)ay\sqrt{W},$$

or writing as before $a - y, b - y, c - y, d - y = a_1, b_1, c_1, d_1$, and also $a - w, b - w, c - w, d - w = a_3, b_3, c_3, d_3$,

If in these formulae w is indefinitely nearly $= a$, then a_3 is indefinitely small, so that $\sqrt{a_3 b_3 c_3 d_3}$ may be neglected in comparison with $\sqrt{a_1 b_1 c_1 d_1}$: also $w - y$ may be put $= a_1$; the formulae thus become

$$A = \frac{\sqrt{b_3 c_3 d_3}}{a_1 \sqrt{a_3}}, \quad B = \frac{a \sqrt{b_3 c_3 d_3}}{a_1 \sqrt{a_3}}, \quad C = \frac{ay \sqrt{b_3 c_3 d_3}}{a_1 \sqrt{a_3}},$$

where the values of A , B , C are each of them indefinitely large on account of the factor $\sqrt{a_3}$ in the denominator; the value of C is $C = ayA$, and substituting this value in the equation $C^2 = abcd$ and then considering A as indefinitely large, the equation becomes $ay = a^2 y^2$, that is, $w = a$; so that $w = a$ is a value of w satisfying this equation.

Cambridge, 3 July, 1878.

698. On a Theorem relating to Covariants.[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 82, 83.]

The theorem given by Prof. Sylvester, Crelle, vol. LXXXV., p. 109, may be stated as follows: If for a binary quantic of the order i in the variables, we consider the whole system of covariants of the degree j in the coefficients, then

$$\Sigma n(\theta) = \Pi(j),$$

where θ denotes the number of asyzygetic covariants of the order θ in the variables, the values of θ being ij , $ij - 2$, $ij - 4$, $\dots$, 1 or 0, according as ij is odd or even.

In the case of the binary quintic $(a, \dots | x, y)^5$ ($i = 5$), we have a series of verifications in the Table 88 of my “Ninth Memoir on Quantics,” *Phil. Trans.*, vol. CLXI. (1871), [462]: viz. writing the small letters $a, b, c, \dots, u, v, \dots$ to denote the several terms, the second column contains the number of them which are asyzygetic; for instance, $i = 5$ as above, $j = 6$, an extract from the table is

Order	30	28	26	24	22	20	18	16	14	12	10	8	6	4	2
Number	1	1	1	2	2	3	2	4	3	4	2	4	1	2	2

The covariants of deg-order $6 \cdot 14$ are a^2b^2 , a^3h , acd , bc^2 , ef (5 terms), but the number of asyzygetic covariants being 4, there exists one syzygy among these five terms. $\Pi(5) \cdot \Pi(6) = 31 \cdot 27 \cdot 26 \cdot 46 \cdot 42 \cdot 87 \cdot 34 \cdot 60 \cdot 39 \cdot 44 \cdot 18 \cdot 28 \cdot 5 \cdot 6 = 462 = \Pi(11)$.

where, for instance deg-order $6 \cdot 14$, the covariants are a^2b^2 , a^3h , acd , bc^2 , ef , but the number against these in the third column being (not 5 but) 4, the meaning is that there exists between these five terms one syzygy, making the number of aszygetic covariants of the deg-order $6 \cdot 14$ to be 4. The second column thus in fact contains the several values of $n(\theta)$, and the sum of the numbers in this column is $= 462$, the value of $\Pi(11)$.

699. On the Triple $\wp$ -Functions.[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 134–138.]

There should be in all 64 functions proportional to irrational algebraical functions of three independent variables x, y, z ; there is no difficulty in obtaining the expression of these 64 functions in the case of the system of differential equations connected with the integral

$$\int dx : \sqrt{a - x \cdot b - x \cdot c - x \cdot d - x \cdot e - x \cdot f - x \cdot g - x \cdot h - x};$$

but this is not the general form of the system for the deficiency (Geschlecht) $p = 3$; and I do not know how to deal with the general form: the present note relates therefore exclusively to the above-mentioned special form.

And if we write the single letters A, B, C, D, E, F, G, H for $\text{al}(u, v, w)_1, \text{al}(u, v, w)_2, \text{al}(u, v, w)_3, \dots, \text{al}(u, v, w)_8$ respectively, each of the capital letters thus denoting a function of (u, v, w) , the expressions of these functions in terms of (x, y, z) are

$$A = \sqrt{a - x \cdot b - x \cdot c - x \cdot d - x \cdot e - x \cdot f - x \cdot g - x \cdot h - x},$$

(seven equations).

Similarly, instead of the 21 functions $\text{al}(u, v, w)_{12}$, $\text{al}(u, v, w)_{13}$, $\dots$, writing AB , $\dots$, FG , each of these binary symbols denoting in like manner a function of (u, v, w) , the definition of AB is

$$AB = A\partial B - B\partial A,$$

where

$$\partial = \frac{d}{du} + \frac{d}{dv} + \frac{d}{dw}.$$

Or substituting for A and B their values, and disregarding the constant factor $\sqrt{a-b}$, this is

$$\begin{aligned} AB = & \frac{1}{2}(y-z)\sqrt{a-y\cdot b-y\cdot a-z\cdot b-z\cdot c-x\cdot d-x\cdot e-x\cdot f-x\cdot g-x\cdot h-} \\ & +\frac{1}{2}(z-x)\sqrt{a-z\cdot b-z\cdot a-x\cdot b-x\cdot c-y\cdot d-y\cdot e-y\cdot f-y\cdot g-y\cdot h-y} \\ & +\frac{1}{2}(x-y)\sqrt{a-x\cdot b-x\cdot a-y\cdot b-y\cdot c-z\cdot d-z\cdot e-z\cdot f-z\cdot g-z\cdot h-z}. \end{aligned}$$

We have thus in all 21 equations, which exhibit the form of the Weierstrassian functions $\text{al}(u, v, w)_{12}, \dots, \text{al}(u, v, w)_{78}$.

To complete the system, there should it is clear be 35 new functions $\text{al}(u, v, w)_{123}, \dots, \text{al}(u, v, w)_{567}$, represented by $ABC, \dots, EFG$, viz. the whole number of functions being $8 + 21 + 35 = 64$.

the equations, which define the $\wp$ -functions $A, B, \dots, H, ABC, \dots, FGH$, and the ω -function Ω , are

$$A = \Omega' \sqrt{a}, \quad (8 \text{ equations})$$

$$ABC = \Omega'' \sqrt{abc}, \quad (56 \text{ equations})$$

and one other relation which I have not as yet investigated.

As regards the algebraical relations between the 64 $\wp$ -functions, it is to be remarked that, selecting in a proper manner 8 of the functions, the square of any one of the other functions can be expressed as a linear function of the squares of the 8 selected functions. To explain this somewhat further, observe that, taking any 5 squares such as $(ABC)^2, (ABD)^2, (ABE)^2, (ABF)^2, (ABG)^2$, these are connected by a linear relation, the coefficients of which are functions of the constants (a, b, c, d, e, f, g, h) .

Here the irrational part vanishes and the rational part is found to be

$$\begin{aligned} & \{(d-e)c_1c_2ade + (e-c)d_1d_2e + (c-d)e_1e_2dc \\ & - (d-e)c_1cd_2e_2 - a_3b_3a_4b_4f_1g_1h_2(z-x)^2 + (e-c)d_1dc_2e_2 \\ & + (c-d)e_1cd_2c_2\} \times \{(d-e)cC_1d_2c_2 + (e-c)dd_1c_2e_2 \\ & + (c-d)ee_1d_2c_2\}. \end{aligned}$$

The three terms in $\{\}$ are here $= -(c-d)(d-e)(e-c)$ multiplied by $(z-x)(x-y)$, $(x-y)(y-z)$, $(y-z)(z-x)$ respectively; hence the term in $[\]$ divides by d and the result is

$$(c-d)(d-e)(e-c)\{a_3b_1a_4b_4f_1g_1h_1(y-z) + a_4b_3a_5b_5f_2g_2h_2(z-x) + aba_6b_6f_3g_3h_3(x-y)\}$$

or finally this is multiplied by

$$= -(c-d)(d-e)(e-c)(x-y)(y-z)(z-x).$$

700. On the Tetrahedroid as a Particular Case of the 16-nodal Quartic Surface.[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 161–164.]

In the paper “Sur un cas particulier de la surface du quatrième ordre avec seize points singuliers,” Crelle, t. LXV. (1866), pp. 284–290, [356], I showed how the surface called the Tetrahedroid could be identified as a special form of Kummer’s 16-nodal quartic surface; but I was not then in possession of the simplified form of the equation of the 16-nodal surface given in my paper “Note sur la surface de Kummer,” Proc. Lond. Math. Soc. t. IX. (1878), pp. 80, 81, [694].

where, $\alpha, \beta, \gamma, \delta, \alpha', \beta', \gamma', \delta', \alpha'', \beta'', \gamma'', \delta''$ being connected as above, the number of constants is = 6.

The equations of the 16 singular planes are

$$\begin{aligned}
 X &= 0, & Y &= 0, \quad Z = 0, \quad W = 0, \\
 \alpha(\gamma'\gamma''Y - \beta'\beta''Z) - W &= 0, & \beta(\alpha'\alpha''Z - \gamma'\gamma''X) - W &= 0, \\
 \gamma(\beta'\beta''X - \alpha'\alpha''Y) - W &= 0, & \delta(\alpha''\alpha Z - \gamma''\gamma X) - W &= 0, \\
 \alpha''(\beta\gamma'Y - \beta'\gamma Z) - W &= 0, & \beta''(\alpha\alpha'Z - \gamma\gamma'X) - W &= 0, \\
 \gamma(\beta'\beta''X - \alpha'\alpha''Y) - W &= 0, & & \text{etc.}
 \end{aligned}$$

Writing x, y, z, w as current coordinates, the equation of the tetrahedroid is

$$\sqrt{\alpha x} + \sqrt{\beta y} + \sqrt{\gamma z} + \sqrt{\delta w} = 0.$$

The linear functions of x, y, z, w have to be multiplied in order to reduce them each to the corresponding linear function of X, Y, Z, W , being given by the table

	X	Y	Z	W
x	1	$\frac{1}{2}(la - m\beta)$	$\frac{1}{2}(la - m\beta)$	1
y	$\frac{1}{n}(\beta'\gamma'' - \gamma'\beta'')$	1	$\frac{1}{2}(m\beta - n\gamma)$	$\frac{1}{2}(\gamma''\alpha - \alpha''\gamma)$
z	$\frac{1}{2}(\alpha''\beta - \beta''\alpha)$	$\frac{1}{n}(\gamma'\alpha'' - \alpha'\gamma'')$	1	$\frac{1}{2}(\alpha'\beta'' - \beta'\alpha'')$
w	$\frac{1}{l}(\beta'\gamma'' - \gamma'\beta'')$	$\frac{1}{m}(\gamma'\alpha'' - \alpha'\gamma'')$	$\frac{1}{n}(\alpha'\beta'' - \beta'\alpha'')$	1

For instance, we have

$$\alpha(\gamma'\gamma''Y - \beta'\beta''Z) - W = 2\frac{1}{l}(la - m\beta)(fx - gy - hz),$$

viz. substituting for Y, Z, W their values, the relation is

$$\frac{1}{m\alpha\beta}\gamma'\gamma''\{-nx+lz+gw\}-\frac{1}{m\alpha\beta}\beta'\beta''\{mx-ly+hw\}=\frac{1}{l}(la-m\beta)(fx-gy-hz).$$

The four planes forming the tetrahedroid meet in points

$$\begin{array}{lll}
 0, & -\gamma', \alpha', & \alpha' \beta' - \beta' \gamma', \\
 \alpha'', & 0, -\gamma'', & \alpha'' \beta'' - \beta'' \gamma'', \\
 -\beta'', & \gamma'', 0, & \gamma'' \alpha'' - \alpha'' \beta'', \\
 \beta'' \alpha' - \alpha'' \beta', & \alpha'' \gamma' - \gamma'' \alpha', \gamma'' \beta' - \beta'' \gamma', & 0,
 \end{array}$$

the four points being in fact the vertices of the tetrahedron formed by the four planes of the tetrahedroid.

Observe that, if the singular planes of the 16-nodal surface in their original order are 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, then the planes forming the last-mentioned four systems of planes are

$$(1, 8, 11, 14), \quad (2, 7, 12, 13), \quad (3, 6, 9, 16), \quad (4, 5, 10, 15),$$

viz. they correspond each of them to a term which in the determinant form is a product of four factors.

701. Algorithm for the Characteristics of the Triple $\wp$ -Functions.[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 165–169.]

The characteristics of the triple $\wp$ -functions may be represented, the 28 odd characteristics by the binary symbols or duads, 12, ..., 78, and the 35 even characteristics (other than $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$) by the ternary symbols or triads 123, ..., 567: which triads may be regarded as abbreviations for the double tetrads 1238.4567, ..., 5678.1234, the 8 being always attached to the expressed triad.

The correspondence of the characteristics with the duads and triads is given by the following table:

	000	100	010	110	001	101	011	111
12	100	18	110	14	011	16	010	16
13	101	17	001	18	111	23	010	24
14	111	25	110	26	001	27	101	28
15	011	34	101	35	100	36	011	37
16	111	38	001	45	001	46	110	47
17	010	48	100	56	111	57	011	58
18	101	67	100	68	010	78	110	3

Or, what is the same thing, it is apparent from the table that the characteristics can be arranged in a systematic form.

	000	100	010	110	001	101	011	111
123	101	124	000	125	001	126	110	127
134	010	135	011	136	100	137	000	145
146	110	147	101	156	000	157	100	167
234	110	235	111	236	000	237	100	245
246	010	247	001	256	100	257	000	267
345	000	346	111	347	011	356	110	357
367	010	456	011	457	111	467	000	567
								001

$g = 56 - 2.$

by means of which the two-line-characteristic is at once found when the duad or triad is given. The new algorithm renders unnecessary the Table I. of Weber's memoir "Theorie der Abel'schen Functionen vom Geschlecht 3" (Berlin, 1876). In fact, the system of six pairs corresponding to an odd characteristic such as 12 is

$$13.23, 14.24, 15.25, 16.26, 17.27, 18.28,$$

and that corresponding to an even characteristic such as 123 (= 1238.4567) is

$$12.38, 13.28, 18.23, 45.67, 46.57, 47.56 :$$

so that all the $(28 + 35 =)63$ systems can be at once written down.

and the whole number of the forms of equation is thus = 1260.

The triads of pairs which enter into the same equation may be triads such as

$12 \cdot 34, 13 \cdot 42, 14 \cdot 23,$	No. = 70,
$12 \cdot 34, 13 \cdot 42, 56 \cdot 78,$	No. = 630,
$12 \cdot 34, 56 \cdot 78, 57 \cdot 68,$	No. = 560,

making the whole number = 1260, as already mentioned.

Cambridge, 7 December, 1878.

702. On the Triple $\wp$ -Functions.[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 190–198.]

A quartic curve has the deficiency 3, and depends therefore on the triple $\wp$ -functions: and these, as functions of 3 arguments, should be connected with functions of 3 points on the curve; but it is easy to understand that it is possible, and may be convenient, to introduce a fourth point, and so regard them as functions of 4 points on the curve: thus in the circle, the functions $\cos u$, $\sin u$ may be regarded as functions of one point $\cos u = x$, $\sin u = y$, or as functions of two points, $\cos u = xx_1 + yy_1$, $\sin u = xy_1 - x_1y$.

And accordingly in Weber's memoir "Theorie der Abel'schen Functionen vom Geschlecht 3," (1876), see p. 156, the triple $\wp$ -functions are regarded as functions of 4 points on the curve: viz. it is in effect shown that (disregarding constant factors) each of the 64 functions is proportional to a determinant, the four lines of which are algebraical functions of the coordinates of the four points respectively: the form of this determinant being different according as the characteristic of the $\wp$ -function is odd or even, or say according as the $\wp$ -function is odd or even.

But the geometrical signification of these formulae requires to be developed.

A quartic curve may be touched in six points by a cubic curve: but (Hesse, 1855) there are two kinds of such tangent cubics, according as the six points of contact are on a conic, or are not on a conic; say we have a conic hexad of points on the quartic, and a cubic hexad of points on the quartic. In either case, three points of the hexad may be assumed at pleasure; we can then in 28 different ways determine the remaining three points of the conic hexad, and in 36 different ways the remaining three points of the cubic hexad: or what is the same thing, there are 28 systems of cubics touching in a conic hexad, and 36 systems of cubics touching in a cubic hexad.

The condition in order that four points of the quartic curve may belong to a hexad (conic or cubic) is given by an equation $\Omega = 0$, where Ω is a determinant the four lines of which are algebraical functions of the coordinates of the four points respectively: but the form of such determinant is different according as the condition belongs to a conic hexad, or to a cubic hexad: we have thus 28 conic determinants and 36 cubic determinants, Ω ; and the 64 $\wp$ -functions are proportional to constant multiples of these determinants; viz. the odd functions correspond to the conic determinants, and the even functions to the cubic determinants.

First, as to the conic hexads: the points of a conic hexad lie in a conic with the two points of contact of some one of the bitangents of the quartic curve: so that, given any three points of the hexad, these together with the two points of contact of the bitangent determine a conic which meets the quartic in the remaining three points of the hexad. Suppose that a, b, c, f, g, h are linear functions of the coordinates such that the equation of the quartic curve is

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0;$$

then $a = 0, b = 0, c = 0, f = 0, g = 0, h = 0$ are six of the bitangents of the curve, and the bitangent $a = 0$ touches the curve at the two points of intersection of this line with the conic $bg - ch = 0$.

The general equation of a conic through these two points $a = 0, bg - ch = 0$, may be written

$$bg - ch + a(Ax + By + Cz) = 0,$$

where for x, y, z we may if we please substitute any three of the six linear functions a, b, c, f, g, h , or any other linear functions of the coordinates (x, y, z) : and the equation may also be written

$$af \pm \sqrt{a(bg - ch)} + a(Ax + By + Cz) = 0.$$

Adopting this latter form, and considering the intersections of the conic with the quartic, that is, considering the relation $\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0$ as holding good, we have

$$-af + bg + ch = -2\sqrt{af \cdot bg}, \quad af - bg + ch = -2\sqrt{af \cdot ch},$$

and we thus have at pleasure one or other of the two equations

$$\begin{aligned} -2\sqrt{bg} + \sqrt{a}(Ax + By + Cz) &= 0, \\ -2\sqrt{ch} + \sqrt{a}(Ax + By + Cz) &= 0, \end{aligned}$$

that is,

$$\begin{aligned} -2\sqrt{bg} + \sqrt{a}(Ax + By + Cz) &= 0, \\ -2\sqrt{ch} + \sqrt{a}(Ax + By + Cz) &= 0. \end{aligned}$$

Hence the condition in order that the four points (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) , (x_4, y_4, z_4) , assumed to be points of the quartic, may belong to the conic hexad, may be written

$$\begin{vmatrix} \sqrt{b_1 c_1 g_1 h_1} & a_1 \sqrt{a_1 f_1} & y_1 \sqrt{a_1 f_1} & z_1 \sqrt{a_1 f_1} \\ \sqrt{b_2 c_2 g_2 h_2} & a_2 \sqrt{a_2 f_2} & y_2 \sqrt{a_2 f_2} & z_2 \sqrt{a_2 f_2} \\ \sqrt{b_3 c_3 g_3 h_3} & a_3 \sqrt{a_3 f_3} & y_3 \sqrt{a_3 f_3} & z_3 \sqrt{a_3 f_3} \\ \sqrt{b_4 c_4 g_4 h_4} & a_4 \sqrt{a_4 f_4} & y_4 \sqrt{a_4 f_4} & z_4 \sqrt{a_4 f_4} \end{vmatrix} = 0,$$

or

$$\begin{vmatrix} \sqrt{b_1 c_1 g_1 h_1} & a_1 \sqrt{a_1 f_1} & y_1 \sqrt{a_1 f_1} & z_1 \sqrt{a_1 f_1} \\ \sqrt{b_2 c_2 g_2 h_2} & a_2 \sqrt{a_2 f_2} & y_2 \sqrt{a_2 f_2} & z_2 \sqrt{a_2 f_2} \\ \sqrt{b_3 c_3 g_3 h_3} & a_3 \sqrt{a_3 f_3} & y_3 \sqrt{a_3 f_3} & z_3 \sqrt{a_3 f_3} \\ \sqrt{b_4 c_4 g_4 h_4} & a_4 \sqrt{a_4 f_4} & y_4 \sqrt{a_4 f_4} & z_4 \sqrt{a_4 f_4} \end{vmatrix} = 0.$$

where, as before, the x , y , z may be replaced by any three of the letters a , b , c , f , g , h , or by any other linear functions of (x, y, z) : and, moreover, although in obtaining the condition we have used for the quartic the equation $\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0$, depending upon six bitangents, yet from the process itself it is clear that the condition can only depend upon the particular bitangent $a = 0$: calling the condition $\Omega = 0$, all the forms of condition which belong to the same bitangent $a = 0$, will be essentially identical, that is, the several determinants Ω will differ only by constant factors; or disregarding these constant factors, we have for the bitangent $a = 0$, a single determinant Ω , which may be taken to be any one of the determinants in question. And we have thus 28 determinants Ω , corresponding to the 28 bitangents respectively.

Coming now to the cubic hexads, Hesse showed that the equation of a quartic curve could be (and that in 36 different ways) expressed in the form, symmetrical determinant = 0, or say

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix} = 0,$$

where $(a, b, c, d, f, g, h, l, m, n)$ are linear functions of the coordinates; and from each of these forms he obtains the equation of a cubic

$$\begin{vmatrix} a, & h, & g, & l, & \alpha \\ h, & b, & f, & m, & \beta \\ g, & f, & c, & n, & \gamma \\ l, & m, & n, & d, & \delta \\ \alpha, & \beta, & \gamma, & \delta, & 0 \end{vmatrix} = 0,$$

containing the four constants $\alpha, \beta, \gamma, \delta$, or say the 3 ratios of these constants, touching the quartic in a cubic hexad of points: that the cubic does touch the quartic in six points appears, in fact, from Hesse's identity

$$\begin{vmatrix} a, & h, & g, & l, & \alpha \\ h, & b, & f, & m, & \beta \\ g, & f, & c, & n, & \gamma \\ l, & m, & n, & d, & \delta \\ \alpha, & \beta, & \gamma, & \delta, & 0 \end{vmatrix} \begin{vmatrix} a, & h, & g, & l, & \alpha' \\ h, & b, & f, & m, & \beta' \\ g, & f, & c, & n, & \gamma' \\ l, & m, & n, & d, & \delta' \\ \alpha', & \beta', & \gamma', & \delta', & 0 \end{vmatrix} = \Omega^2,$$

where Ω is an easily calculated function of the second order in $a, b, c, d, f, g, h, l, m, n$, and also of the second order in the determinants $\alpha\beta' - \alpha'\beta$, etc.

We can obtain such a form of the equation of the quartic, from the before-mentioned equation viz. this equation gives

$$\begin{vmatrix} a, & h, & g, & a \\ h, & b, & f, & b \\ g, & f, & c, & c \\ a, & b, & c, & d \end{vmatrix} = 0,$$

which is of the required form, symmetrical determinant = 0; the equation is, in fact,

$$af^2 + bg^2 + ch^2 - 2bcgh - 2cahf - 2abfg = 0,$$

which is the rationalised form of $\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0$, and we hence have the cubic

$$\begin{vmatrix} a, & h, & g, & a, & \alpha \\ h, & b, & f, & b, & \beta \\ g, & f, & c, & c, & \gamma \\ a, & b, & c, & d, & \delta \\ \alpha, & \beta, & \gamma, & \delta, & 0 \end{vmatrix} = 0,$$

the developed form of which is

$$\begin{aligned} & \alpha^2 bcf + \beta^2 cag + \gamma^2 abh + \delta^2 fgh - (\beta\gamma + f\alpha\delta)(-af + bg + ch) \\ & - (\gamma\alpha + g\beta\delta)(af - bg + ch) - (\alpha\beta + h\gamma\delta)(af + bg - ch) = 0. \end{aligned}$$

Considering the intersections with the quartic $\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0$, we have

$$-af + bg + ch, \quad af - bg + ch, \quad af + bg - ch = -2\sqrt{bcgh}, \quad -2\sqrt{cahf}, \quad -2\sqrt{abfg},$$

and the equation thus becomes

$$(\alpha\sqrt{bcf} + \beta\sqrt{cag} + \gamma\sqrt{abh} + \delta\sqrt{fgh})^2 = 0;$$

viz. for the points of the cubic hexad we have

$$\alpha\sqrt{bcf} + \beta\sqrt{cag} + \gamma\sqrt{abh} + \delta\sqrt{fgh} = 0,$$

and hence the condition in order that the four points (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) , (x_4, y_4, z_4) may belong to the cubic hexad is

$$\begin{vmatrix} \sqrt{b_1c_1f_1} & \sqrt{c_1a_1g_1} & \sqrt{a_1b_1h_1} & \sqrt{f_1g_1h_1} \\ \sqrt{b_2c_2f_2} & \sqrt{c_2a_2g_2} & \sqrt{a_2b_2h_2} & \sqrt{f_2g_2h_2} \\ \sqrt{b_3c_3f_3} & \sqrt{c_3a_3g_3} & \sqrt{a_3b_3h_3} & \sqrt{f_3g_3h_3} \\ \sqrt{b_4c_4f_4} & \sqrt{c_4a_4g_4} & \sqrt{a_4b_4h_4} & \sqrt{f_4g_4h_4} \end{vmatrix} = 0,$$

viz. we have thus the form of the determinant Ω which belongs to a cubic hexad.

It is to be observed that the equation $\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0$ remains unaltered by any of the interchanges a and f , b and g , c and h ; but we thus obtain only two cubic hexads; those answering to the equations

$$\alpha\sqrt{bcf} + \beta\sqrt{cag} + \gamma\sqrt{abh} + \delta\sqrt{fgh} = 0,$$

and

$$\alpha\sqrt{agh} + \beta\sqrt{bhf} + \gamma\sqrt{cfg} + \delta\sqrt{abc} = 0,$$

which give distinct hexads.

The whole number of ways in which the equation of the quartic can be expressed in a form such as $\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0$, attending only to the pairs of bitangents, and disregarding the interchanges of the two bitangents of a pair, is = 1260, and hence the number of forms for the determinant Ω of a cubic hexad is the double of this, = 2520, which is = 36×70 : but the number of distinct hexads is = 36, and thus there must be for each hexad, 70 equivalent forms.

To explain this, observe that every even characteristic except $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, and every odd characteristic, can be (and that in 6 ways) expressed as a sum of two different odd characteristics; we have thus (see Weber's Table I.) a system of $(35 + 28 =) 63$ hexpairs; and selecting at pleasure any three pairs out of the same hextriad, we have a system of $(63 \times 20 =) 1260$ tripairs; giving the 1260 representations of the quartic in a form such as $\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0$.

Each hextriad gives 20 tripairs; and the 63 hextriads give $63 \times 20 = 1260$ tripairs, as already mentioned.

ON THE TRIPLE ϑ -FUNCTIONS.

[702

even characteristic ($56 \times 5 =$) 280 tripairs, which are however 70 tripairs each taken 4 times. A tripair contains in all ($2^3 =$) 8 hemi-tripairs, but these divide themselves into two sets each of 4 hemi-tripairs such that for each hemi-tripair of the first set the three characteristics have a given sum, and for each hemi-tripair of the second set the three characteristics have a different given sum. Hence considering the 70 tripairs corresponding as above to a given even characteristic, in any one of the 70 tripairs, there is a set of 4 hemi-tripairs such that in each of them the sum of the three characteristics is equal to the given even characteristic; and taking the bitangents f, g, h to correspond to any one of these hemi-tripairs, the bitangents which correspond to the other three hemi-tripairs will be $b, c, f; c, a, g$ and a, b, h respectively; and we thus obtain from any one of these one and the same representation

$$\alpha\sqrt{bcf} + \beta\sqrt{cag} + \gamma\sqrt{abh} + \delta\sqrt{fgh} = 0$$

of the cubic hexad. And the 70 tripairs give thus the 70 representations of the same cubic hexad.

The whole number of hemi-tripairs is $36 \times 56 = 2016$: it may be remarked that there exists a system of 288 heptads, each of 7 odd characteristics such that selecting at pleasure any 3 characteristics out of the heptad, we obtain always a hemi-tripair: we have thus in all $288 \times 35 = 10080$ hemi-tripairs: this is $= 2016 \times 5$, or we have the 2016 hemi-tripairs each taken 5 times. Weber's Table II. exhibits 36 out of the 288 heptads.

I recall that in the algorithm derived from Hesse's theory the bitangents are represented by the duads 12, 13, . . . , 78 formed with the eight figures 1, 2, 3, 4, 5, 6, 7, 8; these duads correspond to the odd characteristics as shown in the Table, and the table shows also triads corresponding to all the even characteristics except $\begin{smallmatrix} 000 \\ 000 \end{smallmatrix}$.

Top line of characteristic.

	000	100	010	110	001	101	011	111
000		236	345	137	467	156	124	257
100	237	67	136	12	157	48	256	35
010	245	127	23	68	134	357	15	47
110	126	13	78	145	356	25	46	234
001	567	146	125	247	45	17	38	26
101	147	58	246	34	16	123	27	367
011	135	347	14	57	17	36	167	456
111	346	24	56	235	37	267	457	18

Bottom line of characteristic.

See my “Algorithm of the triple ϑ -functions,” Crelle, t. LXXXVII. p. 165, [701].

451. The $(35 + 28 =) 63$ hexpairs then are
35 hexpairs such as
say this is 1234.5678 or for shortness 567 (the 8 going always with
the expressed triad):
that is, 567 denotes the hexpair

$$12.34; 13.24; 14.23; 56.78; 57.68; 58.67 :$$

and 28 hexpairs such as
say this is 12; that is, 12 denotes the hexpair

$$13.32; 14.42; 15.52; 16.62; 17.72; 18.82.$$

It is to be noticed that the odd characteristics, as represented by their duad symbols, can be added by the formulae

$12 + 23 = 13$, etc.,
or, what is the same thing,
 $12 + 13 + 23 = 0$, etc.,
and
 $12 + 34 = 13 + 24 = 14 + 23 = 56 + 78 = 57 + 68 = 58 + 67 = 567$,
etc.

Thus, referring to the table,

$$12 + 23 = 13 \text{ means } \begin{smallmatrix} 100 & 010 \\ 010 & 100 \end{smallmatrix} = \begin{smallmatrix} 110 \\ 110 \end{smallmatrix};$$

and

$$12 + 34 = 567 \text{ means } \begin{smallmatrix} 100 & 101 \\ 010 & 100 \end{smallmatrix} = \begin{smallmatrix} 001 \\ 110 \end{smallmatrix},$$

which are right.

The 288 heptads are
8 heptads such as

$$\begin{array}{cccccccc} 1, & 2, & 3, & 4, & 5, & 6, & 7, & 8 \\ & 12, & 13, & 14, & 15, & 16, & 17, & 18 \end{array}$$

say this is the heptad 1, denoting the seven duads 12, 13, 14, 15, 16, 17, 18;

and 280 heptads such as

$$\begin{array}{ccccccccc} 1, & 2, & 3, & 4, & / & 5, & \backslash & 6, & 7, & 8 \\ & 12, & 13, & 14, & 15, & & 67, & 68, & 78 \end{array}$$

say this is the heptad 1.678, denoting the seven duads 12, 13, 14, 15, 67, 68, 78.

We hence see that the 2016 hemi-tripairs are
(I.) 280 hemi-tripairs

$$12, \quad 13, \quad 23,$$

say this is 12 (hemi-tripair), denoting the three duads 12, 13, 23;

(II.) 1680 hemi-tripairs

$$12, \quad 67, \quad 68,$$

say this is 12.678, denoting the three duads 12, 67, 68;

(III.) 56 hemi-tripairs

$$123, \quad 234, \quad 134,$$

say this is 123, denoting the three duads 123, 234, 134;
in all $280 + 1680 + 56 = 2016$.

We further see how each hemi-tripair may be completed into a tripair in different ways: thus (I.) gives the 5 tripairs

12	13	14	15	16
34	24	23	23	23
56	56	56	46	45
78	78	78	78	78

the last duad being 5, 6, 7 or 8; (II.) gives the 3 tripairs

12	12	12
67	68	78
34	34	34
(/	7)8	6
4	4	4

and the 2 tripairs

12	12
67	68
(/	5
5	5

while (III.) gives the 5 tripairs

123	123	123	123	123
456	456	457	457	467
78	78	68	6	5
			8	8

452

[702

To each even characteristic there belongs a system of 56 tripairs: thus for $\begin{smallmatrix} 000 \\ 000 \end{smallmatrix}$, the characteristic $\begin{smallmatrix} 000 \\ 000 \end{smallmatrix}$, the 56 hemi-tripairs are 123, that is, 12, 13, 23, etc.; and in the tripair, 1234, that is, 12.34; 13.24; 14.23, we have the set of four hemi-tripairs each of which the sum of the three characteristics is = $\begin{smallmatrix} 000 \\ 000 \end{smallmatrix}$:

$$\begin{smallmatrix} 000 \\ 000 \end{smallmatrix} \left\{ (12 + 23 + 13 = \begin{smallmatrix} 000 \\ 000 \end{smallmatrix}, \text{ etc.}) : \right.$$

whence the other 70 hemi-tripairs: for any such even characteristic, the tripairs are

- 2 such as 1234,
 - 16 such as 54(123),
 - 16 such as 15(678),
 - 36 such as (5162)34.78;
- in all $2 + 16 + 16 + 36 = 70$; and in each of these it is easy to select the hemi-tripairs for which the sum of the 3 duads is = 567.

Cambridge, 27 December, 1878.

703.

ON THE ADDITION OF THE DOUBLE
 ϑ -FUNCTIONS.

[From the Journal für die reine und angewandte Mathematik
(Crelle), t. LXXXVIII. (1879), pp. 74—81.]

455. I assume in general

$$\xi = a - \theta.b - \theta.c - \theta.d - \theta.e - \theta.f - \theta,$$

and I consider the variables x, y, z, w, p, q , connected by the equations

$$\begin{aligned} \frac{1}{\sqrt{X}}, \frac{1}{\sqrt{Y}}, \frac{1}{\sqrt{Z}}, \frac{1}{\sqrt{W}}, \frac{1}{\sqrt{P}}, \frac{1}{\sqrt{Q}} &= \frac{x}{\sqrt{X}}, \frac{y}{\sqrt{Y}}, \frac{z}{\sqrt{Z}}, \frac{w}{\sqrt{W}}, \frac{p}{\sqrt{P}}, \frac{q}{\sqrt{Q}} \\ &= \frac{x^2}{\sqrt{X}}, \frac{y^2}{\sqrt{Y}}, \frac{z^2}{\sqrt{Z}}, \frac{w^2}{\sqrt{W}}, \frac{p^2}{\sqrt{P}}, \frac{q^2}{\sqrt{Q}} \\ &= \frac{\sqrt{X'}}{\sqrt{X}}, \frac{\sqrt{Y'}}{\sqrt{Y}}, \frac{\sqrt{Z'}}{\sqrt{Z}}, \frac{\sqrt{W'}}{\sqrt{W}}, \frac{\sqrt{P'}}{\sqrt{P}}, \frac{\sqrt{Q'}}{\sqrt{Q}}, \end{aligned}$$

equivalent to two independent equations, which serve to determine p, q , or say the symmetrical functions $p + q$ and pq , in terms of x, y, z, w .

These equations, it is well known, constitute a particular integral of the differential equations

$$\frac{dx}{\sqrt{X}} = \frac{dy}{\sqrt{Y}} = \frac{dz}{\sqrt{Z}} = \frac{dw}{\sqrt{W}} = \frac{dp}{\sqrt{P}} = \frac{dq}{\sqrt{Q}},$$

or what is the same thing, regarding ξ, ξ' as arbitrary constants, they constitute the general integral of the differential equations

$$\frac{dx}{\sqrt{X}} = \frac{dy}{\sqrt{Y}} = \frac{dz}{\sqrt{Z}} = \frac{dw}{\sqrt{W}} = \frac{x dx}{\sqrt{X}} = \frac{y dy}{\sqrt{Y}} = \frac{z dz}{\sqrt{Z}} = \frac{w dw}{\sqrt{W}}.$$

I attach the numbers 1, 4, 3, 2, 5, 6 to the variables x, y, z, w, p, q , respectively: and write
(six equations),

$$A_{12} = \sqrt{a - x.a - y}, \quad A_{34} = \sqrt{a - z.a - w}, \quad A_{56} = \sqrt{a - p.a - q};$$

(ten equations),

$$AB_{12} = \sqrt{a - x.b - x.c - x.d - x.e - x.f - x.a - y.b - y.c - y.d - y.e - y.f}$$

where it is to be borne in mind that AB is an abbreviation for $ABF.CDE$, and so in all other cases, the letter F belonging always to the expressed duad: there are sixteen functions $A_{12}, A_{34}, A_{56}, AB_{12}, AC_{12}, AD_{12}, AE_{12}, AF_{12}, BC_{12}, BD_{12}, BE_{12}, BF_{12}, CD_{12}, CE_{12}, CF_{12}, DE_{12}$ being functions of x and y , of z and w , and of p and q , according as the suffix is 12, 34, or 56.

It is to be shown that the 16 functions A_{12}, AB_{12} , etc., of x and y , of the addition of the double ϑ -functions. I use when convenient the abbreviated notations

$$a - x = a_1, \quad a - y = a_2, \text{ etc.},$$

$$b_{12} = x - y, \quad c_{31} = z - x, \text{ etc.},$$

$$e_{31} = z - w, \quad f_{12} = p - q;$$

we have of course

$$X = a_1 b_1 c_1 d_1 e_1 f_1,$$

$$A_{12} = \sqrt{a_1 a_2 b_1 b_2 c_1 c_2 d_1 d_2 e_1 e_2 f_1 f_2} = \sqrt{a_1 a_2 b_1 b_2 c_1 c_2 d_1 d_2 e_1 e_2} \cdot f_{12}, \text{ etc.}$$

17. Proceeding to the investigation, the six equations obtained by writing therein $\alpha, \beta, \gamma, \delta, \epsilon$ for the values x, y, z, w, p, q successively; the four equations

$$\alpha x^2 + \beta x + \gamma + \delta/x = \epsilon \sqrt{X},$$

$$\alpha y^2 + \beta y + \gamma + \delta/y = \epsilon \sqrt{Y},$$

$$\alpha z^2 + \beta z + \gamma + \delta/z = \epsilon \sqrt{Z},$$

$$\alpha w^2 + \beta w + \gamma + \delta/w = \epsilon \sqrt{W},$$

serving to determine the ratios $\alpha : \beta : \gamma : \delta : \epsilon$ in terms of x, y, z, w ; and we have then for the determination of p, q the remaining two equations

$$\begin{aligned}\alpha p^2 + \beta p + \gamma + \delta/p &= \epsilon\sqrt{P}, \\ \alpha q^2 + \beta q + \gamma + \delta/q &= \epsilon\sqrt{Q},\end{aligned}$$

which two equations may be replaced by the identity

$$(\alpha\theta^2 + \beta\theta + \gamma + \delta/\theta)^2 - \epsilon^2\xi = a^2 - \epsilon^2 \cdot \theta - x \cdot \theta - y \cdot \theta - z \cdot \theta - w \cdot \theta - p \cdot \theta - q.$$

457

[703

Writing herein $\theta =$ any one of the values a, b, c, d, e, f , for instance $\theta = a$, and taking the square root of each side, we have

$$\alpha a^2 + \beta a + \gamma + \delta/a = \sqrt{a^2 - \xi} \sqrt{a - x \cdot a - y \cdot a - z \cdot a - w \cdot a - p \cdot a - q},$$

or as this may be written

$$\alpha a^2 + \beta a^2 + \gamma a + \delta = \sqrt{a^2 - \xi} \cdot A_{12}A_{34}A_{56},$$

which equation when reduced gives the proportional value of A_{12} .

For the reduction we require the value of $\alpha a^2 + \beta a^2 + \gamma a + \delta$: calling this for the moment Π , we join to the four equations a fifth equation

$$\alpha a^2 + \beta a^2 + \gamma a + \delta = \Pi.$$

Eliminating $\alpha, \beta, \gamma, \delta$, we find

$$\begin{vmatrix} \sqrt{X}, & x^2, & x, & 1, & \epsilon\sqrt{X}/x \\ \sqrt{Y}, & y^2, & y, & 1, & \epsilon\sqrt{Y}/y \\ \sqrt{Z}, & z^2, & z, & 1, & \epsilon\sqrt{Z}/z \\ \sqrt{W}, & w^2, & w, & 1, & \epsilon\sqrt{W}/w \\ \Pi, & a^2, & a, & 1, & 0 \end{vmatrix} = 0,$$

or, what is the same thing,

$$\begin{vmatrix} x^2, & x, & 1, & \epsilon\sqrt{X}/x \\ y^2, & y, & 1, & \epsilon\sqrt{Y}/y \\ z^2, & z, & 1, & \epsilon\sqrt{Z}/z \\ w^2, & w, & 1, & \epsilon\sqrt{W}/w \\ a^2, & a, & 1, & 0 \end{vmatrix} = 0;$$

viz.

$$\begin{aligned}\Pi.x-y.x-z.x-w.y-z.y-w.z-w &= -\epsilon\{\sqrt{X}.y-z.y-w.y-a.z-w.z-a.w-a \\ &+ \sqrt{Y}.z-w.z-a.z-x.w-a.w-x.a-x \\ &+ \sqrt{Z}.w-a.w-x.w-y.a-x.a-y.x-y \\ &+ \sqrt{W}.a-x.a-y.a-z.x-y.x-z.y-z\},\end{aligned}$$

458

[703

or as it may be written

$$\begin{aligned}\Pi.x-z.x-w.y-z.y-w &= -\{y-z.y-w.a_2\sqrt{X}-x-z.x-w.a_1\sqrt{Y}\} \\ &+ \{z-x.z-y.a_3\sqrt{Z}-z-x.z-y.a_4\sqrt{W}\},\end{aligned}$$

an equation for the determination of Π .

Consider first the expression which multiplies $\epsilon.a-z.a-w$; this is

$$= \Im\{y-z.y-w.a_2\sqrt{X}-x-z.x-w.a_1\sqrt{Y}\}$$

we have

$$BE_{12} = \Im\{\sqrt{b_1c_1d_1e_1f_1a_2c_2d_2e_2f_2} - \sqrt{b_2c_2d_2e_2f_2a_1c_1d_1e_1f_1}\},$$

and multiplying this by

$$C_{12} \cdot D_{12} = \sqrt{a_1c_1d_1a_2c_2d_2}$$

we derive

$$BE_{12} \cdot C_{12} \cdot D_{12} = \Im\{c_1d_2\sqrt{X}-c_2d_1\sqrt{Y}\},$$

and similarly two other equations; the system may be written

$$\begin{aligned}BE_{12} \cdot C_{12} \cdot D_{12} &= \Im\{c_1d_2\sqrt{X}-c_2d_1\sqrt{Y}\}, \\ CE_{12} \cdot D_{12} \cdot B_{12} &= \Im\{d_1b_2\sqrt{X}-d_2b_1\sqrt{Y}\}, \\ DE_{12} \cdot B_{12} \cdot C_{12} &= \Im\{b_1c_2\sqrt{X}-b_2c_1\sqrt{Y}\},\end{aligned}$$

the suffixes on the left-hand side being always 12.

The five letters b, c, d, e, f enter symmetrically, for BE is a mere abbreviation for the double triad $BEF.ACD$; and the like for CE , and DE . Any three of the five letters other than a ; and the remaining two letters e, f enter symmetrically into these equations.

Multiplying these equations by

$$\frac{b - z.b - w}{b - c.b - d}, \quad \frac{c - z.c - w}{c - d.c - b}, \quad \frac{d - z.d - w}{d - b.d - c}$$

respectively, and then adding, the right-hand side becomes

$$= \Im\{y - z.y - w.a_2\sqrt{X} - x - z.x - w.a_1\sqrt{Y}\}.$$

Writing

$$\frac{b - z.b - w}{b - c.b - d} = \frac{1}{c - d.d - b.b - c} \cdot \text{etc.},$$

for the left-hand side becomes

$$= \frac{1}{c - d.d - b.b - c} \{c - d.BE_{12} \cdot C_{12} \cdot D_{12} + d - b.CE_{12} \cdot D_{12} \cdot B_{12} + b - c.DE_{12} \cdot B_{12} \cdot C_{12}\},$$

which for shortness may be written

$$= \frac{1}{c - d.d - b.b - c} \sum \{c - d.BE_{12} \cdot C_{12} \cdot D_{12}\},$$

the summation referring to the three terms obtained by the cyclical interchange of the letters b, c, d .

459

[703

The result thus is

$$\begin{aligned} & \Im\{y - z.y - w.a_2\sqrt{X} - x - z.x - w.a_1\sqrt{Y}\} \\ &= \frac{1}{c - d.d - b.b - c} \sum \{c - d.BE_{12} \cdot C_{12} \cdot D_{12}\}. \end{aligned}$$

Interchanging x, y with z, w respectively, we have of course to interchange the suffixes 1, 2 and 3, 4; we thus find

$$\begin{aligned} & \Im\{w - x.w - y.a_4\sqrt{Z} - z - x.z - y.a_3\sqrt{W}\} \\ &= \frac{1}{c - a.a - b.b - c} \sum \{c - d.BE_{34} \cdot C_{34} \cdot D_{34}\}. \end{aligned}$$

But $\Pi = \alpha a^2 + \beta a^2 + \gamma a + \delta$, and we hence find the value of $\Pi.x - z.x - w.y - z.y - w$.

The resulting equation divides by $A_{12}^2 \cdot A_{34}^2$: throwing out this factor, we have

$$\sqrt{a^2 - \xi^2} = A_{56}^{-2} \{x - z.x - w.y - z.y - w\} (c - d.d - b.b - c) A_{12}^{-2} \\ \sum \{c - d.BE_{12} \cdot C_{12} \cdot D_{12}\} + A_{34}^{-2} \sum \{c - d.BE_{34} \cdot C_{34} \cdot D_{34}\},$$

where, as before, the summations refer to the three terms obtained by the cyclical interchange of the letters b, c, d ; and the remaining two letters e, f enter symmetrically into the formula.

The formula gives thus for A_{12} ten values which are of course equal to each other. Writing for a each letter in succession, we obtain formulae for each of the six single-letter functions A_{12} of p and q ; and the factor $\sqrt{a^2 - \xi^2}$ is the same in all; and the factor $(x - z.x - w.y - z.y - w)$ is common to all the formulae.

We require further the expressions for the double-letter functions of p, q . Considering for example the function BE_{12} , which is

$$= \Im \{ \sqrt{b_1 c_1 d_1 e_1 f_1 a_2 c_2 d_2 e_2 f_2} - \sqrt{b_2 c_2 d_2 e_2 f_2 a_1 c_1 d_1 e_1 f_1} \}.$$

460

[703

But the expressions are respectively

$$\sqrt{a^2 - \xi^2} A_{12} A_{34} A_{56}$$

and

$$\frac{1}{\sqrt{a^2 - \xi^2}} A_{12} A_{34} A_{56};$$

the whole equation thus divides by $A_{12}^2 A_{34}^2$; and then multiplying each side by $\frac{\sqrt{a^2 - \xi^2}}{A_{56}}$, throwing out this factor, we find

$$DE_{12} = \frac{1}{b - c.c - a.a - b} \sum \{c - d.C_{12} \cdot C_{34} \cdot C_{56}\},$$

in

$$\sqrt{a^2 - b^2} A_{12} A_{34} A_{56},$$

which formula if we imagine $\frac{\sqrt{a^2 - \xi^2}}{\epsilon}$ each replaced by its value in terms of the xy -functions and zw -functions, we have an equation of the form

$(x - z.x - w.y - z.y - w) DE_{12} =$ a given rational and integral function of the 16 functions

where M is a given rational and integral function of the 16 functions A_{12} , AB_{12} and A_{34} , AB_{34} of x and y and of z and w respectively.

The factor $(x - z.x - w.y - z.y - w)$ on the left-hand side is retained as being the same factor which enters into the equations for A_{12} , etc.: but on the right-hand side $x - z.x - w.y - z.y - w$ should be expressed in terms of the xy - and zw -functions. This can be done by means of the identity

$$x - z.x - w.y - z.y - w = \frac{\begin{vmatrix} 1, & x + y, & xy \\ 1, & z + w, & zw \\ 1, & a + b, & ab \end{vmatrix} \cdot \begin{vmatrix} 1, & x + y, & xy \\ 1, & z + w, & zw \\ 1, & a + c, & ac \end{vmatrix}}{(a - b)^2(a - c)^2},$$

where the summation refers to the three terms obtained by the cyclical interchange of the letters a , b , c .

The first determinant, multiplied by $a - b$, is in fact

$$\begin{vmatrix} a - z.a - w, & a - x.a - y \\ b - z.b - w, & b - x.b - y \end{vmatrix}$$

and the second determinant, multiplied by $a - c$, is

$$\begin{vmatrix} a - z.a - w, & a - x.a - y \\ c - z.c - w, & c - x.c - y \end{vmatrix},$$

so that the formula may also be written

$$x - z.x - w.y - z.y - w = \frac{\begin{vmatrix} a - z.a - w, & a - x.a - y \\ b - z.b - w, & b - x.b - y \end{vmatrix} \cdot \begin{vmatrix} a - z.a - w, & a - x.a - y \\ c - z.c - w, & c - x.c - y \end{vmatrix}}{(a - b)^2(a - c)^2},$$

or, what is the same thing, it is

$$x - z.x - w.y - z.y - w = \frac{\{L + M(a + b) + Nab\}\{L + M(a + c) + Nac\}}{(a - b)^2(a - c)^2},$$

where L , M , N are

$$= (x + y)zw - (z + w)xy, \quad xy - zw, \quad \text{and} \quad -(x + y) + (z + w);$$

reduced this is readily verified.

Cambridge, 12th March, 1879.

I take this opportunity of remarking that, in the double-letter formulae, the sign of the second term is, not as I have in general written $AB = \Im\{\sqrt{abc_1d_1e_1f_1}\}$, but is $+$, introducing a factor which is a function of x and y , the odd and even ϑ -functions are $= \wp\sqrt{a_1a_2}$, etc.; and this being so, when x and y are interchanged, each single-letter function changes only its sign; and each double-letter function remains unaltered; and thus, since this is a function which on the interchange of x, y changes its sign, there is a change of sign as there should be.

Cambridge, 29th July, 1879.

704.

A MEMOIR ON THE SINGLE AND DOUBLE THETA-FUNCTIONS.

[From the Philosophical Transactions of the Royal Society of London, vol. 171, Part III. (1880), pp. 897—1002. Received November 14,— Read November 28, 1879.]

The Theta-Functions, although arising historically from the Elliptic Functions, may be considered as in order of simplicity preceding these, and connecting themselves directly with the exponential function (e^x or) $\exp .x$; viz. they may be defined each of them as a sum of a series of exponentials, singly infinite in the case of the single functions, doubly infinite in the case of the double functions, and so on. The number of the single functions is $= 4$; the number of the double functions is $= 16$ ($4^2 =$); and the quotients of these, or say three of them each divided by the fourth, are the elliptic functions sn , cn , dn ; and the quotients of these, or say fifteen of them each divided by the sixteenth, are the hyper-elliptic functions of two arguments depending on the square root of a sextic function. Generally, the number

of the p -tuple theta-functions is $= 4^p$; and the quotients of these, or say all but one of them each divided by the remaining function, are the Abelian functions of p arguments depending on the irrationality y defined by the equation $F(x, y) = 0$ of a curve of deficiency p .

If, instead of the ratios of the functions, we consider the ratios of the functions connecting the ratios of $(4p - 1)$ dimensions, then we have the sixteen coordinates of a point on a quadri-quadric two-fold locus in 15-dimensional space, the deficiency of this two-fold locus being of course $= 2$.

463

[704

The Memoir relates only to the single and double functions, and the title has been given accordingly. The investigations contained in it extend to the single functions referred to in the First Part; but in the main and with a view to simplicity of notation the investigations are applicable to the general case of the double functions; but in the present Memoir, although the course of the single theory is in fact more complex than that of the double theory; and on the other hand we need the single theory as a guide for the course of the double theory.

I accordingly stop to point out that the First Part of the Memoir relates to the single functions; the Second and the Third Parts to the double functions respectively; and I consider in the first place the course of the single theory; and then that of the double theory. The paragraphs of the Memoir are numbered consecutively.

The general definition adopted for the theta-functions differs somewhat from that ordinarily used. The earlier memoirs on the double theta-functions are the well-known ones

Rosenhain, "Mémoire sur les fonctions de deux variables à quatre périodes, qui sont les inverses des intégrales ultra-elliptiques de la première classe." *Mém. Savans Étrang.*, t. XI. (1851), pp. 361—468.

Gopel, "Theoriae transcendentium Abelianarum primi ordinis adumbratio levis," *Crelle*, t. XXXV. (1847), pp. 277—312.

My first paper—Cayley, "On the Double ϑ -Functions in connexion with a 16-nodal Surface," *Crelle*, t. LXXXIII. (1877), pp. 210—219, [662]—was founded directly upon these, and was immediately followed by Dr Borchardt's paper,

Borchardt, "Ueber die Darstellung der Kummer'schen Fläche vierter Ordnung mit sechzehn Knotenpunkten durch die Gopelsche bi-quadratische Relation zwischen vier Thetafunctionen mit zwei Variabeln," Ditto, pp. 234—244.

My other later papers, [663, 664, 665, 697, 703], are contained in the same Journal.

FIRST PART.—INTRODUCTORY.

Definition of the theta-functions.

1. The p -tuple theta-functions depend upon $\frac{1}{2}p(p+1)$ parameters which are the coefficients of a quadric function of p ultimately disappearing integers, upon p arguments, and upon $2p$ characters, each $= 0$ or 1 , which form the characteristic of the 4^p functions; but it will be sufficient to write down the formulae in the case $p = 2$.

As already mentioned, the adopted definition differs somewhat from that ordinarily used. I use, as will be seen, a quadric function $J(a, h, bm, n)^2$ with even or odd values; and I write the other term $\frac{1}{2}\pi i(mu + nv)$, instead of $mu + nv$; this comes to affecting the arguments u, v with a factor πi , so that the quarter-periods (instead of being πi) are made to be $= 1$.

2. We write

$$\begin{pmatrix} m, & n \\ u, & v \end{pmatrix} = J(a, h, bm, n)^2 + \frac{1}{2}\pi i(mu + nv),$$

and in like manner

$$\begin{pmatrix} m + \alpha, & n + \beta \\ u + \gamma, & v + \delta \end{pmatrix} = J(a, h, bm + \alpha, n + \beta)^2 + \frac{1}{2}\pi i\{(m + \alpha)(u + \gamma) + (n + \beta)(v + \delta)\},$$

and prefixing to either of these the functional symbol $\exp.$ we have the exponential of the function in question, that is, e with the function as an exponent.

We then write, as the definition of the double theta-functions,

$$\vartheta \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (u, v) = \Sigma \exp. \begin{pmatrix} m + \alpha, & n + \beta \\ u + \gamma, & v + \delta \end{pmatrix},$$

where the summation extends to all positive and negative even integer values (zero included) of m and n respectively; $\alpha, \beta, \gamma, \delta$ might denote any quantities whatever, but for the theta-functions they are regarded as denoting positive or negative integers; this being so, it will appear that the only effect of altering each or any of them by an even integer is to reverse (it may be) the sign of the function; and the distinct functions are consequently the ($4^2 =$) 16 functions obtained by giving to each of the quantities $\alpha, \beta, \gamma, \delta$ the two values 0 and 1 successively.

3. We thus have the double theta-functions, depending on the parameters (a, h, b) which determine the quadric function $(a, h, bm, n)^2$ of the disappearing even integers (m, n) , and on the two arguments (u, v) : in the symbol $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, which is called the characteristic, the characters $\alpha, \beta, \gamma, \delta$ are each of them $= 0$ or 1 ; and we thus have the 16 functions.

The parameters (a, h, b) may be real or imaginary, but they must be such that reducing each of them to its real part the resulting function $(a, h, bm, n)^2$ is invariable in its sign, and negative for all real values of m and n : this is, in fact, the condition for the convergency of the series which give the values of the theta-functions.

4. The characteristic $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ is said to be even or odd according as the sum $\alpha\gamma + \beta\delta$ is even or odd.

Allied functions.

5. As already remarked, the definition of

$$\vartheta \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (u, v)$$

is not restricted to the case where the characters $\alpha, \beta, \gamma, \delta$ are integers; and the functions thus obtained are not regarded as theta-functions, and there is in this no occasion to consider functions of the form where they are not integers; but the expression theta-function will consequently not extend to include them.

Properties of the Theta-Functions: Various sub-headings.

Even-integer alteration of characters.

6. If x, y be even integers, then m, n having the several even integer values from $-\infty$ to $+\infty$ respectively, it is obvious that $m + \alpha, n + \beta$ and $m + \alpha + 2x, n + \beta + 2y$ will have the same series of values; and it thence follows that

$$\vartheta \left(\begin{matrix} \alpha + 2x, & \beta + 2y \\ \gamma, & \delta \end{matrix} \right) (u, v) = \vartheta \left(\begin{matrix} \alpha, & \beta \\ \gamma, & \delta \end{matrix} \right) (u, v).$$

Similarly if z, w are integers, then in the function

$$\vartheta \left(\begin{matrix} \alpha, & \beta \\ \gamma + 2z, & \delta + 2w \end{matrix} \right) (u, v)$$

the argument of the exponential function contains the term

$$\frac{1}{2}\pi i\{(m + \alpha)(u + \gamma + 2z) + (n + \beta)(v + \delta + 2w)\}$$

this differs from its original value by

$$\frac{1}{2}\pi i\{(m + \alpha)2z + (n + \beta)2w\} = \pi i(mz + nw) + \pi i(\alpha z + \beta w),$$

and then, m and n being even integers, $mz + nw$ is also an even integer, and the term $\pi i(mz + nw)$ does not affect the value of the exponential: we thus introduce into each term of the series the factor $\exp .\pi i(\alpha z + \beta w)$, which is, in fact, $= (-)^{\alpha z + \beta w}$; and we consequently have

$$\vartheta \left(\begin{matrix} \alpha, & \beta \\ \gamma + 2z, & \delta + 2w \end{matrix} \right) (u, v) = (-)^{\alpha z + \beta w} \vartheta \left(\begin{matrix} \alpha, & \beta \\ \gamma, & \delta \end{matrix} \right) (u, v);$$

or, uniting the two results,

$$\vartheta \left(\begin{matrix} \alpha + 2x, & \beta + 2y \\ \gamma + 2z, & \delta + 2w \end{matrix} \right) (u, v) = (-)^{\alpha z + \beta w} \vartheta \left(\begin{matrix} \alpha, & \beta \\ \gamma, & \delta \end{matrix} \right) (u, v).$$

This sustains the before-mentioned conclusion that the only distinct functions are the 16 functions obtained by giving to the characters $\alpha, \beta, \gamma, \delta$ the values 0 and 1 respectively.

Odd-integer alteration of characters.

7. The effect is obviously to interchange the different functions.

Even and odd functions.

8. It is clear that $-m - \alpha$, $-n - \beta$ have precisely the same series of values as $m + \alpha$, $n + \beta$ respectively; hence considering the function:

$$\vartheta \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (-u, -v)$$

the linear term in the argument of the exponential

$$\frac{1}{2}\pi i \{(-m - \alpha)(-u + \gamma) + (-n - \beta)(-v + \delta)\}$$

is

$$\frac{1}{2}\pi i \{(m + \alpha)(u - \gamma) + (n + \beta)(v - \delta)\}$$

which differs from its original value

$$\frac{1}{2}\pi i \{(m + \alpha)(u + \gamma) + (n + \beta)(v + \delta)\}$$

by $-\pi i \{(m + \alpha)\gamma + (n + \beta)\delta\}$, that is $-\pi i(m\gamma + n\delta) - \pi i(\alpha\gamma + \beta\delta)$; the term $-\pi i(m\gamma + n\delta)$, where m, γ and n, δ are integers, does not affect the value of the exponential; and the term $-\pi i(\alpha\gamma + \beta\delta)$ introduces into each term of the series the factor $(-)^{\alpha\gamma + \beta\delta}$, that is, $+1$ or -1 according as the characteristic is even or odd. Hence for an even characteristic the function is even, and for an odd characteristic the function is odd; that is,

$$\vartheta \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (-u, -v) = \pm \vartheta \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (u, v),$$

the upper or lower sign according as the characteristic is even or odd.

Change of signs of arguments.

9. Changing the signs of the arguments, or changing the signs of the characters γ, δ , it is clear that we have

$$\vartheta \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (-u, -v) = \vartheta \begin{pmatrix} \alpha, & \beta \\ -\gamma, & -\delta \end{pmatrix} (u, v),$$

and moreover, altering γ, δ each by an even integer, the sign is not altered; hence the signs of γ, δ may be simultaneously changed; that is,

$$\vartheta \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (u, v) = \vartheta \begin{pmatrix} \alpha, & \beta \\ -\gamma, & -\delta \end{pmatrix} (-u, -v).$$

Interchange of characters and arguments.

10. Writing in the original formula m, n for n, m respectively, and at the same time interchanging u with v and α with β , and γ with δ , the function is unaltered; that is,

$$\vartheta \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (u, v) = \vartheta \begin{pmatrix} \beta, & \alpha \\ \delta, & \gamma \end{pmatrix} (v, u).$$

Zero arguments.

11. The function $\vartheta \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (0, 0)$, say for shortness $\vartheta \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$, vanishes if the characteristic be odd; but if the characteristic be even, it is in general different from zero. We write

$$A = \vartheta \begin{pmatrix} 0, & 0 \\ 0, & 0 \end{pmatrix}, \quad B = \vartheta \begin{pmatrix} 0, & 0 \\ 1, & 0 \end{pmatrix}, \quad C = \vartheta \begin{pmatrix} 0, & 0 \\ 0, & 1 \end{pmatrix}, \quad D = \vartheta \begin{pmatrix} 0, & 0 \\ 1, & 1 \end{pmatrix}.$$

And similarly in the general case we denote the $2^{p-1}(2^p + 1)$ even theta-functions by A, B , etc., and the $2^{p-1}(2^p - 1)$ odd theta-functions by the same letters.

Product-theorem.

12. The product of any two theta-functions of the same arguments (u, v) and (u', v') respectively is equal to the sum of four products of theta-functions of the arguments $(u + u', v + v')$ and $(u - u', v - v')$ respectively. In fact, taking the product

$$\vartheta \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} (u, v) \vartheta \begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix} (u', v')$$

the exponential of any term hereof is

$$\exp. \left[\begin{pmatrix} m + \alpha, & n + \beta \\ u + \gamma, & v + \delta \end{pmatrix} + \begin{pmatrix} m' + \alpha', & n' + \beta' \\ u' + \gamma', & v' + \delta' \end{pmatrix} \right],$$

or writing $M = m + m'$, $N = n + n'$, $m = \mu + M$, $m' = \mu' + M$, etc., this is

$$\exp. \left[\begin{pmatrix} M + \frac{1}{2}(\alpha + \alpha'), & N + \frac{1}{2}(\beta + \beta') \\ u + u' + \frac{1}{2}(\gamma + \gamma'), & v + v' + \frac{1}{2}(\delta + \delta') \end{pmatrix} \right. \\ \left. + \begin{pmatrix} \mu + \frac{1}{2}(\alpha - \alpha'), & \nu + \frac{1}{2}(\beta - \beta') \\ u - u' + \frac{1}{2}(\gamma - \gamma'), & v - v' + \frac{1}{2}(\delta - \delta') \end{pmatrix} \right],$$

where observe that the even integer values of m, m' give for M, μ even integer values, but these are not independent; we have in fact $M + \mu = m + m'$, $M - \mu = m - m'$, which are both even, that is, M and μ are of the same parity; the terms are consequently M even, μ even, or else M odd, μ odd; the pairs of values are consequently $(M, \mu) = (\text{even}, \text{even})$, or (odd, odd) . And similarly $(N, \nu) = (\text{even}, \text{even})$, or (odd, odd) . The pair of conditions may be satisfied in four different ways; and for each of these the sum extends to all positive and negative even integer values of M, N, μ, ν ; that is, it extends to all positive and negative even integer values of m, n, m', n' .

Hence arranging the terms according to the four cases respectively, and observing that in the four cases respectively the terms of the product contain the four factors

$$(-)^{\frac{1}{2}(\alpha - \alpha')(\gamma - \gamma') + \frac{1}{2}(\beta - \beta')(\delta - \delta')}, \quad (-)^{\frac{1}{2}(\alpha - \alpha')(\gamma - \gamma') + \frac{1}{2}(\beta - \beta')(\delta - \delta') + 1},$$

etc., we obtain the theorem.

13. The product of two functions where the two characteristics are

$$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix}$$

respectively, is equal to the sum of the four products which correspond to the four characteristics

$$\begin{pmatrix} \frac{1}{2}(\alpha + \alpha'), & \frac{1}{2}(\beta + \beta') \\ \frac{1}{2}(\gamma + \gamma'), & \frac{1}{2}(\delta + \delta') \end{pmatrix} \begin{pmatrix} \frac{1}{2}(\alpha - \alpha'), & \frac{1}{2}(\beta - \beta') \\ \frac{1}{2}(\gamma - \gamma'), & \frac{1}{2}(\delta - \delta') \end{pmatrix}$$

with the proper signs. In particular, when the two characteristics are the same, we have the square of a single function equal to the sum of four products.

14. The cases most required are those in which each of the characteristics $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix}$ has its four characters each = 0 or 1; and consequently the four characteristics in the reduced product have their characters = 0 or $\frac{1}{2}$. It is easy to see that if in any one of these we alter any character by an integer (that is, here by 0 or by 1), the effect is to interchange the functions; hence it is only necessary to write down a set of formulae for the cases where the characteristics in the reduced product have their characters each = 0 or $\frac{1}{2}$, but subject to no other condition; the signs are to be determined by the formula.

Resume of the ulterior theory of the single functions.

15. For the single functions, I attend here to the full theory in the sequel. But attending at present to the single functions, I write down four of the 16 equations, viz. these are

$$\begin{aligned} \vartheta \begin{pmatrix} 0 \\ 0 \end{pmatrix} (u + u') \cdot \vartheta \begin{pmatrix} 0 \\ 0 \end{pmatrix} (u - u') &= XX' + YY', \\ \vartheta \begin{pmatrix} 1 \\ 0 \end{pmatrix} (u + u') \cdot \vartheta \begin{pmatrix} 1 \\ 0 \end{pmatrix} (u - u') &= YX' + XY', \\ \vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix} (u + u') \cdot \vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix} (u - u') &= XX' - YY', \\ \vartheta \begin{pmatrix} 1 \\ 1 \end{pmatrix} (u + u') \cdot \vartheta \begin{pmatrix} 1 \\ 1 \end{pmatrix} (u - u') &= -YX' + XY'; \end{aligned}$$

where X, Y denote $\vartheta \begin{pmatrix} 0 \\ 0 \end{pmatrix} (2u), \vartheta \begin{pmatrix} 1 \\ 0 \end{pmatrix} (2u)$ respectively, and X', Y' the same functions of $2u'$ respectively.

Product-theorem for the single theta-functions.

In the other 12 equations we have on the left-hand the product of different theta-functions of $u + u', u - u'$ respectively, and on the right-hand expressions involving other functions, X_1, Y_1, X'_1, Y'_1 , etc., of $2u$ and $2u'$ respectively.

Writing $u' = 0$, we have on the left-hand, squares or products of the theta-functions of u , and on the right-hand expressions containing functions of $2u$; in particular, the above equations show that the squares of the four theta-functions are equal to linear functions of

X, Y ; that is, there exist between the squared functions two linear relations.

471

[704

16. By introducing a variable argument x , the four squared functions may be taken to be proportional to linear functions of x :

$$\begin{array}{ccccccc} A^2u & : & B^2u & : & C^2u & : & D^2u \\ \mathfrak{A}(a-x) & : & \mathfrak{A}(b-x) & : & \mathfrak{C}(c-x) & : & \mathfrak{D}(d-x), \end{array}$$

where a, b, c, d are constants.

This suggests a new notation; we write

$$\begin{aligned} A^2u &= \omega^2 \mathfrak{A}(a-x), \\ B^2u &= \omega^2 \mathfrak{A}(b-x), \\ C^2u &= \omega^2 \mathfrak{C}(c-x), \\ D^2u &= \omega^2 \mathfrak{D}(d-x); \end{aligned}$$

and the result just mentioned then is

$$A^2u : B^2u : C^2u : D^2u = \mathfrak{A}(a-x) : \mathfrak{A}(b-x) : \mathfrak{C}(c-x) : \mathfrak{D}(d-x),$$

which expresses that the four functions are the coordinates of a point on a quadri-quadric curve in ordinary space.

The remaining 12 of the 16 equations then contain on the left-hand products such as

$$A(u+u') \cdot B(u-u');$$

and by suitably combining them we obtain equations such as

$$\frac{A(u+u') \cdot B(u-u')}{C(u+u') \cdot D(u-u') + D(u+u') \cdot C(u-u')} = \text{function of } u',$$

where for brevity the arguments are as written above; viz. the numerator of the fraction is

$$B(u+u') \cdot A(u-u') - A(u+u') \cdot B(u-u'),$$

and its denominator is

$$C(u + u') \cdot D(u - u') + D(u + u') \cdot C(u - u').$$

Admitting the form of the equation, the value of the function of u' is at once found by writing in the equation $u = 0$; it is, as it ought to be, a function vanishing for $u' = 0$.

17. Take u indefinitely small; the resulting equation appears thus:

$$\frac{Au \cdot Bu' - Bu \cdot Au'}{Cu \cdot Du' + Du \cdot Cu'} = \text{const.},$$

where Au , Bu are the derived functions, or dividing each side by u' , the differential coefficients; or, what is the same thing, that the differential coefficient of the same combination is a constant in regard to u .

18. It appears thus that $\frac{Au \cdot Bu}{Cu \cdot Du}$ is a constant multiple of the product of the two quotient-functions p and q . And then substituting for the several quotient-functions their values in terms of x , we obtain a differential relation between x , u and v ; viz.

$$\frac{dx}{\sqrt{a - x.b - x.c - x.d - x}} = M du,$$

and it is thus that the quotient-functions are as obtained in the sequel; viz.

$$\text{sn}Ku = \sqrt{\frac{FAu}{Bu}} = Du \div Cu,$$

$$\text{cn}Ku = \sqrt{\frac{GAu}{Bu}} = Bu \div Cu,$$

$$\text{dn}Ku = \sqrt{\frac{HAu}{Bu}};$$

in fact, the elliptic-functions:

$$\text{snu}, \quad \text{cnu}, \quad \text{dnu}$$

and we thus identify the functions Au , Bu , Cu , Du with the functions of Jacobi.

If in the above-mentioned four equations we write first $u = 0$, and then $u' = 0$, by means of the results eliminate from the original

equations the quantities X, Y, X', Y' which occur therein, we obtain expressions for the four products such as $A(u + u') \cdot A(u - u')$.

19. One of these equations is

$$C(u + u') \cdot C(u - u') = Cu \cdot Cu' - Du \cdot Du',$$

and taking herein u' indefinitely small, we obtain

$$\frac{Cu \cdot C''u - (C'u)^2}{Cu^2} = \frac{d^2}{du^2} \log Cu,$$

where the left-hand side is given in fact in terms of the theta-function; and taking the exponential of each side twice, we obtain Cu in terms of the function.

This, in fact, corresponds to Jacobi's equation

$$\frac{d^2}{du^2} \log \vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix} u = -k^2 \operatorname{sn}^2 u + \text{const.}$$

20. Which contains the double ξ ; hence, integrating

$$\int \operatorname{sn}^2 u \, du,$$

an exponential as the integral of a squared quotient-function.

From the same equation

$$C(u) \cdot C(u + u') \cdot C(u - u') = Cu \cdot C^2 u' - B^2 u \cdot D^2 u',$$

differentiating logarithmically in regard to u' and integrating in regard to u , we obtain an equation containing on the left-hand side a term $\log \frac{B'u}{Au}$, and on the right-hand side an integral in regard to u of u' :

$$\int_0^u \frac{B'u}{Au} \, du = u' \cdot \text{const.} + \log \vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix} (u - a);$$

in fact, this corresponds to Jacobi's equation

$$\int_0^u \operatorname{sn} u \operatorname{cn} u \, du = \operatorname{sn}' u \operatorname{dn}' u.$$

21. It may further be noticed that if, in the other three equations of the system, we introduce the corresponding quantity ξ in place of u' , then the integral in question is in place of an expression such as

$$\int \frac{dx}{\sqrt{a - x.b - x.c - x.d - x}},$$

where T is $= -\xi$, or is $=$ any one of three forms such as

$$\begin{vmatrix} 1, & x, & x^2 \\ 1, & a+b, & ab \\ 1, & c+d, & cd \end{vmatrix}.$$

472

[704

Resume of the ulterior theory of the double functions.

23. As regards the ulterior theory of the double functions, it is to be observed that we have not only the 16 equations leading to linear relations between binary products of the 16 different functions. We have thus between the 16 functions a system of quadric relations, which in fact determine the ratios of the 16 functions in terms of two variable parameters x, y . (The 16 functions are thus the coordinates of a point on a quadri-quadric two-fold locus in 15-dimensional space.) The forms depend upon six constants, a, b, c, d, e, f :

$$\sqrt{A} = \sqrt{ah}, \quad \sqrt{a - x.a - y},$$

$$\sqrt{Ab} = \sqrt{a - x.b - x.c - x.d - x.e - x.f - x.a - y.b - y.c - y.d - y.e - y.f - y}$$

(observe that in the symbols $\sqrt{ab}$, etc., the letter f always accompanies the two expressed letters; or, what is the same thing, the duad ab is really an abbreviation for the double triad $abf.cde$): this suggests that the functions should be represented by the single and double letter notation $A(u, v), \dots, AB(u, v), \dots$; viz. if for shortness the arguments are omitted, then we have

$A, \quad B, \quad C, \quad D, \quad E, \quad F, \quad AB, \quad AC, \quad AD, \quad AE, \quad AF, \quad BC, \quad BD, \quad BE,$

proportional to determinate constant multiples of

$$\sqrt{a}, \dots, \sqrt{ab}, \dots,$$

of x and y ; and the squared relations between the 16 functions also lead to writing for shortness

$$\frac{\sqrt{a - x.b - x.c - x.d - x.e - x.f - x.a - y.b - y.c - y.d - y.e - y.f - y}}{x - y}$$

24. It is interesting to notice why in the expressions for $\sqrt{ab}$, etc., the effect of the interchange of x, y is, in fact, to change (u, v) into $(-u, -v)$; consequently to change the sign of the odd functions, and to leave unaltered those of the even functions: the interchange does in fact leave $\sqrt{a}$, etc., unaltered, while it changes $\sqrt{ab}$, etc., into $-\sqrt{ab}$, etc.; and thus, since only the ratios are attended to, there is a change of sign as there should be.

25. The equations of the product-theorem lead to expressions for

$$\begin{array}{cc} u + u' & u - u' \\ A \cdot B - B \cdot A, & \end{array}$$

where the arguments, written above, are used to denote the two arguments, viz. $u + u'$ to denote $(u + u', v + v')$ and $u - u'$ to denote $(u - u', v - v')$; and where the letters A, B denote each or either of them a single or double letter. These expressions are found in terms of the functions of (u, v) and of (u', v') :

taking any such expression in u', v' each of them indefinitely small, but with their ratio arbitrary, we obtain the value of

$$A \cdot dB - B \cdot dA,$$

(viz. w here stands for the two arguments (u, v) , and $\mathfrak{D}$ denotes total differentiation $dA = du \cdot \frac{d}{du}A(u, v) + dv \cdot \frac{d}{dv}A(u, v)$), as a quadric function of the functions of (u, v) ; or dividing by A^2 , the form of $\mathfrak{D} \cdot \frac{B}{A}$, that we should equal the differentials of the quotient-functions themselves.

Substituting for the quotient-functions their values in terms of x, y , we have the differential relations between dx, dy, du, dv , viz. putting for shortness

$$X = a - x.b - x.c - x.d - x.e - x.f - x,$$

and

$$Y = a - y.b - y.c - y.d - y.e - y.f - y,$$

these are of the form

$$\frac{dx}{\sqrt{X}} = \frac{x dx}{\sqrt{X}}, \quad \frac{dy}{\sqrt{Y}} = \frac{y dy}{\sqrt{Y}},$$

each of them equal to a linear function of du and dv ; and in fact there are 16 equations: so that the 16 functions thus give the hyperelliptic addition-theorem for the quotient-functions belonging to the integrals

$$\int \frac{dx}{\sqrt{X}}, \quad \int \frac{x dx}{\sqrt{X}}$$

in accordance with the theory of these integrals.

473

[704

26. The first of the product-theorem, putting therein first $u = 0$, then $v' = 0$, and using the results to eliminate the functions on the right-hand side, give expressions for $A(u + u', v + v') \cdot B(u - u', v - v')$, etc., in terms of the functions of (u, v) and (u', v') : that is, they give an addition-with-subtraction theorem for the double theta-functions. And we have thence also consequences analogous to those which present themselves in the theory of the single functions.

Remark as to notation.

27. I remark, as regards the single theta-functions, that the characteristics $\vartheta \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \vartheta \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \vartheta \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ might for shortness be represented by a series of current numbers 0, 1, 2, 3; and the functions be accordingly called $\vartheta_0 u, \vartheta_1 u, \vartheta_2 u, \vartheta_3 u$; but that, instead of this, I prefer to use throughout the before-mentioned functional symbols A, B, C, D .

As regards the double functions, I do, however, denote the characteristics

$$\begin{array}{cccc} \begin{pmatrix} 0, & 0 \\ 0, & 0 \end{pmatrix}, & \begin{pmatrix} 0, & 0 \\ 1, & 0 \end{pmatrix}, & \begin{pmatrix} 0, & 0 \\ 0, & 1 \end{pmatrix}, & \begin{pmatrix} 0, & 0 \\ 1, & 1 \end{pmatrix}, \\ \begin{pmatrix} 1, & 0 \\ 0, & 0 \end{pmatrix}, & \begin{pmatrix} 1, & 0 \\ 1, & 0 \end{pmatrix}, & \begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}, & \begin{pmatrix} 1, & 0 \\ 1, & 1 \end{pmatrix}, \\ \begin{pmatrix} 0, & 1 \\ 0, & 0 \end{pmatrix}, & \begin{pmatrix} 0, & 1 \\ 1, & 0 \end{pmatrix}, & \begin{pmatrix} 0, & 1 \\ 0, & 1 \end{pmatrix}, & \begin{pmatrix} 0, & 1 \\ 1, & 1 \end{pmatrix}, \\ \begin{pmatrix} 1, & 1 \\ 0, & 0 \end{pmatrix}, & \begin{pmatrix} 1, & 1 \\ 1, & 0 \end{pmatrix}, & \begin{pmatrix} 1, & 1 \\ 0, & 1 \end{pmatrix}, & \begin{pmatrix} 1, & 1 \\ 1, & 1 \end{pmatrix}, \end{array}$$

by the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 respectively; or say the characteristics are represented by the numbers 1 to 16, and the corresponding functions are $A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P$.

474

[704

28. But the formulae will be written in a more convenient form by using for the characteristics a binary notation; viz. writing $\alpha, \beta, \gamma, \delta$ each = 0 or 1, I represent the characteristic by the symbol

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

and the corresponding function by

$$\vartheta \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (u, v) \quad \text{or} \quad \vartheta \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}.$$

Second Part.—The Single Theta-Functions.

Definition and fundamental properties.

29. The single theta-functions are defined by the equation

$$\vartheta \begin{pmatrix} \alpha \\ \gamma \end{pmatrix} u = \Sigma \exp. \left\{ J(am)^2 + \frac{1}{2}\pi i(m + \alpha)(u + \gamma) \right\},$$

where the summation extends to all positive and negative even integer values (zero included) of m . Here α, γ are each of them = 0 or 1; and we have the four functions

$$\vartheta \begin{pmatrix} 0 \\ 0 \end{pmatrix} u, \quad \vartheta \begin{pmatrix} 1 \\ 0 \end{pmatrix} u, \quad \vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix} u, \quad \vartheta \begin{pmatrix} 1 \\ 1 \end{pmatrix} u,$$

or, as I prefer to write them,

$$Au, \quad Bu, \quad Cu, \quad Du.$$

30. The product-theorem gives at once the 16 equations for the single functions; viz. these are the equations expressing the products $\vartheta \begin{pmatrix} \alpha \\ \gamma \end{pmatrix} (u+u') \cdot \vartheta \begin{pmatrix} \alpha' \\ \gamma' \end{pmatrix} (u-u')$ in terms of the functions of $2u$ and $2u'$. The formulae are most conveniently obtained by first expressing the squares and products of the functions of u in terms of the functions of $2u$.

31. The formulae for the squares are

$$A^2u = A(2u) + B(2u),$$

$$B^2u = A(2u) - B(2u),$$

$$C^2u = C(2u) + D(2u),$$

$$D^2u = C(2u) - D(2u).$$

32. We then have the product-formulae

$$2Au \cdot Bu = A'u + B'u,$$

$$2Cu \cdot Du = C'u + D'u,$$

where A' , B' , C' , D' denote the like functions with a different parameter.

Notation by means of radical-functions.

33. The foregoing formulae suggest a notation by means of radical-functions. Writing

$$a - x = \alpha, \quad b - x = \beta, \quad c - x = \gamma, \quad d - x = \delta,$$

the radical-functions are

$$\sqrt{\alpha}, \quad \sqrt{\beta}, \quad \sqrt{\gamma}, \quad \sqrt{\delta},$$

and the products which correspond to the four functions are

$$\sqrt{\alpha\beta}, \quad \sqrt{\alpha\gamma}, \quad \sqrt{\alpha\delta}, \quad \sqrt{\beta\gamma}, \quad \sqrt{\beta\delta}, \quad \sqrt{\gamma\delta}.$$

34. The formulae of the product-theorem then assume a very elegant form; in particular, the equations for the squares become

$$A^2u = \mathfrak{A}(a - x),$$

$$B^2u = \mathfrak{A}(b - x),$$

$$C^2u = \mathfrak{C}(c - x),$$

$$D^2u = \mathfrak{D}(d - x),$$

where $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ are the radical-functions.

Third Part.—The Double Theta-Functions.

Definition and fundamental properties.

35. As an instance, taking $\begin{smallmatrix} \alpha & \alpha' \\ \gamma & \gamma' \end{smallmatrix} = \begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$, the product-equation is

$$\begin{aligned} \vartheta \begin{pmatrix} 1 \\ 1 \end{pmatrix} (u + u') \cdot \vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix} (u - u') &= \Theta \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} (2u) \cdot \Theta \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} (2u') + \Theta \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} (2u) \cdot \Theta \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} (2u') \\ &= iP \cdot P' \qquad \qquad \qquad - iQ \cdot Q', \end{aligned}$$

which agrees with the before-given value.

36. The following values are not actually required: but I give them to fix the ideas and to show the meaning of the quantities with which we work.

$u = 0$	
$X = \Theta \begin{pmatrix} 0 \\ 0 \end{pmatrix} (2u) = 1 + 2q^{\frac{1}{2}} \cos 2\pi u + 2q^2 \cos 4\pi u + \dots,$	$\alpha = 1 + 2q^{\frac{1}{2}} + 2q^2 + \dots,$
$Y = \Theta \begin{pmatrix} 1 \\ 0 \end{pmatrix} (2u) = 2q^{\frac{1}{4}} \cos \pi u + 2q^{\frac{9}{4}} \cos 3\pi u + \dots,$	$\beta = 2q^{\frac{1}{4}} + 2q^{\frac{9}{4}} + \dots,$
$X_1 = \Theta \begin{pmatrix} 0 \\ 1 \end{pmatrix} (2u) = 1 - 2q^{\frac{1}{2}} \cos 2\pi u + 2q^2 \cos 4\pi u - \dots,$	$\alpha_1 = 1 - 2q^{\frac{1}{2}} + 2q^2 - \dots,$
$Y_1 = \Theta \begin{pmatrix} 1 \\ 1 \end{pmatrix} (2u) = 2q^{\frac{1}{4}} \sin \pi u + 2q^{\frac{9}{4}} \sin 3\pi u - \dots,$	$\beta'_1 = 2\pi(-q^{\frac{1}{4}} + 3q^{\frac{9}{4}} - \dots)$
	$= \frac{d}{du} Y_1 \text{ for } u = 0.$

$$P = \Theta \left(\frac{1}{2} \right) (2u) = q^{\frac{1}{8}} (\cos \frac{1}{2} \pi u + i \sin \frac{1}{2} \pi u) + q^{\frac{9}{8}} (\cos \frac{3}{2} \pi u - i \sin \frac{3}{2} \pi u) \\ + q^{\frac{25}{8}} (\cos \frac{5}{2} \pi u + i \sin \frac{5}{2} \pi u) + \dots,$$

$$Q = \Theta \left(\frac{1}{0} \right) (2u) = q^{\frac{1}{8}} (\cos \frac{1}{2} \pi u - i \sin \frac{1}{2} \pi u) + q^{\frac{9}{8}} (\cos \frac{3}{2} \pi u + i \sin \frac{3}{2} \pi u) \\ + q^{\frac{25}{8}} (\cos \frac{5}{2} \pi u - i \sin \frac{5}{2} \pi u) + \dots,$$

$$P_1 = \Theta \left(\frac{1}{2} \right) (2u) = \frac{1+i}{\sqrt{2}} \{ q^{\frac{1}{8}} (\cos \frac{1}{2} \pi u + i \sin \frac{1}{2} \pi u) - q^{\frac{9}{8}} (\cos \frac{3}{2} \pi u - i \sin \frac{3}{2} \pi u) \\ - q^{\frac{25}{8}} (\cos \frac{5}{2} \pi u + i \sin \frac{5}{2} \pi u) + \dots \},$$

$$Q_1 = \Theta \left(\frac{1}{1} \right) (2u) = \frac{1-i}{\sqrt{2}} \{ q^{\frac{1}{8}} (\cos \frac{1}{2} \pi u - i \sin \frac{1}{2} \pi u) - q^{\frac{9}{8}} (\cos \frac{3}{2} \pi u + i \sin \frac{3}{2} \pi u) \\ - q^{\frac{25}{8}} (\cos \frac{5}{2} \pi u - i \sin \frac{5}{2} \pi u) + \dots \};$$

and therefore also

$$p = q = q^{\frac{1}{8}} + q^{\frac{9}{8}} + q^{\frac{25}{8}} + \dots, \\ p_1 = \frac{1+i}{\sqrt{2}} \{ q^{\frac{1}{8}} - q^{\frac{9}{8}} - q^{\frac{25}{8}} + q^{\frac{49}{8}} + q^{\frac{81}{8}} - \dots \}, \quad q_1 = \frac{1-i}{\sqrt{2}} \{ \text{Do.} \}; \quad p_1 = i q_1.$$

480

[704

The square set, and x-formulae.

37. We use the square-set, in the first instance by writing therein $u' = 0$; the equations become

$$A^2 u = \alpha X + \beta Y, = \omega^2 \mathfrak{A}(a - x), \\ B^2 u = \beta X + \alpha Y, = \omega^2 \mathfrak{A}(b - x), \\ C^2 u = \alpha X - \beta Y, = \omega^2 \mathfrak{C}(c - x), \\ D^2 u = \beta X - \alpha Y, = \omega^2 \mathfrak{D}(d - x);$$

viz. the equations without their last members show that the squared functions are linear functions of X and Y , such that there exist values of α, β, ω ; X, Y may be assumed equal to the squared functions $\mathfrak{A}(a-x)$, etc., ω^2 being then proportional to x ; the functions $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ then are proportional to $\mathfrak{A}(a-x)$, etc., so that the squared functions are proportional to $\mathfrak{A}(a-x)$, etc.

To show the meaning of the factor ω^2 , observe that, from any two of the equations, for instance from the first and second, we have an equation without ω ,

$$A^2u - B^2u = \mathfrak{A}(a-x) - \mathfrak{A}(b-x);$$

and using this to determine x , and then substituting in $\omega^2 = A^2u \div \mathfrak{A}(a-x)$, we find

$$\omega^2 = \frac{A^2u - B^2u}{(a-b)\mathfrak{A}d},$$

where the numerator is a function not in anywise more important than any other linear function of A^2u and B^2u .

481

[704

38. The function Du vanishes for $u = 0$, and we may assume that the corresponding value of x is $= d$. Writing in the other equations $u = 0$, they become

$$A^2(0) = (\alpha^2 + \beta^2) = \omega_0^2 \mathfrak{A}(a-d),$$

$$B^2(0) = 2\alpha\beta = \omega_0^2 \mathfrak{A}(b-d),$$

$$C^2(0) = (\alpha^2 - \beta^2) = \omega_0^2 \mathfrak{C}(c-d),$$

where ω_0^2 is the value of ω^2 for $u = 0$; these equations denote the quantities $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ containing a constant ω_0^2 , not absolute equalities but only equalities of ratios: thus, without the ω factors the last-mentioned equations would denote

$$A^2(0) : B^2(0) : C^2(0) = \mathfrak{A}(a-d) : \mathfrak{A}(b-d) : \mathfrak{C}(c-d).$$

The function Du becomes on writing therein $x = d$.

It is convenient to omit the ω factors and to write; it being understood that, without them, the formulae are

$$A^2(0) : B^2(0) : C^2(0) = \mathfrak{A}(a-d) : \mathfrak{A}(b-d) : \mathfrak{C}(c-d) : \mathfrak{D}(d-d);$$

that is,

$$A^2(0) = \alpha^2 + \beta^2 : 2\alpha\beta : \alpha^2 - \beta^2 = \mathfrak{A}(a-d) : \mathfrak{A}(b-d) : \mathfrak{C}(c-d) : \mathfrak{D}(d-d).$$

The functions $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ only present themselves in the products $\omega\omega'$, etc., and their absolute magnitudes are therefore essentially indeterminate but regarding ω as a constant factor of properly determined value, the absolute values of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ may be regarded as determinate, and this is accordingly done in the sequel.

482

[704

Relations between the constants.

39. The formulae contain the differences of the quantities a, b, c, d ; denoting these differences in the usual manner by f, g, h, a, b, c , so that

$$\begin{aligned} -h + g - a &= 0, & -g + f - b &= 0, & -f + h - c &= 0, \\ a + b + c &= 0, \end{aligned}$$

and also

$$af + bg + ch = 0,$$

and then assuming the absolute value of one of the quantities $\mathfrak{D}$, we have the system of relations

$$\begin{aligned} \mathfrak{A}^2 &= -agh, & \mathfrak{B}^2 &= bhf, & \mathfrak{C}^2 &= -cfg, & \mathfrak{D}^2 &= abc, \\ \mathfrak{B}\mathfrak{C}a &= -\mathfrak{B}\mathfrak{D}g, & \mathfrak{C}\mathfrak{A}b &= -\mathfrak{C}\mathfrak{D}h, & \mathfrak{A}\mathfrak{B}c &= -\mathfrak{A}\mathfrak{D}f, \\ \mathfrak{B}\mathfrak{C}\mathfrak{D}f &= -\mathfrak{B}gh, & \mathfrak{C}\mathfrak{A}\mathfrak{D}g &= -\mathfrak{C}hf, & \mathfrak{A}\mathfrak{B}\mathfrak{D}h &= -\mathfrak{A}fg, \\ \mathfrak{A}\mathfrak{B}\mathfrak{C}\mathfrak{D} &= abefgh, \end{aligned}$$

and

$$\begin{aligned}
 -a^2\mathfrak{A}^2 + b^2\mathfrak{B}^2 + c^2\mathfrak{C}^2 + f^2\mathfrak{D}^2 &= bcf(af + bg + ch) = 0, \\
 a^2\mathfrak{A}^2 - b^2\mathfrak{B}^2 + c^2\mathfrak{C}^2 + g^2\mathfrak{D}^2 &= cag(af + bg + ch) = 0, \\
 a^2\mathfrak{A}^2 + b^2\mathfrak{B}^2 - c^2\mathfrak{C}^2 + h^2\mathfrak{D}^2 &= abh(af + bg + ch) = 0, \\
 -f^2\mathfrak{A}^2 + g^2\mathfrak{B}^2 + h^2\mathfrak{C}^2 + \mathfrak{D}^2 &= fgh(af + bg + ch) = 0.
 \end{aligned}$$

It is to be remarked that, taking c, a, b, d in the order of decreasing magnitude, we have $-a, b, c, f, g, h$ all positive; hence $\mathfrak{A}^2, \mathfrak{B}^2, \mathfrak{C}^2, \mathfrak{D}^2$ all real; and taking as we may do, $\mathfrak{D}$ negative, then $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ may be taken positive; that is, we have $-a, b, c, g, h, \mathfrak{A}, \mathfrak{B}, \mathfrak{C}, -\mathfrak{D}$ all of them positive.

40. We have

$$\begin{aligned}
 A^2(0) &= \alpha^2 + \beta^2 = \mathfrak{A}f, \\
 B^2(0) &= 2\alpha\beta = \mathfrak{B}g, \\
 C^2(0) &= \alpha^2 - \beta^2 = \mathfrak{C}h.
 \end{aligned}$$

The foregoing equations

483

[704

41. We have A^2u, B^2u, C^2u, D^2u proportional to $\mathfrak{A}(a-x), \mathfrak{A}(b-x), \mathfrak{C}(c-x), \mathfrak{D}(d-x)$ respectively; or, what is the same thing, $\sqrt{A}, \sqrt{B}, \sqrt{C}, \sqrt{D}$ are proportional to the radical-functions. And hence conversely, the variable x is determined as a rational function of A^2u, B^2u, C^2u, D^2u ; viz.

$$x = \frac{a\mathfrak{A}^2A^2u + b\mathfrak{B}^2B^2u + c\mathfrak{C}^2C^2u + d\mathfrak{D}^2D^2u}{\mathfrak{A}^2A^2u + \mathfrak{B}^2B^2u + \mathfrak{C}^2C^2u + \mathfrak{D}^2D^2u}.$$

42. The differential relation between x and u is

$$\frac{dx}{\sqrt{a-x.b-x.c-x.d-x}} = M du,$$

where M is a constant; and the integral of this equation, determining u as a function of x , is the inverse function which gives rise to the whole theory.

43. The foregoing formulae establish the connexion between the theta-functions and the elliptic functions; and they serve to express the latter in terms of the former. In particular, we have

$$\operatorname{sn}Ku = \sqrt{\frac{\mathfrak{D}}{\mathfrak{E}}} \cdot \frac{Du}{Cu}, \quad \operatorname{cn}Ku = \sqrt{\frac{\mathfrak{D}}{\mathfrak{B}}} \cdot \frac{Bu}{Cu}, \quad \operatorname{dn}Ku = \sqrt{\frac{\mathfrak{E}}{\mathfrak{A}}} \cdot \frac{Au}{Cu}.$$

The product-equations.

44. The 16 product-equations for the double theta-functions may be written in the form:

For the square-set:

$$\begin{aligned} A(u+u') \cdot A(u-u') &= AA' + BB' + CC' + DD', \\ B(u+u') \cdot B(u-u') &= AB' + BA' + CD' + DC', \\ C(u+u') \cdot C(u-u') &= AC' + BD' + CA' + DB', \\ D(u+u') \cdot D(u-u') &= AD' + BC' + CB' + DA'. \end{aligned}$$

45. And for the other products:

$$AB(u+u') \cdot AB(u-u') = \text{etc.}$$

The addition-theorem.

46. The addition-theorem is obtained from the product-equations by eliminating the functions of $(u-u')$ between any two of the equations. The resulting formula expresses the quotient-functions of $(u+u')$ in terms of the quotient-functions of u and u' respectively.

484–491

[704

Addition-formulae.

56. The addition-theorem of the quotient-functions for the elliptic functions is of course obtained in my paper "On the Double ϑ -Functions" (Crelle, t. LXXXVII. (1879), pp. 74—81, [697]); viz.

$$dx = \frac{dy}{\sqrt{Y}} = \frac{dz}{\sqrt{Z}},$$

for the differential equation

$$\frac{dx}{\sqrt{X}} = \frac{dy}{\sqrt{Y}} = \frac{dz}{\sqrt{Z}},$$

the particular integral of which to obtain in the formulae of the paper just referred to, reduces itself to $z = x$, and writing for shortness $a_1 = a - y$, $b_1 = b - y$, $c_1 = c - y$, $d_1 = d - y$, then when the interchange is made, the formulae become

$$\begin{aligned} & \sqrt{d-b.d-c} \{ \sqrt{adb_1c_1} + \sqrt{a_1d_1bc} \} \\ & \quad = (bc, ad) \sqrt{d-b.d-c} \frac{X-Y}{\sqrt{adb_1c_1} - \sqrt{a_1d_1bc}}, \\ & \sqrt{d-b.d-c} \{ \sqrt{bdc_1a_1} + \sqrt{b_1d_1ca} \} \\ & \quad = (ac, bd) \sqrt{d-b.d-c} \frac{X-Y}{\sqrt{bdc_1a_1} - \sqrt{b_1d_1ca}}, \\ & \sqrt{d-b.d-c} \{ \sqrt{cda_1b_1} + \sqrt{c_1d_1ab} \} \\ & \quad = (ab, cd) \sqrt{d-b.d-c} \frac{X-Y}{\sqrt{cda_1b_1} - \sqrt{c_1d_1ab}}, \end{aligned}$$

where (bc, ad) , (ac, bd) , (ab, cd) denote respectively

$$\begin{vmatrix} 1, & x+y, & xy \\ 1, & b+c, & bc \\ 1, & a+d, & ad \end{vmatrix}, \quad \begin{vmatrix} 1, & x+y, & xy \\ 1, & a+c, & ac \\ 1, & b+d, & bd \end{vmatrix}, \quad \begin{vmatrix} 1, & x+y, & xy \\ 1, & a+b, & ab \\ 1, & c+d, & cd \end{vmatrix}.$$

57. In the foregoing formulae, (bc, ad) , (ac, bd) , and (ab, cd) denote respectively

$$\begin{vmatrix} 1, & x+y, & xy \\ 1, & b+c, & bc \\ 1, & a+d, & ad \end{vmatrix}, \quad \begin{vmatrix} 1, & x+y, & xy \\ 1, & a+c, & ac \\ 1, & b+d, & bd \end{vmatrix}, \quad \begin{vmatrix} 1, & x+y, & xy \\ 1, & a+b, & ab \\ 1, & c+d, & cd \end{vmatrix};$$

and substituting for $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$, $b-x$, etc., we have moreover

$$A^2(u+v) = \sqrt{c-b.c-d.c-a} (a-x)(a-y)(a-z),$$

$$B^2(u+v) = \sqrt{c-a.c-d.a-d} (b-x)(b-y)(b-z),$$

$$C^2(u+v) = \sqrt{a-b.a-d.b-d} (c-x)(c-y)(c-z),$$

$$D^2(u+v) = \sqrt{c-b.c-a.a-b} (d-x)(d-y)(d-z),$$

the constant multipliers being of course the same in the three columns.

According to what precedes, the radical of the fourth line should be taken with the sign $-$.

The functions (bc, ad) , etc., contained in the formulae, require a transformation such as

$$(b-c)(bc, ad) = \begin{vmatrix} b-x, & b-y \\ c-x, & c-y \end{vmatrix} \div \begin{vmatrix} b-a, & b-d \\ c-a, & c-d \end{vmatrix}$$

in order to make them separately homogeneous in the differences $a-x$, $b-x$, $c-x$, $d-x$, and $a-y$, $b-y$, $c-y$, $d-y$, and therefore to make them expressible as linear functions of the squared functions Au , etc., and $A'u$, etc., respectively; the formulae then give the quotient-functions $\frac{A(u+v)}{D(u+v)}$, etc., in terms of the quotient-functions of u and v respectively.

Doubly infinite product-forms.

58. The product-functions Au , Bu , Cu , Du may be expressed each as a doubly infinite product. Writing for shortness

$$\begin{aligned} m + \frac{1}{2} &= (m, n), & m - \frac{1}{2} &= (m, n), \\ n + \frac{1}{2} &= (m, n), & n - \frac{1}{2} &= (m, n), \end{aligned}$$

the formulae are

$$Au = \prod_{m,n} \left(1 - \frac{u}{(m,n)}\right) \left(1 - \frac{u}{(m,n)}\right) \left(1 - \frac{u}{(m,n)}\right) \left(1 - \frac{u}{(m,n)}\right),$$

$Bu = \text{etc.},$

with the proper constant factors.

493–500

[704

59. The several formulae lead to the expression of the functions in terms of the parameters a, b, c, d , and the arguments u, v ; and they establish the connexion with the hyperelliptic functions.

60. The foregoing theory is carried out in detail in the memoir; and it is shown how the functions may be calculated numerically for given values of the parameters and arguments. The numerical calculations are illustrated by examples; and the results are compared with those obtained by other methods.

61. The memoir concludes with some remarks on the extension of the theory to the general case of p arguments; and on the application of the theta-functions to the solution of problems in the theory of Abelian functions.

End of Papers 702, 703, 704

C. X.

A Memoir on the Single and Double Theta-Functions

*[From the Philosophical Transactions, vol. CLXXI (1880),
pp. 897-1002; abstract, Proceedings of the Royal Society, vol. XXX
(1880),
pp. 125-127.]*

Section 73.

The expansions might be carried further; we have for instance

$$\begin{aligned}\text{th}0(u) &= 1 + 2Q^{\frac{1}{4}} \cos \frac{1}{2}\pi u + 2Q^{\frac{9}{4}} \cos \frac{3}{2}\pi u, \\ c_0 &= 1 + 2Q^{\frac{1}{4}} + 2Q^{\frac{9}{4}}, \\ c_1 &= 1 - 2Q^{\frac{1}{4}} + 2Q^{\frac{9}{4}}, \\ c_2 &= 1 + 2Q^{\frac{1}{4}} - 2Q^{\frac{9}{4}}, \\ c_3 &= 1 - 2Q^{\frac{1}{4}} - 2Q^{\frac{9}{4}}.\end{aligned}$$

And

$$\begin{aligned}c_1 &= 2Q \cos \frac{1}{2}\pi u + 2Q^5 \cos \frac{5}{2}\pi u + 2A \cos \frac{1}{2}\pi(u+2v) + 2A' \cos \frac{1}{2}\pi(u-2v), \\ c_4 &= 2Q + 2Q^5 + 2A + 2A', \\ c_5 &= -2Q \sin \frac{1}{2}\pi u + 2Q^5 \sin \frac{5}{2}\pi u - 2A \sin \frac{1}{2}\pi(u+2v) - 2A' \sin \frac{1}{2}\pi(u-2v), \\ c_8 &= 2Q \cos \frac{1}{2}\pi u + 2Q^5 \cos \frac{5}{2}\pi u - 2A \cos \frac{1}{2}\pi(u+2v) - 2A' \cos \frac{1}{2}\pi(u-2v), \\ c_9 &= 2Q + 2Q^5 - 2A - 2A', \\ c_{12} &= -2Q \sin \frac{1}{2}\pi u + 2Q^5 \sin \frac{5}{2}\pi u + 2A \sin \frac{1}{2}\pi(u+2v) + 2A' \sin \frac{1}{2}\pi(u-2v),\end{aligned}$$

in which last formulae

$$A = QR^{-\frac{1}{2}}S^{\frac{1}{2}}, \quad A' = QR^{\frac{1}{2}}S^{-\frac{1}{2}}.$$

74.

In the single-and-double-letter notation we have six letters A, B, C, D, E, F ; and the duads $AB, AC, \dots, BE$ are used as abbreviations for the double triads ABF, CDE , &c., the letter F always accompanying the expressed duad; there are thus in all six single-letter symbols, and 10 double-letter symbols, in all 16 symbols, corresponding to the double-theta functions, as already mentioned in the order

$$\begin{array}{cccccccccccccccc} \text{th}_0 & \text{th}_1 & \text{th}_2 & \text{th}_3 & \text{th}_4 & \text{th}_5 & \text{th}_6 & \text{th}_7 & \text{th}_8 & \text{th}_9 & \text{th}_{10} & \text{th}_{11} & \text{th}_{12} & \text{th}_{13} & \text{th}_{14} \\ BD, & CE, & CD, & BE, & AC, & C, & AB, & B, & BC, & DE, & F, & A, & AD, & D, & E \end{array}$$

where observe that the single letters C, B, F, A, D, E correspond to the odd functions $\text{th}_5, \text{th}_7, \text{th}_{10}, \text{th}_{11}, \text{th}_{13}, \text{th}_{14}$ respectively.

75.

The ground of the notation is as follows:

The algebraical relations between the double theta-functions are such that, introducing six constant quantities a, b, c, d, e, f and two variable quantities x, y , it is allowable to express the 16 functions as proportional to given functions of these quantities $(a, b, c, d, e, f; x, y)$; viz. there are six functions the notations of which depend on the single letters a, b, c, d, e, f respectively, and 10 functions the notations of which depend on the pairs $ab, ac, ad, ae, bc, bd, be, cd, ce, de$ respectively: each of the 16 functions is, in fact, proportional to the product of two factors, viz. a constant factor depending only on (a, b, c, d, e, f) , and a variable factor containing also x and y . Attending in the first instance to the variable factors, and writing for shortness

$$\begin{aligned} a - x, b - x, c - x, d - x, e - x, f - x &= a, b, c, d, e, f; \quad x - y = \theta; \\ a - y, b - y, c - y, d - y, e - y, f - y &= a_1, b_1, c_1, d_1, e_1, f_1; \end{aligned}$$

these are of the forms

$$\sqrt{a} = \sqrt{[aa_1]}, \quad \sqrt{ab} = \frac{1}{\theta} \{ \sqrt{abf} \cdot c_1 d_1 e_1 + \sqrt{a_1 b_1 f_1} \cdot cde \}.$$

I remark that to avoid ambiguity the squares of these expressions are throughout written as $(\sqrt{a})^2$ and $(\sqrt{ab})^2$ respectively.

76.

There is, for the constant factors, a like single-and-double-letter notation which will be mentioned presently; but in the first instance I use for the even functions the before-mentioned 10 letters c , and for the odd ones six letters k . I assume that the values $x, y = \infty, \infty$ (ratio not determined) correspond to the values $u = 0, v = 0$ of the arguments; and that w is a function of (x, y) which, when (x, y) are interchanged, changes only its sign, and which for indefinitely large values of (x, y) becomes $= \frac{x-y}{6\sqrt[4]{xy}}$.

This being so, we write (adding a second column which will be afterwards explained)

$$\begin{array}{ll}
 {}^{\text{th}}0 = BD = c_0\sqrt{bd}, & k_0\sqrt{bd}, \\
 {}^{\text{th}}1 = CE = c_1\sqrt{ce}, & k_1\sqrt{ce}, \\
 {}^{\text{th}}2 = CD = c_2\sqrt{cd}, & k_2\sqrt{cd}, \\
 {}^{\text{th}}3 = BE = c_3\sqrt{be}, & k_3\sqrt{be}, \\
 {}^{\text{th}}4 = AC = c_4\sqrt{ac}, & k_4\sqrt{ac}, \\
 {}^{\text{th}}5 = C = c_5\sqrt{c}, & k_5\sqrt{c}, \\
 {}^{\text{th}}6 = AB = c_6\sqrt{ab}, & k_6\sqrt{ab}, \\
 {}^{\text{th}}7 = B = c_7\sqrt{b}, & k_7\sqrt{b}, \\
 {}^{\text{th}}8 = BC = c_8\sqrt{bc}, & k_8\sqrt{bc}, \\
 {}^{\text{th}}9 = DE = c_9\sqrt{de}, & k_9\sqrt{de}, \\
 {}^{\text{th}}10 = F = c_{10}\sqrt{f}, & k_{10}\sqrt{f}, \\
 {}^{\text{th}}11 = A = c_{11}\sqrt{a}, & k_{11}\sqrt{a}, \\
 {}^{\text{th}}12 = AD = c_{12}\sqrt{ad}, & k_{12}\sqrt{ad}, \\
 {}^{\text{th}}13 = D = c_{13}\sqrt{d}, & k_{13}\sqrt{d}, \\
 {}^{\text{th}}14 = E = c_{14}\sqrt{e}, & k_{14}\sqrt{e}, \\
 {}^{\text{th}}15 = AE = c_{15}\sqrt{ae}, & k_{15}\sqrt{ae};
 \end{array}$$

viz. here, on writing $x, y = \infty, \infty$, each of the functions $\sqrt{ab}$, &c. becomes $= \frac{x-y}{6\sqrt[4]{xy}}$ and each of the functions $\sqrt{a}$, &c., becomes $= \sqrt[4]{xy}$; hence by reason of the assumed form of ${}^{\text{th}}0$, the odd functions each vanish (their evanescent values being proportional to $k_1, k_7, k_{10}, k_{11}, k_{13}, k_{14}$ respectively), while the even functions become equal to $c_0, c_1, c_2, c_3, c_4, c_6, c_8, c_9, c_{12}, c_{15}$ respectively.

Observe further that on interchanging x, y , the even functions remain unaltered, while the odd functions change their sign; that is, the interchange of x, y corresponds to the change u, v into $-u, -v$.

77.

As to the values of the 10 c 's and the six k 's in terms of a, b, c, d, e, f , these are proportional to fourth roots, $\sqrt[4]{a}$, &c., $\sqrt[4]{ab}$, &c.; in $\sqrt[4]{a}$, a is in the first instance regarded as standing for the pentad $bcdef$, and then this is used to denote a product of differences $bc.bd.be.bf.cd.ce.cf.de.df.ef$; similarly ab is in the first instance regarded as standing for the double triad $abf.cde$, and then each of these triads is used to denote a product of differences, $ab.af.bf$ and $cd.ce.de$ respectively.

The order of the letters is always the alphabetical one, viz. the single letters and duads denote pentads and double triads, thus:

$$\begin{array}{ll}
 a = bcdef, & ab = abf.cde, \\
 b = acdef, & ac = acf.bde, \\
 c = abdef, & ad = adf.bce, \\
 d = abcef, & ae = aef.bcd, \\
 e = abcdf, & bc = bcf.ade, \\
 f = abcde, & bd = bdf.ace, \\
 & be = bef.acd, \\
 & cd = cdf.abe, \\
 & ce = cef.abd, \\
 & de = def.abc.
 \end{array}$$

There is no fear of ambiguity in writing (and we accordingly write) the squares of $\sqrt{a}$ and $\sqrt{ab}$ as $\sqrt^2 a$ and $\sqrt^2 ab$ respectively; the fourth powers are written $(\sqrt^2 a)^2$ and $(\sqrt^2 ab)^2$; the double stroke of the radical symbol $\sqrt^2$ is in every case perfectly distinctive.

This being so we have as above $c_0 = \sqrt^2 bd$, &c., $k_1 = \sqrt^2 c$, &c.; it is, however, important to notice that the fourth roots in question do not denote positive values, but they are fourth roots each taken with its proper sign ($+$, $-$, $+i$, $-i$, as the case may be) so as to satisfy the identical relations which exist between the sixteen constants; and it is not easy to determine the signs.

The variables u, v are connected with x, y by the differential relations

$$\sigma du + \tau dv = -\frac{1}{2} \left(\frac{dx}{\sqrt{x}} + \frac{dy}{\sqrt{y}} \right), \quad \rho du + \sigma dv = -\frac{1}{2} \left(\frac{dx}{\sqrt{x}} - \frac{dy}{\sqrt{y}} \right),$$

where $\Delta^2 = abcdef$, $\Delta_1^2 = a_1b_1c_1d_1e_1f_1$; which equations contain the constants $\varpi, \rho, \sigma, \tau$, the values of which will be afterwards connected with the other constants.

78.

The c 's are expressed as functions of four quantities $\alpha, \beta, \gamma, \delta$, and connected with each other, and with the constants a, b, c, d, e, f , by the formulae

$$\begin{aligned} c_0 &= \alpha^2 + \beta^2 + \gamma^2 + \delta^2 = \sqrt^2 bd, \\ c_1 &= 2(\alpha\beta + \gamma\delta) = \sqrt^2 ce, \\ c_2 &= 2(\alpha\gamma + \beta\delta) = \sqrt^2 cd, \\ c_3 &= 2(\alpha\delta + \beta\gamma) = \sqrt^2 be, \\ c_4 &= \alpha^2 - \beta^2 + \gamma^2 - \delta^2 = \sqrt^2 ac, \\ c_6 &= 2(\alpha\beta - \gamma\delta) = \sqrt^2 ab, \\ c_8 &= \alpha^2 + \beta^2 - \gamma^2 - \delta^2 = \sqrt^2 bc, \\ c_9 &= 2(\alpha\gamma - \beta\delta) = \sqrt^2 de, \\ c_{12} &= 2(\alpha\delta - \beta\gamma) = \sqrt^2 ad, \\ c_{15} &= \alpha^2 - \beta^2 - \gamma^2 + \delta^2 = \sqrt^2 ae. \end{aligned}$$

It hence appears that we can form an arrangement

c_0	c_3	c_6	c_9	c_{12}	c_{15}
a	b	c	a'	b'	c'
a''	b''	c''	a	b	c

a system of coefficients in the transformation between two sets of rectangular coordinates.

We have, between the squares of these coefficients of transforma-

tion, a system of 6 + 9 equations

$$\begin{array}{lll}
 a^2 + b^2 + c^2 = 1, & a'^2 + b'^2 + c'^2 = 1, & a''^2 + b''^2 + c''^2 = 1, \\
 a^2 + a'^2 + a''^2 = 1, & b^2 + b'^2 + b''^2 = 1, & c^2 + c'^2 + c''^2 = 1, \\
 b^2 + c^2 = a'^2 + a''^2, & b'^2 + c'^2 = a''^2 + a^2, & b''^2 + c''^2 = a^2 + a'^2, \\
 c^2 + a^2 = b'^2 + b''^2, & c'^2 + a'^2 = b''^2 + b^2, & c''^2 + a''^2 = b^2 + b'^2, \\
 a^2 + b^2 = c'^2 + c''^2, & a'^2 + b'^2 = c''^2 + c^2, & a''^2 + b''^2 = c^2 + c'^2.
 \end{array}$$

78. (continued)

that is,

$$\begin{aligned}
c_{12}^2 + c_1^2 + c_6^2 - c_9^2 &= 0, \\
c_4^2 + c_3^2 - c_2^2 + c_{15}^2 &= 0, \\
c_8^2 - c_{12}^2 + c_9^2 + c_2^2 &= 0, \\
c_1^2 + c_6^2 - c_9^2 - c_3^2 + c_8^2 &= 0, \\
c_{15}^2 + c_8^2 - c_6^2 + c_3^2 + c_{12}^2 &= 0, \\
c_4^2 + c_3^2 + c_8^2 - c_1^2 + c_{15}^2 &= 0, \\
c_6^2 + c_3^2 + c_8^2 - c_{12}^2 - c_9^2 &= 0, \\
c_{15}^2 + c_8^2 + c_2^2 - c_1^2 - c_4^2 &= 0, \\
c_4^2 - c_{12}^2 + c_9^2 - c_3^2 - c_2^2 &= 0, \\
c_8^2 + c_2^2 + c_{15}^2 - c_1^2 - c_6^2 &= 0, \\
c_1^2 + c_9^2 - c_{15}^2 - c_8^2 - c_2^2 &= 0, \\
c_4^2 + c_3^2 + c_2^2 - c_6^2 - c_{12}^2 &= 0, \\
c_{15}^2 + c_3^2 - c_9^2 - c_4^2 + c_8^2 &= 0, \\
c_1^2 + c_6^2 + c_2^2 - c_4^2 - c_{15}^2 &= 0, \\
c_{12}^2 + c_9^2 - c_1^2 - c_4^2 - c_3^2 &= 0,
\end{aligned}$$

and we have, between their products, a system of 6 + 9 equations

$$\begin{aligned}
a'a'' + b'b'' + c'c'' &= 0, & a''a + b''b + c''c &= 0, & aa' + bb' + cc' &= 0, \\
bc' + b'c' + b''c'' &= 0, & ca' + c'a' + c''a'' &= 0, & ab' + a'b' + a''b'' &= 0, \\
a, b, c &= b'c'' - b''c', & a', b', c' &= b''c - bc'', & a'', b'', c'' &= b'c' - b'c', \\
&& \text{etc.} &&
\end{aligned}$$

79.

I form in the two notations the following tables:

Table of the 16 Kummer hexads.

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>B</i>	<i>F</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>A</i>	<i>B</i>	<i>F</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>F</i>
<i>B</i>	<i>AB</i>	<i>AC</i>	<i>AD</i>	<i>AE</i>	<i>AB</i>	<i>BC</i>	<i>BD</i>	<i>BE</i>	<i>AB</i>	<i>CD</i>	<i>CE</i>	<i>AC</i>	<i>DE</i>	<i>A</i>
<i>C</i>	<i>CD</i>	<i>BD</i>	<i>BC</i>	<i>BC</i>	<i>AC</i>	<i>AD</i>	<i>AC</i>	<i>AC</i>	<i>BC</i>	<i>AB</i>	<i>AB</i>	<i>BC</i>	<i>AB</i>	<i>B</i>
<i>D</i>	<i>CE</i>	<i>BE</i>	<i>BE</i>	<i>BD</i>	<i>AD</i>	<i>AE</i>	<i>AE</i>	<i>AD</i>	<i>BD</i>	<i>AE</i>	<i>AD</i>	<i>CD</i>	<i>AC</i>	<i>C</i>
<i>E</i>	<i>DE</i>	<i>DE</i>	<i>CE</i>	<i>CD</i>	<i>AE</i>	<i>DE</i>	<i>CE</i>	<i>CD</i>	<i>BE</i>	<i>BE</i>	<i>BD</i>	<i>CE</i>	<i>BC</i>	<i>D</i>
<i>F</i>	11	11	11	11	7	7	7	7	5	5	5	13	13	1
	11	7	5	13	14	10	5	13	14	10	13	14	10	1
	7	6	4	14	12	6	8	3	6	2	1	4	9	1
	5	2	8	8	4	12	4	4	8	6	6	8	6	3
	1	3	3	3	12	15	15	12	15	12	2	4	2	1
	9	9	1	2	15	9	1	2	3	3	1	8	9	9

80.**Table of the 120 pairs.**

[This table occupies the whole of page 507 in the original, presenting the 120 pairs of theta-functions arranged in a systematic form with their characteristics. The arrangement is by duads and single letters corresponding to the 16 symbols.]

81.

Table of the 60 Göpel tetrads.

<i>A.B.AE.BE</i>	11	<i>A.B.AD.BD</i>	7	<i>A.B.AC.BC</i>	15	<i>C.D.CE.DE</i>
<i>CE.CD.DE</i>	5	<i>CF.AB.DE</i>	13	<i>E.F.AB.CD</i>	1	<i>D.F.AB.CE</i>
<i>D.E.CD.CE</i>	14	<i>AC.BD.AD.BC</i>	10	<i>AC.BE.AE.BC</i>	6	<i>AD.BE.AE.BD</i>
<i>A.C.AE.CE</i>	4	<i>A.C.AD.CD</i>	12	<i>A.C.AB.BC</i>	8	<i>B.D.BE.DE</i>
<i>B.E.BD.DE</i>	6	<i>B.F.AC.DE</i>	14	<i>E.F.AC.BD</i>	2	<i>D.F.AC.BE</i>
<i>D.E.BD.BE</i>	4	<i>AB.CD.AD.BC</i>	12	<i>AB.CE.AE.BC</i>	8	<i>AD.CE.AE.CD</i>
<i>A.D.AE.DE</i>	3	<i>A.D.AC.CD</i>	11	<i>A.D.AB.BD</i>	7	<i>B.C.BE.CE</i>
<i>B.E.BC.CE</i>	9	<i>B.F.AD.CE</i>	1	<i>E.F.AD.BC</i>	9	<i>C.F.AD.BE</i>
<i>C.E.CD.DE</i>	11	<i>AB.CD.AC.BD</i>	7	<i>AB.DE.AE.BD</i>	15	<i>AC.DE.AE.CD</i>
<i>A.E.AD.DE</i>	5	<i>A.E.AC.CE</i>	13	<i>A.E.AB.BE</i>	9	<i>B.C.BD.CD</i>
<i>B.D.BC.CD</i>	11	<i>B.F.AE.CD</i>	3	<i>D.F.AE.BC</i>	13	<i>C.F.AE.BD</i>
<i>C.D.BC.BD</i>	5	<i>AB.CE.AC.BE</i>	13	<i>AB.DE.AD.BE</i>	5	<i>AC.DE.AD.CE</i>
<i>A.F.BC.DE</i>	7	<i>A.F.BD.CE</i>	15	<i>A.F.BE.CD</i>	11	<i>B.C.AB.AC</i>
<i>B.D.AB.AD</i>	6	<i>B.E.AB.AE</i>	2	<i>D.E.AD.AE</i>	12	<i>C.E.AC.AE</i>
<i>C.D.AC.AD</i>	4	<i>BD.CE.BE.CD</i>	14	<i>BC.DE.BE.CD</i>	6	<i>BC.DE.BD.CE</i>

82.

The product-theorem, and its results.

The product-theorem was

$${}^{\text{th}}\alpha\beta(u+u').{}^{\text{th}}\gamma\delta(u-u') = \sum \sum (-)^{p\alpha+q\beta\text{th}} \frac{\gamma+p}{2}, \frac{\delta+q}{2} (2u)^{\text{th}} \frac{\gamma+p}{2}, \frac{\delta+q}{2} (2u'),$$

where only one argument is exhibited, viz. $u+u'$, $u-u'$, $2u$, $2u'$ are written in place of $(u+u', v+v')$, $(u-u', v-v')$, $(2u, 2v)$, $(2u', 2v')$ respectively.

The expression on the right-hand side is always a sum of four terms, corresponding to the values $(0,0)$, $(1,0)$, $(0,1)$, and $(1,1)$ of (p,q) .

For the development of the results it was found convenient to use the following auxiliary diagram.

Upper half of characteristic.

[The original contains an extensive tabular diagram showing the upper halves of characteristics for the 256 equations, with entries for $\frac{1}{2}$, $\frac{3}{2}$, $\frac{5}{2}$, $\frac{7}{2}$, $\frac{9}{2}$, $\frac{11}{2}$, $\frac{13}{2}$, $\frac{15}{2}$ across 16 columns with + and - signs.]

83.

The upper characters of the th's have thus the values 0, 1, $\frac{1}{2}$, $\frac{3}{2}$; the lower characters are originally 2, 1, 0, or -1 , and these have when necessary to be, by the addition or subtraction of 2, reduced to 0 or 1; the effect of this change is either to leave the th unaltered, or to multiply it by -1 or $+i$, as follows:

$${}^{\text{th}}\frac{3}{2} = -i {}^{\text{th}}\frac{1}{2}, \quad {}^{\text{th}}\frac{5}{2} = i {}^{\text{th}}\frac{1}{2},$$

where only the first column of characters is shown, but the same rule applies to the second column; and where we must of course combine the multipliers corresponding to the first and second columns respectively: for instance

$$\begin{aligned} {}^{\text{th}}\frac{7}{2}, \frac{7}{2} &= - {}^{\text{th}}\frac{1}{2}, \frac{1}{2}, & {}^{\text{th}}\frac{7}{2}, \frac{9}{2} &= -i {}^{\text{th}}\frac{1}{2}, \frac{1}{2}, \\ {}^{\text{th}}\frac{9}{2}, \frac{7}{2} &= -i {}^{\text{th}}\frac{1}{2}, \frac{1}{2}, & {}^{\text{th}}\frac{9}{2}, \frac{9}{2} &= + {}^{\text{th}}\frac{1}{2}, \frac{1}{2}, \end{aligned}$$

and so on.

84.

But instead of the th's, we introduce single letters (X, Y, Z, W) , (E, F, G, H) , (I, J, K, L) , (M, N, P, Q) , with the suffixes $(0, 1, 2, 3)$, in all 64 symbols, thus:

th (2 <i>u</i>)	00	10	01	11
th 0	<i>X</i>	<i>Y</i>	<i>Z</i>	<i>W</i>
th 1	<i>X</i> ₁	<i>Y</i> ₁	<i>Z</i> ₁	<i>W</i> ₁
th 2	<i>X</i> ₂	<i>Y</i> ₂	<i>Z</i> ₂	<i>W</i> ₂
th 3	<i>X</i> ₃	<i>Y</i> ₃	<i>Z</i> ₃	<i>W</i> ₃

that is, th0⁰(2*u*) = *X*, th1⁰(2*u*) = *Y*, &c., and

th (2 <i>u</i>)	00	10	01	11
th 0	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
th 1	<i>E</i> ₁	<i>F</i> ₁	<i>G</i> ₁	<i>H</i> ₁
th 2	<i>E</i> ₂	<i>F</i> ₂	<i>G</i> ₂	<i>H</i> ₂
th 3	<i>E</i> ₃	<i>F</i> ₃	<i>G</i> ₃	<i>H</i> ₃

th (2 <i>u</i>)	00	10	01	11
th 0	<i>I</i>	<i>J</i>	<i>K</i>	<i>L</i>
th 1	<i>I</i> ₁	<i>J</i> ₁	<i>K</i> ₁	<i>L</i> ₁
th 2	<i>I</i> ₂	<i>J</i> ₂	<i>K</i> ₂	<i>L</i> ₂
th 3	<i>I</i> ₃	<i>J</i> ₃	<i>K</i> ₃	<i>L</i> ₃

th (2 <i>u</i>)	00	10	01	11
th 0	<i>M</i>	<i>N</i>	<i>P</i>	<i>Q</i>
th 1	<i>M</i> ₁	<i>N</i> ₁	<i>P</i> ₁	<i>Q</i> ₁
th 2	<i>M</i> ₂	<i>N</i> ₂	<i>P</i> ₂	<i>Q</i> ₂
th 3	<i>M</i> ₃	<i>N</i> ₃	<i>P</i> ₃	<i>Q</i> ₃

The functions of (2*u*') are denoted in like manner by accented letters th0⁰(2*u*') = *X*', &c.

85.

To simplify the expression of the results, instead of in each case writing down the suffixes, I have indicated them by means of the column headed "Suff." Thus

Suff.		
1	${}^{\text{th}}8(u + u') {}^{\text{th}}0(u - u') = XX' + YY' + ZZ' + WW'$	00
2		10
		01
		11

means that the equation is to be read

$${}^{\text{th}}8^0(u + u') \cdot {}^{\text{th}}0^0(u - u') = X_1X'_1 + Y_1Y'_1 + Z_1Z'_1 + W_1W'_1.$$

It is hardly necessary to mention that the ${}^{\text{th}}8$ of the left-hand column shows the current numbers of the theta-functions; viz. the left-hand side of the equation is ${}^{\text{th}}8(u + u') \cdot {}^{\text{th}}0(u - u')$.

86.

First set, 64 equations.

Suffixes.					
0	${}^{\text{th}}0(u+u'){}^{\text{th}}0(u-u')$	=	$XX' + YY' + ZZ' + WW'$	0000	
1	${}^{\text{th}}4(u+u'){}^{\text{th}}4(u-u')$	=	$XX' + YY' + ZZ' + WW'$	0010	
2	${}^{\text{th}}8(u+u'){}^{\text{th}}8(u-u')$	=	$XX' + YY' + ZZ' + WW'$	0001	
3	${}^{\text{th}}12(u+u'){}^{\text{th}}12(u-u')$	=	$XX' + YY' + ZZ' + WW'$	0011	
1	${}^{\text{th}}0(u+u'){}^{\text{th}}4(u-u')$	=	$XX' - YY' + ZZ' - WW'$	0000	
3	${}^{\text{th}}4(u+u'){}^{\text{th}}4(u-u')$	=	$XX' - YY' + ZZ' - WW'$	0010	
3	${}^{\text{th}}8(u+u'){}^{\text{th}}8(u-u')$	=	$XX' - YY' + ZZ' - WW'$	0001	
2	${}^{\text{th}}12(u+u'){}^{\text{th}}12(u-u')$	=	$XX' - YY' + ZZ' - WW'$	0011	
2	${}^{\text{th}}0(u+u'){}^{\text{th}}8(u-u')$	=	$XX' + YY' - ZZ' - WW'$	0000	
3	${}^{\text{th}}4(u+u'){}^{\text{th}}4(u-u')$	=	$XX' + YY' - ZZ' - WW'$	0010	
1	${}^{\text{th}}8(u+u'){}^{\text{th}}8(u-u')$	=	$XX' + YY' - ZZ' - WW'$	0001	
1	${}^{\text{th}}12(u+u'){}^{\text{th}}12(u-u')$	=	$XX' + YY' - ZZ' - WW'$	0011	
3	${}^{\text{th}}0(u+u'){}^{\text{th}}12(u-u')$	=	$XX' - YY' - ZZ' + WW'$	0000	
2	${}^{\text{th}}4(u+u'){}^{\text{th}}4(u-u')$	=	$XX' - YY' - ZZ' + WW'$	0010	
1	${}^{\text{th}}8(u+u'){}^{\text{th}}8(u-u')$	=	$XX' - YY' - ZZ' + WW'$	0001	
1	${}^{\text{th}}12(u+u'){}^{\text{th}}12(u-u')$	=	$XX' - YY' - ZZ' + WW'$	0011	

86. (continued)

First set, 64 equations (continued).

Suffixes.					
1	${}^{\text{th}}1(u+u'){}^{\text{th}}1(u-u')$	=	$YX' + XY' + WZ' + ZW'$	1000	
2	${}^{\text{th}}5(u+u'){}^{\text{th}}5(u-u')$	=	$YX' + XY' + WZ' + ZW'$	1010	
3	${}^{\text{th}}9(u+u'){}^{\text{th}}9(u-u')$	=	$YX' + XY' + WZ' + ZW'$	1001	
1	${}^{\text{th}}13(u+u'){}^{\text{th}}13(u-u')$	=	$YX' + XY' + WZ' + ZW'$	1011	
1	${}^{\text{th}}1(u+u'){}^{\text{th}}5(u-u')$	=	$YX' - XY' + WZ' - ZW'$	1000	
2	${}^{\text{th}}5(u+u'){}^{\text{th}}5(u-u')$	=	$-YX' + XY' - WZ' + ZW'$	1010	
3	${}^{\text{th}}9(u+u'){}^{\text{th}}9(u-u')$	=	$YX' - XY' + WZ' - ZW'$	1001	
2	${}^{\text{th}}13(u+u'){}^{\text{th}}13(u-u')$	=	$-YX' + XY' - WZ' + ZW'$	1011	
2	${}^{\text{th}}1(u+u'){}^{\text{th}}9(u-u')$	=	$YX' + XY' - WZ' - ZW'$	1000	
3	${}^{\text{th}}5(u+u'){}^{\text{th}}5(u-u')$	=	$YX' + XY' - WZ' - ZW'$	1010	
1	${}^{\text{th}}9(u+u'){}^{\text{th}}9(u-u')$	=	$YX' + XY' - WZ' - ZW'$	1001	
1	${}^{\text{th}}13(u+u'){}^{\text{th}}13(u-u')$	=	$YX' + XY' - WZ' - ZW'$	1011	
3	${}^{\text{th}}1(u+u'){}^{\text{th}}13(u-u')$	=	$YX' - XY' - WZ' + ZW'$	1000	
2	${}^{\text{th}}5(u+u'){}^{\text{th}}5(u-u')$	=	$-YX' + XY' + WZ' - ZW'$	1010	
1	${}^{\text{th}}9(u+u'){}^{\text{th}}9(u-u')$	=	$YX' - XY' - WZ' + ZW'$	1001	
2	${}^{\text{th}}13(u+u'){}^{\text{th}}13(u-u')$	=	$-YX' + XY' + WZ' - ZW'$	1011	

86. (continued)

Suffixes.					
2	${}^{\text{th}}2(u+u'){}^{\text{th}}2(u-u')$	=	$ZX' + WY' + XZ' + YW'$	0100	
1	${}^{\text{th}}6(u+u'){}^{\text{th}}6(u-u')$	=	$ZX' + WY' + XZ' + YW'$	0110	
2	${}^{\text{th}}10(u+u'){}^{\text{th}}10(u-u')$	=	$ZX' + WY' + XZ' + YW'$	0101	
3	${}^{\text{th}}14(u+u'){}^{\text{th}}14(u-u')$	=	$ZX' + WY' + XZ' + YW'$	0111	
1	${}^{\text{th}}2(u+u'){}^{\text{th}}6(u-u')$	=	$ZX' - WY' + XZ' - YW'$	0100	
3	${}^{\text{th}}6(u+u'){}^{\text{th}}6(u-u')$	=	$ZX' - WY' + XZ' - YW'$	0110	
3	${}^{\text{th}}10(u+u'){}^{\text{th}}10(u-u')$	=	$ZX' - WY' + XZ' - YW'$	0101	
2	${}^{\text{th}}14(u+u'){}^{\text{th}}14(u-u')$	=	$ZX' - WY' + XZ' - YW'$	0111	
2	${}^{\text{th}}2(u+u'){}^{\text{th}}10(u-u')$	=	$ZX' + WY' - XZ' - YW'$	0100	
3	${}^{\text{th}}6(u+u'){}^{\text{th}}6(u-u')$	=	$ZX' + WY' - XZ' - YW'$	0110	
1	${}^{\text{th}}10(u+u'){}^{\text{th}}10(u-u')$	=	$-ZX' - WY' + XZ' + YW'$	0101	
1	${}^{\text{th}}14(u+u'){}^{\text{th}}14(u-u')$	=	$-ZX' - WY' + XZ' + YW'$	0111	
3	${}^{\text{th}}2(u+u'){}^{\text{th}}14(u-u')$	=	$ZX' - WY' - XZ' + YW'$	0100	
2	${}^{\text{th}}6(u+u'){}^{\text{th}}6(u-u')$	=	$ZX' - WY' - XZ' + YW'$	0110	
1	${}^{\text{th}}10(u+u'){}^{\text{th}}10(u-u')$	=	$-ZX' + WY' + XZ' - YW'$	0101	
2	${}^{\text{th}}14(u+u'){}^{\text{th}}14(u-u')$	=	$-ZX' + WY' + XZ' - YW'$	0111	

86. (concluded)

Suffixes.					
3	${}^{\text{th}}3(u+u'){}^{\text{th}}3(u-u')$	=	$WX' + ZY' + YZ' + XW'$	1100	
1	${}^{\text{th}}7(u+u'){}^{\text{th}}7(u-u')$	=	$WX' + ZY' + YZ' + XW'$	1110	
2	${}^{\text{th}}11(u+u'){}^{\text{th}}11(u-u')$	=	$WX' + ZY' + YZ' + XW'$	1101	
3	${}^{\text{th}}15(u+u'){}^{\text{th}}15(u-u')$	=	$WX' + ZY' + YZ' + XW'$	1111	
1	${}^{\text{th}}3(u+u'){}^{\text{th}}7(u-u')$	=	$WX' - ZY' + YZ' - XW'$	1100	
2	${}^{\text{th}}7(u+u'){}^{\text{th}}7(u-u')$	=	$-WX' + ZY' - YZ' + XW'$	1110	
3	${}^{\text{th}}11(u+u'){}^{\text{th}}11(u-u')$	=	$WX' - ZY' + YZ' - XW'$	1101	
2	${}^{\text{th}}15(u+u'){}^{\text{th}}15(u-u')$	=	$-WX' + ZY' - YZ' + XW'$	1111	
2	${}^{\text{th}}3(u+u'){}^{\text{th}}11(u-u')$	=	$WX' + ZY' - YZ' - XW'$	1100	
3	${}^{\text{th}}7(u+u'){}^{\text{th}}7(u-u')$	=	$WX' + ZY' - YZ' - XW'$	1110	
1	${}^{\text{th}}11(u+u'){}^{\text{th}}11(u-u')$	=	$-WX' - ZY' + YZ' + XW'$	1101	
1	${}^{\text{th}}15(u+u'){}^{\text{th}}15(u-u')$	=	$-WX' - ZY' + YZ' + XW'$	1111	
3	${}^{\text{th}}3(u+u'){}^{\text{th}}15(u-u')$	=	$WX' - ZY' - YZ' + XW'$	1100	
2	${}^{\text{th}}7(u+u'){}^{\text{th}}7(u-u')$	=	$-WX' + ZY' + YZ' - XW'$	1110	
1	${}^{\text{th}}11(u+u'){}^{\text{th}}11(u-u')$	=	$-WX' + ZY' + YZ' - XW'$	1101	
1	${}^{\text{th}}15(u+u'){}^{\text{th}}15(u-u')$	=	$WX' - ZY' - YZ' + XW'$	1111	

87.

Second set, 64 equations.

Suffixes.					
1	${}^{\text{th}}1(u+u'){}^{\text{th}}0(u-u')$	=	$EE' + GG' + FF' + HH'$	1000	
1	${}^{\text{th}}5(u+u'){}^{\text{th}}4(u-u')$	=	$EE' + GG' + FF' + HH'$	1010	
2	${}^{\text{th}}9(u+u'){}^{\text{th}}8(u-u')$	=	$EE' + GG' + FF' + HH'$	1001	
3	${}^{\text{th}}13(u+u'){}^{\text{th}}12(u-u')$	=	$EE' + GG' + FF' + HH'$	1011	
1	${}^{\text{th}}1(u+u'){}^{\text{th}}4(u-u')$	=	$-iEE' + iGG' - iFF' + iHH'$	1000	
2	${}^{\text{th}}5(u+u'){}^{\text{th}}0(u-u')$	=	$iEE' - iGG' + iFF' - iHH'$	1010	
3	${}^{\text{th}}9(u+u'){}^{\text{th}}12(u-u')$	=	$-iEE' + iGG' - iFF' + iHH'$	1001	
2	${}^{\text{th}}13(u+u'){}^{\text{th}}8(u-u')$	=	$iEE' - iGG' + iFF' - iHH'$	1011	
2	${}^{\text{th}}1(u+u'){}^{\text{th}}8(u-u')$	=	$EE' + GG' - FF' - HH'$	1000	
3	${}^{\text{th}}5(u+u'){}^{\text{th}}12(u-u')$	=	$EE' + GG' - FF' - HH'$	1010	
1	${}^{\text{th}}9(u+u'){}^{\text{th}}0(u-u')$	=	$EE' + GG' - FF' - HH'$	1001	
1	${}^{\text{th}}13(u+u'){}^{\text{th}}4(u-u')$	=	$EE' + GG' - FF' - HH'$	1011	
3	${}^{\text{th}}1(u+u'){}^{\text{th}}12(u-u')$	=	$-iEE' + iGG' + iFF' - iHH'$	1000	
2	${}^{\text{th}}5(u+u'){}^{\text{th}}8(u-u')$	=	$iEE' - iGG' - iFF' + iHH'$	1010	
1	${}^{\text{th}}9(u+u'){}^{\text{th}}4(u-u')$	=	$-iEE' + iGG' + iFF' - iHH'$	1001	
1	${}^{\text{th}}13(u+u'){}^{\text{th}}0(u-u')$	=	$iEE' - iGG' - iFF' + iHH'$	1011	

87. (continued)

Second set, 64 equations (continued).

Suffixes.					
1	${}^{\text{th}}0(u+u'){}^{\text{th}}1(u-u')$	=	$EG' + GE' + FH' + HF'$	0010	
1	${}^{\text{th}}4(u+u'){}^{\text{th}}5(u-u')$	=	$EG' + GE' + FH' + HF'$	0010	
2	${}^{\text{th}}8(u+u'){}^{\text{th}}9(u-u')$	=	$EG' + GE' + FH' + HF'$	0001	
3	${}^{\text{th}}12(u+u'){}^{\text{th}}13(u-u')$	=	$EG' + GE' + FH' + HF'$	0011	
1	${}^{\text{th}}0(u+u'){}^{\text{th}}5(u-u')$	=	$iEG' - iGE' + iFH' - iHF'$	0010	
2	${}^{\text{th}}4(u+u'){}^{\text{th}}1(u-u')$	=	$iEG' - iGE' + iFH' - iHF'$	0010	
3	${}^{\text{th}}8(u+u'){}^{\text{th}}13(u-u')$	=	$iEG' - iGE' + iFH' - iHF'$	0001	
2	${}^{\text{th}}12(u+u'){}^{\text{th}}9(u-u')$	=	$iEG' - iGE' + iFH' - iHF'$	0011	
2	${}^{\text{th}}0(u+u'){}^{\text{th}}9(u-u')$	=	$EG' + GE' - FH' - HF'$	0010	
3	${}^{\text{th}}4(u+u'){}^{\text{th}}13(u-u')$	=	$EG' + GE' - FH' - HF'$	0010	
1	${}^{\text{th}}8(u+u'){}^{\text{th}}1(u-u')$	=	$EG' + GE' - FH' - HF'$	0001	
1	${}^{\text{th}}12(u+u'){}^{\text{th}}5(u-u')$	=	$EG' + GE' - FH' - HF'$	0011	
3	${}^{\text{th}}0(u+u'){}^{\text{th}}13(u-u')$	=	$iEG' - iGE' - iFH' + iHF'$	0010	
2	${}^{\text{th}}4(u+u'){}^{\text{th}}9(u-u')$	=	$iEG' - iGE' - iFH' + iHF'$	0010	
1	${}^{\text{th}}8(u+u'){}^{\text{th}}5(u-u')$	=	$iEG' - iGE' - iFH' + iHF'$	0001	
1	${}^{\text{th}}12(u+u'){}^{\text{th}}1(u-u')$	=	$iEG' - iGE' - iFH' + iHF'$	0011	

87. (continued)

Suffixes.					
3	${}^{\text{th}}3(u+u'){}^{\text{th}}2(u-u')$	=	$FE' + HG' + EF' + GH'$	1101	
1	${}^{\text{th}}7(u+u'){}^{\text{th}}6(u-u')$	=	$FE' + HG' + EF' + GH'$	1110	
2	${}^{\text{th}}11(u+u'){}^{\text{th}}10(u-u')$	=	$FE' + HG' + EF' + GH'$	1101	
3	${}^{\text{th}}15(u+u'){}^{\text{th}}14(u-u')$	=	$FE' + HG' + EF' + GH'$	1111	
1	${}^{\text{th}}3(u+u'){}^{\text{th}}6(u-u')$	=	$-iFE' + iHG' - iEF' + iGH'$	1101	
2	${}^{\text{th}}7(u+u'){}^{\text{th}}2(u-u')$	=	$iFE' - iHG' + iEF' - iGH'$	1110	
3	${}^{\text{th}}11(u+u'){}^{\text{th}}14(u-u')$	=	$-iFE' + iHG' - iEF' + iGH'$	1101	
2	${}^{\text{th}}15(u+u'){}^{\text{th}}10(u-u')$	=	$iFE' - iHG' + iEF' - iGH'$	1111	
2	${}^{\text{th}}3(u+u'){}^{\text{th}}10(u-u')$	=	$FE' + HG' - EF' - GH'$	1101	
3	${}^{\text{th}}7(u+u'){}^{\text{th}}14(u-u')$	=	$FE' + HG' - EF' - GH'$	1110	
1	${}^{\text{th}}11(u+u'){}^{\text{th}}2(u-u')$	=	$-FE' - HG' + EF' + GH'$	1101	
1	${}^{\text{th}}15(u+u'){}^{\text{th}}6(u-u')$	=	$-FE' - HG' + EF' + GH'$	1111	
3	${}^{\text{th}}3(u+u'){}^{\text{th}}14(u-u')$	=	$-iFE' + iHG' + iEF' - iGH'$	1101	
2	${}^{\text{th}}7(u+u'){}^{\text{th}}10(u-u')$	=	$iFE' - iHG' - iEF' + iGH'$	1110	
1	${}^{\text{th}}11(u+u'){}^{\text{th}}6(u-u')$	=	$iFE' - iHG' - iEF' + iGH'$	1101	
1	${}^{\text{th}}15(u+u'){}^{\text{th}}2(u-u')$	=	$-iFE' + iHG' + iEF' - iGH'$	1111	

87. (concluded)

Suffixes.					
2	${}^{\text{th}}2(u+u'){}^{\text{th}}3(u-u')$	=	$FG'+HE'+EH'+GF'$	0111	
1	${}^{\text{th}}6(u+u'){}^{\text{th}}7(u-u')$	=	$FG'+HE'+EH'+GF'$	0111	
2	${}^{\text{th}}10(u+u'){}^{\text{th}}11(u-u')$	=	$FG'+HE'+EH'+GF'$	0101	
3	${}^{\text{th}}14(u+u'){}^{\text{th}}15(u-u')$	=	$FG'+HE'+EH'+GF'$	0111	
1	${}^{\text{th}}2(u+u'){}^{\text{th}}7(u-u')$	=	$iFG'-iHE'+iEH'-iGF'$	0111	
3	${}^{\text{th}}6(u+u'){}^{\text{th}}3(u-u')$	=	$iFG'-iHE'+iEH'-iGF'$	0111	
3	${}^{\text{th}}10(u+u'){}^{\text{th}}15(u-u')$	=	$iFG'-iHE'+iEH'-iGF'$	0101	
2	${}^{\text{th}}14(u+u'){}^{\text{th}}11(u-u')$	=	$iFG'-iHE'+iEH'-iGF'$	0111	
2	${}^{\text{th}}2(u+u'){}^{\text{th}}11(u-u')$	=	$FG'+HE'-EH'-GF'$	0111	
3	${}^{\text{th}}6(u+u'){}^{\text{th}}15(u-u')$	=	$FG'+HE'-EH'-GF'$	0111	
1	${}^{\text{th}}10(u+u'){}^{\text{th}}3(u-u')$	=	$-FG'-HE'+EH'+GF'$	0101	
1	${}^{\text{th}}14(u+u'){}^{\text{th}}7(u-u')$	=	$-FG'-HE'+EH'+GF'$	0111	
3	${}^{\text{th}}2(u+u'){}^{\text{th}}15(u-u')$	=	$iFG'-iHE'-iEH'+iGF'$	0111	
2	${}^{\text{th}}6(u+u'){}^{\text{th}}11(u-u')$	=	$-iFG'+iHE'+iEH'-iGF'$	0111	
1	${}^{\text{th}}10(u+u'){}^{\text{th}}7(u-u')$	=	$-iFG'+iHE'+iEH'-iGF'$	0101	
1	${}^{\text{th}}14(u+u'){}^{\text{th}}3(u-u')$	=	$iFG'-iHE'-iEH'+iGF'$	0111	

88.

Third set, 64 equations.

Suffixes.					
2	${}^{\text{th}}2(u+u'){}^{\text{th}}0(u-u')$	=	$II' + JJ' + KK' + LL'$	0100	
1	${}^{\text{th}}6(u+u'){}^{\text{th}}4(u-u')$	=	$II' + JJ' + KK' + LL'$	0110	
2	${}^{\text{th}}10(u+u'){}^{\text{th}}8(u-u')$	=	$II' + JJ' + KK' + LL'$	0101	
3	${}^{\text{th}}14(u+u'){}^{\text{th}}12(u-u')$	=	$II' + JJ' + KK' + LL'$	0111	
1	${}^{\text{th}}2(u+u'){}^{\text{th}}4(u-u')$	=	$II' - JJ' + KK' - LL'$	0100	
3	${}^{\text{th}}6(u+u'){}^{\text{th}}0(u-u')$	=	$II' - JJ' + KK' - LL'$	0110	
3	${}^{\text{th}}10(u+u'){}^{\text{th}}12(u-u')$	=	$II' - JJ' + KK' - LL'$	0101	
2	${}^{\text{th}}14(u+u'){}^{\text{th}}8(u-u')$	=	$II' - JJ' + KK' - LL'$	0111	
2	${}^{\text{th}}2(u+u'){}^{\text{th}}8(u-u')$	=	$-iII' - iJJ' + iKK' + iLL'$	0100	
3	${}^{\text{th}}6(u+u'){}^{\text{th}}12(u-u')$	=	$-iII' - iJJ' + iKK' + iLL'$	0110	
1	${}^{\text{th}}10(u+u'){}^{\text{th}}0(u-u')$	=	$iII' + iJJ' - iKK' - iLL'$	0101	
1	${}^{\text{th}}14(u+u'){}^{\text{th}}4(u-u')$	=	$iII' + iJJ' - iKK' - iLL'$	0111	
3	${}^{\text{th}}2(u+u'){}^{\text{th}}12(u-u')$	=	$-iII' + iJJ' + iKK' - iLL'$	0100	
2	${}^{\text{th}}6(u+u'){}^{\text{th}}8(u-u')$	=	$-iII' + iJJ' + iKK' - iLL'$	0110	
1	${}^{\text{th}}10(u+u'){}^{\text{th}}4(u-u')$	=	$iII' - iJJ' - iKK' + iLL'$	0101	
1	${}^{\text{th}}14(u+u'){}^{\text{th}}0(u-u')$	=	$iII' - iJJ' - iKK' + iLL'$	0111	

88. (continued)

Third set, 64 equations (continued).

Suffixes.					
3	${}^{\text{th}}3(u+u'){}^{\text{th}}1(u-u')$	=	$JJ' + IJ' + LK' + KL'$	1101	
1	${}^{\text{th}}7(u+u'){}^{\text{th}}5(u-u')$	=	$JJ' + IJ' + LK' + KL'$	1110	
2	${}^{\text{th}}11(u+u'){}^{\text{th}}9(u-u')$	=	$JJ' + IJ' + LK' + KL'$	1101	
3	${}^{\text{th}}15(u+u'){}^{\text{th}}13(u-u')$	=	$JJ' + IJ' + LK' + KL'$	1111	
1	${}^{\text{th}}3(u+u'){}^{\text{th}}5(u-u')$	=	$JJ' - IJ' + LK' - KL'$	1101	
2	${}^{\text{th}}7(u+u'){}^{\text{th}}1(u-u')$	=	$-JJ' + IJ' - LK' + KL'$	1110	
3	${}^{\text{th}}11(u+u'){}^{\text{th}}13(u-u')$	=	$JJ' - IJ' + LK' - KL'$	1101	
2	${}^{\text{th}}15(u+u'){}^{\text{th}}9(u-u')$	=	$-JJ' + IJ' - LK' + KL'$	1111	
2	${}^{\text{th}}3(u+u'){}^{\text{th}}9(u-u')$	=	$-iJJ' - iIJ' + iLK' + iKL'$	1101	
3	${}^{\text{th}}7(u+u'){}^{\text{th}}13(u-u')$	=	$-iJJ' - iIJ' + iLK' + iKL'$	1110	
1	${}^{\text{th}}11(u+u'){}^{\text{th}}1(u-u')$	=	$iJJ' + iIJ' - iLK' - iKL'$	1101	
1	${}^{\text{th}}15(u+u'){}^{\text{th}}5(u-u')$	=	$iJJ' + iIJ' - iLK' - iKL'$	1111	
3	${}^{\text{th}}3(u+u'){}^{\text{th}}13(u-u')$	=	$-iJJ' + iIJ' + iLK' - iKL'$	1101	
2	${}^{\text{th}}7(u+u'){}^{\text{th}}9(u-u')$	=	$iJJ' - iIJ' - iLK' + iKL'$	1110	
1	${}^{\text{th}}11(u+u'){}^{\text{th}}5(u-u')$	=	$iJJ' - iIJ' - iLK' + iKL'$	1101	
1	${}^{\text{th}}15(u+u'){}^{\text{th}}1(u-u')$	=	$-iJJ' + iIJ' + iLK' - iKL'$	1111	

88. (continued)

Suffixes.					
2	${}^{\text{th}}0(u+u'){}^{\text{th}}2(u-u')$	=	$IK' + JL' + KI' + LJ'$	0010	
1	${}^{\text{th}}4(u+u'){}^{\text{th}}6(u-u')$	=	$IK' + JL' + KI' + LJ'$	0010	
2	${}^{\text{th}}8(u+u'){}^{\text{th}}10(u-u')$	=	$IK' + JL' + KI' + LJ'$	0001	
3	${}^{\text{th}}12(u+u'){}^{\text{th}}14(u-u')$	=	$IK' + JL' + KI' + LJ'$	0011	
1	${}^{\text{th}}0(u+u'){}^{\text{th}}6(u-u')$	=	$IK' - JL' + KI' - LJ'$	0010	
3	${}^{\text{th}}4(u+u'){}^{\text{th}}2(u-u')$	=	$IK' - JL' + KI' - LJ'$	0010	
3	${}^{\text{th}}8(u+u'){}^{\text{th}}14(u-u')$	=	$IK' - JL' + KI' - LJ'$	0001	
2	${}^{\text{th}}12(u+u'){}^{\text{th}}10(u-u')$	=	$IK' - JL' + KI' - LJ'$	0011	
2	${}^{\text{th}}0(u+u'){}^{\text{th}}10(u-u')$	=	$iIK' + iJL' - iKI' - iLJ'$	0010	
3	${}^{\text{th}}4(u+u'){}^{\text{th}}14(u-u')$	=	$iIK' + iJL' - iKI' - iLJ'$	0010	
1	${}^{\text{th}}8(u+u'){}^{\text{th}}2(u-u')$	=	$iIK' + iJL' - iKI' - iLJ'$	0001	
1	${}^{\text{th}}12(u+u'){}^{\text{th}}6(u-u')$	=	$iIK' + iJL' - iKI' - iLJ'$	0011	
3	${}^{\text{th}}0(u+u'){}^{\text{th}}14(u-u')$	=	$iIK' - iJL' - iKI' + iLJ'$	0010	
2	${}^{\text{th}}4(u+u'){}^{\text{th}}10(u-u')$	=	$iIK' - iJL' - iKI' + iLJ'$	0010	
1	${}^{\text{th}}8(u+u'){}^{\text{th}}6(u-u')$	=	$iIK' - iJL' - iKI' + iLJ'$	0001	
1	${}^{\text{th}}12(u+u'){}^{\text{th}}2(u-u')$	=	$iIK' - iJL' - iKI' + iLJ'$	0011	

88. (concluded)

Suffixes.					
1	${}^{\text{th}}1(u+u'){}^{\text{th}}3(u-u')$	=	$JK' + IL' + LI' + KJ'$	1001	
2	${}^{\text{th}}5(u+u'){}^{\text{th}}7(u-u')$	=	$JK' + IL' + LI' + KJ'$	1010	
3	${}^{\text{th}}9(u+u'){}^{\text{th}}11(u-u')$	=	$JK' + IL' + LI' + KJ'$	1001	
3	${}^{\text{th}}13(u+u'){}^{\text{th}}15(u-u')$	=	$JK' + IL' + LI' + KJ'$	1011	
1	${}^{\text{th}}1(u+u'){}^{\text{th}}7(u-u')$	=	$JK' - IL' + LI' - KJ'$	1001	
2	${}^{\text{th}}5(u+u'){}^{\text{th}}3(u-u')$	=	$-JK' + IL' - LI' + KJ'$	1010	
3	${}^{\text{th}}9(u+u'){}^{\text{th}}15(u-u')$	=	$JK' - IL' + LI' - KJ'$	1001	
2	${}^{\text{th}}13(u+u'){}^{\text{th}}11(u-u')$	=	$-JK' + IL' - LI' + KJ'$	1011	
2	${}^{\text{th}}1(u+u'){}^{\text{th}}11(u-u')$	=	$iJK' + iIL' - iLI' - iKJ'$	1001	
3	${}^{\text{th}}5(u+u'){}^{\text{th}}15(u-u')$	=	$iJK' + iIL' - iLI' - iKJ'$	1010	
1	${}^{\text{th}}9(u+u'){}^{\text{th}}3(u-u')$	=	$iJK' + iIL' - iLI' - iKJ'$	1001	
1	${}^{\text{th}}13(u+u'){}^{\text{th}}7(u-u')$	=	$iJK' + iIL' - iLI' - iKJ'$	1011	
3	${}^{\text{th}}1(u+u'){}^{\text{th}}15(u-u')$	=	$iJK' - iIL' - iLI' + iKJ'$	1001	
2	${}^{\text{th}}5(u+u'){}^{\text{th}}11(u-u')$	=	$-iJK' + iIL' + iLI' - iKJ'$	1010	
1	${}^{\text{th}}9(u+u'){}^{\text{th}}7(u-u')$	=	$iJK' - iIL' - iLI' + iKJ'$	1001	
1	${}^{\text{th}}13(u+u'){}^{\text{th}}3(u-u')$	=	$-iJK' + iIL' + iLI' - iKJ'$	1011	

89.

Fourth set, 64 equations.

Suffixes.					
3	${}^{\text{th}}3(u+u'){}^{\text{th}}0(u-u')$	=	$MM' + NN' + PP' + QQ'$	1100	
1	${}^{\text{th}}7(u+u'){}^{\text{th}}4(u-u')$	=	$MM' + NN' + PP' + QQ'$	1110	
2	${}^{\text{th}}11(u+u'){}^{\text{th}}8(u-u')$	=	$MM' + NN' + PP' + QQ'$	1101	
3	${}^{\text{th}}15(u+u'){}^{\text{th}}12(u-u')$	=	$MM' + NN' + PP' + QQ'$	1111	
1	${}^{\text{th}}3(u+u'){}^{\text{th}}4(u-u')$	=	$-iMM' + iNN' - iPP' + iQQ'$	1100	
2	${}^{\text{th}}7(u+u'){}^{\text{th}}0(u-u')$	=	$+iMM' - iNN' + iPP' - iQQ'$	1110	
3	${}^{\text{th}}11(u+u'){}^{\text{th}}12(u-u')$	=	$-iMM' + iNN' - iPP' + iQQ'$	1101	
2	${}^{\text{th}}15(u+u'){}^{\text{th}}8(u-u')$	=	$+iMM' - iNN' + iPP' - iQQ'$	1111	
2	${}^{\text{th}}3(u+u'){}^{\text{th}}8(u-u')$	=	$-iMM' - iNN' + iPP' + iQQ'$	1100	
3	${}^{\text{th}}7(u+u'){}^{\text{th}}12(u-u')$	=	$-iMM' - iNN' + iPP' + iQQ'$	1110	
1	${}^{\text{th}}11(u+u'){}^{\text{th}}0(u-u')$	=	$iMM' + iNN' - iPP' - iQQ'$	1101	
1	${}^{\text{th}}15(u+u'){}^{\text{th}}4(u-u')$	=	$iMM' + iNN' - iPP' - iQQ'$	1111	
3	${}^{\text{th}}3(u+u'){}^{\text{th}}12(u-u')$	=	$-MM' + NN' + PP' - QQ'$	1100	
2	${}^{\text{th}}7(u+u'){}^{\text{th}}8(u-u')$	=	$+MM' - NN' - PP' + QQ'$	1110	
1	${}^{\text{th}}11(u+u'){}^{\text{th}}4(u-u')$	=	$+MM' - NN' - PP' + QQ'$	1101	
1	${}^{\text{th}}15(u+u'){}^{\text{th}}0(u-u')$	=	$-MM' + NN' + PP' - QQ'$	1111	

89. (continued)

Fourth set, 64 equations (continued).

Suffixes.					
2	${}^{\text{th}}2(u+u'){}^{\text{th}}1(u-u')$	=	$MN' + NM' + PQ' + QP'$	0110	
1	${}^{\text{th}}6(u+u'){}^{\text{th}}5(u-u')$	=	$MN' + NM' + PQ' + QP'$	0110	
2	${}^{\text{th}}10(u+u'){}^{\text{th}}9(u-u')$	=	$MN' + NM' + PQ' + QP'$	0101	
3	${}^{\text{th}}14(u+u'){}^{\text{th}}13(u-u')$	=	$MN' + NM' + PQ' + QP'$	0111	
1	${}^{\text{th}}2(u+u'){}^{\text{th}}5(u-u')$	=	$iMN' - iNM' + iPQ' - iQP'$	0110	
3	${}^{\text{th}}6(u+u'){}^{\text{th}}1(u-u')$	=	$iMN' - iNM' + iPQ' - iQP'$	0110	
3	${}^{\text{th}}10(u+u'){}^{\text{th}}13(u-u')$	=	$iMN' - iNM' + iPQ' - iQP'$	0101	
2	${}^{\text{th}}14(u+u'){}^{\text{th}}9(u-u')$	=	$iMN' - iNM' + iPQ' - iQP'$	0111	
2	${}^{\text{th}}2(u+u'){}^{\text{th}}9(u-u')$	=	$-iMN' - iNM' + iPQ' + iQP'$	0110	
3	${}^{\text{th}}6(u+u'){}^{\text{th}}13(u-u')$	=	$-iMN' - iNM' + iPQ' + iQP'$	0110	
1	${}^{\text{th}}10(u+u'){}^{\text{th}}1(u-u')$	=	$iMN' + iNM' - iPQ' - iQP'$	0101	
1	${}^{\text{th}}14(u+u'){}^{\text{th}}5(u-u')$	=	$iMN' + iNM' - iPQ' - iQP'$	0111	
3	${}^{\text{th}}2(u+u'){}^{\text{th}}13(u-u')$	=	$MN' - NM' - PQ' + QP'$	0110	
2	${}^{\text{th}}6(u+u'){}^{\text{th}}9(u-u')$	=	$MN' - NM' - PQ' + QP'$	0110	
1	${}^{\text{th}}10(u+u'){}^{\text{th}}5(u-u')$	=	$-MN' + NM' + PQ' - QP'$	0101	
1	${}^{\text{th}}14(u+u'){}^{\text{th}}1(u-u')$	=	$-MN' + NM' + PQ' - QP'$	0111	

89. (continued)

Suffixes.					
1	${}^{\text{th}}1(u+u'){}^{\text{th}}2(u-u')$	=	$MP' + NQ' + PM' + QN'$	1001	
2	${}^{\text{th}}5(u+u'){}^{\text{th}}6(u-u')$	=	$MP' + NQ' + PM' + QN'$	1010	
3	${}^{\text{th}}9(u+u'){}^{\text{th}}10(u-u')$	=	$MP' + NQ' + PM' + QN'$	1001	
3	${}^{\text{th}}13(u+u'){}^{\text{th}}14(u-u')$	=	$MP' + NQ' + PM' + QN'$	1011	
1	${}^{\text{th}}1(u+u'){}^{\text{th}}6(u-u')$	=	$-iMP' + iNQ' - iPM' + iQN'$	1001	
2	${}^{\text{th}}5(u+u'){}^{\text{th}}2(u-u')$	=	$iMP' - iNQ' + iPM' - iQN'$	1010	
3	${}^{\text{th}}9(u+u'){}^{\text{th}}14(u-u')$	=	$-iMP' + iNQ' - iPM' + iQN'$	1001	
2	${}^{\text{th}}13(u+u'){}^{\text{th}}10(u-u')$	=	$iMP' - iNQ' + iPM' - iQN'$	1011	
2	${}^{\text{th}}1(u+u'){}^{\text{th}}10(u-u')$	=	$iMP' + iNQ' - iPM' - iQN'$	1001	
3	${}^{\text{th}}5(u+u'){}^{\text{th}}14(u-u')$	=	$iMP' + iNQ' - iPM' - iQN'$	1010	
1	${}^{\text{th}}9(u+u'){}^{\text{th}}2(u-u')$	=	$iMP' + iNQ' - iPM' - iQN'$	1001	
1	${}^{\text{th}}13(u+u'){}^{\text{th}}6(u-u')$	=	$iMP' + iNQ' - iPM' - iQN'$	1011	
3	${}^{\text{th}}1(u+u'){}^{\text{th}}14(u-u')$	=	$MP' - NQ' - PM' + QN'$	1001	
2	${}^{\text{th}}5(u+u'){}^{\text{th}}10(u-u')$	=	$-MP' + NQ' + PM' - QN'$	1010	
1	${}^{\text{th}}9(u+u'){}^{\text{th}}6(u-u')$	=	$MP' - NQ' - PM' + QN'$	1001	
1	${}^{\text{th}}13(u+u'){}^{\text{th}}2(u-u')$	=	$-MP' + NQ' + PM' - QN'$	1011	

89. (concluded)

Suffixes.					
2	${}^{\text{th}}0(u+u'){}^{\text{th}}3(u-u')$	=	$MQ' + NP' + PN' + QM'$	0011	
1	${}^{\text{th}}4(u+u'){}^{\text{th}}7(u-u')$	=	$MQ' + NP' + PN' + QM'$	0011	
2	${}^{\text{th}}8(u+u'){}^{\text{th}}11(u-u')$	=	$MQ' + NP' + PN' + QM'$	0001	
3	${}^{\text{th}}12(u+u'){}^{\text{th}}15(u-u')$	=	$MQ' + NP' + PN' + QM'$	0011	
1	${}^{\text{th}}0(u+u'){}^{\text{th}}7(u-u')$	=	$iMQ' - iNP' + iPN' - iQM'$	0011	
3	${}^{\text{th}}4(u+u'){}^{\text{th}}3(u-u')$	=	$iMQ' - iNP' + iPN' - iQM'$	0011	
3	${}^{\text{th}}8(u+u'){}^{\text{th}}15(u-u')$	=	$iMQ' - iNP' + iPN' - iQM'$	0001	
2	${}^{\text{th}}12(u+u'){}^{\text{th}}11(u-u')$	=	$iMQ' - iNP' + iPN' - iQM'$	0011	
2	${}^{\text{th}}0(u+u'){}^{\text{th}}11(u-u')$	=	$iMQ' + iNP' - iPN' - iQM'$	0011	
3	${}^{\text{th}}4(u+u'){}^{\text{th}}15(u-u')$	=	$iMQ' + iNP' - iPN' - iQM'$	0011	
1	${}^{\text{th}}8(u+u'){}^{\text{th}}3(u-u')$	=	$iMQ' + iNP' - iPN' - iQM'$	0001	
1	${}^{\text{th}}12(u+u'){}^{\text{th}}7(u-u')$	=	$iMQ' + iNP' - iPN' - iQM'$	0011	
3	${}^{\text{th}}0(u+u'){}^{\text{th}}15(u-u')$	=	$-MQ' + NP' + PN' - QM'$	0011	
2	${}^{\text{th}}4(u+u'){}^{\text{th}}11(u-u')$	=	$-MQ' + NP' + PN' - QM'$	0011	
1	${}^{\text{th}}8(u+u'){}^{\text{th}}7(u-u')$	=	$-MQ' + NP' + PN' - QM'$	0001	
1	${}^{\text{th}}12(u+u'){}^{\text{th}}3(u-u')$	=	$-MQ' + NP' + PN' - QM'$	0011	

90.

I re-arrange these in sets of 16 equations, the equations of the first or square-set of 16 being taken as they stand, but those of the other sets being combined in pairs by addition and subtraction as will be seen. And I now drop altogether the characteristics, retaining only the current numbers: thus, in the set of equations next written down, the first equation is

$${}^{\text{th}}0(u+u'){}^{\text{th}}0(u-u') = XX' + YY' + ZZ' + WW' :$$

in the second set, the first equation is

$$\frac{1}{2}\{{}^{\text{th}}4(u+u'){}^{\text{th}}4(u-u') + {}^{\text{th}}4(u+u'){}^{\text{th}}4(u-u')\} = X_1X'_1 + Z_1Z'_1,$$

and so in other cases.

First or square-set of 16.

$u+u'$	$u-u'$	$u+u'$	$u-u'$	(Suffixes 0.)
0	0	4	4	$= XX' + YY' + ZZ' + WW'$
8	8	12	12	$= XX' - YY' + ZZ' - WW'$
1	1	5	5	$= YX' + XY' + WZ' + ZW'$
9	9	13	13	$= -YX' + XY' - WZ' + ZW'$
2	2	6	6	$= ZX' + WY' + XZ' + YW'$
10	10	14	14	$= -ZX' - WY' + XZ' + YW'$
3	3	7	7	$= WX' + ZY' + YZ' + XW'$
11	11	15	15	$= -WX' - ZY' + YZ' + XW'$

91.

Second set of 16.

$u + u'$	$u - u'$	$u + u'$	$u - u'$		$u + u'$	$u - u'$	$u + u'$	$u - u'$	
4	4	12	8	$= XX' + ZZ'$	4	4	12	8	$= XX' + ZZ'$
5	1	1	5	$= YX' + WZ'$	5	1	1	5	$= YX' + WZ'$
6	2	2	6	$= ZX' + XZ'$	6	2	2	6	$= ZX' + XZ'$
7	3	3	7	$= WX' + YZ'$	7	3	3	7	$= WX' + YZ'$

Third set of 16. (Suffixes 2.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$			
8	8	$= XX' + YY'$	12	4	4	12	$= XX' - YY'$		
9	1	1	9	$= YX' + XY'$	13	5	5		
10	2	2	10	$= ZX' + WY'$	14	6	6		
11	3	3	11	$= WX' + ZY'$	15	7	7		

$u + u'$	$u - u'$				$u + u'$	$u - u'$			
8	8	$= ZZ' + WW'$	12	4	4	12	$= ZZ' - WW'$		
9	1	1	9	$= WZ' + ZW'$	13	5	5		
10	2	2	10	$= XZ' + YW'$	14	6	6		
11	3	3	11	$= YZ' + XW'$	15	7	7		

93.

Fourth set of 16. (Suffixes 3.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$			
12	12	$= XX' + WW'$	8	4	4	8	$= XX' -$		
13	1	1	13	$= YX' + ZW'$	9	5	5		
14	2	2	14	$= ZX' + YW'$	10	6	6		
15	3	3	15	$= WX' + XW'$	11	7	7		

$u + u'$	$u - u'$				$u + u'$	$u - u'$			
12	12	$= YY' + ZZ'$	8	4	4	8	$= -YY' + Z$		
13	1	1	13	$= XY' + WZ'$	9	5	5		
14	2	2	14	$= WY' + XZ'$	10	6	6		
15	3	3	15	$= ZY' + YZ'$	11	7	7		

94.

Fifth set of 16. (Suffixes 0.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
1	1	$= E + G \cdot E' + G'$	5	4	4	5	$= i(E - G$
9	8	8	9	$= E + G$	13	12	12
3	2	2	3	$= F + H$	7	6	6
11	10	10	11	$= -(F + H)$	15	14	14

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
1	1	$= E - G \cdot E' - G'$	5	4	4	5	$= i(E + G$
9	8	8	9	$= E - G$	13	12	12
3	2	2	3	$= F - H$	7	6	6
11	10	10	11	$= -(F - H)$	15	14	14

95.

Sixth set of 16. (Suffixes 1.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$		
5	5	$= E - iG \cdot E' + iG'$	1	4	4	1	$= -i(E$	
8	13	13	8	$= E - iG$	9	12	12	
7	2	2	7	$= F - iH$	3	6	6	
15	10	10	15	$= -(F - iH)$	11	14	14	

$u + u'$	$u - u'$				$u + u'$	$u - u'$		
5	5	$= E + iG \cdot E' - iG'$	1	4	4	1	$= -i(E$	
8	13	13	8	$= E + iG$	9	12	12	
7	2	2	7	$= F + iH$	3	6	6	
15	10	10	15	$= -(F + iH)$	11	14	14	

96.

Seventh set of 16. (Suffixes 2.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
9	9	$= E + G \cdot E' + G'$	13	4	4	13	$= i(E - G$
1	8	8	1	$= E + G$	5	12	12
11	2	2	11	$= F + H$	15	6	6
3	10	10	3	$= -(F + H)$	7	14	14

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
9	9	$= E - G \cdot E' - G'$	13	4	4	13	$= i(E + G$
1	8	8	1	$= E - G$	5	12	12
11	2	2	11	$= F - H$	15	6	6
3	10	10	3	$= -(F - H)$	7	14	14

97.

Eighth set of 16. (Suffixes 3.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
13	13	$= E - iG \cdot E' + iG'$	9	4	4	9	$= -i(E$
5	8	8	5	$= E - iG$	1	12	12
15	2	2	15	$= F - iH$	11	6	6
7	10	10	7	$= -(F - iH)$	3	14	14

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
13	13	$= E + iG \cdot E' - iG'$	9	4	4	9	$= -i(E$
5	8	8	5	$= E + iG$	1	12	12
15	2	2	15	$= F + iH$	11	6	6
7	10	10	7	$= -(F + iH)$	3	14	14

98.

Ninth set of 16. (Suffixes 0.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$			
2	2	$= I + K \cdot I' + K'$		6	4	4	6	$= I + K$	
10	8	8		10	$= i(I - K)$	14	12	12	14
3	1	1		3	$= J + L$	7	5	5	7
11	9	9		11	$= i(J - L)$	15	13	13	15

$u + u'$	$u - u'$				$u + u'$	$u - u'$			
2	2	$= I - K \cdot I' - K'$		6	4	4	6	$= I - K$	
10	8	8		10	$= i(I + K)$	14	12	12	14
3	1	1		3	$= J - L$	7	5	5	7
11	9	9		11	$= i(J + L)$	15	13	13	15

99.

Tenth set of 16. (Suffixes 1.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$			
6	6	$= I + K \cdot I' + K'$	2	4	4	2	$= I + K$		
14	8	8	14	$= i(I - K)$	10	12	12	10	
7	1	1	7	$= J + L$	3	5	5	3	
15	9	9	15	$= i(J - L)$	11	13	13	11	

$u + u'$	$u - u'$				$u + u'$	$u - u'$			
6	6	$= I - K \cdot I' - K'$	2	4	4	2	$= I - K$		
14	8	8	14	$= i(I + K)$	10	12	12	10	
7	1	1	7	$= J - L$	3	5	5	3	
15	9	9	15	$= i(J + L)$	11	13	13	11	

100.

Eleventh set of 16. (Suffixes 2.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
10	10	$= I - iK \cdot I' + iK'$		14	4	4	14 $= -i(I -$
2	8	8		2	$= -i(I + iK)$	6	12 12
11	1	1		11	$= J - iL$	15	5 5
3	9	9		3	$= -i(J + iL)$	7	13 13

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
10	10	$= I + iK \cdot I' - iK'$		14	4	4	14 $= -i(I -$
2	8	8		2	$= -i(I - iK)$	6	12 12
11	1	1		11	$= J + iL$	15	5 5
3	9	9		3	$= -i(J - iL)$	7	13 13

101.

Twelfth set of 16. (Suffixes 3.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
14	14	$= I - iK \cdot I' + iK'$	10	4	4	10	$= -i(I -$
6	8	8	6	$= -i(I + iK)$	2	12	12
15	1	1	15	$= J - iL$	11	5	5
7	9	9	7	$= -i(J + iL)$	3	13	13

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
14	14	$= I + iK \cdot I' - iK'$	10	4	4	10	$= -i(I -$
6	8	8	6	$= -i(I - iK)$	2	12	12
15	1	1	15	$= J + iL$	11	5	5
7	9	9	7	$= -i(J - iL)$	3	13	13

102.

Thirteenth set of 16. (Suffixes 0.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
3	3	$= M + Q \cdot M' + Q'$	7	4	4	7	$= i(M -$
11	8	8	11	$= i(M - Q)$	15	12	12
2	1	1	2	$= N + P$	6	5	5
10	9	9	10	$= i(N - P)$	14	13	13

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
3	3	$= M - Q \cdot M' - Q'$	7	4	4	7	$= i(M +$
11	8	8	11	$= i(M + Q)$	15	12	12
2	1	1	2	$= N - P$	6	5	5
10	9	9	10	$= i(N + P)$	14	13	13

103.

Fourteenth set of 16. (Suffixes 1.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$
7	7	$= M - iQ \cdot M' + iQ'$	3	4	4	3 $= -i$
15	8	8	15	$= i(M + iQ)$	11	12
6	1	1	6	$= N - iP$	2	5
14	9	9	14	$= +i(N + iP)$	10	13

$u + u'$	$u - u'$				$u + u'$	$u - u'$
7	7	$= M + iQ \cdot M' - iQ'$	3	4	4	3 $= -i$
15	8	8	15	$= -i(M - iQ)$	11	12
6	1	1	6	$= N + iP$	2	5
14	9	9	14	$= +i(N - iP)$	10	13

104.

Fifteenth set of 16. (Suffixes 2.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
11	11	$= M - iQ \cdot M' + iQ'$	15	4	4	15	$= i(\Lambda$
3	8	8	3	$= -i(M + iQ)$	7	12	
10	1	1	10	$= N - iP$	14	5	
2	9	9	2	$= -i(N + iP)$	6	13	

$u + u'$	$u - u'$				$u + u'$	$u - u'$	
11	11	$= M + iQ \cdot M' - iQ'$	15	4	4	15	$= i(\Lambda$
3	8	8	3	$= -i(M - iQ)$	7	12	
10	1	1	10	$= N + iP$	14	5	
2	9	9	2	$= -i(N - iP)$	6	13	

105.

Sixteenth set of 16. (Suffixes 3.)

$u + u'$	$u - u'$				$u + u'$	$u - u'$			
15	15	$= M - Q \cdot M' - Q'$	11	4	4	11	$= -i(M$		
7	8	8	7	$= -i(M + Q)$	3	12	12		
14	1	1	14	$= N - P$	10	5	5		
6	9	9	6	$= -i(N + P)$	2	13	13		

$u + u'$	$u - u'$				$u + u'$	$u - u'$			
15	15	$= M + Q \cdot M' + Q'$	11	4	4	11	$= -i(M$		
7	8	8	7	$= -i(M - Q)$	3	12	12		
14	1	1	14	$= N + P$	10	5	5		
6	9	9	6	$= -i(N - P)$	2	13	13		

106.

In the square set, writing $u' = v' = 0$, and $\alpha, \beta, \gamma, \delta$ for X', Y', Z', W' also slightly altering the arrangement, the system becomes: and further writing herein $u = 0, v = 0$, it becomes

u	X	Y	Z	W
c_0	α	β	γ	δ
$^{th}4$	α	$-\beta$	γ	$-\delta$
$^{th}8$	α	β	$-\gamma$	$-\delta$
$^{th}12$	α	$-\beta$	$-\gamma$	δ

$$\begin{aligned} &= \alpha^2 - \beta^2 + \gamma^2 - \delta^2 \\ &= \alpha^2 - \beta^2 - \gamma^2 - \delta^2 \\ &= \alpha^2 - \beta^2 - \gamma^2 + \delta^2 \end{aligned}$$

	4	8	12		
1	β	δ	γ	β	$2(\alpha\gamma\delta + \beta\gamma\delta)$
5	$-\alpha$	δ	$-\gamma$	β	$2(\alpha\beta\delta - \gamma\delta)$
9	α	$-\delta$	$-\gamma$	β	$2(\alpha\beta\gamma + \beta\gamma\delta)$
13	$-\alpha$	$-\delta$	γ	β	$2(\alpha\beta\gamma - \gamma\delta)$

	1	5	9	13		
2	γ	δ	α	γ	$2(\alpha\gamma + \beta\delta)$	
6	γ	$-\delta$	$-\alpha$	β	$2(\alpha\gamma - \beta\delta)$	
10	$-\gamma$	δ	$-\alpha$	$-\beta$		
14	$-\gamma$	$-\delta$	α	$-\beta$		

	2	6	10	14		
3	δ	γ	β	α	$2(\alpha\delta + \beta\gamma)$	
7	δ	$-\gamma$	β	$-\alpha$	$2(\alpha\delta - \beta\gamma)$	
11	$-\delta$	γ	$-\beta$	$-\alpha$		
15	$-\delta$	$-\gamma$	$-\beta$	α		

	3	7	11	15
	$2(\alpha\delta + \beta\gamma)$	$2(\alpha\delta - \beta\gamma)$		

viz. this last is the before-mentioned system of equations giving the values of the 10 zero-functions c in terms of the four constants $\alpha, \beta, \gamma, \delta$.

107.

The system first obtained is a system of 16 equations

$$\sqrt{(\text{th}(u, v))} = \alpha X + \beta Y + \gamma Z + \delta W, \quad \&c.,$$

showing that the squares of the theta-functions are each of them a linear function of the four quantities X, Y, Z, W .

If the functions on the right-hand side were independent (a syzygetic) linear functions of X, Y, Z, W , it would follow that any four (selected at pleasure) of the squared theta-functions are linearly independent, and that we could in terms of these four express linearly each of the remaining 12 squared functions. But this is not so; the form of the linear functions of (X, Y, Z, W) is such that we can (and that in 16 different ways) select out of the 16 linear functions six functions, such that any four of them are connected by a linear equation; and there are consequently 16 hexads of squared theta-functions, such that any four out of the same hexad are connected by a linear relation.

The hexads are shown by the foregoing “Table of the 16 Kummer hexads.”

108.

The *a posteriori* verification is immediately effected; taking for instance the first column, the equations are

	<i>X</i>	<i>Y</i>	<i>Z</i>	<i>W</i>		
<i>A</i>	<i>B</i>	<i>AB</i>	<i>CD</i>	<i>CE</i>		
<i>BE</i>	11	7	6	2	1	9
	$= \beta$	δ	γ	β		
	$= -\beta$	$-\alpha$	δ	$-\alpha$		
	$= \gamma$	β	$-\alpha$	γ		

viz. it should thence follow that there is a linear relation between any four of the six squared functions 11, 7, 6, 2, 1, 9: and it is accordingly seen that this is so.

It further appears that, in the several linear relations, the coefficients (obtained in the first instance as functions of $\alpha, \beta, \gamma, \delta$) are in fact the 10 constants *c*: the 15 relations connecting the several systems of four out of the six squared functions are given in the following table.

109.

[The original contains an extensive table of 15 linear relations between the squared theta-functions, with coefficients formed from the 10 constants c . The equations are of the form $c_i^{\text{th}}i^2 + c_j^{\text{th}}j^2 - c_k^{\text{th}}k^2 + c_l^{\text{th}}l^2 = 0$, representing the Kummer hexad relations.]

The relations are:

$$\begin{aligned} c_9^{\text{th}}9^2 + c_7^{\text{th}}7^2 - c_6^{\text{th}}6^2 + c_2^{\text{th}}2^2 &= 0, \\ c_9^{\text{th}}9^2 + c_7^{\text{th}}7^2 - c_1^{\text{th}}1^2 + c_2^{\text{th}}2^2 &= 0, \\ c_6^{\text{th}}6^2 + c_{15}^{\text{th}}15^2 - c_{12}^{\text{th}}12^2 + c_4^{\text{th}}4^2 - c_2^{\text{th}}2^2 &= 0, \quad \text{etc.} \end{aligned}$$

Read $c_i^{\text{th}}i^2 + c_j^{\text{th}}j^2 - c_k^{\text{th}}k^2 + c_l^{\text{th}}l^2 = 0$, &c.

110.

The first set of 16 equations is the square-set, which has been already considered. If in each of the other sets of 16 equations we write in like manner $u' = 0$, each set in fact reduces itself to eight equations; sets 2, 3, 4 give thus $8 + 8 + 8 = 24$ equations; sets 5 to 8, 9 to 12, and 13 to 15, give each $8 + 8 + 8 + 8 = 32$ equations; or we have sets of 24, 32, 32, 32, together 120 equations, the number being of course one half of $256 - 16$, the number of equations after deducting the 16 equations of the square set.

111.

The first set, 24 equations.

This is derived from the second, third, and fourth sets, each of 16 equations, by writing therein $u' = 0$.

Taking $\alpha_1, \beta_1, \gamma_1, \delta_1$ for the zero-functions corresponding to X_1, Y_1, Z_1, W_1 , then on writing $u' = 0$, X'_1, Y'_1, Z'_1, W'_1 become $\alpha_1, \beta_1, \gamma_1, \delta_1$.

In the second set of 16 equations, the first equations thus are

$$\begin{aligned} {}^{\text{th}}4u \cdot {}^{\text{th}}0u &= \alpha_1 X_1 + \gamma_1 Z_1, \\ {}^{\text{th}}4u \cdot {}^{\text{th}}8u &= \alpha_1 Z_1 - \gamma_1 Z_1, \end{aligned}$$

$$\begin{aligned} {}^{\text{th}}8u \cdot {}^{\text{th}}0u &= \beta_1 Y_1 + \delta_1 W_1, \\ 0 &= \beta_1 Y_1 - \delta_1 W_1, \end{aligned}$$

viz. the equations of the column require that, and are all satisfied if, $\beta_1 = 0, \delta_1 = 0$; hence the zero-functions are $\alpha_1, 0, \gamma_1, 0$; and this being so we have only the equations of the first column.

And similarly as regards the third and fourth sets; the zero values corresponding to X_2, Y_2, Z_2, W_2 and X_3, Y_3, Z_3, W_3 are $\alpha_2, 0, 0, \delta_2$ and $\alpha_3, \beta_3, 0, 0$ respectively; and we have in all $8 + 8 + 8 = 24$ equations.

These are:

(Suffixes 1.)			(Suffixes 2.)		
${}^{\text{th}}4u \cdot {}^{\text{th}}0u$	X	Z	${}^{\text{th}}8u$	X	T
4	$= \gamma$	α	8	$= \alpha$	γ
12	$= \alpha$	$-\gamma$	12	$= \gamma$	$-\alpha$
5	$= \gamma$	$-\alpha$	9	$= \alpha$	$-\gamma$
13	$= -\gamma$	$-\alpha$	13	$= -\alpha$	γ
6	$= \alpha$	γ	10	$= \alpha$	$-\beta$
14	$= -\alpha$	γ	14	$= \beta$	$-\alpha$
7	$= \gamma$	α	11	$= -\beta$	$-\alpha$
15	$= \gamma$	$-\alpha$	15	$= \beta$	α

(Suffixes 3.)		
th 12 <i>u</i>	<i>X</i>	<i>W</i>
12	$= \alpha$	δ
8	$= \delta$	$-\alpha$
4	$= \alpha$	$-\delta$
0	$= \delta$	α
13	$= \alpha$	δ
9	$= -\delta$	α
5	$= \delta$	α
1	$= \alpha$	$-\delta$

111. (continued)

And:

$$\begin{array}{lll}
{}^{\text{th}}4u \cdot {}^{\text{th}}0u = \alpha^2 + \gamma^2 & {}^{\text{th}}8u = \alpha^2 + \beta^2 & {}^{\text{th}}12u = \alpha^2 + \delta^2 \\
{}^{\text{th}}8u = \alpha^2 - \gamma^2 & {}^{\text{th}}12u = \alpha^2 - \beta^2 & {}^{\text{th}}4u = \alpha^2 - \delta^2 \\
{}^{\text{th}}12u = \alpha^2 - \gamma^2 & {}^{\text{th}}4u = \alpha^2 - \beta^2 & {}^{\text{th}}8u = \alpha^2 - \delta^2 \\
{}^{\text{th}}6u = 2\alpha\gamma & {}^{\text{th}}9u = 2\alpha\beta & {}^{\text{th}}1u = 2\alpha\delta \\
{}^{\text{th}}9u = 2\alpha\gamma & {}^{\text{th}}1u = 2\alpha\beta & {}^{\text{th}}15u = 2\alpha\delta \\
{}^{\text{th}}15u = 2\alpha\gamma & {}^{\text{th}}5u = 2\alpha\beta & {}^{\text{th}}3u = 2\alpha\delta.
\end{array}$$

112.

The second set, 32 equations.

To exhibit these in a convenient form, I alter the notation, viz. I write

$$\begin{array}{llllllll} E + Q, & i(E - G), & (F + H), & i(F - H) = X, & Y, & Z, & W \\ E_1 + iG_1, & E_1 - iG_1, & F_1 + iH_1, & F_1 - iH_1 = X_1, & Y_1, & Z_1, & W_1 \\ E_2 + iG_2, & E_2 - iG_2, & F_2 + iH_2, & F_2 - iH_2 = X_2, & Y_2, & Z_2, & W_2 \\ E_3 + iG_3, & E_3 - iG_3, & F_3 + iH_3, & F_3 - iH_3 = X_3, & Y_3, & Z_3, & W_3 \end{array}$$

so that as regards the present set of equations, X, Y, Z, W , signify as just mentioned.

And, this being so, the corresponding zero-values are

$$\alpha, \quad 0, \quad \gamma, \quad 0 \quad | \quad \alpha_1, \quad 0, \quad \gamma_1, \quad 0 \quad | \quad \alpha_2, \quad 0, \quad 0, \quad \delta_2 \quad | \quad \alpha_3, \quad 0, \quad 0, \quad \delta_3.$$

The equations then are:

(Suf. 0)			(Suf. 1)			(Suf. 2)			(Suf. 3)
th 3u	X	Z	th 3u	X	Z	th 3u	X	W	th 3u
1	= γ	1	4	= $-i\alpha$	$-i\gamma$	9	= γ	9	1
9	8	= α	$-\gamma$	9	12	= $-i\alpha$	+ $i\gamma$	1	8
1	12	= $-i\alpha$	+ $i\delta$						
3	2	= γ	3	6	= $-i\gamma$	$-i\alpha$	15	6	= δ
α	15	2	= δ	α					
11	10	= γ	$-\delta$	11	14	= $-i\gamma$	+ $i\alpha$	7	14
7	10	= $-\delta$	α						

112. (continued)

(Suf. 0)	th 3u	YW	(Suf. 1)	th 3u	YW	(Suf. 2)	th 3u	YZ
4	= γ		5	= γ		13	4	= γ
13	12	= $\alpha - \gamma$	13	8	= $\alpha - \delta$	5	12	= $\alpha - \delta$
7	6	= γ	7	2	= γ	11	2	= $-\delta\alpha$
15	14	= $\gamma - \alpha$	15	10	= $\gamma - \alpha$	3	10	= $\delta\alpha$

And:

th3u : $1 = \alpha^2 + \gamma^2$ $4 = -i(\alpha^2 + \gamma^2)$ $9 = \alpha^2 - \delta^2$ $1 = -i(\alpha^2 + \delta^2)$
 $8 = \alpha^2 - \gamma^2$ $12 = -i(\alpha^2 - \gamma^2)$ $1 = \alpha^2 + \delta^2$ $12 = -i(\alpha^2 - \delta^2)$
 $3 = 2\alpha\gamma$ $6 = -2i\alpha\gamma$ $15 = 2\alpha\delta$ $15 = 2\alpha\delta.$

113.

Third set, 32 equations.

We again change the notation, writing

$$\begin{array}{ccccccc} I + K, & i(I - K), & J + L, & i(J - L) = X, & Y, & Z, & W \\ I_1 + iK_1, & I_1 - iK_1, & J_1 + iL_1, & J_1 - iL_1 = X_1, & Y_1, & Z_1, & W_1 \\ I_2 + iK_2, & I_2 - iK_2, & J_2 + iL_2, & J_2 - iL_2 = X_2, & Y_2, & Z_2, & W_2 \\ I_3 + iK_3, & I_3 - iK_3, & J_3 + iL_3, & J_3 - iL_3 = X_3, & Y_3, & Z_3, & W_3 \end{array}$$

the zero values being

$$\alpha, \quad 0, \quad \gamma, \quad 0 \quad | \quad \alpha_1, \quad 0, \quad 0, \quad \delta_1 \quad | \quad \alpha_2, \quad 0, \quad \gamma_2 \quad | \quad \alpha_3, \quad 0, \quad 0, \quad \delta_3.$$

114.

Fourth set, 32 equations.

Again changing the notation, we write

$$\begin{array}{cccccccc} M + Q, & i(M - Q), & N + P, & i(N - P) = X, & Y, & Z, & W \\ M_1 + iQ_1, & M_1 - iQ_1, & N_1 + iP_1, & N_1 - iP_1 = X_1, & Y_1, & Z_1, & W_1 \\ M_2 + iQ_2, & M_2 - iQ_2, & N_2 + iP_2, & N_2 - iP_2 = X_2, & Y_2, & Z_2, & W_2 \\ M_3 + iQ_3, & M_3 - iQ_3, & N_3 + iP_3, & N_3 - iP_3 = X_3, & Y_3, & Z_3, & W_3 \end{array}$$

the zero values being

$$\alpha, \quad 0, \quad \gamma, \quad 0 \quad | \quad \alpha_1, \quad 0, \quad 0, \quad \delta_1 \quad | \quad \alpha_2, \quad 0, \quad \gamma_2, \quad 0 \quad | \quad \alpha_3, \quad 0, \quad 0, \quad \delta_3.$$

115.

It will be noticed that the pairs of theta-functions which present themselves in these equations are the same as in the foregoing "Table of the 120 pairs." And the equations show that the four products, each of a pair of theta-functions, belonging to the upper half or to the lower half of any column of the table, are such that any three of the four products are connected by a linear equation.

The coefficients of these linear relations are, in fact, functions such as the $\alpha^2 + \delta^2$, $\alpha^2 - \delta^2$, $2\alpha\delta$ written down at the foot of the several systems of eight equations, and they are consequently products each of two zero-functions c .

116.

In the left-hand four of these, omitting successively the first, second, third, and fourth equation, and from the remaining three eliminating the X_3 and W_3 , we write down, almost mechanically,

${}^{\text{th}}3u$	${}^{\text{th}}4u$	${}^{\text{th}}8u$	${}^{\text{th}}12u$
3	15	7	11
$-\delta^2 - \alpha^2$	$\alpha^2 - \delta^2$	$-\delta^2 + \alpha^2$	$\alpha^2 + \delta^2$
$2\alpha\delta$	$+2\alpha\delta$	$-2\alpha\delta$	$-\alpha^2 - \delta^2$
$\delta^2 + \alpha^2$	$-2\alpha\delta$		

and thence derive the first of the next following system of equations; read

$$\begin{aligned} c_9 {}^{\text{th}}9 {}^{\text{th}}3 + c_0 {}^{\text{th}}12 {}^{\text{th}}0 + c_4 {}^{\text{th}}4 {}^{\text{th}}8 &= 0, \\ -c_3 {}^{\text{th}}3 {}^{\text{th}}15 + c_7 {}^{\text{th}}7 {}^{\text{th}}11 + c_1 {}^{\text{th}}1 {}^{\text{th}}5 &= 0, \\ c_6 {}^{\text{th}}6 {}^{\text{th}}2 + c_2 {}^{\text{th}}2 {}^{\text{th}}10 + c_0 {}^{\text{th}}0 {}^{\text{th}}14 &= 0, \end{aligned}$$

where the theta-functions have the arguments u, v .

Observe that, on writing herein $u = 0, v = 0$, the first three equations become each of them identically $= 0$; the fourth equation becomes

$$-c_4 c_8 + c_0 c_{12} - c_3 c_{15} = 0,$$

which is one of the relations between the c 's and serves as a verification.

But in the right-hand system, on writing $u = v = 0$, each of the four equations becomes identically $= 0$.

117.

The equations are:

th_9	th_4th_8	$\text{th}_0\text{th}_{12}$	$\text{th}_3\text{th}_{15}$	$\text{th}_7\text{th}_{11}$
th_e	$\text{th}_3\text{th}_{15}$	$-\text{th}_0\text{th}_{12}$	th_4th_8	$-\text{th}_3\text{th}_{15}$
	$-\text{th}_4\text{th}_8$	$-\text{th}_0\text{th}_{12}$	$\text{th}_0\text{th}_{12}$	$-\text{th}_4\text{th}_8$
	$\text{th}_3\text{th}_{15}$	$-\text{th}_4\text{th}_8$	$\text{th}_0\text{th}_{12}$	$-\text{th}_3\text{th}_{15}$
			$= 0,$	

th_9	th_5th_9	$\text{th}_1\text{th}_{13}$	$\text{th}_2\text{th}_{14}$	$\text{th}_6\text{th}_{10}$
th_e	$\text{th}_3\text{th}_{15}$	$-\text{th}_0\text{th}_{12}$	th_4th_8	$-\text{th}_3\text{th}_{15}$
	th_4th_8	$-\text{th}_0\text{th}_{12}$	$\text{th}_0\text{th}_{12}$	$-\text{th}_4\text{th}_8$
	$\text{th}_3\text{th}_{15}$	$-\text{th}_4\text{th}_8$	$-\text{th}_0\text{th}_{12}$	$-\text{th}_3\text{th}_{15}$
			$= 0,$	

th_9	th_7th_9	$\text{th}_3\text{th}_{13}$	$\text{th}_0\text{th}_{14}$	$\text{th}_4\text{th}_{10}$
th_e	$\text{th}_1\text{th}_{15}$	$-\text{th}_2\text{th}_{12}$	th_6th_8	$-\text{th}_1\text{th}_{15}$
	th_6th_8	$-\text{th}_2\text{th}_{12}$	$\text{th}_2\text{th}_{12}$	$-\text{th}_6\text{th}_8$
	$\text{th}_1\text{th}_{15}$	$-\text{th}_6\text{th}_8$	$\text{th}_2\text{th}_{12}$	$-\text{th}_1\text{th}_{15}$
			$= 0,$	

117. (continued)

th_9	$\text{th}_0^{\text{th}6}$	$\text{th}_2^{\text{th}4}$	$\text{th}_9^{\text{th}15}$	$\text{th}_{11}^{\text{th}13}$
th_e	$\text{th}_9^{\text{th}15}$	$-\text{th}_2^{\text{th}4}$	$\text{th}_0^{\text{th}6}$	$-\text{th}_9^{\text{th}15}$
	$\text{th}_0^{\text{th}6}$	$-\text{th}_2^{\text{th}4}$	$\text{th}_2^{\text{th}4}$	$-\text{th}_0^{\text{th}6}$
	$\text{th}_9^{\text{th}15}$	$-\text{th}_0^{\text{th}6}$	$\text{th}_2^{\text{th}4}$	$-\text{th}_9^{\text{th}15}$
			$= 0,$	

th_9	$\text{th}_1^{\text{th}7}$	$\text{th}_3^{\text{th}5}$	$\text{th}_8^{\text{th}14}$	$\text{th}_{10}^{\text{th}12}$
th_e	$\text{th}_9^{\text{th}15}$	$-\text{th}_2^{\text{th}4}$	$\text{th}_0^{\text{th}6}$	$-\text{th}_9^{\text{th}15}$
	$\text{th}_0^{\text{th}6}$	$-\text{th}_2^{\text{th}4}$	$\text{th}_2^{\text{th}4}$	$-\text{th}_0^{\text{th}6}$
	$\text{th}_9^{\text{th}15}$	$-\text{th}_0^{\text{th}6}$	$\text{th}_2^{\text{th}4}$	$-\text{th}_9^{\text{th}15}$
			$= 0,$	

118.

The foregoing equations may be verified, and it is interesting to verify them, by means of the approximate values of the functions: thus, for one of the equations, we have

$$\begin{aligned}
 & (2A + 2A')(-2A + 2A') \cdot 1 \cdot 1 \\
 & \quad - 1 \cdot 1 \cdot 2A \cos \frac{1}{2}\pi(u + v) + 2A' \cos \frac{1}{2}\pi(u - v) \\
 & \quad \cdot -2A \cos \frac{1}{2}\pi(u + v) + 2A' \cos \frac{1}{2}\pi(u - v) \\
 & \quad + 1 \cdot 1 \cdot -2A \sin \frac{1}{2}\pi(u + v) - 2A' \sin \frac{1}{2}\pi(u - v) \\
 & \quad \cdot -2A \sin \frac{1}{2}\pi(u + v) + 2A' \sin \frac{1}{2}\pi(u - v) = 0.
 \end{aligned}$$

viz. the equation to be verified is here

$$\begin{aligned}
 & -4A^2 + 4A'^2 + 4A^2 \cos^2 \frac{1}{2}\pi(u + v) - 4A'^2 \cos^2 \frac{1}{2}\pi(u - v) \\
 & \quad + 4A^2 \sin^2 \frac{1}{2}\pi(u + v) - 4A'^2 \sin^2 \frac{1}{2}\pi(u - v) = 0,
 \end{aligned}$$

which is right.

119.

In the equation

$$\begin{vmatrix} \text{th}3 & \text{th}2 & \text{th}Q & \text{th}1 \\ 2Q & 1 & 2Q \cos \frac{1}{2}\pi u & 1 \\ -2Q & 1 & 2Q \cos \frac{1}{2}\pi u & 1 \\ = 0; \end{vmatrix}$$

this is right, but there is no verification as to the term $c_8c_9^{\text{th}8\text{th}9}$; taking the more approximate values, the term in question taken negatively, that is, $-c_8c_9^{\text{th}8\text{th}9}$, is

$$= -(2A + 2A') \cdot 2S \sin \frac{1}{2}\pi u \cdot -2A \sin \frac{1}{2}\pi(u + v) + 2A' \sin \frac{1}{2}\pi(u - v),$$

which is

$$= -8S^2(A + A')A \cos \frac{1}{2}\pi u + 8S^2(A + A')A \cos \frac{1}{2}\pi(u + 2v) + 8S^2(A + A')A' \cos \frac{1}{2}\pi(u - 2v),$$

and this ought therefore to be the value of the first two terms, that is, of

$$\begin{aligned} & (2Q + 2Q^5 - 2A - 2A')(1 - 2Q^4 - 2S^4) \\ & \times \{2Q \cos \frac{1}{2}\pi u + 2Q^5 \cos \frac{5}{2}\pi u + 2A \cos \frac{1}{2}\pi(u + 2v) + 2A' \cos \frac{1}{2}\pi(u - 2v)\} (1 - 2Q^4 \cos \pi u + 2 \\ & \quad - (2Q + 2Q^5 + 2A + 2A')(1 - 2Q^4 + 2S^4) \\ & \times \{2Q \cos \frac{1}{2}\pi u + 2Q^5 \cos \frac{5}{2}\pi u - 2A \cos \frac{1}{2}\pi(u + 2v) - 2A' \cos \frac{1}{2}\pi(u - 2v)\} (1 - 2Q^4 \cos \pi u - 2 \end{aligned}$$

which to the proper degree of approximation is

$$\begin{aligned} & = (2Q - 4Q^5 - 4QS^4 + 2Q^5 - 2A - 2A') \{2Q \cos \frac{1}{2}\pi u - 4Q^5 \cos \frac{1}{2}\pi u \cos \pi u \\ & + 4QS^4 \cos \frac{1}{2}\pi u \cos \pi v + 2Q^5 \cos \frac{5}{2}\pi u + 2A \cos \frac{1}{2}\pi(u + 2v) + 2A' \cos \frac{1}{2}\pi(u - 2v)\} \\ & - (2Q - 4Q^5 + 4QS^4 + 2Q^5 + 2A + 2A') \{2Q \cos \frac{1}{2}\pi u - 4Q^5 \cos \frac{1}{2}\pi u \cos \pi u \\ & - 4QS^4 \cos \frac{1}{2}\pi u \cos \pi v + 2Q^5 \cos \frac{5}{2}\pi u - 2A \cos \frac{1}{2}\pi(u + 2v) - 2A' \cos \frac{1}{2}\pi(u - 2v)\}. \end{aligned}$$

119. (continued)

This is

$$(2M_0 - 2n)(2M + 2n) - (2M_0 + 2n)(2M - 2n) = 8(M_0n - Mn_0),$$

if for a moment

$$M = Q \cos \frac{1}{2}\pi u - 2Q^5 \cos \frac{1}{2}\pi u \cos \pi u + Q^5 \cos \frac{5}{2}\pi u,$$

$$M_0 = Q - 2Q^5 + Q^5,$$

$$n = 2QS^4 \cos \frac{1}{2}\pi u \cos \pi v + A \cos \frac{1}{2}\pi(u + 2v) + A' \cos \frac{1}{2}\pi(u - 2v),$$

$$n_0 = 2QS^4 + A + A',$$

or substituting and reducing, the value of $8(M_0n - Mn_0)$ to the proper degree of approximation is found to be

$$= -8Q(2QS^4 + A + A') \cos \frac{1}{2}\pi u + 8(Q^5S^4 + SQA) \cos \frac{1}{2}\pi(u + 2v) + 8(Q^5S^4 + SQA') \cos \frac{1}{2}\pi(u - 2v)$$

which in virtue of the relations

$$QA = A'^2S^4, \quad QA' = A^2S^4, \quad Q^5S^4 = AA',$$

is equal to the foregoing value of $c_8c_9^{\text{th}}8^{\text{th}}9$.

I have thought it worth while to give this somewhat elaborate verification.

120.

Résumé of the foregoing results.

In what precedes we have all the quadric relations between the 16 double theta-functions: or say we have the linear relations between squares (squared functions) and the linear relations between pairs (products of two functions): the number of the asyzygetic linear relations between squares is obviously $= 12$; and that of the asyzygetic linear relations between pairs is $= 60$ (since each of the 30 tetrads of pairs gives two asyzygetic relations): there are thus in all $12+60 = 72$, asyzygetic linear relations.

But these constitute only a 13-fold relation between the functions, viz. they are such as to give for the ratios of the 16 functions expressions depending upon two arbitrary parameters, x, y .

Or taking the 16 functions as the coordinates of a point in 15-dimensional space, these coordinates are connected by a 13-fold relation (expressed by means of the foregoing system of 72 quadric equations), and the locus is thus a 2-fold, or two-dimensional, locus in 15-dimensional space.

Hence, taking any four of the functions, these are connected by a single equation that is, regarding the four functions as the coordinates of a point in ordinary space, the locus of the point is a surface.

In particular, the four functions may be any four functions belonging to a hexad: by what precedes there is then a linear relation between the squares of the four functions: or the locus is a quadric surface. Each hexad gives 15 such surfaces, or the number of quadric surfaces is $(16 \times 15 =) 240$.

121.

The 16-nodal quartic surfaces.

If the four functions are those contained in any two pairs out of a tetrad of pairs (see the foregoing "Table of the 120 pairs"), then the locus is a quartic surface, which is, in fact, a Kummer's 16-nodal quartic surface.

For if for a moment x, y and z, w are two pairs out of a tetrad, and r, s be either of the remaining pairs of the tetrad; then we have rs a linear function of xy and zw . Squaring, r^2s^2 is a linear function of $x^2y^2, xyzw, z^2w^2$; but we then have r^2 and s^2 each of them a linear function of x^2, y^2, z^2, w^2 ; or substituting we have an equation of the fourth order, containing terms of the second order in (x^2, y^2, z^2, w^2) , and also a term in $xyzw$.

It is clear that, if instead of r, s we had taken the remaining pair of the tetrad, we should have obtained the same quartic equation in (x, y, z, w) .

And moreover it appears by inspection that, if xy and zw are pairs in a tetrad, then xz and yw are pairs in a second tetrad, and xw and yz are pairs in a third tetrad; we obtain in each case the same quartic equation.

We have from each tetrad of pairs six sets of four functions (x, y, z, w) : and the number of such sets is thus $(6 \cdot 30 =) 60$: these are shown in the foregoing "Table of the 60 Göpel tetrads," viz. taking as coordinates of a point the four functions in any tetrad of this table, the locus is a 16-nodal quartic surface.

122.

To exhibit the process I take a tetrad 4, 7, 8, 11 containing two odd functions; and representing these for convenience by x, y, z, w , viz. writing

$${}^{\text{th}}4(u), {}^{\text{th}}7(u), {}^{\text{th}}8(u), {}^{\text{th}}11(u) = x, y, z, w,$$

we have then X, Y, Z, W linear functions of the four squares, viz. it is easy to obtain

$$\begin{aligned}\alpha(x^2 + z^2) - \beta(y^2 + w^2) &= 2(\alpha^2 - \beta^2)X, \\ \beta(x^2 + z^2) - \alpha(y^2 + w^2) &= 2(\alpha^2 - \beta^2)W, \\ -\gamma(x^2 - z^2) + \delta(y^2 - w^2) &= 2i(\gamma^2 - \delta^2)Y, \\ -\delta(x^2 - z^2) + \gamma(y^2 - w^2) &= 2i(\gamma^2 - \delta^2)Z.\end{aligned}$$

Also considering two other functions ${}^{\text{th}}0(u)$ and ${}^{\text{th}}12(u)$, or as for shortness I write them, ${}^{\text{th}}0$ and ${}^{\text{th}}12$, we have

$$\begin{aligned}{}^{\text{th}}0^2 &= \alpha X + \beta Y + \gamma Z + \delta W, \\ {}^{\text{th}}12^2 &= \alpha X - \beta Y - \gamma Z + \delta W,\end{aligned}$$

and substituting the foregoing values of X, Y, Z, W , we find

$$\begin{aligned}M^{{}^{\text{th}}0^2} &= Ax^2 + By^2 + Cz^2 + Dw^2, \\ M^{{}^{\text{th}}12^2} &= Cx^2 + Dy^2 + Az^2 + Bw^2,\end{aligned}$$

where, writing down the values first in terms of $\alpha, \beta, \gamma, \delta$ and then in terms of the c 's, we have

$$\begin{aligned}M &= (\alpha^2 - \beta^2)(\gamma^2 - \delta^2) = \frac{1}{4}(c_4^2 - c_6^2), \\ A &= \beta\gamma - \alpha\delta = \frac{1}{2}(-c_1c_9), \\ B &= -\alpha\beta(\gamma^2 - \delta^2) + \beta\gamma(\alpha^2 - \delta^2) = \frac{1}{2}(c_3c_8^2 - c_1c_{12}^2), \\ C &= \alpha\gamma - \beta\delta = \frac{1}{2}(c_1c_3), \\ D &= [\text{similar expression}].\end{aligned}$$

and we then have further that is, whence equating the two values of ${}^{\text{th}}0^2{}^{\text{th}}12^2$, we have the required quartic equation in x, y, z, w .

123.

But the reduction is effected more simply if instead of the c 's we introduce the rectangular coefficients a , b , c , &c.

We then have

$$M = (c''^2 - b''^2), \quad A = -a''c, \quad C = a'b, \quad B = -b'c'' - b''c'', \quad D = b'b'' + c'c'' = a'a''$$

and the equations become

$$\begin{aligned} (c''^2 - b''^2)x^2 &= -a''cx^2 + bcy^2 + a'bz^2 - a'a''w^2, \\ (c''^2 - b''^2)y^2 &= a'bx^2 - a'a''y^2 - a''cz^2 + bcw^2, \\ \sqrt{b'c''}z &= \sqrt{-a''c}xz + \sqrt{a'by}\sqrt{w}, \end{aligned}$$

so that the elimination gives

$$\begin{aligned} b'c''(-a''cx^2 + bcy^2 + a'bz^2 - a'a''w^2)(a'bx^2 - a'a''y^2 - a''cz^2 + bcw^2) \\ = (c''^2 - b''^2)^2 \{a^2x^2z^2 - b''c''y^2w^2 + 2\sqrt{-aa''bc'b''c''}xyzw\}, \end{aligned}$$

viz. this is

$$\begin{aligned} -a'a''bb'c''(x^4 + y^4 + z^4 + w^4) \\ + a'^2b'c''(a''^2 + b^2)(x^2w^2 + z^2w^2) \\ + \{b'c''(a''b' + a'c'') - a'(b'' - c''^2)^2\}cw^2 \\ + \{b'c''(a'a'' + b'c'') + b''c'(b'' - c''^2)^2\}cw^2 \\ - a''bb'c''(a''^2 + c^2)(x^2w^2 + y^2z^2) \\ - 2(b'' - c''^2)^2\sqrt{-ab''c'}xyzw = 0. \end{aligned}$$

124.

In this equation the coefficients of x^2z^2 and y^2w^2 are each $= c_0^4 a'' bc(b'^2 + c''^2)$, as at once appears from the identities

$$\begin{aligned}\sqrt{a'b.bc'' - c''.a''c} &= a\sqrt{b'^2 - c''^2}, \\ \sqrt{a'b.c'' - b'.a''c} &= \sqrt{b'^2 - c''^2}, \\ \sqrt{a'a''.b' - c''.bc} &= -b''\sqrt{b'^2 - c''^2}, \\ \sqrt{a'a''.c - b'.bc} &= c'\sqrt{b'^2 - c''^2},\end{aligned}$$

by multiplying together in each pair the left-hand and the right-hand sides respectively.

Substituting and dividing by $-a'a''bb'cc''$, we have

$$x^4 + y^4 + z^4 + w^4 - \frac{a''^2 + b^2}{a'a''} \cdot \frac{b'^2 + c''^2}{bb'} x^2 z^2 - \dots - \frac{2(b''^2 - c''^2)^2 \sqrt{-ab'c'}}{a'a''bb'cc''} xyzw = 0.$$

or, if we herein restore the c 's in place of the rectangular coefficients, this is

$$\sqrt{(x^2 + y^2 + z^2 + w^2)},$$

which is the equation of the 16-nodal quartic surface. Substituting for x, y, z, w their values $\xi_4, \xi_7, \xi_8, \xi_{11}$ (u), we have the equation connecting the four theta-functions 4, 7, 8, 11 of a Göpel tetrad. And there is an equation of the like form between the four functions of any other Göpel tetrad: for obtaining the actual equations some further investigation would be necessary.

The xy -expressions of the theta-functions.

125. The various quadric relations between the theta-functions, admitting that they constitute a 13-fold relation, show that the theta-functions may be expressed as proportional to functions of two arbitrary parameters x, y ; and two of these functions being assumed at pleasure the others of them would be determinate; we have of course (though it would not be easy to arrive at it in this manner) such a system in the foregoing expressions of the 16 functions in terms of x, y ; and conversely these expressions must satisfy identically the quadric relations between the theta-functions.

126. To show that this is so as to the general form of the equations, consider first the xy -factors $\sqrt{k}, \sqrt{ab}, \&c.$ As regards the squared functions $\sqrt{aby}$, we have for instance

$$(\sqrt{ab})^2 = \frac{1}{2}\{ab_1c_1d_1e_1 + a_1b_1cde + 2\sqrt{\mathfrak{X}\mathfrak{Y}}\},$$

$$(\sqrt{cdf})^2 = \frac{1}{2}\{cd_1f_1a_1b_1e_1 + c_1d_1f_1abe + 2\sqrt{\mathfrak{X}\mathfrak{Y}}\},$$

each of these contains the same irrational part $\sqrt{\mathfrak{X}\mathfrak{Y}}$, and the difference is therefore rational: and it is moreover integral, for we have

$$\sqrt{aby} - cd = \frac{1}{2}(abc_1d_1 - a_1b_1cd)(f_1e - fe_1),$$

where each factor divides by θ , and consequently the product by θ^2 ; the value is in fact

$$= (e - f) \begin{vmatrix} 1, & x + y, & xy \\ 1, & a + b, & ab \\ 1, & c + d, & cd \end{vmatrix}.$$

a linear function of $1, x + y, xy$. This is the case as regards the difference of any two of the squares $(\sqrt{ab})^2, (\sqrt{ac})^2$, &c.; hence selecting any one of these squares, for instance $(\sqrt{de})^2$, any other of the squares is of the form $\lambda + \mu(x + y) + \nu xy + \rho(\sqrt{de})^2$ ($\rho = 1$); and obviously, the other squares $(\sqrt{a})^2$, &c., are of the like form, the last coefficient ρ being $= 0$. We hence have the theorem that each square can be expressed as a linear function of any four (properly selected) squares.

127. But we have also the theorem of the 16 Kummer hexads. Obviously the six squares $(\sqrt{a})^2, (\sqrt{b})^2, (\sqrt{c})^2, (\sqrt{d})^2, (\sqrt{e})^2, (\sqrt{f})^2$ are a hexad, viz. each of these is a linear function of $1, x + y, xy$; and therefore selecting any three of them, each of the remaining three can be expressed as a linear function of these. But farther the squares $(\sqrt{a})^2, (\sqrt{bf})^2, (\sqrt{abf})^2, (\sqrt{cdf})^2, (\sqrt{ce})^2, (\sqrt{def})^2$ form a hexad. For reverting to the expression obtained for $\sqrt{aby} - cd$, the determinant contained therein is a linear function of aa_1 and bb_1 , that is, of $(\sqrt{a})^2$ and $(\sqrt{b})^2$; we, in fact, have

$$(a-b) \begin{vmatrix} 1, & x+y, & xy \\ 1, & a+b, & ab \\ 1, & c+d, & cd \end{vmatrix} = (b-c)(b-d)(a-x)(a-y) - (a-c)(a-d)(b-x)(b-y).$$

Hence $\sqrt{aby} - cd$ is a linear function of $(\sqrt{a})^2, (\sqrt{b})^2$; and by a mere interchange of letters $bc - ce, \sqrt{aby} - de$, are each of them also a linear function of $(\sqrt{a})^2$ and $(\sqrt{b})^2$; whence the theorem. And we have thus all the remaining 15 hexads.

128. We have a like theory as regards the products of pairs of functions. A tetrad of pairs is of one of the two forms $\sqrt{a}\sqrt{b}, \sqrt{ac}\sqrt{bc}, \sqrt{ad}\sqrt{bd}, \sqrt{ae}\sqrt{be}$, and $\sqrt{f}\sqrt{ab}, \sqrt{c}\sqrt{de}, \sqrt{d}\sqrt{ce}, \sqrt{e}\sqrt{cd}$; in the first case the terms are

$$\sqrt{aa_1}\sqrt{bb_1}, \quad \frac{1}{2}\{(ab_1 + a_1b)\sqrt{cdefc_1d_1e_1f_1} + (cd_1e_1f_1 + c_1d_1ef)\sqrt{\mathfrak{X}\mathfrak{Y}}\},$$

$$\frac{1}{2}\{(df_1c_1e_1 + d_1f_1ce)\sqrt{\mathfrak{X}\mathfrak{Y}}\}, \quad \frac{1}{2}\{(ef_1c_1d_1 + e_1f_1cd)\sqrt{\mathfrak{X}\mathfrak{Y}}\},$$

and as regards the last three terms the difference of any two of them is a mere constant multiple of $\sqrt{aa_1}\sqrt{bb_1}$; for instance, the second term - the third term is $\frac{1}{2}(cd_1 - c_1d)(fe_1 - f_1e)\sqrt{aa_1}\sqrt{bb_1} = (c - d)(f - e)\sqrt{aa_1}\sqrt{bb_1}$.

we have thus a tetrad such that, selecting any two terms, each of the remaining terms is a linear function of these. In the second case, the terms are

$$\frac{1}{2}\{f_1\sqrt{abcd_1e_1f_1} + f\sqrt{a_1bcdef}\},$$

$$\frac{1}{2}\{c\cdots + c_1\cdots\}, \quad \frac{1}{2}\{d\cdots + d_1\cdots\}, \quad \frac{1}{2}\{e\cdots + e_1\cdots\},$$

whence clearly the four terms are a tetrad as above. And it may be added that any linear function of the four terms is of the form $\frac{1}{2}\{(\lambda + \mu x)\sqrt{abcd_1e_1f_1} + (\lambda + \mu y)\sqrt{a_1bcdef}\}$.

129. Considering next the actual equations between the squared theta-functions, take as a specimen that is,

$$c_2(\sqrt{ab})^2 - c_2(\sqrt{cd})^2 + c_5(\sqrt{ce})^2 - c_6(\sqrt{de})^2 = 0,$$

where $c_2, c_2, c_5, c_6 = \sqrt{ab}, \sqrt{cd}, \sqrt{ce}, \sqrt{de}$ respectively. Since the functions $\sqrt{aby}$, &c., contain the same irrational term $\sqrt{\mathfrak{X}\mathfrak{Y}}$, it is clear that the equation can only be true if $c_2 - c_2 + c_5 - c_6 = 0$ and, this being so, it will be true if

$$c_2^2\{(\sqrt{ab})^2 - (\sqrt{cd})^2\} - c_5^2\{(\sqrt{ab})^2 - (\sqrt{ce})^2\} + c_6^2\{(\sqrt{ab})^2 - (\sqrt{de})^2\} = 0,$$

where, by what precedes, each of the terms in $\{\}$ is a linear function of $(\sqrt{a})^2$ and $(\sqrt{b})^2$. Attending first to the term in $(\sqrt{ab})^2$, the coefficient hereof is

$$ef \cdot bc \cdot bd \cdot c_2^2 - df \cdot bc \cdot be \cdot c_5^2 + cf \cdot bd \cdot be \cdot c_6^2,$$

where for shortness bc, bd , &c., are written to denote the differences $b-c, b-d$, &c. substituting for c_2^2 its value $(\sqrt{cd})^2 = cd \cdot cf \cdot df \cdot ab \cdot ae \cdot be$, and similarly for c_5^2 and c_6^2 their values, $= ce \cdot cf \cdot ef \cdot ab \cdot ad \cdot bd$, and $de \cdot df \cdot ef \cdot ab \cdot ac \cdot bc$ respectively, the whole expression contains the factor $ab \cdot bc \cdot bd \cdot be \cdot cf \cdot df \cdot ef$, and throwing this out, the equation to be verified becomes

$$cd \cdot ae - ce \cdot ad + de \cdot ac = 0,$$

$$c_2^2 - c_5^2 + c_6^2 - c_2^2 = 0;$$

which is true identically. The verification is thus made to depend upon that of $c_2^2 - c_5^2 + c_6^2 - c_2^2 = 0$; and similarly for the other relations between the squared functions, the verification depends upon relations containing the fourth powers, or the products of squares, of the constants c and k .

130. Among these are included the before-mentioned system of equations involving the fourth powers or the products of squares of only the constants c ; and it is interesting to show how these are satisfied identically by the values $c_0 = \sqrt{bd}$, &c. Thus one of these equations is

$$c_{14}^2 + c_{15}^2 + c_{12}^2 = c_{01}^2;$$

substituting the values, there is a factor ce which divides out, and the resulting equation is

$$ad \cdot cf \cdot df \cdot bc \cdot be + cf \cdot ef \cdot ab \cdot ad \cdot bd + ab \cdot af \cdot bf \cdot cd \cdot de - ac \cdot ae \cdot bd \cdot bf \cdot df = 0.$$

There are here terms in a^2 , a , a^0 which should separately vanish; for the terms in a^2 , the equation becomes

$$df \cdot bc \cdot be + bd \cdot cf \cdot ef + bf \cdot cd \cdot de - bd \cdot bf \cdot df = 0,$$

which is easily verified; and the equations in a and a^0 may also be verified. An equation involving products of the squares is $c_{14}c_{15} - c_{12}c_{13} + c_{14}c_{15} = 0$. The term $c_{14}^2c_{15}^2$ is here $\cdot bce \cdot \cdot abc$ which is $= \sqrt{\{(bc)^2(df)^2 \cdot ab \cdot ac \cdot ad \cdot af \cdot bc \cdot ce \cdot de \cdot ef\}}$, which is taken $= bc \cdot df \cdot \sqrt{(ab \cdot ac \cdot ad \cdot af \cdot bc \cdot ce \cdot de \cdot ef)}$; similarly the values of $c_{12}^2c_{13}^2$ and $c_{12}^2c_{14}^2$ are $= bd \cdot cf$ and $bf \cdot cd$ each multiplied by the same radical, and the equation to be verified is

$$bc \cdot df - bd \cdot cf + bf \cdot cd = 0,$$

which is right: the other equations may be verified in a similar manner.

131. Coming next to the equations connecting the pairs of theta-functions, for instance this is

$$c_5c_6c_9c_{10}\sqrt{bd}\sqrt{ad} - \sqrt{be}\sqrt{ae} + c_4c_8c_{12}c_{16}\sqrt{bf}\sqrt{af} = 0,$$

the products $\sqrt{bd}\sqrt{ad}$ and $\sqrt{be}\sqrt{ae}$ contain besides a common term the terms $\frac{1}{2}(df_1e_1 + d_1f_1ce)\sqrt{aa_1}\sqrt{bb_1}$, and $\frac{1}{2}(ef_1c_1d_1 + e_1f_1cd)\sqrt{aa_1}\sqrt{bb_1}$; hence their difference contains $\frac{1}{2}(de_1 - d_1e)(fc_1 - f_1c)\sqrt{aa_1}\sqrt{bb_1}$, which is $= de \cdot fc\sqrt{aa_1}\sqrt{bb_1}$, that is, $de \cdot fc\sqrt{a}\sqrt{b}$: hence the equation to be verified is

$$ae \cdot bc \cdot c_5c_6c_9c_{10} + c_4c_8c_{12}c_{16} = 0.$$

$$c_5c_6c_9c_{10} = \sqrt{bef} \cdot acd\sqrt{arf} \cdot bcd\sqrt{bdf} \cdot ace\sqrt{adf} \cdot bce,$$

where under the fourth root we have 24 factors, which are, in fact, 12 factors twice repeated; and if we write $\Pi = ab \cdot ac \cdot ad \cdot ae \cdot af \cdot bc \cdot bd \cdot be \cdot bf \cdot cd \cdot ce \cdot cf \cdot de \cdot df \cdot ef$ for the product of all the 15 factors, then the 12 factors are in fact all those of Π , except ab, cf, de ; viz. we have

$$c_5 c_6 c_9 c_{10} = \sqrt{\Pi^2 (ab)^2 (cf)^2 (de)^2}.$$

Again, $c_4c_8c_{12}c_{16} = \sqrt{acf} \cdot bde\sqrt{bcf} \cdot ade\sqrt{acdef}\sqrt{bcdef}$, is a fourth root of a product of 32 factors, which are in fact 16 factors twice repeated, and in the 16 factors, ab does not occur, cf and de occur each twice, and the other 12 factors each once: we thus have

$$c_4c_8c_{12}c_{16} = \sqrt{\Pi^2(cf)^2(de)^2(ab)^2},$$

and the relation to be verified assumes the form

$$fc \cdot de\sqrt{\Pi^2(cf)^2(de)^2} + \sqrt{\Pi^2(cf)^2(de)^2} = 0,$$

which, taking $fc \cdot de = -\sqrt{\{(cf)^2(de)^2\}}$, is right. And so for the other equations. It will be observed that, in the equation $de \cdot fc \cdot c_5c_6c_9c_{10} + c_4c_8c_{12}c_{16} = 0$, and in the other equations upon which the verifications depend, there is no ambiguity of sign: the signs of the radicals have to be determined consistently with all the equations which connect the c 's and the k 's: that this is possible appears evident *a priori*, but the actual verification presents some difficulty. I do not here enter further into the question.

Further results of the product-theorem, the $u \pm u'$ formulæ.

132. Recurring now to the equations in $u + u'$, $u - u'$, by putting therein $u' = 0$, we can express X, Y, Z, W in terms of four of the squared functions of u , and by putting $u = 0$ we can express X', Y', Z', W' in terms of four of the squared functions of u' ; and, substituting in the original equations, we have the products $\vartheta(u+u')$, $\vartheta(u-u')$ in terms of the squared functions of u and u' . Selecting as in a former investigation the functions 4, 7, 8, 11, which were called X, Y, Z, W , it is more convenient to use single letters

p, q, r, s for the squared functions of u ; and similarly p', q', r', s' for those of u' ; viz. we write

$$4^2, 7^2, 8^2, 11^2 = p, q, r, s, \quad \text{and} \quad 4'^2, 7'^2, 8'^2, 11'^2 = p', q', r', s'.$$

Also $\alpha, \beta, \gamma, \delta$ being the constants belonging to the Göpel tetrad (4, 7, 8, 11), we have

$$\alpha = c_4c_7c_8c_{11}, \quad \beta = c_4c_7c_8c_{11}, \quad \gamma = c_4c_7c_8c_{11}, \quad \delta = c_4c_7c_8c_{11};$$

and P, Q, R, S being the products which belong to the tetrad, viz.

$$P = \vartheta_4(u+u')\vartheta_4(u-u'), \quad Q = \vartheta_7(u+u')\vartheta_7(u-u'), \quad \&c.,$$

then the formulæ are

$$P = \alpha(pp' + qq' + rr' + ss') + \beta(pp' + qq' - rr' - ss') + \gamma(pp' - qq' + rr' - ss') + \delta(pp' - qq' - rr' + ss')$$

$$R = \alpha(pp' - qq' + rr' - ss') + \beta(pp' - qq' - rr' + ss') + \gamma(pp' + qq' + rr' + ss') + \delta(pp' + qq' - rr' - ss')$$

$$Q = \alpha(pq' - qp' + rs' - sr') + \beta(pq' - qp' - rs' + sr') + \gamma(pq' + qp' + rs' + sr') + \delta(pq' + qp' - rs' - sr')$$

$$S = \alpha(pr' - rp' + qs' - sq') + \beta(pr' + rp' - qs' - sq') + \gamma(pr' - rp' - qs' + sq') + \delta(pr' + rp' + qs' - sq')$$

By means of these values, we have

$$4(\alpha^2 - \beta^2)X'X = \alpha^2(p+r)(p'+r') + \beta^2(q+s)(q'+s') - \alpha\beta[(p+r)(q'+s') + (p'+r')(q+s)],$$

$$4(\alpha^2 - \beta^2)W'W = \alpha^2(p+r)(p'+r') + \beta^2(q+s)(q'+s') - \alpha\beta[\dots],$$

$$4(\beta^2 - \gamma^2)Y'Y = \beta^2(p-r)(p'-r') + \gamma^2(q-s)(q'-s') - \beta\gamma[(p-r)(q'-s') + (p'-r')(q-s)],$$

$$4(\beta^2 - \gamma^2)Z'Z = \gamma^2(p-r)(p'-r') + \beta^2(q-s)(q'-s') - \beta\gamma[\dots].$$

Hence

$$4(\alpha^2 - \beta^2)(X'X - W'W) = (p+r)(p'+r') - (q+s)(q'+s'),$$

$$4(\beta^2 - \gamma^2)(Y'Y - Z'Z) = (p-r)(p'-r') - (q-s)(q'-s'),$$

and substituting in the expressions for P and R ,

$$4(\alpha^2 - \beta^2)(\beta^2 - \gamma^2)P = (\beta^2 - \gamma^2)[(p+r)(p'+r') - (q+s)(q'+s')] - (\alpha^2 - \beta^2)[(p-r)(p'-r')]$$

$$4(\alpha^2 - \beta^2)(\beta^2 - \gamma^2)R = \dots$$

Similarly

$$4(\alpha^2 - \beta^2)W'X = \alpha\beta[(p+r)(p'+r') + (q+s)(q'+s')] - \alpha^2(p+r)(q'+s') - \beta^2(q+s)(p'+r')$$

$$4(\beta^2 - \gamma^2)X'Y = \dots$$

$$4(\beta^2 - \gamma^2)Y'Z = \dots$$

whence

$$4(\alpha^2 - \beta^2)(W'X - X'W) = -[(p+r)(q'+s') - (p'+r')(q+s)],$$

$$4(\beta^2 - \gamma^2)(Z'Y - Y'Z) = -[(p-r)(q'-s') - (p'-r')(q-s)],$$

and substituting in the expressions for Q and S ,

$$4(\alpha^2 - \beta^2)(\beta^2 - \gamma^2)Q = -(\beta^2 - \gamma^2)[(p+r)(q'+s') - (p'+r')(q+s)] + (\alpha^2 - \beta^2)[(p-r)(q'-s) - (p'-r)(q-s)]$$

$$4(\alpha^2 - \beta^2)(\beta^2 - \gamma^2)S = \dots$$

133. Hence, collecting and reducing,

$$4P = -(\alpha^2 - \beta^2 + \gamma^2 - \delta^2)(pp' - qq' + rr' - ss') + (\alpha^2 + \beta^2 - \gamma^2 - \delta^2)(pr' + p'r - qs' - q's),$$

$$4R = (\alpha^2 - \beta^2 + \gamma^2 - \delta^2)(pp' - qq' + rr' - ss') + (\alpha^2 + \beta^2 - \gamma^2 - \delta^2)(pr' + p'r - qs' - q's),$$

$$4Q = (\alpha^2 + \beta^2 - \gamma^2 - \delta^2)(pq' - p'q + rs' - r's) - (\alpha^2 - \beta^2 + \gamma^2 - \delta^2)(ps' - p's + qr' - q'r),$$

$$4S = -(\alpha^2 + \beta^2 - \gamma^2 - \delta^2)(ps' - p's + qr' - q'r) + (\alpha^2 - \beta^2 + \gamma^2 - \delta^2)(pq' - p'q + rs' - r's).$$

we have

$$\rho_0(=c_4^2) = \alpha^2 - \beta^2 + \gamma^2 - \delta^2, \quad \sigma_0(=c_8^2) = \alpha^2 + \beta^2 - \gamma^2 - \delta^2,$$

and thence

$$\sigma_0^2 - \rho_0^2 = 4(\alpha^2 - \delta^2)(\beta^2 - \gamma^2);$$

the equations hence become

$$(\sigma_0^2 - \rho_0^2)P = -\rho_0(pp' - qq' + rr' - ss') + \sigma_0(pr' + p'r - qs' - q's),$$

$$(\sigma_0^2 - \rho_0^2)R = \rho_0(pp' - qq' + rr' - ss') + \sigma_0(pr' + p'r - qs' - q's),$$

$$(\sigma_0^2 - \rho_0^2)Q = \sigma_0(pq' - p'q + rs' - r's) - \rho_0(ps' - p's + qr' - q'r),$$

$$(\sigma_0^2 - \rho_0^2)S = -\sigma_0(ps' - p's + qr' - q'r) + \rho_0(pq' - p'q + rs' - r's).$$

On writing in the equations $u' = 0$, then $P, Q, R, S, p', q', r', s'$ become $= p, q, r, s, p, q, r, s$; and the equations are (as they should be) true identically. The equations may be written

$$4 \cdot 4 = -4(4 \cdot 4 - 7 \cdot 7 + 8 \cdot 8 - 11 \cdot 11) + 8(4 \cdot 8 + 8 \cdot 4 - 7 \cdot 11 - 11 \cdot 7),$$

$$4 \cdot 8 = +8(\dots) - 4(\dots),$$

$$4 \cdot 7 = +4(4 \cdot 7 - 7 \cdot 4 + 8 \cdot 11 - 11 \cdot 8) - 8(4 \cdot 11 - 11 \cdot 4 + 8 \cdot 7 - 7 \cdot 8),$$

$$4 \cdot 11 = -8(\dots) + 4(\dots).$$

There is of course such a system for each of the 60 Göpel tetrads.

Differential relations connecting the theta-functions with the quotient-functions.

134. Imagine p, q, r, s , &c., changed into x, y, z, w , &c.; that is, let x, y, z, w represent the theta-functions 4, 7, 8, 11 of u, v ; and similarly x', y', z', w' those of u', v' ; and x_0, y_0, z_0, w_0 those of 0, 0. Let u', v' be each of them indefinitely small; and take $9 = u' \frac{d}{du} + v' \frac{d}{dv}$, as the symbol of total differentiation in regard to u, v , the infinitesimals u' and v' being arbitrary: then, as far as the second order, we have in general

$$\vartheta(u + u', v + v') = \vartheta(u, v) + 9\vartheta(u, v) + \frac{1}{2}9^2\vartheta(u, v).$$

and hence

$$P = (x + dx + \frac{1}{2}d^2x)(x - dx + \frac{1}{2}d^2x) = x^2 + \{xd^2x - (dx)^2\},$$

and similarly for Q, R, S . Moreover, observing that x' and z' are even functions, y' and w' are odd functions, of u', v' , we have

$$x' = x_0 + 9x_0, \quad y' = y_0 + 9y_0, \quad z' = z_0 + 9z_0, \quad w' = w_0 + 9w_0,$$

where $9x_0, 9y_0, \&c.$, are what $9x, 9y, \&c.$, become on writing therein $u = 0, v = 0$; $9y_0, 9w_0$ are of course linear functions, $9^2x_0, 9^2z_0$ quadric functions of u' and v' . The values of x'^2, y'^2, z'^2, w'^2 are thus $x_0^2 + x_0 9x_0, (9y_0)^2, z_0^2 + z_0 9z_0, (9w_0)^2$; and we have

$$x^2 + x'^2 - y^2 - y'^2 - z^2 - z'^2 + w^2 + w'^2 = 2(x^2 - y^2 - z^2 + w^2) + x_0 9x_0 - z_0 9z_0,$$

$$x^2 - x'^2 + y^2 - y'^2 - z^2 + z'^2 - w^2 + w'^2 = 2(x^2 + y^2 - z^2 - w^2) - x_0 9x_0 + z_0 9z_0,$$

$$-x^2 + x'^2 - y^2 + y'^2 - z^2 + z'^2 + w^2 - w'^2 = 2(-x^2 - y^2 - z^2 + w^2) + x_0 9x_0 + z_0 9z_0,$$

$$-x^2 + x'^2 + y^2 - y'^2 + z^2 - z'^2 - w^2 + w'^2 = 2(-x^2 + y^2 + z^2 - w^2) - x_0 9x_0 - z_0 9z_0.$$

135. On substituting these values, the constant terms (or terms independent of u', v') disappear of themselves; and the equations, transposing the second and third of them, become

$$\begin{aligned}(z_0^2 - x_0^2)\{xd^2x - (dx)^2\} &= (-x_0y_0 + y_0w_0) + (x_0y_0 - z_0w_0) + (-x_0z_0 + z_0x_0) + (x_0w_0 - y_0z_0) \\[yd^2y - (dy)^2] &= -(x_0y_0 - y_0w_0) - (-x_0x_0 + z_0z_0) - (x_0w_0 - z_0y_0) - (-x_0z_0 + z_0x_0) \\[zd^2z - (dz)^2] &= (-x_0y_0 + y_0w_0) + (x_0w_0 - z_0y_0) + (-x_0z_0 + z_0x_0) + (x_0w_0 - y_0z_0) \\[wd^2w - (dw)^2] &= -(x_0w_0 - w_0y_0) - (-x_0z_0 + z_0x_0) - (x_0y_0 - y_0w_0) - (-x_0z_0 + z_0x_0)\end{aligned}$$

where it will be recollected that x, y, z, w mean $\vartheta_4(u), \vartheta_7(u), \vartheta_8(u), \vartheta_{11}(u)$; x_0 is $\vartheta_4(0)$, that is c_4 , and z_0 is $\vartheta_8(0)$, that is c_8 . But the formulae contain also

$$9x_0 = (c_4'', c_4''')(u', v'), \quad 9y_0 = (c_7'', c_7''')(u', v'),$$

$$9z_0 = (c_8'', c_8''')(u', v'), \quad 9w_0 = (c_{11}'', c_{11}''')(u', v').$$

The formulae may be written

$$\begin{aligned}(c_4^2 - c_8^2)\{44\} &= (-4 \cdot 4 + 8 \cdot 8) & + (4 \cdot 7 - 8 \cdot 11) & + (-4 \cdot 8 + 8 \cdot 4) & + (4 \cdot 11 - 8 \cdot 7) \\(c_4^2 - c_8^2)\{77\} &= -(4 \cdot 7 - 8 \cdot 11) & - (-4 \cdot 4 + 8 \cdot 8) & - (4 \cdot 11 - 8 \cdot 7) & - (-4 \cdot 8 + 8 \cdot 4) \\(c_4^2 - c_8^2)\{88\} &= (-4 \cdot 8 + 8 \cdot 4) & + (4 \cdot 11 - 8 \cdot 7) & + (-4 \cdot 4 + 8 \cdot 8) & + (4 \cdot 7 - 8 \cdot 11) \\(c_4^2 - c_8^2)\{11 \ 11\} &= -(4 \cdot 11 - 8 \cdot 7) & - (-4 \cdot 8 + 8 \cdot 4) & - (4 \cdot 7 - 8 \cdot 11) & + (-4 \cdot 4 + 8 \cdot 8)\end{aligned}$$

where $d^2c_4, d^2c_8, dc_7, dc_{11}$ are written in place of $9^2x_0, 9^2z_0, 9y_0, 9w_0$. There is of course a like system of equations for each of the Göpel tetrads.

136. Observe that, dividing the first equation by $\vartheta_4^2(u)$, or say by x^2 , the left-hand side is a mere constant multiple of $d^2 \log \vartheta_4$; and the right-hand side depends only on the quotient-functions $\vartheta_7 \div \vartheta_4$, $\vartheta_8 \div \vartheta_4$, $\vartheta_{11} \div \vartheta_4$; each side is a quadric function of u, v . Equating the terms in $u'^2, u'v', v'^2$ respectively, we have $\frac{d^2}{du^2} \log \vartheta_4, \frac{d^2}{dudv} \log \vartheta_4, \frac{d^2}{dv^2} \log \vartheta_4$, each of them expressed as a linear function of the squares of the quotient-functions $\vartheta_7 \div \vartheta_4, \vartheta_8 \div \vartheta_4, \vartheta_{11} \div \vartheta_4$. The formula is thus a second-derivative formula serving for the expression of a double theta-function by means of three quotient-functions.

Differential relations of the theta-functions.

137. In “The second set of 16,” selecting the eight equations which contain Y' and W' , these are

$u + \nu'$	$\nu - \nu'$	$\nu + \nu'$	$\nu - \nu'$
Y	W		
$4 - 4 = Y' + W',$ $1 + 1 = X' + Z',$	$12 - 12 = Y' - W',$ $13 + 9 = X' - Z',$	$6 - 6 = W' + Y',$ $7 + 3 = Z' + X',$	$14 - 10 = W' - Y',$ $15 + 11 = Z' - X'.$

Then, considering any line in the upper half and any two lines in the lower half, we can from the three equations eliminate Y' and W' , thus obtaining an equation such as

$$\begin{vmatrix} \vartheta_4 & \vartheta_0 & X' & Z' \\ \vartheta_{12} & \vartheta_{15} & Y' & W' \\ \vartheta_6 & \vartheta_3 & Z' & X' \end{vmatrix} = 0,$$

viz. this is

$$-4X'Z'(\vartheta_4\vartheta_0-\vartheta_0\vartheta_4)+(Z'Y'+Y'X')(\vartheta_{12}\vartheta_{15}+\vartheta_{15}\vartheta_{12})+(-Z'Y'+Y'X')(\vartheta_6\vartheta_3+\vartheta_3\vartheta_6)$$

where the arguments of the theta-functions are as above $u + u', u - u'$; and the suffixes of the X', Y', Z', W' are all = 1.

138. Suppose that in this equation u' becomes indefinitely small. If u were $= 0$, the values of X', Y', Z', W' would be $\alpha, 0, \gamma, 0$; hence u' being indefinitely small, we take them to be $\alpha, 9\beta, \gamma, 9\delta$, where

$$9 = (u' \frac{d}{du} + v' \frac{d}{dv}), \quad 9\alpha = (\alpha' u' + \alpha'' v'), \quad \&c.,$$

are, in fact, linear functions of u' and v' . We have

$$\vartheta_0(u+u')\vartheta_4(u-u') - \vartheta_4(u+u')\vartheta_0(u-u'), \quad \text{standing for} \quad 9_0(u+u')9_4(u-u') - 9_4(u+u')$$

and here

$$\vartheta_4(u \pm u') = \vartheta_4 \pm 9\vartheta_4, \quad \vartheta_0(u \pm u') = \vartheta_0 \pm 9\vartheta_0;$$

the function in question is thus

$$(\vartheta_4 + 9\vartheta_4)(\vartheta_0 - 9\vartheta_0) - (\vartheta_4 - 9\vartheta_4)(\vartheta_0 + 9\vartheta_0) = 2(\vartheta_0 9\vartheta_4 - \vartheta_4 9\vartheta_0),$$

where the arguments are u, v , and the 9 denotes $u' \frac{d}{du} + v' \frac{d}{dv}$. Also $\alpha\gamma_0 + \gamma\alpha_0$, that is, $9\vartheta_4(u+u')9\vartheta_4(u-u') + 9\vartheta_8(u+u')9\vartheta_8(u-u')$, becomes simply $= 2\vartheta_4 9\vartheta_8$; and similarly $\beta\delta_0 + \delta\beta_0$ becomes $= 2\vartheta_{12} 9\vartheta_0$; and the equation thus is

$$-2\alpha\gamma(\vartheta_0 9\vartheta_4 - \vartheta_4 9\vartheta_0) + (\alpha\delta 9\beta_0 + \gamma\beta 9\delta_0)\vartheta_4\vartheta_8 + (-\alpha\delta 9\beta_0 + \gamma\beta 9\delta_0)\vartheta_{12}\vartheta_0 = 0,$$

where the proper suffix 1 is restored to the α, β, γ , and δ .

139. The equation shows that the differential combination $\vartheta_0 9\vartheta_4 - \vartheta_4 9\vartheta_0$ is a linear function of $\vartheta_5\vartheta_1$ and $\vartheta_{13}\vartheta_1$, the coefficients of these products being of course linear functions of u and v . Writing the equation

$$\vartheta_0 9\vartheta_4 - \vartheta_4 9\vartheta_0 = A\vartheta_5\vartheta_1 + B\vartheta_{13}\vartheta_1,$$

we can if we please determine the coefficients in terms of the constants $c, c'', c''', c^{\text{IV}}, c^{\text{V}}$; viz. taking u, v indefinitely small, we have

$$\vartheta_1 = c_0, \quad 9\vartheta_1 = u'(c_1''u + c_1'''v) + v'(c_1^{\text{IV}}u + c_1^{\text{V}}v),$$

$$\vartheta_4 = c_4, \quad 9\vartheta_0 = u'(c_0''u + c_0'''v) + v'(c_0^{\text{IV}}u + c_0^{\text{V}}v),$$

$$\vartheta_1 = c_1, \quad 9\vartheta_8 = c_8'u + c_8''v,$$

or substituting, and equating the coefficients of u' and v' respectively, we have

$$c_0(c_4''u' + c_4'''v') - c_4(c_0''u' + c_0'''v') = Ac_1c_8' + Bc_1c_8'',$$

$$c_0(c_4^{\text{IV}}u' + c_4^{\text{V}}v') - c_4(c_0^{\text{IV}}u' + c_0^{\text{V}}v') = -Ac_1c_8'' + Bc_1c_8',$$

which equations give the values of A, B .

140. Disregarding the values of the coefficients, and attending only to the form of the equation $\vartheta_0 9 \vartheta_4 - \vartheta_4 9 \vartheta_0 = A \vartheta_5 \vartheta_1 + B \vartheta_{13} \vartheta_1$, this is one of a system of 120 equations; viz. referring to the foregoing table of the 120 pairs, it in fact appears that taking any pair such as $\vartheta_0 \vartheta_4$ out of the upper compartment or the lower compartment of any column of the table, the corresponding differential combination $\vartheta_0 9 \vartheta_4 - \vartheta_4 9 \vartheta_0$ is a linear function of any two of the four pairs in the other compartment of the same column.

Differential relation of u, v and x, y .

141. We have, as before, in the two notations, the pairs

	A . B	C . DE	D . CE	E . CD
	11.7	5.9	13.1	14.2
F . AB		10.6	...	...

From the expressions given above for the four pairs below the line, it is clear that any linear function of these four pairs may be represented by

$$(a - b) \, 9 \{ (\lambda + \mu y) \sqrt{cdef} a_1 b_1 + (\lambda + \mu x) \sqrt{c_1 d_1 e_1 f_1} ab \},$$

where λ, μ are constant coefficients: the factor $(a - b)$ has been introduced for homogeneity. And we hence see that the differential combination $9(\vartheta_0 \vartheta_4)$ is a linear function of the four pairs in the upper compartment of the same column; viz. the form of the relation is the same as if the constants a, b , &c. were ordinary algebraical quantities.

142. But if in the equation we substitute for the pairs their values in terms of x, y , the differential combination becomes a function of x, y, dx, dy, d^2x, d^2y ; and the equation then is a differential relation between u, v and x, y .

The hyperelliptic case.

143. In the hyperelliptic case, the constants c are connected by the relations $c_0^2 + c_4^2 + c_8^2 + c_{12}^2 = c_1^2 + c_5^2 + c_9^2 + c_{13}^2$, $c_2^2 + c_6^2 + c_{10}^2 + c_{14}^2 = c_3^2 + c_7^2 + c_{11}^2 + c_{15}^2$, and $c_0c_2 + c_4c_6 + c_8c_{10} + c_{12}c_{14} = c_1c_3 + c_5c_7 + c_9c_{11} + c_{13}c_{15}$. We have in this case the equation $x^2 + y^2 + z^2 + w^2 = 0$, which is satisfied identically by the values $x : y : z : w = \vartheta_4 : \vartheta_7 : \vartheta_8 : \vartheta_{11}$; and the quotient-functions $\vartheta_7 \div \vartheta_4, \vartheta_8 \div \vartheta_4, \vartheta_{11} \div \vartheta_4$ are connected by a quadric relation, which is in fact the equation of the 16-nodal quartic surface. The theory of the differential relations connecting u, v with x, y is thus intimately connected with the geometry of this surface.

144. I do not here enter further into the hyperelliptic case, or the general theory of the differential relations; but what precedes is sufficient to show that the theory of the double theta-functions presents a most interesting field of research, in which much remains to be done.

[The memoir is followed by a table of the 120 pairs of theta-functions, arranged in 15 columns and 8 rows, with the upper and lower compartments as described in the text.]

[705 705. PROBLEMS AND SOLUTIONS. rvith their Solutions frovi the Educ

705] distance of centres, or what is the same thing, squared difference

[705 or $\wedge$ -s_a»g» = 3(a'- $\wedge$ ='), that is, :3a''-4 $\wedge$ •' $\wedge$ -aV = 0; viz. this i

705] (so that $a+h-g = 0$, &c., $a+b + c = 0$, $af+ bg + ch = 0$), the six p

[705 the same point ; and similarly the lines PR , QS , pr , qs , and the

705] Solution by the Proposer. Let the numbers be 1, 2, 3, 4, 5, 6, 7,

[705 as one out of many ways of effecting the identification. Observe

705] Combining AC with 8, 9, we have the triads 8 (24, 37, 56) and 9 (

[705 Observe that there is in the system a single triad of triads Aaa'

705] system of all the substitutions (or say the entire group), or els

[705 XVII., p. 60.] 3677. (Proposed by Profe-ssor C.A.YLEY.)-Find at a

705] PROBLEMS AND SOLUTIONS. we get " '» +,-^=< <i), (-^.) (-^) (- «" :

[705 Equating the discriminant to zero, this is = $\hat{=}$ $V = 4 (ilG - £\gg)$ ']

705] The expression of the function F is in effect given in my paper,

[705 Cunningham in a paper in the last number of the Quarterly Journal

705] XXII., July to December, 1874, pp. 20, 21.] 4354. (Proposed by R.

[705 XXII., pp. 60-64.] 4458. (Proposed by Professor Cayley.)-Find (1)

705] 58S viz. substituting for p its values, this is $\sqrt{(a + 6 - 2\sqrt{-2/i})^2}$.

[705 the transformations depending on the identity $8V|oz^*) = a6-(\backslash + //$

+ 1-65 + 0-94, + 1-22 + 213, - - 3-4.9. 705] PROBLEMS AKD SOLUTIONS. 2

PROBLEMS AND SOLUTIONS. [705 lying wholly between the two asymptotes y

705] Table I., pp. 3-106 gives, for all numbers 1 to 1000, the denomin.

[705 When $i=2$, if a is drawn, the adventurer will be satisfied or he w

705] Remark by Professor Cayley. There is a serious difficulty in the

[705 XXIV.. pp. 72-74.] 4793. (Proposed by Professor Wolstenholme, M.A.

705] PROBLEMS AND SOLUTIONS. thus $\{x dx - ny$ consists of a series of te

[705 Hence, in general, where observe that the last term is $= x dx^{\wedge y}$. F

705] and we thence at once deduce the converse relation $X : y : z = x'$

[705 whcncG -F{A,B,C)=-{a*- fi'Y + 2 (a» Xf (a« + 3a>0 - 8a-6^c= (a* -

705] Solution by Professor Cayley. Mr Woolhouse's elegant theorem depe

[705 or, reducing each equation by those which precede it, these becom

705] PROBLEMS AND SOLUTIONS. Solution by the Proposer. Consider the eq

[705 XXVII., January to June, 1877, p. 20.] 5130. (Proposed by Profess

705] have $a = \sqrt[3]{7 + 8 + \sqrt{}}$; and since $n = a + \sqrt[3]{7 + y + S}$, we have th

[705 It is very noticeable how close the agreement is between the proo

PROBLEMS AND SOLUTIONS.

[705]

Problems 4354–4458 [Vol. XXII., July to December, 1874]

4354. (Proposed by R. Tucker, M.A.)—Solve the equations

$$-x^2 + xy + xz = a = -20, \quad (20)$$

$$-y^2 + xy + yz = b = -8, \quad (21)$$

$$-z^2 + xz + yz = c = -4. \quad (22)$$

[By Professor Cayley. Note on Question 4354.] A question of simple algebra such as this, becomes more interesting when interpreted geometrically: thus, writing the equations in the form

$$-x^2 + xy + xz = aw^2,$$

$$zx + zy - z^2 = cw^2,$$

$$yx - y^2 + yz = bw^2,$$

and then putting for shortness

$$\alpha = -a + b + c, \quad \beta = a - b + c, \quad \gamma = a + b - c,$$

the solutions obtained are

$$x : y : z : w = \alpha a : \beta b : \gamma c : (\alpha\beta\gamma)^{\frac{1}{2}},$$

and

$$x : y : z : w = \alpha a : \beta b : \gamma c : -(\alpha\beta\gamma)^{\frac{1}{2}};$$

say these are

$$[\alpha a, \beta b, \gamma c, (\alpha\beta\gamma)^{\frac{1}{2}}]$$

and

$$[\alpha a, \beta b, \gamma c, -(\alpha\beta\gamma)^{\frac{1}{2}}].$$

But the equations are also satisfied by

$$(x = 0, y = z, w = 0), \quad \text{or} \quad (y = 0, z = x, w = 0), \quad \text{or} \quad (z = 0, x = y, w = 0),$$

that is, $(0, 1, 1, 0)$, $(1, 0, 1, 0)$, $(1, 1, 0, 0)$; and the three having in common the line $x = y = z, w = 0$, these are points; the last mentioned two points each once, and the first twice: $2 + 3 \cdot 2 = 8$.

The three equations represent quadric surfaces, each two of them intersecting in a proper curve; the three have in common the line $x = y = z, w = 0$. To verify this, observe that, at each of the three points, the surfaces have a common tangent line; this is made up of the tangent planes of each two of the surfaces at the point, and the curve of intersection of any two of the surfaces therefore touches the third surface; wherefore the point counts for two intersections.

In fact, taking (X, Y, Z, W) as current coordinates, the equations of the tangent planes at (x, y, z, w) are

$$\begin{aligned} X(2x - y - z) - Y \cdot x - Z \cdot x + 2aWw &= 0, \\ -X \cdot y + Y(-x + 2y - z) - Z \cdot y + 2bWw &= 0, \\ -X \cdot z - Y \cdot z + Z(-x - y + 2z) + 2cWw &= 0; \end{aligned}$$

hence at the point $(0, 1, 1, 0)$ these equations are

$$-X + Y - Z = 0, \quad -X - Y + Z = 0, \quad -2Z = 0,$$

which three planes meet in the line $X = 0, Y - Z = 0$; and similarly for the other two of the three points.

4458. (Proposed by Professor Cayley.)—[Vol. XXII., pp. 60–64.]

Find the intersections of the two quartic curves

$$(\lambda ab - xy)^2 = abx(a - y)(b - y), \quad (23)$$

$$(\mu ab - xy)^2 = aby(a - x)(b - x); \quad (24)$$

the curves are the trace of the surfaces in some particular cases; for instance, when $a = 1$, $b = 2$, $\lambda = \mu = -2$.

[Solution by the Proposer.] 1. The 16 intersections are made up as follows: 5 points at infinity on the line $x = 0$, 5 at infinity on the line $y = 0$, the two points $(x = a, y = b)$, $(x = b, y = a)$, and 4 other points; $16 = 5 + 5 + 2 + 4$. As to the points at infinity, observe that, as regards the first curve, this line is the same for the point at infinity on the line $x = 0$, which is a flecnodal branch; this point counts as $2 + 3 = 5$ intersections; and, similarly, as regards the second curve, the point at infinity on the line $y = 0$.

It remains to find the coordinates of the 4 points of intersection. Assume $xy = ab\omega$, then the equations become

$$\lambda(1 - \omega) \cdot ab = ax + y - (a + b)x, \quad \mu(1 - \omega) \cdot ab = by + x - (a + b)y,$$

hence, eliminating successively x and y , the factor $1 - \omega$ divides out, this for $(x = a, y = b)$ for which obviously $\omega = 1$; and the equations become

$$(\lambda - \lambda\omega)(1 - \omega) + (a + b)\omega = (1 + \omega)x, \quad (\mu - \mu\omega)(1 - \omega) + (a + b)\omega = (1 + \omega)y.$$

Multiplying these two equations together, and substituting for xy its value $ab\omega$, we find

$$\{2\lambda\mu(1 - \omega) + (a + b)\}(1 - \omega)^2 + (a + b)^2\omega^2 - (1 + \omega)^2ab = 0.$$

Write, for shortness,

$$p = (\lambda + \mu)(a + b) - \lambda\mu,$$

then, dividing by ω and writing $\omega + \frac{1}{\omega} = \theta$, the equation is

$$(2\lambda\mu + p)(\theta - 2) + (a + b)^2 - ab(\theta + 2) = 0;$$

viz. this is a quadric equation for θ . But, instead of this equation thus for θ , it is convenient to introduce $\epsilon = \frac{\theta-1}{\theta+1}$, or

$$\theta = \frac{1+\epsilon}{1-\epsilon};$$

the equation becomes

$$\{2\lambda\mu(1-\epsilon) + p(1+\epsilon)\}(1-\epsilon) + (a+b)^2(1-\epsilon)^2 - 4ab(1-\epsilon) = 0,$$

or

$$\theta^2\{(a+b)^2 - 4(p-2\lambda\mu)\} + \theta\{-2a^2 - 2b^2 + 4(p+2\lambda\mu)\} + (a-b)^2 = 0;$$

viz. substituting for p its value, this is

$$\theta^2(a+b-2\lambda-2\mu)^2 + 2\theta\{-a^2 - b^2 + 2(\lambda+\mu)(a+b) - 2(\lambda-\mu)^2\} + (a-b)^2 = 0;$$

or if we write

$$A = a^2 - 2a(\lambda + \mu) + (\lambda - \mu)^2, \quad B = b^2 - 2b(\lambda + \mu) + (\lambda - \mu)^2,$$

this is

$$\theta^2(a+b-2\lambda-2\mu)^2 - 2(A+B)\theta + (a-b)^2 = 0,$$

whence

$$\{(a-b)^2 - (A+B)\theta\}^2 = \theta^2\{(A+B)^2 - (a-b)^2(a+b-2\lambda-2\mu)^2\} = \theta^2\{(A+B)^2 - (A-B)^2\},$$

viz. taking for convenience the sign $-$ on the right-hand side, this is

$$(a-b)^2 - (A+B)\theta = -2\theta\sqrt{AB};$$

and we have thus

$$\theta = \frac{(a-b)^2}{(\sqrt{A} - \sqrt{B})^2},$$

that is,

$$\frac{\theta-1}{\theta+1} = \frac{1}{a-b} \cdot \frac{A-B}{\sqrt{A} + \sqrt{B}} = \frac{\sqrt{A} - \sqrt{B} + a-b}{a-b}.$$

We may write

$$x = \frac{1}{\omega} \{\epsilon(\omega-1) + \frac{1}{2}(a+b) + \frac{1}{2}(a+b-2\lambda-2\mu)\theta\},$$

$$y = \frac{1}{\omega} \{ \epsilon(\omega - 1) + \frac{1}{2}(a + b) + \frac{1}{2}(a + b - 2\lambda - 2\mu)\theta \};$$

whence also $x - y = (\mu - \lambda)(\omega - 1)$, as is seen at once from the original equations; then we have

$$\frac{1}{2}(a - b) \{ \frac{1}{2}(a + b - 2\lambda - 2\mu) \} \theta = \frac{1}{2}(\sqrt{A} - \sqrt{B})(\sqrt{A} + \sqrt{B} - \frac{a - b}{2\epsilon})$$

and the values are

$$x = \frac{(a - b)(\mu - \lambda) + b\sqrt{A} - a\sqrt{B}}{2\lambda(a - b)}, \quad y = \frac{(a - b)(\lambda - \mu) + b\sqrt{A} - a\sqrt{B}}{2\mu(a - b)},$$

which may be expressed in the more simple form

$$x = \frac{1}{\lambda}(a + \lambda - \mu + \sqrt{A})(b + \lambda - \mu + \sqrt{B}),$$

$$y = \frac{1}{\mu}(a - \lambda + \mu + \sqrt{A})(b - \lambda + \mu + \sqrt{B}),$$

the transformations depending on the identity

$$\sqrt{A}\sqrt{B} - a + b = ab - (\lambda + \mu)(a + b) + (\lambda - \mu)^2 + \sqrt{A}(b - \lambda - \mu) + \sqrt{B}(a - \lambda - \mu) + \sqrt{AB},$$

which is easily verified. Of course, since the signs of $\sqrt{A}$, $\sqrt{B}$ are arbitrary, we have 4 systems of values of (x, y) , which is right.

In the original equations, for a, b, λ, μ, x, y , write $k^{-2}, 1, \lambda x^2, -\mu y^2, x^2, -y^2$; then the equations become

$$\lambda x^2(1 + k^2 y^2) = x^2(1 + y^2)(1 + k^2 y^2), \quad \mu y^2(1 + k^2 x^2) = y^2(1 - x^2)(1 - k^2 x^2),$$

and we thence have

$$x^2 + y^2 = \frac{\lambda \sqrt{(1 + y^2)(1 + k^2 y^2)} + \mu y \sqrt{(1 - x^2)(1 - k^2 x^2)}}{1 + k^2 x^2 y^2} = \text{sn}(\alpha + \beta i);$$

viz. if $x = \text{sn } \alpha$ (sinam α), $iy = \text{sn } \beta$, this is $x^2 + y^2 = \text{sn}(\alpha + \beta i)$; viz. the problem is (for a given modulus k , less than 1, as usual, to be assumed positive, and the quantity λ/μ the form $\text{sn}(\alpha + \beta i)$). The proper solution is that in which the signs of the radicals are each +; it may be shown that in this case the value of x^2 is positive and less than 1, viz. the value of y^2 is positive.

The values thus are

$$x^2 = \frac{1}{2}(A' - B' - \sqrt{A'}\sqrt{B'}), \quad y^2 = \frac{1}{2}(A' - B' + \sqrt{A'}\sqrt{B'}),$$

where

$$A' = 1 - 2\lambda^2 + 2\mu^2 + (\lambda^2 + \mu^2)^2, \quad B' = \frac{1}{2} - \frac{1}{2}\lambda^2 + \frac{1}{2}\mu^2 + (\lambda^2 + \mu^2)^2.$$

The solution is really equivalent to that given by Richelot (Crelle, t. XLV., 1853, p. 225). To verify this partially, observe that, writing a, r for Richelot's $\tan^2 \phi, \tan^2 \psi$, we have

$$\frac{x^2}{y^2} = \frac{a}{r}; \quad \lambda = \frac{1}{\sqrt{kr}}, \quad \mu = \frac{1}{\sqrt{ka}},$$

giving

$$\frac{1}{2}(\lambda + \mu)\sqrt{A'}, \quad \frac{1}{2}(\lambda - \mu)\sqrt{B'};$$

whence

$$2\sigma\lambda = 1 + \lambda^2 + \mu^2 - \sqrt{A'}, \quad 2\tau\mu = 1 + \lambda^2 + \mu^2 - \sqrt{B'};$$

or the above value of x^2 is $\frac{\sigma}{\sigma+\tau}$, agreeing with his; the value of y^2 is, however, presented under a somewhat different form.

2. The curves are

$$(2 - xy)^2 = 2x(1 - y)(2 - y) \quad \text{and} \quad (2 - xy)^2 = y(1 - x)(2 - x),$$

each passing through the points $(1, 2), (2, 1)$; the four points of intersection found by the foregoing general theory are all real, viz.

$$x = \frac{1}{4}(2 + \sqrt{3})(5 + \sqrt{17}) = \text{say } 17.00 \text{ and } 1.65,$$

$$y = -\frac{1}{4}(-1 + \sqrt{3})(-1 + \sqrt{17}) = \text{say } +0.94 \text{ and } +2.13,$$

and

$$x = \frac{1}{4}(2 - \sqrt{3})(5 - \sqrt{17}) = \text{say } -0.12 \text{ and } -3.49,$$

$$y = -\frac{1}{4}(-1 - \sqrt{3})(-1 - \sqrt{17}) = \text{say } +1.22 \text{ and } -0.57.$$

The equation of the curve (1) may also be written in the forms

$$x(y^2 - 2x) + 2yx - 4x + 4 = 0, \quad xy^2 + x(-2y^2 + 2y - 4) + 4 = 0.$$

The original form shows that, if x is negative (but by a further examination it appears that between these limits of y , y being outside these limits, then x is positive; in fact, the whole curve lies on the positive side of the axis of y . And then the inspection of the first

quadric equation shows that the lines $x = \infty$ and $x = 2$ are each an asymptote. The point at infinity on the axis of y is a flecnode, the tangent to the flecnodal branch being $x = 0$, and that of the ordinary branch $x = 2$. Similarly, from the second quadric equation, the point at infinity on the axis of x is in fact a flecnode, having the line $y = 0$ for the tangent to the flecnodal branch, and $y = 2$ for that of the other branch. It further appears that the line $y = 0$ is in fact a cusp, having the axis x for the cuspidal tangent.

The equation of the curve (2) may also be written in the forms

$$x^2y + (x^2 - 7x + 2)y + 4 = 0, \quad y^2x + (y^2 - 7y + 2)x + 4 = 0.$$

The original form shows that, if y is between 1 and 2, the points at $x = 2$ and $\frac{1}{2}$, and the easy to see that the between these limits, $y = 1$ and $y = 2$, x is negative; and if x between 1 and 2, is positive; in fact, the existence of an oval, meeting the line $y = 1$ and $y = 2$ in the points $x = 2$ and $x = \frac{1}{2}(25 + \sqrt{113}) = \text{say } 2.2$ and 0.9 . The remainder of the curve lies wholly below the line $y = 0$. The first quadric equation shows the asymptote $y = 0$; the point at infinity on the axis of x is a cusp, having the axis y for the cuspidal tangent. The second quadric equation shows the asymptotes $y = 0$, $y = -1$; the point at infinity on the axis of x is in fact a flecnode, having the line $y = 0$ for the tangent to the flecnodal branch, and $y = 1$ for that of the other branch. It is further seen that there are two vertical tangents $x = \frac{1}{2}(11 \pm \sqrt{113}) = 10.8$ or 0.2 ; the former of these touches a branch between the two asymptotes $y = 0$, $y = -1$; the latter one of the branches belonging to the cuspidal asymptote $x = 2$; this last branch cuts the asymptote $y = 0$ at $y = -2$, and then, cutting the asymptote $y = -1$ and $x = -\frac{1}{3}(= -0.3)$, goes on to touch at infinity the asymptote $y = 0$. It is now easy to trace the curve.

The figure shows the two curves. The curve (1) is shown by a continuous line, the curve (2) by a thick dotted line; the points 1, 2, 3, 4 show the above mentioned four intersections of the curves; the point 1 and the dotted branch through it are drawn considerably out of their true positions; viz. as above appearing, the x -coordinate of 1 is $= 17.00$, and the equation of the vertical tangent to the branch is $x = 10.8$.

4620. (Proposed by A. B. Evans, M.A.)—[Vol. XXII., pp. 78, 79.]

Find the least integral values of x and y that will satisfy the equation $x^2 - 953y^2 = -1$.

[Solution by Professor Cayley.] The values are given in Degen's Tables, viz.

$$x = 2746864744198985747130, \quad y = 88979677488830275367615376.$$

The work referred to is entitled "Canon Pellianus, sive Tabula simplicissimam aequationis $y' = ax^2 + 1$ solutionem pro singulis numeri dati valoribus ab 1 usque ad 1000 in numeris rationalibus iisdemque integris exhibens. Auctore C. F. Degen, Hafniae (Copenhagen), 1817."

Table I., pp. 3—106 gives, for all numbers 1 to 1000, the denominators, (η) the quotients of the convergent fraction of $\sqrt{a}$, and also the least values of x, y which will satisfy the equation $x^2 - ay^2 = +1$. Thus

$$a = 953, \quad x = 2746864744198985747130, \quad y = 88979677488830275367615376.$$

Table II., pp. 109—112, is described as giving for all those values of a between 1 and 1000, for which there exists a solution of the equation $x^2 - ay^2 = -1$, the least values of x and y which will satisfy this equation; thus $a = 953$, x and y as above. It is, however, to be noticed that the values $a = \beta^2 + 1$, $x = \beta$, $y = 1$, for which there is the obvious solution, are omitted from the table. The reason for this appears, but the heading should have been different.

Problems 4528–4793 [Vol. XXIII.–XXIV., 1875–1876]

4528. (Proposed by Professor Cayley.)—[Vol. XXIII., pp. 18, 19.]

A lottery is arranged as follows: there are n tickets representing $a, b, c, \dots$ pounds respectively. A person draws once, looks at his ticket; and if he pleases, draws again (out of the remaining $n - 1$ tickets), looks at his ticket, and if he pleases draws again (out of the remaining $n - 2$ tickets); and so on, drawing in all not more than k times; and he receives the value of the last drawn ticket. Supposing that he regulates his drawings in the manner most advantageous to him according to the theory of probabilities, what is the value of his expectation?

[Solution by the Proposer.] Let the expression “ a or a ” signify “ a or a , whichever of the two is greatest,” and let $M_1(a, b, c, \dots)$ denote the mean of the quantities $(a, b, c, \dots)$, viz. their sum divided by the number of them.

To fix the ideas, consider five quantities a, b, c, d, e , and write

$$M_1(a, b, c, d, e) = M_1(a, b, c, d, e),$$

$$M_2(a, b, c, d, e) = M_1\{a \text{ or } M_1(b, c, d, e), b \text{ or } M_1(a, c, d, e), \dots, e \text{ or } M_1(a, b, c, d)\},$$

$$M_3(a, b, c, d, e) = M_1\{a \text{ or } M_2(b, c, d, e), b \text{ or } M_2(a, c, d, e), \dots, e \text{ or } M_2(a, b, c, d)\},$$

and so on. And the like in the case of any number of quantities $a, b, c, \dots$.

Then the value of the expectation is $= M_k(a, b, c, \dots)$.

For, when $k = 1$, the value is obviously $= M_1(a, b, c, \dots)$.

When $k = 2$, if a is drawn, the adventurer will be satisfied or he will draw again, according as a or $M_1(b, c, \dots)$ is greatest; viz. in this case the value of the expectation is “ a or $M_1(b, c, \dots)$.” So if b is drawn, the adventurer will be satisfied or he will draw again, as b or $M_1(a, c, \dots)$ is greatest; viz. “ b or $M_1(a, c, \dots)$ ”; and so on: and the value of the total expectation, the several cases being equally probable, is

$$M_1\{a \text{ or } M_1(b, c, \dots), b \text{ or } M_1(a, c, \dots), \dots\} = M_2(a, b, c, \dots) :$$

and the like for $k = 3, k = 4$, &c.

For instance, $a, b, c, d = 1, 2, 3, 4$,

$$M_1(1, 2, 3, 4) = \frac{5}{2},$$

$$M_2(1, 2, 3, 4) = M_1(1 \text{ or } \frac{9}{2}, 2 \text{ or } \frac{7}{2}, 3 \text{ or } \frac{5}{2}, 4 \text{ or } \frac{3}{2}) = M_1(\frac{9}{2}, \frac{7}{2}, 3, 4) = \frac{15}{4},$$

$$M_3(1, 2, 3, 4) = M_1(1 \text{ or } 4, 2 \text{ or } \frac{15}{4}, 3 \text{ or } \frac{7}{2}, 4 \text{ or } \frac{5}{2}) = M_1(4, \frac{15}{4}, \frac{7}{2}, 4) = \frac{49}{16},$$

$$M_4(1, 2, 3, 4) = M_1(1 \text{ or } \frac{49}{16}, 2 \text{ or } 4, 3 \text{ or } \frac{15}{4}, 4 \text{ or } \frac{7}{2}) = M_1(\frac{49}{16}, 4, \frac{15}{4}, 4) = \frac{177}{64}.$$

Or finally $M_1, M_2, M_3, M_4 = \frac{5}{2}, \frac{15}{4}, \frac{49}{16}, \frac{177}{64} = M_1, M_2, M_3, M_4$.

Cor.—If the $a, b, c, \dots$ denote penalties instead of prizes, then the solution is the same, except that “ a or a ” must now denote “ a or a , whichever of them is least.”

4581. (Proposed by the Rev. M. M. U. Wilkinson.)—
[Vol. XXIII., pp. 47, 48.]

A witness, whose statement is what he opines once in m times, and whose opinion is correct once in n times, asserted that the number of a note, issued by a bank universally known to have issued notes numbered from B to $B + A - 1$ inclusive, was $B + P$, where P is either $0, 1, 2, \dots$, or $A - 1$. Prove (1) that the probability that the note in question was that note is

$$\frac{(m-1)(n-1)}{mn} \left(\frac{1}{A-1} \right),$$

and (2) the probability that this last statement was correct.

Remark (Remark by Professor Cayley.). *There is a serious difficulty in this question. Try the answer in the case $m = 10$, $n = 10$. The witness is what he opines once out of 10 times—he is an atrocious liar once out of 10 times, that is, wrongly 9 times out of 10; he is in what he thinks, 9 times out of 10; and he opines rightly once out of 10 times. The witness says that the note was signed by X , and the chance of this being so is found to be $\frac{1+9}{10+9} = \frac{10}{19}$, or more than $\frac{1}{2}$; the larger part $\frac{9}{19}$ of this obtained as follows: the witness having said that the note was signed by X , the chances are 9 out of 10 that he opines, and, thinking the reverse, the chance is 9 out of 10 that he thought wrongly, viz. that the note was signed by X . But can the statement of such a witness create any probability in favour of the event?*

The fallacy seems to consist in the assumption that the opinion is of that nature that the note was a definite number forming a series. But if the opinion be that the note was not 99, such opinions, the witness of 10 times out of 10 opinion true it is quite conceivable; in 9 times out of no amount of ingenuity would be able to know the number of the note, from the mere fact that the witness opines that the note is not a named number.

Suppose there are 500 notes, and that n can have a determinate value irrespective of the event. I think that in calling the probability $= p_1 p_2$, we mean the probability of A 's making the statement, " B says that the event did happen" to be regarded as a simple statement, and the probability of the truth of the statement to be taken to be $= p_1 p_2$. Mr Martin seems to introduce into his solution a hypothesis contradictory to the assumptions of the question.

I remark further that in my solution I assume that the event is of such a nature that, when there is any testimony in regard to it, the probability is determined by that testimony, irrespectively of the antecedent probability. This is quite consistent with the antecedent probability being, not zero, but k (as it may very well be) indefinitely small; so if k were $= 0$, the whole probability $= p_1 p_2$. But there is no absolutely reason for assigning any determinate value for k ; and so the solutions which assume respectively $\beta = 1$ and $\beta = 0$, seem to me on this ground erroneous.

4638. (Proposed by Professor Cayley.)—[Vol. XXIII., p. 58.]

Find the equation of the surface which is the envelope of the quadric surface $ax^2 + by^2 + cz^2 + dw^2 = 0$, where a, b, c, d are variable parameters connected by the equation $Abc + Bca + Cab + Fad + Gbd + Hcd = 0$; and in the particular case in which the constants A, B, C, F, G, H satisfy the condition $(AF)^{\frac{1}{2}} + (BG)^{\frac{1}{2}} + (CH)^{\frac{1}{2}} = 0$.

4694. (Proposed by Professor Cayley.)—[Vol. XXIV., July to December, 1875, p. 41.]

Taking F, F' a pair of reciprocal points with respect to a circle, centre O ; then if F, F' are centres of force, each force varying as (distance) $^{-2m}$, prove that (1) the resultant force upon any point P on the circle is in the direction of a fixed point S on the axis OFF' ; and if, moreover, the forces at the unit of distance are as $(OF)^{2m-2}$ to $(OF')^{2m-2}$, then (2) the resultant force is proportional to $(SP)^{-2m} \cdot (PV)^{-2m+2}$, where PV is the chord through S .

**4793. (Proposed by Professor Wolstenholme, M.A.)—
[Vol. XXIV., pp. 72–74.]**

If $y = x^n(\log x)^r$, where n and r are integers, prove that

$$\frac{d^{n+r}y}{dx^{n+r}} = \frac{r(r-1)\cdots(r-n+1)}{x^r} \cdot y - \frac{r(r-1)(r-2)\cdots(r-n+2)}{x^{r-1}} \cdot \frac{dy}{dx} + \cdots$$

the coefficients being

$$\frac{r(r-1)\cdots(r-s+1)}{s!} \cdot \frac{(n+r)!}{(n+s)!}$$

for the s -th term, so that the result may be symbolically written

$$\frac{d^{n+r}y}{dx^{n+r}} = \left\{ \frac{d}{dx} \cdot \left(x \frac{d}{dx} - n \right) \right\}^r y,$$

where D denotes $\frac{d}{dx}$ and operates on $\frac{d^{n+r}y}{dx^{n+r}}$ only, and Δ operates on all terms after the r -th, vanishing since $\Delta^r x^n = 0$ when r is an integer $> n$.

[Solution by Professor Cayley.] Since $y = x^n(\log x)^r$, therefore $(x \frac{d}{dx} - n)y = rx^n(\log x)^{r-1}$; repeating the same operation, we have $(x \frac{d}{dx} - n)^r y = [r]^r x^n$, whence $\frac{d^n}{dx^n} (x \frac{d}{dx} - n)^r y = [r]^r [n]^r$.

Now, for any value whatever of the function y , we have

$$\frac{d^n}{dx^n} (x \frac{d}{dx} - n)^r y = Ax^n \frac{d^{n+r}y}{dx^{n+r}} + Bx^{n-1} \frac{d^{n+r-1}y}{dx^{n+r-1}} + Cx^{n-2} \frac{d^{n+r-2}y}{dx^{n+r-2}} + \&c.,$$

the coefficients $A, B, C, \dots$ being functions, presumably of r, n , of the form of the function y . It will, however, appear that $A, B, C, \dots$ are, in fact, functions of r only.

To see how this is, observe that $(x \frac{d}{dx} - n)^r$ consists of a set of terms $(x \frac{d}{dx})^s$, ($s = 0$ to r), where $(x \frac{d}{dx})^s$ denotes s repetitions of the operation $x \frac{d}{dx}$; by a well-known theorem, this is $= [x \frac{d}{dx} + s - 1]^s$, where, after expansion of the factorial, $(x \frac{d}{dx})^t$ is to be replaced by $x^t \frac{d^t}{dx^t}$; thus

$$(x \frac{d}{dx})^2 = (x \frac{d}{dx} + 1)^2 = x^2 \frac{d^2}{dx^2} + x \frac{d}{dx}, \quad (x \frac{d}{dx})^3 = [x \frac{d}{dx} + 2]^3 = x^3 \frac{d^3}{dx^3} + 3x^2 \frac{d^2}{dx^2} + 2x \frac{d}{dx}.$$

Thus $(x \frac{d}{dx} - n)^r$ consists of a series of terms $x^t \frac{d^{t+s}}{dx^{t+s}}$, where $t + s = r$, or $t = r - s$; and the term is $x^{r-s} \frac{d^r}{dx^r}$, or $x^{r-s} \frac{d^r}{dx^r}$. Observing that

$\frac{d^n}{dx^n}(x\frac{d}{dx} - n)^r$ consists of a series of terms of the form $\frac{d^n}{dx^n}x^{r-s}\frac{d^r}{dx^r}$, where the unaccented symbol operates on the x and the accented symbol on the y ; and putting $x^{r-s}\frac{d^r}{dx^r} = x^{r-s}\frac{d^{r-s}}{dx^{r-s}} \cdot \frac{d^s}{dx^s}$, that is, $x^{r-s}\frac{d^r}{dx^r} = x^{r-s}\frac{d^{r-s}}{dx^{r-s}} \cdot \frac{d^s}{dx^s}$; so that the term is $x^{r-s}\frac{d^{n+r-s}}{dx^{n+r-s}}y$.

To understand how the coefficients can be independent of n , take the particular case $r = 2$; then we have here

$$\frac{d^n}{dx^n}(x\frac{d}{dx} - n)^2y = Ax^n\frac{d^{n+2}y}{dx^{n+2}} + Bx^{n+1}\frac{d^{n+1}y}{dx^{n+1}} + Cx^n\frac{d^ny}{dx^n}.$$

The right-hand side is

$$\frac{d^n}{dx^n}\{x^2\frac{d^2}{dx^2} - (2n-1)x\frac{d}{dx} + n^2\}y = \{x^2\frac{d^{n+2}}{dx^{n+2}} + 2nx\frac{d^{n+1}}{dx^{n+1}} + (n^2-n)\frac{d^n}{dx^n} - (2n-1)x\frac{d^n}{dx^n}\}y$$

hence

$$A = 1, \quad B = 2n - (2n - 1) = 1, \quad C = (n^2 - n) - n(2n - 1) + n^2 = 0;$$

and we thus see also how in this particular case the last coefficient is $= 0$, viz. that we have

$$\frac{d^n}{dx^n}(x\frac{d}{dx} - n)^2y = x^2\frac{d^{n+2}y}{dx^{n+2}} + x\frac{d^{n+1}y}{dx^{n+1}},$$

without any term in $\frac{d^ny}{dx^n}$.

To find the coefficients $A, B, C, \dots$ generally, write $y = x^{n+r+\theta}$, then $x\frac{d}{dx} - n = r + \theta$, and consequently

$$\frac{d^n}{dx^n}(x\frac{d}{dx} - n)^ry = (r + \theta)^rx^{n+\theta} = (r + \theta)^r[r + \theta + n]^nx^{n+\theta};$$

whence

$$(r + \theta)^r[r + \theta + n]^n = A[r + \theta + n]^{n+r} + B[r + \theta + n]^{n+r-1} + \dots$$

or, what is the same thing,

$$(r + \theta)^r = A[r + \theta]^r + B[r + \theta]^{r-1} + \dots$$

Since every term on the left-hand side, and every term on the right-hand side, contains the factor $r + \theta$; it is not on the right-hand side any term $[r + \theta]^r$; dividing the equation by $r + \theta$, then becomes

$$(r + \theta)^{r-1} = A[r + \theta - 1]^{r-1} + B[r + \theta - 1]^{r-2} + \dots$$

and we thus have the series

$$Ux = (1+x)^r, \quad \text{where } r + \theta = 1 + x,$$

taking the terms in the reverse order,

$$Ux = u_0 + \frac{x}{1} \Delta u_0 + \frac{x(x-1)}{1 \cdot 2} \Delta^2 u_0 + \&c.,$$

the well-known one.

Hence, in general,

$$\frac{d^{n+r}y}{dx^{n+r}} = \left\{ \frac{d}{dx} \left(x \frac{d}{dx} - n \right) \right\}^r y,$$

where observe that the last term is $= x \frac{d^{n+1}y}{dx^{n+1}}$.

For the function $y = x^n(\log x)^r$, the value of each side is $= [r]^r [n]^r$.

4752. (Proposed by Professor Cayley.)—[Vol. XXIV., pp. 89–91.]

Mr Wolstenholme's Question 3067 may evidently be stated as follows:
If (a, b, c) are the coordinates of a point on the cubic curve

$$a^3 + b^3 + c^3 = (b + c)(c + a)(a + b),$$

and if

$$x : y : z = a(-a + b + c) : b(a - b + c) : c(a + b - c);$$

then (x, y, z) are the coordinates of a point on the same cubic curve. This being so, it is required to find the geometrical relation of the two points to each other.

[Solution by Professor Cayley.] 1. On referring to Wolstenholme's Question 3067 (Reprint, Vol. XIII., p. 70), it appears that the coordinates (x, y, z) of the point may be expressed in the more simple form

$$x : y : z = a(-a + b + c) : b(a - b + c) : c(a + b - c);$$

viz. the given relation between (a, b, c) being equivalent to

$$a^3 + b^3 + c^3 - (b + c)(c + a)(a + b) = 0,$$

we have

$$\frac{x}{a(-a + b + c)} = \frac{y}{b(a - b + c)} = \frac{z}{c(a + b - c)};$$

and thence

$$\frac{-4abc}{x^2/a - (y^2/b + z^2/c)} = \dots;$$

and consequently

$$x : y : z = x'(-x' + y' + z') : y'(x' - y' + z') : z'(x' + y' - z'),$$

whence the transformation in question.

2. In place of (x, y, z) for the coordinates of the second point, writing (x', y', z') , and (x, y, z) for those of the first point, the two points are connected by the relation

$$x' : y' : z' = x(-x + y + z) : y(x - y + z) : z(x + y - z);$$

and we thence at once deduce the converse relation

$$x : y : z = x'(-x' + y' + z') : y'(x' - y' + z') : z'(x' + y' - z').$$

Hence, writing

$$\begin{aligned} (-x + y + z, x - y + z, x + y - z) &= (\xi, \eta, \zeta), \\ (-x' + y' + z', x' - y' + z', x' + y' - z') &= (\xi', \eta', \zeta'), \end{aligned}$$

and similarly

$$x' : y' : z' = x\xi' : y\eta' : z\zeta' = x^2\xi : y^2\eta : z^2\zeta;$$

so that $\xi/\xi' = \eta/\eta' = \zeta/\zeta'$; the coordinates of the two points, we see that

$$x : y : z = x'\xi' : y'\eta' : z'\zeta';$$

regarding these as inverse points one of the other in regard to the triangle $\xi = 0, \eta = 0, \zeta = 0$.

To complete the solution, we must introduce these new coordinates into the equation of the cubic curve. Writing this under the form

$$Hxyz + 2(-x + y + z)(x - y + z)(x + y - z) = 0,$$

and observing that the equation is $(\xi + \eta)(\eta + \zeta)(\zeta + \xi) + 2\xi\eta\zeta = 0$; viz. this is a cubic curve inverting into itself. And the two points in question are thus any two inverse points on this cubic curve.

3. In regard to the original form, that the point (x, y, z) defined by the equations

$$\frac{x}{-a^2 + b^2 + c^2} = \frac{y}{a^2 - b^2 + c^2} = \frac{z}{a^2 + b^2 - c^2}$$

lies on the cubic curve

$$a^3 + b^3 + c^3 - (b + c)(c + a)(a + b) = 0,$$

Professor Sylvester proceeds as follows:—Writing

$$(x, y, z) = (a^4 - (b^2 - c^2)^2, b^4 - (c^2 - a^2)^2, c^4 - (a^2 - b^2)^2) = (A, B, C), \quad \text{suppose;}$$

and

$$F(a, b, c) = a^3 + b^3 + c^3 - (b + c)(c + a)(a + b),$$

he observes that the truth of the theorem depends on the identity

$$F(A, B, C) + F(a, b, c) \cdot F(a, -b, c) \cdot F(a, b, -c) \cdot F(a, -b, -c) = 0,$$

and that, in order to prove the identity generally, it is sufficient to prove it for the three cases $a^2 = 0$, $a^2 = b^2 + c^2$, $a^2 = b^2$, which may be effected without difficulty.

But, for a general proof of the identity, write

$$\lambda = b^2 + c^2, \quad \mu = b^2 - c^2,$$

so that

$$A = a^4 - \mu^2, \quad B = (a^2 + \mu)(-a^2 + \lambda), \quad C = (-a^2 + \lambda)(a^2 - \mu).$$

Whence

$$\begin{aligned} -F(A, B, C) &= -(a^4 - \mu^2)^3 + 2(a^2 \lambda \mu)^2 (a^2 + 3a^2 \mu - 8a^2 b^2 c^2) (a^4 - \mu^2) (a^2 - \lambda) \\ &= a^{12} - 6\lambda a^{10} + (6\lambda^2 + 9\mu^2 - 8b^2 c^2) a^8 + \lambda(-2\lambda^2 - 18\mu^2 + 8b^2 c^2) a^6 \\ &\quad + \mu^2(18\lambda^2 - 3\mu^2 + 8b^2 c^2) a^4 + \lambda \mu^2(-6\lambda^2 - 8b^2 c^2) a^2 + \mu^6. \end{aligned}$$

Moreover

$$F(a, b, c) = a\{a^2 - (b + c)^2\} - (b + c)\{a^2 - (b - c)^2\},$$

therefore

$$F(a, -b, -c) = a\{a^2 - (b + c)^2\} + (b + c)\{a^2 - (b - c)^2\};$$

whence

$$\begin{aligned} F(a, b, c) \cdot F(a, -b, -c) &= a^2\{a^2 - (b + c)^2\}^2 - (b + c)^2\{a^2 - (b - c)^2\}^2 \\ &= a^4 - 3\gamma^2 a^2 + \gamma^2(\gamma^2 + 2\delta^2), \end{aligned}$$

if $\gamma = b + c$, $\delta = b - c$. By changing the sign of c , we interchange γ and δ , and we thus have

$$F(a, b, -c) \cdot F(a, -b, c) = a^4 - 3\delta^2 a^2 + \delta^2(2\gamma^2 + \delta^2);$$

and the identity to be verified is thus

$$\begin{aligned} [a^4 - 3\gamma^2 a^2 + \gamma^2(\gamma^2 + 2\delta^2)] \cdot [a^4 - 3\delta^2 a^2 + \delta^2(2\gamma^2 + \delta^2)] \\ = a^{12} - 6\lambda a^{10} + \dots \end{aligned}$$

the values of λ , μ in terms of γ , δ being

$$\lambda = \frac{1}{2}(\gamma^2 + \delta^2), \quad \mu = \gamma\delta;$$

substituting these values on the right-hand side, the verification can be completed without difficulty.

Problems 4946–5208 [Vol. XXV.–XXVII., 1876–1877]

4946. (Proposed by Professor Cayley.)—[Vol. XXV., January to July, 1876, p. 82.]

Show that the attraction of an indefinitely thin double convex lens on a point at the centre of one of its faces is equal to that of the infinite plate included between the tangent plane at the point and the parallel tangent plane of the other face of the lens.

5020. (Proposed by W. S. B. Woolhouse, F.R.A.S.)—[Vol. XXVI., July to December, 1876, pp. 41, 42.]

Let $v = \frac{1}{2}(n+1)$, $v' = \frac{1}{2}n(n-1)$, $v'' = \frac{1}{2}n(n-1)(n-2)$, &c. be the maximum coefficient and the sums of the first, second, third, ... differences of the coefficients of the expansion of the binomial $(1+x)^n$ taken as the central or maximum coefficient, and also let $S_1, S_2, S_3, \dots, S_n$ be the coefficients; then show that the algebraic function

$$x^n - S_1x^{n-1} + S_2x^{n-2} - S_3x^{n-3} + \&c.$$

is divisible by $(x-1)^n$ without remainder; and that the sum of the numerical coefficients of the quotient is equal to $1 \cdot 3 \cdot 5 \cdots (2n-1)$.

[See Solution to Question 1894, Reprint, vol. v., p. 113.]

[Solution by Professor Cayley.] Mr Woolhouse's elegant theorem depends ultimately on the property of triangular numbers $\phi(-n)$, so that, writing down the series $\phi(n) = \frac{1}{2}(n^2+n)$, $\phi(n+1) = \cdots$ we have, in fact,

$$\dots, 10, 6, 3, 1, 0, 0, 1, 3, 6, 10, \dots$$

a single continuous series, obtained by giving to n the different negative and positive integer values, zero included.

Thus a particular case is

$$(1+x)^6 - 5(1+x)^3 + 9(1+x) - 5 = 1 \cdot 3 \cdot 5x^2 + \&c. x^3 + \dots,$$

where, on the left-hand side, the exponents are the triangular numbers $\frac{1}{2}n(n+1)$, and the coefficients, after the first, are the differences of the binomial coefficients of the power $2n$ (in the particular case, $n=3$); viz. the binomial coefficients being 1, 6, 15, 20, 15, 6, 1, the differences taken as far as they are positive are 5, 9, 5.

Expanding the several terms and writing down only the coefficients, we have a diagram:

$$\begin{array}{ccccccc} 1, & 6, & 15, & 20, & 15, & 6, & 1, \\ -5 & 1, & 3, & 3, & 1, & & \\ +9 & 1, & 1, & & & & \\ -5 & 1, & & & & & \end{array}$$

The theorem in the particular case depends on the identities

$$1 - 5 + 9 - 5 = 0, \quad 6 - 15 + 9 = 0, \quad 15 - 20 + 5 = 0,$$

or writing, as above, h, g, f, e, d, c, b, a to denote the triangular numbers

$$\begin{aligned} h^2 - 5g^2 + 9f^2 - 5e^2 &= 0, & h^3 - 5g^3 + 9f^3 - 5e^3 &= 0, \\ h^4 - 5g^4 + 9f^4 - 5e^4 &= 0, & h^5 - 5g^5 + 9f^5 - 5e^5 &= 1 \cdot 2 \cdot 3 \cdot 1 \cdot 3 \cdot 5. \end{aligned}$$

These may be replaced by

$$\begin{aligned} h^2 - 5g^2 + 9f^2 - 5e^2 &= 0, \\ h^3 - 5g^3 + 9f^3 - 5e^3 &= 0, \\ h^4 - 5f^4 + 9f^4 - 5e^4 &= 0, \\ h^5 - 5g^5 + 9f^5 - 5e^5 &= 1 \cdot 2 \cdot 3 \cdot 1 \cdot 3 \cdot 5. \end{aligned}$$

Consider any one of these, for instance the third; the function on the left-hand is

$$\frac{1}{24}h^4 - (6-1)\frac{1}{6}g^4 + (15-6)\frac{1}{2}f^4 - (20-15)e^4,$$

or, introducing the values a, b, c, d, e, f, g, h which are any successive triangular numbers, this is in fact

$$\frac{1}{24}(\xi+6)^4 - 6 \cdot \frac{1}{6}(\xi+5)^4 + 15 \cdot \frac{1}{2}(\xi+4)^4 - 20(\xi+3)^4 + 15(\xi+2)^4 - 6(\xi+1)^4 + 1 \cdot \xi^4 = 0,$$

an immediate consequence of the well-known theorem

$$(\xi+6)^m - 6(\xi+5)^m + 15(\xi+4)^m - 20(\xi+3)^m + 15(\xi+2)^m - 6(\xi+1)^m + \xi^m = 0,$$

for any value of m up to $m = 5$, and $= 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6$ for $m = 6$.

We have thus all the equations except the last; and as regards the last equation, observe that the equation to be verified is

$$\frac{1}{6!} \left[\frac{1}{2}(e+6)(e+5) \right]^2 - 6 \left[\frac{1}{2}(e+5)(e+4) \right]^2 + \dots = 1 \cdot 3 \cdot 5;$$

viz. this may be replaced by

$$\frac{1}{2}(e+6)^2 - 6(e+5)^2 + \dots = 2\mu \cdot 2 \cdot 3 \cdot 1 \cdot 3 \cdot 5 = 2 \cdot 4 \cdot 6 \cdot 1 \cdot 3 \cdot 5 = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6,$$

which is right.

It is clear that the proof, although worked out on a particular case, is perfectly general; and Mr Woolhouse's theorem is thus proved.

5079. (Proposed by Professor Cayley.)—[Vol. XXVI., pp. 77, 78.]

Show that the curve

$$\{(\beta-\gamma)^2-\delta^2\}\{(x-\beta)^2+y^2\}+\{(\beta+\gamma)^2-\delta^2\}\{(x+\beta)^2+y^2\}=\{(1-\xi)^2+\eta^2\}\{(x-\gamma-\delta)^2+y^2\}$$

where $t = (\sqrt{-1})$ as usual, $\xi = \frac{\beta^2+\gamma^2-\delta^2}{(\beta+\gamma)^2-\delta^2}$, $\eta = \frac{-2\gamma\delta}{(\beta+\gamma)^2-\delta^2}$, is a real bicircular quartic having the axial foci β , $-\beta$, $\gamma + \delta$, $\gamma - \delta$.

[Solution by the Proposer.] Consider the equation

$$(1+m\sqrt{-1})\{(x-\beta)^2+y^2\}+q(1-m\sqrt{-1})\{(x+\beta)^2+y^2\}=(\lambda+\mu\sqrt{-1})\{(x-\gamma-\delta)^2+y^2\}$$

This is

$$\begin{aligned} & (1+m\sqrt{-1})\{x^2+y^2-\beta^2-2\beta x\sqrt{-1}\} \\ & + q(1-m\sqrt{-1})\{x^2+y^2-\beta^2+2\beta x\sqrt{-1}\} \\ & - (\lambda+\mu\sqrt{-1})\{x^2+y^2-\beta^2+\beta^2+\gamma^2-\delta^2-2\gamma x-2(\beta-\gamma)\delta\sqrt{-1}\}=0, \end{aligned}$$

where, putting the imaginary part equal to zero,

$$\begin{aligned} & m(1-q)(x^2+y^2-\beta^2)-2\beta(1+q)\sqrt{-1} \\ & - \mu\{x^2+y^2-\beta^2+(\beta^2+\gamma^2-\delta^2)-2\gamma x\}+2\lambda(\beta-\gamma)\delta=0, \end{aligned}$$

which will be true identically if

$$\begin{aligned} m(1-q)-\mu &= 0, \\ -2\beta(1+q)+\mu\gamma+\lambda\delta &= 0, \\ -\mu(\beta^2+\gamma^2-\delta^2)-2\lambda\gamma\delta &= 0. \end{aligned}$$

The last gives

$$\lambda = \varrho(\beta^2 + \gamma^2 - \delta^2), \quad \mu = -2\varrho\gamma\delta, \quad \varrho \text{ arbitrary};$$

and then

$$\begin{aligned} 2\beta(1-q)\beta &= 2\varrho\gamma\delta(\beta^2 + \gamma^2 - \delta^2) = 2\varrho\delta(\beta^2 - \gamma^2 - \delta^2), \\ m(1-q) &= -2\varrho\gamma\delta; \end{aligned}$$

so that, putting $\varrho\delta = (1-q)\xi$, or $\varrho = (1-q)\xi/\delta$, we have

$$\begin{aligned} 1 &= \varrho(\beta^2 + \gamma^2 - \delta^2) = (1-q)\xi \frac{\beta^2 + \gamma^2 - \delta^2}{\delta}, \\ m &= -2\varrho\gamma = (1-q)\xi \frac{-2\gamma}{\delta}, \\ \mu &= (1-q)\xi(-2\gamma), \\ \lambda &= (1-q)\xi(\beta^2 + \gamma^2 - \delta^2). \end{aligned}$$

Therefore

$$1 \pm m\sqrt{-1} = (1 - q) \left\{ \frac{\beta^2 + \gamma^2 - \delta^2}{\delta} \mp \frac{2\gamma\sqrt{-1}}{\delta} \right\},$$

$$\lambda \pm \mu\sqrt{-1} = (1 - q)\xi\{[\beta^2 + (\gamma + \delta)^2] \mp [\beta^2 + (\gamma - \delta)^2]\sqrt{-1}\};$$

and the equation is

$$\{(\beta - \gamma)^2 - \delta^2\}\{(x - \beta)^2 + y^2\} + \{(\beta + \gamma)^2 - \delta^2\}\{(x + \beta)^2 + y^2\} = \{(1 - \xi)^2 + \eta^2\}\{(x - \gamma - \delta)^2$$

which is a real curve having the axial foci β , $-\beta$, $\gamma + \delta$, $\gamma - \delta$; viz. β being a focus, and the curve being real, it is clear that $-\beta$ is also a focus.

5130. (Proposed by Professor Cayley.)—[Vol. XXVII., January to June, 1877, p. 20.]

Show that the envelope of a variable circle, having its centre on a given conic and cutting at right angles a given circle, is a bicircular quartic; which, when the given conic and the circle have double contact, becomes a pair of circles; and, by means of the last-mentioned particular case of the theorem, connect together the porisms arising out of the two problems:

- (i) Given two conics, to find a polygon of n sides inscribed in the one and each side touching the circumscribed about the other.
- (ii) Given two circles, to find a closed series of n circles, each touching the two circles and the two adjacent circles of the series.

5208. (Proposed by Professor Sylvester.)—[Vol. XXVII., pp. 81–83.]

Let the magnitude of any ramification signify the number of its branches, and let its partial magnitudes in respect to any node signify the magnitudes of the ramifications which come together at that node. If at any node the largest magnitude exceeds by k the sum of the other magnitudes, let the node be called *superior by k* , or be said to be of *superiority k* ; but if no magnitude exceeds the sum of the other magnitudes, let the node be called *subequal*. Then the theorem is, in any ramification, either there is one and only one subequal node; or else there are two and only two nodes each superior by unity, these two nodes being contiguous.

[Solution by Professor Cayley.] The proof consists in showing that (1) there cannot be more than one subequal node; (2) there cannot be more than two nodes each superior by unity; and if there is one such node, then there is, contiguous to it, another such node; (3) starting from a node which is superior by more than unity, there is always a contiguous node which is either of smaller superiority, or else subequal; for, these theorems holding good, we can, by (3), always arrive at a node which is either subequal or else superior by unity; in the former case, by (1), the subequal node thus arrived at is unique; in the latter case, by (2), we have, contiguous to the node arrived at, a second node superior by unity; and we have thus a unique pair of nodes each superior by unity.

I will prove only (3), as it is easy to see that the like process applies to the proof of (1) and (2).

Let the whole magnitude be n ; and suppose at a node P which is superior by k , the largest magnitude is α , and that the other magnitudes are, say β, γ, δ . We have

$$\alpha = \beta + \gamma + \delta + k;$$

and since $n = \alpha + \beta + \gamma + \delta$, we have thence $n = 2\alpha - k$, or

$$\alpha = \frac{1}{2}(n + k), \quad \beta + \gamma + \delta = \frac{1}{2}(n - k);$$

clearly k is even or odd, according as n is even or odd.

Suppose now that we pass from P , along the branch of magnitude α , to a node Q ; and let the magnitudes for Q be $\alpha', \beta', \gamma', \delta', \epsilon$, where α' denotes the magnitude for the branch QP . We have $\alpha' = \beta + \gamma + \delta + 1$ for the contiguous ramification: the branch QP and of the ramifications of magnitudes β, γ, δ which meet in P . We have thus

$$\alpha' = \frac{1}{2}(n - k) + 1 = \frac{1}{2}n - \frac{1}{2}(k - 2)$$

and thence

$$\beta' + \gamma' + \delta' + \epsilon' = \frac{1}{2}n + \frac{1}{2}(k - 2).$$

Supposing here that k is greater than 1, viz. that it is $=$ or > 2 , $k - 2$ is positive; and if α' be the greatest magnitude belonging to the node Q , this is a subequal node. But it may be that α' is not the greatest magnitude; supposing then that the greatest magnitude is β' , we have

$$\beta' = \frac{1}{2}n + \frac{1}{2}(k - 2) - \gamma' - \delta' - \epsilon',$$

$$\alpha' + \gamma' + \delta' + \epsilon' = \frac{1}{2}n - \frac{1}{2}(k - 2) + \gamma' + \delta' + \epsilon',$$

and thence

$$\beta' - (\alpha' + \gamma' + \delta' + \epsilon') = k - 2 - 2(\gamma' + \delta' + \epsilon');$$

viz. $k - 2$ that is, if the greatest branch of the node Q is either subequal, or else, being superior, the superiority is at most $= k - 2$. That is, from the node P , of superiority $=$ or > 2 , we pass along the branch of greatest magnitude to the contiguous node Q , this is either subequal, or else of superiority less than that of P ; which is the foregoing proposition (3).

The subequal node, and the two nodes of superiority 1, may be termed respectively the *centre* and the *bicentre*; the theorem thus is, every ramification has either a centre or else a bicentre.

But the centre and the bicentre here considered, due (as remarked by Professor Sylvester) to M. Camille Jordan, and which we here consider, not the magnitude, but the length of a ramification, such distance: viz. length being measured by the number of branches to be travelled over in order to reach the most distant terminal node. The ramification has either a centre or else a bicentre of distance: viz. if, disregarding the other ramifications, it is a node such that, for two or more of the branches which proceed from it, the lengths are equal; and the bicentre a pair of contiguous nodes such that, from the two nodes, there are from each of them two or more ramifications the lengths of which are equal to each other and superior to those of the other ramifications.

It is very noticeable how close the agreement is between the proofs for the two kinds of centre. Say, as regards distance, if at any node the length of the longest branch exceeds by k the length of the next longest branch or branches, then the node is superior by k , or is of the superiority k ; but, if there are two or more longest branches, then the node is subequal. And say next, in regard to magnitude, if at any node the largest magnitude exceeds by k the sum of all the other magnitudes, the node is superior by k , or has a superiority k ; but if the largest magnitude does not exceed the sum of the other magnitudes, then the node is subequal. Then, whether we attend to distance or to magnitude, the three propositions hold good: (1) there cannot be more than one subequal node; (2) there cannot be more than two nodes each superior by unity; and if there is one such node, there is contiguous to it another such node; (3) starting from a node which is superior by more than unity, there is always a contiguous node which is of smaller superiority or else subequal; whence, as in the solution just referred to, there is always, as regards distance, a centre or a bicentre; and there is always, as regards magnitude, a centre or a bicentre.

On Mr Artemas Martin's First Question in Probabilities [Vol. XXVII., pp. 89, 90]

BY PROFESSOR CAYLEY.

The question was, "A says that B says that a certain event took place; required the probability that the event did take place, p_1 and p_2 being A's and B's respective probabilities of speaking the truth."

The solutions, referred to or given on pp. 77—79 of volume xxvii. of the Reprint, give the following values for the probability in question:

Todhunter's Algebra	$\frac{p_1 p_2}{p_1 p_2 + (1 - p_1)(1 - p_2)},$
Artemas Martin	$\frac{p_1 p_2}{p_1 p_2 + (1 - p_1)(1 - p_2)},$
American Mathematicians and Woolhouse	$p_1 p_2.$

It seems to me that the true answer cannot be expressed in terms of p_1 and p_2 only, but that it involves two other constants β and k ; and my value is

$$\text{Cayley } \frac{p_1 p_2}{p_1 p_2 + \beta(1 - p_1)(1 - p_2) + k(1 - \beta)(1 - p_1)}.$$

In obtaining this I introduce, but I think of necessity, elements which Mr Woolhouse calls extraneous and imperfect.

B told A that the event happened, or he did not tell A this; the only evidence is A's statement that B told him that the event happened; and the chances are p_1 and $1 - p_1$. But, in the latter case, either B told A that the event did not happen, or he did not tell him at all; the chances (on the supposition of the incorrectness of A's statement) are β and $1 - \beta$; and the chances of the three cases are thus p_1 , $\beta(1 - p_1)$, and $(1 - \beta)(1 - p_1)$. On the supposition of the first and second cases respectively, the chances for the event having happened are p_2 and $1 - p_2$; on the supposition of the third case (viz. here there is no information as to the event) the chance is the antecedent probability k , and the whole chance in favour of the event is

$$p_1 p_2 + \beta(1 - p_1)(1 - p_2) + k(1 - \beta)(1 - p_1).$$

If $\beta = 1$, we have Todhunter's solution; if $\beta = 0$, and also $k = 0$, we have the solution preferred by Woolhouse; but we do not (otherwise than by establishing between k and β a relation which is quite arbitrary) obtain Martin's solution.

The error in A says that B told him as to the event, and says further that the event did happen: the probability of the truth of this compound statement is p_1 , where, in calling the probability $= p_1$, we mean that if A is speaking the truth, the event took place; this is to be regarded as a simple statement, and the probability of the truth of the statement to be taken to be $= p_1 p_2$. Mr Martin seems to introduce into his solution a hypothesis contradictory to the assumptions of the question.

I remark further that in my solution I assume that the event is of such a nature that, when there is any testimony in regard to it, the probability is determined by that testimony, irrespectively of the antecedent probability. This is quite consistent with the antecedent probability being, not zero, but k (as it may very well be) indefinitely small; so if k were $= 0$, the whole probability $= p_1 p_2$. But there is no absolutely reason for assigning any determinate value for k ; and so the solutions which assume respectively $\beta = 1$ and $\beta = 0$, seem to me on this ground erroneous.

Problems 4870–5625 [Vol. XXIX., January to June, 1878]

4870. (Proposed by Professor Cayley.)—[Vol. XXIX., p. 20.]

If $\alpha, \beta, \gamma, \delta; \alpha_1, \beta_1, \gamma_1, \delta_1$ are such that

$$(\alpha_1 - \delta_1)(\beta_1 - \gamma_1) = (\alpha - \delta)(\beta - \gamma), \quad (\beta_1 - \delta_1)(\gamma_1 - \alpha_1) = (\beta - \delta)(\gamma - \alpha), \quad (\gamma_1 - \delta_1)(\alpha_1 - \beta_1) = (\gamma - \delta)(\alpha - \beta)$$

show that the three equations

$$\frac{dx}{\sqrt{(x - \alpha)(x - \beta)(x - \gamma)(x - \delta)}} = \frac{dx_1}{\sqrt{(x_1 - \alpha_1)(x_1 - \beta_1)(x_1 - \gamma_1)(x_1 - \delta_1)}}$$

such are equivalent relations between the constants, and show also that the equations in the foregoing differences $\alpha_1 - \delta_1, \beta_1 - \delta_1, \gamma_1 - \delta_1$ may be so constituted with the particular integral of the differential equation.

5625. (Proposed by Professor Cayley.)—[Vol. XXIX., pp. 96, 97.]

Given three conics passing through the four points; and on the first a point A , on the second a point B , and on the third a point C . It is required to find on the first a point A' , on the second a point B' , and on the third a point C' , such that the intersections of the lines $B'C'$ and BC , $C'A'$ and CA , $A'B'$ and AB lie on the first conic; $C'A'$ and CA , $A'B'$ and BA , $B'C'$ and BC lie on the second conic; and $A'B'$ and CB , $C'B'$ and CA lie on the third conic.

5306. (Proposed by Professor Cayley.)—[Vol. XXXI., January to June, 1879, p. 38.]

The equation

$$\{(x + y + z)^2 - yz - zx - xy\}^2 = 4(2q + 1)xyz(x + y + z)$$

represents a trinodal quartic curve having the lines $x = 0, y = 0, z = 0, x + y + z = 0$ for its four bitangents; $X = 0, Y = 0, Z = 0$ represent the sides of the triangle formed by the three nodes.

5387. (Proposed by Professor Cayley.)—[Vol. XXXII., July to December, 1879, p. 35.]

Show that a cubic surface has at most 4 conical points, and a quartic surface at most 16 conical points.

5927. (Proposed by Professor Cayley.)—[Vol. XXXIII., January to July, 1880, p. 17.]

If $\{a + \beta + \gamma + \dots\}^n$ denote the expansion of $(a + \beta + \gamma + \dots)^n$, retaining those terms only $Na^pb^qc^r \dots$ in which $b + c + d + \dots \geq p - 1$, $c + d + \dots \geq p - 2$, &c. &c.; prove that

$$a^n = (x+a)^n - \{a\}^1(x+a+\beta)^{n-1} + \{a+\beta\}^2(x+a+\beta+\gamma)^{n-2} - \{a+\beta+\gamma\}^3(x+a+\beta+\gamma+\dots)^{n-3} + \dots$$

Problems 6155–6800 [Vol. XXXVI.–XXXVII., 1881–1882]

6155. (Proposed by Professor Cayley.)—[Vol. XXXVI., 1881, p. 21.]

Given, by means of their metrical coordinates, any two lines; it is required to find their inclination, shortest distance, and the coordinates of the line of shortest distance.

N.B.—If λ, μ, ν are the inclinations of a line to three rectangular axes, and α, β, γ the coordinates to the same axes of a point on the line, then the metrical coordinates of the line are

$$a, b, c = \cos \lambda, \cos \mu, \cos \nu,$$

$$f, g, h = \beta \cos \nu - \gamma \cos \mu, \gamma \cos \lambda - \alpha \cos \nu, \alpha \cos \mu - \beta \cos \lambda,$$

satisfying identically the relations

$$a^2 + b^2 + c^2 = 1, \quad af + bg + ch = 0.$$

6470. (Proposed by Professor Cayley.)—[Vol. XXXVI., p. 64.]

It is required, by a real linear transformation, to express the equation of a given cubic curve in the form

...

6766. (Proposed by Professor Cayley.)—[Vol. XXXVI., pp. 106, 107.]

Find the stationary and the double tangents of the curve $x^4 + y^4 + z^4 = 0$.

[Solution by the Proposer.] Take l, m and n fourth roots of $+1$; then the 28 double tangents are the lines $x = ly, x = lz, y = lz, (4 + 4 + 4 =) 12$ lines; and the lines $x + my + nz = 0, 16$ lines; and the first 12 of these, each counted twice, are the 24 stationary tangents.

In fact, any one of the 12 lines is an osculating tangent, or line meeting the curve in 4 coincident points; it counts therefore once as a double tangent, and twice as a stationary tangent. There should consequently be 16 other double tangents; and it only needs to be

shown that these are the 16 lines $x + my + nz = 0$. Consider any one line $x + my + nz = 0$; for its intersections with the curve $x^4 + y^4 + z^4 = 0$, we have

$$(my + nz)^4 + y^4 + z^4 = 0,$$

or, as this may be written,

$$(my + nz)^4 + m^4 y^4 + n^4 z^4 = 0,$$

viz. this is

$$2(1, 1, 1, my, nz)^4 = 0,$$

or, what is the same thing,

$$2[(1, 1, 1, my, nz)^4] = 0;$$

so that the line is a double tangent, the two points of contact being given by means of $(1, 1, 1, my, nz)^2 = 0$; viz. ω being an imaginary cube root of unity, we have $nz = \omega my$ or $\omega^2 my$; and thence

$$x : y : z = -m : 1 : -\frac{n}{\omega m} \quad \text{or} \quad -m : 1 : -\frac{n}{\omega^2 m},$$

for the points of contact, for

$$x + my + nz = 0, \quad x^4 + y^4 + z^4 = 0.$$

6800. (Proposed by W. J. C. Miller, B.A.)—
[Vol. XXXVII., 1882, p. 74.]

Prove that, if

$$\frac{ayz}{y^2 + z^2} = \frac{bzx}{z^2 + x^2} = \frac{cxy}{x^2 + y^2},$$

then

$$a^2 + b^2 + c^2 = abc + 4.$$

[By Professor Cayley. Note on Question 6800.] The identity given by the solution is a very interesting one. Writing (a, b, c, d) as the coordinates of a point in space, and (x, y, z) for a given system of values of (x, y, z) ; and considering (a, b, c, d) as coordinates of a point in the plane, it may be shown without difficulty that, to a given point (x, y, z) , there correspond two systems of values of (a, b, c, d) satisfying

$$4d^2 - d(a^2 + b^2 + c^2) + abc = 0;$$

to a given system of values (a, b, c, d) there corresponds, it is clear, a single point (x, y, z) in the plane. We have thus a correspondence between the points of the cubic surface $4d^2 - d(a^2 + b^2 + c^2) + abc = 0$, and the points of the plane; the plane and cubic surface have thus a (1, 2) correspondence with each other.

Problems 5244–5689 [Vol. XXXVIII.–XXXIX., 1883]

5244. (Proposed by Professor Cayley.)—[Vol. XXXVIII., 1883, pp. 87–89.]

Writing for shortness

$$\begin{aligned} A &= a^2 + \beta^2 + \gamma^2 + \delta^2, \\ G &= \beta^2 + \gamma^2 + \delta^2 - a^2, \\ H &= \gamma^2 + \delta^2 + a^2 - \beta^2, \\ F &= \delta^2 + a^2 + \beta^2 - \gamma^2, \\ L &= a^2 + \beta^2 - \gamma^2 - \delta^2, \\ M &= a^2 - \beta^2 + \gamma^2 - \delta^2, \\ N &= a^2 - \beta^2 - \gamma^2 + \delta^2, \end{aligned}$$

show that the equation

$$\begin{aligned} &LMN(x^2 + y^2 + z^2 + w^2)^2 \\ &\quad - MN(F^2 + 2L)(y^2z^2 + x^2w^2) + NL(G^2 + 2M)(z^2x^2 + y^2w^2) \\ &\quad + LM(H^2 + 2N)(x^2y^2 + z^2w^2) - 2a^2\beta^2\gamma^2\delta^2 FGHAxyzw = 0 \end{aligned}$$

belongs to a 16-nodal quartic surface, having the nodes

	x	y	z	w
1	a	β	γ	δ
2	a	β	γ	$-\delta$
3	a	β	$-\gamma$	δ
4	a	β	$-\gamma$	$-\delta$
5	a	$-\beta$	γ	δ
6	a	$-\beta$	γ	$-\delta$
7	a	$-\beta$	$-\gamma$	δ
8	a	$-\beta$	$-\gamma$	$-\delta$
9	$-a$	β	γ	δ
10	$-a$	β	γ	$-\delta$
11	$-a$	β	$-\gamma$	δ
12	$-a$	β	$-\gamma$	$-\delta$
13	$-a$	$-\beta$	γ	δ
14	$-a$	$-\beta$	γ	$-\delta$
15	$-a$	$-\beta$	$-\gamma$	δ
16	$-a$	$-\beta$	$-\gamma$	$-\delta$

and the sixteen singular tangent planes represented by the equations

$$(a, \beta, \gamma, \delta)(x, y, z, w) = 0, \quad \&c.$$

**7190. (Proposed by Professor Wolstenholme, M.A.)—
[Vol. XXXVIII., pp. 87–89.]**

If x, y, z be three quantities satisfying the two symmetrical equations

$$yz + zx + xy = 0, \quad x^2 + y^2 + z^2 + 3xyz = 0,$$

prove that (1) they will satisfy also one of the two pairs of semi-symmetrical expressions

$$yz^2 + z^2x + xy^2 = (y - z)(z - x)(x - y) = +xyz,$$

$$yz^2 + z^2x + xy^2 = (y - z)(z - x)(x - y) = -xyz;$$

and (2) one set of the following equations will also be satisfied:

$$x^2 + yz - z^2 = 0, \quad y^2 + zx - x^2 = 0, \quad z^2 + xy - y^2 = 0;$$

$$x^2 + yz - y^2 = 0, \quad y^2 + zx - z^2 = 0, \quad z^2 + xy - x^2 = 0.$$

[Solution by Professor Cayley.] The two symmetrical equations represent a conic and a cubic respectively; and if we denote by α a root of the equation

$$u^3 + \nu^2 - 2\nu - 1 = 0,$$

then the other two roots of this equation are β, γ ; viz. if $\alpha^3 + \alpha^2 - 2\alpha - 1 = 0$, then we have

$$(u - \alpha)(u + 1 + \frac{1}{\alpha})(u + \frac{1}{\alpha} - \alpha) = u^3 + u^2 - 2u - 1,$$

an identity which is easily verified.

It then may be remarked that, if the last left-hand side of the six mentioned equation which remains unaltered when α is changed into $\frac{1}{\alpha}$ or $-\frac{1}{\alpha} - 1$ is

$$(u - \alpha)(u - \phi\alpha)(u - \psi\alpha),$$

then the coordinates can be expressed indifferently in terms of any one of the roots (α, β, γ) , viz. the coordinates are

$$(\alpha^2 - 1, -\alpha, -1), \quad (-1, \alpha^2 - 1, -\alpha), \quad (-\alpha, -1, \alpha^2 - 1), \quad \dots (1, 2, 3),$$

or they are

$$(\alpha^2-1, -1, -\alpha), \quad (-\alpha, \alpha^2-1, -1), \quad (-1, -\alpha, \alpha^2-1), \quad \dots (4, 5, 6);$$

the coordinates of the points equal to the like expressions in β and γ respectively, as shown by the attached numbers.

Thus, writing $x, y, z = \alpha^2 - 1, -\alpha, -1$, we find

$$\begin{aligned} yz + zx + xy &= -\alpha - \alpha^2 + 1 - \alpha^2 + \alpha = -(\alpha^3 + \alpha^2 - 2\alpha - 1) = 0, \\ x^2 + y^2 + z^2 + 4xyz &= (\alpha^2 - 1)^2 + \alpha^2 + 1 + 4\alpha(\alpha^2 - 1) \\ &= \alpha^6 - 3\alpha^4 + 3\alpha^2 + 3\alpha - 4\alpha - 2 = (\alpha^3 + \alpha^2 - 2\alpha - 1)(\alpha^3 - \alpha^2) \end{aligned}$$

which verifies the formulas for the six points of intersection.

Take, again, $x, y, z = \alpha^2 - 1, -\alpha, -1$; then we find

$$\begin{aligned} y^2z + z^2x + xy^2 &= -\alpha - (\alpha^2 - 1)^2 + \alpha^2(\alpha^2 - 1) = \alpha^4 - \alpha - 1, \\ y^2z + z^2x + x^2y &= +(\alpha^2 - 1) - \alpha(\alpha^2 - 1)^2 = -\alpha^5 + 2\alpha^3 - \alpha - 1. \end{aligned}$$

Or, since $\alpha^3 = -\alpha^2 + 2\alpha + 1$, and thence $\alpha^4 = 3\alpha^2 - \alpha - 1$, $\alpha^5 = -4\alpha^2 + 5\alpha + 3$, the last equation becomes

$$y^2z + z^2x + x^2y = 2\alpha^2 - 2\alpha - 2.$$

We have also

$$xyz = \alpha^3 - \alpha = -\alpha^2 + \alpha + 1;$$

hence the point in question is situate on each of the cubics

$$yz^2 + z^2x + xy^2 + xyz = 0, \quad y^2z + z^2x + x^2y + 2xyz = 0,$$

and

$$y^2z + z^2x + x^2y - 2(yz^2 + z^2x + xy^2) = 0;$$

and this, of course, shows that each of the three cubics; and the points 4, 5, 6 are all of them situate on each of the three cubics $yz^2 + z^2x + xy^2 + 2xyz = 0$, $y^2z + z^2x + x^2y + xyz = 0$, $y^2z + z^2x + x^2y - 2(yz^2 + z^2x + xy^2) = 0$.

Again, from the values $x, y, z = \alpha^2 - 1, -\alpha, -1$, we have

$$x^2 + yz - y^2 = 0, \quad z^2 + xy - x^2 = 0, \quad y^2 + zx - z^2 = 0;$$

on each of these conics; similarly the point 2 lies on each of the same conics; and the point 3 lies on each of the same conics that is, the conics in question have in common the points 1, 2, 3.

In like manner, the conics $x^2 + yz - z^2 = 0$, $y^2 + zx - x^2 = 0$, $z^2 + xy - y^2 = 0$ have in common the points 4, 5, 6.

The general result is that the given conic and the cubic meet in six points forming two groups of points (1, 2, 3) and (4, 5, 6); through the points (1, 2, 3) we have three cubics and three conics; and through the points (4, 5, 6) we have three cubics and three conics.

If in the equation $u^3 + u^2 - 2u - 1 = 0$, whose roots are $(\alpha), \beta, \gamma$, we put $u = 2 \cos \theta$, the equation becomes

$$2(3 \cos \theta + \cos 3\theta) + 2(1 + \cos 2\theta) - 4 \cos \theta - 2 = 0,$$

or

$$2 \cos 3\theta + 2 \cos 2\theta + 2 \cos \theta = 0;$$

or $\frac{\sin \frac{3\theta}{2}}{\sin \frac{\theta}{2}} = 0$; or the three roots are $2 \cos \frac{2}{7}\pi$, $2 \cos \frac{4}{7}\pi$, $2 \cos \frac{6}{7}\pi$.

The two equations $yz + zx + xy = 0$, $x^2 + y^2 + z^2 + 3xyz = 0$, are satisfied if $x : y : z =$ these three roots in any order, giving the six solutions.

The semi-symmetrical systems are satisfied, the one by $x : y : z = \cos \frac{2}{7}\pi : \cos \frac{4}{7}\pi : \cos \frac{6}{7}\pi$, or $y : z : x$, or $z : x : y$; and the other by $x : y : z = \cos \frac{2}{7}\pi : \cos \frac{6}{7}\pi : \cos \frac{4}{7}\pi$, or $y : z : x$, or $z : x : y$, or $x : z : y$, or $y : x : z$, or $z : y : x$.

5689. (Proposed by Professor Cayley.)—[Vol. XXXIX., 1883, p. 31.]

Show that (1) the apparent contour of a Steiner's surface ($2xyz + y^2z^2 + z^2x^2 + x^2y^2 = 0$), as seen from an exterior point on a nodal line (say the axis of z), projected on the plane of the other two nodal lines, is an ellipse passing through the four points $(\pm 1, 0)$ and $(0, \pm 1)$; and (2) find the surface-contour, or curve of contact, of the cone and surface.

4722. (Proposed by Professor Cayley.)—[Vol. XXXIX., p. 49.]

Show that

1. the conditions in regard to the reality of the roots of the equation

$$(x^2 - a)^2 + 16A(x - m) = 0,$$

are, if

$$(4m^2 - 3a)^2 - (8m^2 - 9ma - 27A)^2 = -,$$

then the roots are two real, two imaginary; but if

$$(4m^2 - 3a)^2 - (8m^2 - 9ma - 27A)^2 = +,$$

then, if simultaneously $A = +$ and $A(ma - 9A) = +$, the roots are all real, but otherwise they are imaginary.

2. If the roots of the foregoing equation are all imaginary, then for any value whatever of y , the roots of the equation

$$(x^2 + y^2 - a)^2 + 16A(x - my) = 0$$

are all imaginary.

Using the term "Cassinian" to denote a bi-circular quartic having four foci in a right line; show that the equation of a Cassinian having its four foci at the points $x = a$, $x = b$, $x = c$, $x = d$ on the axis of X , may be written in the four equivalent forms

$$\begin{aligned} \tau(d - c) \cdot A_1 + \rho(b - d) \cdot B_1 + \sigma(c - b) \cdot C_1 + \tau(d - c) \cdot D_1 &= 0, \\ \tau(c - d) \cdot A_1 + \rho(d - a) \cdot C_1 + \sigma(a - c) \cdot D_1 &= 0, \quad \&c., \&c., \end{aligned}$$

where A_1, B_1, C_1, D_1 are the distances from the four foci and ρ, σ, τ are connected by the equation

$$\rho^2(a-d)(b-c) + \sigma^2(b-d)(c-a) + \tau^2(c-d)(a-b) = 0.$$

Show also that the curve has, at right angles to the axis of x , two double tangents, the equation whereof is any one of the three equivalent forms

$$(a+d-2x)(b+c-2x) : (b+d-2x)(c+a-2x) : (c+d-2x)(a+b-2x) = \rho^2 : \sigma^2 : \tau^2.$$

Problems 7376–5421 [Vol. XL.–XLI., 1884]**7376. (Proposed by Professor Cayley.)—[Vol. XL., 1884, p. 32.]**

Show how the construction of a regular heptagon may be made to depend on the trisection of the angle $\cos^{-1} \left(\frac{\sqrt{7}-1}{2\sqrt{2}} \right)$.

7352. (Proposed by Professor Cayley.)—[Vol. XL., p. 110.]

Denoting by $x, y, z, \xi, \eta, \zeta$ homogeneous linear functions of four coordinates, such that identically

$$x + y + z + \xi + \eta + \zeta = 0,$$

where $af = bg = ch = 1$; show that

$$\sqrt{(x\xi)} + \sqrt{(y\eta)} + \sqrt{(z\zeta)} = 0$$

is the equation of a quartic surface having the sixteen singular tangent planes (each touching it along a conic)

$$\begin{aligned} ax + by + cz + f\xi + g\eta + h\zeta &= 0, & x + y + z &= 0, & x + \eta + \zeta &= 0, \\ ax + by + cz &= 0, & \xi + y + \zeta &= 0, & x + y + \xi &= 0, & ax + g\eta + cz &= 0, \\ x = 0, & y = 0, & z = 0, & \xi = 0, & \eta = 0, & \zeta = 0, \\ \frac{1-bc}{1-ca}, & \frac{1-ca}{1-ab}, & \frac{1-ab}{1-fg}, & \frac{1-gh}{1-hf}, & \frac{1-hf}{1-fg}. \end{aligned}$$

5421. (Proposed by Professor Cayley.)—[Vol. XLI., 1884, p. 37.]

Suppose

$$\delta x = m_1(x - a_1), m_2(x - a_2), m_3(x - a_3), m_4(x - a_4),$$

where, for any given value of x , we write $+$, $-$ or 0 , according as the linear function is positive, negative, or zero; and where the order of the terms is not attended to.

If x is any one of the values a_1, a_2, a_3, a_4 , the corresponding δ is $+++-, ++-+, +-++, -+++$; and if I denote indifferently the

first or the second form, and R denote indifferently the third or the fourth form, then it is to be shown that the four δ 's are R, R, R, R or else R, R, I, I .

Problems 8340–5271 [Vol. XLIV.–XLVII., 1886–1887]

8340. (Proposed by F. Morley, B.A.)—[Vol. XLIV., 1886, p. 109.]

Show that (1) on a chess-board the number of squares visible is 204, and the number of rectangles (including squares) visible is 1,296; and (2) on a similar board, with n squares in each side, the number of visible squares is the sum of the first n square numbers, and the number of rectangles (including squares) is the sum of the first n cube numbers.

[Solution by Professor Cayley.] In a board of n^2 squares, the number of squares of breadth $n - r + 1$ is $= r$; from each other of pairs of vertical lines at a distance $n - r + 1$, and the number of pairs of horizontal lines at a distance $n - s + 1$ is $= s$. Hence the number of rectangles of breadth $n - r + 1$ and depth $n - s + 1$, or say the number of $(n - r + 1)(n - s + 1) = rs$.

For instance, $n = 4$, the number of rectangles 4^2 , $4 \cdot 3$, $3 \cdot 4$, &c., is shown in the diagram; hence the whole number of rectangles is $(1 + 2 + 3 + 4)^2 = 1^3 + 2^3 + 3^3 + 4^3$, and so for any value of n .

The same diagram shows that the whole number is $= 1^2 + 2^2 + 3^2 + 4^2$ and so for any value of n .

8636. (Proposed by Mahendra Nath Ray, M.A., LL.B.)—[Vol. XLVII., 1887, pp. 49, 50.]

Show that the following equations are satisfied by the same value of x , and find this value:

$$ax(x^2 - a^2)^{\frac{1}{2}} + bx(x^2 - b^2)^{\frac{1}{2}} + cx(x^2 - c^2)^{\frac{1}{2}} = 2abc,$$

$$2(x^2 - a^2)^{\frac{1}{2}}(x^2 - b^2)^{\frac{1}{2}}(x^2 - c^2)^{\frac{1}{2}} = x(a^2 + b^2 + c^2 - 2x^2).$$

[Solution by Professor Cayley.] The second equation rationalised gives

$$4(x^2 - a^2)(x^2 - b^2)(x^2 - c^2) = x^2(a^2 + b^2 + c^2 - 2x^2)^2,$$

that is,

$$4x^6 - 4x^4(a^2 + b^2 + c^2) + 4x^2(b^2c^2 + c^2a^2 + a^2b^2) - 4a^2b^2c^2 = 4x^6 - 4x^4(a^2 + b^2 + c^2) + x^2(a^2 + b^2 + c^2)^2 - 4x^4(a^2 + b^2 + c^2) + 4x^2(a^2b^2 + a^2c^2 + b^2c^2) - 4a^2b^2c^2$$

if, for shortness,

$$V = -a^4 - b^4 - c^4 + 2b^2c^2 + 2c^2a^2 + 2a^2b^2.$$

We thence find

$$V(x^2 - a^2) = a^2(-a^2 + b^2 + c^2)^2,$$

$$V(x^2 - b^2) = b^2(a^2 - b^2 + c^2)^2,$$

$$V(x^2 - c^2) = c^2(a^2 + b^2 - c^2)^2;$$

and therefore

$$V\sqrt{a^2b^2c^2(x^2 - a^2)} = 2a^2b^2c(-a^2 + b^2 + c^2), \quad \&c.$$

Or, assuming the sign of the square roots,

$$V\sqrt{ax(x^2 - a^2)} = 2abc(-a^2 + b^2 + c^2)^{\frac{1}{2}},$$

$$V\sqrt{bx(x^2 - b^2)} = 2abc(+a^2b^2 - b^4 + b^2c^2)^{\frac{1}{2}},$$

$$V\sqrt{cx(x^2 - c^2)} = 2abc(c^2a^2 + b^2c^2 - c^4)^{\frac{1}{2}};$$

whence, adding, the whole divides by V and we have

$$ax(x^2 - a^2)^{\frac{1}{2}} + bx(x^2 - b^2)^{\frac{1}{2}} + cx(x^2 - c^2)^{\frac{1}{2}} = 2abc.$$

Observe that the second equation rationalised gives an equation of the form $(x^2, 1)^4 = 0$, that is, the foregoing value of x^2 is one of the four values of x^2 .

5271. (Proposed by Professor Cayley.)—[Vol. XLVII., 1887, p. 141.]

If ω be an imaginary cube root of unity, show that, if

$$\frac{(1 - \omega^2)x + \omega^2(1 - \omega x^2)}{(1 - \omega^2) - \omega(1 - \omega x^2)} \cdot \frac{dy}{dx} = \frac{(1 - \omega)(1 + \omega x^2)}{(1 - y^2)(1 + \omega y^2)},$$

then

$$\frac{\frac{dy}{dx}}{\sqrt{(1 - y^2)(1 + \omega y^2)}} = \frac{\frac{dx}{dy}}{\sqrt{(1 - x^2)(1 + \omega x^2)}};$$

and explain the general theory.

Problem 3105 [Vol. L., 1889, p. 189]

3105. (Proposed by Professor Cayley.)—[Vol. L., 1889, p. 189.]

The following singular problem of partitions arises out of the geometrical theory given in Cremona’s Memoir, “Sulle trasformazioni geometriche delle figure piane,” Mem. di Bologna, tom. v. (1865).

A number is to be partitioned on the same type $1 \cdot 11 + 8 \cdot 2$ as follows: for instance, $27 = 1 \cdot 11 + 8 \cdot 2$.

(Observe that the partitionment is to be made up in any manner with the literal factors $a, b, c, d, e, f, g, h, i$; always the positive integers as the product of 27.) Forming, then, symmetrical products, we have $a^9(bcdefghi)$.

But, to take another example, $37 = 0 \cdot 11 + 3 \cdot 8 + 1 \cdot 5 + 4 \cdot 2 = 1 \cdot 11 + 0 \cdot 8 + 4 \cdot 5 + 3 \cdot 2$.

The first of these (symmetrically as above) gives on its own type, and the second gives on its own type; though it may be on the second type; viz. the partitions of the two products respectively are:

First product on second type: $abcdefgh, ab, ac, ad, ae, af, ag, ah, ai, bc, bd, be, bf, bg, bh, bi$.

Second product on first type: $a^2bcdegh, a^2bcdefh, a^2bcdefg, abcde, abcdf, abcdg, abcdh, abcde, abcdf, abcdg, abcdh$.

so that in the first example the type is sibi-reciprocal, but in the second example there are two conjugate types. Other examples are:

Parts	55	53	48	64	55
	2	14	3	1	3
	5	2	3	6	8
	3	2	6	5	7
	1	5	2	3	2
	56	2	6	5	7
	1	5	2	3	2
No.	1	3	4	1	1
	13	1	17	1	20

viz. the first four columns give each of them a sibi-reciprocal type, but the last two double columns give conjugate types. It is required to investigate the general solution.

Problems 3304–3481 [Vol. L., 1889, pp. 191, 192]

3304. (Proposed by Professor Cayley.)—[Vol. L., p. 191.]

The coordinates x, y, z being proportional to the perpendicular distances from the sides of an equilateral triangle, it is required to trace the curve

$$(y - z)\sqrt{x} + (z - x)\sqrt{y} + (x - y)\sqrt{z} = 0.$$

[Prof. Cayley remarks that the curve which presents itself in the following theorem, communicated to him (with a demonstration) several years ago by Mr J. Griffiths, is a particular case of a more general curve: The locus of a point (x, y, z) such that its pedal circle (that is, the circle which passes through the feet of the perpendiculars drawn from the point in question upon the sides of the triangle of reference) touches the nine-point circle, is the sextic curve

$$y^2 z^2 \cos^2 A (x \cos A)^2 + z^2 x^2 \cos^2 B (y \cos B)^2 + x^2 y^2 \cos^2 C (z \cos C)^2 \\ - 2yz \cos A (x \cos A)(y \cos B)(z \cos C) - \dots = 0.$$

To trace this more general curve would be an interesting problem.]

3481. (Proposed by Professor Cayley.)—[Vol. L., p. 192.]

Find, in the Hamiltonian form, the equations for the motion of a particle acted on by a central force in a plane.

Problems 10716–3186 [Vol. LXI., 1894, pp. 122–124]

10716. (Proposed by Professor Cayley.)—[Vol. LXI., 1894, pp. 122, 123.]

In a hexahedron $ABCD A'B'C'D'$ the edges AA' , BB' , CC' , DD' intersect in four points, say AA' , DD' in α ; BB' , CC' in β ; CC' , DD' in γ ; AA' , BB' in δ : that is, starting with the duad of lines $\alpha\beta$, $\gamma\delta$, which join the extremities of the edges, the four edges are the lines joining the extremities of a duad. The question arises, "Given two tetrads $ABCD$, $A'B'C'D'$, when are the lines joining the eight of the twelve edges of a hexahedron?" Given the tetrahedron $\alpha\beta\gamma\delta$ and the two tetrads of joining lines, so that the duads $\alpha\beta$, $\gamma\delta$; $\alpha\gamma$, $\beta\delta$; $\alpha\delta$, $\beta\gamma$ which join the extremities of the edges are the four duads; and the lines AB , CD , $A'B'$, $C'D'$ are the edges AD , BC , $A'D'$, $B'C'$.

The duad $\alpha\beta$, $\gamma\delta$ is considered which determines the relative position of the two finite lines $\alpha\beta$ and $\gamma\delta$.

3162. (Proposed by Professor Cayley.)—[Vol. LXI., p. 123.]

By a proper determination of the coordinates, the skew cubic through any six given points may be taken to be

$$x : y : z : w = 1 : t : t^2 : t^3,$$

or, what is the same thing, the coordinates of the six points may be taken to be $(1, t_i, t_i^2, t_i^3)$, $i = 1, 2, \dots, 6$. Assuming this, it is required to show that if

$$V = 9xyzw - 4xz^3 - 4yw^3 + 3z^2w^2 - x^2w^2,$$

then the equation of the Jacobian surface of the six points is

$$\begin{aligned} & (2xp_2 - yp_1 - 2w)\partial_x V \\ & + (3xp_3 - 2yp_2 - zp_1 + w)\partial_y V \\ & + (2xp_4 - yp_3 - 2zp_2 + wp_1)\partial_z V \\ & + (xp_4 - 2yp_3 + zp_2 + 2w)\partial_w V = 0, \end{aligned}$$

where $p_1 = \sum t_i$, $p_2 = \sum t_i t_j$, &c.

3186. (Proposed by Professor Cayley.)—[Vol. LXI., p. 124.]

An unclosed polygon of $(m+1)$ vertices is constructed as follows: the abscissae of the several vertices are $0, 1, 2, \dots, m$; and to the abscissa k the ordinate is the chance of $m+k$ heads in $2m$ tosses of a coin; then the areas of any very large and small portions of the polygon, what information is afforded in regard to the general results of the Theory of Probabilities?

3229. (Proposed by Professor Cayley.)—[Vol. LXI., p. 124.]

It is required to find the value of the elliptic integral $F(c, \theta)$ when c is very nearly $= 1$ and θ very nearly $= \frac{1}{2}\pi$; that is, the value of

$$\int_0^{\frac{\pi}{2}-b} \frac{d\theta}{\sqrt{\sin^2(a) + \cos^2(a) \sin^2(\theta)}}$$

where a, b are each of them indefinitely small.

N.B.—Observe that, for $a = 0$, b small, the value is $= \log \cot \frac{1}{2}b$; and for $b = 0$, a small, the value is equal $\log \frac{4}{a}$, and for $a = 0$, $b = 0$, the value is $= \infty$.

Index of Problems [pp. 615–616]

IN THE FOLLOWING CONTENTS, THE PROBLEMS ARE REFERRED
TO EACH BY ITS NUMBER AND THE PAGE; AN ASTERISK SHEWS
THAT THERE IS NO NUMBER; A DASH SHEWS THAT NO SOLUTION
IS GIVEN.

	PAGE	
3002	Collins	566 Systems of circles
3144	Cayley	568 Locus in piano
3120		569 Jacobian of quadric surfaces
3249		569 Sets of four points on a conic
3206		570 Arrangements of points and lines
3278		570 Arrangements of numbers
3329		Substitutions and permutations
3356		574 Roots of quartic equation
3507		575 Quadric cones through given points
3536		Particle under central forces
3591		Relation between areas of triangles
2652		Parallel surfaces of an ellipsoid
3677		576 Angle between normal and bisector of chord
3564		Minimum circle enclosing three points
3875		Description of curves
3430		578 Negative pedals of ellipse
4298	Glaisher	Quadrilateral inscribable in circle
4392	Roberts	Symmetrical determinant
4354	Tucker	581 Geometrical interpretation (note)
4458	Cayley	582 Two quartic curves
4520	Miller	
4620	Evans	586 Use of Degen's tables
4528	Cayley	587 Expectation
4581	Wilkinson	588 A question in chances
4638	Cayley	589 Envelope of family of quadrics
4694		Resultant of forces
4793	Wolstenholme	590 Relation among derivatives of a function
4752	Cayley	592 Cubic curve
4946		594 Attraction of lens-shaped body
5020	Woolhouse	Algebraical theorem
5079	Cayley	596 Bicircular quartic
5130		597 Bicircular quartic
5208	Sylvester	598 Trees
5306	Cayley	600 A question in probabilities
4870		601 A system of equations
5625		602 Conics and lines
5387		Conical points
5927		603 Algebraical theorem
6155		Coordinates of a line
6470		Equation of a cubic curve
6766	Miller	Singular tangents of a quartic
6800	Cayley	604 Geometrical interpretation (note)
5244		16 nodal quartic surface
7190	Wolstenholme	605 A Steiner's surface

END OF VOL. X

CAMBRIDGE: PRINTED BY J. AND C. F. CLAY, AT THE
UNIVERSITY PRESS.